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Lawford BJ, Hall M, Hinman RS, Van der Esch M, Harmer AR, Spiers L, Kimp A, Dell'Isola A, Bennell KL

Lawford BJ, Hall M, Hinman RS, Van der Esch M, Harmer AR, Spiers L, Kimp A, Dell'Isola A, Bennell KL.
Exercise for osteoarthritis of the knee.
Cochrane Database of Systematic Reviews 2024, Issue 12. Art. No.: CD004376.
DOI: 10.1002/14651858.CD004376.pub4.

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[Intervention Review]

Exercise for osteoarthritis of the knee

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Editorial group: Cochrane Central Editorial Service.
Publication status and date: New search for studies and content updated (conclusions changed), published in Issue 12, 2024.

Citation: Lawford BJ, Hall M, Hinman RS, Van der Esch M, Harmer AR, Spiers L, Kimp A, Dell'Isola A, Bennell KL. Exercise for osteoarthritis of the knee. *Cochrane Database of Systematic Reviews* 2024, Issue 12. Art. No.: CD004376. DOI: 10.1002/14651858.CD004376.pub4.

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ABSTRACT

Background

Knee osteoarthritis (OA) is a major public health issue causing chronic pain, impaired physical function, and reduced quality of life. As there is no cure, self-management of symptoms via exercise is recommended by all current international clinical guidelines. This review updates one published in 2015.

Objectives

We aimed to assess the effects of land-based exercise for people with knee osteoarthritis (OA) by comparing:

1) exercise versus attention control or placebo;

2) exercise versus no treatment, usual care, or limited education;

3) exercise added to another co-intervention versus the co-intervention alone.

Search methods

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase, and two trial registries (ClinicalTrials.gov and World Health Organisation International Clinical Trials Registry Platform), together with reference lists, from the date of the last search (1st May 2013) until 4 January 2024, unrestricted by language.

Selection criteria

We included randomised controlled trials (RCTs) that evaluated exercise for knee OA versus a comparator listed above. Our outcomes of interest were pain severity, physical function, quality of life, participant-reported treatment success, adverse events, and study withdrawals.

Data collection and analysis

We used the standard methodological procedures expected by Cochrane for systematic reviews of interventions.

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Main results

We included 139 trials (12,468 participants): 30 (3065 participants) compared exercise to attention control or placebo; 60 (4834 participants) compared exercise with usual care, no intervention or limited education; and 49 (4569 participants) evaluated exercise added to another intervention (e.g. weight loss diet, physical therapy, detailed education) versus that intervention alone. Interventions varied substantially in duration, ranging from 2 to 104 weeks. Most of the trials were at unclear or high risk of bias, in particular, performance bias (94% of trials), detection bias (94%), selective reporting bias (68%), selection bias (57%), and attrition bias (48%).

Exercise versus attention control/placebo

Compared with attention control/placebo, low-certainty evidence indicates exercise may result in a slight improvement in pain immediately post-intervention (mean 8.70 points better (on a scale of 0 to 100), 95% confidence interval (CI) 5.70 to 11.70; 28 studies, 2873 participants). Moderate-certainty evidence indicates exercise likely results in an improvement in physical function (mean 11.27 points better (on a scale of 0 to 100), 95% CI 7.64 to 15.09; 24 studies, 2536 participants), but little to no improvement in quality of life (mean 6.06 points better (on a scale of 0 to 100), 95% CI −0.13 to 12.26; 6 studies, 454 participants). There was moderate-certainty evidence that exercise likely increases participant-reported treatment success (risk ratio (RR) 1.46, 95% CI 1.11 to 1.92; 2 studies 364 participants), and likely does not increase study withdrawals (RR 1.08, 95% CI 0.92 to 1.26; 29 studies, 2907 participants). There was low-certainty evidence that exercise may not increase adverse events (RR 2.02, 95% CI 0.62 to 6.58; 11 studies, 1684 participants).

Exercise versus no treatment/usual care/limited education

Compared with no treatment/usual care/limited education, low-certainty evidence indicates exercise may result in an improvement in pain immediately post-intervention (mean 13.14 points better (on a scale of 0 to 100), 95% CI 10.36 to 15.91; 56 studies, 4184 participants). Moderate-certainty evidence indicates exercise likely results in an improvement in physical function (mean 12.53 points better (on a scale of 0 to 100), 95% CI 9.74 to 15.31; 54 studies, 4352 participants) and a slight improvement in quality of life (mean 5.37 points better (on a scale of to 100), 95% CI 3.19 to 7.54; 28 studies, 2328 participants). There was low-certainty evidence that exercise may result in no difference in participant-reported treatment success (RR 1.33, 95% CI 0.71 to 2.49; 3 studies, 405 participants). There was moderate-certainty evidence that exercise likely results in no difference in study withdrawals (RR 1.03, 95% CI 0.88 to 1.20; 53 studies, 4408 participants). There was low-certainty evidence that exercise may increase adverse events (RR 3.17, 95% CI 1.17 to 8.57; 18 studies, 1557 participants).

Exercise added to another co-intervention versus the co-intervention alone

Moderate-certainty evidence indicates that exercise when added to a co-intervention likely results in improvements in pain immediately post-intervention compared to the co-intervention alone (mean 10.43 points better (on a scale of 0 to 100), 95% CI 8.06 to 12.79; 47 studies, 4441 participants). It also likely results in a slight improvement in physical function (mean 9.66 points better, 95% CI 7.48 to 11.97 (on a 0 to 100 scale); 44 studies, 4381 participants) and quality of life (mean 4.22 points better (on a 0 to 100 scale), 95% CI 1.36 to 7.07; 12 studies, 1660 participants) immediately post-intervention. There was moderate-certainty evidence that exercise likely increases participant-reported treatment success (RR 1.63, 95% CI 1.18 to 2.24; 6 studies, 1139 participants), slightly reduces study withdrawals (RR 0.82, 95% CI 0.70 to 0.97; 41 studies, 3502 participants), and slightly increases adverse events (RR 1.72, 95% CI 1.07 to 2.76; 19 studies, 2187 participants).

Subgroup analysis and meta-regression

We did not find any differences in effects between different types of exercise, and we found no relationship between changes in pain or physical function and the total number of exercise sessions prescribed or the ratio (between exercise group and comparator) of real-time consultations with a healthcare provider.

Clinical significance of the findings

To determine whether the results found would make a clinically meaningful difference to someone with knee OA, we compared our results to established 'minimal important difference' (MID) scores for pain (12 points on a 0 to 100 scale), physical function (13 points), and quality of life (15 points). We found that the confidence intervals of mean differences either did not reach these thresholds or included both a clinically important and clinically unimportant improvement.

Authors' conclusions

We found low- to moderate-certainty evidence that exercise probably results in an improvement in pain, physical function, and quality of life in the short-term. However, based on the thresholds for minimal important differences that we used, these benefits were of uncertain clinical importance. Participants in most trials were not blinded and were therefore aware of their treatment, and this may have contributed to reported improvements.

PLAIN LANGUAGE SUMMARY

Is exercise an effective therapy to treat knee osteoarthritis?

Key messages

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- Exercise may lead to improvement in pain and physical function when compared to 'attention' control (i.e. involving more than one interaction with study personnel or a care provider) or placebo, and improvement in pain, physical function, and quality of life when compared to no treatment, usual care (e.g. routine treatment offered by family doctor) or limited education (e.g. provision of a basic information pamphlet or link to information online). Exercise may also lead to improvements in pain, physical function, and quality of life when added to another treatment when compared to that treatment alone. However, the benefits were of uncertain clinical importance, meaning that they may not result in a change in symptoms that makes a noticeable difference to patients.

- Despite the large number of studies, many involved a small number of participants and were of poor quality (e.g. participants were aware of which treatment group they were in), which decreased our confidence in the conclusions.

How might exercise help people with knee osteoarthritis?

Knee osteoarthritis is a common cause of ongoing pain and disability. Exercise aims to increase muscle strength and improve joint function and range of motion, which may reduce osteoarthritis pain and joint dysfunction and help people return to their usual daily activities. Exercise programmes include a wide range of different exercise types, durations, and delivery methods.

What did we want to find out?

This review is an update of one last published in 2015. We wanted to know whether exercise improves pain, physical function, and quality of life for people with knee osteoarthritis.

What did we do?

We searched for studies of adults with knee osteoarthritis that assessed the effects of land-based exercise on pain, physical function, and quality of life. The studies had to compare a group of people doing exercise to a group who received attention or a placebo (fake) intervention, or to a group receiving no treatment, usual care, or limited education. We also included studies comparing exercise plus another intervention versus that intervention on its own.

We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and consistency of results.

What did we find?

We found 139 studies with a total of 12,468 people. Study participants were, on average, aged between 41 and 81 years. Exercise programs lasted between 2 and 104 weeks. Outcomes were measured right after the end of the exercise program, with some studies also collecting outcomes in the weeks and months after the end of the program.

Thirty trials (22%) compared exercise to attention control or placebo. Sixty trials (43%) compared exercise to no treatment, usual care, or limited education. Forty-nine trials (35%) added exercise to another type of treatment and compared that to the other treatment on its own. To determine whether benefits would make a meaningful difference to someone with knee osteoarthritis, we compared our findings to established 'minimal important difference' scores for pain (12 points on a 0 to 100 point scale), physical function (13 points), and quality of life (15 points).

<u>Exercise compared with attention control or placebo</u>

- Exercise may result in a slight improvement in pain (8.7 points better on a 0 to 100 point scale).

- Exercise likely results in an improvement in physical function (11.3 points better on a 0 to 100 point scale).

- Exercise likely results in little to no improvement in quality of life (6.1 points better on a 0 to 100 point scale).

In terms of adverse events at final follow-up, we found that exercise may not increase adverse events.

<u>Exercise compared with no treatment, usual care, or limited education</u>

- Exercise may result in an improvement in pain (13.1 points better on a 0 to 100 point scale).

- Exercise likely results in an improvement in physical function (12.5 points better on a 0 to 100 point scale).

- Exercise likely results in a slight improvement in quality of life (5.4 points better on a 0 to 100 point scale).

In terms of adverse events at final follow-up, we found that exercise may increase adverse events.

<u>Exercise added to another treatment compared to the other treatment alone</u>

- Exercise likely results in an improvement in pain (10.4 points better on a 0 to 100 point scale).

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- Exercise likely results in a slight improvement in physical function (9.7 points better on a 0 to 100 point scale).

- Exercise likely results in a slight improvement in quality of life (4.2 points better on a 0 to 100 point scale).

In terms of adverse events at final follow-up, we found that exercise likely increases adverse events slightly.

What are the limitations of the evidence?

We have low to moderate confidence in our results. Many studies are of poor quality with small sample sizes. Therefore, some studies may have made the benefits of exercise seem greater than they are.

How up to date is this evidence?

We searched for studies up to 4 January 2024.

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SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Exercise compared to attention control/placebo for adults with knee osteoarthritis

Exercise compared to attention control/placebo for adults with knee osteoarthritis

Patient or population: adults with knee osteoarthritis
Setting: any non-perioperative setting in any country
Intervention: exercise
Comparison: attention control/placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with attention control/placebo	Risk with exercise				
Pain (immediately post-intervention) Scale from: 0 to 100 (0 represents no pain) follow-up: range 4 weeks to 104 weeks	The mean pain was **44.40** points	mean **8.7 points lower** (11.7 lower to 5.7 lower)	-	2873 (28 RCTs)	⊕⊕⊖⊖ Low[a]	Exercise may result in a slight improvement in pain. However, based on the minimal important differences that we used, the confidence interval of the mean difference does not meet the threshold for a clinically important improvement. [b,c]
Physical function (immediately post-intervention) Scale from: 0 to 100 (0 represents no physical disability) follow-up: range 4 weeks to 104 weeks	The mean physical function was **35.80** points	mean **11.27 points lower** (15.09 lower to 7.64 lower)	-	2536 (24 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results in an improvement in physical function. However, the confidence interval of the mean difference includes both a clinically important and clinically unimportant improvement. [b,d]
Quality of life (immediately post-intervention) Scale from: 0 to 100 (100 is maximum quality of life) follow-up: range 4 weeks to 104 weeks	The mean quality of life was **48.40** points	mean **6.06 points higher** (0.13 lower to 12.26 higher)	-	454 (6 RCTs)	⊕⊕⊕⊖ Moderate[e]	Exercise likely results in little to no improvement in quality of life. Based on the minimal important differences that we used, the confidence interval of the mean difference does not meet the threshold for a clinically important difference. [b,f]
Participant-reported treatment success (immediately post-intervention)	306 per 1000	**446 per 1000** (339 to 587)	**RR 1.46** (1.11 to 1.92)	364 (2 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely increases participant-reported treatment success.

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follow-up: range 4 weeks to 104 weeks						
Study withdrawals (at final follow-up) follow-up: range 4 weeks to 104 weeks	171 per 1000	**185 per 1000** (158 to 216)	**RR 1.08** (0.92 to 1.26)	2907 (29 RCTs)	⊕⊕⊕⊖ Moderate[a]	Exercise likely does not increase study withdrawals.
Adverse events (at final follow-up) follow-up: range 4 weeks to 104 weeks	69 per 1000	**140 per 1000** (43 to 457)	**RR 2.02** (0.62 to 6.58)	1684 (11 RCTs)	⊕⊕⊖⊖ Low[a,g]	Exercise may not increase adverse events.

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_445511163621360046.

[a] Downgraded as > 50% of trials were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic as trials with unblinded assessment of subjective outcomes are estimated to exaggerate the treatment benefit. Correcting for these biases would likely drive the improvements seen with exercise towards null.
[b] Standardised mean difference (SMD) back-translated to 0 to 100 scale by multiplying by the weighted standard deviation (SD) of the comparison groups at baseline (using data from all studies that used a 0 to 100 scale). Mean (SD) baseline knee pain: 44.4 (15.0). Mean (SD) baseline physical function: 35.8 (19.1). Mean (SD) baseline quality of life: 48.4 (12.9).
[c] The 95% confidence intervals do not include a clinically important change (pre-defined as 12 points on a 0 to 100 point pain scale).
[d] The 95% confidence intervals include a clinically important change (pre-defined as 13 points on a 0 to 100 point pain scale).
[e] Publication bias was not assessed due to the small number of studies (< 10 studies).
[f] The 95% confidence intervals do not include a clinically important change (pre-defined as 15 points on a 0 to 100 point pain scale).
[g] Downgraded for imprecision as the ratio of the upper and lower 95% confidence intervals of the risk ratio is > 3.0.

Summary of findings 2. Summary of findings table - Exercise compared to no treatment/usual care/limited education for adults with knee osteoarthritis

Exercise compared to no treatment/usual care/limited education for adults with knee osteoarthritis

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Patient or population: adults with knee osteoarthritis
Setting: any non-perioperative setting in any country
Intervention: exercise
Comparison: no treatment/usual care/limited education

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no treatment/usual care/limited education	Risk with exercise				
Pain (immediately post-intervention) Scale from: 0 to 100 (0 represents no pain) follow-up: range 2 weeks to 52 weeks	The mean pain was **51.10** points	mean **13.14 points lower** (15.91 lower to 10.36 lower)	-	4184 (56 RCTs)	⊕⊕⊖⊖ Low	Exercise may result in an improvement in pain. However, based on the minimal important differences that we used, the confidence interval of the mean difference includes both a clinically important and clinically unimportant improvement. [a,b]
Physical function (immediately post-intervention) Scale from: 0 to 100 (0 represents no physical disability) follow-up: range 2 weeks to 52 weeks	The mean physical function was **42.50** points	mean **12.53 points lower** (15.31 lower to 9.74 lower)	-	4352 (54 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results in an improvement in physical function. However, based on the minimal important differences that we used, the confidence interval of the mean difference includes both a clinically important and clinically unimportant improvement. [b,c]
Quality of life (immediately post-intervention) Scale from: 0 to 100 (100 is maximum quality of life) follow-up: range 2 weeks to 52 weeks	The mean quality of life was **54.30** points	mean **5.37 points higher** (3.19 higher to 7.54 higher)	-	2328 (28 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results in a slight improvement in quality of life. However, based on the minimal important differences that we used, the confidence interval of the mean difference does not meet the threshold for a clinically important difference. [b,d]
Participant-reported treatment success (immediately post-intervention) follow-up: range 2 weeks to 52 weeks	325 per 1000	**432 per 1000** (231 to 810)	**RR 1.33** (0.71 to 2.49)	405 (3 RCTs)	⊕⊕⊖⊖ Low	Exercise may result in no difference in participant-reported treatment success.
Study withdrawals (at final follow-up)	119 per 1000	**123 per 1000** (105 to 143)	**RR 1.03** (0.88 to 1.20)	4408 (53 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results in no difference in study withdrawals.

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follow-up: range 2 weeks to 104 weeks						
Adverse events (at final follow-up) follow-up: range 2 weeks to 104 weeks	77 per 1000	**244 per 1000** (90 to 659)	**RR 3.17** (1.17 to 8.57)	1557 (18 RCTs)	⊕⊕⊖⊖ Low[e,f]	Exercise may increase adverse events.

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_446127721125345136.

[a] The 95% confidence intervals include a clinically important change (pre-defined as 12 points on a 0 to 100 point pain scale).
[b] Standardised mean difference (SMD) back-translated to 0 to 100 scale by multiplying by the weighted standard deviation (SD) of the comparison groups at baseline (using data from all studies that used a 0 to 100 scale). Mean (SD) baseline knee pain: 51.1 (18.5). Mean (SD) baseline physical function: 42.2 (17.4). Mean (SD) baseline quality of life: 54.3 (14.5).
[c] The 95% confidence intervals include a clinically important change (pre-defined as 13 points on a 0 to 100 point pain scale).
[d] The 95% confidence intervals do not include a clinically important change (pre-defined as 15 points on a 0 to 100 point pain scale).
[e] Downgraded as > 50% of trials were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic as trials with unblinded assessment of subjective outcomes are estimated to exaggerate the treatment benefit. Correcting for these biases would likely drive the improvements seen with exercise towards null.
[f] Downgraded for imprecision as the ratio of the upper and lower 95% confidence intervals of the risk ratio is > 3.0.

Summary of findings 3. Summary of findings table - Exercise when added to another co-intervention compared to the co-intervention alone for adults with knee osteoarthritis

Exercise when added to another co-intervention compared to the co-intervention alone for adults with knee osteoarthritis

Patient or population: adults with knee osteoarthritis
Setting: any non-perioperative setting in any country
Intervention: exercise when added to another co-intervention
Comparison: the co-intervention alone

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Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with the co-intervention alone	Risk with exercise when added to another co-intervention				
Pain (immediately post-intervention) Scale from: 0 to 100 (0 represents no pain) follow-up: range 2 weeks to 104 weeks	The mean pain was **57.90** points	mean **10.43 points lower** (12.79 lower to 8.06 lower)	-	4441 (47 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results in an improvement in pain. However, based on the minimal important differences that we used, the confidence interval of the mean difference includes both a clinically important and clinically unimportant improvement. [a,b]
Physical function (immediately post-intervention) Scale from: 0 to 100 (0 represents no physical disability) follow-up: range 2 weeks to 104 weeks	The mean physical function was **50.60** points	mean **9.66 points lower** (11.97 lower to 7.48 lower)	-	4381 (44 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results in a slight improvement in physical function. However, based on the minimal important differences that we used, the confidence interval of the mean difference does not meet the threshold for a clinically important difference. [a,c]
Quality of life (immediately post-intervention) Scale from: 0 to 100 (100 is maximum quality of life) follow-up: range 2 weeks to 104 weeks	The mean quality of life was **64.20** points	mean **4.22 points higher** (1.36 higher to 7.07 higher)	-	1660 (12 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely results a slight improvement in quality of life. However, based on the minimal important differences that we used, the confidence interval of the mean difference does not meet the threshold for a clinically important difference. [a,d]
Participant-reported treatment success (immediately post-intervention) follow-up: range 2 weeks to 104 weeks	400 per 1000	**653 per 1000** (472 to 897)	**RR 1.63** (1.18 to 2.24)	1139 (6 RCTs)	⊕⊕⊕⊖ Moderate[e]	Exercise likely increases participant-reported treatment success.
Study withdrawals (at final follow-up) follow-up: range 2 weeks to 104 weeks	139 per 1000	**114 per 1000** (98 to 135)	**RR 0.82** (0.70 to 0.97)	3502 (41 RCTs)	⊕⊕⊕⊖ Moderate	Exercise likely reduces study withdrawals slightly.
Adverse events (at final follow-up)	42 per 1000	**73 per 1000** (45 to 117)	**RR 1.72** (1.07 to 2.76)	2187 (19 RCTs)	⊕⊕⊕⊖ Moderate[f]	Exercise likely increases adverse events slightly.

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follow-up: range 2 weeks to 104 weeks

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the __relative effect__ of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_446145842223879248.

[a] Standardised mean difference (SMD) back-translated to 0 to 100 scale by multiplying by the weighted standard deviation (SD) of the comparison groups at baseline (using data from all studies that used a 0 to 100 scale). Mean (SD) baseline knee pain: 57.9 (13.9). Mean (SD) baseline physical function: 50.6 (13.6). Mean (SD) baseline quality of life: 64.2 (13.6).
[b] The 95% confidence intervals include a clinically important change (pre-defined as 12 points on a 0 to 100 point pain scale).
[c] The 95% confidence intervals include a clinically important change (pre-defined as 13 points on a 0 to 100 point pain scale).
[d] The 95% confidence intervals do not include a clinically important change (pre-defined as 15 points on a 0 to 100 point pain scale).
[e] Publication bias was not assessed due to the small number of studies (< 10 studies).
[f] Downgraded as > 50% of trials were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic as trials with unblinded assessment of subjective outcomes are estimated to exaggerate the treatment benefit. Correcting for these biases would likely drive the improvements seen with exercise towards null.

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BACKGROUND

Description of the condition

Osteoarthritis is a condition that may affect all tissues within a synovial joint, with loss of articular cartilage, alteration to subchondral bone, degradation of the menisci, synovial inflammatory response, and bone and cartilage osteophytes, resulting in structural and functional failure of the joint (Hunter 2019). People with knee OA typically have joint pain, experience difficulties performing activities of daily living, and have reduced quality of life (Hunter 2019). In addition, OA is commonly associated with a number of comorbidities, such as obesity, cardiovascular disease, diabetes mellitus, and mental health impairments (Veronese 2016).

Knee osteoarthritis (OA) is common, affecting more than 654 million people worldwide (approximately 23% of people aged 40 years and over) (Cui 2020). Osteoarthritis is the 15th highest cause of years lived with disability (Steinmetz 2023) and caused a 63% increase in years lived with disability from 1990 to 2016 (GBD 2017).

Diagnosis and classification of OA may be based on radiographic findings, symptoms or clinical presentation, or self-report. Radiographic classifications typically rely on features present on images of the joint (Dahaghin 2005; Kellgren 1957). However, there is considerable discordance between radiographic findings and symptoms (Finan 2013; Zhang 2010a). As such, contemporary clinical guidelines advocate diagnosis based on clinical presentation, including being 45 years or over, reporting joint pain during activity, and reporting no joint stiffness during the morning or stiffness that lasts for less than 30 minutes (NICE 2022).

Description of the intervention

Exercise is a subcategory of physical activity that is planned, structured, and repetitive (Caspersen 1985). Exercise covers a range of physical activities that aim to improve muscle strength and function, balance, joint range of motion, and/or aerobic fitness. Exercise interventions are typically heterogeneous in design, dose (e.g. duration, frequency, intensity), delivery format (e.g. clinician supervised or not, individual or group, in-person or remote), specific exercise types (e.g. muscle strengthening, aerobic), and may be combined with other conservative treatments (e.g. education, physical therapy).

How the intervention might work

All current clinical guidelines advocate regular ongoing exercise as a cornerstone in the management of knee OA, irrespective of age, comorbidity, pain severity, or disability (Bannuru 2019; Kolasinski 2020; Moseng 2024; NICE 2022; RACGP 2018). Reviews and mediation analyses have identified potential physical and psychosocial mechanisms by which exercise affects clinical outcomes, including pain and physical function (Beckwée 2013; Lima 2022; Runhaar 2015).

Physical mechanisms of action

Exercise has been shown to potentially affect symptom improvement via various physical mechanisms (Lima 2022; Runhaar 2015; Runhaar 2023). Muscle weakness is common in people with OA, and often evident in muscles surrounding, and distal to, the local joint (Hinman 2010; Muraki 2015; Vårbakken 2019). Increased muscle strength may be one of the mechanisms by which exercise training can lead to improvements in pain and physical function (Bandak 2019; Hall 2018; Runhaar 2015; Ribeiro 2024). However, more recent evidence suggests that improvement in muscle strength mediates only a small proportion of the effect on pain and physical function (Runhaar 2023). Aerobic exercise is also important as it can reduce various inflammatory cytokines (Bricca 2019; Gleeson 2011; Zhang 2013), which can contribute to reduced pain (Paley 2016). Exercise can help maintain a healthy weight and reduce visceral adipose tissue, which is another potential mechanism for improved symptoms (Fantuzzi 2005; Klyne 2020). Exercise is also hypothesised to regulate various metabolic pathways in healthy cartilage (Blazek 2016) and has been shown to contribute to changes in systemic inflammation in those with overweight or obesity (Runhaar 2019), potentially contributing to improvement in symptoms.

Psychological mechanisms of action

Exercise has also been shown to affect symptoms via various psychosocial mechanisms (Lima 2022; Ribeiro 2024; Runhaar 2015). Previous research has demonstrated that increased self-efficacy (belief in one's ability to complete a task or achieve a goal) regarding climbing stairs mediated stair climbing performance following exercise (Rejeski 1998) and that increased arthritis self-efficacy and reduced fear of movement mediated improvements in pain and physical function (Lawford 2023). Exercise may also help reduce OA fatigue (Casilda-López 2017; Park 2017; Power 2008; Wyatt 2014), which is a key component of disruptions to social, leisure, and daily activities in people with OA (Power 2008). Moreover, exercise may contribute to improvements in other psychosocial outcomes such as depression and social function (Hurley 2018).

Why it is important to do this review

Exercise is a common approach to the treatment of OA and is recommended as first-line management by all current international guidelines for knee OA (Bannuru 2019; Kolasinski 2020; Moseng 2024; NICE 2022; RACGP 2018). This is the third update to this Cochrane review evaluating the effectiveness of exercise for knee OA (Fransen 2002; Fransen 2008; Fransen 2015). The previous update of this review included 54 trials with any active non-specific exercise interventions or no treatment as comparator groups (Fransen 2015). Since that review, many new RCTs investigating the effectiveness of exercise for knee OA have been published. In addition, in the analyses in the prior version of this review, all comparator groups were considered together (exercise versus no exercise comparator group), rather than the analyses being stratified by comparator type. This is important as the effects of exercise have been shown to differ depending on the kind of comparator group used (Pedersen 2023). Given that there is increasing evidence that clinical improvements associated with exercise may be largely driven by contextual factors and a placebo response (Bandak 2022; Englund 2023; Henriksen 2024), it is important to evaluate the effectiveness of exercise when compared to placebo or attention control interventions (which are designed to account for contextual factors and placebo response). This updated review also examined the effects of exercise in some subgroups that were not included in the previous review, including number of exercise sessions prescribed and number of synchronous contact occasions (i.e. consultations that happen live/in real-time) with a healthcare provider.

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OBJECTIVES

We aimed to assess the effects of land-based exercise for people with knee osteoarthritis (OA) by comparing:

1) exercise versus attention control or placebo;

2) exercise versus no treatment, usual care, or limited education;

3) exercise plus another co-intervention versus the same co-intervention alone.

METHODS

Criteria for considering studies for this review

Types of studies

We conducted this review according to the guidelines recommended by the Cochrane Musculoskeletal Editorial Board (Ghogomu 2014). We included randomised controlled trials (RCTs). Studies reported as full-text, those published as abstract only, and unpublished data were eligible for inclusion. There were no language restrictions.

In contrast to the previous review, we did not include quasi-randomised controlled trials (defined as trials that use a group allocation method that is not truly random) due to the considerable volume of relevant RCTs.

Types of participants

We included male and female adults with an established diagnosis of knee OA, as defined in the study, which could have been according to accepted clinical criteria (Altman 1991a; NICE 2022; Zhang 2010b), or self-report on the basis of chronic joint pain (with or without radiographic confirmation).

We also included trials that involved participants with OA in other joints, provided that the outcomes for people with OA of the knee were presented separately (or could be obtained from the trial authors) or if more than 80% of the participants in the trial had OA of the knee.

Types of interventions

We included any land-based exercise regimens (defined as 'planned, structured, and repetitive activity'). Exercise interventions eligible for inclusion could:

- incorporate education or behaviour change strategies designed to help participants adhere to the exercise programme (e.g. information about the benefits of exercise, short-message services [SMS] or mobile app to maximise adherence);
- be supervised or unsupervised; and
- include other non-surgical co-interventions alongside the exercise intervention, as long as these were permitted or provided in a similar way in any comparator groups.

Exercise interventions were ineligible if they:

- were perioperative (i.e. specifically recruited all participants from surgical waiting lists and/or evaluated outcomes post-surgery);
- included whole-body vibration, gait aid prescription, or gait retraining as the intervention (however, backward/retro walking prescribed as a therapeutic exercise program was eligible); and
- included any co-interventions (apart from education and behaviour change strategies) that were not equally included in any comparator groups (e.g. pain coping skills training).

Comparator (control) groups eligible for inclusion were:

1. placebo, sham, or attention control (any intervention that was designed to control for contextual/placebo effects and described by the authors as 'placebo' or 'sham', and/or an 'attention control' involving more than one bout of synchronous interaction with study personnel/care provider (excluding contact to obtain outcome measures));
2. no treatment (no defined allocated intervention), usual care (where the authors stated that participants could receive 'normal/usual care', but this was not controlled by the trial), or limited education (provided with a one-off information resource); and
3. any non-surgical intervention that was also offered or provided, equally, as a co-intervention in the exercise group (e.g. weight loss, manual therapy, insoles or footwear, pain coping skills, physical therapy).

In contrast to the previous review, we amended comparator 3 to include non-exercise (and non-surgical) active intervention as the comparator only if it was also and equally provided in the exercise group. This allowed evaluation of the effect of exercise when added to another non-surgical intervention or interventions.

Types of outcome measures

Major outcomes

In accordance with international consensus regarding the core set of outcome measures for phase III clinical trials in OA (Smith 2019), each RCT was required to include assessment of at least one of the following.

1. Knee pain severity
2. Self-reported physical function
3. Quality of life

Time points

We assessed clinical outcomes (pain, physical function, quality of life) at three time points: immediately at the end of the treatment (post-treatment), up to 12 months after cessation of the intervention, and more than 12 months after cessation of the intervention. Each included study was required to report measurement of outcomes for at least one of these time periods.

In contrast to the previous review, we did not assess outcomes at 2 to 6 months after treatment because of the volume of research we anticipated finding, as well as our interest in longer-term outcomes.

Secondary outcomes

- Participant-reported treatment success, as defined by the trialists, at post-intervention
- Number of study dropouts/withdrawals at final follow-up (for any reason, related or unrelated to the intervention)

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- Number and nature of adverse events (mild or serious in nature that were deemed by trial authors to be related to the intervention), as well as whether withdrawals from adverse events were reported, at final follow-up

Search methods for identification of studies

Electronic searches

We searched the reference list of the previous review for relevant trials. We then conducted a new search from the search date of that review (May 2013) to 4 January 2024. No language restrictions were applied. We searched the following databases.

- MEDLINE OvidSP
- Embase OvidSP
- Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Library
- ClinicalTrials.gov (www.ClinicalTrials.gov) for unpublished completed trials
- World Health Organization (WHO) trials portal (https://trialsearch.who.int/) for ongoing studies

The search strategies used are shown in Appendix 1. Since the last review, we have updated the search terms for Embase based on the most recent Cochrane Highly Sensitive Search Strategy (Lefebvre 2024). In addition, we did not include CINAHL in this review update as we expected the four databases above to sufficiently cover all publications.

Searching other resources

We searched the reference lists of the included studies. We did not search for grey literature.

Data collection and analysis

Selection of studies

Two review authors (who were paired at random into four teams, i.e. they were not consistently matched with the same other author) independently screened retrieved titles and abstracts to identify potentially relevant studies. All review authors were involved in this process. We retrieved the full-text report or published record of any study that either of the two review authors considered potentially eligible. We translated studies published in languages other than English using Google Translate before we assessed them for inclusion. Two review authors independently screened the full texts, identified studies for inclusion, and recorded reasons for exclusion of ineligible studies. If agreement between the two authors about inclusion or exclusion was not achieved at any stage, a third review author adjudicated. Review authors were not involved in screening of any trials that they had authored.

We excluded duplicates and collated multiple reports of the same study under a single reference ID so that each study, rather than each report, is the unit of interest in the review. We recorded the selection process in sufficient detail to complete a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram (Page 2021).

We managed the results of our search using Covidence software (Covidence).

Data extraction and management

We used a data collection form for study characteristics and outcome data, which we piloted on at least two eligible studies. We extracted the following study characteristics.

1. Methods: location, study setting, withdrawals, and date of study
2. Participants: N (number), mean age, sex, diagnostic criteria, body mass index (BMI), and baseline pain and physical function
3. Interventions: description of exercise intervention and comparison, including duration of intervention, setting, type and content of exercise, mode of treatment delivery (supervised or unsupervised, delivered in-person or remotely, individual or group), number of contact occasions with healthcare provider, number of exercise sessions, frequency of exercise programme, individual session duration, and adherence. We extracted characteristics of exercise intervention based on Consensus on Exercise Reporting Template (CERT) criteria (Slade 2016)
4. Outcomes: major outcomes (as listed above), scales used, and time points reported (post-intervention, up to 12 months, and more than 12 months)
5. Characteristics of the design of the trial as outlined below in the Assessment of risk of bias in included studies section
6. Notes: trial funding, notable declarations of interest of trial authors
7. Information needed to assess the certainty of the evidence using GRADE (e.g. baseline risk in the control group for key outcomes) and to formulate summary of findings tables

Teams of two review authors (who were paired at random, i.e. they were not consistently matched with the same other author) independently extracted outcome data from the included studies. All review authors were included in this process. Review authors were not involved in data extraction of any trials that they had authored. We extracted the means, standard deviations, and number of participants per group for continuous outcomes. We noted in the Characteristics of included studies table if outcome data were not reported in a usable way and when we transformed or estimated data from a figure. We resolved disagreements by consensus or by involving a third review author. One review author (BJL) transferred data to RevMan Web (Review Manager Web). We double-checked that data were entered correctly by comparing the data presented in the systematic review with individual study reports, and a third author double-checked that data from new additional studies were entered correctly.

If a trial provided data from more than one pain scale, we extracted data from the pain scale that was highest on the list below, according to the hierarchy of pain-related outcomes used in the previous review (Fransen 2015).

1. Global pain (visual analogue scale [VAS] or numeric rating scale (NRS))
2. Pain on walking (VAS or NRS)
3. Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain subscale
4. Composite pain scores other than WOMAC
5. Pain on activities other than walking (VAS or NRS)
6. Pain at rest or pain during the night (VAS or NRS)

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We considered the Knee Injury and Osteoarthritis Outcome Score (KOOS) pain subscale to be equivalent to the WOMAC pain subscale. Where pain subscales of composite scales were not presented, we did not use total scores and attempted to obtain the data from the authors. If the same scale was used with slightly different questions (e.g. participants being asked to use a VAS to rate their average pain over the past week or to rate their average pain over the past two weeks), we took the outcome that was closest to "over the past week".

When a trial reported more than one physical function scale, we extracted data for one scale according to the hierarchy presented below.

1. Global disability score
2. Walking disability
3. WOMAC function subscale
4. Composite disability scores other than WOMAC
5. Disability other than walking
6. WOMAC global scale
7. Lequesne Osteoarthritis Index global score
8. Other algofunctional scale

We considered the KOOS ADL subscale to be equivalent to the WOMAC function subscale. Where function subscales of composite scales were not presented, we used the overall scores for function.

If data on more than one quality of life scale were reported in a trial, we extracted data according to the hierarchy presented below.

1. Short Form (SF)-36, Mental Component Summary (MCS)
2. SF-12 MCS
3. EuroQol
4. Sickness Impact Profile (SIP)
5. Nottingham Health Profile (NHP)
6. Other quality of life scales

We also collected the following data from each trial.

- Participant-reported treatment success, as defined by the trialists (e.g. Osteoarthritis Research Society International (OARSI) Standing Committee for Clinical Trials Response Criteria Initiative and the Outcome Measures in Rheumatology (OMERACT); global rating of change), at post-intervention
- Number of study dropouts/withdrawals at final follow-up
- Number and nature of adverse events (non-serious or serious in nature that were deemed by trial authors to be related to the intervention), as well as whether withdrawals due to adverse events were reported, at final follow-up

To prevent selective inclusion of data based on the results, we used the following a priori defined decision rules to select data from trials.

- Where trials reported outcomes at multiple time points, we extracted data from the time point furthest from randomisation within each of the time periods of interest.
- Where trialists reported both end-score and change-from-baseline values for the same outcome, we extracted the end-score values.
- Where trialists reported both unadjusted and adjusted values for the same outcome, we extracted adjusted values.
- We extracted data analysed by intention-to-treat (ITT) for outcomes assessing benefits (pain, function, quality of life, participant-reported treatment success), if reported. For outcomes assessing adverse events, we extracted data according to the group to which a participant was randomised.
- Review authors who were named authors of the included studies did not extract data from their own studies or assess their risk of bias.

Assessment of risk of bias in included studies

Review authors BJL and MH independently assessed the risk of bias for each new study using the RoB 1 tool and the criteria outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2022). For trials authored by BJL, AD completed the risk of bias assessment. We resolved any disagreements by discussion or by involving another review author. We assessed the risk of bias according to the following domains.

1. Random sequence generation
   a. Criteria for judgement of 'low' risk: a random (i.e. unpredictable) assignment sequence was used (e.g. coin toss, rolling a die, drawing of ballots with the study group labels from a dark bag, computer-generated random sequence, pre-ordered sealed envelopes, telephone call to a central office, or pre-ordered list of treatment assignments).
2. Allocation concealment
   a. Criteria for judgement of 'low' risk: the person responsible for randomising participants had no influence on the assignment sequence or on the decision about the eligibility of the patient. The randomisation schedule was concealed via sealed opaque envelope or stored on locked folder/server.
3. Blinding of participants and personnel
   a. Criteria for judgement of 'low' risk: exercise and control groups were indistinguishable for participants. Limited disclosure was used to blind participants to group components and study hypotheses.
4. Blinding of outcome assessment
   a. Criteria for judgement of 'low' risk: If participants were blinded, the blinding procedure was adequate for outcome assessment.
5. Incomplete outcome data
   a. Criteria for judgement of 'low' risk: the study described and provided reasons for the number of participants who were included in the study but did not complete the observation period or were not included in the analysis. The percentage of withdrawals and dropouts did not exceed 20% (based on a threshold used in a prior Cochrane review (Hayden 2021)). All randomised participants were analysed and reported in the group to which they were allocated for outcome measurement, irrespective of non-compliance.
6. Selective outcome reporting
   a. Criteria for judgement of 'low' risk: all results from all prespecified outcomes were adequately reported in the published report of the trial. We obtained this information by comparing the protocol or trial registry record and the report.

We graded each potential source of bias as high, low, or unclear risk, and provided a justification for our judgement in the risk of

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bias table. We presented the figures generated by the RoB 1 tool to provide summary assessments of the risk of bias.

Measures of treatment effect

We used Cochrane statistical software, Review Manager (RevMan; Review Manager 2020), to perform data analysis. We analysed dichotomous data as risk ratio (RR), with 95% confidence intervals (CI). We analysed continuous data as standardised mean difference (SMD), with 95% CI. We entered data with a consistent direction of effect across studies (i.e. with higher scores indicating worse pain or physical function).

Where different scales were used to measure the same conceptual outcome, we calculated SMD, with corresponding 95% CIs. All SMDs were back-translated to mean differences (0 to 100 scale). We did this for each comparator group by multiplying by the mean weighted standard deviation (SD) of the comparison group at baseline in all trials that used a 0 to 100 scale for each outcome (Higgins 2022).

To interpret the clinical importance of results, we compared our mean differences to established minimal important differences (MIDs) on 0 to 100 point scales for pain, function, and quality of life. These represent the magnitude of changes in symptoms that are meaningful for patients with OA, and they are frequently used to aid clinical interpretation of effects of OA exercise interventions. We used MIDs from a systematic review that established the most credible MIDs in people with knee pain for each of the outcome instruments (Devji 2017). The most credible MID was the median of all MIDs. For pain, we used an MID of 12 points on a 0 to 100 point scale (based on the MID for WOMAC pain instrument). For function, we used an MID of 13 on a 0 to 100 point scale (based on the MID for the WOMAC function instrument). For quality of life, we used an MID of 15 on a 0 to 100 point scale (based on the MID for the EQ-5D quality of life scale).

For dichotomous outcomes, we calculated the absolute percentage change from the difference in the risks between the intervention and control groups using GRADEpro (GRADEpro GDT). For continuous outcomes, we calculated the absolute benefit as the improvement in the intervention group minus the improvement in the control group (mean difference), on a 0 to 100 scale.

Unit of analysis issues

We identified the level at which randomisation occurred (e.g. individuals, clusters) and addressed the following when they arose.

Where multiple arms were reported per participant in a single trial, we included only the relevant arms. If a three-arm trial was relevant to the same meta-analysis (e.g. exercise A versus exercise B versus no intervention), we pooled the exercise groups for that trial.

For cluster-randomised trials, we used an intraclass correlation coefficient (Hurley 2007) to estimate the design effect and adjusted the sample size and mean variance accordingly (Higgins 2024).

Dealing with missing data

We contacted investigators as needed to verify key study characteristics and obtain missing numerical outcome data (e.g. when data were not available for all participants). If we received no response from investigators after two attempts (at least one month apart), we considered the data to be irretrievable, and we classified the study as "awaiting classification". For continuous outcomes, we calculated the SMD based on the number of participants randomised at baseline.

Where possible, we calculated missing SDs from other statistics, such as standard errors, confidence intervals, or P values, according to the methods recommended in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2022). Where data were calculated or imputed, we reported this in the notes section of the Characteristics of included studies table.

Assessment of heterogeneity

To determine whether a meta-analysis was appropriate, we assessed clinical and methodological diversity in the included studies in terms of participants, interventions, outcome measurement tools, and study characteristics. We assessed statistical heterogeneity by visually inspecting forest plots to check for obvious differences in results between studies, and when there were 10 or more studies in a meta-analysis, we used the I^2 and Chi^2 tests to aid our interpretation.

As recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2022), we interpreted an I^2 statistic for heterogeneity of 0 to 40% as 'might not be important'; 30 to 60% may represent 'moderate' heterogeneity; 50 to 90% may represent 'substantial' heterogeneity; and 75 to100% may represent 'considerable' heterogeneity. We kept in mind that the observed value of I^2 depends on the magnitude and direction of effects, and the strength of evidence for heterogeneity (e.g. P value from the Chi^2 test, or a CI for the I^2 statistic (uncertainty in the value of the I^2 statistic is substantial when the number of studies is small)).

For the Chi^2 test, we interpreted P (probability) equal to or greater than 0.10 as indicating evidence of statistical heterogeneity. Where we identified heterogeneity, we indicated it in the review, and we investigated possible causes by following the recommendations in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2022). This included removing outliers with extreme results (defined as having an SMD greater than plus or minus two).

Assessment of reporting biases

We evaluated between-study publication bias with Egger's test and inspection of funnel plots (Egger 1997). To assess outcome reporting bias, we checked trial protocols against published reports or registries. For studies published after 1 July 2005, we searched the Clinical Trial Register at the International Clinical Trials Registry Platform of the World Health Organization (www.who.int/clinical-trials-registry-platform) for the a priori trial protocol.

To assess small-study effects, we generated funnel plots for meta-analyses that included at least 10 trials of varying size. If we detected asymmetry in the funnel plot, we reviewed the characteristics of the trials to assess whether the asymmetry was likely due to publication bias or other factors, such as the methodological or clinical diversity of the trials.

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Data synthesis

We pooled outcomes that assessed the benefits of exercise across studies with a similar comparator and stratified the primary analysis by follow-up time point, as follows:

- exercise versus attention control or placebo;
- exercise versus no treatment, usual care, or limited education;
- exercise plus another co-intervention versus the same other co-intervention alone.

As we expected some differences in the effect of the intervention across studies, we used the random-effects model for meta-analysis.

Subgroup analysis and investigation of heterogeneity

We performed the following subgroup analyses to assess if there were differences in pain and function immediately post-intervention for our three comparisons.

1. Type of exercise
   a. Strengthening (including neuromuscular)
   b. Aerobic
   c. Mind-body (e.g. yoga, Tai Chi)
   d. Stretching
   e. Balance/proprioceptive
   f. Mixed (programmes that included more than one type of exercise)
2. Total number of prescribed exercise sessions (number per week (as specified by authors) multiplied by programme duration in weeks)
3. Ratio of number of synchronous contact occasions (in-person or via videoconferencing or telephone) with healthcare provider between exercise and comparison groups. Only studies with at least one synchronous contact occasion in exercise and comparator groups were included. For example, a ratio of 1 means an equal number of synchronous contact occasions was provided in the exercise and comparator group. A ratio greater than 1 means there was a greater number in the exercise group.

For subgroup 1, we used the formal test for subgroup interactions in RevMan. We also compared the magnitude of the effects between the subgroups by means of assessing the overlap of the confidence intervals of the summary estimates. Non-overlap of the confidence intervals indicated statistical significance. For subgroups 2 and 3, we performed a meta-regression in SPSS (IBM SPSS Statistics for Windows). We deemed meta-regression to be most appropriate for these subgroups as they were measured on continuous scales. We used regression coefficients and corresponding P values for interpretation.

Sensitivity analysis

We performed sensitivity analyses for our three comparisons to investigate the robustness of the effect of potential selection, attrition, and detection bias on outcomes immediately post-intervention. We decided to conduct these analyses for outcomes of pain and function only, given the number of analyses already included in the review, as well as the fact that we anticipated fewer studies to include quality of life outcomes.

We also examined estimates when only including trials that had a low risk of bias on three or more of the six risk-of-bias domains and were fully-powered (i.e. for pain, a sample size of 33 or more per arm (which was calculated based on the pooled weighted SD of 15.0 and our predefined MID of 12)and for function, a sample size of 48 or more per arm (which was calculated based on the pooled weighted SD of 19.6 and our predefined MID of 13).

Summary of findings and assessment of the certainty of the evidence

We created summary of findings tables using GRADEpro software (GRADEpro GDT) for each comparison. We reported pain, function, quality of life, and participant-reported treatment success immediately post-intervention, and dropouts/withdrawals and adverse events at final follow-up.

Two review authors (BJL and MH) assessed the certainty of the evidence across all studies contributing to the meta-analysis for each outcome, using the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness, and publication bias), as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Schünemann 2022). Our GRADE assessment was based on our GRADE Quality Assessment Rubric (Table 1; adapted from OA clinical guidelines (Bannuru 2019)). We reported the certainty of evidence as high, moderate, low, or very low. We justified, documented, and incorporated judgements into our reporting of results for each outcome. For any findings involving trials authored by review author BJL, another review author, AD, assessed the certainty of the evidence.

RESULTS

Description of studies

Results of the search

We updated the search for the current review on 4 January 2024, searching for studies published since the date of the search in the first version of the review (May 2013).

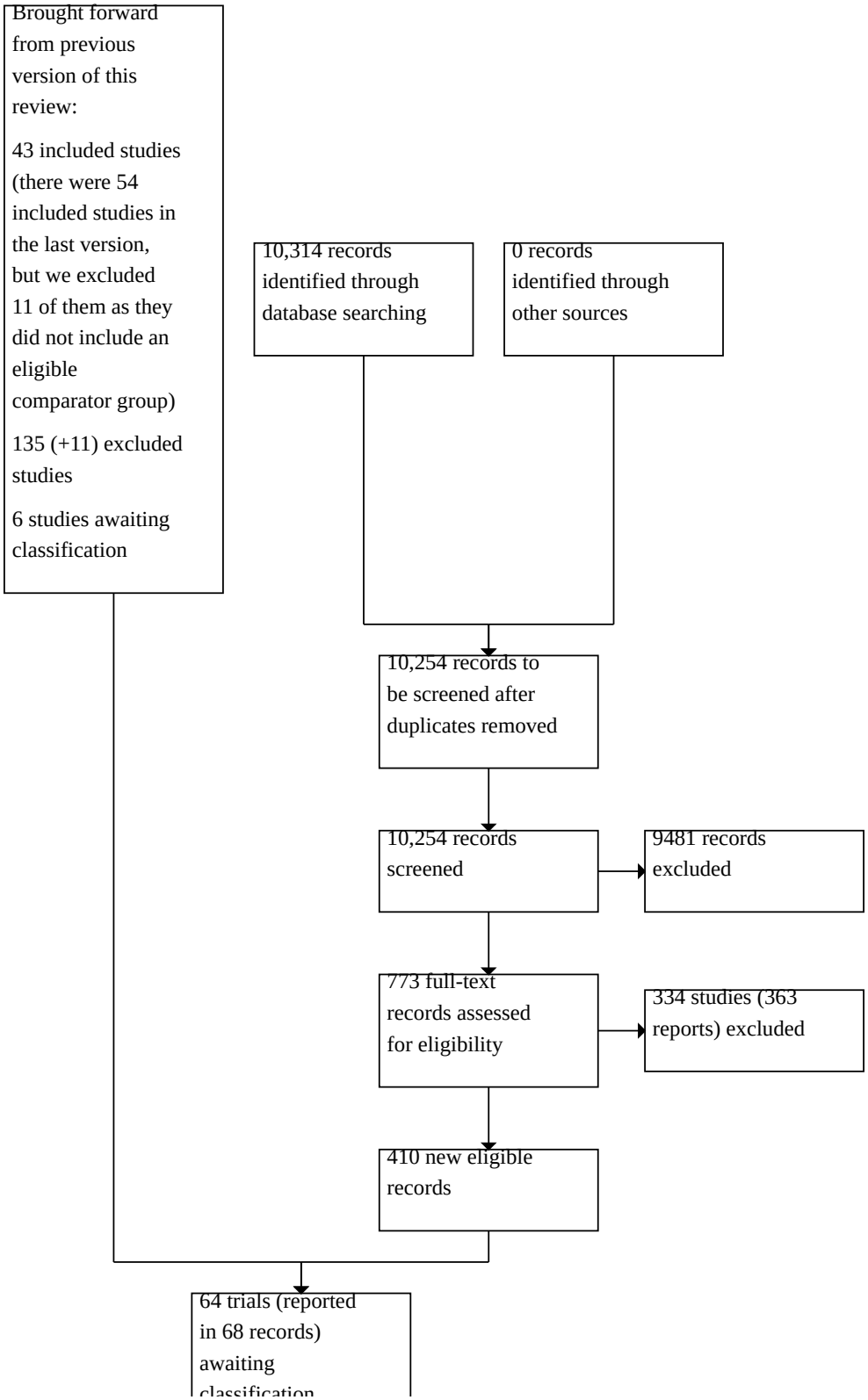
Using the PRISMA template, we present a flow diagram of the study selection process in Figure 1. The updated search strategy identified 10,314 new citations (10,254 after de-duplication). Of these, we assessed 773 in full text and included 96 new trials in the review (Alghadir 2019; Allen 2018; Anadkat 2015; Anwer 2014; Assar 2020; Bandak 2022; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Braghin 2018; Chen 2014; Chen 2021; Cheung 2014; Cheung 2017; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Almeida 2020; De Matos 2019; De Rooij 2017; DeVita 2018; Dincer 2016; Drummen 2023; Dwyer 2015; Gohir 2021; Gondhalekar 2013; Guo 2022; Henriksen 2014; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Hu 2020; Imoto 2012; Janyacharoen 2018; Jorge 2015; Joshi 2019; Kang 2022; Karadag 2019; Kholvadia 2019; Kilic 2020; Kim 2013; Kim 2020; Knoop 2013; Krishnan 2021; Kumar 2013; Kuntz 2018; Lai 2021; Lee 2019; Lee 2023; Li 2020; Liu 2019; Lu 2017; Marconcin 2018; Mazloum 2018; Mehwish 2023; Messier 2013; Messier 2021; Mete 2022; Multanen 2014; Nahayatbin 2018; Nelligan 2021; Oh 2020; Øiestad 2023; Ozlu 2023; Pazit 2018; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rego 2023; Rakhshani 2022; Robbins 2022; Rosadi 2023; Rosedale 2014; Sadeghi 2023; Samuel 2014; Samut 2015; Santana 2022; Schlenk 2020; Sedaghatnezhad 2021; Shabbir 2022; Shahine 2020; Shen 2022; Shin 2017; Song 2020; Takacs 2017; Tsai 2013; Vassao 2021; Vincent 2020; Wallis

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2017; Wortley 2013; Xiao 2021; Ye 2020). Of the 54 trials included in the previous version of this review, we excluded 11 as they no longer met the inclusion criteria, i.e. did not include an eligible comparator (Bennell 2005; Bezalel 2010; Deyle 2000; Doi 2008; Foroughi 2011; Hughes 2004; Kao 2012; Mikesky 2006; Minor 1989; Quilty 2003; Yip 2007a). Thus, in total, in this update we included 139 trials with 12,468 participants (Abbott 2013; Alghadir 2019; Allen 2018; An 2008; Anadkat 2015; Anwer 2014; Assar 2020; Baker 2001; Bandak 2022; Bautch 1997; Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Braghin 2018; Brismée 2007; Bruce-Brand 2012; Chang 2012; Chen 2014; Chen 2021; Cheung 2014; Cheung 2017; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Almeida 2020; De Matos 2019; De Rooij 2017; DeVita 2018; Dincer 2016; Drummen 2023; Dwyer 2015; Ettinger 1997; Foley 2003; Fransen 2001; Fransen 2007; Gohir 2021; Gondhalekar 2013; Guo 2022; Gur 2002; Hay 2006; Henriksen 2014; Hernandez 2019; Hinman 2020; Holm 2020; Hopman-Rock 2000; Hsu 2021; Hu 2020; Huang 2003; Huang 2005; Hurley 2007; Imoto 2012; Jan 2008; Jan 2009; Janyacharoen 2018; Jenkinson 2009; Jorge 2015; Joshi 2019; Kang 2022; Karadag 2019; Keefe 2004; Kholvadia 2019; Kilic 2020; Kim 2013; Kim 2020; Knoop 2013; Kovar 1992; Krishnan 2021; Kumar 2013; Kuntz 2018; Lai 2021; Lee 2009; Lee 2019; Lee 2023; Li 2020; Lim 2008; Lin 2009; Liu 2019; Lu 2017; Lund 2008; Marconcin 2018; Maurer 1999; Mazloum 2018; Mehwish 2023; Messier 2004; Messier 2013; Messier 2021; Mete 2022; Multanen 2014; Nahayatbin 2018; Nelligan 2021; O'Reilly 1999; Oh 2020; Øiestad 2023; Ozlu 2023; Pazit 2018; Peloquin 1999; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rakhshani 2022; Rego 2023; Robbins 2022; Rogind 1998; Rosadi 2023; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Salli 2010; Samuel 2014; Samut 2015; Santana 2022; Schilke 1996; Schlenk 2020; Sedaghatnezhad 2021; Sekir 2005; Shabbir 2022; Shahine 2020; Shen 2022; Shin 2017; Simao 2012; Song 2003; Song 2020; Takacs 2017; Talbot 2003; Thomas 2002; Thorstensson 2005; Topp 2002; Tsai 2013; Van Baar 1998; Vassao 2021; Vincent 2020; Wallis 2017; Wang 2011; Wortley 2013; Xiao 2021; Ye 2020).

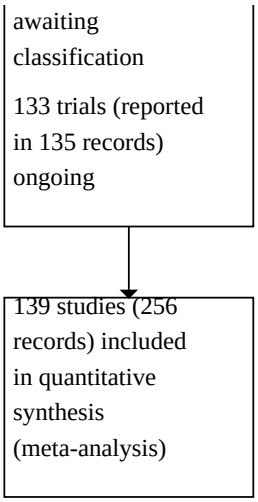
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Figure 1. Study selection process



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Figure 1. (Continued)



We classified a further 64 studies as awaiting classification because they were either available only in abstract form (with insufficient information to include or exclude), were trial registry records (where the trial was marked as complete more than 24 months ago, and the publication could not be found), or we could not extract data (the authors did not respond to request for data, or the publication was in a language other than English and Google translation was insufficient to allow extraction). There are 133 ongoing studies (including abstracts reporting trial outcomes within the prior 24 months, registry records that were not marked as complete or were marked as complete in the prior 24 months, and published trial protocols).

Included studies

We provide a full description of all the included trials in the Characteristics of included studies table, and we provide a summary of trial and participant characteristics in Table 2. We contacted 43 trial authors to request or clarify data, with 21 (49%) responding.

Study design and setting

The 139 included studies were all randomised trials of parallel design; one trial was cluster-randomised (Hurley 2007). One hundred and fourteen studies had two arms that were of relevance to this review, 21 had three (Alghadir 2019; Cheung 2017; De Almeida 2020; Ettinger 1997; Gur 2002; Hurley 2007; Jan 2008; Jan 2009; Kuntz 2018; Mazloum 2018; Messier 2021; Nahayatbin 2018; Pazit 2018; Rafiq 2021b; Rosadi 2023; Salli 2010; Samut 2015; Topp 2002; Vincent 2020; Wortley 2013; Øiestad 2023), and four had four (Huang 2003; Lim 2008; Liu 2019; Sadeghi 2023). When a study had more than one exercise group, we combined exercise groups into one group for our analysis.

The trials were conducted in the following countries.

- Argentina (Hernandez 2019)
- Australia (Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Drummen 2023; Foley 2003; Fransen 2001; Fransen 2007; Hinman 2020; Lim 2008; Nelligan 2021; Pazit 2018; Wallis 2017)
- Brazil (Braghin 2018; Da Silva 2015; De Almeida 2020; De Matos 2019; Imoto 2012; Jorge 2015; Rego 2023; Robbins 2022; Santana 2022; Simao 2012; Vassao 2021)
- Canada (Chopp-Hurley 2017; Kuntz 2018; Peloquin 1999; Rosedale 2014; Takacs 2017)
- China (An 2008; Chen 2021; Guo 2022; Hu 2020; Kang 2022; Lai 2021; Li 2020; Liu 2019; Lu 2017; Qiu 2023; Shen 2022; Song 2020; Xiao 2021; Ye 2020)
- Denmark (Bandak 2022; Christensen 2015; Henriksen 2014; Holm 2020; Lund 2008; Rogind 1998)
- Egypt (Shahine 2020)
- Finland (Multanen 2014)
- India (Anadkat 2015; Bhagat 2017; Gondhalekar 2013; Joshi 2019; Kumar 2013; Samuel 2014)
- Indonesia (Rosadi 2023)
- Iran (Assar 2020; Mazloum 2018; Nahayatbin 2018; Rakhshani 2022; Sadeghi 2023; Sedaghatnezhad 2021)
- Ireland (Bruce-Brand 2012)
- Japan (Kim 2013)
- Malaysia (Krishnan 2021)
- Netherlands (De Rooij 2017; Hopman-Rock 2000; Knoop 2013; Van Baar 1998)
- New Zealand (Abbott 2013)
- Norway (Øiestad 2023)
- Pakistan (Mehwish 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Shabbir 2022)
- Portugal (Marconcin 2018)
- Saudi Arabia (Alghadir 2019; Anwer 2014)
- South Africa (Kholvadia 2019)
- South Korea (Kim 2020; Lee 2009; Lee 2019; Lee 2023; Oh 2020; Shin 2017; Song 2003)
- Sweden (Thorstensson 2005)
- Taiwan (Chang 2012; Chen 2014; Hsu 2021; Huang 2003; Huang 2005; Jan 2008; Jan 2009; Lin 2009; Wang 2011)
- Thailand (Janyacharoen 2018)

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- Turkey (Dincer 2016; Gur 2002; Karadag 2019; Kilic 2020; Mete 2022; Ozlu 2023; Salli 2010; Samut 2015; Sekir 2005)
- the UK (Gohir 2021; Hay 2006; Hurley 2007; Jenkinson 2009; O'Reilly 1999; Thomas 2002)
- the USA (Allen 2018; Baker 2001; Bautch 1997; Brismée 2007; Cheung 2014; Cheung 2017; DeVita 2018; Ettinger 1997; Keefe 2004; Kovar 1992; Maurer 1999; Messier 2004; Messier 2013; Messier 2021; Salacinski 2012; Schilke 1996; Schlenk 2020; Talbot 2003; Topp 2002; Tsai 2013; Vincent 2020; Wortley 2013)
- One study was multicentre, with centres across the USA and South Africa (Dwyer 2015).

Participants

We summarise pertinent characteristics of participants in Table 2.

Participants were aged between 41 and 81 years and had a mean BMI ranging from 22.4 to 37.9 kg/m^2. Diagnosis of knee OA was made using American College of Rheumatology (ACR) criteria (Abbott 2013; Allen 2018; An 2008; Anwer 2014; Assar 2020; Baker 2001; Bandak 2022; Bautch 1997; Bennell 2010; Bennell 2016; Bhagat 2017; Brismée 2007; Chen 2021; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Almeida 2020; De Rooij 2017; DeVita 2018; Dincer 2016; Drummen 2023; Dwyer 2015; Ettinger 1997; Fransen 2001; Fransen 2007; Gohir 2021; Gondhalekar 2013; Guo 2022; Hopman-Rock 2000; Huang 2003; Huang 2005; Imoto 2012; Jan 2008; Jan 2009; Janyacharoen 2018; Jorge 2015; Kang 2022; Karadag 2019; Knoop 2013; Lee 2019; Lim 2008; Liu 2019; Lu 2017; Lund 2008; Marconcin 2018; Maurer 1999; Mazloum 2018; Ozlu 2023; Pazit 2018; Qiu 2023; Rakhshani 2022; Rogind 1998; Salli 2010; Samut 2015; Schlenk 2020; Sedaghatnezhad 2021; Sekir 2005; Shen 2022; Simao 2012; Talbot 2003; Topp 2002; Van Baar 1998; Vassao 2021; Vincent 2020; Wortley 2013; Xiao 2021; Ye 2020); National Institute for Health and Care Excellence clinical criteria (Bennell 2022a; Bennell 2022b; Hinman 2020; Nelligan 2021); self-report (Cheung 2017; Hay 2006; Jenkinson 2009; O'Reilly 1999; Schilke 1996; Thomas 2002); radiographic classification (Anadkat 2015; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chang 2012; De Matos 2019; Foley 2003; Gur 2002; Henriksen 2014; Holm 2020; Hsu 2021; Hu 2020; Jan 2008; Jan 2009; Kilic 2020; Kim 2020; Kovar 1992; Krishnan 2021; Lai 2021; Lee 2009; Lee 2023; Lin 2009; Mehwish 2023; Messier 2004; Messier 2013; Messier 2021; Mete 2022; Multanen 2014; Nahayatbin 2018; Oh 2020; Øiestad 2023; Peloquin 1999; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rego 2023; Robbins 2022; Rogind 1998; Rosadi 2023; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Salli 2010; Samuel 2014; Shabbir 2022; Shin 2017; Simao 2012; Song 2003; Takacs 2017; Thorstensson 2005; Wallis 2017); or clinician diagnosis (Anadkat 2015; Henriksen 2014; Hernandez 2019; Hurley 2007; Hu 2020; Joshi 2019; Kovar 1992; Lai 2021; Li 2020; Santana 2022; Shahine 2020; Song 2020; Tsai 2013; Wang 2011).

Interventions

Characteristics of the exercise and comparison groups are summarised in Table 3. A CERT (consensus on exercise reporting template) checklist summary for each study can be found in a supplementary table (https://doi.org/10.6084/m9.figshare.27012928.v1).

Exercise interventions varied substantially in terms of type of exercise used, frequency, intensity, and duration prescribed, how exercise was delivered (e.g. in a group or individual setting), and length of intervention (Table 3). Most exercise interventions were delivered in person, though 6% were delivered entirely via digital technology (Allen 2018; Bennell 2022a; Bennell 2022b; Gohir 2021; Guo 2022; Hinman 2020; Li 2020; Nelligan 2021).

Types of comparators varied widely across trials. Thirty studies compared intervention with an attention control or placebo group (e.g. home visits or telephone calls, regular health education sessions, or saline injection) (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Jenkinson 2009; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Lai 2021; Liu 2019; Lu 2017; Maurer 1999; Messier 2004; Messier 2021; Peloquin 1999; Rego 2023; Santana 2022; Schlenk 2020; Shen 2022; Simao 2012; Song 2003; Song 2020; Tsai 2013; Vincent 2020; Ye 2020); 60 studies compared exercise with no treatment or usual care or limited education (e.g. routine care offered by family doctor, wait list, education pamphlet) (Abbott 2013; Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Hay 2006; Henriksen 2014; Hopman-Rock 2000; Hu 2020; Hurley 2007; Jan 2009; Janyacharoen 2018; Jorge 2015; Karadag 2019; Keefe 2004; Kim 2020; Krishnan 2021; Lee 2009; Lee 2023; Li 2020; Lim 2008; Lin 2009; Lund 2008; Mazloum 2018; Multanen 2014; Øiestad 2023; O'Reilly 1999; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Schilke 1996; Sekir 2005; Shahine 2020; Takacs 2017; Thorstensson 2005; Topp 2002; Vassao 2021; Wallis 2017; Wang 2011; Wortley 2013; Xiao 2021); and 49 studies evaluated exercise when added to another co-intervention compared to that co-intervention alone (e.g. co-interventions included weight loss diet, physical therapy, pain coping skills training, amongst others) (Alghadir 2019; Anadkat 2015; Anwer 2014; Assar 2020; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Gondhalekar 2013; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Huang 2003; Huang 2005; Jan 2008; Joshi 2019; Kholvadia 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rakhshani 2022; Rosadi 2023; Salli 2010; Samuel 2014; Sedaghatnezhad 2021; Shabbir 2022; Shin 2017; Talbot 2003; Thomas 2002; Van Baar 1998).

Outcomes

The outcome measures that were reported in each study are presented in Table 4. Follow-up ranged from 2 weeks to 24 months post-randomisation.

Pain

Pain was measured by all but four studies (Jan 2009; Kholvadia 2019; Kim 2020; Rafiq 2021a). Pain was measured on a VAS or NRS in 65 studies (Alghadir 2019; Anadkat 2015; Anwer 2014; Assar 2020; Bautch 1997; Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Brismée 2007; Chen 2014; Chen 2021; Cheung 2017; Christensen 2015; De Almeida 2020; De Rooij 2017; Dincer 2016; Drummen 2023; Gohir 2021; Gondhalekar 2013; Guo 2022; Gur 2002; Hernandez 2019; Hinman 2020; Hopman-Rock 2000; Hu 2020; Huang 2003; Huang 2005; Imoto 2012; Jorge 2015; Joshi 2019; Kang 2022; Karadag 2019; Kilic 2020; Kim 2013; Knoop 2013; Kumar 2013; Lai 2021; Lee 2019; Lee 2023; Lund 2008; Mehwish 2023; Mete 2022;

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Nelligan 2021; Øiestad 2023; Ozlu 2023; Qiu 2023; Rakhshani 2022; Robbins 2022; Rogind 1998; Rosadi 2023; Rosedale 2014; Sadeghi 2023; Salli 2010; Samut 2015; Sedaghatnezhad 2021; Sekir 2005; Shabbir 2022; Shen 2022; Shin 2017; Takacs 2017; Van Baar 1998; Vincent 2020; Wallis 2017). Sixty-two studies measured pain using the Knee Osteoarthritis Outcome Scale (KOOS) or Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) (Abbott 2013; Allen 2018; An 2008; Baker 2001; Bandak 2022; Braghin 2018; Bruce-Brand 2012; Chang 2012; Cheung 2014; Chopp-Hurley 2017; De Matos 2019; DeVita 2018; Dwyer 2015; Foley 2003; Fransen 2001; Fransen 2007; Hay 2006; Henriksen 2014; Holm 2020; Hsu 2021; Hurley 2007; Jan 2008; Janyacharoen 2018; Jenkinson 2009; Krishnan 2021; Kuntz 2018; Lee 2009; Li 2020; Lim 2008; Lin 2009; Liu 2019; Lu 2017; Marconcin 2018; Maurer 1999; Messier 2004; Messier 2013; Messier 2021; Multanen 2014; Nahayatbin 2018; O'Reilly 1999; Oh 2020; Pazit 2018; Rafiq 2021b; Rafiq 2021c; Rego 2023; Salacinski 2012; Samuel 2014; Santana 2022; Schlenk 2020; Shahine 2020; Simao 2012; Song 2003; Song 2020; Thomas 2002; Thorstensson 2005; Topp 2002; Tsai 2013; Vassao 2021; Wang 2011; Wortley 2013; Xiao 2021; Ye 2020). Three studies measured pain using the Arthritis Impact Measurement Scale (AIMS; Keefe 2004; Kovar 1992; Peloquin 1999). Two studies used the Lequesne Index (Da Silva 2015; Mazloum 2018). One study used a 1 to 6 scale (Ettinger 1997), one used the McGill Questionnaire (Talbot 2003), and one used the Osteoarthritis Screening Index (OASI; Schilke 1996).

Physical function

Physical function was measured by all but 13 studies (Assar 2020; Keefe 2004; Lai 2021; Lee 2023; Mazloum 2018; Rakhshani 2022; Sedaghatnezhad 2021; Sekir 2005; Shen 2022; Song 2020; Talbot 2003; Vincent 2020; Xiao 2021). One hundred and eight studies measured function using the KOOS ADL or WOMAC physical function subscale (Abbott 2013; Alghadir 2019; Allen 2018; An 2008; Anadkat 2015; Anwer 2014; Baker 2001; Bandak 2022; Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Braghin 2018; Brismée 2007; Bruce-Brand 2012; Chang 2012; Chen 2021; Cheung 2014; Cheung 2017; Chopp-Hurley 2017; De Almeida 2020; De Matos 2019; De Rooij 2017; DeVita 2018; Dincer 2016; Drummen 2023; Dwyer 2015; Foley 2003; Fransen 2001; Fransen 2007; Gohir 2021; Gondhalekar 2013; Guo 2022; Hay 2006; Henriksen 2014; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Hu 2020; Hurley 2007; Jan 2008; Jan 2009; Janyacharoen 2018; Jenkinson 2009; Jorge 2015; Joshi 2019; Kang 2022; Karadag 2019; Kholvadia 2019; Kilic 2020; Kim 2020; Knoop 2013; Krishnan 2021; Kumar 2013; Kuntz 2018; Lee 2009; Lee 2019; Li 2020; Lim 2008; Lin 2009; Liu 2019; Lu 2017; Lund 2008; Marconcin 2018; Maurer 1999; Mehwish 2023; Messier 2004; Messier 2013; Messier 2021; Mete 2022; Multanen 2014; Nahayatbin 2018; Nelligan 2021; Øiestad 2023; O'Reilly 1999; Oh 2020; Ozlu 2023; Pazit 2018; Qiu 2023; Rafiq 2021a; Rafiq 2021c; Rego 2023; Rosadi 2023; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Salli 2010; Samuel 2014; Samut 2015; Santana 2022; Schlenk 2020; Shabbir 2022; Shahine 2020; Shin 2017; Simao 2012; Song 2003; Takacs 2017; Thomas 2002; Thorstensson 2005; Topp 2002; Tsai 2013; Vassao 2021; Wallis 2017; Wang 2011; Wortley 2013; Ye 2020). Five studies used the Lequesne Index (Chen 2014; Da Silva 2015; Huang 2003; Huang 2005; Robbins 2022). Three studies used the AIMS (Bautch 1997; Kovar 1992; Peloquin 1999). Two studies used global disability score (Christensen 2015; Kim 2013), and two used the Impact of Rheumatic Diseases on General Health and Lifestyle scale (IRGL; Hopman-Rock 2000; Van Baar 1998). One study used the Algofunctional Index (AFI) (Rogind 1998), one used a 1 to 6 scale (Ettinger 1997), one used an NRS (Gur 2002), one used the OASI (Schilke 1996), one used a patient-specific functional scale (Rafiq 2021b), and one used the Short Form-36 (SF-36) (Imoto 2012).

Quality of life

Quality of life was measured in 50 studies (An 2008; Bandak 2022; Bennell 2016; Bennell 2022a; Bennell 2022b; Bruce-Brand 2012; Cheung 2014; Cheung 2017; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; Drummen 2023; Foley 2003; Fransen 2001; Fransen 2007; Gohir 2021; Hay 2006; Henriksen 2014; Hinman 2020; Holm 2020; Hurley 2007; Janyacharoen 2018; Jorge 2015; Kang 2022; Keefe 2004; Kim 2020; Krishnan 2021; Kuntz 2018; Lee 2009; Li 2020; Liu 2019; Lu 2017; Lund 2008; Marconcin 2018; Mehwish 2023; Messier 2013; Nahayatbin 2018; Nelligan 2021; Øiestad 2023; O'Reilly 1999; Pazit 2018; Rego 2023; Rosadi 2023; Rosedale 2014; Salacinski 2012; Salli 2010; Thorstensson 2005; Wallis 2017; Wang 2011; Xiao 2021). Sixteen studies used the SF-36 (An 2008; Bruce-Brand 2012; Christensen 2015; Da Silva 2015; Fransen 2001; Jorge 2015; Kang 2022; Lee 2009; Lu 2017; Messier 2013; O'Reilly 1999; Rego 2023; Rosadi 2023; Salli 2010; Thorstensson 2005; Xiao 2021). Fourteen studies used the KOOS Quality of Life (QoL) scale (Bandak 2022; Chopp-Hurley 2017; Henriksen 2014; Janyacharoen 2018; Krishnan 2021; Kuntz 2018; Li 2020; Liu 2019; Lund 2008; Mehwish 2023; Nahayatbin 2018; Rosedale 2014; Salacinski 2012; Wang 2011). Six studies measured quality of life using the Assessment of Quality of Life (AQoL) (Bennell 2016; Bennell 2022a; Bennell 2022b; Hinman 2020; Nelligan 2021; Pazit 2018). Five studies used the EuroQoL (Drummen 2023; Holm 2020; Marconcin 2018; Øiestad 2023; Wallis 2017). Four studies used the SF-12 (Cheung 2014; Cheung 2017; Foley 2003; Fransen 2007). Two studies used the EuroQol-5D (Hurley 2007; Kim 2020). One study used the AIMS (Keefe 2004), one used the Hospital Anxiety and Depression Scale (HADS) (Hay 2006), and one study used the Arthritis Research UK Musculoskeletal Health Questionnaire (MSK-HQ) (Gohir 2021).

Participant-reported treatment success

Participant-reported treatment success was measured in 14 studies (Allen 2018; Bandak 2022; Bennell 2016; Bennell 2022a; Bennell 2022b; Christensen 2015; De Rooij 2017; Hay 2006; Hinman 2020; Jenkinson 2009; Jorge 2015; Knoop 2013; Nelligan 2021; Rafiq 2021c). Nine studies measured treatment success via global rating of change (Allen 2018; Bennell 2016; Bennell 2022a; Bennell 2022b; De Rooij 2017; Hinman 2020; Jorge 2015; Knoop 2013; Nelligan 2021); three studies used the Outcome Measures in Rheumatology - Osteoarthritis Research Society International (OMERACT-OARSI) criteria (Bandak 2022; Christensen 2015; Hay 2006); one study classified treatment success as achieving 30% or greater improvement in pain relief (Jenkinson 2009); and one study used satisfaction measured on NRS (Rafiq 2021c).

Adverse events

Adverse events were reported in 48 studies (Abbott 2013; Alghadir 2019; Allen 2018; An 2008; Bandak 2022; Bennell 2016; Bennell 2022a; Bennell 2022b; Chen 2021; Cheung 2014; Cheung 2017; Christensen 2015; De Rooij 2017; Drummen 2023; Dwyer 2015; Ettinger 1997; Fransen 2007; Gohir 2021; Hay 2006; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Knoop 2013; Kovar 1992; Li 2020; Liu 2019; Mehwish 2023; Messier 2013; Messier 2021; Multanen 2014; Nelligan 2021; Øiestad 2023; Pazit 2018; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rego 2023; Rogind 1998;

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Schlenk 2020; Shabbir 2022; Takacs 2017; Tsai 2013; Wallis 2017; Ye 2020). Details of adverse events are described in Table 5.

Excluded studies

We brought forward 135 excluded studies from the last version of the review and excluded 11 previously included studies. We excluded a further 334 studies (363 reports) from the updated searches after full-text assessment (Figure 1). We excluded studies for the following reasons: did not include an eligible comparator intervention, did not include an eligible exercise intervention, did not measure self-reported pain/function/quality of life, were not randomised, did not include people with knee OA, or were perioperative.

Sixty-four studies (68 records) are awaiting classification (six of these were brought forward from the last version): 34 were conference abstracts from more than 24 months ago (with no apparent full-text publication), 11 were records of a registry of trial marked as 'complete' more than 24 months ago (without a retrievable publication), three were trial protocols published more than eight years ago, four were trials with mixed populations (i.e. not just knee OA) where authors did not respond to our request for data or were unable to provide data, and four were trials without data (and authors did not respond to our request for data); one study was not in English (Turkish), and the Google Translate version was insufficient to allow data to be accurately extracted; and six studies were records from the previous review, where the publication was irretrievable, or the record was not in English (and the Google translated version was insufficient to allow data to be accurately extracted).

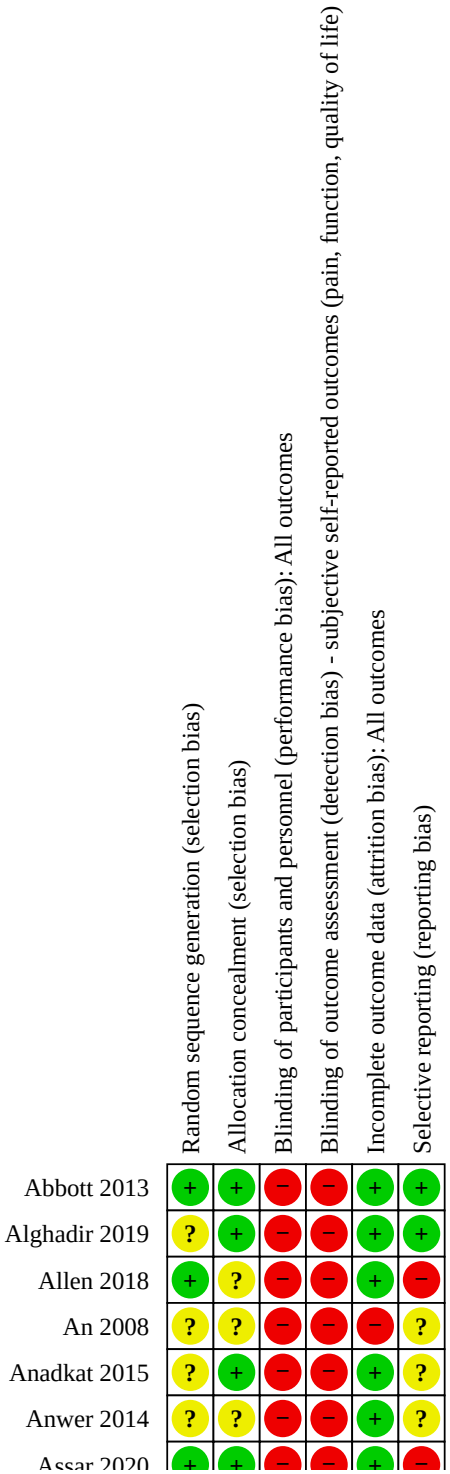
There were 135 records (i.e. trial registry records, published protocols, conference abstracts) of 133 ongoing trials.

Risk of bias in included studies

The risk of bias assessment for each study is reported in the Characteristics of included studies table and summarised in Figure 2.

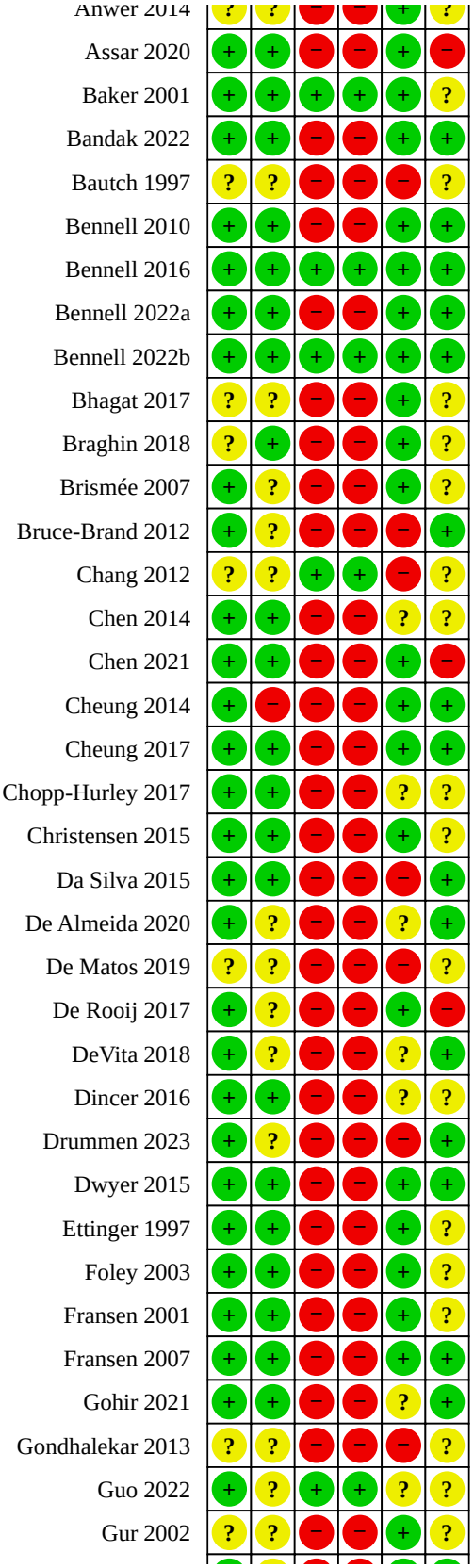
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Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study



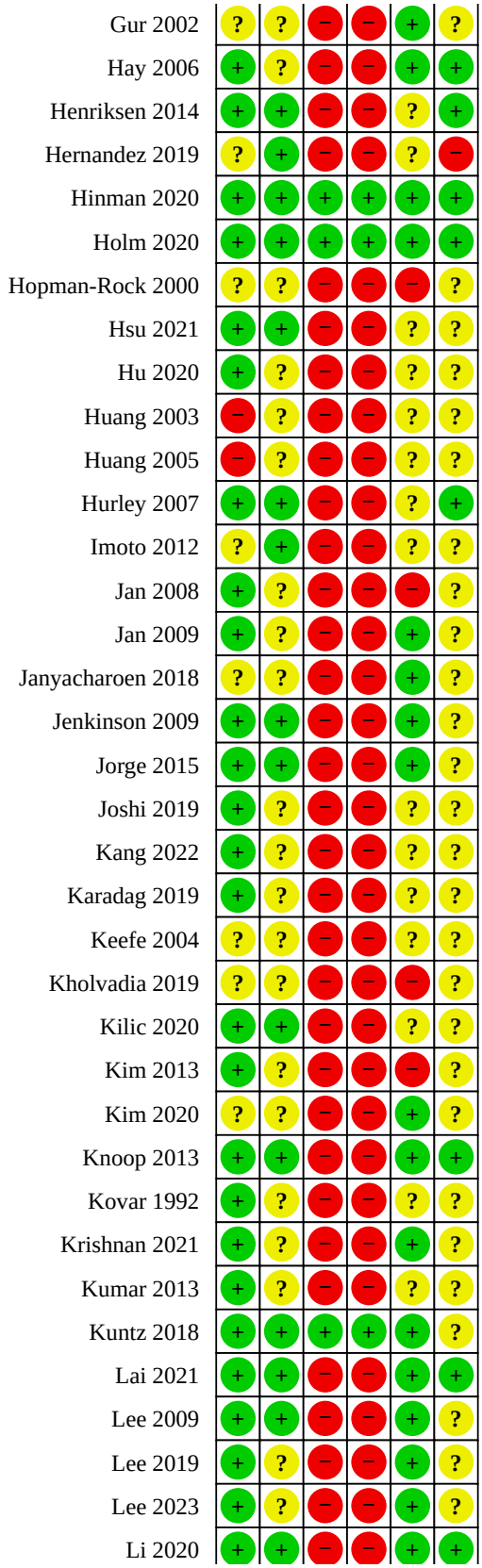
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Figure 2. (Continued)



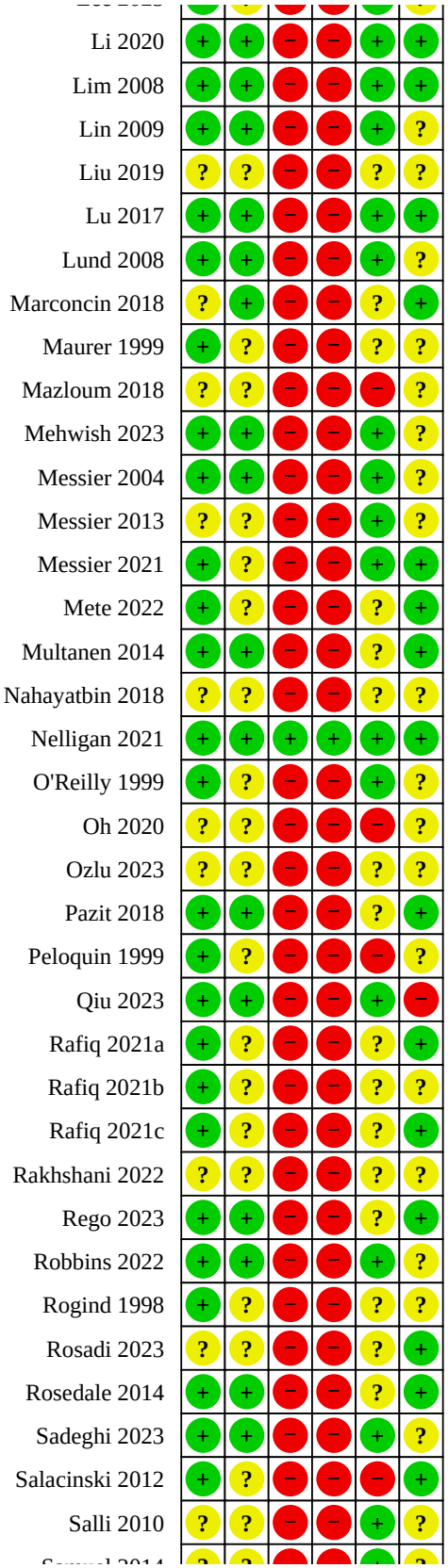
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Figure 2. (Continued)



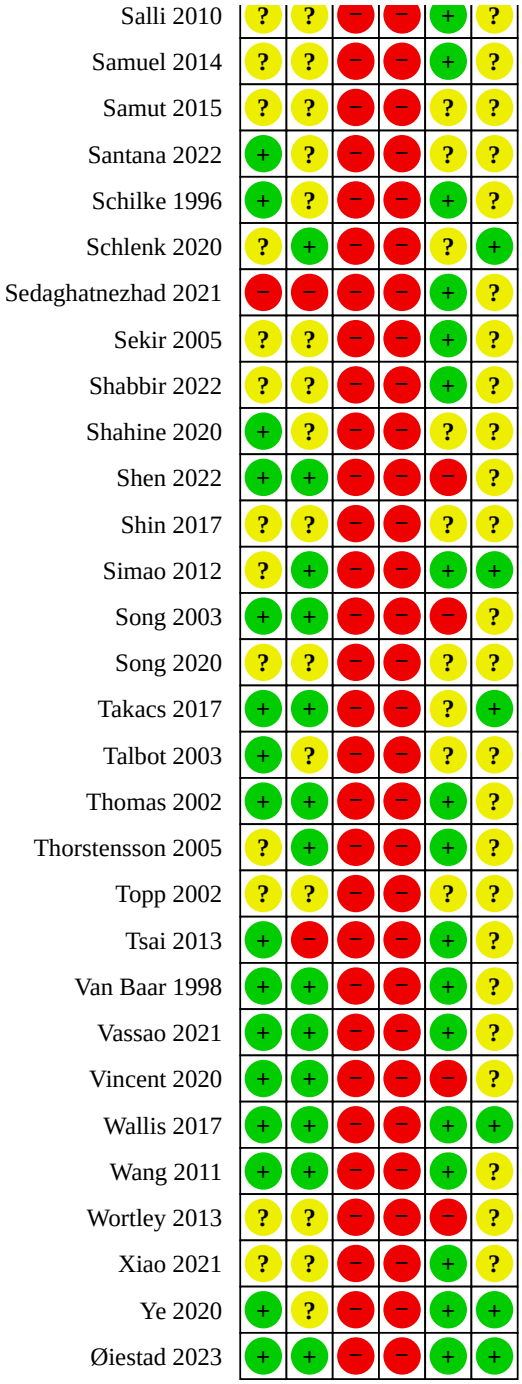
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Figure 2. (Continued)



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Figure 2. (Continued)



Allocation

Fifty-eight (42%) studies adequately generated a random sequence and concealed the sequence until allocation, thus we considered these studies at low risk of selection bias (Abbott 2013; Assar 2020; Baker 2001; Bandak 2022; Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Chen 2014; Chen 2021; Cheung 2017; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; Dincer 2016; Dwyer 2015; Ettinger 1997; Foley 2003; Fransen 2001; Fransen 2007; Gohir 2021; Henriksen 2014; Hinman 2020; Holm 2020; Hsu 2021; Hurley 2007; Jenkinson 2009; Jorge 2015; Kilic 2020; Knoop 2013; Kuntz 2018; Lai 2021; Lee 2009; Li 2020; Lim 2008; Lin 2009; Lu 2017; Lund 2008; Mehwish 2023; Messier 2004; Multanen 2014; Nelligan 2021; Øiestad 2023; Pazit 2018; Qiu 2023; Rego 2023; Robbins 2022; Rosedale 2014; Sadeghi 2023; Shen 2022; Song 2003; Takacs 2017; Thomas 2002; Van Baar 1998; Vassao 2021; Vincent 2020; Wallis 2017; Wang 2011). Eighty (57%) studies did not report their method of randomisation, and we judged their risk of selection bias as unclear (Alghadir 2019; Allen 2018; An 2008; Anadkat 2015; Anwer 2014; Bautch 1997; Bhagat 2017; Braghin 2018; Brismée 2007; Bruce-Brand 2012; Chang 2012; Cheung 2014; De Almeida

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2020; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Gondhalekar 2013; Guo 2022; Gur 2002; Hay 2006; Hernandez 2019; Hopman-Rock 2000; Hu 2020; Huang 2003; Huang 2005; Imoto 2012; Jan 2008; Jan 2009; Janyacharoen 2018; Joshi 2019; Kang 2022; Karadag 2019; Keefe 2004; Kholvadia 2019; Kim 2013; Kim 2020; Kovar 1992; Krishnan 2021; Kumar 2013; Lee 2019; Lee 2023; Liu 2019; Marconcin 2018; Maurer 1999; Mazloum 2018; Messier 2013; Messier 2021; Mete 2022; Nahayatbin 2018; O'Reilly 1999; Oh 2020; Ozlu 2023; Peloquin 1999; Rakhshani 2022; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rogind 1998; Rosadi 2023; Salacinski 2012; Salli 2010; Samuel 2014; Samut 2015; Santana 2022; Schilke 1996; Schlenk 2020; Sekir 2005; Shabbir 2022; Shahine 2020; Shin 2017; Simao 2012; Song 2020; Talbot 2003; Thorstensson 2005; Topp 2002; Tsai 2013; Wortley 2013; Xiao 2021; Ye 2020). One (< 1%) study reported inadequate methods, and we judged it to be at high risk of bias (Sedaghatnezhad 2021).

Blinding

Nine (6%) studies adequately blinded participants and investigators to their treatment by using limited disclosure or an adequate comparison intervention that enabled blinding (Baker 2001; Bennell 2016; Bennell 2022b; Chang 2012; Guo 2022; Hinman 2020; Holm 2020; Kuntz 2018; Nelligan 2021). Thus, we considered these studies at low risk of performance bias. The nature of the interventions in the remaining 131 (94%) studies precluded blinding of participants to treatment allocation, and therefore we judged them at high risk of performance bias.

All outcomes were self-reported by participants, and only eight (6%) studies adequately blinded outcome assessors via limited disclosure (Baker 2001; Bennell 2016; Bennell 2022b; Chang 2012; Guo 2022; Hinman 2020; Holm 2020; Kuntz 2018; Nelligan 2021). We considered these studies to have a low risk of detection bias. As participants were unblinded in the remaining 131 (94%) studies, we considered them to be at unclear or high risk of detection bias.

Incomplete outcome data

We judged attrition bias to be low in 72 (52%) studies, as the number of dropouts was small (less than 20%) and consistent across interventions, or the trial authors used intention-to-treat analyses, or both (Abbott 2013; Alghadir 2019; Allen 2018; Anadkat 2015; Anwer 2014; Assar 2020; Baker 2001; Bandak 2022; Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Braghin 2018; Brismée 2007; Chen 2021; Cheung 2014; Cheung 2017; Christensen 2015; De Rooij 2017; Dwyer 2015; Ettinger 1997; Foley 2003; Fransen 2001; Fransen 2007; Gur 2002; Hay 2006; Hinman 2020; Holm 2020; Jan 2009; Janyacharoen 2018; Jenkinson 2009; Jorge 2015; Kim 2020; Knoop 2013; Krishnan 2021; Kuntz 2018; Lai 2021; Lee 2009; Lee 2019; Lee 2023; Li 2020; Lim 2008; Lin 2009; Lu 2017; Lund 2008; Mehwish 2023; Messier 2004; Messier 2013; Messier 2021; Nelligan 2021; Øiestad 2023; O'Reilly 1999; Qiu 2023; Robbins 2022; Sadeghi 2023; Salli 2010; Samuel 2014; Schilke 1996; Sedaghatnezhad 2021; Sekir 2005; Shabbir 2022; Simao 2012; Thomas 2002; Thorstensson 2005; Tsai 2013; Van Baar 1998; Vassao 2021; Wallis 2017; Wang 2011; Xiao 2021; Ye 2020). There was an unclear or high risk of bias in 67 (48%) studies as the number of dropouts was large (more than 20%) or authors did not use intention-to-treat analyses, or both (An 2008; Bautch 1997; Bruce-Brand 2012; Chang 2012; Chen 2014; Chopp-Hurley 2017; Da Silva 2015; De Almeida 2020; De Matos 2019; DeVita 2018; Dincer 2016; Drummen 2023; Gohir 2021; Gondhalekar 2013; Guo 2022; Henriksen 2014; Hernandez 2019; Hopman-Rock 2000; Hsu 2021; Hu 2020; Huang 2003; Huang 2005; Hurley 2007; Imoto 2012; Jan 2008; Joshi 2019; Kang 2022; Karadag 2019; Keefe 2004; Kholvadia 2019; Kilic 2020; Kim 2013; Kovar 1992; Kumar 2013; Liu 2019; Marconcin 2018; Maurer 1999; Mazloum 2018; Mete 2022; Multanen 2014; Nahayatbin 2018; Oh 2020; Ozlu 2023; Pazit 2018; Peloquin 1999; Rakhshani 2022; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rego 2023; Rogind 1998; Rosadi 2023; Rosedale 2014; Salacinski 2012; Samut 2015; Santana 2022; Schlenk 2020; Shahine 2020; Shen 2022; Shin 2017; Song 2003; Song 2020; Takacs 2017; Talbot 2003; Topp 2002; Vincent 2020; Wortley 2013).

Selective reporting

We judged reporting bias to be low in 45 (32%) studies, as these were prospectively registered and results for all outcomes in the trial registry were reported (Abbott 2013; Alghadir 2019; Bandak 2022; Bennell 2010; Bennell 2016; Bennell 2022a; Bennell 2022b; Bruce-Brand 2012; Cheung 2014; Cheung 2017; Da Silva 2015; De Almeida 2020; DeVita 2018; Drummen 2023; Dwyer 2015; Fransen 2007; Gohir 2021; Hay 2006; Henriksen 2014; Hinman 2020; Holm 2020; Hurley 2007; Knoop 2013; Lai 2021; Li 2020; Lim 2008; Lu 2017; Marconcin 2018; Messier 2021; Mete 2022; Multanen 2014; Nelligan 2021; Øiestad 2023; Pazit 2018; Rafiq 2021a; Rafiq 2021c; Rego 2023; Rosadi 2023; Rosedale 2014; Salacinski 2012; Schlenk 2020; Simao 2012; Takacs 2017; Wallis 2017; Ye 2020). Reporting bias was unclear in 89 (64%) studies as they were either not registered or it was not possible to access the trial registry (An 2008; Anadkat 2015; Anwer 2014; Baker 2001; Bautch 1997; Bhagat 2017; Braghin 2018; Brismée 2007; Chang 2012; Chen 2014; Chopp-Hurley 2017; Christensen 2015; De Matos 2019; Dincer 2016; Ettinger 1997; Foley 2003; Fransen 2001; Gondhalekar 2013; Guo 2022; Gur 2002; Hopman-Rock 2000; Hsu 2021; Hu 2020; Huang 2003; Huang 2005; Imoto 2012; Jan 2008; Jan 2009; Janyacharoen 2018; Jenkinson 2009; Jorge 2015; Joshi 2019; Kang 2022; Karadag 2019; Keefe 2004; Kholvadia 2019; Kilic 2020; Kim 2013; Kim 2020; Kovar 1992; Krishnan 2021; Kumar 2013; Kuntz 2018; Lee 2009; Lee 2019; Lee 2023; Lin 2009; Liu 2019; Lund 2008; Maurer 1999; Mazloum 2018; Mehwish 2023; Messier 2004; Messier 2013; Nahayatbin 2018; O'Reilly 1999; Oh 2020; Ozlu 2023; Peloquin 1999; Qiu 2023; Rakhshani 2022; Rafiq 2021b; Robbins 2022; Rogind 1998; Sadeghi 2023; Salli 2010; Samuel 2014; Samut 2015; Santana 2022; Schilke 1996; Sedaghatnezhad 2021; Sekir 2005; Shabbir 2022; Shahine 2020; Shen 2022; Shin 2017; Song 2003; Song 2020; Talbot 2003; Thomas 2002; Thorstensson 2005; Topp 2002; Tsai 2013; Van Baar 1998; Vassao 2021; Vincent 2020; Wang 2011; Wortley 2013; Xiao 2021). Reporting bias was high in five (4%) studies as they were unregistered or did not report all outcomes listed in their registry (Allen 2018; Assar 2020; Chen 2014; De Rooij 2017; Hernandez 2019).

Other potential sources of bias

We did not assess any other potential sources of bias.

Effects of interventions

See: **Summary of findings 1** Summary of findings table - Exercise compared to attention control/placebo for adults with knee osteoarthritis; **Summary of findings 2** Summary of findings table - Exercise compared to no treatment/usual care/limited education for adults with knee osteoarthritis; **Summary of findings 3** Summary of findings table - Exercise when added to another co-intervention compared to the co-intervention alone for adults with knee osteoarthritis

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Exercise versus placebo/attention control

Thirty studies with 3065 participants assessed exercise compared with placebo/attention control (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Jenkinson 2009; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Lai 2021; Liu 2019; Lu 2017; Maurer 1999; Messier 2004; Messier 2021; Peloquin 1999; Rego 2023; Santana 2022; Schlenk 2020; Shen 2022; Simao 2012; Song 2003; Song 2020; Tsai 2013; Vincent 2020; Ye 2020). Findings are summarised in Summary of findings 1.

Pain

Twenty-eight of 30 studies (2873 participants) reported pain immediately post-intervention (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Jenkinson 2009; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Lai 2021; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Schlenk 2020; Shen 2022; Simao 2012; Song 2003; Song 2020; Tsai 2013; Vincent 2020; Ye 2020). There was low-certainty evidence that exercise may result in a slight improvement in pain immediately post-intervention (Analysis 1.1). With attention control/placebo, the mean pain score was 44.4 points on a 0 to 100 point scale (where 0 represents no pain). With exercise, pain was 8.70 points better (95% CI 5.70 to 11.70). The confidence interval of the mean difference did not meet the threshold (12 points) for a clinically important effect. We downgraded the evidence as there was high risk of bias (more than 50% of trials were at high risk of performance and detection bias) and risk of publication bias (asymmetrical plot and Egger's test P < 0.05 (supplementary file https://doi.org/10.6084/m9.figshare.27012928.v1), in addition to the large volume of unpublished trials awaiting classification).

We conducted a series of separate sensitivity analyses. After removing four outliers with extreme results (De Almeida 2020; Rego 2023; Shen 2022; Song 2020), the mean was 5.25 points better with exercise (95% CI 3.30 to 7.20). When including only the 13 trials at low risk of selection bias, the mean was 8.10 points better with exercise (95% CI 4.20 to 12.15; Analysis 3.1). When including only the 14 trials at low risk of attrition bias, the mean was 5.40 points better with exercise (95% CI 3.30 to 7.65; Analysis 3.3). As only two trials were at low risk of detection bias (Baker 2001; Kuntz 2018), we did not perform sensitivity analysis for this variable. When including only the six trials that were likely to be fully powered and had low risk of bias (Bandak 2022; Cheung 2017; Ettinger 1997; Foley 2003; Jenkinson 2009; Messier 2021), the mean was 4.20 points better with exercise (95% CI 1.65 to 6.75).

Two of 28 studies (388 participants) reported pain at longer-term follow-up (from 36 to 52 weeks post-intervention) (Bandak 2022; Schlenk 2020). The mean pain score was 1.80 points better with exercise (95% CI -1.20 to 4.80; Analysis 2.1).

As only one of 30 studies reported pain more than 12 months post-intervention (Messier 2004), we did not perform meta-analysis for this comparison group.

Subgroup analysis: type of exercise

Effects were similar across strengthening exercise (mean 3.45 points better with exercise, 95% CI -0.15 to 7.05; 9 studies, 1159 participants; Analysis 4.1), mind-body exercise (mean 7.08 points better with exercise, 95% CI 3.45 to 12.15; 5 studies, 294 participants), and mixed exercise (mean 4.95 points better with exercise, 95% CI 1.95 to 7.95; 5 studies, 566 participants). We did not include stretching and aerobic exercise in the analysis due to the low number of studies in these subgroups (2 and 1, respectively). No studies used balance/proprioceptive exercise.

Meta-regression analysis: total number of prescribed exercise sessions

All 28 studies that reported pain were included in the meta-regression (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Jenkinson 2009; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Lai 2021; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Schlenk 2020; Shen 2022; Simao 2012; Song 2003; Song 2020; Tsai 2013; Vincent 2020; Ye 2020). The total number of prescribed exercise sessions within the intervention was not associated with change in pain immediately post-intervention (regression coefficient = 0.08, 95% CI -0.05 to 0.18, points on a 0 to 100 scale; P = 0.235).

Meta-regression analysis: ratio (between exercise and control) of synchronous (live) contact occasions with intervention provider

All 28 studies that reported pain were included in the meta-regression (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Jenkinson 2009; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Lai 2021; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Schlenk 2020; Shen 2022; Simao 2012; Song 2003; Song 2020; Tsai 2013; Vincent 2020; Ye 2020). The ratio of synchronous contact occasions between exercise and comparison was not associated with change in pain immediately post-intervention (regression coefficient = 0.92, 95% CI -0.63 to 2.46, points on a 0 to 100 scale; P = 0.232).

Physical function

Twenty-four of 30 studies (2536 participants) reported physical function immediately post-intervention (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Santana 2022; Schlenk 2020; Simao 2012; Song 2003; Tsai 2013; Ye 2020). There was moderate-certainty evidence that exercise likely results in an improvement in physical function immediately post-intervention (Analysis 1.2). With attention control/placebo, the mean physical function score was 35.8 points on a 0 to 100 point scale (where 0 represents no physical disability). With exercise, function was 11.27 points better (95% CI 7.64 to 15.09). The confidence interval of the mean difference includes both clinically important improvement (13 points) and clinically unimportant improvement. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

We conducted a series of separate sensitivity analyses. When three outliers with extreme results were removed (De Almeida 2020; Rego 2023; Santana 2022), the mean was 8.40 points better with exercise (95% CI 5.54 to 11.27). When including only the nine trials at low risk of selection bias, the mean was 12.03 points better with exercise (95% CI 6.49 to 17.57; Analysis 3.2). When including only the 12 trials at low risk of attrition bias, the mean was 8.60 points better with

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exercise (95% CI 4.58 to 12.61; Analysis 3.4). As only two trials were at low risk of detection bias (Baker 2001; Kuntz 2018), sensitivity analysis for this variable was not performed. When only including four trials that were likely to be fully powered and had low risk of bias (Bandak 2022; Cheung 2017; Ettinger 1997; Messier 2021), the mean was 3.44 (95% CI -0.38 to 7.26).

Two of 24 studies (388 participants) reported physical function up to 12-months post-intervention (Bandak 2022; Schlenk 2020). The mean was -0.19 points worse with exercise (95% CI -4.01 to 3.63; Analysis 2.2). Only one of 30 studies (Messier 2004) reported physical function more than 12 months post-intervention, and thus meta-analysis was not possible for this comparison group for long-term follow-up.

Subgroup analysis: type of exercise

Effects were similar across strengthening exercise (mean 4.97 points better with exercise, 95% CI -0.76 to 10.70; 6 studies, 860 participants; Analysis 4.2), mind-body exercise (mean 12.80 points better with exercise, 95% CI 7.07 to 18.34; 6 studies, 294 participants), and mixed exercise (mean 15.28 points better with exercise, 95% CI 6.69 to 23.88; 6 studies, 600 participants). Aerobic exercise was not included in the analysis due to the low number of studies in this subgroup (1 study). No studies used stretching or balance/proprioceptive exercise.

Meta-regression analysis: total number of prescribed exercise sessions

All 24 studies that reported physical function were included in the meta-regression (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Santana 2022; Schlenk 2020; Simao 2012; Song 2003; Tsai 2013; Ye 2020). The total number of prescribed exercise sessions within the intervention was not associated with change in function immediately post-intervention (regression coefficient = 0.11, 95% CI -0.02 to 0.25, points on a 0 to 100 point scale; P = 0.067).

Meta-regression analysis: ratio (between exercise and control) of synchronous (live) contact occasions with healthcare provider

All 24 studies that reported physical function were included in the meta-regression (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Santana 2022; Schlenk 2020; Simao 2012; Song 2003; Tsai 2013; Ye 2020). The ratio of synchronous contact occasions between exercise and comparison was not associated with change in function immediately post-intervention (regression coefficient = 1.68, 95% CI -0.06 to 3.44, points on a 0 to 100 point scale; P = 0.057).

Quality of life

Six of 30 studies (454 participants) reported quality of life immediately post-intervention (Bandak 2022; Cheung 2017; Foley 2003; Kuntz 2018; Lu 2017; Rego 2023). There was moderate-certainty evidence that exercise likely results in little to no improvement in quality of life immediately post-intervention (Analysis 1.3). With attention control/placebo, the mean quality of life score was 48.4 points on a 0 to 100 point scale (where 100 represents maximum quality of life). With exercise, quality of life was 6.06 points better (95% CI -0.13 to 12.26). The confidence interval of the mean difference does not meet the threshold (15 points) for a clinically important effect. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias). When we removed an outlier with extreme results (Rego 2023), the mean was 2.84 points better with exercise (95% CI -0.90 to 6.71).

Only one of six studies reported quality of life up to 12 months post-intervention (Bandak 2022), and thus meta-analysis was not possible for this comparison group at follow-up. No studies reported quality of life more than 12 months post-intervention.

Participant-reported treatment success

Two of 30 studies (364 participants) measured participant-reported treatment success (Bandak 2022; Jenkinson 2009). There was moderate-certainty evidence that exercise likely increases participant-reported treatment success. With attention control/placebo, the risk was 306 per 1000. With exercise, the risk was 446 per 1000 (95% CI 339 to 587; RR 1.46, 95% CI 1.11 to 1.92; Analysis 1.4). We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Study withdrawals

Twenty-nine of 30 studies (2907 participants) reported study withdrawals (Baker 2001; Bandak 2022; Brismée 2007; Cheung 2017; De Almeida 2020; Ettinger 1997; Foley 2003; Imoto 2012; Jenkinson 2009; Kang 2022; Kim 2013; Kovar 1992; Kuntz 2018; Lai 2021; Liu 2019; Lu 2017; Maurer 1999; Messier 2021; Peloquin 1999; Rego 2023; Santana 2022; Schlenk 2020; Shen 2022; Simao 2012; Song 2003; Song 2020; Tsai 2013; Vincent 2020; Ye 2020). There was moderate-certainty evidence that exercise likely does not increase study withdrawals. With attention control/placebo, the risk was 171 per 1000. With exercise, the risk was 185 per 1000 (95% CI 158 to 216; RR 1.08, 95% CI 0.92 to 1.26; Analysis 1.5). We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Adverse events

Table 5 summarises the number and nature of reported adverse events in each study. Eleven of 30 studies (1684 participants) reported adverse events (Bandak 2022; Cheung 2017; Ettinger 1997; Kovar 1992; Kuntz 2018; Liu 2019; Messier 2021; Rego 2023; Schlenk 2020; Tsai 2013; Ye 2020). There was low-certainty evidence that exercise may not increase adverse events. With attention control/placebo, the risk was 69 per 1000. With exercise, the risk was 140 per 1000 (95% CI 43 to 457; RR 2.02, 95% CI 0.62 to 6.58; Analysis 1.6). We downgraded the certainty of the evidence as there was high risk of bias (more than 50% of trials were at high risk of performance and detection bias) and imprecision (the ratio of the upper and lower 95% confidence intervals of the RR was > 3.0).

Exercise versus no treatment/usual care/limited education

Sixty studies with 4834 participants assessed exercise compared with no treatment/usual care/limited education (Abbott 2013; Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Hay 2006; Henriksen 2014; Hopman-Rock 2000; Hu 2020; Hurley 2007; Jan 2009; Janyacharoen 2018; Jorge 2015; Karadag 2019;

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Keefe 2004; Kim 2020; Krishnan 2021; Lee 2009; Lee 2023; Li 2020; Lim 2008; Lin 2009; Lund 2008; Mazloum 2018; Multanen 2014; Øiestad 2023; O'Reilly 1999; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Schilke 1996; Sekir 2005; Shahine 2020; Takacs 2017; Thorstensson 2005; Topp 2002; Vassao 2021; Wallis 2017; Wang 2011; Wortley 2013; Xiao 2021). Findings are summarised in Summary of findings 2.

Pain

Fifty-six of 60 studies (4184 participants) reported pain immediately post-intervention (Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Hay 2006; Henriksen 2014; Hopman-Rock 2000; Hu 2020; Janyacharoen 2018; Jorge 2015; Karadag 2019; Keefe 2004; Krishnan 2021; Lee 2009; Lee 2023; Li 2020; Lim 2008; Lin 2009; Lund 2008; Mazloum 2018; Øiestad 2023; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Schilke 1996; Sekir 2005; Shahine 2020; Takacs 2017; Thorstensson 2005; Topp 2002; Vassao 2021; Wallis 2017; Wang 2011; Wortley 2013; Xiao 2021). There was low-certainty evidence that exercise may result in an improvement in pain (Analysis 1.1). With no treatment/usual care/limited education, the mean pain score was 51.1 on a 0 to 100 point scale (where 0 represents no pain). With exercise, pain was 13.14 points better (95% CI 10.36 to 15.91). The confidence interval of the mean difference includes both clinically important (12 points) and clinically unimportant improvements. We downgraded the certainty of the evidence as there was high risk of bias (more than 50% of trials were at high risk of performance and detection bias) and some evidence of publication bias (asymmetrical plot and Egger's test P < 0.05 (supplementary file DOI: 10.6084/m9.figshare.27012928), plus a large volume of unpublished trials awaiting classification).

We conducted a series of separate sensitivity analyses. When we removed five outliers with extreme results (Drummen 2023; Gur 2002; Janyacharoen 2018; Karadag 2019; Shahine 2020), the mean was 10.18 points better with exercise (95% CI 8.33 to 12.03). When including only the 23 trials at low risk of selection bias, the mean was 9.99 points better with exercise (95% CI 7.22 to 12.77; Analysis 3.1). When including only the 28 trials at low risk of attrition bias, the mean was 10.92 points better with exercise (95% CI 7.40 to 14.25; Analysis 3.3). As only one trial was at low risk of detection bias (Guo 2022), we did not perform sensitivity analysis for this variable. When including only 14 trials that were likely to be fully powered and had low risk of bias (Bennell 2010; Christensen 2015; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Hay 2006; Lim 2008; Lin 2009; Multanen 2014; Øiestad 2023; Robbins 2022; Rosedale 2014; Sadeghi 2023), the mean was 9.99 points better with exercise (95% CI 6.66 to 13.51).

Thirteen of 56 studies (1652 participants) reported pain up to 12 months post-intervention (Abbott 2013; Allen 2018; Bruce-Brand 2012; Chen 2014; De Rooij 2017; Hay 2006; Hurley 2007; Lund 2008; Øiestad 2023; Rogind 1998; Rosedale 2014; Thorstensson 2005). The mean was 5.92 points better with exercise (95% CI 0.74 to 11.10; Analysis 2.1). No studies reported pain more than 12 months post-intervention.

Subgroup analysis: type of exercise

Effects were similar across strengthening exercise (mean 14.06 points better with exercise, 95% CI 8.70 to 19.24; 19 studies, 1303 participants; Analysis 4.3), aerobic exercise (mean 20.35 points better with exercise, 95% CI 2.59 to 38.30; 6 studies, 320 participants), mind-body exercise (19.43 points better with exercise, 95% CI 8.51 to 30.34; 7 studies, 621 participants), and mixed exercise (mean 9.44 points better with exercise, 95% CI 7.03 to 12.03; 18 studies, 1621 participants). We did not include stretching and balance/proprioceptive exercise in the analysis due to the low number of studies in these subgroups (1 study in each).

Meta-regression analysis: total number of prescribed exercise sessions

Fifty-two of 56 studies that reported pain were included in the meta-regression (Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Henriksen 2014; Hopman-Rock 2000; Hu 2020; Janyacharoen 2018; Jorge 2015; Karadag 2019; Keefe 2004; Krishnan 2021; Lee 2009; Lee 2023; Lim 2008; Lin 2009; Lund 2008; Mazloum 2018; Øiestad 2023; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Schilke 1996; Sekir 2005; Takacs 2017; Thorstensson 2005; Topp 2002; Vassao 2021; Wang 2011; Wortley 2013; Xiao 2021; we did not include the other four as the total number of exercise sessions was unclear or unspecified (Hay 2006; Li 2020; Shahine 2020; Wallis 2017). There was no relationship between the total number of prescribed exercise sessions within the intervention and change in pain immediately post-intervention (regression coefficient = 0.04, 95% CI -0.11 to 0.17, points on a 0 to 100 point scale; P = 0.629).

Meta-regression analysis: ratio (between exercise and control) of synchronous (live) contact occasions with healthcare provider

Six of 56 studies (509 participants) that reported pain were included in the meta-regression (Da Silva 2015; Hay 2006; Hu 2020; Janyacharoen 2018; Multanen 2014; Wortley 2013); we did not include the other 50 as their exercise or comparison group involved no synchronous contact occasions. There was no relationship between the ratio of synchronous contact occasions between exercise and comparison and change in pain immediately post-intervention (regression coefficient = -0.19, 95% CI -2.98 to 2.61, points on a 0 to 100 point scale; P = 0.858).

Physical function

Fifty-four of 60 studies (4352 participants) reported physical function immediately post-intervention (Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Hay 2006; Henriksen 2014; Hopman-Rock 2000; Hu 2020; Hurley 2007; Jan 2009; Janyacharoen 2018; Jorge 2015; Karadag 2019; Kim 2020; Krishnan 2021; Lee 2009; Li 2020; Lim 2008; Lin 2009; Lund 2008; Multanen 2014; Øiestad 2023; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Schilke 1996; Shahine 2020; Takacs 2017; Thorstensson 2005; Topp 2002; Vassao 2021; Wallis 2017; Wang 2011; Wortley 2013). There was moderate-certainty evidence that exercise likely results in an improvement in physical function (Analysis 1.2). With no treatment/usual care/limited education, the mean physical function score

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was 42.5 points on a 0 to 100 point scale (where 0 represents no physical disability). With exercise, function was 12.53 points better (95% CI 9.74 to 15.31). The confidence interval of the mean difference includes both clinically important (13 points) and clinically unimportant improvements. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

We conducted a series of separate sensitivity analyses. When we removed six outliers with extreme results (Chen 2014; Drummen 2023; Gur 2002; Janyacharoen 2018; Karadag 2019; Shahine 2020), the mean was 9.05 points better with exercise (95% CI 6.96 to 10.96). When including only the 25 trials that were at low risk of selection bias, the mean was 10.61 points better with exercise (95% CI 7.48 to 13.57; Analysis 3.2). When including only the 27 trials at low risk of attrition bias, the mean was 11.14 points better with exercise (95% CI 7.48 to 14.62; Analysis 3.4). As only one trial was at low risk of detection bias (Guo 2022), we did not perform sensitivity analysis for this variable. When we included only 11 trials that were likely to be fully powered and had low risk of bias (Christensen 2015; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Hay 2006; Hurley 2007; Lim 2008; Øiestad 2023; Rosedale 2014; Sadeghi 2023), the mean was 8.87 points better with exercise (95% CI 5.57 to 12.01).

Twelve of 54 studies (1604 participants) reported physical function up to 12 months post-intervention (Abbott 2013; Allen 2018; Bruce-Brand 2012; Chen 2014; De Rooij 2017; Hay 2006; Hurley 2007; Lund 2008; Øiestad 2023; Rogind 1998; Rosedale 2014; Thorstensson 2005). The mean was 5.67 points better with exercise (95% CI 2.44 to 8.70; Analysis 2.2). No studies reported physical function more than 12 months post-intervention.

Subgroup analysis: type of exercise

Effects were similar across strengthening exercise (mean 15.49 points better with exercise, 95% CI 9.40 to 21.40; 18 studies, 1324 participants; Analysis 4.4), aerobic exercise (mean 17.92 points better with exercise, 95% CI 1.57 to 34.28; 6 studies, 320 participants), mind-body exercise (20.18 points better with exercise, 95% CI 7.48 to 33.06; 6 studies, 337 participants), and mixed exercise (mean 7.48 points better with exercise, 95% CI 5.57 to 9.57, 19 studies, 2074 participants). We did not include stretching exercise in the analysis as only one study was in this subgroup. No studies used balance/proprioceptive exercise.

Subgroup analysis: total number of prescribed exercise sessions

Fifty of 54 studies that reported physical function were included in the meta-regression (Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Henriksen 2014; Hopman-Rock 2000; Hu 2020; Hurley 2007; Jan 2009; Janyacharoen 2018; Jorge 2015; Karadag 2019; Kim 2020; Krishnan 2021; Lee 2009; Lim 2008; Lin 2009; Lund 2008; Multanen 2014; Øiestad 2023; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Schilke 1996; Takacs 2017; Thorstensson 2005; Topp 2002; Vassao 2021; Wang 2011; Wortley 2013). We did not include the other four studies as the total number of exercise sessions was unclear or unspecified (Hay 2006; Li 2020; Shahine 2020; Wallis 2017). There was no relationship between the total number of prescribed exercise sessions within the intervention and change in function immediately post-intervention (regression coefficient = 0.03, 95% CI -0.07 to 0.16, points on a 0 to 100 point scale; P = 0.464).

Meta-regression analysis: ratio (between exercise and control) of synchronous (live) contact occasions with healthcare provider

Six of 54 studies that reported physical function were included in the meta-regression (Da Silva 2015; Hay 2006; Hu 2020; Janyacharoen 2018; Multanen 2014; Wortley 2013). We did not include the remaining 48 as their exercise and/or comparison group involved no synchronous contact occasions. There was no relationship between the ratio of synchronous contact occasions between exercise and comparison and change in function immediately post-intervention (regression coefficient = -0.38, 95% CI -1.74 to 0.97, points on a 0 to 100 point scale; P = 0.481).

Quality of life

Twenty-eight of 60 studies (2328 participants) reported quality of life immediately post-intervention (Bruce-Brand 2012; Cheung 2014; Chopp-Hurley 2017; Christensen 2015; Da Silva 2015; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Hay 2006; Henriksen 2014; Janyacharoen 2018; Jorge 2015; Keefe 2004; Krishnan 2021; Lee 2009; Li 2020; Lund 2008; Øiestad 2023; Pazit 2018; Rosedale 2014; Salacinski 2012; Thorstensson 2005; Wallis 2017; Wang 2011; Xiao 2021). There was moderate-certainty evidence that exercise likely results in a slight improvement in quality of life (Analysis 1.3). With no treatment/usual care/limited education, the mean quality of life score was 54.3 on a 0 to 100 point scale (where 100 is maximum quality of life). With exercise, quality of life was 5.37 points better (95% CI 3.19 to 7.54). The confidence interval of the mean difference does not include the threshold (15 points) for a clinically important effect. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Six of 28 studies (906 participants) reported quality of life up to 12 months post-intervention (Bruce-Brand 2012; Hurley 2007; Lund 2008; Øiestad 2023; Rosedale 2014; Thorstensson 2005). The mean was 0.73 points better with exercise (95% CI -2.90 to 1.31 Analysis 2.3). No studies reported quality of life more than 12 months post-intervention.

Participant-reported treatment success

Three of 60 studies (405 participants) measured participant-reported treatment success (Christensen 2015; Hay 2006; Jorge 2015). There was low-certainty evidence that exercise may result in no difference in participant-reported treatment success. With no treatment/usual care/limited education, the risk was 325 per 1000. With exercise, the risk was 432 per 1000 (95% CI 231 to 810; RR 1.33, 95% CI 0.71 to 2.49; Analysis 1.4). We downgraded the certainty of the evidence as there was high risk of bias (more than 50% of trials were at high risk of performance and detection bias) and imprecision (the ratio of the upper and lower 95% confidence intervals of the RR was > 3.0).

Study withdrawals

Fifty-three of 60 studies (4408 participants) reported study withdrawals (Allen 2018; An 2008; Bennell 2010; Braghin 2018; Bruce-Brand 2012; Chen 2014; Cheung 2014; Christensen 2015; Da Silva 2015; De Matos 2019; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2001; Fransen 2007; Gohir 2021; Guo 2022; Gur 2002; Hay 2006; Henriksen 2014; Hu 2020; Hurley 2007; Jan 2009;

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Janyacharoen 2018; Jorge 2015; Karadag 2019; Keefe 2004; Kim 2020; Krishnan 2021; Lee 2009; Lee 2023; Li 2020; Lim 2008; Lin 2009; Lund 2008; Mazloum 2018; Multanen 2014; Øiestad 2023; O'Reilly 1999; Pazit 2018; Robbins 2022; Rogind 1998; Rosedale 2014; Sadeghi 2023; Salacinski 2012; Samut 2015; Sekir 2005; Takacs 2017; Thorstensson 2005; Vassao 2021; Wallis 2017; Wang 2011; Wortley 2013; Xiao 2021). There was moderate-certainty evidence that exercise likely does not increase study withdrawals. With no treatment/usual care/limited education, the risk was 119 per 1000. With exercise, the risk was 123 per 1000 (95% CI 105 to 144; RR 1.03, 95% CI 0.88 to 1.20; Analysis 1.5). We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Adverse events

Table 5 summarises the number and nature of reported adverse events in each study. Eighteen of 60 studies (1557 participants) reported adverse events (Abbott 2013; Allen 2018; An 2008; Cheung 2014; Christensen 2015; De Rooij 2017; DeVita 2018; Drummen 2023; Fransen 2007; Gohir 2021; Hay 2006; Li 2020; Multanen 2014; Øiestad 2023; Pazit 2018; Rogind 1998; Takacs 2017; Wallis 2017). There was low-certainty evidence that exercise may increase adverse events. With no treatment/usual care/limited education, the risk was 77 per 1000. With exercise, the risk was 244 per 1000 (95% CI 90 to 659; RR 3.17, 95% CI 1.17 to 8.57; Analysis 1.6). We downgraded the certainty of the evidence as there was high risk of bias (more than 50% of trials were at high risk of performance and detection bias) and imprecision (the ratio of the upper and lower 95% confidence intervals of the RR was > 3.0).

Exercise added to another co-intervention compared to that co-intervention alone

Forty-nine studies (4569 participants) evaluated exercise when added to another co-intervention (compared to that co-intervention alone) (Alghadir 2019; Anadkat 2015; Anwer 2014; Assar 2020; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Gondhalekar 2013; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Huang 2003; Huang 2005; Jan 2008; Joshi 2019; Kholvadia 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rakhshani 2022; Rosadi 2023; Salli 2010; Samuel 2014; Sedaghatnezhad 2021; Shabbir 2022; Shin 2017; Talbot 2003; Thomas 2002; Van Baar 1998). Findings are summarised in Summary of findings 3.

Pain

Forty-seven of 49 studies (4441 participants) reported pain immediately post-intervention (Alghadir 2019; Anadkat 2015; Anwer 2014; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Gondhalekar 2013; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Jan 2008; Joshi 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rakhshani 2022; Rosadi 2023; Salli 2010; Samuel 2014; Sedaghatnezhad 2021; Shabbir 2022; Shin 2017; Talbot 2003; Van Baar 1998). There was moderate-certainty evidence that exercise likely results in an improvement in pain immediately post-intervention (Analysis 1.1). With the comparison group, the mean pain score was 57.9 points on a 0 to 100 point scale (where 0 represents no pain). With exercise, pain was 10.43 points better (95% CI 8.06 to 12.79). The confidence interval of the mean difference includes both clinically important (12 points) and clinically unimportant improvements. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

We conducted a series of separate sensitivity analyses. When we removed four outliers with extreme results (Anwer 2014; Joshi 2019; Mete 2022; Samuel 2014), the mean was 8.06 points better with exercise (95% CI 6.26 to 10.01). When including only the 17 trials at low risk of selection bias, the mean was 5.42 points better with exercise (95% CI 3.20 to 7.51; Analysis 3.1). When including only the seven trials at low risk of detection bias, the mean was 8.76 points better with exercise (95% CI 4.17 to 13.34; Analysis 3.5). When including only the 23 trials at low risk of attrition bias, the mean was 9.31 points better with exercise (95% CI 6.39 to 12.37; Analysis 3.3). When we included 10 trials that were likely to be fully powered and had low risk of bias (Alghadir 2019; Bennell 2016; Bennell 2022a; Bennell 2022b; Hinman 2020; Holm 2020; Knoop 2013; Nelligan 2021; Thomas 2002; Van Baar 1998), the mean was 4.73 points better with exercise (95% CI 3.48 to 6.12).

Thirteen of 47 studies (1517 participants) reported pain up to 12-months post-intervention (Bennell 2016; Bennell 2022a; Bennell 2022b; Hernandez 2019; Hinman 2020; Huang 2003; Huang 2005; Knoop 2013; Nahayatbin 2018; Ozlu 2023; Salli 2010; Sedaghatnezhad 2021; Talbot 2003). The mean was 10.84 points better with exercise (95% CI 5.56 to 16.26; Analysis 2.1). No studies reported pain more than 12 months post-intervention.

Subgroup analysis: type of exercise

Effects were similar across strengthening exercise (mean 12.09 points better with exercise, 95% CI 8.06 to 16.12; 23 studies, 2475 participants; Analysis 4.5), aerobic exercise (mean 12.51 points better with exercise, 95% CI 5.70 to 19.32; 9 studies, 362 participants), and mixed exercise (mean 5.84 points better with exercise, 95% CI 3.34 to 8.48; 12 studies, 1300 participants). We did not include mind-body and balance/proprioceptive exercise in the analysis due to the low number of studies in these subgroups (1 in both). No studies used stretching exercise.

Meta-regression analysis: total number of prescribed exercise sessions

All but one of the 47 studies that reported pain were included in the meta-regression (Alghadir 2019; Anadkat 2015; Anwer 2014; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Gondhalekar 2013; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Jan 2008; Joshi 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rakhshani 2022; Rosadi 2023; Salli 2010; Samuel 2014; Sedaghatnezhad 2021; Shabbir 2022; Shin 2017; Van Baar 1998). We did not include the final study as the total number of exercise sessions was unclear or unspecified (Talbot 2003). There was no relationship between the total number of prescribed exercise sessions within the intervention and change in pain immediately post-intervention (regression coefficient = 0.01, 95% CI -0.01 to 0.04, points on a 0 to 100 point scale; P = 0.348).

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Meta-regression analysis: ratio (between exercise and control) of synchronous (live) contact occasions with healthcare provider

Thirty-four of 47 studies that reported pain were included in the meta-regression (Alghadir 2019; Anadkat 2015; Anwer 2014; Bautch 1997; Bennell 2016; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rakhshani 2022; Rosadi 2023; Samuel 2014; Sedaghatnezhad 2021; Shabbir 2022; Van Baar 1998). We did not include the other 13 studies as their exercise or comparison group involved no synchronous occasions (Assar 2020; Bennell 2022a; Bennell 2022b; Gondhalekar 2013; Huang 2003; Huang 2005; Jan 2008; Joshi 2019; Nelligan 2021; Rafiq 2021b; Rafiq 2021c; Salli 2010; Shin 2017). There was no relationship between the ratio of synchronous contact occasions between exercise and comparison and change in pain immediately post-intervention (regression coefficient = -0.19, 95% CI -1.32 to 0.92, points on a 0 to 100 point scale; P = 0.716).

Physical function

Forty-four of 49 studies (4381 participants) reported physical function immediately post-intervention (Alghadir 2019; Anadkat 2015; Anwer 2014; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Gondhalekar 2013; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Huang 2003; Huang 2005; Jan 2008; Joshi 2019; Kholvadia 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rosadi 2023; Salli 2010; Samuel 2014; Shabbir 2022; Shin 2017; Van Baar 1998). There was moderate-certainty evidence that exercise likely results in a slight improvement in physical function immediately post-intervention (Analysis 1.2). With the comparator, the mean physical function scale was 50.6 points on a 0 to 100 point scale (where 0 is no physical disability). With exercise, function was 9.66 points better (95% CI 7.48 to 11.97). The confidence interval of the estimate does not include the threshold (13 points) for a clinically important effect. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

We conducted a series of separate sensitivity analyses. When we removed three outliers with extreme results (Anwer 2014; Joshi 2019; Rosadi 2023), the mean was 8.16 points better with exercise (95% CI 6.26 to 10.20). When including only the 16 trials at low risk of selection bias, the mean was 4.90 points better with exercise (95% CI 2.72 to 7.21; Analysis 3.2). When including only the seven trials at low risk of detection bias, the mean was 8.57 points better with exercise (95% CI 3.81 to 13.46; Analysis 3.6). When including only the 22 trials at low risk of attrition bias, the mean was 7.48 points better with exercise (95% CI 4.09 to 9.93; Analysis 3.4). When only including nine trials that were likely to be fully powered and had low risk of bias (Bennell 2016; Bennell 2022a; Bennell 2022b; Hinman 2020; Holm 2020; Knoop 2013; Nelligan 2021; Thomas 2002; Van Baar 1998), the mean was 5.03 points better with exercise (95% CI 2.86 to 7.07).

Twelve of 44 studies (1525 participants) reported physical function up to 12 months post-intervention (Bennell 2016; Bennell 2022a; Bennell 2022b; Hernandez 2019; Hinman 2020; Huang 2003; Huang 2005; Kholvadia 2019; Knoop 2013; Nahayatbin 2018; Ozlu 2023; Salli 2010). The mean was 8.30 points better with exercise (95% CI 3.94 to 12.65; Analysis 2.2). No studies reported physical function more than 12 months post-intervention.

Subgroup analysis: type of exercise

Effects were similar across strengthening exercise (mean 9.66 points better with exercise, 95% CI 6.26 to 13.19; 21 studies, 2407 participants; Analysis 4.6), aerobic exercise (mean 13.87 points better with exercise, 95% CI 8.43 to 19.45; 7 studies, 298 participants), and mixed exercise (mean 7.75 points better with exercise, 95% CI 3.67 to 11.83; 13 studies, 1372 participants). We did not include mind-body and balance/proprioceptive exercise in the analysis due to the low number of studies in these subgroups (1 in each). No studies used stretching exercise.

Meta-regression analysis: total number of prescribed exercise sessions

All 44 studies that reported physical function were included in the meta-regression (Alghadir 2019; Anadkat 2015; Anwer 2014; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Gondhalekar 2013; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Jan 2008; Joshi 2019; Kholvadia 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rosadi 2023; Salli 2010; Samuel 2014; Shabbir 2022; Shin 2017; Van Baar 1998). There was no relationship between the total number of prescribed exercise sessions within the intervention and change in physical function immediately post-intervention (regression coefficient = 0.01, 95% CI -0.01 to 0.04, points on the 0 to 100 point scale; P = 0.348).

Meta-regression analysis: ratio (between exercise and control) of synchronous (live) contact occasions with healthcare provider

Thiry-two of 44 studies (2993 participants) that reported physical function were included in the meta-regression (Alghadir 2019; Anadkat 2015; Anwer 2014; Bautch 1997; Bennell 2016; Bhagat 2017; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Kholvadia 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nahayatbin 2018; Oh 2020; Ozlu 2023; Qiu 2023; Rosadi 2023; Samuel 2014; Shabbir 2022; Van Baar 1998) - 12 were not included as their exercise and/or comparison group involved 0 synchronous occasions (Bennell 2022a; Bennell 2022b; Gondhalekar 2013; Huang 2003; Huang 2005; Jan 2008; Joshi 2019; Nelligan 2021; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Shin 2017). There was no relationship between the ratio of synchronous contact occasions between exercise and comparison and change in physical function immediately post-intervention (regression coefficient = -0.03, 95% CI -1.1 to 1.03, points on a 0 to 100 point scale; P = 0.950).

Quality of life

Twelve of 49 studies (1660 participants) reported quality of life immediately post-intervention (Bennell 2016; Bennell 2022a; Bennell 2022b; Hinman 2020; Holm 2020; Marconcin 2018; Mehwish 2023; Messier 2013; Nahayatbin 2018; Nelligan 2021; Rafiq 2021a; Rosadi 2023). There was moderate-certainty evidence that exercise likely results in a slight improvement in quality of life immediately post-intervention (Analysis 1.3). With the comparator, the mean quality of life score was 64.2 points on a 0 to 100 point scale (where

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100 represents maximum quality of life). With exercise, the mean quality of life score was 4.22 points better (95% CI 1.36 to 7.07). The confidence interval of the estimate does not include the threshold (15 points) for a clinically important effect. We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Six of 12 studies (897 participants) reported quality of life up to 12 months post-intervention (Bennell 2016; Bennell 2022a; Bennell 2022b; Hinman 2020; Nahayatbin 2018; Rosadi 2023). The mean was 5.30 points better with exercise (95% CI 1.36 to 9.25; Analysis 2.3). No studies reported quality of life > 12 months post-intervention.

Participant-reported treatment success

Six of 49 studies (1139 participants) measured participant-reported treatment success (Bennell 2016; Bennell 2022a; Hinman 2020; Knoop 2013; Nelligan 2021). There was moderate-certainty evidence that exercise likely increases participant-reported treatment success. With the comparator, the risk was 400 per 1000. With exercise, the risk was 653 per 1000 (95% CI 472 to 897; RR 1.63, 95% CI 1.18 to 2.24; Analysis 1.4). We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Study withdrawals

Forty-one of 49 studies (3502 participants) reported study withdrawals (Alghadir 2019; Anwer 2014; Assar 2020; Bautch 1997; Bennell 2016; Bennell 2022a; Bennell 2022b; Chang 2012; Chen 2021; Dincer 2016; Dwyer 2015; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Jan 2008; Joshi 2019; Kilic 2020; Knoop 2013; Kumar 2013; Lee 2019; Marconcin 2018; Mehwish 2023; Messier 2013; Mete 2022; Nelligan 2021; Oh 2020; Ozlu 2023; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Rosadi 2023; Salli 2010; Samuel 2014; Sedaghatnezhad 2021; Shabbir 2022; Talbot 2003; Van Baar 1998). There was moderate-certainty evidence that exercise likely reduces study withdrawals slightly. With the comparator, the risk was 139 per 1000. With exercise, the risk was 114 per 1000 (95% CI 98 to 135; RR 0.82, 95% CI 0.70 to 0.97; Analysis 1.5). We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

Adverse events

Table 5 summarises the number and nature of reported adverse events in each study. Nineteen of 49 trials (2187 participants) reported adverse events (Alghadir 2019; Bennell 2016; Bennell 2022a; Bennell 2022b; Chen 2021; Dwyer 2015; Hernandez 2019; Hinman 2020; Holm 2020; Hsu 2021; Knoop 2013; Mehwish 2023; Messier 2013; Nelligan 2021; Qiu 2023; Rafiq 2021a; Rafiq 2021b; Rafiq 2021c; Shabbir 2022). There was moderate-certainty evidence that exercise likely increases adverse events slightly. With the comparator, the risk was 42 per 1000. With exercise, the risk was 73 per 1000 (95% CI 45 to 117; RR 1.72, 95% CI 1.07 to 2.76; Analysis 1.6). We downgraded the certainty of the evidence as there was a high risk of bias (more than 50% of trials were at high risk of performance and detection bias).

DISCUSSION

Summary of main results

A large number of randomised controlled trials of land-based exercise for people with knee osteoarthritis have been published since the previous version of this Cochrane review (Fransen 2015). The review now includes 139 trials (96 of which are new), with 12,468 participants. Thirty of the trials compared exercise to attention control/placebo; 60 compared exercise to no treatment/usual care/limited education; and 49 evaluated exercise when added to another co-intervention (compared to that co-intervention alone). Exercise interventions ranged from 2 to 104 weeks in duration.

Compared to attention control/placebo, our findings demonstrate that exercise may result in a slight improvement in pain, and likely results in an improvement in physical function, but likely results in little to no improvement in quality of life immediately post-intervention (Summary of findings 1). However, based on the minimal important differences (MIDs) that we used, benefits for pain and function were of uncertain clinical importance. Further, when we removed trials with an unclear or high risk of bias and inadequate sample size, mean differences were smaller (and therefore clinically important thresholds were still not met). Benefits may not be sustained in the longer-term (up to 12-months post-intervention), though only two trials evaluated longer-term effects. We found that exercise likely increases participant-reported treatment success immediately post-intervention, likely does not increase study withdrawals at final follow-up, and may not increase adverse events at final follow-up. We have low to moderate confidence in these effect estimates. We downgraded the certainty of the evidence due to the number of trials that were at unclear or high risk of bias, some evidence of publication bias, and some imprecision.

Compared to no treatment, usual care or limited education, our findings demonstrate that exercise may result in an improvement in pain, likely results in an improvement in physical function, and likely results in a slight improvement in quality of life immediately post-intervention (Summary of findings 2). However, based on the MIDs that we used, the benefits were of uncertain clinical importance. Benefits were sustained in the longer-term, though the magnitude of the effects were smaller. We found that exercise may result in little to no difference in participant-reported treatment success immediately post-intervention, likely results in no difference in study withdrawals at final follow-up, and may increase adverse events at final follow-up. We have low to moderate confidence in these effect estimates. We downgraded the certainty of the evidence due to the number of trials that were at unclear or high risk of bias, some evidence of publication bias, and some imprecision.

When exercise was added to another co-intervention (compared to that co-intervention alone), our findings demonstrate that exercise likely results in an improvement in pain, a slight improvement in physical function, and a slight improvement in quality of life immediately post-intervention (Summary of findings 3). However, based on the MIDs that we used, the benefits were of uncertain clinical importance. Effects were sustained in the longer-term. We found that exercise likely increases participant-reported treatment success immediately post-intervention, reduces withdrawals slightly, and increases

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adverse events slightly. We have moderate confidence in the effect estimates. We downgraded the certainty of the evidence due to the number of trials that were at unclear or high risk of bias.

Subgroup analysis did not suggest any differences between different types of exercise, and there was no relationship between changes in pain or physical function and the total number of exercise sessions prescribed or ratio (between exercise group and comparator) of live/real-time consultations with a healthcare provider.

Overall completeness and applicability of evidence

Overall, the trials included participants whose results are largely generalisable to people with knee OA across different countries and settings. Trials were conducted across 28 different countries, most frequently the USA, Australia, Canada, and the UK, as well as low-middle income countries including Argentina, India, Iran, and Pakistan. Participants were, on average, aged between 41 and 81 years, and had a BMI ranging from an average of 22.4 to 37.9 kg/m^2. Given that there is a poor association between radiographic changes and symptoms in knee OA (Kinds 2011), we did not extract data relating to the radiographic severity of OA amongst study participants. However, the range of samples included in our review, with varying symptomatic severity at baseline, suggests that our results can be broadly generalised.

The exercise interventions included in our review varied substantially in terms of type, dosage, delivery, and length. For example, more than 10 different types of exercise were prescribed, and interventions ranged from two to 78 weeks in length. Many exercise interventions included elements of behaviour change support to maximise adherence, though others did not, which may have contributed to inconsistency in results. This heterogeneity in exercise interventions likely explains the high I^2 statistics we observed in many of our meta-analyses.

There was also heterogeneity in the comparison groups. Around a quarter of the trials compared exercise to attention control/placebo, though only two of those were described by the authors as placebo/sham. Of those that used attention control as comparison, this most frequently involved education sessions, regular telephone contact, or home visits to provide attention, though the number and duration of contact sessions varied between studies. Just over 40% of trials compared exercise to no treatment/usual care/limited education. There was some variability in 'usual care' interventions, with some trials stating that participants received routine care offered by their family doctor, while others provided participants with information about community services and resources that were available to them. Around a third of the trials evaluated exercise when added to another co-intervention (compared to that co-intervention alone); there was substantial variability in type of comparison intervention, for example, ultrasound, routine physiotherapy, education, pain coping skills training, and weight loss diet. This heterogeneity in the comparison intervention likely also contributed to some inconsistency and imprecision in our effect estimates.

Most of the trials assessed the effects of exercise on pain and physical function. Fewer trials measured quality of life (32%) or participant-reported treatment success (10%). In addition, two-thirds of the studies (66%) failed to report adverse events. As such, we are less certain of the applicability of results for these outcomes.

Longer-term outcomes, beyond the length of the intervention, were only measured in 20% of trials. In particular, only one trial (1%; Messier 2004) measured outcomes more than 12 months post-intervention. We are therefore less certain about the sustainability of exercise effects in the longer-term after the intervention has stopped.

A substantial number of included studies had small sample sizes (mean 40 participants per study; SD = 35.8). Only 21% of included trials had an overall low risk of bias and had a sample size large enough to provide at least 80% power to detect a clinically important difference in pain. We believe that additional small trials of exercise for knee OA may further inflate our effect estimates and uncertainty in the evidence.

Overall, we found that there is a crucial need for better reporting of the design and conduct of trials. The CONSORT statement provides a checklist for trial authors to use when presenting results of a clinical trial. Exercise intervention details were often poorly reported, including specific information relating to exercise dosage and delivery. The Consensus on Exercise Reporting Template and the TIDieR checklist provide trial authors with helpful tools to use when describing exercise interventions. There is also a need for journal editors to ensure trial manuscripts meet the recommendations of these tools.

Exercise versus attention control/placebo

We downgraded the evidence for all outcomes as more than 50% of the trials in this comparison were at high risk of performance and detection bias. While recognising that blinding of participants and personnel is difficult to achieve in these types of trials, failure to blind is problematic as trials with unblinded assessment of subjective outcomes are estimated to exaggerate the treatment benefit (Wood 2008). We hypothesise that correcting for these biases would likely drive improvements towards null. We further downgraded the evidence for pain as there was some evidence of publication bias (asymmetrical plot and Egger's test $P < 0.05$, as well as the fact that we identified a substantial number of records relating to unpublished trials). For pain, function, and quality of life, although there was considerable heterogeneity ($I^2 > 75\%$), this was likely explained by the significant variation in intervention and comparison types and length. Although the 95% CI of the SMD for quality of life was > 0.5, we did not downgrade for imprecision as the number of participants (N = 454) exceeded the suggested optimal information size (N = 400; as recommended by Zeng 2022). However, we did downgrade for imprecision for adverse events, as the ratio of the upper and lower 95% CI for the RR was > 3.0 (as recommended by Zeng 2022).

Exercise versus no intervention/usual care/limited education

We downgraded the evidence for all outcomes as more than 50% of the trials in this comparison were at high risk of performance and detection bias. We further downgraded the evidence for pain as there was some evidence of publication bias (asymmetrical plot and Eggers test $P < 0.05$, as well as the fact that we identified a substantial number of records relating to unpublished trials). For pain, function, and quality of life, although there was considerable heterogeneity ($I^2 > 50\%$), this was likely explained by the significant variation in intervention and comparison types and length. We downgraded for imprecision for participant-reported treatment

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success and adverse events as the ratio of the upper and lower 95% CI of the RRs was > 3.0 (Zeng 2022).

Exercise (when added to another co-intervention) compared to that co-intervention alone

We downgraded the evidence for all outcomes as more than 50% of trials were at high risk of performance and detection bias. For pain, function, quality of life, and participant-reported treatment success, there was considerable heterogeneity (I^2 > 75%), but this was likely explained by the significant variation in intervention and comparison types and length. For quality of life, although the 95% CI of the SMD was > 0.2, we did not downgrade for imprecision as the number of participants (N = 1660) exceeded the optimal information size (Zeng 2022).

Potential biases in the review process

To minimise bias, we used up to nine independent assessors to screen and select studies. To ensure consistency in judgement and decisions, only two review authors independently performed risk of bias judgement for all trials and also independently performed GRADE assessment for all comparisons (other than for trials authored by BJL, for which AD conducted risk of bias and GRADE assessments).

To the best of our knowledge, we identified all relevant trials meeting the review's eligibility criteria through a comprehensive search of major electronic databases and trial registries without language restrictions. However, it is possible that we may have missed some trials that were published in journals that are not indexed in the databases that we searched. We were unable to obtain data from 14 trials (four were mixed samples and authors were unable to provide data for knee OA only; five authors did not respond to requests for data; five were not in English and the Google translated versions were inadequate to comprehend). In addition, we identified records of 50 other trials that were likely completed but unpublished. There were another 133 records of ongoing trials. If a significant volume of these trials are of a high-quality (i.e. low risk of bias, appropriately powered), we expect that this would likely drive estimates for symptom improvements with exercise towards null.

Agreements and disagreements with other studies or reviews

The previous version of this Cochrane review, Fransen 2015, included 54 trials and concluded that there was high-quality evidence that exercise provides small to moderate short-term benefits for pain, function, and quality of life compared to any or no intervention. Our findings are largely similar for exercise compared to no intervention/usual care/limited education, and exercise compared to any other intervention, though the standardised mean differences (SMDs) we found in this review were larger than in the previous review (e.g. our SMD for pain was -0.71, 95% CI -0.86 to -0.56 (exercise versus no treatment/usual care/limited education), compared to -0.49, 95% CI -0.59 to -0.39 in the previous review (exercise versus no exercise (no intervention or any non-exercise intervention)). This increase was likely driven by the substantial volume of new trials, which were small and often of poor quality. Indeed, our GRADE ratings were generally lower than those in the previous review due to the overall high risk of bias in the included studies, as well as the likelihood of publication bias. When we excluded trials that were at unclear or high risk of bias or were likely to be underpowered (or both), the SMDs were similar to the prior review (e.g. for pain in this review, -0.54, 95% CI -0.73 to -0.36 (exercise versus no treatment/usual care/limited education) compared to -0.49, 95% CI -0.59 to -0.39 in the previous review (exercise versus no exercise (no intervention or any non-exercise intervention)).

A new addition to this updated review was that we analysed results according to the comparator, whereas the prior review combined results from all studies irrespective of the comparator. Compared to attention control/placebo, we found that exercise may lead to improvements in pain and physical function, but the benefits were of uncertain clinical importance. A recent systematic review of exercise with or without other physical therapy interventions compared to placebo for OA identified four trials in knee OA (Dean 2021). Their SMDs were larger than ours (they reported SMD of -0.9, 95% CI -1.8 to 0.0, for pain, in favour of exercise, and -2.2, 95% CI -5.0 to 0.6, for function), though their confidence intervals were wide and included our estimates. In addition, they included exercise trials that involved co-interventions (e.g. knee taping, manual therapy), which we excluded.

Our findings can also be compared to other systematic reviews evaluating the effectiveness of exercise for knee OA (Holden 2023; Goh 2019; Sasaki 2022; Si 2023). Holden 2023 included 31 trials and an individual participant data (IPD) meta-analysis, reporting evidence of small, positive effects of questionable clinical importance for pain and physical function. Mean differences reported within their review were marginally smaller than ours (e.g. they reported a 6.36-point improvement (95% CI 4.27 to 8.45) in pain, compared to our finding of 8.7 (95% CI 5.7 to 11.7) when compared to attention control/placebo). This contrast in findings may be explained by the fact that they included both hip and knee OA and analysed only a small proportion of the studies included here, as well as their use of IPD meta-analysis methods.

Other recent systematic reviews compared exercise to placebo or 'pure control' (Sasaki 2022), no intervention (Si 2023), or usual care (Goh 2019), collectively reporting small improvements in pain, physical function, and quality of life, which decreased in the longer term. Their SMDs were generally smaller than those found in our review (e.g. for pain: -0.25 (95% CI -0.36 to -0.14) in Sasaki 2022; -0.32 (95% CI -0.41 to -0.22) in Si 2023; and -0.56 (95% CI -0.68 to -0.44) in Goh 2019), which may be explained by the fact that those reviews included only a small proportion of the studies included here, some included both knee and hip OA (Sasaki 2022), and there were some differences in methods (e.g. narrower review questions and selection criteria).

We found that exercise may increase adverse events. However, the vast majority of reported adverse events were not serious. This reflects the findings of a recent systematic review and meta-analysis examining adverse events of exercise therapy (not specifically in OA), which found that exercise increases the risk of non-serious adverse events, but not serious adverse events (Niemeijer 2020).

We found that effects on pain and physical function were similar for different types of exercise. This reflects the subgroup analyses in the previous version of this review (Fransen 2015), which also found effects were similar between exercise types (including quadriceps strengthening programmes, lower

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limb strengthening programmes, combination strengthening and aerobic programmes, walking programmes, and 'other' programmes). A recent network meta-analysis found no difference between exercise types in knee OA (Mo 2023), although another network meta-analysis found that mixed-exercise programmes were less effective than aerobic and mind-body programmes for people with hip or knee OA (Goh 2019). Further work is warranted to discern if effects differ between types of exercise programmes.

We found that the number of prescribed exercise sessions was not associated with the magnitude of change in pain or physical function. A prior review evaluated whether a dose-response relationship exists between resistance exercise and outcomes in people with knee OA (Turner 2020). It concluded that 20 to 36 exercise sessions were associated with large effect sizes (24 weeks was the commonest intervention length). However, they acknowledged that there was heterogeneity amongst the included studies, making it difficult to determine the optimal dosage with certainty. Furthermore, they only included a subset (12 trials) of the trials included in our review.

Finally, we found that the ratio (exercise versus comparator) of synchronous contact occasions with a healthcare provider was not associated with the magnitude of change in pain or physical function. To our knowledge, no other reviews have evaluated the ratio of contact occasions as we did. One review found that the number of supervised exercise sessions in trials for knee OA was not associated with change in outcomes (Juhl 2014), which reflects our findings. However, their method of analysis (i.e. counting the number of supervised sessions delivered, rather than the difference in contact between exercise and comparator groups) would not have fully accounted for the contextual factors associated with provider supervision or consultation.

AUTHORS' CONCLUSIONS

Implications for practice

We found low- to moderate-certainty evidence that exercise may provide a benefit for symptoms in the treatment of knee osteoarthritis (OA). However, when comparing our mean differences (on a 0 to 100 scale) to the established minimally important differences (MIDs) that we used, these benefits were of uncertain clinical importance. There are a number of important considerations relating to the use of MIDs to interpret the clinical importance of outcomes (Dekker 2023; Dworkin 2009). The MIDs we selected for our review were based on established MIDs for the WOMAC (Western Ontario and McMaster Universities Osteoarthritis Index) pain and function instrument, yet we applied them to pooled results from studies that also used other pain or function instruments (e.g. visual analogue scale (VAS), numerical rating scale (NRS), Knee Injury and Osteoarthritis Outcome Score (KOOS)). Further, MIDs are known to be context-specific, depending on factors like the time between outcome assessments, severity of participant symptoms at baseline, as well as participant age and other baseline characteristics (Concoff 2019; King 2011). Indeed, in OA, a wide variety of different MID thresholds have been used for the same symptom or outcome (Concoff 2019). The clinical relevance of our findings may therefore vary depending on which MIDs are used. It has therefore been recommended that MIDs only be used as a supplementary tool when making treatment recommendations or treatment decisions (Bannuru 2014; Dekker 2023). Other factors including cost, convenience, and safety of exercise interventions for people with knee OA should also be considered (Bannuru 2014; Dekker 2023).

We focused on four measures of effectiveness - pain, physical function, quality of life, and participant-reported treatment success. However, exercise is known to be important for other more general health outcomes that are not OA-specific (e.g. sedentary behaviour, cardiovascular health, mental health, and risk and impact of comorbidities) (Ruegsegger 2018). As such, this review potentially underestimates the broader benefits of exercise for people with knee OA.

We found that the benefits of exercise declined in the longer-term, though only 20% of trials evaluated longer-term effects. It is not clear whether participants in our included trials were encouraged to continue exercising beyond the intervention. Declining long-term benefits may reflect cessation of exercise or declining adherence, which is a well-established challenge in exercise interventions for OA (Pisters 2007; Holden 2021).

We found that all types of exercise included in our subgroup analysis were similarly effective, and there was no evidence of a relationship between the number of prescribed exercise sessions or the ratio (exercise versus comparator) of synchronous healthcare provider contact occasions and change in pain or function. In light of this, and the heterogeneity of exercise interventions included in this review, we are not able to draw any conclusions regarding the most effective type or dosage of exercise. Future publications by this author team will further address these topics.

Implications for research

We identified a large number of randomised controlled trials on exercise for knee OA. However, most of these were of poor quality, did not report methods or interventions well, and had small sample sizes. We expect that these poor quality studies may have led to our results overestimating the effects of exercise, and our sensitivity analyses also reflected this. Future trials of exercise should use robust designs (including blinding of participants) and use large sample sizes that are sufficient to ensure adequate power to detect a minimal clinically important difference in outcome. Further, there is a need for improved reporting of exercise and comparator interventions, intervention protocols, and outcomes.

More trials comparing exercise to no treatment, usual care, or limited education are unlikely to change our conclusions (Verhagen 2019). However, given the paucity of placebo-controlled trials evaluating the effectiveness of exercise for knee OA, there is a need for robust, high-quality placebo-controlled trials in this area. This recommendation is supported by recent work that attempted to better understand whether a causal relationship exists between exercise and change in pain, and concluded that the current evidence base is not sufficient to infer or rule out causality (Henriksen 2024). Designing a placebo intervention for exercise trials is challenging. A range of placebo interventions have been used - including sham topical creams, saline injection, and inactive ultrasound. How to address the lack of blinding of participants is also an important issue.

Less than half of the trials in our review measured quality of life or adequately reported adverse events. Future trials should include these outcome measures, as well as other outcomes related to overall health, comorbidities, cardiovascular health, and sedentary

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behaviour. Very few trials in our review included longer-term follow-up of outcomes after intervention completion. Future trials should endeavour to collect outcomes beyond the end of the intervention.

There are a range of additional research questions that are important to evaluate in the future. Further work is needed to develop a better understanding of the differential response to exercise in order to identify who might benefit most, including studies of people with varying levels of disease severity. It is also important to determine the best type of exercise and the optimal dosage and delivery of exercise (e.g. type of exercise, group or individual exercise sessions, number of different exercises prescribed), as well as whether certain interventions combined with exercise are associated with greater benefits for symptom alleviation. Our team's future publications will address some of these questions.

Recommendations for future systematic review on this topic

The size of this review required a significant amount of time and resources for one team to complete. Given the number of different systematic reviews of exercise for OA that have been conducted in the last five years, better co-ordination of efforts between review teams would help streamline the process; for example, by sharing extracted data and risk of bias assessments for exercise trials, potentially via a central accessible repository for any review authors to access.

Future updates of this review may be considered if further placebo-controlled trials are published that are likely to increase the certainty of effect estimates of harms. Further trials comparing exercise with no intervention, usual care or limited education are likely to be of limited value, and we are unlikely to include such studies in future updates.

ACKNOWLEDGEMENTS

Dr Renea Johnston, Managing Editor, Australian Editorial Base, Cochrane Musculoskeletal Review Group, for overall guidance and expert advice.

Diane Horrigan, Cochrane Musculoskeletal Review Group, for designing the updated search strategy and running the search.

Associate Professor Milena Simic for her input on the review protocol and her assistance screening titles and abstracts.

Ms Kathryn Oliver and Mr Jesse Pardo for assisting with data extraction.

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Editorial and peer reviewer contributions

Cochrane Back and Neck supported the review authors in the development of this Cochane review. The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Walter R. Frontera, MD, PhD, FRCP Department of Physical Medicine, Rehabilitation, and Sports Medicine Department of Physiology University of Puerto Rico School of Medicine
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Hannah Payne, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer reviewer comments and supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald and Denise Mitchell, Cochrane Central Production service
- Peer reviewers (provided comments and recommended an editorial decision): Professor Toby Smith, Warwick Medical School, University of Warwick, UK (clinical/content review); Hans Lund, Western Norway University of Applied Sciences, Bergen Campus, Norway (clinical/content review); Helen P French, School of Physiothrapy, RCSI University of Medicine and Health Sciences, Dublin, Ireland (clinical/content review); Brian Duncan (consumer review); Jennifer Hilgart, Cochrane Evidence Production and Methods Directorate (methods review); Ina Monsef, Cochrane Haematology, Institute of Public Health, Faculty of Medicine and University Hospital Cologne, University of Cologne, Germany (search review). One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

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REFERENCES

References to studies included in this review

Abbott 2013 *{published and unpublished data}*

Abbott JH, Chapple C, Pinto D, Wright A, Theis JC. Exercise therapy and/or manual therapy for hip or knee osteoarthritis: 2-year follow-up of a randomized controlled trial. *Arthritis & Rheumatology* 2014;**10**:S1266-7. [DOI: 10.1002/art.38914]

Abbott JH, Chapple C, Pinto D, Wright AA, De La Barra SL, Baxter GD, et al. Exercise therapy, manual therapy, or both, for management of osteoarthritis of the hip or knee: 2-year follow-up of a randomized clinical trial. *Osteoarthritis and Cartilage* 2014;**1**:S51.

Abbott JH, Chapple CM, Fitzgerald GK, Fritz JM, Childs JD, Harcombe H, et al. The incremental effects of manual therapy or booster sessions in addition to exercise therapy for knee osteoarthritis: a randomized clinical trial. *Journal of Orthopaedic and Sports Physical Therapy* 2015;**45**(12):975-83. [DOI: 10.2519/jospt.2015.6015]

Abbott JH, Fitzgerald G, Fritz JM, Childs J, Chapple CM, Harcombe H. Do manual therapy or booster sessions in addition to exercise therapy for knee osteoarthritis provide additional benefits? A randomized clinical trial. *Osteoarthritis and Cartilage* 2015;**23**:2015-04.

* Abbott JH, Robertson MC, Chapple C, Pinto D, Wright AA, Leon de la Barras S, et al. Manual therapy, exercise therapy, or both, in addition to usual care for osteoarthritis of the hip or knee: a randomized controlled trial. I: clinical effectiveness. *Osteoarthritis and Cartilage* 2013;**21**:525-34.

Abbott JH, Wilson R, Pinto D, Chapple CM, Wright AA. Incremental clinical effectiveness and cost effectiveness of providing supervised physiotherapy in addition to usual medical care in patients with osteoarthritis of the hip or knee: 2-year results of the MOA randomised controlled trial. *Osteoarthritis and Cartilage* 2019;**27**(3):424-34. [DOI: 10.1016/j.joca.2018.12.004]

Abbott JH, Wilson R, Pinto D, The MOATrial Team. Long-term cost-effectiveness of exercise therapy and/or manual therapy for hip or knee osteoarthritis: randomized controlled trial and computer simulation modelling. *Osteoarthritis and Cartilage* 2019;**27 Suppl 1**:S36. [DOI: 10.1016/j.joca.2019.02.052]

Bove AM, Smith KJ, Bise CG, Fritz JM, Childs JD, Brennan GP, et al. Exercise, manual therapy, and booster sessions in knee osteoarthritis: cost-effectiveness analysis from a multicenter randomized controlled trial. *Physical Therapy* 2018;**98**(1):16-27. [DOI: 10.1093/ptj/pzx104]

Fitzgerald GK, Fritz J, Childs J, Brennan GP, Landsittel DP, Neilson B, et al. Exercise, manual therapy, and use of booster sessions in physical therapy for knee OA: a multi-center randomized clinical trial. *Arthritis & Rheumatology* 2014;**10)**:S393. [DOI: 10.1002/art.38914]

Fitzgerald GK, Fritz JM, Childs JD, Brennan GP, Talisa V, Gil AB, et al. Exercise, manual therapy, and use of booster sessions in physical therapy for knee osteoarthritis: a multi-center, factorial randomized clinical trial. *Osteoarthritis and Cartilage* 2016;**24**(8):1340-9. [DOI: 10.1016/j.joca.2016.03.001]

Pinto D, Robertson MC, Abbott JH, Hansen P, Campbell AJ, Team MOATrial. Manual therapy, exercise therapy, or both, in addition to usual care, for osteoarthritis of the hip or knee. 2: economic evaluation alongside a randomized controlled trial. *Osteoarthritis and Cartilage* 2013;**Vol.21**(10):1504-13p. [DOI: 10.1016/j.joca.2013.06.014]

Alghadir 2019 *{published data only}*

Alghadir A, Anwer S. Effect of retro and forward walking on quadriceps muscle strength, pain, function, and mobility in patients with knee osteoarthritis: a protocol for a randomized controlled trial. *BMC Musculoskeletal Disorders* 2016;**17**(1):(no pagination). [DOI: 10.1186/s12891-016-1021-z]

* Alghadir AH, Anwer S, Sarkar B, Paul AK, Anwar D. Effect of 6-week retro or forward walking program on pain, functional disability, quadriceps muscle strength, and performance in individuals with knee osteoarthritis: a randomized controlled trial (retro-walking trial). *BMC Musculoskeletal Disorders* 2019;**20**(1):(no pagination). [DOI: 10.1186/s12891-019-2537-9]

Allen 2018 *{published data only}*

Allen K, Arbeeva L, Callahan LF, Golightly YM, Goode AP, Heiderscheit B, et al. Physical therapy vs. internet-based exercise training for patients with knee osteoarthritis: results of a randomized controlled trial. *Arthritis and Rheumatology. Conference: American College of Rheumatology/Association of Rheumatology Health Professionals Annual Scientific Meeting, ACR/ARHP* 2017;**69**:Supplement 10.

* Allen KD, Arbeeva L, Callahan LF, Golightly YM, Goode AP, Heiderscheit BC, et al. Physical therapy vs internet-based exercise training for patients with knee osteoarthritis: results of a randomized controlled trial. *Osteoarthritis and Cartilage* 2018;**26(3)**:383-96. [DOI: 10.1016/j.joca.2017.12.008]

Anderson ML, Allen KD, Golightly YM, Arbeeva LS, Goode A, Huffman KM, et al. Fall risk and utilization of balance training for adults with symptomatic knee osteoarthritis: secondary analysis from a randomized clinical trial. *Journal of Geriatric Physical Therapy (2001)* 2019;**42**(2):E39-E44. [DOI: 10.1519/JPT.0000000000000213]

Coffman CJ, Arbeeva L, Schwartz TA, Callahan LF, Golightly YM, Goode AP, et al. Application of heterogeneity of treatment-effects methods: exploratory analyses of a trial of exercise-based interventions for knee osteoarthritis. *Arthritis Care & Research* 2022;**74**(8):1359-68. [DOI: 10.1002/acr.24564]

NCT02312713. Physical therapy versus internet-based exercise training for patients with knee osteoarthritis. https://clinicaltrials.gov/study/NCT02312713 (first posted 9 December 2014).

Williams QI, Gunn AH, Beaulieu JE, Benas BC, Buley B, Callahan LF, et al. Physical therapy vs. internet-based exercise training (PATH-IN) for patients with knee osteoarthritis: study protocol of a randomized controlled trial. *BMC*

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Musculoskeletal Disorders 2015;**16**(1):(no pagination). [DOI: 10.1186/s12891-015-0725-9]

An 2008 *{published data only}*

An B, Dai K, Zhu Z, Wang Y, Hao Y, Tang T, et al. Baduanjin alleviates the symptoms of knee osteoarthritis. *Journal of Alternative and Complementary Medicine* 2008;**14**(2):167-74.

Anadkat 2015 *{published data only}*

Anadkat H, Ajith S, Dhanesh Kumar KU. Effectiveness of retro walking treadmill training on pain and disability in knee osteoarthritis: a randomized controlled trail [sic]. *International Journal of Pharma and Bio Sciences* 2015;**6(4)**:B43-50.

Anwer 2014 *{published data only}*

Anwer S, Alghadir A. Effect of isometric quadriceps exercise on muscle strength, pain, and function in patients with knee osteoarthritis: a randomized controlled study. *Journal of Physical Therapy Science* 2014;**26**(5):745-8. [DOI: 10.1589/jpts.26.745]

Assar 2020 *{published data only}*

Assar S, Gandomi F, Mozafari M, Sohaili F. The effect of Total resistance exercise vs. aquatic training on self-reported knee instability, pain, and stiffness in women with knee osteoarthritis: A randomized controlled trial. *BMC Sports Science, Medicine and Rehabilitation* 2020;**12**(1):(no pagination). [DOI: 10.1186/s13102-020-00175-y]

Irct20181222042070n1. TRX training vs. aquatic therapy for patients with kneeosteoarthritis. Irct20181222042070n1 2019.

Baker 2001 *{published data only}*

Baker KR, Nelson ME, Felson DT, Layne JE, Sarno R, Roubenoff R. The efficacy of home based progressive strength training in older adults with knee osteoarthritis: a randomized controlled trial. *Journal of Rheumatology* 2001;**28**:1655-65.

Bandak 2022 *{published data only}*

* Bandak E, Christensen R, Overgaard A, Kristensen LE, Ellegaard K, Guldberg-Moller J, et al. Exercise and education versus saline injections for knee osteoarthritis: a randomised controlled equivalence trial. *Annals of the Rheumatic Diseases* 2022;**81(4)**:537-43. [DOI: 10.1136/annrheumdis-2021-221129]

Bandak E, Christensen R, Overgaard AF, Kristensen L, Ellegaard K, Guldberg-Moller J, et al. Exercise and education vs. saline injections for knee osteoarthritis: a randomised controlled equivalence trial. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S53-4. [DOI: 10.1016/j.joca.2022.02.060]

Bandak E, Overgaard AF, Kristensen LE, Ellegaard K, Guldberg-Moller J, Bartholdy C, et al. Exercise therapy and patient education versus intra-articular saline injections in the treatment of knee osteoarthritis: an evidence-based protocol for an open-label randomised controlled trial (the DISCO trial). *Trials* 2021;**22**(1):(no pagination). [DOI: 10.1186/s13063-020-04952-5]

Henriksen M, Christensen R, Kristensen LE, Bliddal H, Bartholdy C, Boesen M, et al. Exercise and education vs intra-articular saline for knee osteoarthritis: a 1-year follow-up of a randomized trial. *Osteoarthritis and Cartilage.* 2023;**31**(5):627-35. [DOI: 10.1016/j.joca.2022.12.011]

Henriksen M, Nielsen SM, Christensen R, Kristensen L, Bliddal H, Bartholdy C, et al. Who are likely to benefit from the good life with osteoarthritis in Denmark (glad) exercise and education program? An effect modifier analysis from a randomized trial. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S52-3. [DOI: 10.1016/j.joca.2022.02.059]

Henriksen M, Nielsen SM, Christensen R, Kristensen LE, Bliddal H, Bartholdy C, et al. Who are likely to benefit from the Good Life with osteoArthritis in Denmark (GLAD) exercise and education program? An effect modifier analysis of a randomised controlled trial. *Osteoarthritis and Cartilage* 2023;**31**(1):106-14. [DOI: 10.1016/j.joca.2022.09.001]

Bautch 1997 *{published data only}*

Bautch JC, Malone DG, Vailas AC. Effects of exercise on knee joints with osteoarthritis: a pilot study of biologic markers. *Arthritis Care & Research* 1997;**10**:48-55.

Bennell 2010 *{published data only}*

Bennell KL, Hunt MA, Wrigley TV, Hunter DJ, McManus FJ, Hodges PW, et al. Hip strengthening reduces symptoms but not knee load in people with medial knee osteoarthritis and varus malalignment: a randomised controlled trial. *Osteoarthritis and Cartilage* 2010;**18**(5):621-8.

Bennell 2016 *{published data only}*

Bennell KL, Ahamed Y, Bryant C, Jull G, Hunt M, Kenardy J, et al. Physiotherapist-delivered exercise and pain coping skills training is more effective than either intervention alone in knee osteoarthritis. *Osteoarthritis and Cartilage* 2014;**1**:S192-3.

* Bennell KL, Ahamed Y, Jull G, Bryant C, Hunt MA, Forbes AB, et al. Physical therapist-delivered pain coping skills training and exercise for knee osteoarthritis: randomized controlled trial. *Arthritis Care & Research* 2016;**68**(5):590-602. [DOI: 10.1002/acr.22744]

Bennell 2022a *{published data only}*

Bennell KL, Keating C, Lawford BJ, Kimp AJ, Egerton T, Brown C, et al. Better Knee, Better Me: effectiveness of two scalable health care interventions supporting self-management for knee osteoarthritis-protocol for a randomized controlled trial. *BMC Musculoskeletal Disorders* 2020;**21**(1):(no pagination). [DOI: 10.1186/s12891-020-3166-z]

Bennell KL, Lawford BJ, Keating C, Brown C, Kasza J, Mackenzie D, et al. Comparing video-based telehealth-delivered exercise and weight loss programs with online education on outcomes of knee osteoarthritis: a randomized trial. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S55. [DOI: 10.1016/j.joca.2022.02.062]

* Bennell KL, Lawford BJ, Keating C, Brown C, Kasza J, Mackenzie D, et al. Comparing video-based, telehealth-delivered exercise and weight loss programs with online education on outcomes of knee osteoarthritis. *Annals of Internal Medicine* 2022;**175**(2):198-209. [DOI: 10.7326/M21-2388]

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Bennell 2022b *{published data only}*

ACTRN12618000930280. Better Knee, Better Me: effectiveness of two scalable health care interventions supporting self-management for knee osteoarthritis; a randomised controlled trial. https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=375067&isReview=true (registered 1 June 2018).

Actrn12620000012976. Comparing effects of an online exercise plus education program to online education only in people with knee osteoarthritis: randomised controlled trial: the COPE Trial. Actrn12620000012976 2020.

* Bennell KL, Schwartz S, Teo PL, Hawkins S, Mackenzie D, McManus F, et al. Effectiveness of an unsupervised online yoga program on pain and function in people with knee osteoarthritis a randomized clinical trial. *Annals of Internal Medicine* 2022;**175**(10):1345-55. [DOI: 10.7326/M22-1761]

Bhagat 2017 *{published data only}*

Bhagat P, Jagtap V, Poovishnu Devi T. Effect of circuit training in osteoarthritis of knee. *Asian Journal of Pharmaceutical and Clinical Research* 2017;**10**(7):333-5. [DOI: 10.22159/ajpcr.2017.v10i7.18322]

Braghin 2018 *{published data only}*

Braghin R, Libardi EC, Junqueira C, Nogueira - Barbosa MH, de Abreu DC. Exercise on balance and function for knee osteoarthritis: a randomized controlled trial. *Journal of Bodywork and Movement Therapies* 2018;**22**(1):76-82. [DOI: 10.1016/j.jbmt.2017.04.006]

Brismée 2007 *{published data only}*

Brismée J, Paige RL, Chyu M, Boatright JD, Hagar JM, McCaleb JA, et al. Group and home-based Tai Chi in elderly subjects with knee osteoarthritis: a randomized controlled trial. *Clinical Rehabilitation* 2007;**21**(2):99-111.

Bruce-Brand 2012 *{published data only}*

Bruce-Brand RA, Walls RJ, Ong JC, Emerson BS, O'Byrne JM, Moyna NM. Effects of home-based resistance training and neuromuscular electrical stimulation in knee osteoarthritis: a randomized controlled trial. *BMC Musculoskeletal Disorders* 2012;**13**:118.

Chang 2012 *{published data only}*

Chang TF, Liou TH, Chen CH, Huang YC, Chang KH. Effects of elastic-band exercise on lower-extremity function among female patients with osteoarthritis of the knee. *Disability and Rehabilitation* 2012;**34**(20):1727-35.

Chen 2014 *{published data only}*

Chen TW, Lin CW, Lee CL, Chen CH, Chen YJ, Lin TY, et al. The efficacy of shock wave therapy in patients with knee osteoarthritis and popliteal cyamella. *Kaohsiung Journal of Medical Sciences* 2014;**30**(7):362-70. [DOI: 10.1016/j.kjms.2014.03.006]

Chen 2021 *{published data only}*

Chen Z, Ye X, Wang Y, Shen Z, Wu J, Chen W, et al. The efficacy of backward walking on static stability, proprioception, pain, and physical function of patients with knee osteoarthritis: a randomized controlled trial. *Evidence-Based Complementary and Alternative Medicine* 2021;**11**:(no pagination). [DOI: 10.1155/2021/5574966]

Cheung 2014 *{published data only}*

Cheung C, Wyman JF, Resnick B, Savik K. Yoga for managing knee osteoarthritis in older women: a pilot randomized controlled trial. *BMC Complementary and Alternative Medicine* 2014;**14**:(no pagination). [DOI: 10.1186/1472-6882-14-160]

* Cheung C, Wyman JF, Savik K. Adherence to a yoga program in older women with knee osteoarthritis. *Journal of Aging and Physical Activity* 2016;**24**(2):181-8. [DOI: 10.1123/japa.2015-0048]

NCT01832155. Yoga for managing knee osteoarthritis in older women: a feasibility studya pilot study testing a hatha yoga exercise program in older women with knee osteoarthritis. https://clinicaltrials.gov/study/NCT01832155 (first posted 15 April 2013).

Cheung 2017 *{published data only}*

* Cheung C, Wyman JF, Bronas U, McCarthy T, Rudser K, Mathiason MA. Managing knee osteoarthritis with yoga or aerobic/strengthening exercise programs in older adults: a pilot randomized controlled trial. *Rheumatology International* 2017;**37**(3):389-98. [DOI: 10.1007/s00296-016-3620-2]

Cheung CK, Wyman J, Bronas U, McCarthy T, Switzer J, Mathiason M. Is yoga better than aerobic/strengthening exercises for managing knee osteoarthritis in older adults? *Osteoarthritis and Cartilage* 2016;**1**:S484.

Nct02525341. Yoga versus exercises for managing OA [Yoga versus aerobic and strengthening exercises for managing knee osteoarthritis]. https://clinicaltrials.gov/study/NCT02525341 (first posted 17 August 2015).

Chopp-Hurley 2017 *{published and unpublished data}*

* Chopp-Hurley JN, Brenneman EC, Wiebenga EG, Bulbrook B, Keir PJ, Maly MR. Randomized controlled trial investigating the role of exercise in the workplace to improve work ability, performance, and patient-reported symptoms among older workers with osteoarthritis. *Journal of Occupational and Environmental Medicine* 2017;**59**(6):550-6. [DOI: 10.1097/JOM.0000000000001020]

Marriott K, Chopp-Hurley J, Loukov D, Karampatos S, Kuntz AB, Wiebenga EG, et al. Muscle strength gains after strengthening exercise explained by reductions in serum inflammation in women with knee osteoarthritis. *Clinical Biomechanics (Bristol, Avon)* 2021;**86**:(no pagination). [DOI: 10.1016/j.clinbiomech.2021.105381]

Christensen 2015 *{published data only}*

Christensen P, Frederiksen R, Bliddal H, Riecke BF, Bartels EM, Henriksen M, et al. Comparison of three weight maintenance programs on cardiovascular risk, bone and vitamins in sedentary older adults. *Obesity (Silver Spring, Md.)* 2013;**21**(10):1982-90. [DOI: 10.1002/oby.20413]

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

* Christensen R, Henriksen M, Leeds AR, Gudbergsen H, Christensen P, Sorensen TJ, et al. Effect of weight maintenance on symptoms of knee osteoarthritis in obese patients: A twelve-month randomized controlled trial. *Arthritis Care & Research* 2015;**67**(5):640-50. [DOI: 10.1002/acr.22504]

Da Silva 2015 *{published data only}*

* Da Silva FS, De Melo FE, Do Amaral MM, Caldas VV, Pinheiro IL, Abreu BJ, et al. Efficacy of simple integrated group rehabilitation program for patients with knee osteoarthritis: Single-blind randomized controlled trial. *Journal of Rehabilitation Research and Development* 2015;**52**(3):309-22.

NCT01850862. Progressive collective-exercise program on the knee osteoarthritis. https://clinicaltrials.gov/study/NCT01850862 (first posted 10 May 2013).

De Almeida 2020 *{published data only}*

* De Almeida AC, Aily JB, Pedroso MG, Goncalves GH, De Carvalho Felinto J, Ferrari RJ, et al. A periodized training attenuates thigh intermuscular fat and improves muscle quality in patients with knee osteoarthritis: results from a randomized controlled trial. *Clinical Rheumatology* 2020;**39**(4):1265-75. [DOI: 10.1007/s10067-019-04892-9]

De Almeida AC, Pedroso MG, Aily JB, Goncalves GH, Pastre CM, Mattiello SM. Influence of a periodized circuit training protocol on intermuscular adipose tissue of patients with knee osteoarthritis: protocol for a randomized controlled trial. *BMC Musculoskeletal Disorders* 2018;**19**(1):(no pagination). [DOI: 10.1186/s12891-018-2325-y]

NCT02761590. Circuit training on knee osteoarthritis patients [Influence of a circuit trining protocol on intermuscular adipose tissue of knee osteoarthritis patients: a randomized controlled trial]. https://clinicaltrials.gov/study/NCT02761590 (first posted 4 May 2016).

De Matos 2019 *{published data only}*

De Matos Brunelli Braghin R, Libardi EC, Junqueira C, Rodrigues NC, Nogueira-Barbosa MH, Renno AC, et al. The effect of low-level laser therapy and physical exercise on pain, stiffness, function, and spatiotemporal gait variables in subjects with bilateral knee osteoarthritis: a blind randomized clinical trial. *Disability and Rehabilitation* 2019;**41**(26):3165-72. [DOI: 10.1080/09638288.2018.1493160]

De Rooij 2017 *{published data only}*

De Rooij M, Van Der Leeden M, Cheung J, Van Der Esch M, Arja Hakkinen A, Haverkamp D, et al. Efficacy of tailored exercise therapy in patients with knee osteoarthritis and comorbidity: a randomized controlled trial. *Annals of the Rheumatic Diseases* 2016;**75 Suppl 2**:78. [DOI: 10.1136/annrheumdis-2016-eular.3855]

* De Rooij M, Van der Leeden M, Cheung J, Van der Esch M, Hakkinen A, Haverkamp D, et al. Efficacy of tailored exercise therapy on physical functioning in patients with knee osteoarthritis and comorbidity: a randomized controlled trial. *Arthritis Care & Research* 2017;**69**(6):807-16. [DOI: 10.1002/acr.23013]

DeVita 2018 *{published data only}*

Aaboe J, Henriksen M, Bartholdy C, Leonardis J, Rider P, Jorgensen L, et al. The effect of quadriceps-strengthening exercise on quadriceps and knee biomichanics during walking in adults with knee osteoarthritis: a randomized controlled trial. *Osteoarthritis and Cartilage* 2014;**1**:S80-1.

* DeVita P, Aaboe J, Bartholdy C, Leonardis JM, Bliddal H, Henriksen M. Quadriceps-strengthening exercise and quadriceps and knee biomechanics during walking in knee osteoarthritis: a two-centre randomized controlled trial. *Clinical Biomechanics (Bristol, Avon)* 2018;**59**:199-206. [DOI: 10.1016/j.clinbiomech.2018.09.016]

Dincer 2016 *{published data only}*

Dincer U, Ariba S, Saygin H, Incedayi M, Rodop O. The effects of closed kinetic chain exercise on articular cartilage morphology: myth or reality? A randomized controlled clinical trial. *Turkiye Fiziksel Tip ve Rehabilitasyon Dergisi* 2016;**62**(1):28-36. [DOI: 10.5606/tftrd.2016.93899]

Drummen 2023 *{published data only}*

ACTRN12618001097235. WALK - A pilot randomised controlled trial evaluating community walking for knee osteoarthritis. https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=374968 (first registered 2 July 2018).

Aitken D, Drummen SJ, Balogun S, Lahham A, Bennell K, Hinman R, et al. Walk - a pilot randomized controlled trial evaluating community walking for knee osteoarthritis. *Osteoarthritis and Cartilage* 2021;**29 Suppl 1**:S387. [DOI: 10.1016/j.joca.2021.02.503]

* Drummen SJ, Balogun S, Lahham A, Bennell K, Hinman RS, Callisaya M, et al. A pilot randomized controlled trial evaluating outdoor community walking for knee osteoarthritis: walk. *Clinical Rheumatology.* 2023;**42**(5):1409-21. [DOI: 10.1007/s10067-022-06477-5]

Dwyer 2015 *{published data only}*

Dwyer L, Parkin-Smith GF, Brantingham JW, Korporaal C, Cassa TK, Globe G, et al. Manual and manipulative therapy in addition to rehabilitation for osteoarthritis of the knee: assessor-blind randomized pilot trial. *Journal of Manipulative and Physiological Therapeutics* 2015;**38(1)**:1-21. [DOI: 10.1016/j.jmpt.2014.10.002]

Ettinger 1997 *{published data only}*

* Ettinger WH, Burns R, Messier SP, Applegate W, Rejeski WJ, Morgan T, et al. A randomized trial comparing aerobic exercise and resistance exercise with a health education program in older adults with knee osteoarthritis. The Fitness Arthritis and Seniors Trial (FAST). *JAMA* 1997;**277**:25-31.

Messier SP, Royer TD, Craven TE, O'Toole ML, Burns R, Ettinger WH. Long-term exercise and its effect on balance in older, osteoarthritis adults: results from the Fitness, Arthritis and Seniors Trial (FAST). *Journal of the American Geriatric Society* 2000;**48**:131-8.

Messier SP, Thompson CD, Ettinger WH Jr. Effects of long-term aerobic or weight training regimes on gait in an older,

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

osteoarthritic population. *Journal of Applied Biomechanics* 1997;**13**:205-25.

Rejeski WJ, Ettinger WH, Martin K, Morgan T. Treating disability in knee osteoarthritis with exercise therapy: a central role for self-efficacy and pain. *Arthritis Care & Research* 1998;**11**:94-101.

Foley 2003 *{published data only}*

Foley A, Halbert J, Hewitt T, Crotty M, Halbert J, Hewitt T, et al. Does hydrotherapy improve strength and physical function in patients with osteoarthritis-a randomised controlled trial comparing a gym based and a hydrotherapy based strengthening program. *Annals of the Rheumatic Diseases* 2003;**62**:1162-7.

Fransen 2001 *{published data only}*

Fransen M, Crosbie J, Edmonds J. Physical therapy is effective for patients with osteoarthritis of the knee: a randomized controlled trial. *Journal of Rheumatology* 2001;**28**:156-64.

Fransen 2007 *{published data only}*

Fransen M, Nairn L, Winstanley J, Lam P, Edmonds J. The Physical Activity for Osteoarthritis Management (PAFORM) study. A randomised controlled clinical trial evaluating hydrotherapy and Tai Chi classes. *Arthritis Care & Research* 2007;**57**:407-14.

Gohir 2021 *{published data only}*

Gohir SA, Eek F, Kelly A, Abhishek A, Valdes AM. Effectiveness of internet-based exercises aimed at treating knee osteoarthritis: the ibeat-OA randomized clinical trial. *JAMA Network Open* 2021;**4**(2):(no pagination). [DOI: 10.1001/jamanetworkopen.2021.0012]

Gondhalekar 2013 *{published data only}*

Gondhalekar GA, Deo MV. Retrowalking as an adjunct to conventional treatment versus conventional treatment alone on pain and disability in patients with acute exacerbation of chronic knee osteoarthritis: a randomized clinical trial. *North American Journal of Medical Sciences* 2013;**5**(2):108-12. [DOI: 10.4103/1947-2714.107527]

Guo 2022 *{published data only}*

Guo D, Ma S, Zhao Y, Dong J, Guo B, Li X. Self-administered acupressure and exercise for patients with osteoarthritis: a randomized controlled trial. *Clinical Rehabilitation* 2022;**36**(3):350-8. [DOI: 10.1177/02692155211049155]

Gur 2002 *{published data only}*

Gur H, Cakin N, Akova B, Okay E, Kucukoglu S. Concentric versus combined concentric-eccentric isokinetic training: effects on functional capacity and symptoms in patients with osteoarthrosis of the knee. *Archives of Physical and Medical Rehabilitation* 2003;**83**:308-16.

Hay 2006 *{published data only}*

Hay EM, Foster NE, Thomas E, Peat G, Phelan M, Yates HE, et al. Effectiveness of community physiotherapy and enhanced pharmacy review for knee pain in people aged over 55 presenting to primary care: pragmatic randomised trial. *BMJ (Clinical Research Ed.)* 2006;**333**:995.

Henriksen 2014 *{published data only}*

Bandak E, Boesen M, Bilddal H, Daugaard C, Hangaard S, Bartholdy C, et al. The effects of exercise on synovitis and bone marrow lesions in knee osteoarthritis: secondary outcomes from a randomized controlled trial. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S315-6.

Bandak E, Boesen M, Bliddal H, Daugaard C, Hangaard S, Bartholdy C, et al. The effect of exercise therapy on inflammatory activity assessed by MRI in knee osteoarthritis: secondary outcomes from a randomized controlled trial. *Knee* 2021;**28**:256-65. [DOI: 10.1016/j.knee.2020.12.022]

Bandak E, Boesen M, Bliddal H, Riis RG, Nielsen SM, Klokker L, et al. Exercise-induced pain changes associate with changes in muscle perfusion in knee osteoarthritis: exploratory outcome analyses of a randomised controlled trial. *BMC Musculoskeletal Disorders* 2019;**20**(1):(no pagination). [DOI: 10.1186/s12891-019-2858-8]

Bartholdy C, Klokker L, Bliddal H, Henriksen M. Standardized rescue training as part of an exercise program for knee osteoarthritis: proof of concept. *Osteoarthritis and Cartilage* 2014;**1**:S52.

Henriksen M, Klokker L, Bartholdy C, Schjodt Jorgensen T, Bandak E, Bliddal H. The effects of exercise therapy on knee joint biomechanics during walking in patients with knee osteoarthritis: secondary outcome analyses from a randomized controlled trial. *Osteoarthritis and Cartilage* 2014;**1**:S44.

Henriksen M, Klokker L, Bartholdy C, Schjoedt-Jorgensen T, Bandak E, Bliddal H. No effects of functional exercise therapy on walking biomechanics in patients with knee osteoarthritis: exploratory outcome analyses from a randomised trial. *BMJ Open Sport & Exercise Medicine* 2016;**2**(1):e000230. [DOI: 10.1136/bmjsem-2017-000230]

Henriksen M, Klokker L, Graven-Nielsen T, Bartholdy C, Joergensen TS, Bandak E, et al. Exercise therapy reduces pain sensitivity in patients with knee osteoarthritis: a randomized controlled trial. *Arthritis and Rheumatism* 2013;**10**:S775. [DOI: 10.1002/art.38216]

* Henriksen M, Klokker L, Graven-Nielsen T, Bartholdy C, Schjodt Jorgensen T, Bandak E, et al. Association of exercise therapy and reduction of pain sensitivity in patients with knee osteoarthritis: a randomized controlled trial. *Arthritis Care & Research* 2014;**66**(12):1836-43. [DOI: 10.1002/acr.22375]

Hernandez 2019 *{published and unpublished data}*

Hernandez D, Dimaro M, Navarro E, Dorado J, Accoce M, Salzberg S, et al. Efficacy of core exercises in patients with osteoarthritis of the knee: a randomized controlled clinical trial. *Journal of Bodywork and Movement Therapies* 2019;**23**(4):881-7. [DOI: 10.1016/j.jbmt.2019.06.002]

Hinman 2020 *{published data only}*

ACTRN12616000054415. Telephone-delivered support and advice for people with symptomatic knee osteoarthritis: telecare Study. Effect of telephone-delivered advice on pain and function in people with knee osteoarthritis: telecare trial.

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https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=369204&isReview=true (registered 20 January 2016).

* Hinman RS, Campbell PK, Lawford BJ, Briggs AM, Gale J, Bills C, et al. Does telephone-delivered exercise advice and support by physiotherapists improve pain and/or function in people with knee osteoarthritis? Telecare randomised controlled trial. *British Journal of Sports Medicine* 2020;**54**(13):790-7. [DOI: 10.1136/bjsports-2019-101183]

Hinman RS, Lawford BJ, Campbell PK, Briggs AM, Gale J, Bills C, et al. Telephone-delivered exercise advice and behavior change support by physical therapists for people with knee osteoarthritis: protocol for the telecare randomized controlled trial. *Physical Therapy* 2017;**97(5)**:524-36. [DOI: 10.1093/ptj/pzx021]

Holm 2020 *{published data only}*

Holm PM, Aagaard P, Schroder H, Skou ST. High-intensity strength training in addition to standard non-surgical treatment in patients with knee osteoarthritis E a randomized controlled trial. *Osteoarthritis and Cartilage* 2017;**25 Suppl 1**:S443-4.

Holm PM, Petersen KK, Wernbom M, Schroder HM, Arendt-Nielsen L, Skou ST. Strength training in addition to neuromuscular exercise and education in individuals with knee osteoarthritis-the effects on pain and sensitization. *European Journal of Pain (United Kingdom)* 2021;**25**(9):1898-911. [DOI: 10.1002/ejp.1796]

* Holm PM, Schroder H, Wernbom M, St S. Low-dose strength training in addition to neuromuscular exercise and education in patients with knee osteoarthritis in secondary care - a randomized controlled trial. *Osteoarthritis and Cartilage* 2020;**13**:744-54. [DOI: 10.1016/j.joca.2020.02.839]

NCT03215602. EXercise as TReatment for osteoArthritis. https://clinicaltrials.gov/study/NCT03215602 (first posted 12 July 2017).

Hopman-Rock 2000 *{published data only}*

Hopman-Rock M, Westhoff M. The effects of a health educational and exercise program for older adults with osteoarthritis of the hip or knee. *Journal of Rheumatology* 2000;**27**:1947-54.

Hsu 2021 *{published data only}*

* Hsu YI, Chen YC, Lee CL, Chang NJ. Effects of diet control and telemedicine-based resistance exercise intervention on patients with obesity and knee osteoarthritis: a randomized control trial [Effects of diet control and telemedicine-based resistance exercise intervention on patients with obesity and knee osteoarthritis: a randomized control trial]. *International Journal of Environmental Research and Public Health* 2021;**18**(15):(no pagination). [DOI: 10.3390/ijerph18157744]

NCT03973463. Effects of diet control and resistance exercise training on obesity adults with knee osteoarthritis. https://clinicaltrials.gov/study/NCT03973463 (first posted 4 June 2019).

Hu 2020 *{published data only}*

Hu X, Lai Z, Wang L. Effects of Taichi exercise on knee and ankle proprioception among individuals with knee osteoarthritis. *Research in Sports Medicine (Print)* 2020;**28**(2):268-78. [DOI: 10.1080/15438627.2019.1663520]

Huang 2003 *{published data only}*

Huang M-H, Lin Y-H, Lee C-L, Yang R-C. Use of ultrasound to increase effectiveness of isokinetic exercise for knee osteoarthritis. *Archives of Physical and Medical Rehabilitation* 2005;**86**:1545-51.

Huang M-H, Lin Y-S, Yang R-C, Lee C-L. A comparison of various therapeutic exercises on the functional status of patients with knee osteoarthritis. *Seminars in Arthritis and Rheumatism* 2003;**32**:398-406.

Huang 2005 *{published data only}*

Huang M-H, Yang R-C, Lee C-L, Chen T-W, Wang M-C. Preliminary results of integrated therapy for patients with knee osteoarthritis. *Arthritis Care & Research* 2005;**53**:812-20.

Hurley 2007 *{published data only}*

Hurley M, Walsh N, Mitchell H. Sustained improvement in physical functioning following an integrated rehabilitation programme for chronic knee pain. *Physiotherapy (United Kingdom)* 2011;**1**:eS520. [DOI: 10.1016/j.physio.2011.04.002]

Hurley MV, Walsh NE, Mitchell H, Nicholas J, Patel A. Long-term outcomes and costs of an integrated rehabilitation program for chronic knee pain: a pragmatic, cluster randomized, controlled trial. *Arthritis Care & Research* 2012;**64**(2):238-47.

* Hurley MV, Walsh NE, Mitchell HL, Pimm TJ, Patel A, Williamson E, et al. Clinical effectiveness of a rehabilitation program integrating exercise, self-management, and active coping strategies for chronic knee pain: a cluster randomized trial. *Arthritis and Rheumatism* 2007;**57**(7):1211-9.

Hurley MV, Walsh NE, Mitchell HL, Pimm TJ, Williamson E, Jones RH, et al. Economic evaluation of a rehabilitation program integrating exercise, self-management, and active coping strategies for chronic knee pain. *Arthritis and Rheumatism* 2007;**57**(7):1220-9.

Imoto 2012 *{published data only}*

Imoto AM, Peccin MS, Trevisani VF. Quadriceps strengthening exercises are effective in improving pain, function and quality of life in patients with osteoarthritis of the knee. *Acta Ortopedica Brasileira* 2012;**20**(3):174-9. [DOI: 10.1590/S1413-78522012000300008]

Jan 2008 *{published data only}*

* Jan M, Lin J, Liau J, Lin Y, Lin D. Investigation of clinical effects of high- and low-resistance training for patients with knee osteoarthritis: a randomized controlled trial. *Physical Therapy* 2008;**88**(4):427-36.

Jan M, Tang P, Lin J, Teng S, Lin Y, Lin D. Efficacy of a target-matching foot-stepping exercise on proprioception and function in patients with knee osteoarthritis. *Journal of Orthopaedic and Sports Physical Therapy* 2008;**38**(1):19-25.

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Jan 2009 *{published data only}*

Jan M, Lin C, Lin Y, Lin J, Lin D. Effects of weight-bearing versus nonweight-bearing exercise on function, walking speed, and position sense in participants with knee osteoarthritis: a randomized controlled trial. *Archives of Physical Medicine and Rehabilitation* 2009;**90**(6):897-904.

Janyacharoen 2018 *{published data only}*

Janyacharoen T, Yonglitthipagon P, Nakmareong S, Katiyajan N, Auvichayapat P, Sawanyawisuth K. Effects of the applied ancient boxing exercise on leg strength and quality of life in patients with osteoarthritis. *Journal of Exercise Rehabilitation* 2018;**14**(6):1059-66. [DOI: 10.12965/jer.1836392.196]

Jenkinson 2009 *{published data only}*

Jenkinson CM, Doherty M, Avery AJ, Read A, Taylor MA, Sach TH, et al. Effects of dietary intervention and quadriceps strengthening exercises on pain and function in overweight people with knee pain: randomised controlled trial. *BMJ (Clinical Research Ed.)* 2009;**339**:b3170.

Jorge 2015 *{published data only}*

Jorge RT, Souza MC, Chiari A, Jones A, Fernandes Ada R, Lombardi Junior I, et al. Progressive resistance exercise in women with osteoarthritis of the knee: a randomized controlled trial. *Clinical Rehabilitation* 2015;**29**(3):234-43. [DOI: 10.1177/0269215514540920]

Joshi 2019 *{published data only}*

Joshi S, Singh SK, Vij JS. Effect of retrowalking, a non-pharmacological treatment on pain, disability, balance and gait in knee osteoarthritis: a randomized controlled trial. *Indian Journal of Public Health Research and Development* 2019;**10**(2):214-9. [DOI: 10.5958/0976-5506.2019.00288.2]

Kang 2022 *{published data only}*

Kang N, Wang Y, Chen G, Guo C, Zhang Z, Mei D, et al. Functional outcomes of Tai Chi exercise prescription in women with knee osteoarthritis. *Sports Medicine and Health Science* 2022;**4**(4):239-44. [DOI: 10.1016/j.smhs.2022.10.001]

Karadag 2019 *{published data only}*

Karadag S, Tasci S, Dogan N, Demir H, Kilic Z. Application of heat and a home exercise program for pain and function levels in patients with knee osteoarthritis: A randomized controlled trial [Application of heat and a home exercise program for pain and function levels in patients with knee osteoarthritis: a randomized controlled trial]. *International Journal of Nursing Practice* 2019;**25(5)**:e12772. [DOI: 10.1111/ijn.12772]

Keefe 2004 *{published data only}*

Keefe FJ, Blumenthal J, Baucom D, Affleck G, Waugh R, Caldwell DS, et al. Effects of spouse-assisted coping skills training and exercise training in patients with osteoarthritic knee pain: a randomized controlled study. *Pain* 2004;**110**:539-49.

Kholvadia 2019 *{published data only}*

Kholvadia A, Constantinou D, Gradidge PJ. Exploring the efficacy of low-level laser therapy and exercise for knee osteoarthritis. *South African Journal of Sports Medicine* 2019;**31**(1):v31i1a6058. [DOI: 10.17159/2078-516X/2019/v31i1a6058]

Kilic 2020 *{published data only}*

Daskapan A, Kilic F, Keskin ED. The effects of aerobic exercise training on pain and disability from osteoarthritis of the knee in postmenopausal women. *Annals of the Rheumatic Diseases* 2018;**77 Suppl 2**:1846-7. [DOI: 10.1136/annrheumdis-2018-eular.6325]

* Kilic F, Demirguc A, Arslan SA, Keskin ED, Aras M. The effect of aerobic exercise training on postmenopausal patients with knee osteoarthritis. *Journal of Back and Musculoskeletal Rehabilitation.* 2020;**26**:794062. [DOI: 10.3233/BMR-191712]

Kim 2013 *{published data only}*

Kim H, Suzuki T, Saito K, Kim M, Kojima N, Ishizaki T, et al. Effectiveness of exercise with or without thermal therapy for community-dwelling elderly Japanese women with non-specific knee pain: a randomized controlled trial. *Archives of Gerontology and Geriatrics* 2013;**57**(3):352-9.

Kim 2020 *{published data only}*

Kim JS, Kim CJ. Effect of a physical activity promoting program based on the IMB model on obese-metabolic health outcomes among obese older adults with knee osteoarthritis. *Journal of Korean Academy of Nursing* 2020;**50**(2):271-85. [DOI: 10.4040/jkan.2020.50.2.271]

Knoop 2013 *{published data only}*

* Knoop J, Dekker J, Van der Leeden M, Van der Esch M, Thorstensson CA, Gerritsen M, et al. Knee joint stabilization therapy in patients with osteoarthritis of the knee: a randomized, controlled trial. *Osteoarthritis and Cartilage* 2013;**21**(8):1025-34.

Knoop J, Steultjens MP, Roorda LD, Lems WF, Van der Esch M, Thorstensson CA, et al. Improvement in upper leg muscle strength underlies beneficial effects of exercise therapy in knee osteoarthritis: Secondary analysis from a randomised controlled trial [Improvement in upper leg muscle strength underlies beneficial effects of exercise therapy in knee osteoarthritis: secondary analysis from a randomised controlled trial]. *Physiotherapy (United Kingdom)* 2015;**101**(2):171-7. [DOI: 10.1016/j.physio.2014.06.002]

Kovar 1992 *{published data only}*

Kovar PA, Allegrante JP, MacKenzie CR, Peterson MG, Gutin B, Charlson ME. Supervised fitness walking in patients with osteoarthritis of the knee. *Annals of Internal Medicine* 1992;**116**:529-34.

Krishnan 2021 *{published data only}*

Krishnan K, Abadi FH, Zainudin FF, Barati AH, Elumalai G. Comparison of the effect of aquatic and Thera-band exercise on pain and quality of life in obese people with knee osteoarthritis. *Journal of Pain Management* 2021;**14**(3):221-9.

Kumar 2013 *{published data only}*

Kumar S, Kumar A, Kumar R. Proprioceptive training as an adjunct in osteoarthritis of knee. *Journal of Musculoskeletal Research* 2013;**16**(1):(no pagination).

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Kuntz 2018 *{published data only}*

* Kuntz AB, Chopp-Hurley JN, Brenneman EC, Karampatos S, Wiebenga EG, Adachi JD, et al. Efficacy of a biomechanically-based yoga exercise program in knee osteoarthritis: a randomized controlled trial. *PloS One* 2018;**13**(4):(no pagination). [DOI: 10.1371/journal.pone.0195653]

Kuntz AB, Karampatos S, Brenneman E, Chopp-Hurley JN, Wiebenga E, Adachi J, et al. Can a biomechanically-designed yoga exercise program yield superior clinical improvements than traditional exercise in women with knee osteoarthritis? *Osteoarthritis and Cartilage* 2016;**1**:S445-6.

Lai 2021 *{published and unpublished data}*

ChiCTR-IOR-16009234. Effects of whole body vibration exercise on neuromuscular function for individuals with knee osteoarthritis. ChiCTR-IOR-16009234 2016.

* Lai Z, Lee S, Chen Y, Wang L. Comparison of whole-body vibration training and quadriceps strength training on physical function and neuromuscular function of individuals with knee osteoarthritis: a randomised clinical trial. *Journal of Exercise Science and Fitness* 2021;**19**(3):150-7. [DOI: 10.1016/j.jesf.2021.01.003]

Lai Z, Wang X, Lee S, Hou X, Wang L. Effects of whole body vibration exercise on neuromuscular function for individuals with knee osteoarthritis: study protocol for a randomized controlled trial. *Trials* 2017;**18**(1):(no pagination). [DOI: 10.1186/s13063-017-2170-6]

Lee 2009 *{published data only}*

Lee H, Park H, Chae Y, Kim S, Kim J, Yin C. Tai Chi Qigong for the quality of life of patients with knee osteoarthritis: a pilot, randomized, waiting list controlled trial. *Clinical Rehabilitation* 2009;**23**(6):504-11.

Lee 2019 *{published data only}*

Lee B-Y, Shin W-Y, An M-J, Choe Y, . Effect of dynamic balance exercise in elderly patients with unilateral knee osteoarthritis. *Clinical Pain* 2019;**18**(1):16-23. [DOI: 10.0000/cp.2019.18.1.16]

Lee 2023 *{published data only}*

* Lee EL, Jang MH, Lee BJ, Han SH, Lee HM, Choi SU, et al. Home-based remote rehabilitation leads to superior outcomes for older women with knee osteoarthritis: a randomized controlled trial. *Journal of the American Medical Directors Association* 2023;**24**(10):1555-61. [DOI: 10.1016/j.jamda.2023.08.013]

Li 2020 *{published data only}*

Li LC, Feehan LM, Xie H, Lu N, Shaw CD, Gromala D, et al. Effects of a 12-week multifaceted wearable-based program for people with knee osteoarthritis: randomized controlled trial. *JMIR MHealth and UHealth* 2020;**8**(7):e19116. [DOI: 10.2196/19116]

Nct02585323. Supra: using wearable activity trackers with a new application to improve physical activity in knee osteoarthritis [Supporting physical activity and reducing sedentary behavior in arthritis]. Nct02585323 2015.

Lim 2008 *{published data only}*

Hall M, Hinman RS, Wrigley TV, Kasza J, Lim BW, Bennell KL. Knee extensor strength gains mediate symptom improvement in knee osteoarthritis: secondary analysis of a randomised controlled trial. *Osteoarthritis and Cartilage* 2018;**26**(4):495-500. [DOI: 10.1016/j.joca.2018.01.018]

Kean CO, Hinman RS, Wrigley TV, Lim BW, Bennell KL. Impact loading following quadriceps strength training in individuals with medial knee osteoarthritis and varus alignment. *Clinical Biomechanics (Bristol, Avon)* 2017;**42**:20-4. [DOI: 10.1016/j.clinbiomech.2017.01.002]

* Lim BW, Hinman RS, Wrigley TV, Sharma L, Bennell KL. Does knee malalignment mediate the effects of quadriceps strengthening on knee adduction moment, pain, and function in medial knee osteoarthritis? A randomized controlled trial. *Arthritis Care & Research* 2008;**59**(7):943-51.

Lin 2009 *{published data only}*

* Lin D, Lin CJ, Lin Y, Jan M. Efficacy of 2 non-weight-bearing interventions, proprioception training versus strength training, for patients with knee osteoarthritis: a randomized clinical trial. *Journal of Orthopaedic and Sports Physical Therapy* 2009;**39**(6):450-7.

Lin DH, Lin YF, Chai HM, Han YC, Jan MH. Comparison of proprioceptive functions between computerized proprioception facilitation exercise and closed kinetic chain exercise in patients with knee osteoarthritis. *Clinical Rheumatology* 2007;**26**(4):520-8.

Liu 2019 *{published data only}*

ChiCTR-IOR-16009308. Explore Tai Chi exercise in improving knee osteoarthritis symptoms based on fMRI: a clinical study. ChiCTR-IOR-16009308 2016.

* Liu J, Chen L, Chen X, Hu K, Tu Y, Lin M, et al. Modulatory effects of different exercise modalities on the functional connectivity of the periaqueductal grey and ventral tegmental area in patients with knee osteoarthritis: a randomised multimodal magnetic resonance imaging study. *British Journal of Anaesthesia* 2019;**123**(4):506-18. [DOI: 10.1016/j.bja.2019.06.017]

Lu 2017 *{published data only}*

Lu J, Huang L, Wu X, Fu W, Liu Y. Effect of Tai Ji Quan training on self-reported sleep quality in elderly Chinese women with knee osteoarthritis: a randomized controlled trail. *Sleep Medicine* 2017;**33**:70-5. [DOI: 10.1016/j.sleep.2016.12.024]

Lund 2008 *{published data only}*

Lund H, Weile U, Christensen R, Rostock B, Downey A, Bartels EM, et al. A randomized controlled trial of aquatic and land-based exercise in patients with knee osteoarthritis. *Journal of Rehabilitation Medicine* 2008;**40**(2):137-44.

Marconcin 2018 *{published data only}*

* Marconcin P, Espanha M, Teles J, Bento P, Campos P, Andre R, et al. A randomized controlled trial of a combined self-management and exercise intervention for elderly people with osteoarthritis of the knee: the PLE^{2}NO

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program. *Clinical Rehabilitation* 2018;**32**(2):223-32. [DOI: 10.1177/0269215517718892]

Marconcin P, Espanha M, Yazigi F, Campos P. The PLE2NO self-management and exercise program for knee osteoarthritis: study protocol for a randomized controlled trial. *BMC Musculoskeletal Disorders* 2016;**17**(1):(no pagination). [DOI: 10.1186/s12891-016-1115-7]

Marconcin P, Yazigi F, Teles J, Campos P, Espanha M. The effectiveness of a randomised clinical trial of PLE2 NO self-management and exercise programme for knee osteoarthritis to improve self-efficacy. *Musculoskeletal Care* 2022;**20**(1):137-44. [DOI: 10.1002/msc.1573]

NCT02562833. The PLE2NO self-management and exercise program for knee osteoarthritis. https://clinicaltrials.gov/study/NCT02562833 (first posted 29 September 2015).

Maurer 1999 *{published data only}*

Maurer BT, Stern AG, Kinossian B, Cook KD, Schumacher HR. Osteoarthritis of the knee: isokinetic quadriceps exercise versus an educational intervention. *Archives of Physical Medicine and Rehabilitation* 1999;**80**:1293-9.

Mazloum 2018 *{published data only}*

Mazloum V, Rabiei P, Rahnama N, Sabzehparvar E. The comparison of the effectiveness of conventional therapeutic exercises and Pilates on pain and function in patients with knee osteoarthritis. *Complementary Therapies in Clinical Practice* 2018;**31**:343-8. [DOI: 10.1016/j.ctcp.2017.10.008]

Mehwish 2023 *{published data only}*

Mehwish B, Ali SS, Mirza Baig AA. Effect of hip joint mobilisations and strength training on pain, physical function and dynamic balance in patients with knee osteoarthritis: a randomized controlled trial. *Journal of Pakistan Medical Association* 2023;**73**(4):749-54. [DOI: 10.47391/JPMA.6223]

Messier 2004 *{published data only}*

* Messier SP, Loeser RF, Miller GD, Morgan TM, Rejeski WJ, Sevick MA, et al. Exercise and dietary weight loss in overweight and obese older adults with knee osteoarthritis. *Arthritis and Rheumatism* 2004;**50**:1501-10.

Miller GD, Nicklas BJ, Davis CC, Legault C, Messier SP. Basal growth hormone concentration increased following a weight loss focused dietary intervention in older overweight and obese women. *Journal of Nutrition, Health & Aging* 2012;**16**(2):169-74.

Nicklas BJ, Ambrosius W, Messier SP, Miller GD, Penninx BW, Loeser RF, et al. Diet-induced weight loss, exercise, and chronic inflammation in older, obese adults: a randomized controlled clinical trial. *American Journal of Clinical Nutrition* 2004;**79**:544-51.

Sevick MA, Miller GD, Loeser RF, Williamson JD, Messier SP. Cost-effectiveness of exercise and diet in overweight and obese adults with knee osteoarthritis. *Medicine and Science in Sports and Exercise* 2009;**41**(6):1167-74.

Messier 2013 *{published data only}*

Bay-Jensen AC, Loeser R, Frederiksen P, Beaver D, Karsdal MA, Nicklas BJ, et al. Effects of dietary weight loss with and without exercise on fibrosis biomarkers in adults with knee osteoarthritis. *Osteoarthritis and Cartilage* 2021;**29 Suppl 1**:S157-8. [DOI: 10.1016/j.joca.2021.02.220]

Beavers KM, Beavers DP, Newman JJ, Anderson AM, Loeser RF, Nicklas BJ, et al. Effects of total and regional fat loss on plasma CRP and IL-6 in overweight and obese, older adults with knee osteoarthritis. *Osteoarthritis and Cartilage* 2015;**23**(2):249-56. [DOI: 10.1016/j.joca.2014.11.005]

DeVita P, Beavers D, Loeser RF, Hunter DJ, Legault C, Newmank J, et al. Predicting knee joint loads in adults with knee osteoarthritis: the intensive diet and exercise for arthritis trial (IDEA). *Osteoarthritis and Cartilage* 2013;**1**:S85.

Messier SP, Beavers D, Mihalko SL, Miller GD, Nicklas BJ, Lyles M, et al. Does long-term intensive diet and exercise reduce the biomechanical burden in overweight and obese adults with knee osteoarthritis? The intensive diet and exercise for arthritis (IDEA) trial. *Osteoarthritis and Cartilage* 2016;**1**:S47-8.

Messier SP, Beavers DP, Mihalko SL, Miller GD, Lyles MF, Hunter DJ, et al. The effects of intensive dietary weight loss and exercise on gait in overweight and obese adults with knee osteoarthritis. The Intensive Diet and Exercise for Arthritis (IDEA) trial. *Journal of Biomechanics* 2019;**1**:109477. [DOI: 10.1016/j.jbiomech.2019.109477]

* Messier SP, Mihalko SL, Legault C, Miller GD, Nicklas BJ, DeVita P, et al. Effects of intensive diet and exercise on knee joint loads, inflammation, and clinical outcomes among overweight and obese adults with knee osteoarthritis: the IDEA randomized clinical trial. *JAMA* 2013;**310**(12):1263-73.

Messier SP, Newman JJ, Scarlett MJ, Mihalko SL, Miller GD, Nicklas BJ, et al. Changes in body weight and knee pain in adults with knee osteoarthritis three-and-a-half years after completing diet and exercise interventions: follow-up study for a single-blind, single-center, randomized controlled trial. *Arthritis Care & Research* 2022;**74**(4):607-16. [DOI: 10.1002/acr.24765]

Munugoda IP, Beavers D, Wirth W, Aitken DA, Guermazi A, Hunter DJ, et al. The effect of weight loss on the progression of meniscal extrusion and size in knee osteoarthritis: a post-hoc analysis of the intensive diet and exercise for arthritis (IDEA) trial. *Osteoarthritis and Cartilage* 2019;**27 Suppl 1**:S29-S30. [DOI: 10.1016/j.joca.2019.02.045]

Pogacnik Murillo AL, Wirth W, Beavers D, Loeser RF, Nicklas BJ, Miller GD, et al. Impact of diet and/or exercise intervention on infrapatellar fat pad morphology-data from the intensive diet and exercise for arthritis (IDEA) trial. *Osteoarthritis and Cartilage* 2015;**2**:A389-90.

Messier 2021 *{published data only}*

Hunter DJ, Beavers DP, Eckstein F, Guermazi A, Loeser RF, Nicklas BJ, et al. The Intensive Diet and Exercise for Arthritis (IDEA) trial: 18-month radiographic and MRI outcomes.

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Osteoarthritis and Cartilage 2015;**23**(7):1090-8. [DOI: 10.1016/j.joca.2015.03.034]

Messier S, Mihalko S, Beavers D, Nicklas B, DeVita P, Carr J, et al. Is long-term high-intensity strength training beneficial or harmful for knee osteoarthritis patients. The strength training and arthritis trial (START). *Arthritis & Rheumatology* 2019;**71 Suppl 10**:1573-4. [DOI: 10.1002/art.41108]

Messier SP, Mihalko S, Beavers D, Nicklas B, DeVita P, Hunter DJ, et al. Is long-term strength training more effective for knee osteoarthritis patients with low- versus high-baseline muscle strength? The strength training and arthritis trial (START). *Osteoarthritis and Cartilage* 2020;**28 Suppl 1**:S355-S356. [DOI: 10.1016/j.joca.2020.02.554]

Messier SP, Mihalko SL, Beavers DP, Nicklas BJ, Devita P, Carr JJ, et al. Effect of high-intensity strength training on knee pain and knee joint compressive forces among adults with knee osteoarthritis: the START randomized clinical trial. *JAMA - Journal of the American Medical Association* 2021;**325**(7):646-57. [DOI: 10.1001/jama.2021.0411]

Messier SP, Mihalko SL, Beavers DP, Nicklas BJ, Devita P, Carr JJ, et al. Strength training for arthritis trial (START): design and rationale. *BMC Musculoskeletal Disorders* 2013;**14**:(no pagination).

Mete 2022 *{published data only}*

Mete E, Sari Z. The efficacy of exergaming in patients with knee osteoarthritis: A randomized controlled clinical trial. *Physiotherapy Research International* 2022;**27**(3):e1952. [DOI: 10.1002/pri.1952]

Multanen 2014 *{published data only}*

Heinonen A, Janhunen E, Multanen J, Jamsa T, Kujala U, Nieminen M, et al. Effects of high-impact training on femoral neck structure in postmenopausal women with mild osteoarthritis: 12-month randomized controlled exercise intervention. *Journal of Bone and Mineral Research. Conference* 2012;**27 Suppl 1**:(no pagination).

Heinonen A, Multanen J, Kujala U, Ojala R, Selanne H, Koli J, et al. High-impact training to improve physical performance in postmenopausal women with early osteoarthritis. *Physiotherapy (United Kingdom)* 2011;**1**:eS470-1. [DOI: 10.1016/j.physio.2011.04.002]

Koli J, Multanen J, Kujala UM, Hakkinen A, Nieminen MT, Kautiainen H, et al. Effects of exercise on patellar cartilage in women with mild knee osteoarthritis. *Medicine and Science in Sports and Exercise* 2015;**47**(9):1767-74. [DOI: 10.1249/MSS.0000000000000629]

Multanen J, Hakkinen A, Kujala U, Ojala R, Selanne H, Koli J, et al. The effects of bone exercise on knee pain, stiffness and function among women with mild knee osteoarthritis: randomized, controlled trial. *Physiotherapy (United Kingdom)* 2011;**1**:eS852-3. [DOI: 10.1016/j.physio.2011.04.002]

Multanen J, Nieminen MT, Hakkinen A, Kujala U, Jamsa T, Kautiainen H, et al. High-impact bone exercise does not have controversial effects on articular cartilage: a randomized controlled quantitative MRI study. *Journal of Bone and Mineral Research. Conference* 2012;**27 Suppl 1**:(no pagination).

* Multanen J, Nieminen MT, Hakkinen A, Kujala UM, Jamsa T, Kautiainen H, et al. Effects of high-impact training on bone and articular cartilage: 12-month randomized controlled quantitative MRI study. *Journal of Bone and Mineral Research* 2014;**29**(1):192-201. [DOI: 10.1002/jbmr.2015]

Nahayatbin 2018 *{published data only}*

IRCT2014010215936N1. Comparing the effect of Taichi [sic] and close kinetic chain exercise on the knee osteoarthritis. https://irct.behdasht.gov.ir/trial/15046 (first registered 26 January 2014).

* Nahayatbin M, Ghasemi M, Rahimi A, Khademi-Kalantari K, Naimi SS, Tabatabaee SM, et al. The effects of routine physiotherapy alone and in combination with either Tai Chi or closed kinetic chain exercises on knee osteoarthritis: a comparative clinical trial study. *Iranian Red Crescent Medical Journal* 2018;**20**(4):(no pagination). [DOI: 10.5812/ircmj.62600]

Nelligan 2021 *{published data only}*

ACTRN12618001167257. My Knee Exercise: a 6 month electronically delivered intervention to support self-management for people with knee osteoarthritis: a pragmatic randomised controlled trial. https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=375219 (first registered 13 July 2018).

Nelligan R, Hinman R, Kasza J, Crofts S, Bennel K. Effects of self-directed web-based strengthening exercise and physical activity supported by automated sms on pain and function in people with knee osteoarthritis: A randomised controlled trial. *Internal Medicine Journal* 2021;**51(SUPPL 2)**:9-10. [DOI: 10.1111/imj.15302]

Nelligan RK, Hinman RS, Kasza J, Bennell KL. Effectiveness of internet-delivered education and home exercise supported by behaviour change SMS on pain and function for people with knee osteoarthritis: a randomised controlled trial protocol. *BMC Musculoskeletal Disorders* 2019;**20**(1):(no pagination). [DOI: 10.1186/s12891-019-2714-x]

Nelligan RK, Hinman RS, Kasza J, Crofts SJ, Bennell KL. Effects of a self-directed web-based strengthening exercise and physical activity program supported by automated text messages for people with knee osteoarthritis: a randomized clinical trial. *JAMA Internal Medicine* 2021;**181**(6):776-85. [DOI: 10.1001/jamainternmed.2021.0991]

Nelligan RK, Hinman RS, McManus F, Lamb KE, Bennell KL. Moderators of the effect of a self-directed digitally delivered exercise program for people with knee osteoarthritis: exploratory analysis of a randomized controlled trial. *Journal of Medical Internet Research* 2021;**23**(10):(no pagination). [DOI: 10.2196/30768]

O'Reilly 1999 *{published data only}*

O'Reilly SC, Muir KR, Doherty M. Effectiveness of home exercise on pain and disability from osteoarthritis of the knee: a randomised controlled trial. *Annals of the Rheumatic Diseases* 1999;**58**:15-9.

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Oh 2020 *{published data only}*

Oh SL, Kim DY, Bae JH, Lim JY. Effects of rural community-based integrated exercise and health education programs on the mobility function of older adults with knee osteoarthritis. *Aging Clinical and Experimental Research.* 2021;**33**(11):3005-14. [DOI: 10.1007/s40520-020-01474-7]

Ozlu 2023 *{published data only}*

Ozlu A, Unver G, Tuna HI, Menekseoglu AK. The effect of a virtual reality-mediated gamified rehabilitation program on pain, disability, function, and balance in knee osteoarthritis: a prospective randomized controlled study. *Games for Health Journal* 2023;**12**(2):118-24. [DOI: 10.1089/g4h.2022.0130]

Pazit 2018 *{published data only}*

ACTRN12616001382460. High speed resistance training and balance training for people with knee osteoarthritis to reduce falls risk. https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=371590&isReview=true (registered 6 October 2016).

Levinger P, Dunn J, Bifera N, Butson M, Elias G, Hill KD. High-speed resistance training and balance training for people with knee osteoarthritis to reduce falls risk: study protocol for a pilot randomized controlled trial. *Trials* 2017;**18**(1):(no pagination). [DOI: 10.1186/s13063-017-2129-7]

* Pazit L, Dunn J, Nancy B, Butson M, George E, Hill KD, et al. Safety and feasibility of high speed resistance training with and without balance exercises for knee osteoarthritis: A pilot randomised controlled trial. *Physical Therapy in Sport* 2018;**34**:154-63. [DOI: 10.1016/j.ptsp.2018.10.001]

Peloquin 1999 *{published data only}*

Peloquin L, Bravo G, Gauthier P, Lacombe G, Billiard J-S. Effects of a cross-training exercise program in persons with osteoarthritis of the knee. A randomized controlled trial. *Journal of Clinical Rheumatology* 1999;**5**:126-36.

Qiu 2023 *{published data only}*

* Qiu J, Zhou T, Jin H, Pan Y, Qian T, Xue C, et al. Effect of adding hip exercises to general rehabilitation treatment of knee osteoarthritis on patients' physical functions: a randomized clinical trial. *BMC Sports Science, Medicine and Rehabilitation* 2023;**15**(1):158. [DOI: 10.1186/s13102-023-00772-7]

Rafiq 2021a *{published data only}*

Rafiq MT, Hamid MS, Hafiz E. Effect of progressive resistance strength training on body mass index, quality of life and functional capacity in knee osteoarthritis: A randomized controlled trial. *Journal of Multidisciplinary Healthcare* 2021;**14**:2161-8. [DOI: 10.2147/JMDH.S317896]

Rafiq 2021b *{published data only}*

Irct20191221045846n1. Role of advising exercises using mobile health in treatment of knee osteoarthritis [Effectiveness of lower limb rehabilitation protocol using mobile health on physical disability and exercise adherence among knee osteoarthritis (OA) patients who are overweight]. Irct20191221045846n1 2020.

Rafiq MT, Abdul Hamid MS, Hafiz E. The effect of rehabilitation protocol using mobile health in overweight and obese patients with knee osteoarthritis: a clinical trial. *Advances in Rheumatology* 2021;**61**(1):(no pagination). [DOI: 10.1186/s42358-021-00221-4]

Rafiq 2021c *{published data only}*

Rafiq MT, Hamid MS, Hafiz E. Short-term effects of strengthening exercises of the lower limb rehabilitation protocol on pain, stiffness, physical function, and body mass index among knee osteoarthritis participants who were overweight or obese: a clinical trial. *Scientific World Journal* 2021;**2021**:(no pagination). [DOI: 10.1155/2021/6672274]

Rakhshani 2022 *{published data only}*

Rakhshani M, Ashtiani AR, Payandeh A, Akbari A, Ghiasi F, Hosseinifar M, et al. Effect of proximal exercises of lower extremity on static balance parameters during quiet standing. *Journal of Family Medicine and Primary Care* 2022;**11**(11):6978-82. [DOI: 10.4103/jfmpc.jfmpc_362_22]

Rego 2023 *{published data only}*

Rego TA, Ferreira AP, Villela DW, Shirahige L, Xavier AB, Braz RR, et al. Effects of mat pilates on older adult women with knee osteoarthritis: a randomized controlled trial. *Journal of Bodywork and Movement Therapies* 2023;**33**:136-41. [DOI: 10.1016/j.jbmt.2022.02.007]

Robbins 2022 *{published data only}*

Ferreira De Meneses SR, Hunter DJ, Young Docko E, Pasqual Marques A. Effect of low-level laser therapy (904 nm) and static stretching in patients with knee osteoarthritis: a protocol of randomised controlled trial. *BMC Musculoskeletal Disorders* 2015;**16**(1):(no pagination). [DOI: 10.1186/s12891-015-0709-9]

Melo L, De Meneses SF, Johnson VL, Deveza LA, Hunter DJ. Stepped care approach for medial tibiofemoral osteoarthritis. *Osteoarthritis and Cartilage* 2017;**25 Suppl 1**:S176.

Meneses SF, Marques AP, Hunter DJ. Is knee range of motion related to pain? *Osteoarthritis and Cartilage* 2016;**1**:S447-8.

* Robbins SR, Alfredo PP, Junior WS, Marques AP. Low-level laser therapy and static stretching exercises for patients with knee osteoarthritis: a randomised controlled trial. *Clinical Rehabilitation* 2022;**36**(2):204-13. [DOI: 10.1177/02692155211047017]

Rogind 1998 *{published data only}*

Rogind H, Bibow-Nielsen B, Jensen B, Moller HC, Frimodt-Moller H, Bliddal H. The effects of a physical training program on patients with osteoarthritis of the knees. *Archives of Physical Medicine and Rehabilitation* 1998;**79**:1421-7.

Rosadi 2023 *{published data only}*

Isrctn17933625. Effectiveness of a combined exercise and physiotherapy intervention in improving functional activity of patients with knee osteoarthritis in Muslim prayer movements. Isrctn17933625 2021.

Rosadi R, Wu PT, You YL, Kuo LC, Lin CF. Effectiveness of combined exercise and physiotherapy intervention in improving

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functional activity of patients with knee osteoarthritis in Muslim pray movements. *Journal of Medical and Biological Engineering* 2023;**98**(7):560-70. [DOI: 10.1007/s40846-023-00778-z]

Rosedale 2014 *{published data only}*

Rosedale R, Rastogi R, May S, Chesworth BM, Filice F, Willis S, et al. Efficacy of exercise intervention as determined by the McKenzie System of Mechanical Diagnosis and Therapy for knee osteoarthritis: a randomized controlled trial. *Journal of Orthopaedic and Sports Physical Therapy* 2014;**44**(3):173-81, A1-6. [DOI: 10.2519/jospt.2014.4791]

Sadeghi 2023 *{published data only}*

Sadeghi A, Rostami M, Khanlari Z, Zeraatchi A, Jalili N, Karimi Moghaddam A, et al. Effectiveness of muscle strengthening exercises on the clinical outcomes of patients with knee osteoarthritis: a randomized four-arm controlled trial. *Caspian Journal of Internal Medicine* 2023;**14**(3):433-42. [DOI: 10.22088/cjim.14.3.433]

Salacinski 2012 *{published data only}*

Salacinski AJ, Krohn K, Lewis SF, Holland ML, Ireland K, Marchetti G. The effects of group cycling on gait and pain-related disability in individuals with mild-to-moderate knee osteoarthritis: a randomized controlled trial. *Journal of Orthopaedic and Sports Physical Therapy* 2012;**42**(12):985-95.

Salli 2010 *{published data only}*

Salli A, Sahin N, Baskent A, Ugurlu H. The effect of two exercise programs on various functional outcome measures in patients with osteoarthritis of the knee: a randomized controlled clinical trial. *Isokinetics and Exercise Science* 2010;**18**(4):201-9.

Samuel 2014 *{published data only}*

Samuel Sundar Doss D, Rekha K, Prathap S. Effects of non weight bearing strength training for knee osteoarthritis. *International Journal of Research in Pharmaceutical Sciences* 2014;**5**(3):188-92.

Samut 2015 *{published data only}*

Samut G, Dincer F, Ozdemir O. The effect of isokinetic and aerobic exercises on serum interleukin-6 and tumor necrosis factor alpha levels, pain, and functional activity in patients with knee osteoarthritis. *Modern Rheumatology* 2015;**25**(6):919-24. [DOI: 10.3109/14397595.2015.1038425]

Santana 2022 *{published data only}*

Santana ET, Da Cunha Machado S, Brandao Lima VN, DeSantana Filho VJ, Dos Santos Maciel LY, De Farias Neto JP, et al. Comparison between exercise therapy and non-hydrolyzed collagen (UC-II) in functionality and quality of life in women with knee osteoarthritis: A randomized controlled clinical trial. *Wiener Klinische Wochenschrift* 2022;**135**:291-300. [DOI: 10.1007/s00508-022-02037-8]

Schilke 1996 *{published data only}*

Schilke JM, Johnson GO, Housh TJ, O'Dell JR. Effects of muscle-strength training on the functional status of patients with osteoarthritis of the knee joint. *Nursing Research* 1996;**45**:68-72.

Schlenk 2020 *{published data only}*

Schlenk EA, Fitzgerald GK, Rogers JC, Kwoh CK, Sereika SM. Promoting physical activity in older adults with knee osteoarthritis and hypertension: a randomized controlled trial. *Journal of Aging and Physical Activity* 2020;**29**(2):207-18. [DOI: 10.1123/japa.2019-0498]

Sedaghatnezhad 2021 *{published data only}*

Irct20171115034920n1. The effect of uphill walking on knee osteoarthritis [The effect of uphill walking on pain, range of motion, stride length, walking speed and disability in patients with knee osteoarthritis]. Irct20171115034920n1 2017.

* Sedaghatnezhad P, Shams M, Karimi N, Rahnama L. Uphill treadmill walking plus physical therapy versus physical therapy alone in the management of individuals with knee osteoarthritis: a randomized clinical trial. *Disability and Rehabilitation* 2021;**43**(18):2541-9. [DOI: 10.1080/09638288.2019.1703146]

Sekir 2005 *{published and unpublished data}*

Sekir U, Gur H. A multi-station proprioceptive exercise program in patients with bilateral knee osteoarthrosis: functional capacity, pain and sensoriomotor function. A randomized controlled trial. *Journal of Sports Science & Medicine* 2005;**4**(4):590-603.

Shabbir 2022 *{published data only}*

NCT04546659. Retro-walking in knee osteoarthritis [Effects of retro-walking on pain, functional disability and performance in knee osteoarthritis]. https://clinicaltrials.gov/study/NCT04546659 (first posted 14 September 2020).

* Shabbir S, Hashim M, Sajjad R, Kayani MS, Syed FA, Ibrahim M, et al. Effects of retro-walking on pain, functional disability and performance in knee osteoarthritis. *Pakistan Journal of Medical and Health Sciences* 2022;**16**(2):38-41. [DOI: 10.53350/pjmhs2216238]

Shahine 2020 *{published data only}*

Shahine NF, El Ashri NI, Senna MK, Abd Elhameed SH. Effect of a pedometer based aerobic walking program on pain and function among elderly patients with knee osteoarthritis. *European Journal of Molecular and Clinical Medicine* 2020;**7**(9):790-9.

Shen 2022 *{published data only}*

Shen P, Li L, Song Q, Sun W, Zhang C, Fong DT, et al. Proprioceptive neuromuscular facilitation improves symptoms among older adults with knee osteoarthritis during stair ascending: a randomized controlled trial. *American Journal of Physical Medicine & Rehabilitation* 2022;**101**(8):753-60. [DOI: 10.1097/PHM.0000000000001906]

Shin 2017 *{published data only}*

Shin YA, Suk MH, Jang HS, Choi HJ. Short-term effects of Theracurmin dose and exercise type on pain, walking ability, and muscle function in patients with knee osteoarthritis. *Journal of Exercise Rehabilitation* 2017;**13**(6):684-92. [DOI: 10.12965/jer.1735064.532]

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Simao 2012 *{published data only}*

Simao AP, Avelar NC, Tossige-Gomes R, Neves CD, Mendonca VA, Miranda AS, et al. Functional performance and inflammatory cytokines after squat exercises and whole-body vibration in elderly individuals with knee osteoarthritis. *Archives of Physical Medicine and Rehabilitation* 2012;**93**(10):1692-700.

Song 2003 *{published data only}*

Song R, Lee E-O, Lam P, Bae S-C. Effects of Tai Chi exercise on pain, balance, muscle strength, and perceived difficulties in physical functioning in older women with osteoarthritis: a randomized clinical trial. *Journal of Rheumatology* 2003;**30**:2039-44.

Song 2020 *{published data only}*

Song Q, Shen P, Mao M, Sun W, Zhang C, Li L. Proprioceptive neuromuscular facilitation improves pain and descending mechanics among elderly with knee osteoarthritis. *Scandinavian Journal of Medicine & Science in Sports* 2020;**30**(9):1655-63. [DOI: 10.1111/sms.13709]

Takacs 2017 *{published data only}*

Takacs J, Krowchuk NM, Garland SJ, Carpenter MG, Hunt MA. Dynamic balance training improves physical function in individuals with knee osteoarthritis: a pilot randomized controlled trial. *Archives of Physical Medicine and Rehabilitation* 2017;**98**(8):1586-93. [DOI: 10.1016/j.apmr.2017.01.029]

Talbot 2003 *{published data only}*

* Talbot LA, Gaines JM, Huynh TN, Metter EJ. A home-based pedometer-driven walking program to increase physical activity in older adults with osteoarthritis of the knee: a preliminary study. *Journal of the American Geriatric Society* 2003;**51**:387-92.

Thomas 2002 *{published data only}*

Thomas KS, Muir KR, Doherty M, Jones AC, O'Reilly SC, Bassey EJ. Home based exercise programme for knee pain and knee osteoarthritis: randomised controlled trial. *BMJ (Clinical Research Ed)* 2002;**325**:752-7.

Thorstensson 2005 *{published data only}*

Thorstensson CA, Roos EM, Petersson IF, Ekdahl C. Six-week high-intensity exercise program for middle-aged patients with knee osteoarthritis: a randomized controlled trial. *BMC Musculoskeletal Disorders* 2005;**6**:27.

Topp 2002 *{published data only}*

Topp R, Woolley S, Horuyak J, Khuder S, Kahaleh B. The effect of dynamic versus isometric resistance training on pain and functioning among adults with osteoarthritis of the knee. *Archives of Physical and Medical Rehabilitation* 2002;**83**:1187-95.

Tsai 2013 *{published data only}*

Tsai PF, Chang JY, Beck C, Kuo YF, Keefe FJ, Rosengren K. A supplemental report to a randomized cluster trial of a 20-week Sun-style Tai Chi for osteoarthritic knee pain in elders with cognitive impairment. *Complementary Therapies in Medicine* 2015;**23**(4):570-6. [DOI: 10.1016/j.ctim.2015.06.001]

* Tsai PF, Chang JY, Beck C, Kuo YF, Keefe FJ. A pilot cluster-randomized trial of a 20-week Tai Chi program in elders with cognitive impairment and osteoarthritic knee: effects on pain and other health outcomes. *Journal of Pain and Symptom Management* 2013;**45**(4):660-9. [DOI: 10.1016/j.jpainsymman.2012.04.009]

Van Baar 1998 *{published data only}*

Van Baar ME, Dekker J, Oostendorp RA, Bijl D, Voorn TB, Bijlsma JW. Effectiveness of exercise in patients with osteoarthritis of hip or knee: nine months' follow up. *Annals of the Rheumatic Diseases* 2001;**60**:1123-30.

* Van Baar ME, Dekker J, Oostendorp RA, Bijl D, Voorn TB, Lemmens JA, et al. The effectiveness of exercise therapy in patients with osteoarthritis of the hip or knee: a randomized clinical trial. *Journal of Rheumatology* 1998;**25**:2432-9.

Vassao 2021 *{published data only}*

Vassao PG, De Souza AC, Da Silveira Campos RM, Garcia LA, Tucci HT, Renno AC. Effects of photobiomodulation and a physical exercise program on the expression of inflammatory and cartilage degradation biomarkers and functional capacity in women with knee osteoarthritis: a randomized blinded study. *Advances in Rheumatology* 2021;**61**(1):(no pagination). [DOI: 10.1186/s42358-021-00220-5]

Vincent 2020 *{published data only}*

Vincent HK, Vasilopoulos T, Vincent KR. Comparative efficacy of eccentric and concentric-focused resistance exercise on pain, leg strength and walking endurance in knee osteoarthritis. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S337.

Vincent HK, Vincent KR. Collagen biomarker responses and pain in obese and nonobese individuals with knee OA after resistance exercise training. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S193-4.

Vincent KR, Montero C, Vincent HK. Progressive enhanced eccentric or concentric resistance exercise training for knee osteoarthritis. *Osteoarthritis and Cartilage* 2013;**1**:S275-6.

* Vincent KR, Vincent HK. Concentric and eccentric resistance training comparison on physical function and functional pain outcomes in knee osteoarthritis: a randomized controlled trial. *American Journal of Physical Medicine & Rehabilitation* 2020;**99**(10):932-40. [DOI: 10.1097/PHM.0000000000001450]

Wasser J, Vincent KR, Vincent HK. Unanticipated pain relief effect with progressive resistance exercise in patients with knee OA and comorbid low back pain. *Physical Medicine and Rehabilitaion* 2018;**10 Suppl 9**:S42.

Wallis 2017 *{published data only}*

ACTRN12615000015549. Prescribing the maximum tolerated dose of walking for people with severe knee osteoarthritis: a phase II, randomised controlled trial. https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?ACTRN=12615000015549 (registered on 14 January 2015).

* Wallis JA, Webster KE, Levinger P, Singh PJ, Fong C, Taylor NF. A walking program for people with severe knee osteoarthritis did not reduce pain but may have benefits for cardiovascular health: a phase II randomised controlled trial. *Osteoarthritis and Cartilage* 2017;**25**(12):1969-79. [DOI: 10.1016/j.joca.2016.12.017]

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Wang 2011 *{published data only}*

Wang TJ, Lee SC, Liang SY, Tung HH, Wu SF, Lin YP. Comparing the efficacy of aquatic exercises and land-based exercises for patients with knee osteoarthritis. *Journal of Clinical Nursing* 2011;**20**(17-18):2609-22.

Wortley 2013 *{published data only}*

Wortley M, Zhang S, Paquette M, Byrd E, Baumgartner L, Klipple G, et al. Effects of resistance and Tai Ji training on mobility and symptoms in knee osteoarthritis patients. *Journal of Sport and Health Science* 2013;**2**(4):209-14. [DOI: 10.1016/j.jshs.2013.01.001]

Xiao 2021 *{published data only}*

Xiao Z, Li G. The effect of Wuqinxi exercises on the balance function and subjective quality of life in elderly, female knee osteoarthritis patients. *American Journal of Translational Research* 2021;**13**(6):6710-6.

Ye 2020 *{published data only}*

ChiCTR-IOR-16010042. Biomechanics effects of Tai Chi in patients with knee osteoarthritis [Effect of qigong in elderly with knee osteoarthritis: a randomized controlled trial]. ChiCTR-IOR-16010042 2016.

* Ye J, Simpson MW, Liu Y, Lin W, Zhong W, Cai S, et al. The effects of Baduanjin Qigong on postural stability, proprioception, and symptoms of patients with knee osteoarthritis: a randomized controlled trial. *Frontiers in Medicine* 2020;**6**:(no pagination). [DOI: 10.3389/fmed.2019.00307]

Øiestad 2023 *{published data only}*

* Oiestad BE, Aaroen A, Rotterud J, Osteras N, Jarstad E, Risberg M. The effect of strength training or cycling on quality of life and knee function in patients with knee osteoarthritis. A randomised controlled trial. *Osteoarthritis and Cartilage* 2023;**31 Suppl 1**:S151. [DOI: 10.1016/j.joca.2023.01.114]

Øiestad BE, Årøen A, Røtterud JH, Østerås N, Jarstad E, Grotle M, et al. The efficacy of strength or aerobic exercise on quality of life and knee function in patients with knee osteoarthritis. A multi-arm randomized controlled trial with 1-year follow-up. *BMC Musculoskeletal Disorders* 2023;**24**(4):714. [DOI: 10.1186/s12891-023-06831-x]

References to studies excluded from this review

Actrn12615000227594 *{published data only}*

Actrn12615000227594. Stepped Care Approach for Medial Tibiofemoral Osteoarthritis (STrEAMline)STrEAMline: the remission of symptoms in tibiofemoral osteoarthritis following an adaptive treatment strategy in overweight/obese participants. Actrn12615000227594 2015.

Actrn12616000273482 *{published data only}*

Actrn12616000273482. Feasibility and potential benefits of high-intensity interval cycling for knee osteoarthritic patients: a randomised control feasibility trial? Is high-intensity interval cycling feasible and more beneficial than continuous cycling for knee osteoarthritic patients? Actrn12616000273482 2016.

Actrn12618001280291 *{published data only}*

Actrn12618001280291. The effect of a virtual reality-based therapy on functions in older adult with knee osteoarthritis. Actrn12618001280291 2018.

Actrn12620001281987 *{published data only}*

Actrn12620001281987. Modifying activity behaviours in osteoarthritis. The best activity is the next activity: the effect of replacing sitting with moving on pain in people with osteoarthritis. Actrn12620001281987 2020.

Actrn12621000535875 *{published data only}*

Actrn12621000535875. Is a digital knee brace helpful for patients with osteoarthritis? Evaluating a digital knee brace and mobile-based CBT application for the treatment of knee pain in adults aged 65 or younger with knee osteoarthritis: a pilot trial. Actrn12621000535875 2021.

Actrn12622000461796 *{published data only}*

Actrn12622000461796. Exercise and weight loss for hip osteoarthritis: a clinical trial. Effectiveness of a telehealth-delivered clinician-supported exercise and weight loss program (Better Hip) for hip osteoarthritis: a randomised controlled trial. Actrn12622000461796 2022.

Actrn12622001231730 *{published data only}*

Actrn12622001231730. How should exercise and education be delivered in exercise physiology interventions in people with knee osteoarthritis? Feasibility of 'exercising into pain' in patients with symptomatic knee osteoarthritis. Actrn12622001231730 2022.

Ageberg 2010 *{published data only}*

Ageberg E, Link A, Roos EM. Feasibility of neuromuscular training in patients with severe hip or knee OA: the individualized goal-based NEMEX-TJR training program. *BMC Musculoskeletal Disorders* 2010;**11**:126.

Aglamis 2008 *{published data only}*

Aglamis B, Toraman NF, Yaman H. Change of quality of life due to exercise training in knee osteoarthritis: SF-36 and WOMAC. *Journal of Back and Musculoskeletal Rehabilitation* 2009;**22**(1):43-8.

Aglamis B, Toraman NF, Yaman H. The effect of a 12-week supervised multicomponent exercise program on knee OA in Turkish women. *Journal of Back and Musculoskeletal Rehabilitation* 2008;**21**(2):121-8.

Ahmad 2023 *{published data only}*

Ahmad MA, Yusof A, Hamid MS, Amin FH, Kamsan SS, Daud DM, et al. Effects of self-management program as adjunctive to usual rehabilitation exercise on pain and functional outcomes in knee osteoarthritis: a randomized controlled trial. *Journal of Research in Health Sciences* 2023;**23**(1):(no pagination). [DOI: 10.34172/jrhs.2023.104]

Akyol 2010 *{published data only}*

Akyol Y, Durmus D, Alayli G, Tander B, Bek Y, Canturk F, et al. Does short-wave diathermy increase the effectiveness of isokinetic exercise on pain, function, knee muscle strength,

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quality of life, and depression in the patients with knee osteoarthritis? A randomized controlled clinical study. *European Journal of Physical and Rehabilitation Medicine* 2010;**46**(3):325-36.

Alfredo 2012 *{published data only}*

Alfredo PP, Bjordal JM, Dreyer SH, Ferreira, Zaguetti G, Ovanessian V, et al. Efficacy of low level laser therapy associated with exercises in knee osteoarthritis: a randomized double-blind study. *Clinical Rehabilitation* 2012;**26**(6):523-33.

Alkhawajah 2019 *{published data only}*

Alkhawajah HA, Alshami AM. The effect of mobilization with movement on pain and function in patients with knee osteoarthritis: a randomized double-blind controlled trial. *BMC Musculoskeletal Disorders* 2019;**20**(1):(no pagination). [DOI: 10.1186/s12891-019-2841-4]

Allen 2021 *{published data only}*

Allen KD, Bongiorni D, Caves K, Coffman CJ, Floegel TA, Greysen HM, et al. STepped exercise program for patients with knee OsteoArthritis (STEP-KOA): protocol for a randomized controlled trial. *BMC Musculoskeletal Disorders* 2019;**20**(1):(no pagination). [DOI: 10.1186/s12891-019-2627-8]

Allen KD, Woolson S, Hoenig HM, Bongiorni D, Byrd J, Caves K, et al. Stepped exercise program for patients with knee osteoarthritis a randomized controlled trial. *Annals of Internal Medicine* 2021;**174**(3):298-307. [DOI: 10.7326/M20-4447]

Kaufman B, Allen KD, Coffman CJ, Woolson S, Bongiorni D, Caves K, et al. Cc4 cost-effectiveness analysis of the stepped exercise program for patients with knee osteoarthritis (step-koa) trial. *Value in Health* 2020;**23 Suppl 1**:S2. [DOI: 10.1016/j.jval.2020.04.011]

Kaufman BG, Allen KD, Coffman CJ, Woolson S, Caves K, Hall K, et al. Cost and quality of life outcomes of the stepped exercise program for patients with knee osteoarthritis trial. *Value in Health* 2022;**25**(4):614-21. [DOI: 10.1016/j.jval.2021.09.018]

Nct02653768. Stepped exercise program for knee osteoarthritis [Stepped exercise program for knee osteoarthritis (STEP-KOA)]. Nct02653768 2016.

Anjum 2023 *{published data only}*

Anjum N, Sheikh RK, Omer A, Anwar K, Khan MM, Aftab A, et al. Comparison of instrument-assisted soft tissue mobilization and proprioceptive neuromuscular stretching on hamstring flexibility in patients with knee osteoarthritis. *PeerJ* 2023;**11**:(no pagination). [DOI: 10.7717/peerj.16506]

Anwer 2011 *{published data only}*

Anwer S, Quddus N, Miraj M, Equebal A. Effectiveness of electromyographic biofeedback training on quadriceps muscle strength in osteoarthritis of knee. *Hong Kong Physiotherapy Journal* 2011;**29**(2):86-93.

Aoki 2009 *{published data only}*

Aoki O, Tsumura N, Kimura A, Okuyama S, Takikawa S, Hirata S. Home stretching exercise is effective for improving knee range of motion and gait in patients with knee osteoarthritis. *Journal of Physical Therapy Science* 2009;**21**(2):113-9.

Aree-Ue 2017 *{published data only}*

Aree-Ue S, Saraboon Y, Belza B. Long-term adherence and effectiveness of a multicomponent intervention for community-dwelling overweight Thai older adults with knee osteoarthritis: 1-year follow up. *Journal of Gerontological Nursing* 2017;**43**(4):40-8. [DOI: 10.3928/00989134-20170111-09]

Armagan 2015 *{published data only}*

Armagan O, Yilmazer S, CCalisir, Ozgen M, Tascioglu F, Oner S, et al. Comparison of the symptomatic and chondroprotective effects of glucosamine sulphate and exercise treatments in patients with knee osteoarthritis. *Journal of Back and Musculoskeletal Rehabilitation* 2015;**28**(2):287-93. [DOI: 10.3233/BMR-140516]

Atamaz 2006 *{published data only}*

Atamaz F, Kirazli Y, Akkoc Y, Atamaz F, Kirazli Y, Akkoc Y. A comparison of two different intra-articular hyaluronan drugs and physical therapy in the management of knee osteoarthritis. *Rheumatology International* 2006;**26**:873-8.

Atamaz 2012 *{published data only}*

Atamaz FC, Durmaz B, Baydar M, Demircioglu OY, Iyiyapici A, Kuran B, et al. Comparison of the efficacy of transcutaneous electrical nerve stimulation, interferential currents, and shortwave diathermy in knee osteoarthritis: a double-blind, randomized, controlled, multicenter study. *Archives of Physical Medicine and Rehabilitation* 2012;**93**(5):748-56.

Ay 2013 *{published data only}*

Ay S, Koldas Dogan S, Evcik D. Is there an effective way to prescribe a home-based exercise program in patients with knee osteoarthritis? A randomized controlled study [Diz osteoartriti olan hastalarda ev egzersiz programi onermenin etkili bir yolu var midir? Randomize kontrollu bir calısma]. *Turkiye Fiziksel Tip ve Rehabilitasyon Dergisi* 2013;**59**(1):1-6. [DOI: 10.4274/tftr.70894]

Bell 2023c *{published data only}*

Bell E, O'Halloran P, Wallis J, Crossley K, Gibbs A, Lee A, et al. Using SUpported Motivational InTerviewing (SUMIT) to improve physical activity for people with knee osteoarthritis. A pilot, feasibility randomised controlled trial. *Journal of Science and Medicine in Sport* 2023;**26 Suppl 2**:S168. [DOI: 10.1016/j.jsams.2023.08.073]

Bell EC, O'Halloran P, Wallis JA, Crossley KM, Gibbs A, Annemarie L, et al. Supported motivational interviewing (SUMIT) to improve physical activity for people with knee osteoarthritis. A pilot, feasibility randomised controlled trial. *Osteoarthritis and Cartilage* 2023;**31(5)**:714-5. [DOI: 10.1016/j.joca.2023.02.068]

Bell EC, O'Halloran P, Wallis JA, Crossley KM, Gibbs AJ, Lee A, et al. Using SUpported Motivational InTerviewing (SUMIT) to increase physical activity for people with knee osteoarthritis: a pilot, feasibility randomised controlled trial. *BMJ Open* 2023;**13**(11):(no pagination). [DOI: 10.1136/bmjopen-2023-075014]

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Bennell 2005 *{published data only}*

Bennell KL, Himan RS, Metcalf BR, Buchbinder R, McConnell J, McColl G, et al. Efficacy of physiotherapy management of knee joint osteoarthritis: a randomised, double blind, placebo controlled trial. *Annals of the Rheumatic Diseases* 2005;**64**:906-12.

Bezalel 2010 *{published data only}*

Bezalel T, Carmeli E, Katz-Leurer M. The effect of a group education programme on pain and function through knowledge acquisition and home-based exercise among patients with knee osteoarthritis: a parallel randomised single-blind clinical trial. *Physiotherapy* 2010;**96**(2):137-43.

Bhalekar 2017 *{published data only}*

Bhalekar PA, Bagade VK. The effect of core strengthening exercises along with conventional treatment on knee joint proprioception, balance, and functional disability in patellofemoral osteoarthritis. *Indian Journal of Physiotherapy & Occupational Therapy* 2017;**11**(3):90-5. [DOI: 10.5958/0973-5674.2017.00080.6]

Bhore 2023 *{published data only}*

Bhore P, Shinde S. Effect of multi-component exercises program on pain-related gait adaptations among individuals with osteoarthritis of the knee joint. *Journal of Education & Health Promotion* 2023;**12**:138. [DOI: 10.4103/jehp.jehp_1628_22]

Boocock 2009 *{published data only}*

Boocock M, McNair P, Cicuttini F, Stuart A, Sinclair T. The short-term effects of running on the deformation of knee articular cartilage and its relationship to biomechanical loads at the knee. *Osteoarthritis and Cartilage* 2009;**17**(7):883-90.

Borjesson 1996 *{published data only}*

Borjesson M, Robertson E, Weidenhielm L, Mattsson E, Olsson E. Physiotherapy in knee osteoarthrosis effect on pain and walking. *Physiotherapy Research International* 1996;**1**:89-97.

Brosseau 2012a *{published data only}*

Brosseau L, Wells G, Kenny G, Reid R, Maetzel A, Tugwell P, et al. The impact of a community-based aerobic walking program for older individuals with mild to moderate knee osteoarthritis. *Physiotherapy (United Kingdom)* 2011;**1**:eS161. [DOI: 10.1016/j.physio.2011.04.002]

Brosseau L, Wells GA, Kenny GP, Reid R, Maetzel A, Tugwell P, et al. The implementation of a community-based aerobic walking program for mild to moderate knee osteoarthritis (OA): a knowledge translation (KT) randomized controlled trial (RCT): Part I: The Uptake of the Ottawa Panel clinical practice guidelines (CPGs). *BMC Public Health* 2012;**12**:871.

Brosseau 2012b *{published data only}*

Brosseau L, Wells GA, Kenny GP, Reid R, Maetzel A, Tugwell P, et al. The implementation of a community-based aerobic walking program for mild to moderate knee osteoarthritis (OA): a knowledge translation (KT) randomized controlled trial (RCT): Part I: The Uptake of the Ottawa Panel clinical practice guidelines (CPGs). *BMC Public Health* 2012;**12**:871.

Bulthuis 2007 *{published data only}*

Bulthuis Y, Drossaers-Bakker KW, Taal E, Rasker J, Oostveen J, Van't Pad Bosch P, et al. Arthritis patients show long-term benefits from 3 weeks intensive exercise training directly following hospital discharge. *Rheumatology* 2007;**46**(11):1712-7.

Bulthuis 2008 *{published data only}*

Bulthuis Y, Mohammad S, Braakman-Jansen LM, Drossaers-Bakker KW, Van de Laar M. Cost-effectiveness of intensive exercise therapy directly following hospital discharge in patients with arthritis: results of a randomized controlled clinical trial. *Arthritis and Rheumatism* 2008;**59**(2):247-54.

Cagnin 2020 *{published data only}*

Cagnin A, Choiniere M, Bureau NJ, Durand M, Mezghani N, Gaudreault N, et al. A multi-arm cluster randomized clinical trial of the use of knee kinesiography in the management of osteoarthritis patients in a primary care setting. *Postgraduate Medicine* 2020;**132**(1):91-101. [DOI: 10.1080/00325481.2019.1665457]

Cagnin A, Choiniere M, Bureau NJ, Durand M, Mezghani N, Gaudreault N, et al. Targeted exercises can improve biomechanical markers in individuals with knee osteoarthritis: a secondary analysis from a cluster randomized controlled trial. *Knee* 2023;**40**:122-34. [DOI: 10.1016/j.knee.2022.10.008]

Callaghan 1995 *{published data only}*

Callaghan MJ, Oldham JA, Hunt J. An evaluation of exercise regimes for patients with osteoarthritis of the knee: a single-blind randomized controlled trial. *Clinical Rehabilitation* 1995;**9**:213-8.

Centor 2023 *{published data only}*

Centor RM, Perez PL. Web exclusive. Annals on call - exercise for knee osteoarthritis: is more better? *Annals of Internal Medicine* 2023;**176**(3):eA220006. [DOI: 10.7326/A22-0006]

Cetin 2008 *{published data only}*

Cetin N, Aytar A, Atalay A, Akman MN. Comparing hot pack, short-wave diathermy, ultrasound, and TENS on isokinetic strength, pain, and functional status of women with osteoarthritic knees: a single-blind, randomized, controlled trial. *American Journal of Physical Medicine & Rehabilitation* 2008;**87**(6):443-51.

Chaipinyo 2009 *{published data only}*

Chaipinyo K, Karoonsupcharoen O. No difference between home-based strength training and home-based balance training on pain in patients with knee osteoarthritis: a randomised trial. *Australian Journal of Physiotherapy* 2009;**55**(1):25-30.

Chamberlain 1982 *{published data only}*

Chamberlain MA, Care G, Harfield B. Physiotherapy in osteoarthrosis of the knees. A controlled trial of hospital versus home exercises. *International Rehabilitation Medicine* 1982;**4**:101-6.

Chang 2019 *{published data only}*

Chang AH, Lee J, Song J, Price LL, Lee AC, Reid KF, et al. Association between pre-intervention physical activity level and

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

treatment response to exercise therapy in persons with knee osteoarthritis-an exploratory study. *ACR Open Rheumatology* 2019;**1(2)**:104-12. [DOI: 10.1002/acr2.1013]

Chao 2021 *{published data only}*

Chao J, Jing Z, Xuehua B, Peilei Y, Qi G. Effect of systematic exercise rehabilitation on patients with knee osteoarthritis: a randomized controlled trial. *Cartilage* 2021;**13**:1734S-1740S. [DOI: 10.1177/1947603520903443]

Chen 2021b *{published data only}*

Chen PY, Song CY, Yen HY, Lin PC, Chen SR, Lu LH, et al. Impacts of tai Chi exercise on functional fitness in community-dwelling older adults with mild degenerative knee osteoarthritis: a randomized controlled clinical trial. *BMC Geriatrics* 2021;**21(1)**:449. [DOI: 10.1186/s12877-021-02390-9]

ChiCTR1800015203 *{published data only}*

ChiCTR1800015203. The effect of YIJINJING on knee stability in knee osteoarthritis. ChiCTR1800015203 2018.

ChiCTR2000031887 *{published data only}*

ChiCTR2000031887. Effects of exercise combined with compound amino acid glucose intervention on middle-aged and elderly osteoarthritis. ChiCTR2000031887 2020.

ChiCTR2000033370 *{published data only}*

ChiCTR2000033370. The biomechanical effectiveness of the Taichi exercise in the management of knee OA: a randomized controlled trial. ChiCTR2000033370 2020.

ChiCTR2000033397 *{published data only}*

ChiCTR2000033397. The feasibility and effectiveness of internet-based rehabilitation for patients with knee osteoarthritis: a randomized controlled trial in the community setting. ChiCTR2000033397 2020.

ChiCTR2000037083 *{published data only}*

ChiCTR2000037083. Clinical application study of Yijinjing in preventing and treating knee osteoarthritis. ChiCTR2000037083 2020.

ChiCTR2000037256 *{published data only}*

ChiCTR2000037256. Study on the mechanism of regulating the balance of tendon and bone strength in knee osteoarthritis with Yijinjing. ChiCTR2000037256 2020.

ChiCTR2000037303 *{published data only}*

ChiCTR2000037303. The clinical study of weight-loss gait training on improving lower limb joint function in elderly patients with osteoarthritis. ChiCTR2000037303 2020.

ChiCTR2000038158 *{published data only}*

ChiCTR2000038158. Effects of a WeChat-based transtheoretical model-based home-exercise intervention for older adults with knee osteoarthritis. ChiCTR2000038158 2020.

ChiCTR2000039005 *{published data only}*

ChiCTR2000039005. The effectiveness of leg swing exercise versus quadriceps strengthening in treating patients with knee osteoarthritis: a randomized controlled trial. ChiCTR2000039005 2020.

ChiCTR2100041978 *{published data only}*

ChiCTR2100041978. Video-linked by physiotherapists and supported by family, exercise and education for community-dwelling older patients with knee osteoarthritis. ChiCTR2100041978 2021.

ChiCTR2100044586 *{published data only}*

ChiCTR2100044586. Clinical observation on the improvement of lower limb function of patients with knee osteoarthritis by South Shaolin kung [Application study of South Shaolin kung to improve lower limb function in patients with knee osteoarthritis]. ChiCTR2100044586 2021.

ChiCTR2100045654 *{published data only}*

ChiCTR2100045654. Study on the effect of kneeling on balance ability of knee osteoarthritis. ChiCTR2100045654 2021.

ChiCTR2100052416 *{published data only}*

ChiCTR2100052416. Research on the combined effect of systemic nutrition and rhythmic exercise on people with abnormal metabolism of motor system. ChiCTR2100052416 2021.

ChiCTR2100053574 *{published data only}*

ChiCTR2100053574. To explore the effect mechanism of Taijiquan in the treatment of knee osteoarthritis from the perspective of rehabilitation omics. ChiCTR2100053574 2021.

ChiCTR2300067530 *{published data only}*

ChiCTR2300067530. Effects of home-based heel-kicking and stretching exercise on pain and joint function in the acute phase of moderate to severe knee osteoarthritis. ChiCTR2300067530 2023.

ChiCTR2300067546 *{published data only}*

ChiCTR2300067546. A randomized controlled trial of modified eight-stage taijiquan for intervention in early to mid-stage knee osteoarthritis. ChiCTR2300067546 2023.

ChiCTR2300070376 *{published data only}*

Electroacupuncture versus exercise in patients with knee osteoarthritis: a randomized controlled trial. ChiCTR2300070376 2023.

ChiCTR2300070872 *{published data only}*

ChiCTR2300070872. Internet plus continuous rehabilitation nursing combined with lower limb quadriceps femoris muscle strength training for knee osteoarthritis: a randomized controlled trial. ChiCTR2300070872 2023.

ChiCTR2300071181 *{published data only}*

ChiCTR2300071181. Research on remote rehabilitation treatment model of knee osteoarthritis. ChiCTR2300071181 2023.

ChiCTR2300072820 *{published data only}*

ChiCTR2300072820. Effect of blood flow restriciton training in patients with knee OA. ChiCTR2300072820 2023.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ChiCTR2300073480 *{published data only}*

ChiCTR2300073480. Construction and effect evaluation of progressive phased resistance exercise intervention scheme for obese subjects with knee osteoarthritis. ChiCTR2300073480 2023.

ChiCTR2300073688 *{published data only}*

ChiCTR2300073688. Effect of the telemedicine-supported multicomponent exercise therapy in patients with knee osteoarthritis: a randomized controlled trial [Digital exercise therapy for the prevention and treatment of knee osteoarthritis: a clinical study protocol]. ChiCTR2300073688 2023.

ChiCTR-IIR-16009240 *{published data only}*

ChiCTR-IIR-16009240. Two methods of treatment for knee osteoarthritis. Manual therapy combined with cryotherapy versus kinesiotherapy combined with cryotherapy for knee osteoarthritis treatment: a randomized trial. ChiCTR-IIR-16009240 2016.

ChiCTR-INR-17014012 *{published data only}*

ChiCTR-INR-17014012. Effect of three different physical therapy approaches on function and disability of the knee joint in patients with knee osteoarthritis. ChiCTR-INR-17014012 2017.

ChiCTR-IOR-16009124 *{published data only}*

ChiCTR-IOR-16009124. The effect of hip muscle training on knee osteoarthritis. ChiCTR-IOR-16009124 2016.

ChiCTR-IOR-17013324 *{published data only}*

ChiCTR-IOR-17013324. Study on the clinical intervention of hip joint training in the treatment of knee osteoarthritis. ChiCTR-IOR-17013324 2017.

Cho 2015 *{published data only}*

Cho Y, Kim M, Lee W. Effect of proprioceptive training on foot posture, lower limb alignment, and knee adduction moment in patients with degenerative knee osteoarthritis: a randomized controlled trial. *Journal of Physical Therapy Science* 2015;**27**(2):371-4. [DOI: 10.1589/jpts.272.371]

Ciolac 2011 *{published data only}*

Ciolac EG, Greve J. Muscle strength and exercise intensity adaptation to resistance training in older women with knee osteoarthritis and total knee arthroplasty. *Clinics (Sao Paulo, Brazil)* 2011;**66**(12):2079-84.

Crotty 2009 *{published data only}*

Crotty M, Prendergast J, Battersby MW, Rowett D, Graves SE, Leach G, et al. Self-management and peer support among people with arthritis on a hospital joint replacement waiting list: a randomised controlled trial. *Osteoarthritis and Cartilage* 2009;**17**(11):1428-33.

Ctri/2014/01/004270 *{published data only}*

Ctri/2014/01/004270. Trial of 2 conservative therapies for knee arthritis patients [Comparative impact of two packages of non pharmacological interventions on quality of life of knee osteoarthritis patients reporting at a tertiary care institution - a randomized control trial]. Ctri/2014/01/004270 2014.

Ctri/2015/01/005485 *{published data only}*

Ctri/2015/01/005485. To check the effect of strengthening exercises and manual therapy in subjects with knee osteoarthritis. Ctri/2015/01/005485 2015.

Ctri/2018/03/012620 *{published data only}*

Ctri/2018/03/012620. Long term benefits of specific physiotherapy for patients with osteoarthritis of knee [Long term efficacy of Mulligan mobilization with movement on pain and functional status in patients with osteoarthritis of knee joint. A randomized clinical trial]. Ctri/2018/03/012620 2018.

Ctri/2019/01/017049 *{published data only}*

Ctri/2019/01/017049. To compare the effect of routienly done strengthening exercises versus joint position sence trining in patients with early osteoarthritis [Effectiveness of proprioceptive training Vversus conventional exercises on postural sway in patients with early knee osteoarthritis]. Ctri/2019/01/017049 2019.

Ctri/2019/09/021331 *{published data only}*

Ctri/2019/09/021331. Using a type of machine with current to reduce pain in patients with knee pain effects of wireless electric stimulation on pain and disability in patient with osteoarthritis knee: a randomized controlled trial. Ctri/2019/09/021331 2019.

Ctri/2020/03/023737 *{published data only}*

Ctri/2020/03/023737. Effectiveness of specific physiotherapy evaluation and treatment in decreasing pain and improving joint function among individuals with mechanical knee pain [Effectiveness of Mckenzie method of evaluation and treatment for mechanical disorders of knee joint]. Ctri/2020/03/023737 2020.

Ctri/2020/03/024311 *{published data only}*

Ctri/2020/03/024311. Knee School for patients suffering from knee osteoarthritis [Pilot the concept of knee school in outpatient department settings at PGIMER, Chandigarh for improving the quality of life of patients suffering from knee osteoarthritis through a randomised controlled trial]. Ctri/2020/03/024311 2020.

Ctri/2020/06/025912 *{published data only}*

Ctri/2020/06/025912. Application of selected yoga asana on treatment of osteoarthritis in knee [Comparison between effects of select yoga asana with exercise therapy in osteoarthritis of knee joint]. Ctri/2020/06/025912 2020.

Ctri/2020/06/025973 *{published data only}*

Ctri/2020/06/025973. Effect of core stability and hip strengthening exercise to reduce pain, improve function and to enhance daily life activity in patients with osteoarthritis of knee. Ctri/2020/06/025973 2020.

Ctri/2020/07/026397 *{published data only}*

Ctri/2020/07/026397. Hip strengthening exercise in knee osteoarthritis [Effectiveness of hip strengthening exercise in patients with knee osteoarthritis]. Ctri/2020/07/026397 2020.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Ctri/2020/11/028915 *{published data only}*

Ctri/2020/11/028915. Effect of active vs passive exercise in knee pain [Effect of active vs passive movement with mobilization on pain & range of motion in patients with chronic osteoarthritis knee]. Ctri/2020/11/028915 2020.

Ctri/2020/11/029138 *{published data only}*

Ctri/2020/11/029138. There is scarcity of literature as whether isokinetic exercise or electromyography biofeedback along with conventional physiotherapy is more effective to strengthen vastus medialis oblique in osteoarthritis knee to reduce pain and increase range of motion. Ctri/2020/11/029138 2020.

Ctri/2021/01/030519 *{published data only}*

Ctri/2021/01/030519. Effect of combined kinetic chain exercises on vastus medialis oblique muscle thickness measured with ultrasound in patients with knee osteoarthrits [Effect of open kinetic chain and closed kinetic chain exercises on vastus medialis oblique muscle volume and]. Ctri/2021/01/030519 2021.

Ctri/2021/04/033203 *{published data only}*

Ctri/2021/04/033203. Effect of muscle strengthening on walking angle in osteoarthritis knee [Effect of core muscle and antigravity muscle strengthening on foot progression angle in patients with osteoarthritis knee: a randomized control trial]. Ctri/2021/04/033203 2021.

Ctri/2021/09/036724 *{published data only}*

Ctri/2021/09/036724. Effects of pilates and egoscue exercises in subjects with knee osteoarthritis [Comparative effective of pilates and egoscue exercises on pain, function, proprioception and peak muscle torque in subjects with knee osteoarthritis- a randomized parallel group]. Ctri/2021/09/036724 2021.

Ctri/2022/03/041074 *{published data only}*

Ctri/2022/03/041074. Effect of weight resistance exercises versus band exercises on knee pain [Effect of resistance training using weight cuff versus theraband resistance training of knee extensors on knee Q angle and gait in knee osteoarthritis. A randomized control trail]. Ctri/2022/03/041074 2022.

Ctri/2022/05/042506 *{published data only}*

Ctri/2022/05/042506. Effect of alternate heat and cold treatment by water and device to reduce pain and improve knee movement and physical activities in people with knee pain due to osteoarthritis. Ctri/2022/05/042506 2022.

Ctri/2022/10/046835 *{published data only}*

Ctri/2022/10/046835. Effects of activity modification with weight bearing exercises on pain and quality of life in patients with knee osteoarthritis â?? A Pilot Randomized Controlled Trial. Ctri/2022/10/046835 2022.

Ctri/2022/12/048269 *{published data only}*

Ctri/2022/12/048269. To know the effect of Unani massage and exercise in knee osteoarthritis [Clinical evaluation of efficacy of Dalk with Roghan e Ahmar jadeed and Riyazat (knee strengthening Eexercise) in knee osteoarthritis]. Ctri/2022/12/048269 2022.

Ctri/2023/01/049164 *{published data only}*

Ctri/2023/01/049164. Efficacy of floss band technique along with conventional approach in management of knee osteoarthritis [Ctri/2023/01/049164]. Ctri/2023/01/049164 2023.

Ctri/2023/04/051550 *{published data only}*

Ctri/2023/04/051550. A clinical trial to find out the effect of resisted exercises in improving walking pattern in oa knee subjects [Efficacy of progressive resisted training of lower extremity muscles and core muscles in improving early stage osteoarthritis of knee]. Ctri/2023/04/051550 2023.

Ctri/2023/04/051625 *{published data only}*

Ctri/2023/04/051625. Effectiveness of a new exercise (wall touch single limb stance) in patients with osteoarthritis knee. Ctri/2023/04/051625 2023.

Ctri/2023/05/053101 *{published data only}*

Ctri/2023/05/053101. Effect of pain education and cognition targeted exercises on pain, disability and quality of life in knee osteoarthritis patients. Ctri/2023/05/053101 2023.

Ctri/2023/06/053585 *{published data only}*

Ctri/2023/06/053585. Cycling for muscle strength in Knee Osteoarthritis [Effect of stationary cycling on quality of life, pain & isokinetic strength in patients with knee osteoarthritis]. Ctri/2023/06/053585 2023.

Ctri/2023/07/054805 *{published data only}*

Ctri/2023/07/054805. Effectivness of core exercise training in patients diagnosed with knee osteoarthritis - a randomized controlled trial. Ctri/2023/07/054805 2023.

Ctri/2023/07/054846 *{published data only}*

To increase power of muscles to decline the pain and correct balance with virtual reality exercise in osteoarthritis patients of knee. Cochrane Database of Systematic Reviews 2023.

Ctri/2023/07/055413 *{published data only}*

Ctri/2023/07/055413. The effectiveness of structured exercise based rehabilitation program in patients with early osteoarthritis knee. Ctri/2023/07/055413 2023.

Ctri/2023/07/055542 *{published data only}*

Ctri/2023/07/055542. Effect of 6-week exercise programme on gait patterns, muscle activation and balance in patients with osteoarthritis of knee. Ctri/2023/07/055542 2023.

Ctri/2023/08/056055 *{published data only}*

Ctri/2023/08/056055. Impact of polyvagal and knee strengthning execises on pain and quality of life in female paitents with knee osteoarthritis. Ctri/2023/08/056055 2023.

Ctri/2023/08/056749 *{published data only}*

Ctri/2023/08/056749. A study to compare the effectiveness of nirgundi patra upanaha and koladi upanaha in janusandhigata vata (knee osteoarthritis). Ctri/2023/08/056749 2023.

Ctri/2023/08/056896 *{published data only}*

Ctri/2023/08/056896. Physiotherapy for knee osteoarthritis. Ctri/2023/08/056896 2023.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Ctri/2023/08/057102 *{published data only}*

Ctri/2023/08/057102. Effectiveness of Better Management for DegenerativeJoint Disease for Knee Program on pain and functions in patients with knee osteoarthritis: a randomized controlled trial. Ctri/2023/08/057102 2023.

Ctri/2023/08/057124 *{published data only}*

Ctri/2023/08/057124. Effects of exercise on arthritic knee cartilage [Impact of exercises on cartilage health, physical performance and quality of life in subjects with knee osteoarthritis]. Ctri/2023/08/057124 2023.

Ctri/2023/10/058883 *{published data only}*

Ctri/2023/10/058883. Evaluating the impact of fast paced and slow paced exercises n strength in patients with knee osteoarthritis. Ctri/2023/10/058883 2023.

Cui 2023 *{published data only}*

Cui D, Lei Y. Diet and exercise and knee pain in patients with osteoarthritis and overweight or obesity. *JAMA* 2023;**329**(15):1317-8. [DOI: 10.1001/jama.2023.2533]

Dantas 2023 *{published data only}*

Dantas G, Sacco IC, Dos Santos AF, Watari R, Matias AB, Serrao P, et al. Effects of a foot-ankle strengthening programme on clinical aspects and gait biomechanics in people with knee osteoarthritis: protocol for a randomised controlled trial. *BMJ Open* 2020;**10**(9):e039279. [DOI: 10.1136/bmjopen-2020-039279]

Dantas GA, Sacco IC, Ferrari AV, Matias AB, Watari R, Oliveira LV, et al. Effects of a foot-ankle muscle strengthening program on pain and function in individuals with knee osteoarthritis: a randomized controlled trial. *Brazilian Journal of Physical Therapy* 2023;**27**(4):100531. [DOI: 10.1016/j.bjpt.2023.100531]

Darlow 2023 *{published data only}*

Darlow B, Brown M, Stanley J, Abbott JH, Briggs AM, Clark J, et al. Reducing the burden of knee osteoarthritis through community pharmacy: protocol for a randomised controlled trial of the Knee Care for Arthritis through Pharmacy Service. *Musculoskeletal Care* 2023;**21**(4):1053-67. [DOI: 10.1002/msc.1785]

De Oliveira 2020 *{published data only}*

De Oliveira Silva D, Pazzinatto MF, Crossley KM, Azevedo FM, Barton CJ. Novel stepped care approach to provide education and exercise therapy for patellofemoral pain: feasibility study. *Journal of Medical Internet Research* 2020;**22(7)**:e18584. [DOI: 10.2196/18584]

De Olivera 2017 *{published data only}*

De Oliveira Gondim IT, Da Costa Torres A, Baudel de Lacerda AT, DQ Kuhni Fernandes, Costa do Couto M, Carneiro Pedrosa MA. Effects of a therapeutic exercises program associated with pompage technique on pain, balance and strength in elderly women with knee osteoarthritis. *Fisioterapia em Movimento* 2017;**30**:S11p. [DOI: 10.1590/1980-5918.030.S01.AO01]

Deyle 2000 *{published data only}*

Deyle GD, Henderson NE, Matekel RL, Ryder MG, Garber MB, Allison SC. Effectiveness of manual physical therapy and exercise in osteoarthritis of the knee. *Annals of Internal Medicine* 2000;**132**(3):173-81.

Deyle 2005 *{published data only}*

Deyle GD, Allison SC, Matekel RL, Ryder MG, Stang JM, Gohdes DD, et al. Physical therapy treatment effectiveness for osteoarthritis of the knee: a randomized comparison of supervised clinical exercise and manual therapy procedures versus a home exercise program. *Physical Therapy* 2005;**12**:1301-17.

Dias 2003 *{published data only}*

Dias RC, Ramos LR, Dias JM. Impact of an exercise and walking protocol on the quality of life of the elderly with osteoarthritis of the knee. *Physiotherapy Research International* 2003;**8**:121-30.

Diracoglu 2005 *{published data only}*

Diracoglu D, Aydin R, Baskent A, Celik A, Diracoglu D, Aydin R, et al. Effects of kinesthesia and balance exercises in knee osteoarthritis. *Journal of Clinical Rheumatology* 2005;**11**:303-10.

Doi 2008 *{published data only}*

Doi T, Akai M, Fujino K, Iwaya T, Kurosawa H, Hayashi K, et al. Effect of home exercise of quadriceps on knee osteoarthritis compared with nonsteroidal antiinflammatory drugs: a randomized controlled trial. *American Journal of Physical Medicine & Rehabilitation* 2008;**87**(4):258-69.

Drks00009257 *{published data only}*

Drks00009257. Effectiveness and efficiency of a supervised strength-machine based hip and knee training for patients with hip or knee osteoarthritis: a controlled study in the context of health service research. Drks00009257 2015.

Drks00024160 *{published data only}*

Drks00024160. Effectiveness of an additional web-based app-intervention to traditional physiotherapy in patients with musculoskeletal pain [Effectiveness of an additional web-based app-intervention to traditional physiotherapy in patients with musculoskeletal pain - Heal]. Drks00024160 2021.

Drks00025033 *{published data only}*

Drks00025033. Individualized exercise promotion program for people with multiple manifested risk factors to increase physical exercise in the long term (MultiPill-Exercise). Drks00025033 2021.

Duman 2012 *{published data only}*

Duman I, Taskaynatan MA, Mohur H, Tan AK. Assessment of the impact of proprioceptive exercises on balance and proprioception in patients with advanced knee osteoarthritis. *Rheumatology International* 2012;**32**(12):3793-8.

Durmus 2007 *{published data only}*

Durmus D, Alayli G, Canturk F. Effects of quadriceps electrical stimulation program on clinical parameters in the patients with knee osteoarthritis. *Clinical Rheumatology* 2007;**26**:674-8.

Durmus 2012 *{published data only}*

Durmus D, Alayli G, Bayrak IK, Canturk F. Assessment of the effect of glucosamine sulfate and exercise on knee cartilage

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using magnetic resonance imaging in patients with knee osteoarthritis: a randomized controlled clinical trial. *Journal of Back and Musculoskeletal Rehabilitation* 2012;**25**(4):275-84.

Ebnezar 2012a *{published data only}*

Ebnezar J, Nagarathna R, Yogitha B, Nagendra HR. Effects of an integrated approach of hatha yoga therapy on functional disability, pain, and flexibility in osteoarthritis of the knee joint: a randomized controlled study. *Journal of Alternative and Complementary Medicine (New York, N.Y.)* 2012;**18**(5):463-72.

Ebnezar 2012b *{published data only}*

Ebnezar J, Nagarathna R, Yogitha B, Nagendra HR. Effect of integrated yoga therapy on pain, morning stiffness and anxiety in osteoarthritis of the knee joint: a randomized control study. *International Journal of Yoga* 2012;**5**(1):28-36.

Evcik 2002 *{published data only}*

Evcik D, Sonel B. Effectiveness of a home-based exercise therapy and walking program on osteoarthritis of the knee. *Rheumatology International* 2002;**22**:103-6.

Evgeniadis 2008 *{published data only}*

Evgeniadis G, Beneka A, Malliou P, Mavromoustakos S, Godolias G. Effects of pre- or postoperative therapeutic exercise on the quality of life, before and after total knee arthroplasty for osteoarthritis. *Journal of Back and Musculoskeletal Rehabilitation* 2008;**21**(3):161-9.

Eyigor 2004 *{published data only}*

Eyigor S, Hepguler S, Capaci K. A comparison of muscle training methods in patients with knee osteoarthritis. *Clinical Rheumatology* 2004;**23**:109-15.

Falck 2018 *{published data only}*

Falck RS, Best JR, Li LC, Chan PC, Feehan LM, Liu-Ambrose T. Can we improve cognitive function among adults with osteoarthritis by increasing moderate-to-vigorous physical activity and reducing sedentary behaviour? Secondary analysis of the MONITOR-OA study. *BMC Musculoskeletal Disorders* 2018;**19**(1):447. [DOI: 10.1186/s12891-018-2369-z]

Fan 2023 *{published data only}*

Fan T, Zeng M, Zhu Z. Diet and exercise and knee pain in patients with osteoarthritis and overweight or obesity. *JAMA* 2023;**329**(15):1316-7. [DOI: 10.1001/jama.2023.2536]

Farr 2010 *{published data only}*

Farr JN, Going SB, McKnight PE, Kasle S, Cussler EC, Cornett M. Progressive resistance training improves overall physical activity levels in patients with early osteoarthritis of the knee: a randomized controlled trial. *Physical Therapy* 2010;**90**(3):356-66.

Feinglass 2012 *{published data only}*

Feinglass J, Song J, Semanik P, Lee J, Manheim L, Dunlop D, et al. Association of functional status with changes in physical activity: insights from a behavioral intervention for participants with arthritis. *Archives of Physical Medicine and Rehabilitation* 2012;**93**(1):172-5.

Feng 2023 *{published data only}*

Feng Y, Wu Y, Liu H, Bao T, Wang C, Wang Z, et al. Effect of the telemedicine-supported multicomponent exercise therapy in patients with knee osteoarthritis: study protocol for a randomized controlled trial. *Trials* 2023;**24**(1):(no pagination). [DOI: 10.1186/s13063-023-07749-4]

Fitzgerald 2011 *{published data only}*

Fitzgerald GK, Piva SR, Gil AB, Wisniewski SR, Oddis CV, Irrgang JJ. Agility and perturbation training techniques in exercise therapy for reducing pain and improving function in people with knee osteoarthritis: a randomized clinical trial. *Physical Therapy* 2011;**91**(4):452-69.

Focht 2022 *{published data only}*

Focht BC, Rejeski WJ, Hackshaw K, Ambrosius WT, Groessl E, Chaplow ZL, et al. The collaborative lifestyle intervention program in knee osteoarthritis patients (CLIP-OA) trial: design and methods. *Contemporary Clinical Trials* 2022;**115**:(no pagination). [DOI: 10.1016/j.cct.2022.106730]

Forestier 2010 *{published data only}*

Forestier R, Desfour H, Tessier JM, Francon A, Foote AM, Genty C, et al. Spa therapy in the treatment of knee osteoarthritis: a large randomised multicentre trial. *Annals of the Rheumatic Diseases* 2010;**69**(4):660-5.

Foroughi 2011 *{published data only}*

Foroughi N, Smith RM, Lange AK, Baker MK, Fiatarone Singh MA, Vanwanseele B. Lower limb muscle strengthening does not change frontal plane moments in women with knee osteoarthritis: a randomized controlled trial. *Clinical Biomechanics (Bristol, Avon)* 2011;**26**(2):167-74.

Foroughi N, Smith RM, Lange AK, Singh MA, Vanwanseele B. Progressive resistance training and dynamic alignment in osteoarthritis: a single-blind randomised controlled trial. *Clinical Biomechanics (Bristol, Avon)* 2011;**26**(1):71-7.

Foster 2007 *{published data only}*

Foster NE, Thomas E, Barlas P, Hill JC, Young J, Mason E, et al. Acupuncture as an adjunct to exercise based physiotherapy for osteoarthritis of the knee: randomised controlled trial. *BMJ (Clinical Research Ed.)* 2007;**335**(7617):436.

Foster 2023 *{published data only}*

Foster NE, Nicholls E, Holden MA, Healey EL, Hay EM. Improving the effectiveness of exercise therapy for adults with knee osteoarthritis: a pragmatic randomized controlled trial (BEEP trial). *Archives of Rehabilitation Research and Clinical Translation* 2023;**5(2)**:(no pagination). [DOI: 10.1016/j.arrct.2023.100266]

Foster NE, Nicholls E, Holden MA, Healey EL, Tooth S, Hay EM. Improving the effectiveness of exercise therapy for older adults with knee pain: a pragmatic randomised controlled trial (the beep trial). *Physiotherapy (United Kingdom)* 2015;**1)**:eS404-eS405. [DOI: 10.1016/j.physio.2015.03.633]

Quicke JG, Foster NE, Croft PR, Ogollah RO, Holden MA. Change in physical activity level and clinical outcomes in older adults with knee pain: a secondary analysis from a randomised

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controlled trial. *BMC Musculoskeletal Disorders* 2018;**19**(1):(no pagination). [DOI: 10.1186/s12891-018-1968-z]

Gaal 2008 *{published data only}*

Gaal J, Varga J, Szekanecz Z, Kurko J, Ficzere A, Bodolay E, et al. Balneotherapy in elderly patients: effect on pain from degenerative knee and spine conditions and on quality of life. *Israel Medical Association Journal* 2008;**10**(5):365-9.

Gaudreault 2011 *{published data only}*

Gaudreault N, Mezghani N, Turcot K, Hagemeister N, Boivin K, de Guise JA. Effects of physiotherapy treatment on knee osteoarthritis gait data using principal component analysis. *Clinical Biomechanics (Bristol, Avon)* 2011;**26**(3):284-91.

Ghandali 2017 *{published data only}*

Ghandali E, Moghadam ST, Hadian MR, Olyaei G, Jalaie S, Sajjadi E. The effect of Tai Chi exercises on postural stability and control in older patients with knee osteoarthritis. *Journal of Bodywork and Movement Therapies* 2017;**21**(3):594-8. [DOI: 10.1016/j.jbmt.2016.09.001]

Gill 2009 *{published data only}*

Gill SD, McBurney H, Schulz DL. Land-based versus pool-based exercise for people awaiting joint replacement surgery of the hip or knee: results of a randomized controlled trial. *Archives of Physical Medicine and Rehabilitation* 2009;**90**(3):388-94.

Gohir 2020 *{published data only}*

Gohir S, Abhishek A, Kelly A, Valdes AM. A randomised controlled trial evaluating the efficacy of internet-based exercises aimed at treating knee osteoarthritis (ibeat-OA). *Arthritis & Rheumatology* 2020;**72 Suppl 10**:2997-9. [DOI: 10.1002/art.41538]

Gohir SA, Greenhaff P, Abhishek A, Valdes AM. Evaluating the efficacy of Internet-Based Exercise programme Aimed at Treating knee Osteoarthritis (iBEAT-OA) in the community: a study protocol for a randomised controlled trial. *BMJ Open* 2019;**9**(10):(no pagination). [DOI: 10.1136/bmjopen-2019-030564]

Nct03545048. Effects of web-based exercises on the population with knee arthritis [Randomised controlled trial evaluating the efficacy of web-based exercises on knee pain due to knee osteoarthritis]. Nct03545048 2018.

Green 1993 *{published data only}*

Green J, McKenna F, Redfern EJ, Chamberlain MA. Home exercises are as effective as outpatient hydrotherapy for osteoarthritis of the hip. *British Journal of Rheumatology* 1993;**32**:812-5.

Gremion 2009 *{published data only}*

Gremion G, Gaillard D, Leyvraz PF, Jolles BM. Effect of biomagnetic therapy versus physiotherapy for treatment of knee osteoarthritis: a randomized controlled trial. *Journal of Rehabilitation Medicine [Stiftelsen Rehabiliteringsinformation]* 2009;**41**(13):1090-5.

Griessmann 2014 *{published data only}*

Griessmann S, Dannhauer T, Hudelmaier M, Ruhdorfer A, Wirth W, Himmer M, et al. Effects of strength and endurance training intervention on thigh intermuscular and subcutaneous adipose tissue in sarkopenic women. *Osteoarthritis and Cartilage* 2014;**1**:S317.

Gustavsson 2023 *{published data only}*

Gustavsson L, Mohaddes M, Samsson K, Beischer S. No major difference in perceived quality of care in patients with hip or knee osteoarthritis assessed in a physical therapy-led triage compared with standard care: a randomized controlled trial. *BMC Musculoskeletal Disorders* 2023;**24**(1):(no pagination). [DOI: 10.1186/s12891-023-06659-5]

Hammami 2022 *{published data only}*

Hammami N, Jdidi H, Khezami MA, Ghidaoui L, Talbi A, Hannachi C, et al. Isokinetic strengthening and neuromuscular electrical stimulation protocol impact on physical performances, functional status and quality of life in knee osteoarthritis overweight/obese women. *Knee* 2022;**39**:106-15. [DOI: 10.1016/j.knee.2022.09.004]

Haslam 2001 *{published data only}*

Haslam R. A comparison of acupuncture with advice and exercises on the symptomatic treatment of osteoarthritis of the hip-a randomised controlled trial. *Acupuncture Medicine* 2001;**19**:19-26.

Helmark 2010 *{published data only}*

Helmark IC, Mikkelsen UR, Borglum J, Rothe A, Petersen MC, Andersen O, et al. Exercise increases interleukin-10 levels both intraarticularly and peri-synovially in patients with knee osteoarthritis: a randomized controlled trial. *Arthritis Research & Therapy* 2010;**12**(4):R126.

Helmark 2012 *{published data only}*

Helmark IC, Petersen MC, Christensen HE, Kjaer M, Langberg H. Moderate loading of the human osteoarthritic knee joint leads to lowering of intraarticular cartilage oligomeric matrix protein. *Rheumatology International* 2012;**32**(4):1009-14.

Hinman 2007 *{published data only}*

Hinman RS, Heywood SE, Day AR. Aquatic physical therapy for hip and knee osteoarthritis: results of a single-blind randomized controlled trial. *Physical Therapy* 2007;**87**:32-43.

Hiyama 2012 *{published data only}*

Hiyama Y, Yamada M, Kitagawa A, Tei N, Okada S. A four-week walking exercise programme in patients with knee osteoarthritis improves the ability of dual-task performance: a randomized controlled trial. *Clinical Rehabilitation* 2012;**26**(5):403-12.

Hoeksma 2004 *{published data only}*

Hoeksma HL, Dekker J, Ronday HK, Heering A, Van der Lubbe N, Vel C, et al. Comparison of manual therapy and exercise therapy in osteoarthritis of the hip: a randomized clinical trial. *Arthritis and Rheumatism* 2004;**51**:722-9.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Holsgaard-Larsen 2017 *{published data only}*

Clausen B, Holsgaard-Larsen A, Sondergaard J, Christensen R, Andriacchi TP, Roos EM. The effect on knee-joint load of instruction in analgesic use compared with neuromuscular exercise in patients with knee osteoarthritis: study protocol for a randomized, single-blind, controlled trial (the EXERPHARMA trial). *Trials* 2014;**15**(1):(no pagination). [DOI: 10.1186/1745-6215-15-444]

Holsgaard-Larsen A, Christensen R, Clausen B, Asondergaard J, Andriacchi TP, Roos EM. The effect of neuromuscular exercise compared with instruction in analgesic use on knee function in patients with early knee osteoarthritis-one year outcome of the exerpharma randomized trial. *Osteoarthritis and Cartilage* 2017;**25 Suppl 1**:S49.

Holsgaard-Larsen A, Clausen B, Sondergaard J, Christensen R, Andriacchi TP, Roos E. The effect on knee-joint load of instruction in analgesic use compared with neuromuscular exercise in patients with early knee osteoarthritisea randomized, singleblind, controlled trial. *Osteoarthritis and Cartilage* 2016;**1**:S497-S498.

Holsgaard-Larsen A, Clausen B, Sondergaard J, Christensen R, Andriacchi TP, Roos EM. The effect of instruction in analgesic use compared with neuromuscular exercise on knee-joint load in patients with knee osteoarthritis: a randomized, single-blind, controlled trial. *Osteoarthritis and Cartilage* 2017;**25(4)**:470-80. [DOI: 10.1016/j.joca.2016.10.022]

Huang 2018 *{published data only}*

Huang L, Guo B, Xu F, Zhao J. Effects of quadriceps functional exercise with isometric contraction in the treatment of knee osteoarthritis. *International Journal of Rheumatic Diseases* 2018;**21**(5):952-9. [DOI: 10.1111/1756-185X.13082]

Huang 2021 *{published data only}*

Huang CC, Wang HH, Chen KC, Yang KJ, Chang LY, Shiang TY, et al. Effects of a dynamic combined training on impulse response for middle-aged and elderly patients with osteoporosis and knee osteoarthritis: a randomized control trial. *Aging Clinical and Experimental Research* 2021;**33**(1):115-23. [DOI: 10.1007/s40520-020-01508-0]

Hunt 2013 *{published data only}*

Hunt MA, Pollock CL, Kraus VB, Saxne T, Peters S, Huebner JL, et al. Relationships amongst osteoarthritis biomarkers, dynamic knee joint load, and exercise: results from a randomized controlled pilot study. *BMC Musculoskeletal Disorders* 2013;**14**:115. [DOI: 10.1186/1471-2474-14-115]

Hunter 2023 *{published data only}*

Bowden JL, Hunter DJ, Hinman RS, Egerton T, Briggs AM, Bunker SJ, et al. Optimising management of knee osteoarthritis in primary care (partner): a cluster randomised controlled trial. *Osteoarthritis and Cartilage* 2023;**31 Suppl 1**:S29-S30. [DOI: 10.1016/j.joca.2023.01.519]

Hunter DJ, Bowden JL, Hinman RS, Egerton T, Briggs AM, Bunker SJ, et al, Team Partner Study. Effectiveness of a new service delivery model for management of knee osteoarthritis in primary care: a cluster randomized controlled trial. *Arthritis Care & Research* 2023;**75**(6):1320-2. [DOI: 10.1002/acr.25037]

Hurley 1998 *{published data only}*

Hurley MV, Scott DL. Improvements in quadriceps sensorimotor function and disability of patients with knee osteoarthritis following a clinically practicable exercise regime. *British Journal of Rheumatology* 1998;**37**:1181-7.

Hurley 2020 *{published data only}*

Hurley DA, Jeffares I, Hall AM, Keogh A, Toomey E, McArdle D, et al. Feasibility cluster randomised controlled trial evaluating a theory-driven group-based complex intervention versus usual physiotherapy to support self-management of osteoarthritis and low back pain (SOLAS). *Trials* 2020;**21**(1):(no pagination). [DOI: 10.1186/s13063-020-04671-x]

Irct20100811004549n10 *{published data only}*

Irct20100811004549n10. The effect of exercise on the balance of patients with knee osteoarthritis [Comparison of the effect of anaerobic strengthening exercises and aerobic exercise (walking) on the balance of patients with knee osteoarthritis]. Irct20100811004549n10 2018.

Irct20101017004952n6 *{published data only}*

Irct20101017004952n6. The effects of MWM techniques on the center of pressure sway changes in knee osteoarthritis. Irct20101017004952n6 2019.

Irct20101228005486n6 *{published data only}*

Irct20101228005486n6. Therapeutic exercise in knee osteoarthritis. Irct20101228005486n6 2019.

Irct20131225015932n18 *{published data only}*

Irct20131225015932n18. The evaluation of the priming effect of transcranial direct current stimulation over primary motor cortex and primary somatosensory cortex on the effect of exercise training on pain, function, balance, kinesiophobia and quality of life in patients with knee osteoarthritis. Irct20131225015932n18 2023.

Irct2014093019346n1 *{published data only}*

Irct2014093019346n1. Effect of quadriceps muscle strengthening exercises on the fiber angle ratio of vastus medialis obliquus to vastus lateralis in patient wit patellofemoral pain syndrome. Irct2014093019346n1 2014.

Irct2015010120526n1 *{published data only}*

Irct2015010120526n1. Self-management program for patients with osteoarthritis. Irct2015010120526n1 2015.

Irct20150519022320n33 *{published data only}*

Irct20150519022320n33. Comparison of effect of Swedish massage and hip strengthening exercises on pain, range of motion and daily life function in the elderly with knee osteoarthritis. Irct20150519022320n33 2023.

Irct20150602022539n11 *{published data only}*

Irct20150602022539n11. Effect of whole body vibration training, on the lower limb muscles’ electromyography and

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

timing of activation in patients with knee osteoarthritis during functional activity. Irct20150602022539n11 2021.

Irct2015081123593n1 *{published data only}*

Irct2015081123593n1. The effect of patellar taping and patellar bracing on pain and fuction in females with knee pain [Comparison of patellar taping versus patellar bracing with exercise therapy on pain and function in females with patellofemoral pain syndrome]. Irct2015081123593n1 2015.

Irct201604041552n6 *{published data only}*

Irct201604041552n6. Comparison of the effects of routine physiotherapy and Pilates exercise on pain, function and quality of life in women with knee osteoarthritis. Irct201604041552n6 2016.

Irct2016061428406n2 *{published data only}*

Irct2016061428406n2. Effects of resistance exercise training on postmenopausal women with knee osteoarthritis. Irct2016061428406n2 2016.

Irct2016070727359n3 *{published data only}*

Irct2016070727359n3. The effect of isokinetic exercise in patieuts with knee osteoarthritis [Evaluation of the effect of isokinetic exercise with conventional physiotherapy on power in patieuts with knee osteoarthritis]. Irct2016070727359n3 2017.

Irct201608044641n12 *{published data only}*

Effectiveness of extra corporal shock wave therapy in the treatment of knee arthritis. Irct201608044641n12 2017.

Irct2016120415932n12 *{published data only}*

Irct2016120415932n12. The effect of core training using unstable sitting in patients with patrllofemoral pain syndrome [The effect of postural control training using unstable sitting on knee joint pain and function in patients with patellofemoral pain syndrome]. Irct2016120415932n12 2017.

Irct20161208031300n2 *{published data only}*

Irct20161208031300n2. Reviews of the effectiveness of the learning-based mobile application (mobile app) to improve physical function of patients with osteoarthritis of the knee ladies. Irct20161208031300n2 2019.

Irct2017013032251n2 *{published data only}*

Irct2017013032251n2. Knee osteoarthritis rehabilitation [Effect of tele rehabilitation compared with rehabilitation at the clinic in patients with osteoarthritis of the knee]. Irct2017013032251n2 2017.

Irct20180226038870n2 *{published data only}*

Irct20180226038870n2. Evaluation if weight bearing exercises on patients with knee osteoarthritis treated by stromal cells [Evaluation of weight bearing exercises on pain and physical function of patients with knee osteoarthritis treated by implantation of autologous bone marrow]. Irct20180226038870n2 2021.

Irct20180314039092n1 *{published data only}*

Irct20180314039092n1. The effect of Tai Chi exercise and glucosamine supplementation on prevention of fall in the elderly women whit knee osteoarthritis [Comparison the effects of Tai Chi exercise and glucosamine supplementation on selected physical factors effective on balance]. Irct20180314039092n1 2018.

Irct20180627040251n5 *{published data only}*

Irct20180627040251n5. Comparison between MSI and routine physiotherapy treatments in knee pain patients [The comparison of treatment using Movement System Impairment method with routine physical therapy in improving pain and function in patents with knee pain: a randomized clinical trial]. Irct20180627040251n5 2023.

Irct20181017041371n1 *{published data only}*

Irct20181017041371n1. Magnet therapy,acupuncture and isometric exercise in treatment knee osteoarthritis [Comparing the effectiveness of magnet therapy,acupuncture and isometric exercise in patients with chronic and moderate knee osteoarthritis]. Irct20181017041371n1 2019.

Irct20190116042386n1 *{published data only}*

Irct20190116042386n1. Effectiveness of a home-based exercise program via the mobile application (Varzeshyar) on pain and disability of knee osteoarthritis [Comparing the effectiveness of a home-based exercise program via the mobile application (Varzeshyar) and a routine home-bas]. Irct20190116042386n1 2019.

Irct20190222042801n1 *{published data only}*

Irct20190222042801n1. The effect of physiotherapy on knee paint [The effect of treatment based on movement system impairment classification theory in people with knee pain (a randomized control trial)]. Irct20190222042801n1 2020.

Irct20200203046360n1 *{published data only}*

Irct20200203046360n1. Effect of whole body vibration on muscle inhibition [Study effect of whole body vibration on arthrogenic muscle inhibition of quadriceps muscle in knee osteoarthritis patients]. Irct20200203046360n1 2020.

Irct20200825048523n2 *{published data only}*

Irct20200825048523n2. The effect of strengthening exercises on anterior knee pain syndrome [The efficacy of of purposeful strengthening exercises based on biomechanical defects on ultrasonographic parameters and kinematic variables in people with Patellofemoral Pain Syndrome]. Irct20200825048523n2 2022.

Irct20201031049207n2 *{published data only}*

Patello femoral pain syndrome [Effects of Mulligan pain release phenomenon technique in the management of patellofemoral pain syndrome; a randomized controlled trial]. Irct20201031049207n2 2020.

Irct20210121050099n1 *{published data only}*

Irct20210121050099n1. Effects of Mulligan and muscle energy techniques on knee osteoarthritis [Comparison of effects of Mulligan techniques and muscle energy technique (MET) on pain and function in knee osteoarthritis]. Irct20210121050099n1 2021.

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Irct20210321050752n4 *{published data only}*

Irct20210321050752n4. Effects of virtual reality and routine physical therapy on balance and postural control [Effects of virtual reality and routine physical therapy on balance and postural control in adulthood population with risk of fall]. Irct20210321050752n4 2023.

Irct20210417050997n1 *{published data only}*

Irct20210417050997n1. The effect of neruomuscular exercises in patients with knee osteoarthritis [The effect of neruomuscular exercises in combination with cognitive-motivational and mindfulness interventions on pain, function, neurofeedback factores, kinesiophobia, and self-rep]. Irct20210417050997n1 2021.

Irct20210815052194n1 *{published data only}*

Irct20210815052194n1. Effect of stretching in patient with knee osteoarthritis [Comparative effects of Mulligan traction straight leg raise versus muscle energy technique on pain intensity and hamstring tightness in patient with knee osteoarthritis]. Irct20210815052194n1 2021.

Irct20220216054038n1 *{published data only}*

Irct20220216054038n1. The effect of tele-rehabilitation nursing on knee osteoarthritis [The effect of tele-rehabilitation nursing on the pain, joint stiffness, physical functions of knees and Health locus of control (HLC) among persons with knee osteoarthritis]. Irct20220216054038n1 2022.

Irct20220830055831n1 *{published data only}*

Irct20220830055831n1. The effect of exercise therapy and drug in relieving pain and disability caused by osteoarthritis of the knee. Irct20220830055831n1 2022.

Irct20221115056509n1 *{published data only}*

Irct20221115056509n1. The effect of high power laser and quadriceps strengthening exercises on osteoarthritis [Comparison of the effect of high-level laser and quadriceps muscle strengthening exercises using biofeedback on pain and function in patients with knee osteoarthritis]. Irct20221115056509n1 2023.

Irct20230101057015n1 *{published data only}*

Irct20230101057015n1. The effect of body balance exercises on knee osteoarthritis of elderly women [The effect of eight-weeks of body balance exercises on posture control, balance, lower limb muscle strength and proprioception of knee and ankle joints in elderly women with knee osteoarthritis]. Irct20230101057015n1 2023.

Irct20230202057310n2 *{published data only}*

Irct20230202057310n2. Mulligan and propriceptive exercises in knee osteoarthritis [Comparative analysis of Mulligan and propriceptive exercises in knee osteoarthritis]. Irct20230202057310n2 2023.

Irct20230306057633n2 *{published data only}*

Irct20230306057633n2. Effects of calisthenic exercises and isometric exercises on diabetic patients with knee osteoarthritis [Comparative effects of calisthenic exercises and isometric exercises on pain, balance and functional disability in diabetic patients with knee osteoarthritis]. Irct20230306057633n2 2023.

Irct20230427058007n1 *{published data only}*

Irct20230427058007n1. Effect of weight bearing functional exercises and non-weight bearing quadriceps strengthening along with breathing exercises in people with knee osteoarthritis and obesity. Irct20230427058007n1 2023.

Irct20230606058395n1 *{published data only}*

Irct20230606058395n1. Comparative effects of PIR technique and Mulligan TSLR technique alng with retrowalking in knee OA [Comparative effects of post isometric relaxation technique and Mulligan traction straight leg raise technique along with retro walking on hamstring flexibility]. Irct20230606058395n1 2023.

Irct20231015059728n1 *{published data only}*

Irct20231015059728n1. The effect and follow up of neuromuscular exercise in water on complications of knee OA [The effect and follow up of neuromuscular exercise in water on strength, co-contraction, quality of life, proprioception and balance of the knee in women with grade 2&3]. Irct20231015059728n1 2023.

Isaramalai 2018 *{published data only}*

Isaramalai SA, Hounsri K, Kongkamol C, Wattanapisitkul P, Tangadulrat N, Kaewmanee T, et al. Integrating participatory ergonomic management in non-weight-bearing exercise and progressive resistance exercise on self-care and functional ability in aged farmers with knee osteoarthritis: a clustered randomized controlled trial. *Clinical Interventions in Aging* 2018;**13**:101-8. [DOI: 10.2147/CIA.S144288]

Tctr20160219001. Effects of progressive resistance, non-weight bearing exercise program and participatory ergonomic self-care management on muscle strength and functional ability of aged para rubber tappers with knee osteoarthritis. Tctr20160219001 2016.

Isrctn13307390 *{published data only}*

Isrctn13307390. A 12-week digital care program for chronic knee pain involving exercise, education, and psychosocial support. Isrctn13307390 2017.

Jan 1991 *{published data only}*

Jan M-H, Lai J-S. The effects of physiotherapy on osteoarthritic knees of females. *Journal of the Formosan Medical Association / Taiwan Yi Zhi* 1991;**90**:1008-13.

Jessep 2009 *{published data only}*

Jessep SA, Walsh NE, Ratcliffe J, Hurley MV. Long-term clinical benefits and costs of an integrated rehabilitation programme compared with outpatient physiotherapy for chronic knee pain. *Physiotherapy* 2009;**95**(2):94-102.

Jimenez 2014 *{published data only}*

Jimenez SC, Fernandez GR, Zurita OF, Linares GD, Farias MA. Effects of education and strength training on functional tests among older people with osteoarthritis. *Revista Medica de Chile* 2014;**142**(4):436-42. [DOI: 10.4067/S0034-98872014000400004]

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Jing 2019 *{published data only}*

Jing C, Bai XH, Jing YZ. Efficacy of exercise rehabilitation in patients with knee osteoarthritis. *Chung Hua I Hsueh Tsa Chih* 2019;**99**(41):3255-9. [DOI: 10.3760/cma.j.issn.0376-2491.2019.41.011]

Jinnouchi 2023 *{published data only}*

Jinnouchi H, Kitamura A, Matsudaira K, Kakihana H, Oka H, Yamagishi K, et al. Brief self-exercise education for adults with chronic knee pain: a randomized controlled trial. *Modern Rheumatology* 2023;**33**(2):408-15. [DOI: 10.1093/mr/roac009]

Jma-Iia00110 *{published data only}*

Jma-Iia00110. Effectiveness of exercise with or without thermal therapy for community-dwelling elderly Japanese women with knee pain: a randomized controlled trial. Jma-Iia00110 2013.

Ju 2015 *{published data only}*

Ju SB, Park GD, Kim SS. Effects of proprioceptive circuit exercise on knee joint pain and muscle function in patients with knee osteoarthritis. *Journal of Physical Therapy Science* 2015;**27**(8):2439-41. [DOI: 10.1589/jpts.27.2439]

Kao 2012 *{published data only}*

Kao MJ, Wu MP, Tsai MW, Chang WW, Wu SF. The effectiveness of a self-management program on quality of life for knee osteoarthritis (OA) patients. *Archives of Gerontology and Geriatrics* 2012;**54**(2):317-24.

Karagulle 2007 *{published data only}*

Karagulle M, Karagulle MZ, Karagulle O, Donmez A, Turan M. A 10-day course of SPA therapy is beneficial for people with severe knee osteoarthritis. A 24-week randomised, controlled pilot study. *Clinical Rheumatology* 2007;**26**(12):2063-71.

Karimi 2021 *{published data only}*

Karimi N, Dehkordi KJ, Rizi RM. Effects of Pilates training vs. suspension training on quality of life in women with knee osteoarthritis: a randomized controlled trial. *Journal of Bodywork and Movement Therapies* 2021;**27**:737-45. [DOI: 10.1016/j.jbmt.2021.04.002]

Karp 2016 *{published data only}*

Karp JF, Dew MA, Wahed AS, Fitzgerald K, Bolon CA, Weiner DK, et al. Challenges and solutions for depression prevention research: methodology for a depression prevention trial for older adults with knee arthritis and emotional distress. *American Journal of Geriatric Psychiatry* 2016;**24(6)**:433-43. [DOI: 10.1016/j.jagp.2015.10.012]

Karp 2019 *{published data only}*

Karp JF, Zhang J, Wahed AS, Anderson S, Dew MA, Fitzgerald GK, et al. Improving patient reported outcomes and preventing depression and anxiety in older adults with knee osteoarthritis: results of a sequenced multiple assignment randomized trial (SMART) study. *American Journal of Geriatric Psychiatry* 2019;**27**(10):1035-45. [DOI: 10.1016/j.jagp.2019.03.011]

Kawasaki 2008 *{published data only}*

Kawasaki T, Kurosawa H, Ikeda H, Kim SG, Osawa A, Takazawa Y, et al. Additive effects of glucosamine or risedronate for the treatment of osteoarthritis of the knee combined with home exercise: a prospective randomized 18-month trial. *Journal of Bone and Mineral Metabolism* 2008;**26**(3):279-87.

Kawasaki 2009 *{published data only}*

Kawasaki T, Kurosawa H, Ikeda H, Takazawa Y, Ishijima M, Kubota M, et al. Therapeutic home exercise versus intraarticular hyaluronate injection for osteoarthritis of the knee: 6-month prospective randomized open-labelled trial. *Journal of Orthopaedic Science* 2009;**14**(2):182-91.

Kct0006037 *{published data only}*

Kct0006037. Exercise combined with electromyostimulation improves knee joint function in elderly women with knee osteoarthritis [Impact of isometric exercise combined with electromyostimulation on inflammatory cytokine levels, muscle strength, and knee joint function]. Kct0006037 2021.

Khachian 2020 *{published data only}*

Khachian A, Seyedoshohadaei M, Haghani H, Amiri F. Effect of self-management program on outcome of adult knee osteoarthritis. *International Journal of Orthopaedic and Trauma Nursing* 2020;**39**:100797. [DOI: 10.1016/j.ijotn.2020.100797]

Khan 2023 *{published data only}*

Khan SA, Parasher P, Ansari MA, Parvez S, Fatima N, Alam I. Effect of an integrated physiotherapy protocol on knee osteoarthritis patients: a preliminary study. *Healthcare (Basel, Switzerland)* 2023;**11**(4):14. [DOI: 10.3390/healthcare11040564]

King 2008 *{published data only}*

King LK, Birmingham TB, Kean CO, Jones IC, Bryant DM, Giffin JR. Resistance training for medial compartment knee osteoarthritis and malalignment. *Medicine and Science in Sports and Exercise* 2008;**40**(8):1376-84.

Kletskova 2021 *{published data only}*

Kletskova O, Rusanov A, Rusanova O, Gaowgzeh R, Nikanorov A. Physical therapy program in the treatment of osteoarthrosis in patients with obesity. *Georgian Medical News* 2021;**315**:103-8.

Kloek 2016a *{published data only}*

Kloek CJ, Bossen D, De Bakker DH, Dekker J, Veenhof C. Blended physical activity intervention with reduced face-to-face contact and usual physical therapy show similar effectiveness in patients with knee and hip osteoarthritis: a randomized controlled trial. *Annals of the Rheumatic Diseases* 2016;**75 Suppl 2**:827-8. [DOI: 10.1136/annrheumdis-2016-eular.2729]

Kloek 2018 *{published data only}*

Kloek CJ, Bossen D, Spreeuwenberg PM, Dekker J, De Bakker DH, Veenhof C. Effectiveness of a blended physical therapist intervention in people with hip osteoarthritis, knee osteoarthritis, or both: a cluster-randomized controlled trial. *Physical Therapy* 2018;**98**(7):560-70. [DOI: 10.1093/ptj/pzy045]

Knoop 2023 *{published data only}*

Knoop J, Dekker J, vVn der Leeden M, Van der Esch M, Ostelo Rwjg. High- versus low-dose exercise therapy for knee osteoarthritis. *Annals of Internal Medicine* 2023;**176**(7):eL230140. [DOI: 10.7326/L23-0140]

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Konishi 2009 *{published data only}*

Konishi I, Tanabe N, Seki N, Suzuki H, Okamura T, Shinoda K, et al. Physiotherapy program through home visits for community-dwelling elderly Japanese women with mild knee pain. *Tohoku Journal of Experimental Medicine* 2009;**219**(2):91-9.

Kotteeswaran 2021 *{published data only}*

Kotteeswaran K, Shanmugasundaram N, Shalini S, Sowmya MV. Effect of multidirectional and unidirectional wobble board lateral step-up exercise in pain, symptom and ADL among unilateral medial compartmental osteoarthrosis of knee. *Biomedicine (India)* 2021;**41**(3):669-77. [DOI: 10.51248/.v41i3.1205]

Kreindler 1989 *{published data only}*

Kreindler H, Lewis CB, Rush S, Schaefer K. Effects of three exercise protocols on strength of persons with osteoarthritis of the knee. *Topics in Geriatriatric Rehabilitation* 1989;**4**:32-9.

Kumar 2017 *{published data only}*

Kumar D, Beavers D, DeVita P, Messier SP. Effects of weight-loss on patellofemoral loading in overweight and obese adults with patellofemoral osteoarthritis: secondary analysis from the IDEA randomized trial. *Osteoarthritis and Cartilage* 2017;**25 Suppl 1**:S171-S172.

Kuptniratsaikul 2002 *{published data only}*

Kuptniratsaikul V, Orchatara T, Nilganuwong S, Visanu T. The efficacy of a muscle exercise program to improve functional performance of the knee in patients with osteoarthritis. *Chotmaihet Thangphaet [Journal of the Medical Association of Thailand]* 2002;**85**:33-9.

Lai 2018 *{published data only}*

Lai Z, Zhang Y, Lee S, Wang L. Effects of strength exercise on the knee and ankle proprioception of individuals with knee osteoarthritis. *Research in Sports Medicine (Print)* 2018;**26**(2):138-46. [DOI: 10.1080/15438627.2018.1431541]

Lall 2023 *{published data only}*

Lall S, Prem V, Karvannan H. Comparison of neuromuscular joint facilitation and quadriceps strengthening exercise in knee osteoarthritis: a randomized controlled trial. *International Journal of Therapeutic Massage & Bodywork* 2023;**16**(3):10-19. [DOI: 10.3822/ijtmb.v16i3.811]

Landsmeer 2016 *{published data only}*

Landsmeer ML, Runhaar J, Van der Plas P, Van Middelkoop M, Vroegindeweij D, Koes B, et al. Reducing progression of knee OA features assessed by MRI in overweight and obese women: Secondary outcomes of a preventive RCT. *Osteoarthritis and Cartilage* 2016;**24**(6):982-90. [DOI: 10.1016/j.joca.2015.12.016]

Lankhorst 1982 *{published data only}*

Lankhorst GJ, Van de Stadt RJ, Van der Korst JK, Hinlopen-Bonrath E, Griffioen FM, De Boer W. Relationship of isometric knee extension torque and functional variables in osteoarthrosis of the knee. *Scandinavian Journal of Rehabilitation Medicine* 1982;**14**:7-10.

Lee 2017a *{published data only}*

Lee A, Harvey WF, Price LL, Han X, Driban JB, Iversen MD, et al. Dose-response effects of Tai Chi and physical therapy exercise interventions in symptomatic knee osteoarthritis. *Arthritis and Rheumatology. Conference: American College of Rheumatology/ Association of Rheumatology Health Professionals Annual Scientific Meeting, ACR/ARHP* 2017;**69**:Supplement 10.

Lee 2018 *{published data only}*

Lee AC, Harvey WF, Price LL, Han X, Driban JB, Iversen MD, et al. Dose-response effects of tai Chi and physical therapy exercise interventions in symptomatic knee osteoarthritis. *Physical Medicine and Rehabilitation* 2018;**10**(7):712-23. [DOI: 10.1016/j.pmrj.2018.01.003]

Li 2017a *{published data only}*

Li LC, Sayre EC, Xie H, Clayton C, Feehan LM. A community-based physical activity counselling program for people with knee osteoarthritis: feasibility and preliminary efficacy of the track-OA study. *JMIR MHealth and UHealth* 2017;**5**(6):e86. [DOI: 10.2196/mhealth.7863]

Liang 2019 *{published data only}*

Liang J, Guo Y, Zheng Y, Lang S, Chen H, You Y, et al. The lower body positive pressure treadmill for knee osteoarthritis rehabilitation. *Journal of Visualized Experiments* 2019;**149**(07):22. [DOI: 10.3791/59829]

Lim 2002 *{published data only}*

Lim BW. A comparative study of open and closed kinetic chain exercise regimes in patients with knee osteoarthritis. *Physiotherapy Singapore* 2002;**5**:34-40.

Lim 2010 *{published data only}*

Lim JY, Tchai E, Jang SN. Effectiveness of aquatic exercise for obese patients with knee osteoarthritis: a randomized controlled trial. *Physical Medicine & Rehabilitation* 2010;**2**(8):723-31.

Lin 2004 *{published data only}*

Lin SY, Davey RC, Cochrane T. Community rehabilitation for older adults with osteoarthritis of the lower limb: a controlled clinical trial. *Clinical Rehabilitation* 2004;**18**:92-101.

Lin 2020 *{published data only}*

Lin YT, Lee WC, Hsieh RL. Active video games for knee osteoarthritis improve mobility but not WOMAC score: a randomized controlled trial. *Annals of Physical and Rehabilitation Medicine.* 2020;**22**:(no pagination). [DOI: 10.1016/j.rehab.2019.11.008]

Liu 2008 *{published data only}*

Liu Y, Guo L, Ma S. Treatment of 256 cases of osteoarthritis of knee joint with Guo Jianhua's four-step therapy. *Journal of Traditional Chinese Medicine / Chung I Tsa Chih Ying Wen Pan* 2008;**28**(2):114-7.

Liu 2023 *{published data only}*

Liu Y, Chen R, Zhang Y, Wang Q, Ren JL, Wang CX, et al. Clinical value of ankle flexion and extension exercises combined with a

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

psychological intervention in knee osteoarthritis. *World Journal of Psychiatry* 2023;**13**(10):743-52. [DOI: 10.5498/wjp.v13.i10.743]

Lorbeer 2023 *{published data only}*

Lorbeer N, Knoll N, Keller J, Domke A, Di Maio S, Armbrecht G, et al. Enhancing physical activity and reducing symptoms of patients with osteoarthritis of the knee: a randomized controlled trial of the PrevOP-Psychological Adherence Program. *BMC Musculoskeletal Disorders* 2023;**24**(1):(no pagination). [DOI: 10.1186/s12891-023-06661-x]

Losciale 2023 *{published data only}*

Losciale JM, Truong LK, Zhang K, Silvester-Lee T, Miciak M, Pajkic A, et al. Preliminary results of the stop osteoarthritis (SOAR) proof-of-concept randomized delay-controlled trial. *Osteoarthritis and Cartilage* 2023;**31 Suppl 1**:S198-S199. [DOI: 10.1016/j.joca.2023.01.173]

Mahalle 2022 *{published data only}*

Mahalle AP, Walke R. Interferential therapy and strengthening exercises in management of knee osteoarthritis. *Journal of Pharmaceutical Negative Results* 2022;**13**:2856-9. [DOI: 10.47750/pnr.2022.13.S06.366]

Mahmoud 2023 *{published data only}*

Mahmoud WS, Osailan A, Elnaggar RK, Alhailiy AB. Radiological and clinical outcomes of concurrent hamstring stretching with quadriceps strengthening in patients with knee osteoarthritis: a randomized clinical trial. *Isokinetics and Exercise Science* 2023;**31**(2):137-47. [DOI: 10.3233/IES-220056]

Mangione 1999 *{published data only}*

Mangione KK, McCully K, Gloviak A, Lefebvre I, Hofmann M, Craik R. The effects of high-intensity and low-intensity cycle ergometry in older adults with knee osteoarthritis. *Journal of Gerontology* 1999;**54A**:M184-90.

Marra 2012 *{published data only}*

Marra CA, Cibere J, Grubisic M, Grindrod KA, Gastonguay L, Thomas JM, et al. Pharmacist-initiated intervention trial in osteoarthritis: a multidisciplinary intervention for knee osteoarthritis. *Arthritis Care & Research* 2012;**64**(12):1837-45.

Mascarin 2012 *{published data only}*

Mascarin NC, Vancini RL, Andrade ML, Magalhaes Ede P, de Lira CA, Coimbra IB. Effects of kinesiotherapy, ultrasound and electrotherapy in management of bilateral knee osteoarthritis: prospective clinical trial. *BMC Musculoskeletal Disorders* 2012;**13**:182.

Mat 2016 *{published data only}*

Mat S, Ng CT, Tan MP. Effect of the Otago exercises on postural balance and fear of falling among older fallers with knee osteoarthritis. *Osteoarthritis and Cartilage* 2016;**1**:S66.

Mat 2018 *{published data only}*

Mat S, Ng CT, Tan PJ, Ramli N, Fadzli F, Rozalli FI, et al. Effect of modified Otago exercises on postural balance, fear of falling, and fall risk in older fallers with knee osteoarthritis and impaired gait and balance: a secondary analysis. *Physical Medicine and Rehabilitation* 2018;**10**(3):254-62. [DOI: 10.1016/j.pmrj.2017.08.405]

McCarthy 2004 *{published data only}*

McCarthy CJ, Mills PM, Pullen R, Roberts C, Silman A, Oldham JA. Supplementing a home exercise programme with a class-based exercise programme is more effective than home exercise alone in the treatment of knee osteoarthritis. *Rheumatology* 2004;**43**:880-6.

McKnight 2010 *{published data only}*

McKnight PE, Kasle S, Going S, Villanueva I, Cornett M, Farr J, et al. A comparison of strength training, self-management, and the combination for early osteoarthritis of the knee. *Arthritis Care & Research* 2010;**62**(1):45-53.

McQuade 2011 *{published data only}*

McQuade KJ, De Oliveira AS. Effects of progressive resistance strength training on knee biomechanics during single leg step-up in persons with mild knee osteoarthritis. *Clinical Biomechanics (Bristol, Avon)* 2011;**26**(7):741-8.

Meiling 2023 *{published data only}*

Meiling Z, Yingli H, Wen S, Li L. Impacts of muscle visual feedback intervention combined with continuous nursing on exercise ability of patients with knee osteoarthritis [Chinese]. *Chinese Journal of Practical Nursing* 2023;**39**(30):2335-40. [DOI: 10.3760/cma.j.cn211501-20230105-00043]

Mesa-Castrillon 2023 *{published data only}*

Mesa-Castrillon CI, Simic M, Ferreira ML, Bennell KL, Luscombe GM, Gater K, et al. Effectiveness of an eHealth-delivered program to EMpower People with musculoskeletal pain in Rural Australia (EMPoweR): a randomised controlled trial. *Arthritis Care & Research.* 2023;**20**:(no pagination). [DOI: 10.1002/acr.25272]

Mesa-Castrillon CI, Simic M, Ferreira ML, Hatswell K, Luscombe G, de Gregorio AM, et al. EHealth to empower patients with musculoskeletal pain in rural Australia (EMPoweR) a randomised clinical trial: study protocol. *BMC Musculoskeletal Disorders* 2021;**22**(1):(no pagination). [DOI: 10.1186/s12891-020-03866-2]

Messier 2000a *{published data only}*

Messier SP, Loeser RF, Mitchell MN, Valle G, Morgan TP, Rejeski WJ, et al. Exercise and weight loss in obese older adults with knee osteoarthritis: a preliminary study. *Journal of the American Geriatric Society* 2000;**48**:1062-72.

Messier 2007 *{published data only}*

Messier SP, Mihalko S, Loeser RF, Legault C, Jolla J, Pfruender J, et al. Glucosamine/chondroitin combined with exercise for the treatment of knee osteoarthritis: a preliminary study. *Osteoarthritis and Cartilage* 2007;**15**(11):1256-66.

Messier 2008 *{published data only}*

Messier SP, Legault C, Lenz ME, Thonar EJ, Loeser RF. Effect of an exercise and dietary intervention on serum biomarkers in overweight and obese adults with osteoarthritis of the knee. *Osteoarthritis and Cartilage* 2008;**16**(9):1047-53.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Messier 2017 *{published data only}*

Messier SP, Callahan LF, Beavers DP, Queen K, Mihalko SL, Miller GD, et al. Weight-loss and exercise for communities with arthritis in North Carolina (WE-CAN): design and rationale of a pragmatic, assessor-blinded, randomized controlled trial. *BMC Musculoskeletal Disorders* 2017;**18**(1):(no pagination). [DOI: 10.1186/s12891-017-1441-4]

Mikesky 2006 *{published data only}*

Mikesky AE, Mazzuca SA, Brandt KD, Perkins SM, Damush T, Lane KA. Effects of strength training on the incidence and progression of knee osteoarthritis. *Arthritis Care & Research* 2006;**55**:690-9.

Minor 1989 *{published data only}*

Minor MA, Hewett JE, Webel RR, Anderson SK, Kay DR. Efficacy of physical conditioning exercise in patients with rheumatoid arthritis and osteoarthritis. *Arthritis and Rheumatism* 1989;**32**:1396-405.

Miyagawa 2016 *{published data only}*

Miyagawa H, Ikemoto T, Honjo H, Akao M, Tsujimoto T, Ushida T. One-year results of voluntary-based supervised exercise or treatment at orthopedic clinic for radiographic severe knee osteoarthritis. *Journal of Physical Therapy Science* 2016;**28**(3):906-10. [DOI: 10.1589/jpts.28.906]

Modarresi 2023 *{published data only}*

Modarresi S, Pearson N, Madden K, Bennell KL, Fahnestock M, Neogi T, et al. Pain informed movement for people with knee osteoarthritis: protocol for a pilot randomized controlled trial. *Osteoarthritis and Cartilage Open* 2023;**5**(4):(no pagination). [DOI: 10.1016/j.ocarto.2023.100402]

Mohammedsadiq 2023 *{published data only}*

Mohammedsadiq HA, Rasool MT. Effectiveness of home-based conventional exercise and cryotherapy on daily living activities in patients with knee osteoarthritis: a randomized controlled clinical trial. *Medicine (United States)* 2023;**102**(18):E33678. [DOI: 10.1097/MD.0000000000033678]

Moss 2007 *{published data only}*

Moss P, Sluka K, Wright A. The initial effects of knee joint mobilization on osteoarthritic hyperalgesia. *Manual Therapy* 2007;**12**:109-18.

Mulla 2018 *{published data only}*

Mulla DM, Wiebenga EG, Chopp-Hurley JN, Kaip L, Jarvis RS, Stephens A, et al. The effects of lower extremity strengthening delivered in the workplace on physical function and work-related outcomes among desk-based workers: a randomized controlled trial. *Journal of Occupational and Environmental Medicine / American College of Occupational and Environmental Medicine* 2018;**60**(11):1005-14. [DOI: 10.1097/JOM.0000000000001408]

Murphy 2008 *{published data only}*

Murphy SL, Strasburg DM, Lyden AK, Smith DM, Koliba JF, Dadabhoy DP, et al. Effects of activity strategy training on pain and physical activity in older adults with knee or hip osteoarthritis: a pilot study. *Arthritis and Rheumatism* 2008;**59**(10):1480-7.

Murphy 2016 *{published data only}*

Murphy SL, Kratz AL, Kidwell K, Lyden AK, Geisser ME, Williams DA. Brief time-based activity pacing instruction as a singular behavioral intervention was not effective in participants with symptomatic osteoarthritis. *Pain* 2016;**157**(7):1563-73. [DOI: 10.1097/j.pain.0000000000000549]

Nathani 2020 *{published data only}*

Nathani S, Tank KD. Effect of PNF stretching on proprioception and physical function in individual with knee osteoarthritis: an experimental study. *Indian Journal of Public Health Research and Development* 2020;**11**(7):779-84.

Nct01945749 *{published data only}*

Nct01945749. Exercise and steroid in knee osteoarthritis [Combined intra articular corticosteroid and physiotherapeutic exercise in patients with osteoarthritis of the knee: a randomised clinical trial]. Nct01945749 2013.

Nct02024126 *{published data only}*

Nct02024126. Exercise therapy in patients with knee osteoarthritis. Nct02024126 2013.

Nct02043613 *{published data only}*

Nct02043613. Context effects in exercise therapy for knee and/or hip pain [Effect of physical surroundings on effects from exercise as treatment for hip or knee pain: a double-blind randomized controlled clinical trial]. Nct02043613 2014.

Nct02098096 *{published data only}*

Nct02098096. Negative work exercise for the treatment of knee arthritis [Development of a negative work exercise regimen as an intervention for posttraumatic osteoarthritis of the knee]. Nct02098096 2014.

Nct02350387 *{published data only}*

Nct02350387. Efficacy of eccentric resistance training in persons with knee osteoarthritis. Nct02350387 2015.

Nct02454374 *{published data only}*

Nct02454374. Joint loading vs normal physiotherapy care in degenerative knees (V1). Nct02454374 2015.

Nct02598804 *{published data only}*

Nct02598804. Evaluation of the effectiveness of self management program on changes in physical activity level of knee osteoarthritis patients in spa therapy. Nct02598804 2015.

Nct02600702 *{published data only}*

Nct02600702. A study of siriraj home exercises protocol for primary osteoarthritis of the knee [A randomized control study of siriraj home exercises protocol for primary knee osteoarthritis patients versus usual orthopedic outpatient consultation in primary knee osteoarthritis]. Nct02600702 2015.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Nct02698072 *{published data only}*

Nct02698072. Improvement in pain and function following a physiotherapy program in older adults with knee osteoarthritis. Nct02698072 2016.

Nct02901964 *{published data only}*

Nct02901964. Effect of strengthening the hip abductor in patients with knee osteoarthritis: randomized controlled trial. Nct02901964 2016.

Nct03050320 *{published data only}*

Nct03050320. Improving resilience and longevity for workers through exercise. Nct03050320 2017.

Nct03167502 *{published data only}*

Nct03167502. Performance in walking in osteoarthritis patients [Performance in walking in osteoarthritis patients: the role of fatigue and the effect of an eccentric muscle building program]. Nct03167502 2017.

Nct03171168 *{published data only}*

The effect of apostherapy on knee pain [The effect of apostherapy on pain and function in knee osteoarthritis population: a randomized controlled trial]. Nct03171168 2017.

Nct03379623 *{published data only}*

Nct03379623. ESCAPE-pain programme in Malaysiaa feasibility randomised controlled trial of the implementation in Malaysian healthcare context of the ESCAPE-pain programme for patients with knee osteoarthritis. Nct03379623 2017.

Nct03470090 *{published data only}*

Nct03470090. The effects of neuromuscular exercises training on physical activity, functionality and balance in knee osteoarthritis. Nct03470090 2018.

Nct03571698 *{published data only}*

Nct03571698. The efficacy of exercise and alternative applications of NMES on pain and function in patients with knee OA Istanbul University, Faculty of Health Science, Division of Physiotherapy and Rehabilitation. Nct03571698 2018.

Nct03621631 *{published data only}*

Nct03621631. Effect of tai Chi exercise on mechanical joint loading in knee osteoarthritis [Real time biofeedback tai Chi training for knee osteoarthritis: a feasibility study]. Nct03621631 2018.

Nct03666585 *{published data only}*

Nct03666585. Comparison of the accuracy in rehabilitation exercise between mobile application actuated rehabilitation exercise guidance and routine rehabilitation exercise guidance for the primary osteoarthritis knee patients. Nct03666585 2018.

Nct03676790 *{published data only}*

Nct03676790. Ultrasound and exercises in knee osteoarthritis [Effect of ultrasound associated to exercises in patients with knee osteoarthritis]. Nct03676790 2018.

Nct03750279 *{published data only}*

Nct03750279. Inflammatory targeted laser treatment of knee osteoarthritis [Inflammatory targeted low-level laser treatment of knee osteoarthritis - a randomized clinical trial]. Nct03750279 2018.

Nct03798002 *{published data only}*

Nct03798002. Neuro-muscular exercise training verses quadriceps training in mild to moderate knee osteoarthritis [Effects of neuro-muscular exercise training verses isolated quadriceps training program in patients with mild to moderate knee osteoarthritis]. Nct03798002 2019.

Nct03893981 *{published data only}*

Nct03893981. Strengthening exercises versus proprioception and balance exercises in the treatment of knee osteoarthritis [Comparison of the effect of strengthening exercises versus proprioception and balance exercises in the treatment of knee osteoarthritis]. Nct03893981 2019.

Nct03921801 *{published data only}*

Nct03921801. Effects of improved calf muscle function on gait, balance and joint loading in older adults [Improving plantarflexor muscle function to alleviate decline in mobility, loss of balance and detrimental knee joint loading in older adults]. Nct03921801 2019.

Nct03951506 *{published data only}*

Nct03951506. Multi-modal exercise program in older aduls with knee osteoarthritis [Effects a multi-modal exercise program in older aduls with knee osteoarthritis]. Nct03951506 2019.

Nct04058002 *{published data only}*

Nct04058002. Project arthritis recovering quality of life through education 70+. Nct04058002 2019.

Nct04099394 *{published data only}*

Nct04099394. Pain & aging: combined interventions for fitness in the community study [Randomized controlled trial of group-based exercise and behavioral health skills training for older adults with painful knee osteoarthritis]. Nct04099394 2019.

Nct04138784 *{published data only}*

Nct04138784. Individualized comprehensive rehabilitation program in chronic knee osteoarthritis [Effectiveness of an individualized comprehensive rehabilitation program in disabled chronic knee osteoarthritis women]. Nct04138784 2019.

Nct04183933 *{published data only}*

Nct04183933. The effect of exercise training on balance and functional status in individuals with osteoarthritis'the effect of different exercise training on balance and functional status in individuals with osteoarthritis'. Nct04183933 2019.

Nct04198467 *{published data only}*

Nct04198467. To evaluate the optimum dose of platelet rich plasma in knee osteoarthritis and compare efficacy with hyaluronic acid [Randomized controlled study to compare safety and efficacy of intra articular administration of platelet rich plasma versus hyaluronic aci]. Nct04198467 2019.

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Nct04225182 *{published data only}*

Nct04225182. Evaluation of the effectiveness of a muscular strengthening protocol with an instrumented orthosis for gonarthrosis patients. Nct04225182 2020.

Nct04323267 *{published data only}*

Nct04323267. Digital home-exercise therapy application for patients with non-surgical knee injuries [Efficacy of digital home-exercise therapy application for patients with non-surgical knee injuries: a randomized, controlled trial]. Nct04323267 2020.

Nct04390932 *{published data only}*

Nct04390932. The impact of therapeutic alliance in the rehabilitation of knee osteoarthritis [the impact of therapeutic alliance on clinical outcomes in patients with symptomatic knee osteoarthritis. A randomized clinical trial protocol]. Nct04390932 2020.

Nct04423887 *{published data only}*

Nct04423887. Effects of creatine supplementation in addition to resistance exercise training in patients with knee osteoarthritis. Nct04423887 2020.

Nct04445350 *{published data only}*

Nct04445350. Motor learning in knee osteoarthritis [Motor learning in knee osteoarthritis therapy - a new rehabilitation approach]. Nct04445350 2020.

Nct04528953 *{published data only}*

Nct04528953. Investigating the effect of interval walking and qigong on people with knee osteoarthritis: pilot study [Investigating the effect of interval walking and qigong on some clinical and biological outcomes of people with knee osteoarthritis]. Nct04528953 2020.

Nct04536519 *{published data only}*

Nct04536519. Foot wear modification along with physical therapy in knee osteoarthritis [Effects of foot wear modification along with physical therapy on functional status in knee osteoarthritis]. Nct04536519 2020.

Nct04540341 *{published data only}*

Nct04540341. Effectiveness of Mulligan mobilization technique and core stabilization exercise with knee osteoarthritis. Nct04540341 2020.

Nct04542655 *{published data only}*

Nct04542655. The effect of exercise on serum levels of collagen and proteoglycan in patients with knee osteoarthritis (EFEX-OA-01). Nct04542655 2020.

Nct04574700 *{published data only}*

Nct04574700. Mobilization effect on hamstring length [Effect of knee joint mobilization on hamstring muscle length]. Nct04574700 2020.

Nct04600661 *{published data only}*

Nct04600661. Low load resistance blood flow restriction training versus neuromuscular training in knee osteoarthritis. Nct04600661 2020.

Nct04654871 *{published data only}*

Nct04654871. Effects of glucosamine and chondroitin sulfate supplementation in addition to resistance exercise training and manual therapy in patients with knee osteoarthritis. Nct04654871 2020.

Nct04678999 *{published data only}*

Nct04678999. Mind-body therapy (remotely delivered) for knee osteoarthritis [Neurobiological mechanisms of mind-body therapy (remotely delivered) for knee osteoarthritis]. Nct04678999 2020.

Nct04747444 *{published data only}*

Nct04747444. Positive pressure treadmill walking for individuals with knee pathology [The effects of anti-gravity treadmill walking in individuals with knee osteoarthritis]. Nct04747444 2021.

Nct04819672 *{published data only}*

Nct04819672. Effect of low intensity training combined with partial restriction of blood flow in individuals with knee osteoarthritis. Nct04819672 2021.

Nct04857879 *{published data only}*

Nct04857879. Implementation of GLA: d program in West Virginians with osteoarthritis of the knee [A pilot randomized controlled trial to assess the implementation and outcomes of Good Life with Osteoarthritis (GLA: D) program in West Virginians suffering from the osteoarthritis]. Nct04857879 2021.

Nct04925895 *{published data only}*

Nct04925895. Effects of dynamic soft tissue mobilization in reducing hamstring tightness in knee osteoarthritis. Nct04925895 2021.

Nct04980300 *{published data only}*

Nct04980300. Delaware Physical Exercise and Activity for Knee osteoarthritis (PEAK). Nct04980300 2021.

Nct05026385 *{published data only}*

Nct05026385. The POMELO (Prevention of Muscle Loss in Osteoarthritis) trial [Targeted knee osteoarthritis care for adults with a BMI ≥5 kg/m2: the prevention of muscle loss in osteoarthritis (P.O.M.E.L.O.) Feasibility and pilot randomized trial]. Nct05026385 2021.

Nct05131048 *{published data only}*

Nct05131048. Home based hip strengthening for knee OA patients [Effects of home-based hip strengthening exercises on pain, range of motion and functional disability in knee osteoarthritis patients: a randomized controlled trial]. Nct05131048 2021.

Nct05169606 *{published data only}*

Nct05169606. Effect of tissue flossing, AIS, PNF on hamstring among knee OA patients [Effect of tissue flossing, active isolated stretch, proprioceptive neuromuscular facilitation on hamstring among knee osteoarthritis patients]. Nct05169606 2021.

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Nct05209893 *{published data only}*

Nct05209893. The effect of telerehabilitation-based exercise and training on outcome measures in patients with knee osteoarthritis. Nct05209893 2022.

Nct05274932 *{published data only}*

Nct05274932. Exercise with blood flow restriction in knee osteoarthritis [Neuromuscular responses, pain intensity and perceived exertion of resistance exercise with blood flow restriction in patients with severe knee osteoarthritis]. Nct05274932 2022.

Nct05403645 *{published data only}*

Nct05403645. Comparison of exercise therapy vs mobilization with movement among patients with knee osteoarthritis [Comparison of exercise therapy vs mobilization with movement among patients with knee osteoarthritis in household females]. Nct05403645 2022.

Nct05601947 *{published data only}*

Nct05601947. Investigation of the effectiveness of Wii-based exercise play therapy in knee osteoarthritis rehabilitation. Nct05601947 2022.

Nct05777421 *{published data only}*

Nct05777421. Efficacy of virtual reality based exercise regimes on kinesiophobia, pain and functional disability in patients with knee osteoarthritis. Nct05777421 2023.

Nct05810376 *{published data only}*

Nct05810376. Effects of digital-based physical activity intervention in individuals with knee osteoarthritis. Nct05810376 2023.

Nct05839535 *{published data only}*

Nct05839535. Enhancing modulation effects of baduanjin through neuromodulation for knee OA [Enhancing modulation effects of baduanjin through non-invasive neuromodulation for knee osteoarthritis]. Nct05839535 2023.

Nct05850793 *{published data only}*

Nct05850793. Comparison of progressive motor imagery and sensorimotor exercises in knee osteoarthritis [Comparison of progressive motor imagery and sensorimotor exercises in knee osteoarthritis: a randomized controlled trial]. Nct05850793 2023.

Nct05877703 *{published data only}*

Nct05877703. Open versus close chain exercise effects with lifestyle modification and education in knee osteoarthritis [Effects of open versus close kinetic chain exercise training program with lifestyle modification and patient's education in grade 3 knee osteoarthriti]. Nct05877703 2023.

Nct05878587 *{published data only}*

Nct05878587. Burger Allen exercises in knee OA with type II diabetes [Effects of Burger Allen exercises on pain, range of motion, and disability in knee osteoarthritis with type II diabetes]. Nct05878587 2023.

Nct05889598 *{published data only}*

Nct05889598. Resistance exercise training to improve bone and articular cartilage health in women [Effects of a resistance exercise programme on risk of osteoporosis and osteoarthritis in females]. Nct05889598 2023.

Nct05893017 *{published data only}*

Nct05893017. Comparison of the effects of different treatment modalities in individuals with knee osteoarthritis [Comparison of the effects of different treatment modalities on pain, quality of life, and musculoskeletal system in individuals with knee osteoarthritis]. Nct05893017 2023.

Nct05941117 *{published data only}*

Nct05941117. Effect of exercises on dynamic balance in knee osteoarthritis patients [Effect of weight bearing versus non-weight bearing strengthening exercises on dynamic balance in knee osteoarthritis patients]. Nct05941117 2023.

Nct05955300 *{published data only}*

Nct05955300. Nutrition and movement to improve quality of life with knee osteoarthritis [Evidence based training program combined with nutrition therapy for patients with knee osteoarthritis]. Nct05955300 2023.

Nct05955430 *{published data only}*

Nct05955430. Percutaneous electrical stimulation, education and exercise program in patients with knee osteoarthritis [Efficacy of percutaneous electrical stimulation compared to transcutaneous electrical stimulation in combination with an education and exercise program]. Nct05955430 2023.

Nct05958212 *{published data only}*

Nct05958212. Innovative gait training for people with knee osteoarthritis using dual belt treadmill [The efficacy of unilateral speed-upscaling gait training in promoting pain and functional recovery in people with total knee replacement: a randomised controlled trial]. Nct05958212 2023.

Nct05995210 *{published data only}*

Nct05995210. Comparing the effectiveness of kinesio taping and knee orthosis in patients with patellofemoral pain syndrome [Comparing the effectiveness of kinesio taping and knee orthosis in patients with patellofemoral pain syndrome: randomized controlled study]. Nct05995210 2023.

Nct05997966 *{published data only}*

Nct05997966. Investigation of the effects of progressive resistance exercises in individuals with knee osteoarthritis. Nct05997966 2023.

Nct06090097 *{published data only}*

Nct06090097. Knee biofeedback rehabilitation through game therapy [Knee-biofeedback rehabilitation interface for game-based home therapy]. Nct06090097 2023.

Nct06105788 *{published data only}*

Nct06105788. Upper and lower extremity exercise and exercise-induced hypoalgesia in knee osteoarthritis [Comparing effects of upper extremity versus lower extremity exercise on exercise-

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induced hypoalgesia in people with knee osteoarthritis: a pilot cross-over study]. Nct06105788 2023.

Neves 2011 *{published data only}*

Neves M, Gualano B, Roschel H, Fuller R, Benatti FB, Pinto A, et al. Beneficial effect of creatine supplementation in knee osteoarthritis. *Medicine and Science in Sports and Exercise* 2011;**43**(8):1538-43.

Ng 2010 *{published data only}*

Ng NT, Heesch KC, Brown WJ. Efficacy of a progressive walking program and glucosamine sulphate supplementation on osteoarthritic symptoms of the hip and knee: a feasibility trial. *Arthritis Research & Therapy* 2010;**12**(1):R25.

Ntr6996 *{published data only}*

Ntr6996. Walking alteration for knee osteoarthritis - the WALK study. Ntr6996 2017.

O'Brien 2016 *{published data only}*

O'Brien KM, Williams A, Wiggers J, Wolfenden L, Yoong S, Campbell E, et al. Effectiveness of a healthy lifestyle intervention for low back pain and osteoarthritis of the knee: protocol and statistical analysis plan for two randomised controlled trials. *Brazilian Journal of Physical Therapy* 2016;**20**(5):477-89. [DOI: 10.1590/bjpt-rbf.2014.0189]

Onwunzo 2021 *{published data only}*

Onwunzo CN, Igwe SE, Umunnah JO, Uchenwoke CI, Ezugwu UA. Effects of isometric strengthening exercises on pain and disability among patients with knee osteoarthritis. *Cureus* 2021;**13**(10):e18972. [DOI: 10.7759/cureus.18972]

Osteras 2019 *{published data only}*

Moseng T, Dagfinrud H, Andreassen O, Dziedzic K, Hagen K, Hansen J, et al. Implementing international osteoarthritis guidelines in primary care: results on symptom-related outcomes in secondary analyses from a randomized controlled study. *Osteoarthritis and Cartilage* 2019;**27 Suppl 1**:S26-S27. [DOI: 10.1016/j.joca.2019.02.041]

Moseng T, Dagfinrud H, Van Bodegom-Vos L, Dziedzic K, Hagen KB, Natvig B, et al. Low adherence to exercise may have influenced the proportion of OMERACT-OARSI responders in an integrated osteoarthritis care model: secondary analyses from a cluster-randomised stepped-wedge trial. *BMC Musculoskeletal Disorders* 2020;**21**(1):236. [DOI: 10.1186/s12891-020-03235-z]

Osteras N, Moseng T, Van Bodegom-Vos L, Dziedzic K, Andreassen O, Fenstad A, et al. Higher quality of care and less surgery after implementing osteoarthritis guidelines in primary care - long-term results from a cluster-randomized controlled trial. *Osteoarthritis and Cartilage* 2020;**28 Suppl 1**:S439-S440. [DOI: 10.1016/j.joca.2020.02.684]

Osteras N, Moseng T, Van Bodegom-Vos L, Dziedzic K, Andreassen O, Fenstad AM, et al. Higher quality of care and less surgery after implementing osteoarthritis guidelines in primary care: long-term results from a cluster randomized controlled trial. *Scandinavian Journal of Rheumatology* 2021;**50 Suppl 130**:5-6. [DOI: 10.1080/03009742.2021.1951065]

Osteras N, Moseng T, Van Bodegom-Vos L, Dziedzic K, Mdala I, Natvig B, et al. Implementing a structured model for osteoarthritis care in primary healthcare: a stepped-wedge cluster-randomised trial. *PLoS Medicine* 2019;**16**(10):(no pagination). [DOI: 10.1371/journal.pmed.1002949]

Osteras 2023 *{published data only}*

Osteras N, Aas E, Moseng T, Van Bodegom-Vos L, Dziedzic K, Natvig B, et al. Longer-term quality of care, effectiveness, and cost-effectiveness of implementing a model of care for osteoarthritis: a cluster-randomized controlled trial. *Osteoarthritis and Cartilage* 2023;**1**:1. [DOI: 10.1016/j.joca.2023.10.003]

Osugi 2014 *{published data only}*

Osugi T, Iwamoto J, Yamazaki M, Takakuwa M. Effect of a combination of whole body vibration exercise and squat training on body balance, muscle power, and walking ability in the elderly. *Therapeutics and Clinical Risk Management* 2014;**10**(1):131-8. [DOI: 10.2147/TCRM.S57806]

Ozdincler 2005 *{published data only}*

Ozdincler AR, Yeldan I, Kinali P. The effects of closed kinetic chain exercise on pain and functional performance of patients with knee osteoarthritis. *Pain Clinic* 2005;**1**:107-15.

Ozudogru 2023 *{published data only}*

Ozudogru A, Gelecek N. Effects of closed and open kinetic chain exercises on pain, muscle strength, function, and quality of life in patients with knee osteoarthritis. *Revista da Associacao Medica Brasileira* 2023;**69**(7):e20230164. [DOI: 10.1590/1806-9282.20230164]

Pactr201711002482359 *{published data only}*

Pactr201711002482359. Efficacy of static stretching exercises versus proprioceptive neuromuscular facilitation stretching techniques on pain severity and functional perform. Pactr201711002482359 2017.

Pactr201804003266300 *{published data only}*

Pactr201804003266300. Cost-effectiveness of an outpatient self-management program for individuals with knee osteoarthritis; a randomised controlled trial [Clinical effectiveness of an outpatient self-management program for individuals with knee osteoarthritis]. Pactr201804003266300 2018.

Pactr202002763876912 *{published data only}*

Pactr202002763876912. The effects of chiropractic manipulation, mobilisation and exercise therapy on knee joint position sense in the treatment of osteoarthritis of the knee. Pactr202002763876912 2020.

Pactr202003828737019 *{published data only}*

Pactr202003828737019. Efficacy of rectus femoris stretching on pain, range of motion and spatiotemporal gait parameters in patients with knee osteoarthritis. Pactr202003828737019 2020.

Pactr202010869767272 *{published data only}*

Pactr202010869767272. Effect of oscillatory whole body vibration versus neuromuscular exercise on static and

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

dynamic balance in patients with knee osteoarthritis. Pactr202010869767272 2020.

Pactr202202919624090 *{published data only}*

Pactr202202919624090. Osteoarthritis in Ghana: understanding patient's needs and implementation of consequential intervention [Effects of an African dance-based exercise program in patients with knee osteoarthritis: a randomized controlled trial]. Pactr202202919624090 2022.

Paolillo 2018 *{published data only}*

Paolillo FR, Paolillo AR, Joao JP, Frasca D, Duchene M, Joao HA, et al. Ultrasound plus low-level laser therapy for knee osteoarthritis rehabilitation: a randomized, placebo-controlled trial. *Rheumatology International* 2018;**38**(5):785-93. [DOI: 10.1007/s00296-018-4000-x]

Park 2021 *{published data only}*

Park S, Min S, Park SH, Yoo J, Jee YS. Influence of isometric exercise combined with electromyostimulation on inflammatory cytokine levels, muscle strength, and knee joint function in elderly women with early knee osteoarthritis. *Frontiers in Physiology* 2021;**12**:(no pagination). [DOI: 10.3389/fphys.2021.688260]

Paterson 2022 *{published data only}*

Paterson G, Gaboury I, Bernick J, Wells GA, Tugwell P, Toupin-April K. A pragmatic pilot randomized controlled trial of the OA go away among individuals with osteoarthritis of the hip or knee. *Physiotherapy Canada* 2022;**74**(4):396-403. [DOI: 10.3138/ptc-2020-0110]

Peng 2021 *{published data only}*

Peng L, Yan Y, Hou X, Niu D, Wei J, Wang J. Evaluation of the effect of a three-color ladder management model for knee osteoarthritis in the community. *American Journal of Translational Research* 2021;**13**(4):3074-83.

Penninx 2001 *{published data only}*

Penninx B, Messier SP, Rejeski WJ, Williamson JD, DiBari M, Cavazzini C, et al. Physical exercise and prevention of disability in activities of daily living in older persons with osteoarthritis. *Archives of Internal Medicine* 2001;**19**:2309-16.

Penninx 2002 *{published data only}*

Penninx BW, Rejeski WJ, Pandya J, Miller ME, Di Bari M, Applegate WB, et al. Exercise and depressive symptoms: a comparison of aerobic and resistance exercise effects on emotional and physical function in older persons with high and low depressive symptomatology. *Journal of Gerontology* 2002;**57B**:P127-32.

Pereira 2011 *{published data only}*

Pereira AP, Tossige-Gomes R, Cunha N, Rocha-Vieira E, Coimbra CC, et al. The effect of adding whole-body vibration to squat training on the functional performance and self-report of disease status in elderly patients with knee osteoarthritis: a randomized, controlled clinical study. *Journal of Alternative and Complementary Medicine (New York, N.Y.)* 2011;**17**(12):1149-55.

Petersen 2010 *{published data only}*

Petersen SG, Saxne T, Heinegard D, Hansen M, Holm L, Koskinen S, et al. Glucosamine but not ibuprofen alters cartilage turnover in osteoarthritis patients in response to physical training. *Osteoarthritis and Cartilage* 2010;**18**(1):34-40.

Petersen 2011 *{published data only}*

Petersen SG, Beyer N, Hansen M, Holm L, Aagaard P, Mackey AL, et al. Nonsteroidal anti-inflammatory drug or glucosamine reduced pain and improved muscle strength with resistance training in a randomized controlled trial of knee osteoarthritis patients. *Archives of Physical Medicine and Rehabilitation* 2011;**92**(8):1185-93.

Peterson 1993 *{published data only}*

Peterson MG, Kovar-Toledano PA, Otis JC, Allegrante JP, Mackenzie CR, Gutin B, et al. Effect of a walking program on gait characteristics in patients with osteoarthritis. *Arthritis Care & Research* 1993;**6**:11-6.

Petrella 2000 *{published data only}*

Petrella RJ, Bartha C. Home based exercise therapy for older patients with knee osteoarthritis: a randomized clinical trial. *Journal of Rheumatology* 2000;**27**:2215-21.

Pietrosimone 2010 *{published data only}*

Pietrosimone BG, Saliba SA, Hart JM, Hertel J, Kerrigan DC, Ingersoll CD. Effects of disinhibitory transcutaneous electrical nerve stimulation and therapeutic exercise on sagittal plane peak knee kinematics and kinetics in people with knee osteoarthritis during gait: a randomized controlled trial. *Clinical Rehabilitation* 2010;**24**(12):1091-101.

Pietrosimone 2012 *{published data only}*

Pietrosimone BG, Saliba SA. Changes in voluntary quadriceps activation predict changes in quadriceps strength after therapeutic exercise in patients with knee osteoarthritis. *Knee* 2012;**19**(6):939-43.

Pietrosimone 2020 *{published data only}*

Pietrosimone B, Luc-Harkey BA, Harkey MS, Davis-Wilson HC, Pfeiffer SJ, Schwartz TA, et al. Using TENS to enhance therapeutic exercise in individuals with knee osteoarthritis. *Medicine and Science in Sports and Exercise* 2020;**52**(10):2086-95. [DOI: 10.1249/MSS.0000000000002353]

Pisters 2010a *{published data only}*

Pisters MF, Veenhof C, de Bakker DH, Schellevis FG, Dekker J. Behavioural graded activity results in better exercise adherence and more physical activity than usual care in people with osteoarthritis: a cluster-randomised trial. *Journal of Physiotherapy* 2010;**56**(1):41-7.

Pisters 2010b *{published data only}*

Pisters MF, Veenhof C, Schellevis FG, De Bakker DH, Dekker J. Long-term effectiveness of exercise therapy in patients with osteoarthritis of the hip or knee: a randomized controlled trial comparing two different physical therapy interventions. *Osteoarthritis and Cartilage* 2010;**18**(8):1019-26.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Piva 2011 *{published data only}*

Piva SR, Fitzgerald GK. Effects of impairment-based exercise on performance of specific self-reported functional tasks in individuals with knee osteoarthritis. *Physical Therapy* 2011;**91**(12):1752-65.

Piyakhachornrot 2011 *{published data only}*

Piyakhachornrot N, Aree-Ue S, Putwatana P, Kawinwonggowit V. Impact of an integrated health education and exercise program in middle-aged Thai adults with osteoarthritis of the knee. *Orthopaedic Nursing* 2011;**30**(2):134-42.

Quicke 2016 *{published data only}*

Quicke JG, Foster NE, Croft PR, Ogollah RO, Holden MA. Is change in physical activity level associated with future clinical outcomes in terms of pain and physical function in older adults with knee pain? *Osteoarthritis and Cartilage* 2016;**1**:S497.

Quilty 2003 *{published data only}*

Quilty B, Tucker M, Campbell R, Dieppe P. Physiotherapy, including quadriceps exercises and patellar taping, for knee osteoarthritis with predominant patello-femoral joint involvement: randomized controlled trial. *Journal of Rheumatology* 2003;**30**:1311-7.

Quirk 1985 *{published data only}*

Quirk AS, Newman RJ, Newman KJ. An evaluation of interferential therapy, shortwave diathermy and exercise in the treatment of osteoarthrosis of the knee. *Physiotherapy* 1985;**71**:55-7.

Rafiq 2021d *{published data only}*

Rafiq MT, Hamid MS, Hafiz E, Rashid K, Chaudhary FA. The role of rehabilitation exercises on weight, functional strength, and exercise adherence in knee osteoarthritis patients. *Current Rheumatology Reviews* 2021;**17**(4):397-403. [DOI: 10.2174/1573397117666210719101551]

Raj 2018 *{published data only}*

Raj NB, Shokri AA, Saha S, Razali H, Othman YN, Mahadeva Rao US. Effect of isokinetic training on quadriceps muscle strength in osteoarthritis of knee. *Research Journal of Pharmacy and Technology* 2018;**11**(6):2517-24. [DOI: 10.5958/0974-360X.2018.00465.1]

Rathwa 2019 *{published data only}*

Rathwa A, Prajapati N, Deepak AN. To compare the effectiveness of proprioceptive circuit exercises versus open kinematics chain exercises on pain and improve muscle strength and physical function in osteoarthritis knee patients. *Indian Journal of Physiotherapy & Occupational Therapy* 2019;**13**(1):88-92. [DOI: 10.5958/0973-5674.2019.00017.0]

Rattanachaiyanont 2008 *{published data only}*

Rattanachaiyanont M, Kuptniratsaikul V. No additional benefit of shortwave diathermy over exercise program for knee osteoarthritis in peri-/post-menopausal women: an equivalence trial. *Osteoarthritis and Cartilage* 2008;**16**(7):823-8.

Ravaud 2004 *{published data only}*

Ravaud P, Giraudeau B, Logeart I, Larguier JS, Rolland D, et al. Management of osteoarthritis with an unsupervised home based exercise programme and/or patient administered assessment tools. A cluster randomised controlled trial with a 2x2 factorial design. *Annals of the Rheumatic Diseases* 2004;**63**:703-8.

RBR-10j4bw25 *{published data only}*

RBR-10j4bw25. Effect of balance training and walking with a flat, flexible sole on elderly women with knee arthrosis. RBR-10j4bw25 2021.

RBR-2twzy65 *{published data only}*

RBR-2twzy65. The effect of different exercise models on elderly people with knee osteoarthritis [Effects of different exercise models with and without blood flow restriction on strength, muscle mass, functionality and pain perception in patients with knee osteoarthritis]. RBR-2twzy65 2021.

RBR-4pzsfxw *{published data only}*

RBR-4pzsfxw. Effects of adding gait retraining to lower limb muscle strengthening on clinical, functional and kinematic parameters of individuals with knee arthrosis. RBR-4pzsfxw 2023.

RBR-7p8wzyr *{published data only}*

RBR-7p8wzyr. Use of video games and conventional physical therapy to improve balance and postural control in patients with knee arthrosis: a randomized clinical trial. RBR-7p8wzyr 2022.

RBR-7t6nzr *{published data only}*

Effects of laser and physical exercise program in patients with knee arthritisPhototherapy and physical exercise program in patients with knee osteoarthritis: controlled, randomized and triple blinded study. RBR-7t6nzr 2018.

RBR-8qby2wt *{published data only}*

RBR-8qby2wt. Effects of a remotely supervised home exercise program, during the COVID-19 pandemic, on the functional capacity and cognitive function of elderly community members: a randomized clinical trial. RBR-8qby2wt 2021.

RBR-9d7c7b *{published data only}*

RBR-9d7c7b. Effects of electrical stimulation over the skull and tight together with exercises on knee osteoarthritis. RBR-9d7c7b 2016.

Reid 2010 *{published data only}*

Reid DA, McNair PJ. Effects of an acute hamstring stretch in people with and without osteoarthritis of the knee. *Physiotherapy* 2010;**96**(1):14-21.

Reid 2011 *{published data only}*

Reid DA, McNair PJ. Effects of a six week lower limb stretching programme on range of motion, peak passive torque and stiffness in people and without osteoarthritis of the knee. *New Zealand Journal of Physiotherapy* 2011;**39**(1):5-12.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Ren 2020 *{published data only}*

Ren Y, Xue H, Li A. Effect of microscope intervention on taijiquan on clinical rehabilitation of elderly patients with knee arthritis. *Investigacion Clinica (Venezuela)* 2020;**61**(1):312-20.

Rizvi 2023 *{published data only}*

Rizvi MR, Sharma A, Hasan S, Ahmad F, Asad MR, Iqbal A, et al. Exploring the impact of integrated polyvagal exercises and knee reinforcement in females with grade II knee osteoarthritis: a randomized controlled trial. *Scientific Reports* 2023;**13**(1):18964. [DOI: 10.1038/s41598-023-45908-4]

Robbins 2021 *{published data only}*

Robbins SR, Melo LR, Urban H, Deveza LA, Asher R, Johnson V, et al. Effectiveness of stepped-care intervention in overweight and obese patients with medial tibiofemoral osteoarthritis: a randomized controlled trial. *Arthritis Care & Research* 2021;**73(4)**:520-30. [DOI: 10.1002/acr.24148]

Robson 2022 *{published data only}*

Robson E, Kamper SJ, Lee H, Palazzi K, O'Brien KM, Williams A, et al. Compliance with telephone-based lifestyle weight loss programs improves low back pain but not knee pain outcomes: complier average causal effects analyses of 2 randomised trials. *Pain* 2022;**163**(7):E862-E868. [DOI: 10.1097/j.pain.0000000000002506]

Runhaar 2019 *{published data only}*

Runhaar J, Beavers D, Miller G, Nicklas B, Loeser R, Bierma-Zeinstra S, et al. Inflammatory cytokines mediate the effects of diet and exercise on pain and function in knee osteoarthritis, independent of BMI. *Osteoarthritis and Cartilage* 2019;**27 Suppl 1**:S453. [DOI: 10.1016/j.joca.2019.02.492]

Sabharwal 2022 *{published data only}*

Sabharwal J, Joshi S. Neuromuscular exercise with neuromuscular electrical stimulation in knee osteoarthritis: a randomised controlled pilot trial. *Advances in Rehabilitation* 2022;**36**(4):8-16. [DOI: 10.5114/areh.2022.121561]

Saji 2023 *{published data only}*

Saji VT, Kotteeswaran K. Feasibility of an exercise protocol on stable and unstable surfaces on lower extremity function, pain and balance in postmenopausal women with OA knee. - A pilot study. *Journal of Population Therapeutics and Clinical Pharmacology* 2023;**30**(17):2133-8. [DOI: 10.53555/jptcp.v30i17.2950]

Sandal 2016 *{published data only}*

Sandal LF, Roos EM, Bogesvang SJ, Thorlund JB. Pain trajectory and exercise-induced pain flares during 8 weeks of neuromuscular exercise in individuals with knee and hip pain. *Osteoarthritis and Cartilage* 2016;**24**(4):589-92. [DOI: 10.1016/j.joca.2015.11.002]

Sayers 2012 *{published data only}*

Sayers SP, Gibson K, Cook CR. Effect of high-speed power training on muscle performance, function, and pain in older adults with knee osteoarthritis: a pilot investigation. *Arthritis Care & Research* 2012;**64**(1):46-53.

Schlenk 2011 *{published data only}*

Schlenk EA, Lias JL, Sereika SM, Dunbar-Jacob J, Kwoh CK. Improving physical activity and function in overweight and obese older adults with osteoarthritis of the knee: a feasibility study. *Rehabilitation Nursing Journal* 2011;**36**(1):32-42.

Schmid 2013 *{published data only}*

Schmid A, McAlindon T, Schmid CH, Wang C. The influence of tai Chi exercise on proprioception in patients with knee osteoarthritis: results from a pilot randomized controlled trial. *International Journal of Integrative Medicine* 2013;**1**:no pagination. [DOI: 10.5772/57137]

Scopaz 2009 *{published data only}*

Scopaz KA, Piva SR, Gil AB, Woollard JD, Oddis CV, Fitzgerald GK. Effect of baseline quadriceps activation on changes in quadriceps strength after exercise therapy in subjects with knee osteoarthritis. *Arthritis & Rheumatism: Arthritis Care & Research* 2009;**61**(7):951-7.

Selfe 2008 *{published data only}*

Selfe TK, Bourguignon C, Taylor AG. Effects of noninvasive interactive neurostimulation on symptoms of osteoarthritis of the knee: a randomized, sham-controlled pilot study. *Journal of Alternative and Complementary Medicine (New York, N.Y.)* 2008;**14**(9):1075-81.

Sen 2004 *{published data only}*

Sen A, Gocen Z, Unver B, Karatosun V, Gunal I. The frequency of visits by the physiotherapist of patients receiving home-based exercise therapy for knee osteoarthritis. *Knee* 2004;**11**:151-3.

Shakoor 2007 *{published data only}*

Shakoor MA, Taslim MA, Hossain MS. Effects of activity modification on the patients with osteoarthritis of the knee. *Bangladesh Medical Research Council Bulletin* 2007;**33**(2):55-9.

Shakoor 2010 *{published data only}*

Shakoor MA, Rahman MS, Azad AK, Islam MS. Effects of isometric quadriceps muscle strengthening exercise on chronic osteoarthritis of the knee. *Bangladesh Medical Research Council Bulletin* 2010;**36**(1):20-2.

Shaughnessy 2023 *{published data only}*

Shaughnessy AF. Tailored exercise program improves function and reduces pain due to knee osteoarthritis. *American Family Physician* 2023;**108**(3):Online-.

Shen 2008 *{published data only}*

Shen CL, James CR, Chyu MC, Bixby WR, Brismée JM, Zumwalt MA, et al. Effects of Tai Chi on gait kinematics, physical function, and pain in elderly with knee osteoarthritis-a pilot study. *American Journal of Chinese Medicine* 2008;**36**(2):219-32.

Silva 2008 *{published data only}*

Silva LE, Valim V, Pessanha AP, Oliveira LM, Myamoto S, Jones A, et al. Hydrotherapy versus conventional land-based exercise for the management of patients with osteoarthritis of the knee: a randomized clinical trial. *Physical Therapy* 2008;**88**(1):12-21.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Silva 2018 *{published data only}*

Silva C, Amaro A, Pinho A, Goncalves RS, Rodrigues M, Ribeiro F. Feasibility of a home-based therapeutic exercise program in individuals with knee osteoarthritis. *Archives of Rheumatology* 2018;**33**(3):295-301. [DOI: 10.5606/ArchRheumatol.2018.6633]

Simao 2019 *{published data only}*

Simao AP, Mendonca VA, Avelar NC, Fonseca SF, Santos JM, Oliveira AC, et al. Whole body vibration training on muscle strength and brain-derived neurotrophic factor levels in elderly woman with knee osteoarthritis: a randomized clinical trial study. *Frontiers in Physiology* 2019;**10**(Jun):(no pagination). [DOI: 10.3389/fphys.2019.00756]

Sled 2010 *{published data only}*

Sled EA, Khoja L, Deluzio KJ, Olney SJ, Culham EG. Effect of a home program of hip abductor exercises on knee joint loading, strength, function, and pain in people with knee osteoarthritis: a clinical trial. *Physical Therapy* 2010;**90**(6):895-904.

Smith 2023 *{published data only}*

Smith JL, Innes AQ, Burns DS, Deniszczyc D, Selfe J, MacConville S, et al. A scalable 12-week exercise and education programme reduces symptoms and improves function and wellbeing in people with hip and knee osteoarthritis. *Frontiers in Rehabilitation Sciences* 2023;**4**:1147938. [DOI: 10.3389/fresc.2023.1147938]

Song 2010 *{published data only}*

Song R, Roberts BL, Lee E, Lam P, Bae S. A randomized study of the effects of t'ai Chi on muscle strength, bone mineral density, and fear of falling in women with osteoarthritis. *Journal of Alternative and Complementary Medicine (New York, N.Y.)* 2010;**16**(3):227-33.

Song 2022 *{published data only}*

ChiCTR2000040721. Effects of Tai Chi exercise on patients with early osteoarthritis: a follow-up study [Effect of Tai Chi exercise on elderly patients with osteoarthritis]. ChiCTR2000040721 2020.

Song J, Wei L, Cheng K, Lin Q, Xia P, Wang X, et al. The effect of modified tai Chi exercises on the physical function and quality of life in elderly women with knee osteoarthritis. *Frontiers in Aging Neuroscience* 2022;**14**:(no pagination). [DOI: 10.3389/fnagi.2022.860762]

Soni 2012 *{published data only}*

Soni A, Joshi A, Mudge N, Wyatt M, Williamson L. Supervised exercise plus acupuncture for moderate to severe knee osteoarthritis: a small randomised controlled trial. *Acupuncture in Medicine* 2012;**30**(3):176-81.

Stitik 2007a *{published data only}*

Stitik TP, Blacksin MF, Stiskal DM, Kim JH, Foye PM, Schoenheer L, et al. Efficacy and safety of hyaluronan treatment in combination therapy with home exercise for knee osteoarthritis pain. *Archives of Physical and Medical Rehabiltation* 2007;**88**:135-41.

Stitik 2007b *{published data only}*

Stitik TP, Blacksin MF, Stiskal DM, Kim JH, Foye PM, Schoenherr L, et al. Efficacy and safety of hyaluronan treatment in combination therapy with home exercise for knee osteoarthritis pain. *Archives of Physical Medicine and Rehabilitation* 2007;**88**(2):135-41.

Sudhesh 2013 *{published data only}*

Sudhesh P. Effect of taping and closed kinetic chain versus traditional approach in osteoarthritis knee. *International Journal of Pharma and Bio Sciences* 2013;**4**(4):B1156-B1165.

Sullivan 1998 *{published data only}*

Sullivan T, Allegrante JP, Peterson MG, Kovar PA, MacKenzie CR. One-year followup of patients with osteoarthritis of the knee who participated in a program of supervised fitness walking and supportive patient education. *Arthritis Care & Research* 1998;**11**:228-33.

Swank 2011 *{published data only}*

Swank AM, Kachelman JB, Bibeau W, Quesada PM, Nyland J, Malkani A, et al. Prehabilitation before total knee arthroplasty increases strength and function in older adults with severe osteoarthritis. *Journal of Strength and Conditioning Research* 2011;**25**(2):318-25.

Sylvester 1989 *{published data only}*

Sylvester KL. Investigation of the effect of hydrotherapy in the treatment of osteoarthritic hips. *Clinical Rehabilitation* 1989;**4**:223-8.

Tan 2020 *{published data only}*

Tan BY, Ding BT, Pereira MJ, Skou ST, Thumboo J, Car J. Collaborative model of care between orthopaedics and allied healthcare professionals trial (CONNACT) - a feasibility study in patients with knee osteoarthritis using a mixed method approach. *BMC Musculoskeletal Disorders* 2020;**21**(1):592. [DOI: 10.1186/s12891-020-03611-9]

Tan 2023 *{published data only}*

Tan B, Pereira M, Yang SY, Lim C, Tan C, Woon E, et al. Collaborative model of care between orthopaedics and allied healthcare professionals (connact) for knee osteoarthritis: 12-months follow-up of an effectiveness-implementation hybrid trial. *Osteoarthritis and Cartilage* 2023;**31 Suppl 1**:S16-S17. [DOI: 10.1016/j.joca.2023.01.502]

Tctr20170705003 *{published data only}*

Tctr20170705003. Effect of elastic band exercise on muscle strength and physical function in severe knee osteoarthritis. Tctr20170705003 2017.

Tctr20210903007 *{published data only}*

Tctr20210903007. Efficacy of blood flow restriction with low-load resistance exercise in adults who have risk of symptomatic knee osteoarthritis. (Double-blind randomized controlled trial). Tctr20210903007 2021.

Tctr20221118001 *{published data only}*

Tctr20221118001. The efficacy and safety of interferential current therapy (IFC) plus exercise compared to sham IFC plus

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exercise for pain relief in patients with knee osteoarthritis. Tctr20221118001 2022.

Tctr20230502005 *{published data only}*

Tctr20230502005. The effect of aromatherapy combined with exercise on knee pain of the elderly in institutions. Tctr20230502005 2023.

Teixeira 2011 *{published data only}*

Teixeira PE, Piva SR, Fitzgerald GK. Effects of impairment-based exercise on performance of specific self-reported functional tasks in individuals with knee osteoarthritis. *Physical Therapy* 2011;**91**(12):1752-65.

Thiengwittayaporn 2009 *{published data only}*

Thiengwittayaporn S, Wetpiryakul P, Foosakun Y, Ngamsom T, Vathanavit P, Pintongtun J. Comparison of the accuracy of quadriceps isometric exercise between using quadriceps education device (QED) and not using QED for osteoarthritic knee patients: a randomized controlled trial. *Chotmaihet Thangphaet [Journal of the Medical Association of Thailand]* 2009;**92 Suppl 6**:S33-8.

Thiengwittayaporn 2023 *{published data only}*

Thiengwittayaporn S, Wattanapreechanon P, Sakon P, Peethong A, Ratisoontorn N, Charoenphandhu N, et al. Development of a mobile application to improve exercise accuracy and quality of life in knee osteoarthritis patients: a randomized controlled trial. *Archives of Orthopaedic and Trauma Surgery* 2023;**143**(2):729-38. [DOI: 10.1007/s00402-021-04149-8]

Toda 2001 *{published data only}*

Toda Y. The effect of energy restriction, walking and exercise on lower extremity lean body mass in obese women with osteoarthritis of the knee. *Journal of Orthopaedic Science* 2001;**6**:148-54.

Tok 2011 *{published data only}*

Tok F, Aydemir K, Peker F, Safaz I, Taskaynatan MA, Ozgul A. The effects of electrical stimulation combined with continuous passive motion versus isometric exercise on symptoms, functional capacity, quality of life and balance in knee osteoarthritis: randomized clinical trial. *Rheumatology International* 2011;**31**(2):177-81.

Topp 2009 *{published data only}*

Topp R, Swank AM, Quesada PM, Nyland J, Malkani A. The effect of prehabilitation exercise on strength and functioning after total knee arthroplasty. *PM&R* 2009;**1**(8):729-35.

Torstensen 2023a *{published data only}*

Torstensen TA, Osteras H, LoMartire R, Rugelbak GM, Grooten WJ, Ang BO. High-versus low-dose exercise therapy for knee osteoarthritis a randomized controlled multicenter trial. *Annals of Internal Medicine* 2023;**176**(2):154-66. [DOI: 10.7326/M22-2348]

Torstensen 2023b *{published data only}*

Torstensen TA, Osteras H, LoMartire R, Rugelbak GM, Grooten WJ, Ang BO. High- versus low-dose exercise therapy for knee osteoarthritis. *Annals of Internal Medicine* 2023;**176**(7):eL230141. [DOI: 10.7326/L23-0141]

Tsai 2009 *{published data only}*

Tsai PF, Chang JY, Beck C, Hagen J, Anand KJ, Kuo YF, et al. The feasibility of implementing tai Chi for nursing home residents with knee osteoarthritis and cognitive impairment. *Activities Directors' Quarterly for Alzheimers and Other Dementia Patients* 2009;**10**(1):9-17.

Tsai 2015 *{published data only}*

Tsai LC, Lee SJ, Yang AJ, Ren Y, Press JM, Zhang LQ. Effects of off-axis elliptical training on reducing pain and improving knee function in individuals with patellofemoral pain. *Clinical Journal of Sport Medicine.* 2015;**14**:(no pagination). [DOI: 10.1097/JSM.0000000000000164]

Tsauo 2008 *{published data only}*

Tsauo J, Cheng P, Yang R. The effects of sensorimotor training on knee proprioception and function for patients with knee osteoarthritis: a preliminary report. *Clinical Rehabilitation* 2008;**22**(5):448-57.

Tunay 2010 *{published data only}*

Tunay VB, Baltaci G, Atay AO. Hospital-based versus home-based proprioceptive and strengthening exercise programs in knee osteoarthritis. *Acta Orthopaedica et Traumatologica Turcica* 2010;**44**(4):270-7.

Tung 2022 *{published data only}*

Tung EV, Wilson EC, Maly MR. Muscle capacity utilization during gait was improved by strengthening exercise in women with symptomatic knee osteoarthritis. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S149-S150. [DOI: 10.1016/j.joca.2022.02.189]

Tuzun 2004 *{published data only}*

Tuzun EH, Aytar A, Eker L, Daskapan A. Effectiveness of two different physical therapy programmes in the treatment of knee osteoarthritis. *Pain Clinic* 2004;**16**:379-87.

Umin000016326 *{published data only}*

Umin000016326. Effects of body weight supported walking exersise using lower body positive pressure device on physical performance in elderly with leg pain. Umin000016326 2015.

Umin000022060 *{published data only}*

Prospective comparison study on efficacy between physiotherapy and kinesitherapy for knee osteoarthritis in improvement of clinical symptoms and degeneration. Umin000022060 2016.

Umin000028944 *{published data only}*

Umin000028944. Effect of tibial internal-rotation exercise and insole on osteoarthiritis of the knee. Umin000028944 2017.

Umin000029045 *{published data only}*

Umin000029045. Effect of exercise therapy with tibial internal-rotation on clinical outcomes and knee kinematics of osteoarthritis of the knee: a randomized controlled trial. Umin000029045 2017.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Umin000034385 *{published data only}*

Umin000034385. On the effect of triceps training at home in elderly women with medial knee osteoarthritis. Umin000034385 2019.

Umin000035225 *{published data only}*

Umin000035225. Effectiveness of supervised exercise therapy for community-dwelling people with chronic knee pain: a randomised controlled trial. Umin000035225 2018.

Umin000039579 *{published data only}*

Umin000039579. Randomized controlled study on efficacy of kinesitherapy using therapeutic modality in patients with knee osteoarthritis. Umin000039579 2020.

Umin000043356 *{published data only}*

Umin000043356. Clinical trial for effectiveness of the video-based home exercise on outcomes for subjects with osteoarthritis of the knee. Umin000043356 2021.

Umin000044440 *{published data only}*

Umin000044440. Effect of pain neuroscience education and mind-body exercises in patients with knee osteoarthritis. Umin000044440 2021.

Umin000044884 *{published data only}*

Effect of basic body awareness therapy on geriatric knee osteoarthritis patients with chronic pain and depressionEffect of basic body awareness therapy on geriatric knee osteoarthritis patients with chronic pain and depression - Basic body awareness thera. Umin000044884 2021.

Umin000051857 *{published data only}*

Jprn-Umin000051857. Are there differences between the Ottawa Panel evidence-based clinical guidelines on pain, function, and psychological factors in individuals with knee osteoarthritis? Jprn-Umin000051857 2023.

Van Gool 2005 *{published data only}*

Van Gool CH, Penninx BW, Kempen GI, Rejeski WJ, Miller GD, Van Eijk JT, et al. Effects of exercise adherence on physical function among overweight older adults with knee osteoarthritis. *Arthritis Care & Research* 2005;**53**:24-32.

Veenhof 2007 *{published data only}*

Coupe VM, Veenhof C, Van Tulder MW, Dekker J, Bijlsma JW, Van den Ende CH. The cost effectiveness of behavioural graded activity in patients with osteoarthritis of hip and/or knee. *Annals of the Rheumatic Diseases* 2007;**66**(2):215-21.

Veenhof C, Dekker J, Koke AJ, Oostendorp RA, Bijlsma JW. Which patients with osteoarthritis of hip and/or knee benefit most from behavioral graded activity? *International Journal of Behavioral Medicine* 2007;**14**(2):86-91.

Villadsen 2014 *{published data only}*

Villadsen A, Overgaard S, Holsgaard-Larsen A, Christensen R, Roos EM. Immediate efficacy of neuromuscular exercise in patients with severe osteoarthritis of the hip or knee: a secondary analysis from a randomized controlled trial. *Journal of Rheumatology* 2014;**41**(7):1385-94. [DOI: 10.3899/jrheum.130642]

Walls 2010 *{published data only}*

Walls RJ, McHugh G, O'Gorman DJ, Moyna N, O'Byrne JM. Effects of preoperative neuromuscular electrical stimulation on quadriceps strength and functional recovery in total knee arthroplasty. A pilot study. *BMC Musculoskeletal Disorders* 2010;**11**:119.

Wan 2023 *{published data only}*

Wan Y, McGuigan P, Bilzon J, Wade L. The effectiveness of a 6-week biofeedback gait retraining programme in people with knee osteoarthritis: protocol for a randomised controlled trial. *BMC Musculoskeletal Disorders* 2023;**24**(1):(no pagination). [DOI: 10.1186/s12891-023-07098-y]

Wang 2006 *{published data only}*

Wang T-J, Belza B, Thompson FE, Whitney JD, Bennett K. Effects of aquatic exercise on flexibility, strength and aerobic fitness in adults with osteoarthritis of the hip or knee. *Journal of Advanced Nursing* 2006;**57**:141-52.

Wang 2007a *{published data only}*

Wang T, Belza B, Thompson EF, Whitney JD, Bennett K. Effects of aquatic exercise on flexibility, strength and aerobic fitness in adults with osteoarthritis of the hip or knee. *Journal of Advanced Nursing* 2007;**57**(2):141-52.

Wang 2007b *{published data only}*

Wang X, Miller GD, Messier SP, Nicklas BJ. Knee strength maintained despite loss of lean body mass during weight loss in older obese adults with knee osteoarthritis. *Journals of Gerontology Series A-Biological Sciences & Medical Sciences* 2007;**62**(8):866-71.

Wang 2009 *{published data only}*

Wang C, Schmid CH, Hibberd PL, Kalish R, Roubenoff R, Rones R, et al. Tai Chi is effective in treating knee osteoarthritis: a randomized controlled trial. *Arthritis and Rheumatism* 2009;**61**(11):1545-53.

Weng 2009 *{published data only}*

Weng MC, Lee CL, Chen CH, Hsu JJ, Lee WD, Huang MH, et al. Effects of different stretching techniques on the outcomes of isokinetic exercise in patients with knee osteoarthritis. *Kaohsiung Journal of Medical Sciences* 2009;**25**(6):306-15.

Whitehurst 2011 *{published data only}*

Whitehurst DG, Bryan S, Hay EM, Thomas E, Young J, Foster NE. Cost-effectiveness of acupuncture care as an adjunct to exercise-based physical therapy for osteoarthritis of the knee. *Physical Therapy* 2011;**91**(5):630-41.

Williamson 2007 *{published data only}*

Williamson L, Wyatt MR, Yein K, Melton JT. Severe knee osteoarthritis: a randomized controlled trial of acupuncture, physiotherapy (supervised exercise) and standard management for patients awaiting knee replacement. *Rheumatology* 2007;**46**:1445-9.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Williamson 2007b *{published data only}*

Williamson L, Wyatt MR, Yein K, Melton JT. Severe knee osteoarthritis: a randomized controlled trial of acupuncture, physiotherapy (supervised exercise) and standard management for patients awaiting knee replacement. *Rheumatology (Oxford, England)* 2007;**46**(9):1445-9.

Wong 2017 *{published data only}*

Wong JB, Chung M, Price LL, Wang C. Cost-effectiveness of Tai Chi versus physical therapy for knee osteoarthritis. *Arthritis and Rheumatology. Conference: American College of Rheumatology/ Association of Rheumatology Health Professionals Annual Scientific Meeting, ACR/ARHP* 2017;**69 Suppl 10**:(no pagination).

Wong 2018 *{published data only}*

Wong JB, Chung M, Price L, Wang C. Tai Chi versus physical therapy for knee osteoarthritis: a cost-effectiveness analysis. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S312-S313.

Wyatt 2001 *{published data only}*

Wyatt FB, Milam S, Manske RC, Deere R. The effects of aquatic and traditional exercise programs on persons with knee osteoarthritis. *Journal of Strengthening Research* 2001;**15**:337-40.

Xiao 20201 *{published data only}*

Xiao C, Zhuang Y, Kang Y. Effects of Wu Qin xi Qigong exercise on physical functioning in elderly people with knee osteoarthritis: a randomized controlled trial. *Geriatrics & Gerontology International* 2020;**20**(10):899-903. [DOI: 10.1111/ggi.14007]

Xiao 2020b *{published data only}*

Xiao CM, Li JJ, Kang Y, Zhuang YC. Follow-up of a Wuqinxi exercise at home programme to reduce pain and improve function for knee osteoarthritis in older people: a randomised controlled trial. *Age and Ageing* 2020;**14**:no pagination. [DOI: 10.1093/ageing/afaa179]

Yilmaz 2010 *{published data only}*

Yilmaz OO, Senocak O, Sahin E, Baydar M, Gulbahar S, Bircan C, et al. Efficacy of EMG-biofeedback in knee osteoarthritis. *Rheumatology International* 2010;**30**(7):887-92.

Yip 2007a *{published data only}*

Yip YB, Wit JW, Fung KK, Wong DY, Chong SY, Chung LH, et al. Impact of an arthritis self-management programme with an added exercise component for osteoarthritic knee sufferers on improving pain, functional outcomes, and use of health care services: an experimental study. *Patient Education Counseling* 2007;**65**:113-21.

Yip 2007b *{published data only}*

Yip YB, Sit JW, Fung KK, Wong DY, Chong SY, Chung LH, et al. Effects of a self-management arthritis programme with an added exercise component for osteoarthritic knee: randomized controlled trial. *Journal of Advanced Nursing* 2007;**59**(1):20-8.

Yip 2008 *{published data only}*

Yip Y, Sit JW, Wong DY, Chong SY, Chung L. A 1-year follow-up of an experimental study of a self-management arthritis programme with an added exercise component of clients with osteoarthritis of the knee. *Psychology, Health & Medicine* 2008;**13**(4):402-14.

Yue 2023 *{published data only}*

Yue H, Li Y, Ma J, Xie C, Xie F, Cai J, et al. Effect of Tai Chi on knee pain and muscle strength in middle-aged and older adults with knee osteoarthritis: a randomized controlled trial protocol. *BMC Complementary Medicine and Therapies* 2023;**23**(1):(no pagination). [DOI: 10.1186/s12906-023-04070-0]

Zhang 2020 *{published data only}*

ChiCTR-TRC-13003264. Effects of Tai Chi program on neuromuscular function for patients with knee osteoarthritis [Joint mechanical load and osteoarthritis: biomechanical based innovative Tai Chi rehabilitation strategies]. ChiCTR-TRC-13003264 2013.

Wang XQ, Huang LY, Liu Y, Li JX, Wu X, Li HP, et al. Effects of tai Chi program on neuromuscular function for patients with knee osteoarthritis: study protocol for a randomized controlled trial. *Trials* 2013;**14**(1):(no pagination). [DOI: 10.1186/1745-6215-14-375]

Zhang Z, Huang L, Liu Y, Wang L, Bernardo-Filho M. Effect of tai Chi training on plantar loads during walking in individuals with knee osteoarthritis. *BioMed Research International* 2020;**2020**: (no pagination). [DOI: 10.1155/2020/3096237]

Zhang 2022 *{published data only}*

Zhang Y, Xie S, Wang X, Song K, Wang L, Zhang R, et al. Effects of Internet of things-based power cycling and neuromuscular training on pain and walking ability in elderly patients with KOA: protocol for a randomized controlled trial. *Trials* 2022;**23**(1):(no pagination). [DOI: 10.1186/s13063-022-06946-x]

Zhao 2019 *{published data only}*

Zhao Z, Wang R, Guo Y, Chen L, Wang K, Zhou H, et al. Static low-angle squatting reduces the intra-articular inflammatory cytokines and improves the performance of patients with knee osteoarthritis. *BioMed Research International* 2019;**2019**:9617923. [DOI: 10.1155/2019/9617923]

Zhu 2023 *{published data only}*

Zhu GC, Chen KM, Belcastro F. Comparing different stretching exercises on pain, stiffness, and physical function disability in older adults with knee osteoarthritis. *Archives of Physical Medicine and Rehabilitation.* 2023;**1**:(no pagination). [DOI: 10.1016/j.apmr.2023.07.001]

References to studies awaiting assessment

Abou-Raya 2013 *{published data only}*

Abou-Raya S, Abou-Raya A, Abd Kareem A. Effects of physical activity on inflammation, skeletal muscle strength/function (sarcopenia) and fat infiltration (sarcopenic obesity) in older adults with knee osteoarthritis: a randomized controlled trial. *Annals of the Rheumatic Diseases. Conference: Annual European Congress of Rheumatology of the European League Against Rheumatism, EULAR* 2013;**72 Suppl 3**:1498-503. [DOI: 10.1136/annrheumdis-2013-eular.2039]

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Actrn12617000183381 *{published data only}*

Actrn12617000183381. Exergaming for people with knee osteoarthritis: a feasibility study [Does exergaming increase muscle strength and decrease instability in people with knee osteoarthritis? A feasibility study]. Actrn12617000183381 2017.

Bossen 2013 *{published data only}*

Bossen D, Veenhof C, Van Beek KE, Spreeuwenberg PM, Dekker J, De Bakker DH. Effectiveness of a web-based physical activity intervention in patients with knee and/or hip osteoarthritis: randomized controlled trial. *Journal of Medical Internet Research* 2013;**15**(11):e257. [DOI: 10.2196/jmir.2662]

Carlos 2012 *{published data only}*

Carlos KP, Belli BS, Alfredo PP. Effect of pulsed ultrasound and continuous ultrasound linked to exercise in patients with knee osteoarthritis: pilot study [Portuguese]. *Fisioterapia e Pesquisa* 2012;**19**(3):275-81.

Chang 2017 *{published data only}*

Chang AH, Lee J, Song J, Price L, Lee AC, Reid KF, et al. Associations between pre-intervention physical activity levels and treatment response to exercise therapy in persons with knee osteoarthritis. *Osteoarthritis and Cartilage* 2017;**25 Suppl 1**:S339.

Cheing 2002 *{published data only}*

Cheing GL, Hui-Chan CW, Chan KM. Does four weeks of TENS and/or isometric exercise produce cumulative reduction of osteoarthritic knee pain? *Clinical Rehabilitation* 2002;**16**:749-60.

Cheing GL, Hui-Chan CW. Would the addition of TENS to exercise training produce better physical performance outcomes in people with knee osteoarthritis than either intervention alone? *Clinical Rehabilitation* 2004;**18**:487.

Coleman 2011 *{published data only}*

Coleman S, Briffa NK, Carroll G, Inderjeeth C, Cook N, McQuade J. An osteoarthritis of the knee self-management education program delivered by multidisciplinary health professionals; a randomised controlled trial. *Physiotherapy (United Kingdom)* 2011;**1**:eS1457. [DOI: 10.1016/j.physio.2011.04.003]

De Rooij 2016 *{published data only}*

De Rooij M, Van Der Leeden M, Cheung J, Van Der Esch M, Hakkinen A, Haverkamp D, et al. Effectiveness of tailored exercise therapy in patients with knee osteoarthritis and comorbidity: a randomized controlled trial. *Osteoarthritis and Cartilage* 2016;**1**:S466-S467.

Eungpinichpong 1997 *{published data only}*

Eungpinichpong W. The efficacy of physical exercise programmes for patients with osteoarthritis of the knee as determined by clinical and gait parameters. *New Zealand Journal of Physiotherapy* 1998;**26**:5.

Gallego-Ariza 2013 *{published data only}*

Gallego-Ariza B, Mateos-Toset S, Cabrera-Martos I, Correa-Toledo A, Badillo-Fontalvo M, Valenza M. Impact of a supervised muscular training on perceived quality of sleep and health in a population with knee osteoarthritis. *Sleep Medicine* 2013;**1**:e63-e64. [DOI: 10.1016/j.sleep.2013.11.119]

Gao 2023 *{published data only}*

ChiCTR2100042278. Effects of PNF intervention on joint pain, joint proprioception and lower limb biomechanical characteristics in older adults with knee osteoarthritis during walking. ChiCTR2100042278 2021.

Gao B, Li L, Shen P, Zhou Z, Xu P, Sun W, et al. Effects of proprioceptive neuromuscular facilitation stretching in relieving pain and balancing knee loading during stepping over obstacles among older adults with knee osteoarthritis: a randomized controlled trial. *PloS One* 2023;**18**(2):e0280941. [DOI: 10.1371/journal.pone.0280941]

Ghroubi 2008 *{published data only}*

Ghroubi S, Elleuch H, Kaffel N, Echikh T, Abid M, Elleuch MH. Contribution of exercise and diet in the management of knee osteoarthritis in the obese. *Annales de Readaptation et de Medecine Physique* 2008;**51**(8):663-70.

Gianoudis 2015 *{published data only}*

Gianoudis J, Wang Y, Bailey C, Ebeling P, Nowson C, Sanders K, et al. Effects of a multi-modal exercise program on BMD, muscle function, knee cartilage structure, defects and bone marrow lesions in older adults: an 18 month RCT. *Journal of Bone and Mineral Research. Conference* 2015;**30 Suppl 1**:182-91. [DOI: 10.1002/jbmr.2763]

Guerrero 2015 *{published data only}*

Guerrero Y, Soomro N, Wilson G, Dam Y, Meiklejohn J, Simpson K, et al. Train High Eat Low for Osteoarthritis study (THE LO study): protocol for a randomized controlled trial. *Journal of Physiotherapy* 2015;**61**(4):217. [DOI: 10.1016/j.jphys.2015.05.020]

Hamad 2016 *{published data only}*

Hamad WH, Sghir M, Guedria M, Maraoui M, Said W, Jerbi S, et al. Contribution of therapeutic education in the treatment of knee osteoarthritis: preliminary results. *Annals of Physical and Rehabilitation Medicine* 2016;**59 Suppl**:e158. [DOI: 10.1016/j.rehab.2016.07.352]

Hasan 2018 *{published data only}*

Hasan MK, Stanmore E, Todd C. Feasibility of enabling self-management and coping with arthritic pain using exercise (ESCAPE-pain) programme for knee osteoarthritis in Malaysia. *Arthritis & Rheumatology* 2018;**70 Suppl 9**:429-30. [DOI: 10.1002/art.40700]

Heggannavar 2015 *{published data only}*

Heggannavar A, Gupta R. Quantitative effects of proprioceptive exercises and Mulligan's MWM in subjects with osteoarthritis of knee-a randomized clinical trail. *Physiotherapy (United Kingdom)* 2015;**1**:eS555-eS556. [DOI: 10.1016/j.physio.2015.03.3370]

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Hughes 2004 *{published data only}*

Hughes SL, Seymour RB, Campbell R, Pollak N, Huber G, Sharma L. Impact of the Fit and Strong intervention on older adults with osteoarthritis. *Gerontologist* 2004;**44**:217-28.

Hughes SL, Seymour RB, Campbell RT, Desai P, Huber G, Chang HJ. Fit and Strong! : bolstering maintenance of physical activity among older adults with lower-extremity osteoarthritis. *American Journal of Health Behavior* 2010;**34**(6):750-63.

Hurley 2018 *{published data only}*

Hurley DA, Jeffares I, Segurado R, Matthews J. Evaluation of the behavioural process and secondary outcomes of the self-management of osteoarthritis and low back pain through activity and skills cluster randomised controlled feasibility trial. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S332.

Islam 2018 *{published data only}*

Islam M. Effects of supervised exercise in older population having osteoarthritis of knee: a randomised control trial. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S333.

Keogan 2007 *{published data only}*

Keogan F, Gilsenan C, Hussey J, O'Connell P. Open or closed chain quadriceps exercises in treatment of osteoarthritis of the knee; which is more effective? A blinded randomised controlled trial. *Physiotherapy Ireland* 2007;**28**(1):47-8.

Kloek 2014 *{published data only}*

Kloek CJ, Bossen D, Veenhof C, Van Dongen JM, Dekker J, De Bakker DH. Effectiveness and cost-effectiveness of a blended exercise intervention for patients with hip and/or knee osteoarthritis: study protocol of a randomized controlled trial. *BMC Musculoskeletal Disorders* 2014;**15**(1):(no pagination). [DOI: 10.1186/1471-2474-15-269]

Kloek 2016 *{published data only}*

Kloek CJ, Bossen D, De Bakker DH, Dekker J, Veenhof C. Blended intervention with reduced face-to-face contact and usual physiotherapy show similar effectiveness in patients with osteoarthritis: a randomized controlled trial. *Physiotherapy* 2016;**102**:2016-1. [DOI: 10.1016/j.physio.2016.10.168]

Lange 2009 *{published data only}*

Lange AK, Vanwanseele B, Foroughi N, Baker MK, Shnier R, Smith RM, et al. Resistive Exercise for Arthritic Cartilage Health (REACH): a randomized double-blind, sham-exercise controlled trial. *BMC Geriatrics* 2009;**9**(1):(no pagination). [DOI: 10.1186/1471-2318-9-1]

Lee 2016 *{published data only}*

Lee A, Price LL, Han X, Chung M, Harvey WF, Driban J, et al. Mindfulness predicts treatment response from non-pharmacological therapy in knee osteoarthritis. *Arthritis & Rheumatology* 2016;**68 Suppl 10**:378-81. [DOI: 10.1002/art.39977]

Lee 2017b *{published data only}*

Lee A, Harvey WF, Han X, Price LL, Driban JB, Bannuru RR, et al. Pain and functional trajectories in symptomatic knee osteoarthritis over a 12- week period of non-pharmacological exercise interventions. *Arthritis and Rheumatology. Conference: American College of Rheumatology/Association of Rheumatology Health Professionals Annual Scientific Meeting, ACR/ARHP* 2017;**69 Suppl 10**:(no pagination).

Li 2017 *{published data only}*

Li L, Sayre EC, Grewal N, Chien J, Noonan G, Falck R, et al. Efficacy of a wearable-enabled physical activity counselling program for people with knee osteoarthritis. *Arthritis and Rheumatology. Conference: American College of Rheumatology/Association of Rheumatology Health Professionals Annual Scientific Meeting, ACR/ARHP* 2017;**69 Suppl 10**:(no pagination).

Loew 2015 *{published data only}*

Loew L, Brosseau L, Wells G, De Angelis G. Relevant determinants influencing walking adherence among older individuals with knee osteoarthritis: participant exercise preference (PEP) pilot randomised clinical trial. *Journal of Rheumatology* 2015;**42**(7):1312. [DOI: 10.3899/jrheum.150322]

Manlapaz 2020 *{published data only}*

Manlapaz DG, Escuadra CT, Regino JM, Briones EM, Kuizon RM, Pelonio PG, et al. Effectiveness of exergaming, Tai Chi, and physical therapy in improving dynamic balance and enjoyment of older adults with knee osteoarthritis. *Osteoarthritis and Cartilage* 2020;**28 Suppl 1**:S166-S167. [DOI: 10.1016/j.joca.2020.02.271]

Marcu 2016 *{published data only}*

Marcu IR, Patru S, Bighea AC. Role of physical exercise in patients with knee osteoarthritis. *Osteoporosis International* 2016;**27 Supl 1**:S523. [DOI: 10.1007/s00198-016-3530-x]

Matei 2014 *{published data only}*

Matei D, Bighea A, Patru S, Traistaru R, Popescu RS. The place of kinetotherapy in rehabilitation program of knee osteoarthritis patients. *Osteoporosis International* 2014;**2**:S279. [DOI: 10.1007/s00198-014-2642-4]

Moonaz 2015 *{published data only}*

Moonaz SH, Bingham CO, Wissow L, Bartlett SJ. Yoga in sedentary adults with arthritis: effects of a randomized controlled pragmatic trial. *Journal of Rheumatology* 2015;**42**(7):1194-202. [DOI: 10.3899/jrheum.141129]

Morgan 2013 *{published data only}*

Morgan NL, Ransford GL, Morgan LP, Driban JB, Wang C. Mindfulness is associated with psychological symptoms, self-efficacy, and quality of life among patients with symptomatic knee osteoarthritis. *Osteoarthritis and Cartilage* 2013;**1**:S257-S258.

Nct02560831 *{published data only}*

Nct02560831. Effects of therapeutic exercises in elderly women with knee osteoarthritis. Nct02560831 2015.

Nct02620462 *{published data only}*

Nct02620462. Wearable sensor-based balance training for patients with knee osteoarthritis [Explore the effectiveness of a sensor-based interactive exercise training on functional

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performance in patients with knee osteoarthritis]. Nct02620462 2015.

Nct03421249 *{published data only}*

Nct03421249. Treatment of knee osteoarthrosis using the electromagnetic fields [Treatment of knee osteoarthrosis using the electromagnetic fields: blinded and controlled study]. Nct03421249 2018.

Nct04019366 *{published data only}*

Nct04019366. Balance training using biodex stability system in knee osteoarthritis [Effects of balance training using biodex stability system in patients with symptomatic knee osteoarthritis]. Nct04019366 2019.

Nct04108650 *{published data only}*

Portuguese Fit & Strong! For older adults with osteoarthritis [Portuguese version of the Fit & Strong! program for older adults with osteoarthritis]. Nct04108650 2019.

Nct04186143 *{published data only}*

Nct04186143. Effects of closed and open kinetic chain exercises [Effects of closed and open kinetic chain exercises on pain, muscles strength, function, and quality of life in patients with knee osteoarthritis in patients with knee osteoarthritis]. Nct04186143 2019.

Nct04299568 *{published data only}*

Nct04299568. Effects of anti-gravity treadmill training on knee osteoarthritis in geriatric population. Nct04299568 2020.

Nct04588558 *{published data only}*

Nct04588558. Investigation of the effectiveness of a structured squat-based program in knee osteoarthritis rehabilitation. Nct04588558 (last updated 19 October 2020).

Nejati 2015 *{published data only}*

Nejati P, Farzinmehr A, Moradi-Lakeh M. The effect of exercise therapy on knee osteoarthritis: a randomized clinical trial. *Medical Journal of the Islamic Republic of Iran* 2015;**29**:186.

O'Hanlon 2016 *{published data only}*

O'Hanlon MS, Siddiqi Q, Allen SL, Edwards KL. A randomised controlled trial to investigate walking 6,000 steps per day on pain and function in knee osteoarthritis progression: The walkout study. *Osteoarthritis and Cartilage* 2016;**1**:S486.

Obuchi 2011 *{published data only}*

Obuchi S, Suzuki T. Effect of therapeutic approach for prevention of knee and low back pain. *Physiotherapy (United Kingdom)* 2011;**1**:eS910-eS911. [DOI: 10.1016/j.physio.2011.04.002]

Oida 2008 *{published data only}*

Oida Y, Morozumi K, Nakamura N, Kitabatake Y, Shiozawa S, Sato S, et al. Effectiveness of a community health service program using exercise intervention for elderly people with osteoarthritis of the knees: a randomized controlled trial. *Nippon Koshu Eisei Zasshi - Japanese Journal of Public Health* 2008;**55**(4):228-37.

Paterson 2018 *{published data only}*

Paterson G, Gaboury I, Tugwell P, Toupin-April K. A pragmatic pilot randomized controlled trial of the OA go away, a symptom and exercise tracking self-management behavioural intervention for promoting adherence to physical activity among individuals with OA of the hip or knee. *Journal of Rheumatology* 2018;**45**(7):967-8. [DOI: 10.3899/jrheum.180300]

Peeler 2019 *{published data only}*

Peeler J, Zhang P, Cornish S, Ghavami S. S100A8/A9, exercise, and creatine in knee osteoarthritis. *FASEB Journal* 2019;**33 Suppl 1**:769.4. [DOI: 10.1096/fasebj.2019.33.1_supplement.769.4]

Pisters 2011 *{published data only}*

Pisters M, Veenhof C, Schellevis F, De Bakker D, Dekker J. Long-term effect of exercise therapy in patients with osteoarthritis: a randomized controlled trial comparing two different physiotherapy interventions. *Physiotherapy (United Kingdom)* 2011;**1**:eS1005. [DOI: 10.1016/j.physio.2011.04.002]

Rahela 2012 *{published data only}*

Rahela MI, Patru S, Bighea A, Popescu RS. Efficacy of physical exercice in patients with knee osteoarthritis. *Annals of the Rheumatic Disease. Conference: Annual European Congress of Rheumatology of the European League Against Rheumatism, EULAR* 2012;**71 Suppl 3**:(no pagination). [DOI: 10.1136/annrheumdis-2012-eular.2763]

Raisanen 2021 *{published data only}*

Raisanen AM, Whittaker J, Toomey C, Pasanen K, Emery CA. Does an 8-week exercise intervention impact knee-related symptoms 3-15-years following intra-articular knee injury? A pilot randomized controlled trial. *Osteoarthritis and Cartilage* 2021;**29 Suppl 1**:S393-S394. [DOI: 10.1016/j.joca.2021.02.510]

RBR-4zx26w *{published data only}*

RBR-4zx26w. Measurement of thigh muscle strength in patients with knee arthrosis who will receive a lubricating injection to the knee and who will do bodybuilding. RBR-4zx26w 2017.

Reid 2016 *{published data only}*

Reid KF, Price L, Harvey WF, Driban JB, Fielding RA, Wang C. Changes in leg muscle strength and power after tai Chi exercise in patients with symptomatic knee osteoarthritis [Changes in leg muscle strength and power after tai chi exercise in patients with symptomatic knee osteoarthritis]. *Osteoarthritis and Cartilage* 2016;**1**:S430.

Rezende 2018 *{published data only}*

Rezende MU, Silva JM, Spada TC, Francisco LS, Santos HP, Greve JM, et al. Educational program in subjects with knee osteoarthritis improves adherence, aerobic and functional capacity [Educational program in subjects with knee osteoarthritis improves adherence, aerobic and functional capacity]. *Osteoarthritis and Cartilage* 2018;**26 Suppl 1**:S413-S414.

Rosa 2012 *{published data only}*

Rosa UH, Velasquez Tlapanco J, Lara Maya C, Villarreal Rios E, Martinez Gonzalez L, Vargas Daza ER, et al. Comparison of the

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

effectiveness of isokinetic vs isometric therapeutic exercise in patients with osteoarthritis of knee. *Reumatologia Clinica* 2012;**8**(1):10-4.

Shellington 2019 *{published data only}*

Shellington EM, Gill DP, Shigematsu R, Petrella RJ. Innovative exercise as an intervention for older adults with knee osteoarthritis: a pilot feasibility study. *La Revue Canadienne du Vieillissement [Canadian Journal on Aging]* 2019;**38**(1):111-21. [DOI: 10.1017/S0714980818000454]

Umin000011402 *{published data only}*

Umin000011402. Additive effect of tidal knee irrigation and exercise for osteoarthritis of the knee. Umin000011402 2013.

Umin000043117 *{published data only}*

Umin000043117. Impact of home exercise program on self-efficacy and quality of life among primary knee osteoarthritis patients: a randomized controlled clinical study. Umin000043117 2021.

Unal 2016 *{published data only}*

Unal E, Kysacyk P, Oksuz S, Arin G, Karabulut E, Akdogan A, et al. The effects of cognitive exercise therapy approach on symptoms and antiinflammatory status in patients with osteoarthritis of the knee. *Fizyoterapi Rehabilitasyon* 2016;**27**(3):S32-S33.

Walsh 2020 *{published data only}*

Walsh N, Cramp F, Palmer S, Pollock J, Hampson L, Gooberman-Hill R, et al. Exercise and self-management for people with chronic knee, hip or lower back pain: a cluster randomised controlled trial of clinical and cost-effectiveness. Study protocol. *Physiotherapy (United Kingdom)* 2013;**99**(4):352-7. [DOI: 10.1016/j.physio.2012.09.002]

Walsh N, Jones L, Phillips S, Thomas R, Odondi L, Palmer S, et al. Facilitating Activity and Self-management for people with Arthritic knee, hip or lower back pain (FASA): a cluster randomised controlled trial. *Musculoskeletal Science and Practice* 2020;**50**:(no pagination). [DOI: 10.1016/j.msksp.2020.102271]

Wang 2016 *{published data only}*

Wang C, Price L, Schmid C, Harvey W, Driban J, McAlindon T. A positive response to nonpharmaceutical interventions in adults with knee osteoarthritis. *Osteoarthritis and Cartilage* 2016;**1**:S490-S491.

Wang 2020 *{published data only}*

Wang L, Chen H, Lu H, Chen J, Jin S, Shang S. The trajectory of exercise adherence over time and the effect of theory-lead intervention for knee osteoarthritis in elderly. *International Journal of Rheumatic Diseases* 2020;**23 Suppl 1**:84. [DOI: 10.1111/1756-185X.13984]

Yamada 2011 *{published data only}*

Yamada E, Rodrigues PL, Almeida FR, Nicolau GP, Gomes IA, Zanetti LF. Effectiveness of transcutaneous electrical nerve stimulation, therapeutic exercise and the pursuit of the association of two techniques in knee osteoarthritis. *Physiotherapy (United Kingdom)* 2011;**1**:eS1051-eS1052. [DOI: 10.1016/j.physio.2011.04.002]

Yavuz 2013 *{published data only}*

Yavuz M, Ataoglu S, Ozsahin M, Baki AE, Icmeli C. Compared effects and effectiveness in applications of isokinetic exercise, laser, and diclophenac iontophoresis in primary osteoarthritis of knee. *Duzce Medical Journal* 2013;**15**(3):15-21.

Zaimi 2012 *{published data only}*

Zaimi PE, Zaimi E, Backa Cico T, Zicishti E. NSAI and effectiveness of physiotherapy in the treatment of gonarthrosis. *Annals of the Rheumatic Disease. Conference: Annual European Congress of Rheumatology of the European League Against Rheumatism, EULAR* 2012;**71 Suppl 3**:(no pagination). [DOI: 10.1136/annrheumdis-2012-eular.1458]

References to ongoing studies

Actrn12615001101572 *{published data only}*

Actrn12615001101572. Feasibility of a walking programme for people with symptoms of hip/knee osteoarthritis [Pedometer-driven walking programme to improve physical activity levels in community-dwelling individuals with symptoms of hip/knee osteoarthritis -; a feasibility study]. Actrn12615001101572 2015.

Actrn12618001494224 *{published data only}*

Actrn12618001494224. E-health to Empower patients with Musculoskeletal Pain in Rural Australia (EMPoweR): a pilot study. Actrn12618001494224 2018.

Actrn12619001297112 *{published data only}*

Actrn12619001297112. Exercise for hip osteoarthritis symptoms: the PHOENIX study [Exercise for hip osteoarthritis related pain and physical dysfunction: the PHOENIX study]. Actrn12619001297112 2019.

Actrn12623000780651 *{published data only}*

Actrn12623000780651. Effects of a self-directed online Tai Chi program in people with knee osteoarthritis: randomised controlled trial (The RETREAT Trial). Actrn12623000780651 2023.

Akgul 2023 *{published data only}*

Akgul H, Birtane M, Tonga E. Effects of digital-based physical activity intervention on pain, function and adherence in individuals with knee osteoarthritis: a pilot randomized-controlled trial. *Gait & Posture* 2023;**106 Suppl 1**:S5-S6. [DOI: 10.1016/j.gaitpost.2023.07.011]

Bilzon 2023 *{published data only}*

Bilzon JL, Deere R, Farrow M, Chowdhury E, Thompson D. Effects of energy restriction and exercise on pain and inflammation in patients with knee osteoarthritis. *Osteoarthritis and Cartilage* 2023;**31**(5):708-709. [DOI: 10.1016/j.joca.2023.02.061]

Bozgeylk 1830 *{published data only}*

Bozgeylk S, Kilinc H, Hur IG, Guney Deniz H. The comparison of three different management strategies following platelet rich plasma injection in patients with knee OA. *Annals of*

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

the Rheumatic Diseases 1830;**82**:1830-1. [DOI: 10.1136/annrheumdis-2023-eular.1499]

Cagnin 2022 *{published data only}*

Cagnin A, Mezghani N, Choiniere M, Bureau NJ, Hagemeister N. Home-based neuromuscular exercises can address biomechanical markers linked to knee osteoarthritis: a secondary analysis from a cluster randomized controlled trial. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S136-S137. [DOI: 10.1016/j.joca.2022.02.173]

ChiCTR1800019208 *{published data only}*

ChiCTR1800019208. The efficacy outcome of exercise rehabilitation training in patients with knee osteoarthritis: a single center, randomized controlled trialthe efficacy outcome of exercise rehabilitation training in patients with knee osteoarthritis. ChiCTR1800019208 2018.

ChiCTR2000039878 *{published data only}*

ChiCTR2000039878. The research of the biomechanical mechanism of the combination of manipulation and exercise in the management of knee osteoarthritis based on. ChiCTR2000039878 2020.

ChiCTR2000040769 *{published data only}*

ChiCTR2000040769. The research of the biomechanical mechanism of the exercise in the management of knee osteoarthritis. ChiCTR2000040769 2020.

ChiCTR2100046313 *{published data only}*

ChiCTR2100046313. Virtual reality assisted lower limbs muscle training for patients with knee osteoarthritis: a pilot randomized controlled trial. ChiCTR2100046313 2021.

ChiCTR2100049907 *{published data only}*

Effects of comprehensive exercise therapy on knee osteoarthritis. ChiCTR2100049907 2021.

ChiCTR2100054191 *{published data only}*

A study on the treatment of knee osteoarthritis with hip exercise based on core stability principle. ChiCTR2100054191 2021.

ChiCTR2300067559 *{published data only}*

ChiCTR2300067559. Efficacy and safety of platelet rich plasma combined with Tai Chi for knee osteoarthritis. ChiCTR2300067559 2023.

ChiCTR2300069339 *{published data only}*

ChiCTR2300069339. Tai Chi for treating patients with knee osteoarthritis: a randomized controlled trial. ChiCTR2300069339 2023.

ChiCTR2300071797 *{published data only}*

ChiCTR2300071797. Effect of exercise rehabilitation prescribed by VAS pain score on cardiopulmonary fitness in patients with knee osteoarthritis. ChiCTR2300071797 2023.

ChiCTR-IOR-14005414 *{published data only}*

ChiCTR-IOR-14005414. Combined application of celecoxib and exercise therapy in people with knee osteoarthritis: randomised controlled trial. ChiCTR-IOR-14005414 2014.

ChiCTR-IPR-17013465 *{published data only}*

ChiCTR-IPR-17013465. Managing knee osteoarthritis with Shi's exercise programs in older adults: a pilot randomized controlled trial. ChiCTR-IPR-17013465 2017.

Ctri/2019/02/017422 *{published data only}*

Ctri/2019/02/017422. To check the effect of Yoga in Osteoarthritis [Effectiveness of yoga therapy on quality of life in osteoarthritis of knee joint]. Ctri/2019/02/017422 2019.

Ctri/2019/07/019989 *{published data only}*

Ctri/2019/07/019989. A study to assess the effectiveness of a mobile app based intervention to improve activity, pain and movement of knee in patient with Knee Osteoarthritis attending Orthopaedic OPD, PGIMER, Chandigarh. Ctri/2019/07/019989 2019.

Ctri/2020/04/024415 *{published data only}*

Ctri/2020/04/024415. Physiotherapy treatment for knee osteoarthritis subjectseffect of neuromuscular exercises on articular cartilage in knee osteoarthritis. Ctri/2020/04/024415 2020.

Ctri/2020/05/025097 *{published data only}*

Ctri/2020/05/025097. Benefits of three different physiotherapy treatments for women with knee paineffectiveness of three different approaches in management of osteoarthritis in postmenopausal women. Ctri/2020/05/025097 2020.

Ctri/2021/06/034213 *{published data only}*

Ctri/2021/06/034213. Effect of exercises and electrical stimulation in patients with knee osteoarthritiseffect of neuromuscular exercises and neuromuscular electrical stimulation in patients with knee osteoarthritis. Ctri/2021/06/034213 2021.

Ctri/2021/12/038685 *{published data only}*

Ctri/2021/12/038685. Effects of core exercises on pain and function in knee osteoarthritis patientseffects of core stability exercises on pain and physical function in patients with knee osteoarthritis - a comparative study. Ctri/2021/12/038685 2021.

Ctri/2022/02/040249 *{published data only}*

Ctri/2022/02/040249. To find the effect of core muscle exercise check with pain and dynamic balance in osteoarthritis of knee patients [Effectiveness of core muscles strengthening on pain and dynamic balance in patients with osteoarthritis knee-an interventional study]. Ctri/2022/02/040249 2022.

Ctri/2022/05/042853 *{published data only}*

Ctri/2022/05/042853. In knee osteoarthritis comparison of effectiveness of ankle, hip and core strengthening exercise. strengthening [Effect of core and lower extremity strengthening on the functionality of patient with knee OA: a comparative study]. Ctri/2022/05/042853 2022.

Ctri/2022/12/047849 *{published data only}*

Ctri/2022/12/047849. App Based physiotherapy rehabilitation in community for the patient with osteoarthritis of knee joint. Ctri/2022/12/047849 2022.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Ctri/2023/02/049448 *{published data only}*

Ctri/2023/02/049448. Core vs hip strengthening on knee function in adults with OA knee [Effect of core versus hip strengthening on knee function in adults with osteoarthritis knee: a randomized controlled trial]. Ctri/2023/02/049448 2023.

Ctri/2023/06/054225 *{published data only}*

Ctri/2023/06/054225. Effect of hip abductor strengthening & iliotibial band stretching in patients with knee pain. Ctri/2023/06/054225 2023.

Ctri/2023/08/056593 *{published data only}*

Ctri/2023/08/056593. Effect of addition of backward walking to traditional treatment on pain, balance, walking pattern, strength, flexibility, disability and mobility in patients with knee pain. Ctri/2023/08/056593 2023.

Ctri/2023/10/059073 *{published data only}*

Ctri/2023/10/059073. Importance of lumbar core stabilization exercises on pulmonary functions, pain, dynamic balance and functional activities in patients with knee osteoarthritis. Ctri/2023/10/059073 2023.

Ctri/2023/11/059651 *{published data only}*

Ctri/2023/11/059651. To observe the effect of homoeopathic medicines and yoga therapy in management of knee pain. Ctri/2023/11/059651 2023.

Dieter 2022 *{published data only}*

Dieter V, Haupt G, Janssen P, Krauss I. Randomized-controlled trial to evaluate a 12-week app- and brace-assisted exercise intervention in patients with knee osteoarthritis. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S401-S402. [DOI: 10.1016/j.joca.2022.02.539]

Drks00009677 *{published data only}*

Drks00009677. Preventing the progression of osteoarthritis of the knee grade 2- 3 by mechanical stimulation with high impact physical exercise training device (GALILEO). Drks00009677 2016.

Drks00018847 *{published data only}*

Drks00018847. Feasibility and effectiveness of an individualized, group-based physical exercise program with educational components in people with knee and/or hip OA; a randomized controlled pilot study. Drks00018847 2019.

Drks00023269 *{published data only}*

Drks00023269. Randomized controlled study in a wait-list control design to evaluate a 12-week app- and orthesis-supported training intervention in patients with moderate to severe knee osteoarthritis. Drks00023269 2020.

Drks00027164 *{published data only}*

Drks00027164. The usability and preliminary effectiveness of an app-based training and educational program (Join2Move) in people with hip and/ or knee osteoarthritis-; a randomized controlled pilot study. Drks00027164 2021.

Drks00028477 *{published data only}*

Drks00028477. SmArt-E: smartphone-assisted osteoarthritis training with eduction. Drks00028477 2022.

Drks00030932 *{published data only}*

Drks00030932. Randomized controlled trial to evaluate a 12-week app-guided exercise intervention in patients with knee osteoarthritis (re.flex). Drks00030932 2023.

Fugaru 2022 *{published data only}*

Fugaru O, Kamal D, Kamal C, Rogoveanu O, Traistaru R. Rehabilitation program in osteoarthritis patients during the COVID-19 lockdown. *Osteoporosis International* 2022;**32 Suppl 1**:S128-S129. [DOI: 10.1007/s00198-021-06125-9]

Gholami 2023 *{published data only}*

Gholami Z, Faezi ST, Letafatkar A, Madreseh E. Pain neuroscience education, blended exercises and booster sessions as an effective therapy for pain, functional and psychological factors in patients with knee osteoarthritis: a study protocol for a single-blind randomised controlled trial with 2 2 factorial design during 6-month follow-up. *BMJ Open* 2023;**13**(5):(no pagination). [DOI: 10.1136/bmjopen-2022-070336]

Irct20101017004952n10 *{published data only}*

Irct20101017004952n10. Effect of exercise therapy with and without intra-articular hyaluronic acid in knee osteoarthritis. Irct20101017004952n10 2020.

Irct20101228005486n7 *{published data only}*

Irct20101228005486n7. Comparative study of the efficacy of three protocols of Neuromuscular Electrical Stimulation (NMES), exercise therapy and NMES combined with exercise therapy on pain, quadriceps muscle thickness, and physical function of patients with knee osteoarthritis: a controlled randomized trial with a three-month follow-up. Irct20101228005486n7 2020.

Irct20150131020888n10 *{published data only}*

Irct20150131020888n10. The effect of treadmill exercise in subjects with bilateral knee osteoarthritis [The effect of treadmill exercise with conventional physical therapy on the spatiotemporal parameters of the gait in subjects with bilateral moderate knee osteoarthritis]. Irct20150131020888n10 2018.

Irct20160531028186n4 *{published data only}*

Irct20160531028186n4. The effect of hydrotherapy on balance in patients with grade 2 and 3 knee osteoarthritis: a randomized clinical trial. Irct20160531028186n4 2019.

Irct20161208031300n1 *{published data only}*

Irct20161208031300n1. The effect of resistance training and nano curcumin supplementation on synovial fluid biomarkers [Study of the effect of resistance training and nano curcumin supplementation on synovial fluid biomarkers in patients with knee osteoarthritis]. Irct20161208031300n1 2018.

Irct2017052027741n4 *{published data only}*

Irct2017052027741n4. The effect of exercise on the knee injury and osteoarthritis outcomes of elders [A comparison of the effects of attendance and non attendance exercise training on the knee injury and osteoarthritis outcomes in low literacy elderly dwelling in Mashhad]. Irct2017052027741n4 2017.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Irct20170827035926n2 *{published data only}*

The effect of ”chair yoga” on knee joints [sic] pain & physical function, balance and fear of falling among community-dwelling older adults with knee osteoarthritis. Irct20170827035926n2 2018.

Irct2017100218117n2 *{published data only}*

Irct2017100218117n2. The effect of home-based rehabilitation on the treatment of knee qsteoarthritis [The effect of home-based rehabilitation ttraining on pain, performance, general health and quality of life in patients with knee osteoarthritis]. Irct2017100218117n2 2017.

Irct20190322043099n1 *{published data only}*

Irct20190322043099n1. The effect of proprioceptive exercises in the military elderly with knee osteoarthritis [The effect of proprioceptive exercises on pain, balance and risk of falling in the military elderly with knee osteoarthritis; a randomized clinical trial]. Irct20190322043099n1 2023.

Irct20190816044538n1 *{published data only}*

Irct20190816044538n1. Effect of hip muscles strengthening exercises on anterior knee pain [Effect of hip abductor and external rotator muscles strengthening on patellofemoral pain symptoms, hip muscles strength, and lower extremity kinematics in patients with patellofemoral pain]. Irct20190816044538n1 2019.

Irct20191221045846n2 *{published data only}*

Irct20191221045846n2. Short term effects of exercises among knee osteoarthritis (OA) participants who are overweight or obese [Short term effects of strengthening exercises of lower limb rehabilitation protocol (LLRP) on pain, stiffness, physical function and body mass index (BMI)]. Irct20191221045846n2 2020.

Irct20200107046035n10 *{published data only}*

Irct20200107046035n10. The effect of combined rehabilitation program on pain and function in athletes with knee osteoarthritis [The effect of 6 weeks of combined rehabilitation program (electrical stimulation and isometric contraction) on pain and function in athletes with knee osteoarthritis]. Irct20200107046035n10 2022.

Irct20200204046368n8 *{published data only}*

Irct20200204046368n8. The home-based exercise intervention and women with knee osteoarthritis [The effects of a home-based exercise intervention on women with knee osteoarthritis]. Irct20200204046368n8 2021.

Irct20200807048331n1 *{published data only}*

Irct20200807048331n1. Comparison of the effect of using elastic band and flamingo exercise on knee osteoarthritis [Comparison of the effect of two methods using elastic band and flamingo exercise on functional balance, proprioceptive and osteoarthritis indicators in patients with knee osteoarthritis]. Irct20200807048331n1 2020.

Irct20210129050172n1 *{published data only}*

Irct20210129050172n1. Investigating the effect of exercise program on improving sleep quality and quality of life [The effect of exercise program on improving quality of life and sleep quality in the elderly with osteoarthritis of the knee]. Irct20210129050172n1 2021.

Irct20210529051427n1 *{published data only}*

Irct20210529051427n1. The effect of hip abductor exercises on pain, function and balance parameters in patients with knee osteoarthritis. Irct20210529051427n1 2021.

Irct20220206053950n2 *{published data only}*

Irct20220206053950n2. Effectiveness of muscle strengthening exercises on knee osteoarthritis [Effectiveness of muscle strengthening exercises on pain, physical function and morning stiffness of patients with osteoarthritis of knee: a randomized four-arm controlled trial]. Irct20220206053950n2 2022.

Irct20230124057196n1 *{published data only}*

Irct20230124057196n1. Effects of knee strengthening exercises with and without hip strengthening exercises in women with knee osteoarthritis: randomized controlled trail. Irct20230124057196n1 2023.

Irct20230216057434n3 *{published data only}*

Irct20230216057434n3. Effectiveness of two manual therapy technique in osteo arthritis [Effect of Iliotibial band myofascial release combined with valgus controlled exercises on pain, range of motion, balance and quality of life in patients with grade II knee osteoarthritis]. Irct20230216057434n3 2023.

Isrctn12950684 *{published data only}*

Isrctn12950684. Telerehabilitation versus monitored increased physical activity on pain and functionality of patients with knee osteoarthritis. Isrctn12950684 2020.

Moutzouri M, Gioftsos G. How effective is a blended web-based rehabilitation for improving pain, physical activity, and knee function of patients with knee osteoarthritis? Study protocol for a randomized control trial. *PloS One* 2022;**17**(5):(no pagination). [DOI: 10.1371/journal.pone.0268652]

Isrctn15564385 *{published data only}*

Groves-Williams D, McHugh GA, Bennell KL, Comer C, Hensor EM, Conner M, et al. Evaluation of two electronic-rehabilitation programmes for persistent knee pain: protocol for a randomised feasibility trial. *BMJ Open* 2022;**12**(6):(no pagination). [DOI: 10.1136/bmjopen-2022-063608]

Isrctn15564385. E-rehab for knee pain [Evaluation of electronic-rehabilitation programmes for chronic knee pain]. Isrctn15564385 2020.

Isrctn75983430 *{published data only}*

Isrctn75983430. Tailored exercise management for people aged 80 years or older with hip/knee osteoarthritis [Tailored Exercise Management for People aged 80 years or older with hip/knee Osteoarthritis (TEMPO): a feasibility randomised trial]. Isrctn75983430 2021.

Itmctr2200006721 *{published data only}*

Itmctr2200006721. Effects of Yijinjing exercise on patients with knee osteoarthritis: a randomised controlled trial. Itmctr2200006721 2022.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Jakiela 2022 *{published data only}*

Jakiela JT, Voinier D, Hinman RS, Copson J, Schmitt LA, Leonard TR, et al. Comparing an expanded versus brief telehealth physical therapy intervention for knee osteoarthritis: study protocol for the delaware PEAK randomized controlled trial. *Physical Therapy* 2022;**06**:pzac139. [DOI: 10.1093/ptj/pzac139]

Jprn-Umin000048356 *{published data only}*

Jprn-Umin000048356. An association between central sensitization and autonomic nervous system and an effect of aeobic exercise for patietns with knee and hip osteoarthritis. Jprn-Umin000048356 2022.

jRCTs052200069 *{published data only}*

JRCTs052200069. Pain relief evaluation using transcutaneous electrical nerve stimulation(TENS) and exercise therapy for osteoarthritis of the knee. JRCTs052200069 2020.

Karaborklu 2023 *{published data only}*

Karaborklu Argut S, Celik D, Ergin O, Kilicoglu O. The comparison of PRP alone, supervised exercise alone, and PRP combined with supervised exercise in management of knee osteoarthritis. *Annals of the Rheumatic Diseases* 1818;**82**:1818-19. [DOI: 10.1136/annrheumdis-2023-eular.283]

Kct0006170 *{published data only}*

Kct0006170. Effects of the lower extremity muscle strengthening exercise on knee osteoarthritis using the Biodfeedback system (exoRehab) : a randomized controlled trial(RCT) study. Kct0006170 2021.

Kong 2022 *{published data only}*

Kong J, Wang C. Neurobiological mechanisms of a remote-delivered mind-body intervention for knee osteoarthritis. *Arthritis & Rheumatology* 2022;**74 Suppl 9**:3790-1. [DOI: 10.1002/art.42355]

Kong 2023 *{published data only}*

Kong J, Wang C. Modulatory effects of remote tai Chi on the functional connectivity of dorsolateral prefrontal cortex in patients with knee osteoarthritis pain: a randomized brain-imaging study. *Osteoarthritis and Cartilage* 2023;**31 Suppl 1**:S299. [DOI: 10.1016/j.joca.2023.01.308]

Kumar 2022 *{published data only}*

Kumar D, Costello KE, Shah N, Neogi T, Lee S, Rose MJ, et al. Mindful walking in knee osteoarthritis: a feasibility randomized controlled trial. *Osteoarthritis and Cartilage* 2022;**30 Suppl 1**:S72-S73. [DOI: 10.1016/j.joca.2022.02.087]

Moutzouri 2023 *{published data only}*

Moutzouri MM, Gioftsos GG. Investigation of a blended rehabilitation approach in function and physical activity levels of patients with knee osteoarthritis: a randomised controlled trial implemented in structurally weak urban areas in Greece. *Aging Clinical and Experimental Research* 2023;**35 Suppl 1**:S161. [DOI: 10.1007/s40520-023-02442-7]

Nct02387034 *{published data only}*

Nct02387034. Swedish PAP in osteoarthritis - a RCT study [The effect on Swedish physical activity on prescription (PAP) in patients in primary care with osteoarthritis in knee or hip - a RCT-study]. Nct02387034 2015.

Nct02748291 *{published data only}*

Nct02748291. An investigation into the role of walking in treating the symptoms of knee osteoarthritis: the walkout study. Nct02748291 2016.

Nct03064139 *{published data only}*

Nct03064139. Mindful walking intervention for knee osteoarthritis [Mind Your Walk intervention forcommunity-based management of knee OA: a feasibility study]. Nct03064139 2017.

Nct03363490 *{published data only}*

Nct03363490. The study of neuromuscular exercise therapy and patients self-management program in knee osteoarthritis. Nct03363490 2017.

Nct03431298 *{published data only}*

Nct03431298. The benefit of functional movement control for patients with knee osteoarthritisthe benefit of increasing physical activity and functional movement control for patients with stage II or III knee osteoarthritis. Nct03431298 2018.

Nct03920020 *{published data only}*

Nct03920020. Comparison of the effects of concentric and eccentric isokinetic exercises in patients with knee osteoarthritis. Nct03920020 2019.

Nct04099017 *{published data only}*

Nct04099017. Mulligan manual therapy and trunk stabilization exercises versus isometric knee strengthening on knee osteoarthritis. Nct04099017 2019.

Nct04154059 *{published data only}*

Nct04154059. Strengthening program for foot-ankle muscles in people with knee osteoarthritis [Effects of a strengthening program for foot-ankle muscles in clinical aspects and gait biomechanics of people with knee osteoarthritis: a randomized blinded clinical trial]. Nct04154059 2019.

Nct04308967 *{published data only}*

Nct04308967. Central sensitization in knee osteoarthritis [Effects of balance exercises on central sensitization in patients with knee osteoarthritis]. Nct04308967 2020.

Nct04325334 *{published data only}*

Nct04325334. Osteoarthritis running & cartilage assessment linking biomechanical and imaging outcomes to better understand the effects of running on knee joint health. Nct04325334 2020.

Nct04362618 *{published data only}*

Nct04362618. Exercise therapy for osteoarthritis pain: how does it work? Nct04362618 2020.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Nct04615377 *{published data only}*

Nct04615377. The Kaia knee and hip intervention for self-management of knee and hip pain [The Kaia knee and hip intervention for self-management of knee and hip painthe kaia knee and hip intervention for self-management of knee and hip pain in patients with osteoarthritis - a randomized controlled study]. Nct04615377 2020.

Nct04645732 *{published data only}*

Nct04645732. Personalized exercise therapy and self-management for patients with multimorbidity [Improving health in people with multimorbidity: a paradigm shift in health care from disease-based curative models to personalized exercise therapy and self-management]. Nct04645732 2020.

Nct04669834 *{published data only}*

Nct04669834. Effect of core stability exercise on isolated patellofemoral osteoarthritis [Effect of core stability exercise on temporal activation of core muscles and vasti in isolated patellofemoral osteoarthritis]. Nct04669834 2020.

Nct04678609 *{published data only}*

Nct04678609. OKV versus CKC for biomechanics and function in mild knee OA patients [Effects of open and closed kinetic chain exercise on functional outcomes and lower limb biomechanics in mild knee osteoarthritis patients]. Nct04678609 2020.

Nct04697667 *{published data only}*

Nct04697667. Exercise and PRP vs exercise alone in patients with knee osteoarthritis [The combination of exercise and PRP vs exercise alone in patients with knee osteoarthritis: a randomized controlled clinical trial]. Nct04697667 2021.

Nct04767854 *{published data only}*

Nct04767854. Digital osteoarthritis caredigital osteoarthritis care- DigiOA. Nct04767854 2021.

Nct04909086 *{published data only}*

Nct04909086. Different modes of isotonic exercises for knee osteoarthritis [Effect of different modes of isotonic exercises on clinical outcomes and cartilage biomarkers in knee osteoarthritis]. Nct04909086 2021.

Nct05197010 *{published data only}*

Nct05197010. Clinical efficacy of IoMT-based exercise program for the elderly [Clinical efficacy of IoMT-based exercise program for the elderly with chronic musculoskeletal disorders]. Nct05197010 2022.

Nct05272124 *{published data only}*

Nct05272124. Effects of hip abductors strengthening in knee osteoarthritis [Comparison of hip abductors and quadriceps strengthening with quadriceps strengthening alone in knee osteoarthritis]. Nct05272124 2022.

Nct05272748 *{published data only}*

Nct05272748. The effects of conventional exercise combined with core stabilization exercise by telerehabilitation in patients with hip and/or knee osteoarthritis. Nct05272748 2022.

Nct05287802 *{published data only}*

Nct05287802. The efficacy of balance and proprioception exercises in patients with knee osteoarthritis [The efficacy of balance and proprioception exercises in patients with knee osteoarthritis: a randomized controlled study]. Nct05287802 2022.

Nct05319964 *{published data only}*

Nct05319964. Anti-gravity treadmill training in patients with knee osteoarthritis. Nct05319964 2022.

Nct05334524 *{published data only}*

Nct05334524. Kinesiotaping with and without combined chain exercises in patients with knee osteoarthritis [Effects of kinesiotaping with and without combined chain exercises on pain, range of motion and quality of life in patients with knee osteoarthritis]. Nct05334524 2022.

Nct05518890 *{published data only}*

Nct05518890. Effect of energy restriction and exercise for knee osteoarthritis [Effect of short-term negative energy balance with or without exercise on systemic biomarkers in patients with knee osteoarthritis]. Nct05518890 2022.

Nct05595317 *{published data only}*

Nct05595317. The effects of non-immersive virtual reality exercises on muscle excitability in knee osteoarthritis [The investigation of the effects of non-immersive virtual reality exercises on muscle excitability around the knee and balance in patients with knee osteoarthritis]. Nct05595317 2022.

Nct05670314 *{published data only}*

Nct05670314. lIfestyle iNterventionS for PaIn ReliEf (INSPIRE) molecular signatures of endocannabinoid induced pain relief in humans: lifestyle interventions, systemic and localised changes. Nct05670314 2023.

Nct05827003 *{published data only}*

Nct05827003. Effects of exercise versus topical agents on knee osteoarthritisthe comparison of the effectiveness of exercise and topical agent treatments in knee osteoarthritis: a randomized controlled trial. Nct05827003 2023.

Nct05836792 *{published data only}*

Nct05836792. The effect of cycle ergometer with biofeedback on deep sensation in elderly patients with knee osteoarthritis. Nct05836792 2023.

Nct05839262 *{published data only}*

Nct05839262. Active video games for older adults with knee and/or hip osteoarthritis [Physical exercise guided by active video games for optimizing clinical and psychosocial outcomes in older adults with knee and/or hip osteoarthritis under a community-based rehabilitation]. Nct05839262 2023.

Nct05855681 *{published data only}*

Nct05855681. Effect of strengthening hip rotators and ankle muscles on knee osteoarthritis. Nct05855681 2023.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Nct05857605 *{published data only}*

Nct05857605. The impact of a seated compact elliptical trainer on individuals with knee OA. Nct05857605 2023.

Nct05997862 *{published data only}*

Nct05997862. A hip flexion feedback system for exercise monitoring in individuals with osteoarthritis and obesity. Nct05997862 2023.

Nct05999110 *{published data only}*

Nct05999110. Algarve Active Ageing - Cardiac and Osteoarthritis Rehabilitation (A3-COR) Algarve Active Ageing - Cardiac and Osteoarthritis Rehabilitation: personalized physical exercise protocol in after acute myocardial infarction or cardiovascular risk and knee osteoarthritis. Nct05999110 2023.

Nct06010394 *{published data only}*

Nct06010394. Control of edema in the knee in patients with knee osteoarthritis [Comparison of the effectiveness of exercise and classical massage in the control of edema in the knee in patients with knee osteoarthritis]. Nct06010394 2023.

Nct06084949 *{published data only}*

Nct06084949. Effect of structured exercise in patients with knee osteoarthritis [Effect of Nordic walking and inspiratory muscle training to promote cardiopulmonary function in patients with knee osteoarthritis]. Nct06084949 2023.

Nct06130774 *{published data only}*

Nct06130774. Effects of retro-walking in patients with knee OA [Effects of retro-walking in knee OA along with conventional therapy]. Nct06130774 2023.

Ng 2022 *{published data only}*

Ng WH, Jamaludin NI, Sahabuddin FN, Ab Rahman S, Ahmed Shokri A, Shaharudin S. Comparison of the open kinetic chain and closed kinetic chain strengthening exercises on pain perception and lower limb biomechanics of patients with mild knee osteoarthritis: a randomized controlled trial protocol. *Trials* 2022;**23**(1):(no pagination). [DOI: 10.1186/s13063-022-06153-8]

Nicolson 2023 *{published data only}*

Nicolson PJ, Holden MA, Marian IR, Williamson E, Dutton SJ, Garrett A, et al. Tailored exercise management (TEMPO) versus usual care for people aged 80 years or older with hip/knee osteoarthritis: study protocol for a feasibility randomised controlled trial. *Pilot and Feasibility Studies* 2023;**9**(1):(no pagination). [DOI: 10.1186/s40814-023-01275-5]

Pactr202108619222230 *{published data only}*

Pactr202108619222230. Long-term effects of combined chain exercises plus kinesio taping compared with combined chain exercises or kinesio taping alone in knee osteoarthritis: a randomized clinical trial. Pactr202108619222230 2021.

RBR-29jhy2r *{published data only}*

RBR-29jhy2r. Pilates method in the treatment of people with arthrosis of the knee [The influence of the Pilates method on improving pain and function in elderly with osteoarthritis of the knee]. RBR-29jhy2r 2022.

RBR-2fkzb2s *{published data only}*

RBR-2fkzb2s. Effect of acupuncture with electrical stimuli and therapy with games in elderly with knee arthrosis [Effect of electroacupuncture and Wii therapy on functional performance and inflammatory parameters in elderly with knee osteoarthritis]. RBR-2fkzb2s 2022.

RBR-2hxj58 *{published data only}*

RBR-2hxj58. Effect of physical therapy in the treatment of knee osteoarthritis [Effects of different physical therapy protocols on knee osteoarthritis treatment]. RBR-2hxj58 2015.

RBR-4kryg4s *{published data only}*

RBR-4kryg4s. Effectiveness of an internet-delivered physical exercise program in pain and physical function of women with knee osteoarthritis. RBR-4kryg4s 2023.

RBR-53trrwk *{published data only}*

RBR-53trrwk. Effects of graded exercise in pain processing in patients with knee arthrosis [Effects of graded exercise in pain processing in patients with knee osteoarthritis: a randomized controlled trial]. RBR-53trrwk 2023.

RBR-5g9n5g *{published data only}*

RBR-5g9n5g. Effect of hip muscle strengthening in patients with osteoarthritis of the knee [Efficacy of the strengthening of the postero lateral lateral musculature of the hip in individuals with osteoarthritis of knee]. RBR-5g9n5g 2018.

RBR-5zpphs *{published data only}*

RBR-5zpphs. Impact of hidrotherapy and dry land exercises in daily activities and quality of life of older people with knee osteoarthritis [Impact of physiotherapy aquatic and dry land exercise in functional capacity and quality of life of older adults with knee osteoarthritis]. RBR-5zpphs 2016.

RBR-747wrkx *{published data only}*

RBR-747wrkx. Analysis of the effects of progressive neuromuscular training on pain and neuromuscular performance in elderly women with osteoarthritis of the knee. RBR-747wrkx 2021.

RBR-7s6ymc *{published data only}*

RBR-7s6ymc. Effect of low-level laser and exercise improving balance and functionality in people with knee osteoarthritis. RBR-7s6ymc 2015.

Slctr/2022/022 *{published data only}*

Slctr/2022/022. Finding the effect of a customized exercise program in improving pain and disability of people with knee arthritis compared to usual exercise program: a randomized controlled study. Slctr/2022/022 2022.

Tctr20230130003 *{published data only}*

Tctr20230130003. Short term effects of home-based isometric quadriceps exercise can alleviate pain and improve quality of life in elderly people with knee osteoarthritis: a randomized control trial. Tctr20230130003 2023.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Tctr20230130005 *{published data only}*

Tctr20230130005. Effectiveness of mobile exergaming with sensor-based visual feedback for active range of motion and quadriceps muscle power training in elderly with knee osteoarthritis: a prospective randomized controlled trial. Tctr20230130005 2023.

Umin000039723 *{published data only}*

Umin000039723. Central pain modulation and exercise-induced hypoalgesia in Japanese patients with knee and hip osteoarthritisCentral pain modulation and exercise-induced hypoalgesia in Japanese patients with knee and hip osteoarthritis. Umin000039723 2020.

Wang 2022a *{published data only}*

Wang XY, Xie SH, Zhang YJ, Zhu SY, Zhang RS, Wang L, et al. Effect of IoT-based power cycling and quadriceps training on pain and function in patients with knee osteoarthritis: a randomized controlled trial protocol. *Medicine (United States)* 2022;**101**(50):E31841. [DOI: 10.1097/MD.0000000000031841]

Wang 2022b *{published data only}*

Wang HN, Chen Y, Cheng L, Wang ST, Hu DX, Wang LN, et al. Effect of low-load resistance training with different degrees of blood flow restriction in patients with knee osteoarthritis: study protocol for a randomized trial. *Trials* 2022;**23**(1):(no pagination). [DOI: 10.1186/s13063-021-05946-7]

Weber 2023 *{published data only}*

Weber F, Muller C, Bahns C, Kopkow C, Farber F, Gellert P, et al. Smartphone-assisted training with education for patients with hip and/or knee osteoarthritis (SmArt-E): study protocol for a multicentre pragmatic randomized controlled trial. *BMC Musculoskeletal Disorders* 2023;**24**(1):(no pagination). [DOI: 10.1186/s12891-023-06255-7]

Xie 2018 *{published data only}*

Xie Y, Zhang C, Jiang W, Huang J, Xu L, Pang G, et al. Quadriceps combined with hip abductor strengthening versus quadriceps strengthening in treating knee osteoarthritis: A study protocol for a randomized controlled trial. *BMC Musculoskeletal Disorders* 2018;**19**(1):(no pagination). [DOI: 10.1186/s12891-018-2041-7]

Yang 2023 *{published data only}*

Yang K, Ding Y, Xu H, Lai M, Xu X, Yu X, et al. Efficacy and safety of platelet-rich plasma combined with Tai Chi for knee osteoarthritis: study protocol for a placebo-controlled randomized trial. *Journal of Orthopaedic Surgery and Research* 2023;**18**(1):885. [DOI: 10.1186/s13018-023-04372-6]

Zheng 2021 *{published data only}*

Zheng H, Zhang D, Zhu Y, Wang Q. Effect of Tai Chi exercise on lower limb function and balance ability in patients with knee osteoarthritis: A protocol of a randomized controlled trial. *Medicine (United States)* 2021;**100**(46):E27647. [DOI: 10.1097/MD.0000000000027647]

Additional references

Altman 1991a

Altman RD. Criteria for classification of clinical osteoarthritis. *Journal of Rheumatology. Supplement* 1991;**27**:10-2.

Bandak 2019

Bandak E, Boesen M, Bliddal H, Riis Robert G, Nielsen Sabrina M, Klokker L, et al. Exercise-induced pain changes associate with changes in muscle perfusion in knee osteoarthritis: exploratory outcome analyses of a randomised controlled trial. *BMC Musculoskeletal Disorders* 2019;**20**(1):1-12.

Bannuru 2014

Bannuru RR, Vaysbrot EE, McIntyre LF. Did the American Academy of Orthopaedic Surgeons osteoarthritis guidelines miss the mark? *Arthroscopy: the Journal of Arthroscopic & Related Surgery* 2014;**30**(1):86-9.

Bannuru 2019

Bannuru RR, Osani MC, Vaysbrot EE, Arden NK, Bennell K, Bierma-Zeinstra SM, et al. OARSI guidelines for the non-surgical management of knee, hip, and polyarticular osteoarthritis. *Osteoarthritis and Cartilage* 2019;**27**(11):1578-89.

Beckwée 2013

Beckwee D, Vaes P, Cnudde M, Swinnen E, Bautmans I. Osteoarthritis of the knee: why does exercise work? A qualitative study of the literature. *Ageing Research Reviews* 2013;**12**(1):226-36.

Blazek 2016

Blazek AD, Nam J, Gupta R, Pradhan M, Perera P, Weisleder NL, et al. Exercise-driven metabolic pathways in healthy cartilage. *Osteoarthritis and Cartilage* 2016;**24**(7):1210-22.

Bricca 2019

Bricca A, Struglics A, Larsson S, Steultjens M, Juhl CB, Roos EM. Impact of exercise therapy on molecular biomarkers related to cartilage and inflammation in individuals at risk of, or with established, knee osteoarthritis: a systematic review and meta-analysis of randomized controlled trials. *Arthritis Care & Research* 2019;**71**(11):1504-15.

Casilda-López 2017

Casilda-López J, Valenza Marie C, Cabrera-Martos I, Díaz-Pelegrina A, Moreno-Ramírez MP, Valenza-Demet G. Effects of a dance-based aquatic exercise program in obese postmenopausal women with knee osteoarthritis: a randomized controlled trial. *Menopause* 2017;**24**(7):768-73.

Caspersen 1985

Caspersen CJ, Powell KE, Christenson GM. Physical activity, exercise, and physical fitness: definitions and distinctions for health-related research. *Public Health Reports* 1985;**100**(2):126-31. PMID: 3920711; PMCID: PMC1424733.

Cohen 1998

Cohen J. Statistical power analysis for the behavioral sciences. 2nd edition. Hillsdale, NJ: Erlbaum, 1998.

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Concoff 2019

Concoff A Jeffrey R, Freddie F, Mohit B, Kevin B, Karlsson J, et al. A comparison of treatment effects for nonsurgical therapies and the minimum clinically important difference in knee osteoarthritis: a systematic review. *JBJS Reviews* 2019;**7**(8):e5.

Covidence [Computer program]

Covidence systematic review software. Version 3 March 2020. Melbourne, Australia: Veritas Health Innovation, 2020. Available at www.covidence.org.

Cui 2020

Cui A, Li H, Wang D, Zhong J, Chen Y, Lu H. Global, regional prevalence, incidence and risk factors of knee osteoarthritis in population-based studies. *EClinicalMedicine* 2020;**29**:100587.

Dahaghin 2005

Dahaghin S, Bierma-Zeinstra SMA, Ginai AZ, Pols HAP, Hazes JMW, Koes BW. Prevalence and pattern of radiographic hand osteoarthritis and association with pain and disability (the Rotterdam study). *Annals of the Rheumatic Diseases* 2005;**64**(5):682-7.

De Almeida 2021

De Almeida AC, Aily JB, Pedroso MG, Gonçalves GH, Pastre CM, Mattiello SM. Reductions of cardiovascular and metabolic risk factors after a 14-week periodized training model in patients with knee osteoarthritis: a randomized controlled trial. *Clinical Rheumatology* 2021;**40**(1):303-14. [DOI: 10.1007/s10067-020-05213-1] [PMID: 32514678]

Dean 2021

Dean BJ, Collins J, Thurley N, Rombach I, Bennell K. Exercise therapy with or without other physical therapy interventions versus placebo interventions for osteoarthritis-systematic review. *Osteoarthritis and Cartilage Open* 2021;**3**(3):100195.

Deeks 2022

Deeks JJ, Higgins JP, DG Altman. Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions Version 6.3. Cochrane, updated February 2022.

Dekker 2023

Dekker J, De Boer M, Ostelo R. Minimal important change and difference in health outcome: an overview of approaches, concepts, and methods. *Osteoarthritis and Cartilage* 2023;**32**(1):8-17.

Devji 2017

Devji T, Guyatt GH, Lytvyn L, Brignardello-Petersen R, Foroutan F, Sadeghirad B, et al. Application of minimal important differences in degenerative knee disease outcomes: a systematic review and case study to inform BMJ Rapid Recommendations. *BMJ Open* 2017;**7**(5):e015587.

Dworkin 2009

Dworkin RH, Turk DC, McDermott MP, et al. Interpreting the clinical importance of group differences in chronic pain clinical trials: IMMPACT recommendations. *Pain* 2009;**146**(3):238-44.

Egger 1997

Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;**315**(7109):629-34.

Englund 2023

Englund M, Turkiewicz A. The emperor's new clothes? *Osteoarthritis and Cartilage* 2023;**31**(5):549-51.

Fantuzzi 2005

Fantuzzi G. Adipose tissue, adipokines, and inflammation. *Journal of Allergy and Clinical Immunology* 2005;**115**(5):911-9.

Finan 2013

Finan PH, Luis FB, Sara CB, Shahid H, Raymond JP, et al. Discordance between pain and radiographic severity in knee osteoarthritis: findings from quantitative sensory testing of central sensitization. *Arthritis and Rheumatism* 2013;**65**(2):363-72.

Fransen 2015

Fransen M, McConnell S, Harmer AR, Van der Esch M, Simic M, Bennell KL. Exercise for osteoarthritis of the knee. *Cochrane Database of Systematic Reviews* 2015, Issue 1. Art. No: CD004376. [DOI: 10.1002/14651858.CD004376.pub3]

GBD 2017

GBD 2016 Disease and Injury Incidence and Prevalence Collaborators. Global, regional, and national incidence, prevalence, and years lived with disability for 328 diseases and injuries for 195 countries, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet* 2017;**390**(10100):1211-59.

Ghogomu 2014

Ghogomu EAT, Maxwell LJ, Buchbinder R, Rader T, Pardo JP, Johnston RV, et al. Updated method guidelines for Cochrane musculoskeletal group systematic reviews and metaanalyses. *Journal of Rheumatology* 2014;**41**(2):194-205.

Gleeson 2011

Gleeson M, Bishop NC, Stensel DJ, Lindley MR, Mastana SS, Nimmo MA. The anti-inflammatory effects of exercise: mechanisms and implications for the prevention and treatment of disease. *Nature Reviews Immunology* 2011;**11**(9):607-15.

Goh 2019

Goh S, Persson MS, Stocks J, Hou Y, Lin J, Hall MC, et al. Efficacy and potential determinants of exercise therapy in knee and hip osteoarthritis: a systematic review and meta-analysis. *Annals of Physical and Rehabilitation Medicine* 2019;**62**(5):356-65.

GRADEpro GDT [Computer program]

GRADEpro GDT. Version accessed 3 March 2020. Hamilton (ON): McMaster University (developed by Evidence Prime), 2020. Available at https://www.gradepro.org.

Hall 2018

Hall M, Hinman RS, Wrigley TV, Kasza J, Lim B-W, Bennell KL. Knee extensor strength gains mediate symptom improvement

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in knee osteoarthritis: secondary analysis of a randomised controlled trial. *Osteoarthritis and Cartilage* 2018;**26**(4):495-500.

Hayden 2021

Hayden JA, Ellis J, Ogilvie R, Malmivaara A, Van Tulder MW. Exercise therapy for chronic low back pain. *Cochrane Database of Systematic Reviews* 2021, Issue 9. Art. No: CD009790. [DOI: 10.1002/14651858.CD009790.pub2] [PMID: 34580864]

Henriksen 2024

Henriksen M, Runhaar J, Turkiewicz A, Englund M. Exercise for knee osteoarthritis pain: Association or causation? *Osteoarthritis and Cartilage* 2024;**1**:(no pagination).

Higgins 2022

Higgins JP, Altman DG, Sterne JA. Chapter 8: Assessing risk of bias in included studies. In: In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions Version 6.3 (updated February 2022). Cochrane, 2022.

Higgins 2024

Higgins JPT, Eldridge S, Li T. Chapter 23: Including variants on randomized trials [last updated October 2019]. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.5. Cochrane, 2024. Available from www.training.cochrane.org/handbook.

Hinman 2010

Hinman RS, Hunt MA, Creaby MW, Wrigley TV, McManus FJ, Bennell KL. Hip muscle weakness in individuals with medial knee osteoarthritis. *Arthritis Care & Research* 2010;**62**(8):1190-3.

Holden 2021

Holden MA, Button K, Collins NJ, Henrotin Y, Hinman RS, Larsen JB, et al. Guidance for implementing best practice therapeutic exercise for patients with knee and hip osteoarthritis: what does the current evidence base tell us? *Arthritis Care & Research* 2021;**73**(12):1746-53.

Holden 2023

Holden MA, Hattle M, Runhaar J, Riley RD, Healey EL, Quicke J, et al. Moderators of the effect of therapeutic exercise for knee and hip osteoarthritis: a systematic review and individual participant data meta-analysis. *Lancet Rheumatology* 2023;**5**(7):12.

Hunter 2019

Hunter DJ, Bierma-Zeinstra S. Osteoarthritis. *Lancet* 2019;**393**(10182):1745-59.

Hurley 2018

Hurley M, Kelly D, Hallett R, Grant R, Hauari H, Walsh N, et al. Exercise interventions and patient beliefs for people with hip, knee or hip and knee osteoarthritis: a mixed methods review. *Cochrane Database of Systematic Reviews* 2018, Issue 4. Art. No: CD010842. [DOI: 10.1002/14651858.CD010842.pub2]

IBM SPSS Statistics for Windows [Computer program]

IBM SPSS Statistics for Windows. Armonk, NY: IBM Corp, Released 2023. Version 29.0.2.0.

Juhl 2014

Juhl C, Christensen R, Roos EM, Zhang W, Lund H. Impact of exercise type and dose on pain and disability in knee osteoarthritis: a systematic review and meta-regression analysis of randomized controlled trials. *Arthritis & Rheumatology* 2014;**66**(3):622-36.

Kellgren 1957

Kellgren JH, Lawrence JS. Radiological assessment of osteo-arthrosis. *Annals of the Rheumatic Diseases* 1957;**16**(4):494.

Kinds 2011

Kinds MB, Welsing PM, Vignon EP, Bijlsma JW, Viergever MA, Marijnissen AC, et al. A systematic review of the association between radiographic and clinical osteoarthritis of hip and knee. *Osteoarthritis and Cartilage* 2011;**19**(7):768-78.

King 2011

King, MT. A point of minimal important difference (MID): a critique of terminology and methods. *Expert Review of Pharmacoeconomics & Outcomes Research* 2011;**11**(2):171-84.

Klyne 2020

Klyne DM, Hodges PW. Circulating adipokines in predicting the transition from acute to persistent low back pain. *Pain Medicine* 2020;**21**(11):2975-85.

Kolasinski 2020

Kolasinski SL, Neogi T, Hochberg MC, Oatis C, Guyatt G, Block J, et al. 2019 American College of Rheumatology/Arthritis Foundation Guideline for the Management of Osteoarthritis of the Hand, Hip, and Knee. *Arthritis & Rheumatology* 2020;**72**(2):149-62.

Lawford 2023

Lawford BJ, Hinman RS, McManus FJ, Lamb KE, Egerton T, Keating C, et al. How does exercise, with and without diet, improve pain and function in knee osteoarthritis? A secondary analysis of a randomised controlled trial exploring potential mediators of effects. *Arthritis Care & Research* 2023;**75**(11):2316-27.

Lefebvre 2024

Lefebvre C, Glanville J, Briscoe S, Featherstone R, Littlewood A, Metzendorf M-I, Noel-Storr A, Paynter R, Rader T, Thomas J, Wieland LS. Technical Supplement to Chapter 4: searching for and selecting studies. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.5. Available from www.training.cochrane.org/handbook. Cochrane, 2024.

Lima 2022

Lima YL, Lee H, Klyne DM, Dobson FL, Hinman RS, Bennell KL, et al. How do non-surgical interventions improve pain and physical function in people with osteoarthritis? A scoping

Copyright © 2024 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

review of mediation analysis studies. *Arthritis Care & Research* 2022;**75**(3):467-81.

Mo 2023

Mo L, Jiang B, Mei T, Zhou D. Exercise therapy for knee osteoarthritis: a systematic review and network meta-analysis. *Orthopaedic Journal of Sports Medicine* 2023;**11**(5):23259671231172773.

Moseng 2024

Moseng T, Vlieland TPMV, Battista S, Beckwée D, Boyadzhieva V, Conaghan PG, et al. EULAR recommendations for the non-pharmacological core management of hip and knee osteoarthritis: 2023 update. *Annals of the Rheumatic Diseases* 2024;**83**(6):730-40.

Muraki 2015

Muraki S, Akune T, Teraguchi M, Kagotani R, Asai Y, Yoshida M, et al. Quadriceps muscle strength, radiographic knee osteoarthritis and knee pain: the ROAD study. *BMC Musculoskeletal Disorders* 2015;**16**(1):1-10.

NICE 2022

Osteoarthritis in over 16s: diagnosis and management. National Institute for Health and Care Excellence 2022.

Niemeijer 2020

Niemeijer A, Lund H, Stafne SN, Ipsen T, Goldschmidt CL, Jørgensen CT, et al. Adverse events of exercise therapy in randomised controlled trials: a systematic review and meta-analysis. *British Journal of Sports Medicine* 2020;**54**(18):1073-80.

Page 2021

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Systematic Reviews* 2021;**10**(1):1-11.

Paley 2016

Paley CA, Johnson MI. Physical activity to reduce systemic inflammation associated with chronic pain and obesity. *Clinical Journal of Pain* 2016;**32**(4):365-70.

Park 2017

Park J, McCaffrey R, Newman D, Liehr P, Ouslander JG. A pilot randomized controlled trial of the effects of chair yoga on pain and physical function among community-dwelling older adults with lower extremity osteoarthritis. *Journal of the American Geriatrics Society* 2017;**65**(3):592-7.

Pedersen 2023

Pedersen JR, Sari DM, Juhl CB, Thorlund JB, Skou ST, Roos EM, et al. Variability in effect sizes of exercise therapy for knee osteoarthritis depending on comparator interventions. *Annals of Physical and Rehabilitation Medicine* 2023;**66**(4):101708.

Pisters 2007

Pisters MF, Veenhof C, Van Meeteren NLU, Ostelo RW, De Bakker DH, Schellevis FG, et al. Long-term effectiveness of exercise therapy in patients with osteoarthritis of the hip or knee: a systematic review. *Arthritis Care & Research* 2007;**57**(7):1245-53.

Power 2008

Power JD, Badley EM, French MR, Wall AJ, Hawker GA. Fatigue in osteoarthritis: a qualitative study. *BMC Musculoskeletal Disorders* 2008;**9**(1):1-8.

RACGP 2018

The Royal Australian College of General Practitioners. Guideline for the management of knee and hip osteoarthritis. The Royal Australian College of General Practitioners 2018.

Rejeski 1998

Rejeski WJ, Ettinger WH Jr, Martin K, Morgan T. Treating disability in knee osteoarthritis with exercise therapy: A central role for self-efficacy and pain. *Arthritis & Rheumatology* 1998;**11**(2):94-101.

Ribeiro 2024

Ribeiro DC, Lamb SE, Abbott JH. Mediators of the effects of exercise and manual therapy for people with knee and hip osteoarthritis: a secondary, exploratory analysis of the MOA trial. *Osteoarthritis and Cartilage Open* 2024;**6**(1):100431.

Ruegsegger 2018

Ruegsegger GN, Booth FW. Health benefits of exercise. *Cold Spring Harbor Perspectives in Medicine* 2018;**8**(7):a029694.

Runhaar 2015

Runhaar J, Luijsterburg P, Dekker J, Bierma-Zeinstra S. Identifying potential working mechanisms behind the positive effects of exercise therapy on pain and function in osteoarthritis; [sic] a systematic review. *Osteoarthritis and Cartilage* 2015;**23**(7):1071-82.

Runhaar 2019

Runhaar J, Beavers DP, Miller GD, Nicklas BJ, Loeser RF, Bierma-Zeinstra S, et al. Inflammatory cytokines mediate the effects of diet and exercise on pain and function in knee osteoarthritis independent of BMI. *Osteoarthritis and Cartilage* 2019;**27**(8):1118-23.

Runhaar 2023

Runhaar J, Holden MA, Hattle M, Quicke J, Healey EL, Van der Windt D, et al. Mechanisms of action of therapeutic exercise for knee and hip OA remain a black box phenomenon: an individual patient data mediation study with the OA Trial Bank. *RMD Open* 2023;**9**(3):e003220. [DOI: 10.1136/rmdopen-2023-003220]

Sasaki 2022

Sasaki R, Honda Y, Oga S, Fukushima T, Tanaka N, Kajiwara Y, et al. Effect of exercise and/or educational interventions on physical activity and pain in patients with hip/knee osteoarthritis: a systematic review with meta-analysis. *PloS One* 2022;**17**(11):e0275591.

Schünemann 2022

Schünemann HJ, Vist GE, Higgins JP, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and drawing conclusions. In: Higgins JPT, Thomas J, Chandler J, Cumpston

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M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions Version 6.3 (updated February 2022). Cochrane, 2022.

Si 2023

Si J, Sun L, Li Z, Zhu W, Yin W, Peng L. Effectiveness of home-based exercise interventions on pain, physical function and quality of life in individuals with knee osteoarthritis: a systematic review and meta-analysis. *Journal of Orthopaedic Surgery and Research* 2023;**18**(1):503.

Slade 2016

Slade SC, Dionne CE, Underwood M, Buchbinder R. Consensus on exercise reporting template (CERT): explanation and elaboration statement. *British Journal of Sports Medicine* 2016;**50**(23):1428-37.

Smith 2019

Smith TO, Hawker GA, Hunter DJ, March LM, Boers M, Shea BJ, et al. The OMERACT-OARSI core domain set for measurement in clinical trials of hip and/or knee osteoarthritis. *Journal of Rheumatology* 2019;**46**(8):981-9.

Steinmetz 2023

Steinmetz JD, Culbreth GT, Haile LM, Rafferty Q, Lo J, Fukutaki KG, et al. Global, regional, and national burden of osteoarthritis, 1990-2020 and projections to 2050: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet Rheumatology* 2023;**5**(9):e508-22.

Turner 2020

Turner MN, Hernandez DO, Cade W, Emerson CP, Reynolds JM, Best TM. The role of resistance training dosing on pain and physical function in individuals with knee osteoarthritis: a systematic review. *Sports Health* 2020;**12**(2):200-6.

Verhagen 2019

Verhagen AP, Ferreira M, Reijneveld-van de Vendel EA, Teirlinck CH, Runhaar J, Van Middelkoop M, et al. Do we need another trial on exercise in patients with knee osteoarthritis? : [sic] No new trials on exercise in knee OA. *Osteoarthritis and Cartilage* 2019;**27**(9):1266-9.

Veronese 2016

Veronese N, Cereda E, Maggi S, Luchini C, Solmi M, Smith T, et al. Osteoarthritis and mortality: a prospective cohort study and systematic review with meta-analysis. *Seminars in Arthritis and Rheumatism* 2016;**46**(2):160-7.

Vårbakken 2019

Vårbakken K, Lorås H, Nilsson Kjell G, Engdal M, Stensdotter A. Relative difference in muscle strength between patients with knee osteoarthritis and healthy controls when tested bilaterally and joint-inclusive: an exploratory cross-sectional study. *BMC Musculoskeletal Disorders* 2019;**20**(1):1-13.

Wood 2008

Wood L, Egger M, Gluud LL, Schulz KF, Jüni P, Altman DG, et al. Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ* 2008;**336**(7644):601-5.

Wyatt 2014

Wyatt B, Mingo CA, Waterman MB, White P, Cleveland RJ, Callahan LF. Impact of the arthritis foundation's walk with ease program on arthritis symptoms in african americans. *Preventing Chronic Disease* 2014;**13**(11):E199.

Zeng 2022

Zeng L, Brignardello-Petersen R, Hultcrantz M, Mustafa RA, Murad MH, Alfonso I, et al. GRADE Guidance 34: update on rating imprecision using a minimally contextualized approach. *Journal of Clinical Epidemiology* 2022;**150**:216-24. [DOI: 10.1016/j.jclinepi.2022.07.014]

Zhang 2010a

Zhang W, Nuki G, Moskowitz RW, Abramson S, Altman RD, Arden NK, et al. OARSI recommendations for the management of hip and knee osteoarthritis: Part III: changes in evidence following systematic cumulative update of research published through January. *Osteoarthritis and Cartilage* 2010;**18**(4):476-99.

Zhang 2010b

Zhang W, Doherty M, Peat G, Bierma-Zeinstra MA, Arden NK, Bresnihan B, et al. EULAR evidence-based recommendations for the diagnosis of knee osteoarthritis. *Annals of the Rheumatic Diseases* 2010;**69**(3):483-9.

Zhang 2013

Zhang S, Liu H, Xu X, Zhi J, Geng J, Chen J. Effects of exercise therapy on knee joint function and synovial fluid cytokine levels in patients with knee osteoarthritis. *Molecular Medicine Reports* 2013;**7**(1):183-6.

Zhu 2016

Zhu Q, Huang L, Wu X, Wang L, Zhang Y, Fang M, et al. Effects of Tai Ji Quan training on gait kinematics in older Chinese women with knee osteoarthritis: a randomized controlled trial. *Journal of Sport and Health Science* 2016;**5**(3):297-303.

References to other published versions of this review

Fransen 2002

Fransen M, et al. Therapeutic exercise for people with osteoarthritis of the hip or knee. A systematic review. *Journal of Rheumatology* 2002;**29**:1737-45.

Fransen 2008

Fransen M, McConnell S. Exercise for osteoarthritis of the knee. *Cochrane Database of Systematic Reviews* 2008, Issue 4. Art. No: CD004376. [DOI: 10.1002/14651858.CD004376.pub2]

Fransen 2009

Fransen M, McConnell S. Land-based exercise for osteoarthritis of the knee: a metaanalysis of randomized controlled trials. *Journal of Rheumatology* 2009;**36**:1109-17.

* Indicates the major publication for the study

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CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Abbott 2013

Study characteristics	
Methods	Conducted in New Zealand
Participants	Hip and knee OA recruitment 57 knee OA community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person 1. Exercise (aerobic + strengthening + neuromuscular): 156 sessions (3 x 50 minutes per week over 16 weeks); 9 synchronous contact occasions 2. Usual care; all participants continued to receive routine care offered by their own GP and other healthcare providers. No trial interventions were provided.
Outcomes	At 52 weeks (post-intervention) and 104 weeks Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Outcomes measured only at 1 year Only data related to those with knee OA used for outcomes at 52 weeks (authors unresponsive to request for knee data at 104 weeks)

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Online randomisation service, stratified for hip or knee OA
Allocation concealment (selection bias)	Low risk	Varied block size randomisation, randomisation service kept schedule
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded participants and therapists
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants, self-reported pain and physical function

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Abbott 2013 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up balanced between allocation groups, ITT analysis
Selective reporting (reporting bias)	Low risk	Registered trial

Alghadir 2019

Study characteristics	
Methods	Conducted in Saudi Arabia
Participants	68 people with knee OA ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Routine physiotherapy; 18 synchronous contact occasions 2. Routine physiotherapy + retrowalking; 18 sessions (3 x 10 to 30 minutes per week over 6 weeks); 18 synchronous contact occasions 3. Routine physiotherapy + forward walking; 18 sessions (3 x 10 to 30 minutes per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC global scale) No quality of life measure No participant-reported treatment success measure
Notes	Groups 2 and 3 were pooled for meta-analysis.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information on how the numbers were randomly generated
Allocation concealment (selection bias)	Low risk	Blank folders were numbered from 1 to 68, given concealed codes for the group assignment by an independent therapist, and kept in a safe locker.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded from knowledge of which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported	High risk	Unblinded assessor (participant)

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Alghadir 2019 *(Continued)*

outcomes (pain, function, quality of life)		
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analyses. There was 18% attrition but no reasons were provided.
Selective reporting (reporting bias)	Low risk	Registered trial - measures in the registry appear to be reported.

Allen 2018

Study characteristics	
Methods	Conducted in USA
Participants	Knee OA recruitment 210 community volunteers ACR criteria
Interventions	Home-based, unsupervised individual setting, entirely digitally-delivered 1. Wait list; 0 synchronous contact occasions 2. Exercise (strengthening, mixed aerobic, stretching): 48 sessions (3x per week over 16 weeks); 0 synchronous contact occasions
Outcomes	At 17 weeks (immediately post-intervention) and 52 weeks Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure Participant-reported treatment success (global assessment of change)
Notes	Participant-reported treatment success reported on continuous scale (rather than dichotomous). Not included in meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Unclear risk	No information about concealment other than "participants were given the randomisation assignment via telephone by study co-ordinator"
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.

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Allen 2018 *(Continued)*

Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition and reasons provided. Attrition was 14%. ITT analysis
Selective reporting (reporting bias)	High risk	Registered trial. Not all outcomes in registry appear in publication. Secondary measures Brief Fear of Movement scale, Patient Health Questionnaire, and PROMIS (sleep and fatigue) are in registry but not in primary publication.

An 2008

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 28 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Baduanjin (type of Qigong, less physically demanding than Tai Chi): 40 sessions (5 x 30 minutes per week over 8 weeks); 40 synchronous contact occasions
Outcomes	At 8 weeks Pain (WOMAC) Physical function (WOMAC) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	Poor comparability at baseline for WOMAC pain and physical function. Post-treatment scores indicate non-normal distribution, i.e. mean (SD) not appropriate

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Nothing other than 'patients were randomised'
Allocation concealment (selection bias)	Unclear risk	No information provided

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An 2008 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Not blinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Not disclosed
Incomplete outcome data (attrition bias) All outcomes	High risk	High loss to follow-up: 3 (21%) and 4 (29%). No ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Anadkat 2015

Study characteristics	
Methods	Conducted in India
Participants	Knee OA recruitment 40 community volunteers Diagnosed by clinician; Grade 3 KL grade
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Conventional exercises: 15 synchronous contact occasions 2. Conventional exercises + retrowalking: 51 sessions (5 x 10 minutes per week over 3 weeks); 15 synchronous contact occasions
Outcomes	At 3 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC global scale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information on how the randomisation was generated.
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes

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Anadkat 2015 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition not reported on. All participants included in the analysis.
Selective reporting (reporting bias)	Unclear risk	Not registered

Anwer 2014

Study characteristics	
Methods	Conducted in Saudi Arabia
Participants	Knee OA recruitment 42 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Ultrasound: 6 synchronous contact occasions 2. Ultrasound + exercise (strengthening): 50 sessions (10 x per week over 5 weeks); 50 synchronous contact occasions
Outcomes	At 5 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (reduced WOMAC function scale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No information described on how numbers were randomly generated
Allocation concealment (selection bias)	Unclear risk	No information on methods used to conceal the allocation

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Anwer 2014 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There were no participants lost to follow-up.
Selective reporting (reporting bias)	Unclear risk	Not registered

Assar 2020

Study characteristics	
Methods	Conducted in Iran
Participants	Knee OA recruitment 24 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, entirely in-person 1. Drug regimen: 0 synchronous contact occasions 2. Drug regimen + exercise (Total Resistance eXercises (TRX)): 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (global pain VAS/NRS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number generator software used
Allocation concealment (selection bias)	Low risk	Sequentially numbered, opaque sealed envelopes were used; physiotherapist who was not involved in the data collection and evaluation assigned participants to groups.

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Assar 2020 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition - all participants included in analysis
Selective reporting (reporting bias)	High risk	Trial registered. Not all outcomes (e.g. time up and go) registered were reported.

Baker 2001

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 46 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person 1. Attention control; 7 synchronous contact occasions (home visits, where received general nutrition education) 2. Exercise (strengthening): 48 sessions (3x per week over 16 weeks); 12 synchronous contact occasions
Outcomes	At 16 weeks Pain (WOMAC) Function (WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Generated by independent statistician
Allocation concealment (selection bias)	Low risk	Central allocation

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Baker 2001 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded to the active intervention.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were outcome assessors and were blinded to hypotheses.
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Bandak 2022

Study characteristics	
Methods	Conducted in Denmark
Participants	Knee OA recruitment 206 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, entirely in-person 1. Intra-articular saline injections; 4 synchronous contact occasions 2. Exercise (neuromuscular exercise): 14 sessions (2 x 60 minutes per week over 12 weeks); 14 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 52 weeks Pain (KOOS pain subscale) Physical function (KOOS ADL function subscale) Quality of life (KOOS QoL) Participant-reported treatment success (OMERACT-OARSI)
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Allocation sequence was developed by the trial biostatistician not actively involved in the conduct of the trial.

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Bandak 2022 *(Continued)*

Allocation concealment (selection bias)	Low risk	Allocation was concealed until an investigator pressed randomised in the electronic trial management system.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition and reasons for attrition were reported. ITT analysis. 88% of participants had follow-up data.
Selective reporting (reporting bias)	Low risk	Trial registered. Outcomes in registry match publication

Bautch 1997

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 34 community volunteers ACR criteria
Interventions	Clinic, supervised individual, entirely-in-person 1. Education; 12 synchronous contact occasions 2. Education + exercise (range of motion + walking): 36 sessions (3x per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks Pain (VAS) Function (AIMS) No quality of life measure No participant-reported treatment success measure
Notes	Allocation groups very incomparable baseline pain and BMI, with active treatment group demonstrating more severe disease

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

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Bautch 1997 *(Continued)*

Random sequence generation (selection bias)	Unclear risk	No information
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	High risk	Efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Bennell 2010

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA reruitment 89 community volunteers ACR criteria, KL Grade II+, medial tibiofemoral compartment disease
Interventions	Clinic, mix of supervised/unsupervised individual, entirely in-person 1. Waiting list; 0 synchronous contact occasions 2. Exercise (strengthening): 60 sessions (5 x 15 to 30 minutes per week over 12 weeks); 7 synchronous contact occasions
Outcomes	At 12 weeks 1. Pain (NRS pain on walking) 2. Physical function (WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

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Bennell 2010 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated random numbers, permuted block 4 to 6
Allocation concealment (selection bias)	Low risk	Independent investigator; sealed opaque envelopes; central location
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced loss to follow-up (13% vs 16%), ITT analysis
Selective reporting (reporting bias)	Low risk	Registered trial

Bennell 2016

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 147 community volunteers ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised individual setting, entirely in-person 1. Pain coping skills training; 10 synchronous contact occasions 2. Pain coping skills training + exercise (neuromuscular exercise): 48 sessions (4 x 25 minutes per week over 12 weeks); 10 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 52 weeks Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (AQoL-6D) Participant-reported treatment success (global rating of change)
Notes	
Risk of bias	

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Bennell 2016 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A computer-generated random numbers table generated by trial biostatistician
Allocation concealment (selection bias)	Low risk	Allocations were sealed in consecutively numbered opaque envelopes by an independent person and stored in a locked location.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded to study hypotheses. Personnel (care providers) were not blinded to which intervention a participant received. Outcome assessors collecting and entering participant data were blinded to allocation.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were the outcome assessors and were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition and reasons for attrition were reported. 9% dropout. ITT analysis
Selective reporting (reporting bias)	Low risk	Outcomes reported consistent with registry

Bennell 2022a

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 240 community volunteers NICE clinical criteria
Interventions	Home, unsupervised individual setting, entirely digitally-delivered 1. Education website; 0 synchronous contact occasions 2. Education website + exercise (strengthening, physical activity plan): 78 sessions (3x per week over 26 weeks); 6 synchronous contact occasions
Outcomes	At 26 weeks (immediately post-intervention) and 52 weeks Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (AQoL-8D) Participant-reported treatment success (global rating of change)
Notes	

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Bennell 2022a *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schedule was prepared by a biostatistician.
Allocation concealment (selection bias)	Low risk	Allocation was concealed in password-protected software (REDCap) and revealed by a researcher not involved in recruitment or outcome assessment.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (10%) and reasons for attrition were described. ITT analysis.
Selective reporting (reporting bias)	Low risk	Outcomes in registry are consistent with this report.

Bennell 2022b

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 212 community volunteers NICE clinical criteria
Interventions	Home, unsupervised individual setting, entirely digitally-delivered 1. Education website; 0 synchronous contact occasions 2. Education website + exercise (yoga): 36 sessions (3 x 30 minutes per week over 12 weeks); 0 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 24 weeks Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (AQoL-6D) Participant-reported treatment success (global rating of change)

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Bennell 2022b *(Continued)*

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	An independent biostatistician prepared a computer-generated randomisation schedule with permuted blocks of varying sizes.
Allocation concealment (selection bias)	Low risk	Password-protected software (REDCap) concealed allocation, which was revealed by a researcher not involved in recruitment or outcome assessment.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded to hypotheses. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were the outcome assessors and were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (8%) and reasons for attrition were reported. ITT analysis
Selective reporting (reporting bias)	Low risk	Outcomes in registry consistent with report

Bhagat 2017

Study characteristics

Methods	Conducted in India
Participants	Knee OA recruitment 30 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Interferential therapy; 12 synchronous contact occasions 2. Interferential therapy + exercise (circuit training): 12 sessions (3 x 20 minutes per week over 4 weeks); 12 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (Global pain VAS/NRS) Physical function (WOMAC global scale) No quality of life measure No participant-reported treatment success measure

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Bhagat 2017 *(Continued)*

Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Simple random sampling
Allocation concealment (selection bias)	Unclear risk	Information on the method used to conceal the allocation sequence is not described.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition not reported on. All participants appear to be included in the analysis.
Selective reporting (reporting bias)	Unclear risk	Trial is not registered and there is no protocol to verify outcome reporting.

Braghin 2018

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 31 community volunteers Radiographic OA
Interventions	Clinic, supervised group setting, entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, mixed aerobic, stretching): 16 sessions (2 x 50 to 60 minutes per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure

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Braghin 2018 *(Continued)*

Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Simple randomisation
Allocation concealment (selection bias)	Low risk	Group selection was done by a raffle, using opaque envelopes from which the volunteers drew.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition. ITT analysis.
Selective reporting (reporting bias)	Unclear risk	No trial registry or protocol to verify reported outcomes against

Brismée 2007

Study characteristics

Methods	Conducted in USA
Participants	Knee OA recruitment 41 community volunteers ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised mix of individual/group, entirely in-person 1. Attention control; education programme delivered via 18 synchronous contact occasions 2. Exercise (Tai Chi): 36 sessions (3 x 40 minutes per week over 12 weeks); 18 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (VAS 0 to 10) Physical function (WOMAC 17 to 85) No quality of life measure No participant-reported treatment success measure

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Brismée 2007 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation table stratified by age and sex
Allocation concealment (selection bias)	Unclear risk	Not stated
Blinding of participants and personnel (performance bias) All outcomes	High risk	Not blinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up 9% intervention, 27% control; apparent ITT (data from all participants who did not drop out in the first week)
Selective reporting (reporting bias)	Unclear risk	Not registered

Bruce-Brand 2012

Study characteristics

Methods	Conducted in Ireland
Participants	Knee OA recruitment 27 community volunteers KL Grade III+
Interventions	Home, mixed supervised/unsupervised individual setting, delivered entirely in-person 1. Standard care; unclear number synchronous contact occasions 2. Exercise (strengthening): 18 sessions (2 x 30 minutes per week over 6 weeks); 12 synchronous contact occasions
Outcomes	At week 8 and week 14 Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) Quality of life (SF-36 MCS)

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Bruce-Brand 2012 *(Continued)*

	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers, stratified by age and sex
Allocation concealment (selection bias)	Unclear risk	Investigator with no clinical role in the study
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	High risk	High loss to follow-up (29% and 54%), no ITT analysis
Selective reporting (reporting bias)	Low risk	Registered trial

Chang 2012

Study characteristics

Methods	Conducted in Taiwan
Participants	Knee OA recruitment 60 community volunteers KL Grade II or III, knee flexion > 90 degrees
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. General physiotherapy; 16 synchronous contact occasions 2. General physiotherapy + exercise (strengthening): 16 sessions (2 x 60 minutes per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks 1. Pain (WOMAC pain subscale) 2. Physical function (WOMAC function subscale) No quality of life measure

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Chang 2012 *(Continued)*

	No participant-reported treatment success measure
Notes	All participants 'prohibited' from using Chinese medicine or alternative therapies and non-habitual exercise during the study.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomly assigned but no description of procedure provided
Allocation concealment (selection bias)	Unclear risk	No indication
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants blinded, therapist unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants blinded
Incomplete outcome data (attrition bias) All outcomes	High risk	High loss to follow-up, unbalanced (20% and 44% controls), no ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Chen 2014

Study characteristics

Methods	Conducted in Taiwan
Participants	Knee OA recruitment 60 community volunteers Altman III criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person: 1. No treatment; 0 synchronous contact occasions 2. Exercise (strengthening): 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) and 26 weeks Pain (on activities other than walking) (VAS/NRS) Physical function (Lequesne OA Index global score) No quality of life measure

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Chen 2014 *(Continued)*

	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Opaque sealed envelopes containing treatment allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition not described. Assume all participants were included in the analysis
Selective reporting (reporting bias)	Unclear risk	Not registered and no protocol to verify reporting.

Chen 2021

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 32 community volunteers ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised individual setting, delivered entirely in-person 1. Conventional treatment; 4 synchronous contact occasions 2. Conventional treatment + exercise (retrowalking): 12 sessions (3 x 10 minutes per week over 4 weeks); 5 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure

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Chen 2021 *(Continued)*

	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	The numbers were kept in a locked location in a sealed opaque envelope.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was no attrition and all participants were included in the analysis.
Selective reporting (reporting bias)	High risk	Outcome data are inconsistent with trial registration (e.g FPI and muscle tension).

Cheung 2014

Study characteristics

Methods	Conducted in USA
Participants	Knee OA recruitment 36 community volunteers ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (Yoga): 40 sessions (5 x 30 to 60 minutes per week over 8 weeks); 8 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale)

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Cheung 2014 *(Continued)*

	Quality of life (SF-12 MCS) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	High risk	Those who received an odd number were assigned to the treatment group and those who received an even number were assigned to the wait-list control. This was not quasi-randomised as participants were given an ID number based on their order of enrolment, but it was a separate sequence of computer-generated random numbers that determined to which group participants were allocated.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (3%) and reason reported. ITT analysis.
Selective reporting (reporting bias)	Low risk	Outcome measures are consistent with trial registry.

Cheung 2017

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 83 community volunteers Self-reported medical diagnosis of OA
Interventions	Mix of home/clinic, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person 1. Attention control; education brochures + 8 synchronous contact occasions (telephone calls, asked about symptoms and general health) 2. Exercise (yoga): 40 sessions (5 x 30 to 45 minutes per week over 8 weeks); 8 synchronous contact occasions

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Cheung 2017 *(Continued)*

	3. Exercise (aerobic, strengthening): 64 sessions (8 x 45 minutes per week over 8 weeks); 8 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (SF-12 MCS) No participant-reported treatment success measure
Notes	Group 2 and 3 were pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	An online software programme was used to generate random numbers.
Allocation concealment (selection bias)	Low risk	A sealed envelope
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (4-18% across groups) and reasons are reported. ITT analysis.
Selective reporting (reporting bias)	Low risk	Reporting is consistent with trial registry.

Chopp-Hurley 2017

Study characteristics

Methods	Conducted in Canada
Participants	Hip and knee OA recruitment 18 knee OA community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions

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Chopp-Hurley 2017 *(Continued)*

	2. Exercise (Yoga): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	Only data for those with knee OA used

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were stratified by function and randomised using custom Matlab programme.
Allocation concealment (selection bias)	Low risk	Group allocation was concealed from the blinded assessor.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (17%) occurred but no reasons were reported. Unclear if analysis were ITT.
Selective reporting (reporting bias)	Unclear risk	Trial not registered and do not have protocol to verify.

Christensen 2015

Study characteristics

Methods	Conducted in Denmark
Participants	Knee OA recruitment 128 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised mix of individual/group setting, delivered entirely in-person

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Christensen 2015 *(Continued)*

	1. No intervention; 0 synchronous contact occasions 2. Exercise (neuromuscular): 196 sessions (3 to 4x per week over 52 weeks); 52 synchronous contact occasions
Outcomes	At 52 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (global disability score) Quality of life (SF-36 MCS) Participant-reported treatment success (OARSI OMERACT)
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation list was drawn up by biostatistican
Allocation concealment (selection bias)	Low risk	Allocation concealment was done on the basis of all the participants entering the study at baseline. The random assignment prevented knowledge of forthcoming allocations by study participants and personnel recruiting into the study.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (8%) and reasons for attrition are reported. ITT analysis.
Selective reporting (reporting bias)	Unclear risk	Some secondary outcome measures listed in the trial registry (e.g. MRI, ultrasound) are not detailed in this publication.

Da Silva 2015

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 41 community volunteers ACR criteria

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Da Silva 2015 *(Continued)*

Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Education; 1 synchronous contact occasions 2. Exercise (strengthening, proprioceptive, body mobility): 16 sessions (2 x 60 minutes per week over 8 weeks); 17 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (Lequesne Index Pain subscale) Physical function (Lequesne OA index global scale) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-based random number sequence was generated by a volunteer not involved in the study.
Allocation concealment (selection bias)	Low risk	Sequentially numbered opaque sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (27%) and reasons for attrition were reported. Analysis performed on participants who completed the trial
Selective reporting (reporting bias)	Low risk	Outcomes reported are consistent with outcomes in the trial registry.

De Almeida 2020

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 66 community volunteers

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De Almeida 2020 *(Continued)*

	ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; education sessions, 6 synchronous contact occasions 2. Exercise (circuit training): 42 sessions (3x per week over 14 weeks); 42 synchronous contact occasions 3. Exercise (conventional strength training): 42 sessions (3 x 60 minutes per week over 14 weeks); 42 synchronous contact occasions
Outcomes	At 14 weeks (immediately post-intervention) Pain (Global pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Group 2 and 3 were pooled for meta-analysis Function data obtained from De Almeida 2021 Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomised through a randomisation website (https://www.random.org)
Allocation concealment (selection bias)	Unclear risk	A researcher not involved with the evaluations and intervention session conducted the randomisation and balance distribution.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (8%) and reasons for attrition were reported. Analysis not performed as ITT
Selective reporting (reporting bias)	Low risk	Outcomes are consistent with outcomes listed in the trial registry.

De Matos 2019

Study characteristics

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De Matos 2019 *(Continued)*

Methods	Conducted in Brazil
Participants	Knee OA recruitment 56 community volunteers Other: KL Grade I-III
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, proprioceptive/balance, circuit training, cycling, stretching): 15 sessions (2 x 50 minutes per week over 8 weeks); 15 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information provided
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (> 50%) and reasons, although not in detail, are reported. Analysis only performed on 53.8% of the initial randomised sample
Selective reporting (reporting bias)	Unclear risk	Unable to verify against either trial registry or published protocol

De Rooij 2017

Study characteristics

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De Rooij 2017 *(Continued)*

Methods	Conducted in the Netherlands
Participants	Knee OA recruitment 126 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, proprioceptive/balance, mixed aerobic, stretching): 140 sessions (7 x 30 to 60 minutes per week over 20 weeks); 40 synchronous contact occasions
Outcomes	At 20 weeks (immediately post-intervention) and 32 weeks follow-up Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) Participant-reported treatment success (global change) No quality of life measure
Notes	Global change only reported for intervention group; not included in meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomly assignment to the intervention group or the control group by the web-based programme MagMin.
Allocation concealment (selection bias)	Unclear risk	Participants were randomised by an independent staff member who had no other involvement in the trial.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (11%) and reasons for attrition were reported. ITT analysis
Selective reporting (reporting bias)	High risk	Some secondary outcome measures in the trial registry are not in this publication (e.g. HbAc1 mmol/mol, dyspnoea).

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DeVita 2018

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 31 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation sequence was made before commencement of study activities by computer random number generator.
Allocation concealment (selection bias)	Unclear risk	No details provided on methods used to conceal the allocation sequence.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (3%) and reasons were reported. ITT not used.
Selective reporting (reporting bias)	Low risk	Outcomes reported in the publication are consistent with outcomes in the trial registry.

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Dincer 2016

Study characteristics	
Methods	Conducted in Turkey
Participants	Knee OA recruitment 35 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised mix of individual/group setting, delivered entirely in-person 1. TENS; 15 synchronous contact occasions 2. TENS + exercise (strengthening): 60 sessions (5 x 30 minutes per week over 12 weeks); 15 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	All participants were randomly assigned to groups using computer-generated numerical assignments.
Allocation concealment (selection bias)	Low risk	The sequence numbers were written in closed envelopes and each patient chose one to assign a group.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (14%) and reasons for attrition were reported. Analysis was only performed on those who completed the study.
Selective reporting (reporting bias)	Unclear risk	Unable to verify against a published study protocol or trial registry

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Drummen 2023

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 40 community volunteers ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised mix of individual/group setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (walking): 72 sessions (3 x 60 minutes per week over 24 weeks); 48 synchronous contact occasions
Outcomes	At 24 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (EuroQoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Simple randomisation using computer-generated random numbers via a central randomisation website hosted by University of Tasmania
Allocation concealment (selection bias)	Unclear risk	A blinded staff member with no direct involvement in the study performed the allocation using the description above.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were not unblinded to group allocation.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (17.5%) and reasons for attrition were reported. Analysis completed only on those with complete outcomes
Selective reporting (reporting bias)	Low risk	Outcomes reported appear consistent with outcomes in trial registry.

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Dwyer 2015

Study characteristics	
Methods	Multicentre RCT conducted in USA and South Africa
Participants	Knee OA recruitment 55 community volunteers ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised mix of individual/group setting, entirely in-person 1. MMT; 6 synchronous contact occasions 2. MMT + exercise (strengthening + cycling + stretching): 12 sessions (3 x 20 minutes per week over 4 weeks); 9 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation was conducted using a random number process where a sequence of random numbers with sequence boundaries was generated using computer software.
Allocation concealment (selection bias)	Low risk	Each treatment assigned was placed in a sealed opaque envelope, producing a series of sequentially numbered sealed opaque envelopes.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessor.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (6%) and reasons for attrition are reported. ITT analysis
Selective reporting (reporting bias)	Low risk	Outcomes appear consistent with trial registry

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Ettinger 1997

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 439 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; education, 3 synchronous contact occasions 2. Exercise (aerobic walking): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions 3. Exercise (strengthening): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	Mean score at 12, 36, 72 weeks Pain (range 1 to 6) Function (range 1 to 6) No quality of life measure No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis After cessation of classes, high level of regular telephone monitoring (monthly in past 9 months)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Web-based, central allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants unblinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis, missing data balanced between allocation groups
Selective reporting (reporting bias)	Unclear risk	Not registered

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Foley 2003

Study characteristics	
Methods	Conducted in Australia
Participants	Hip and knee OA recruitment 70 community volunteers Radiographic criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; fortnightly telephone calls to check condition, drug use, or injuries, 3 synchronous contact occasions 2. Exercise (strengthening + range of motion): 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (WOMAC pain) Function (WOMAC function) Quality of life (SF-12 MCS) No participant-reported treatment success measure
Notes	Separate analysis per knee OA only or hip OA; gym-based group versus controls; about 40% on orthopaedic waiting list

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants unblinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Small numbers lost to follow-up, balanced between allocation groups, ITT
Selective reporting (reporting bias)	Unclear risk	Not registered

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Fransen 2001

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 126 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (strengthening + mixed aerobic): 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain) Function (WOMAC function) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes, sequentially numbered
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants unblinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

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Fransen 2007

Study characteristics	
Methods	Conducted in Australia
Participants	Hip and knee OA recruitment 97 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (Tai Chi): 24 sessions (2 x 60 minutes per week for 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (WOMAC pain) Function (WOMAC function) Quality of life (SF-12 MCS) No participant-reported treatment success measure
Notes	Outcome data for knees only included (data provided by authors)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	Central allocation by administrator
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Low risk	Trial registered NCT00123994

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Gohir 2021

Study characteristics	
Methods	Conducted in the UK
Participants	Knee OA recruitment 146 community volunteers ACR criteria
Interventions	Home, unsupervised individual setting, delivered entirely digitally 1. Education; 0 synchronous contact occasions 2. Exercise (strengthening, neuromuscular, proprioceptive/balance): 42 sessions (7x per week over 6 weeks); 0 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (MSK-HQ) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation software
Allocation concealment (selection bias)	Low risk	Sealed envelope, randomisation software
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (22%) and reasons are reported. Described ITT analysis, but Figure 1 suggests otherwise
Selective reporting (reporting bias)	Low risk	Outcomes reported appear consistent with published protocol.

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Gondhalekar 2013

Study characteristics	
Methods	Conducted in India
Participants	Knee OA recruitment 30 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised setting, delivered entirely in-person 1. Conventional treatment; unknown number synchronous contact occasions 2. Conventional treatment + exercise (retrowalking): 63 sessions (3 sessions per day for 21 days (3 weeks), 10 minutes per session); unknown number synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (global VAS/NRS) Physical function (WOMAC global scale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to confirm number synchronous contact occasions but did not respond

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No description provided
Allocation concealment (selection bias)	Unclear risk	No description provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (9%) was reported but no reasons given. Analysis was performed only on those with complete data.
Selective reporting (reporting bias)	Unclear risk	No trial registry or protocol to verify

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Guo 2022

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 111 community volunteers ACR criteria
Interventions	Home, unsupervised individual setting, delivered entirely digitally-delivered 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening, mixed aerobic): 24 sessions (3x per week over 16 weeks); 24 synchronous contact occasions
Outcomes	At 16 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation list with a random block size of 6 was generated by and independent randomiser
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded to hypotheses. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were the outcome assessors and were blinded.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (6%) and reasons for attrition explained. Analysis only performed on those with complete data
Selective reporting (reporting bias)	Unclear risk	No trial registry or protocol available to verify

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Gur 2002

Study characteristics	
Methods	Conducted in Turkey
Participants	Knee OA recruitment 23 community volunteers Radiographic, bilateral KL Grade II-III
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. No treatment; 0 synchronous contact occasions 2. Exercise (concentric strengthening): 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions 3. Exercise (eccentric and concentric strengthening): 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (VAS: 7 items) Function (VAS: 5 items) No quality of life measure No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No information
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants unblinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	No dropouts
Selective reporting (reporting bias)	Unclear risk	Not registered

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Hay 2006

Study characteristics	
Methods	Conducted in the UK
Participants	Knee pain recruitment 217 community volunteers Presenting with persistent knee pain
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Education; 1 synchronous contact occasion 2. Exercise (strengthening + mixed aerobic): unclear number sessions over 10 weeks; 3 to 6 synchronous contact occasions
Outcomes	At 12 weeks and 26 weeks Pain (WOMAC pain) Function (WOMAC physical function) Quality of life (Hospital Anxiety and Depression Scale) Participant-reported treatment success measure (OMERACT-OARSI)
Notes	

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers generator
Allocation concealment (selection bias)	Unclear risk	Small blocks of 6 per practice
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded particpants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up minimal and balanced, ITT analysis
Selective reporting (reporting bias)	Low risk	Registered trial

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Henriksen 2014

Study characteristics	
Methods	Conducted in Denmark
Participants	Knee OA recruitment 60 community volunteers Clinical diagnosis confirmed by radiography
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, neuromuscular, circuit training): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer generated list of random numbers
Allocation concealment (selection bias)	Low risk	The allocation was sealed from study staff enrolling participants by using sequentially numbered opaque sealed envelopes.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (18%) and reasons for attrition were reported. Analysis performed on those who adhered to protocol and who had complete data recordings.
Selective reporting (reporting bias)	Low risk	Outcomes are consistent with trial registry.

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Hernandez 2019

Study characteristics	
Methods	Conducted in Argentina
Participants	Knee OA recruitment 113 community volunteers Confirmed by clinician
Interventions	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person 1. Conventional exercises; 24 synchronous contact occasions 2. Conventional exercises + core stability exercises: 36 sessions (3x per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 24 weeks Pain (global pain VAS/NRS) Physical function (short form WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Function outcomes obtained from authors

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No details provided
Allocation concealment (selection bias)	Low risk	Participants were randomised to group by external physical therapist with allocation concealment by sequentially numbered opaque sealed envelopes.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were blinded assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (42%) and reasons for attrition are detailed. Analysis performed on those with complete data
Selective reporting (reporting bias)	High risk	Not registered

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Hinman 2020

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 175 community volunteers NICE clinical criteria
Interventions	Home, unsupervised individual setting, delivered entirely digitally-delivered 1. Phone advice; 1 synchronous contact occasion 2. Phone advice + exercise (strengthening, physical activity): 78 sessions (3x per week over 26 weeks); 6 to 11 synchronous contact occasions
Outcomes	At 26 weeks (immediately post-intervention) and 52 weeks Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (AQoL-8D) Participant-reported treatment success (global improvement)
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The randomisation schedule was prepared by a biostatistican.
Allocation concealment (selection bias)	Low risk	Allocation was concealed by password-protected software.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Limited disclosure was used to blind participants, who were also the assessors (outcomes participant reported). Participants were informed that the trial was comparing two different forms of telephone-delivered self-management advice. Intervention components, health professionals involved, and hypotheses were not disclosed. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were the outcome assessors and were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (90%) and reasons were reported. ITT analysis.

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Hinman 2020 *(Continued)*

Selective reporting (reporting bias)	Low risk	Outcomes reported appear consistent with trial registry

Holm 2020

Study characteristics	
Methods	Conducted in Denmark
Participants	Knee OA recruitment 90 community volunteers Symptomatic and radiographic (KL2 or more) knee OA
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Education + neuromuscular exercise; 24 synchronous contact occasions 2. Education + neuromuscular exercise + strengthening exercise: 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL subscale) Quality of life (EuroQoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	An external staff member administered the randomisation list.
Allocation concealment (selection bias)	Low risk	Another external staff member put group allocation into sequentially numbers, sealed opaque envelopes, which patient opened after baseline testing
Blinding of participants and personnel (performance bias) All outcomes	Low risk	By exercising on separate days, participants were kept unaware of the content of exercise in the comparator group and therefore did not know if they had been randomised to the intervention or comparator exercises. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were the outcome assessors and were blinded.

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Holm 2020 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (3%) and reasons for attrition are detailed. IITT analysis
Selective reporting (reporting bias)	Low risk	Outcomes appear consistent with trial registry

Hopman-Rock 2000

Study characteristics	
Methods	Conducted in the Netherlands
Participants	Hip and knee OA recruitment 82 community volunteers ACR criteria
Interventions	Mix of clinic/home, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (strengthening + mixed aerobic): 24 sessions (4 x 60 minutes per week for 6 weeks); 6 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (VAS) Function (IRGL mobility) No quality of life measure No participant-reported treatment success measure
Notes	Only 6 treatment occasions Separate analysis for OA knee provided to authors of previous (2015) review. Inconsistency in sample size for pain and function

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No details
Allocation concealment (selection bias)	Unclear risk	No details
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias)	High risk	Unblinded participants

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Hopman-Rock 2000 *(Continued)*

- subjective self-reported outcomes (pain, function, quality of life)		
Incomplete outcome data (attrition bias) All outcomes	High risk	Efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Hsu 2021

Study characteristics	
Methods	Conducted in Taiwan
Participants	Knee OA recruitment 44 community volunteers KL grade III+
Interventions	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered via mix of in-person and digital 1. Diet; 12 synchronous contact occasions 2. Diet + exercise (strengthening): 36 sessions (3x per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random allocation prepared by clinical staff not involved in the enrolment process
Allocation concealment (selection bias)	Low risk	Random allocation concealed by using opaque sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.

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Hsu 2021 *(Continued)*

Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (4%) and reasons for attrition was reported. Analysis on those who completed
Selective reporting (reporting bias)	Unclear risk	Cannot locate trial registration to verify

Hu 2020

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 92 community volunteers Clinical assessment - KL grade 1 or greater plus physical examination
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Education; 1 synchronous contact occasion 2. Exercise (Tai Chi): 72 sessions (3x per week over 24 weeks); 73 synchronous contact occasions
Outcomes	At 24 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.

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Hu 2020 *(Continued)*

Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (10%) and reasons were reported. Analysis performed on those who completed the study
Selective reporting (reporting bias)	Unclear risk	Unable to verify outcomes against trial registration or published protocol

Huang 2003

Study characteristics	
Methods	Conducted in Taiwan
Participants	Knee OA recruitment 132 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Hotpack/ROM; unclear number of contact occasions 2. Hotpack/ROM + exercise (isokinetic strengthening): 24 sessions (3 x 60 minutes per week for 8 weeks); 24 synchronous contact occasions 2. Hotpack/ROM + exercise (isometric strengthening): 24 sessions (3 x 60 minutes per week for 8 weeks); 24 synchronous contact occasions 2. Hotpack/ROM + exercise (isotonic strengthening): 24 sessions (3 x 60 minutes per week for 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks, 1 year Pain (VAS pain) Function (Lequesne function) No quality of life measure No participant-reported treatment success measure
Notes	3 exercise groups (isokinetic, isometric, isotonic) pooled for meta-analysis Authors contacted to request or clarify data

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	High risk	Sequential numbers I-IV (representing treatment allocation)

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Huang 2003 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	8% loss to follow-up, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Huang 2005

Study characteristics	
Methods	Conducted in Taiwan
Participants	Knee OA recruitment 70 community volunteers (in the 2 relevant groups) ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Hotpack/ROM; unclear number of contact occasions 2. Hotpack/ROM + exercise (strengthening): 24 sessions (3 x 60 minutes per week for 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) and 52 weeks Pain (VAS) Function (Lequesne) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	High risk	Sequential numbers I to IV (representing treatment allocation)

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Huang 2005 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded particpants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	9% loss to follow-up, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Hurley 2007

Study characteristics	
Methods	Conducted in the UK
Participants	Knee OA recruitment 418 community volunteers > 50 years of age who consulted a primary care physician for knee pain of > 6 months' duration
Interventions	Clinic, supervised mix of individual/group setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening, balance, aerobic and motor control): 12 sessions (2 x 45 minutes per week over 6 weeks); 12 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) and 26 weeks Pain (WOMAC pain subscale): only 6 months Physical function (WOMAC function subscale) Quality of life (EQ5D 0 to 1): only 6 months No participant-reported treatment success measure
Notes	Group 2 was combined results of 2 exercise-based interventions: individual and class-based for all meta-analyses. WOMAC physical function was declared main outcome, with results provided for 6 weeks and 6 months.

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

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Hurley 2007 *(Continued)*

Random sequence generation (selection bias)	Low risk	Cluster-randomisation: 54 primary care practices were randomly assigned, not participants. Randomisation list was generated by a study co-author at an external location.
Allocation concealment (selection bias)	Low risk	Randomisation list was generated by a study co-author not involved in execution of the study
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded particpants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	18% lost to follow-up, balanced between allocation groups, no ITT, but effect of withdrawal was assessed
Selective reporting (reporting bias)	Low risk	Registered study

Imoto 2012

Study characteristics	
Methods	Conducted in Brazil
Participants	Knee OA recruitment 100 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; biweekly phone calls to encourage health education; 3 synchronous contact occasions 2. Exercise (strengthening): 16 sessions (2 x 30 to 40 minutes per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (SF-36 functional capacity) No quality of life measure No participant-reported treatment success meaure
Notes	Authors contacted to request or clarify data
Risk of bias	

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Imoto 2012 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	A statistician performed the allocation of the participants using a computer programme.
Allocation concealment (selection bias)	Low risk	Aiming to avoid selection bias, a third person numbered and sealed the opaque envelopes.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (19%) and reasons for attrition are reported. Analysis performed on those who completed
Selective reporting (reporting bias)	Unclear risk	Unable to verify against a trial registration or published protocol

Jan 2008

Study characteristics	
Methods	Conducted in Taiwan
Participants	Knee OA recruitment 102 community volunteers ACR criteria, KL Grade < IV
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Education; 0 synchronous contact occasions 2. Exercise (high resistance training): 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions 3. Exercise (low resistance training): 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life assessment No participant-reported treatment success measure
Notes	Participants did not take NSAIDs during study.

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Jan 2008 *(Continued)*

	Groups 2 and 3 combined for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random integer generator used
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	High risk	Unbalanced loss to follow-up: 4 (13%) in control group, 0 in exercise group, no ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Jan 2009

Study characteristics

Methods	Conducted in Taiwan
Participants	Knee OA recruitment 106 community volunteers ACR criteria, KL Grade < IV (most KL II)
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening - weight bearing): 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions 3. Exercise (strengthening - non-weight bearing): 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Physical function (WOMAC function subscale) No pain measure No quality of life measure

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Jan 2009 *(Continued)*

	No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	States random number tables
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	High risk	Not blinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss to follow-up (5 discontinued treatment in the 2 exercise allocations)
Selective reporting (reporting bias)	Unclear risk	NCT9100002377 not found

Janyacharoen 2018

Study characteristics

Methods	Conducted in Thailand
Participants	Knee OA recruitment 40 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Education; 1 synchronous contact occasions 2. Exercise (Sakon Nakhom ancient boxing): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL subscale) Quality of life (KOOS QoL)

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Janyacharoen 2018 *(Continued)*

	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No details provided
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Unclear risk	Unable to verify against a trial registration or published protocol

Jenkinson 2009

Study characteristics

Methods	Conducted in the UK
Participants	Knee OA recruitment 158 community volunteers 45 years of age and older, BMI > 28, knee pain on most days past month
Interventions	Home, unsupervised individual setting, delivered entirely in-person 1. Attention control; home visits, and support telephone calls, 6 synchronous contact occasions 2. Exercise (strengthening): 728 sessions (7x per week over 104 weeks); 6 synchronous contact occasions
Outcomes	At 24 months Pain (WOMAC pain subscale) Physical function (WOMAC function subscale), not estimable as no data for control group Participant-reported treatment success (30% or greater pain relief)

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Jenkinson 2009 *(Continued)*

	No quality of life measure
Notes	Physical function not estimable as no data for control group

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer random sequence generator, 2 × 2 factorial design, blocks of 10 stratified by sex, age, and BMI
Allocation concealment (selection bias)	Low risk	Prepared by trial researcher, kept in locked drawer, opened by co-ordinator in sequential order
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded particpants
Incomplete outcome data (attrition bias) All outcomes	Low risk	High loss to follow-up at 24 months (26% exercise, 14% control), but ITT analysis using multiple imputation methods
Selective reporting (reporting bias)	Unclear risk	Study registered, but outcome measures not provided at time of registration. No justification for why only selected domains of SF-36 (physical function, bodily pain)?

Jorge 2015

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 60 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (strengthening): 24 sessions (2x per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (SF-36 mental health domain)

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Jorge 2015 *(Continued)*

	Participant-reported treatment success (global change)
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A random selection process was electronically carried out allocating the participants to an experimental group and a control group.
Allocation concealment (selection bias)	Low risk	Sealed envelopes were used to ensure the confidentiality of the selection process.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (3%) and reasons for attrition detailed. ITT analysis
Selective reporting (reporting bias)	Unclear risk	Unable to verify against a trial registration or protocol publication

Joshi 2019

Study characteristics

Methods	Conducted in India
Participants	Knee OA recruitment 42 community volunteers Diagnosed by clinician
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Conventional exercise: 0 synchronous contact occasions 2. Conventional exercise + retrowalking: 18 sessions (3 x 10 minutes per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC global scale)

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Joshi 2019 *(Continued)*

	No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computed-generated random number tables
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition is not reported on.
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registration or published protocol

Kang 2022

Study characteristics

Methods	Conducted in China
Participants	Knee OA recruitment 30 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; wellness education: 9 synchronous contact occasions 2. Exercise (Tai Chi): 108 sessions (3 x 60 minutes per week over 36 weeks); 108 synchronous contact occasions
Outcomes	At 36 weeks (immediately post-intervention) Pain (VAS/NRS) Physical function (WOMAC function subscale)

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Kang 2022 *(Continued)*

	Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	QoL data unable to be extracted; authors unresponsive to contact

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Used a random number table method
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (10%) and reasons for attrition reported. Analysis only on those who completed
Selective reporting (reporting bias)	Unclear risk	Not registered

Karadag 2019

Study characteristics	
Methods	Conducted in Turkey
Participants	Knee OA recruitment 35 community volunteers ACR criteria
Interventions	Home, mix of supervised/unsupervised individual setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening): 40 sessions (10 x 10 minutes per week over 4 weeks); 13 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale)

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Karadag 2019 *(Continued)*

	No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was performed with random numbers on the computer.
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (14%) and reasons reported. Analysis performed on those who completed
Selective reporting (reporting bias)	Unclear risk	Unable to verify as no trial registration or published protocol available

Keefe 2004

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 34 community volunteers Persistent knee pain
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Standard care; 0 synchronous contact occasions 2. Exercise (strengthening, mixed aerobic): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (AIMS pain subscales) Quality of Life (AIMS psychological)

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Keefe 2004 *(Continued)*

	No physical function measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only 'randomly allocated'
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded particpants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Only 6% loss to follow-up, balanced between allocations. Efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Kholvadia 2019

Study characteristics

Methods	Conducted in South Africa
Participants	Knee OA recruitment 72 community volunteers Diagnostic criteria unclear
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Low level laser therapy; 12 synchronous contact occasions 2. Low level laser therapy + exercise (strengthening, neuromuscular, stretching): 36 sessions (3x per week over 12 weeks); 12 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 24-month follow-up Physical function (WOMAC global scale) No pain measure No quality of life measure

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Kholvadia 2019 *(Continued)*

	No participant-reported treatment success meaure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Details not provided
Allocation concealment (selection bias)	Unclear risk	Sealed envelopes were used.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (8.5%) occurred but reasons not reported. Analysis performed on those who completed the study
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registration or published protocol

Kilic 2020

Study characteristics

Methods	Conducted in Turkey
Participants	Knee OA recruitment 50 community volunteers KL grade
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Physiotherapy: 30 synchronous contact occasions 2. Physiotherapy + exercise (walking): 40 sessions (5 x 40 minutes per week over 6 weeks); 30 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure

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Kilic 2020 *(Continued)*

	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Online random allocation software
Allocation concealment (selection bias)	Low risk	Randomised to either the treatment or control groups with the sealed envelope method
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participant were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (16%) occurred. Reasons reported without great detail. ITT analysis not performed.
Selective reporting (reporting bias)	Unclear risk	No trial registration or protocol to verify reported outcomes against

Kim 2013

Study characteristics

Methods	Conducted in Japan
Participants	Knee OA recruitment 74 community volunteers with knee OA Aged 75 years and older with knee pain
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; education, 3 synchronous contact occasions 2. Exercise (strengthening, proprioceptive/balance training): 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (global pain VAS) Physical function (global disability score) No quality of life measure

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Kim 2013 *(Continued)*

	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (9%) and reasons reported. Analysis on complete cases
Selective reporting (reporting bias)	Unclear risk	Trial registry not accessible

Kim 2020

Study characteristics

Methods	Conducted in South Korea
Participants	Knee OA recruitment 80 community volunteers with knee OA KL Grade II-IV
Interventions	Mix of home/clinic, supervised/unsupervised individual/group setting, delivered via mix of in-person and digital delivery 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening, stretching, dance): 48 sessions (4 x 60 minutes per week over 12 weeks); 12 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Physical function (WOMAC global scale) Quality of life (EQ-5D Korean version) (outcomes for control missing from manuscript; no response from authors to our request for data)

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Kim 2020 *(Continued)*

	No pain measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Computer programme
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (11%) and reasons reported. ITT analysis performed
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registry or published protocol

Knoop 2013

Study characteristics

Methods	Conducted in the Netherlands
Participants	Knee OA recruitment 159 community volunteers with knee OA ACR criteria
Interventions	Mix of home/clinic, mix of supervised/unsupervised group setting, delivered entirely in-person 1. Exercise without feedback; 24 synchronous contact occasions 2. Exercise with feedback (neuromuscular/neuromotor exercise with personalised feedback from PT) + exercise for knee position: 84 sessions (7 x 420 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 38 weeks Pain (global pain NRS) Physical function (WOMAC function subscale)

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Knoop 2013 *(Continued)*

	Participant-reported treatment success (global perceived effect) No quality of life measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer generated random sequence
Allocation concealment (selection bias)	Low risk	Concealed using opaque sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessor.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (3%) and reasons reported. ITT analysis performed
Selective reporting (reporting bias)	Low risk	Outcomes consistent with trial registry

Kovar 1992

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 103 community volunteers Diagnosis via pain and x-ray
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; weekly telephone call regarding activities of daily living, 8 synchronous contact occasions 2. Exercise (walking, stretching): 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (AIMS)

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Kovar 1992 *(Continued)*

	Function (AIMS) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Missing data balanced between allocation groups, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Krishnan 2021

Study characteristics

Methods	Conducted in Malaysia
Participants	Knee OA recruitment 30 community volunteers KL grade II
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No treatment; 0 synchronous contact occasions 2. Exercise (strengthening): 16 sessions (2 x 60 minutes per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (KOOS pain subscale)

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Krishnan 2021 *(Continued)*

	Physical function (KOOS ADL subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition reported
Selective reporting (reporting bias)	Unclear risk	No trial registry or published protocol to verify reported outcomes against

Kumar 2013

Study characteristics

Methods	Conducted in India
Participants	Knee OA recruitment 44 community volunteers Unclear diagnosis criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Supervised exercise + ultrasound; 12 synchronous contact occasions 2. Supervised exercise + ultrasound + proprioceptive/balance training: 12 sessions (3x per week over 4 weeks); 24 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS)

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Kumar 2013 *(Continued)*

	Physical function (Reduced WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was done by chit box method.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No attrition described
Selective reporting (reporting bias)	Unclear risk	No protocol or trial registration available to verifying the reported outcomes

Kuntz 2018

Study characteristics	
Methods	Conducted in Canada
Participants	Knee OA recruitment 31 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Attention control; meditation 12 synchronous contact occasions 2. Exercise (yoga): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions 3. Exercise (strengthening): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention)

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Kuntz 2018 *(Continued)*

	Pain (KOOS pain subscale) Physical function (KOOS ADL function subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was performed using custom Matlab software.
Allocation concealment (selection bias)	Low risk	Participants received group allocation information in an opaque envelope.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants and personnel (care providers) were blinded to study hypotheses. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Participants were the outcome assessors and were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (3%). ITT analysis not performed, but only 1 individual omitted
Selective reporting (reporting bias)	Unclear risk	Not all outcomes in trial registration are reported in this publication.

Lai 2021

Study characteristics

Methods	Conducted in China
Participants	Knee OA recruitment 54 community volunteers with knee OA Diagnosed by an orthopaedic surgeon based on patient's medical history, x-ray findings and physical assessment
Interventions	Mix of home/clinic, entirely supervised group setting, delivered entirely in-person 1. Attention control; health education, 8 synchronous contact occasions 2. Exercise (strengthening): 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention)

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Lai 2021 *(Continued)*

	Pain (global pain VAS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data; data provided by authors

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (17%) reasons reported. ITT analysis performed
Selective reporting (reporting bias)	Low risk	Consistent with registry

Lee 2009

Study characteristics

Methods	Conducted in South Korea
Participants	Knee OA recruitment 44 community volunteers KL Grade II+
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention control; 0 synchronous contact occasions 2. Exercise (Tai Chi Qigong): 16 sessions (2 x 45 minutes per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale)

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Lee 2009 *(Continued)*

	Physical function (WOMAC function subscale)
	Quality of life (SF-36 MCS)
	No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated, balanced block randomisation (2:1)
Allocation concealment (selection bias)	Low risk	Sealed envelopes with identification number. Opened in order
Blinding of participants and personnel (performance bias) All outcomes	High risk	Not blinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblilnded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up low (n = 3), included in analysis
Selective reporting (reporting bias)	Unclear risk	Study not registered

Lee 2019

Study characteristics

Methods	Conducted in South Korea
Participants	Knee OA recruitment
	30 community volunteers
	ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person
	1. HA injection; 1 synchronous contact occasion
	2. HA injection + exercise (strengthening, proprioceptive, core stability): 20 sessions (5 x 30 minutes per week over 4 weeks); 20 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention)
	Pain (Global VAS/NRS)

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Lee 2019 *(Continued)*

	Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was performed by using computerised random number generator.
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded assessor (participant)
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was no attrition.
Selective reporting (reporting bias)	Unclear risk	Not registered; protocol unavailable.

Lee 2023

Study characteristics	
Methods	Conducted in South Korea
Participants	Knee OA recruitment 36 community volunteers KL Grade II+
Interventions	Home, mix of supervised/unsupervised individual setting, delivered via mix of in-person and remotely 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening): 24 sessions (3x per week over 8 weeks); 5 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (VAS/NRS) No physical function measure

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Lee 2023 *(Continued)*

	No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table prepared in Excel (assume computer generated)
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were aware of group allocation.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (14%) and reasons reported.
Selective reporting (reporting bias)	Unclear risk	Additional outcomes are reported that are not listed in the trial registry (e.g. blood markers). https://trialsearch.who.int/Trial2.aspx?TrialID=KCT0006170

Li 2020

Study characteristics

Methods	Conducted in China
Participants	Knee OA recruitment 51 community volunteers Physician-informed diagnosis
Interventions	Home, mix of supervised/unsupervised individual setting, delivered entirely digitally-delivered 1. Wait list; 0 synchronous contact occasions 2. Exercise (physical activity): variable number of sessions (Variable frequency/duration over 12 weeks); 5 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL subscale)

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Li 2020 *(Continued)*

	Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Data taken from 13-week follow-up only (i.e. longer-term data not used), as control group received exercise programme from week 14 onwards

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Using number generated by SAS version 9.4
Allocation concealment (selection bias)	Low risk	Randomisation done by research staff not involved in the research variable block sizes to ensure allocation concealment
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded – they were outcome assessors of patient-reported data.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were not blinded – they were outcome assessors of patient-reported data.
Incomplete outcome data (attrition bias) All outcomes	Low risk	< 20% attrition and ITT analysis used
Selective reporting (reporting bias)	Low risk	Outcomes are in line with those on the clinical trial registry.

Lim 2008

Study characteristics

Methods	Conducted in Australia
Participants	Knee OA recruitment 107 community volunteers ACR criteria
Interventions	Home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. No intervention (malaligned); 0 synchronous contact occasions 2. No intervention (neutrally aligned); 0 synchronous contact occasions 3. Exercise (strengthening - malaligned): 60 sessions (5x per week over 12 weeks); 7 synchronous contact occasions

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Lim 2008 *(Continued)*

	4. Exercise (strengthening - neutrally aligned): 60 sessions (5x per week over 12 weeks); 7 synchronous contact occasions
Outcomes	At 13 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	2 exercise groups (malaligned and neutrally aligned) pooled; 2 control groups (malaligned and neutrally aligned) pooled

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table used, stratified by alignment in blocks of 6
Allocation concealment (selection bias)	Low risk	Independent researcher randomly assigned participants
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up 4% to 18%, but ITT analysis using last observation carried forward
Selective reporting (reporting bias)	Low risk	Registered study

Lin 2009

Study characteristics

Methods	Conducted in Taiwan
Participants	Knee OA recuitment 72 community volunteers 50 years of age and older, KL Grade < IV, history of knee pain > 6 months
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions

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Lin 2009 *(Continued)*

	2. Exercise (strengthening): 24 sessions (3 x 50 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	Sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Not blinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	6% to 8% loss to follow-up, ITT analysis
Selective reporting (reporting bias)	Unclear risk	Study not registered

Liu 2019

Study characteristics

Methods	Conducted in China
Participants	Knee OA recruitment 140 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; health education, 12 synchronous contact occasions

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Liu 2019 *(Continued)*

	2. Exercise (Tai Chi): 60 sessions (5 x 60 minutes per week over 12 weeks); 60 synchronous contact occasions 3. Exercise (Baduanjin): 60 sessions (5 x 60 minutes per week over 12 weeks); 60 synchronous contact occasions 4. Exercise (cycling): 60 sessions (5 x 60 minutes per week over 12 weeks); 60 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL function subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	Groups 2, 3, 4 pooled for meta-analysis Authors contacted to request or clarify data. QoL data unable to be extracted from publication (provided minimum and maximum only); authors unresponsive to request for data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (23%) and reasons reported. ITT analysis not performed
Selective reporting (reporting bias)	Unclear risk	Trial registry inaccessible

Lu 2017

Study characteristics

Methods	Conducted in China

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Lu 2017 *(Continued)*

Participants	Knee OA recruitment 46 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; health education, 12 synchronous contact occasions 2. Exercise (Tai Ji Quan): 72 sessions (3 x 60 minutes per week over 24 weeks); 72 synchronous contact occasions
Outcomes	At 24 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data. Pain and function extracted from related publication by the authors: Zhu 2016

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Allocation made by a research staff member not directly involved in the trial
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (13%) was reported and the reasons. ITT analysis performed
Selective reporting (reporting bias)	Low risk	This is a secondary report.

Lund 2008

Study characteristics

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Lund 2008 *(Continued)*

Methods	Conducted in Denmark
Participants	Knee OA recruitment 52 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, balance, stretching): 16 sessions (2 x 50 minutes per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) and 20 weeks Pain (VAS during walking) Physical function (KOOS ADL) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Envelope method with blocks of 18
Allocation concealment (selection bias)	Low risk	Envelope method, so screener unaware which will be chosen
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	7% to 20% loss to follow-up at 20 weeks; however, ITT analysis was used
Selective reporting (reporting bias)	Unclear risk	Study not registered

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Marconcin 2018

Study characteristics	
Methods	Conducted in Portugal
Participants	Knee OA recruitment 80 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Education; 6 synchronous contact occasions 2. Education + exercise (strengthening, proprioceptive/balance): 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain subscale) Physical function (KOOS ADL subscale) Quality of life (EuroQoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No details provided
Allocation concealment (selection bias)	Low risk	The randomisation sequence was conducted 1:1 allocation using opaque envelopes containing treatment group allocation.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to group allocation. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (16%) occurred and reasons reported. ITT analysis not performed.
Selective reporting (reporting bias)	Low risk	Appears consistent with the protocol

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Maurer 1999

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 113 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Attention control; health education, 4 synchronous contact occasions 2. Exercise (strengthening): 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Function (WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers generator
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Missing data balanced between groups, not study-related, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

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Mazloum 2018

Study characteristics	
Methods	Conducted in Iran
Participants	Knee OA recruitment 49 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (pilates): 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions 3. Exercise (conventional therapeutic exercise): 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (Lequesne OA Index) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (16%) and reasons reported. ITT analysis not performed
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registration or protocol

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Mehwish 2023

Study characteristics	
Methods	Conducted in Pakistan
Participants	Knee OA recruitment 44 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Knee exercises; 18 synchronous contact occasions 2. Knee exercises + hip strengthening (strengthening): 18 sessions (5 x 30 to 40 minutes per week over 4 weeks); 18 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (KOOS ADL subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	The treatment schedule was concealed using sequentially numbered opaque sealed envelopes.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants aware of group allocation
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (0%)
Selective reporting (reporting bias)	Unclear risk	No registry record or protocol to assess

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Messier 2004

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 158 community volunteers Knee pain on most days of month, pain with activity, KL grade I-III
Interventions	Clinic, mix supervised/unsupervised mix group/individual setting, delivered entirely in-person: 1. Attention control; health education, 6 synchronous contact occasions 2. Exercise (strengthening, walking): 78 sessions (3x per week for 26 weeks); 51 synchronous contact occasions
Outcomes	At 6 months (immediately post-intervention) and 18 months Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Physical function assessed at 18 months, SD of baseline used.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Web-based
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

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Messier 2013

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 304 community volunteers KL grade II or III
Interventions	Mix clinic/home, mix supervised/unsupervised mix individual/group setting, delivered entirely in-person 1. Diet; 54 synchronous contact occasions 2. Diet + exercise (strengthening + mixed aerobic): 234 sessions (3 x 60 minutes per week over 78 weeks); 90 synchronous contact occasions
Outcomes	At 78 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	Serious adverse events reported (unrelated to study); non-serious not reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (16%) and reasons reported. ITT analysis performed
Selective reporting (reporting bias)	Unclear risk	Some outcomes not reported in primary trial (cartilage morphology)

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Messier 2021

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 377 community volunteers KL grade 2 or 3
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; health education, 24 synchronous contact occasions 2. Exercise (high-intensity strengthening): 234 sessions (3 x 60 minutes per week over 78 weeks); 234 synchronous contact occasions 3. Exercise (low-intensity strengthening): 234 sessions (3 x 60 minutes per week over 78 weeks); 234 synchronous contact occasions
Outcomes	At 78 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated permuted block randomisation
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (18%) and reasons reported. ITT analysis (stated in protocol)
Selective reporting (reporting bias)	Low risk	Reported outcomes are consistent with published protocol

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Mete 2022

Study characteristics	
Methods	Conducted in Turkey
Participants	Knee OA recruitment 64 community volunteers KL grade 2 or 3
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Physiotherapy; 30 synchronous contact occasions 2. Physiotherapy + exercise (strengthening): 30 sessions (5 x 20 minutes per week over 6 weeks); 30 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data. Physical function outcomes not included in meta-analysis due to suspicion of typographical error in publication (authors did not respond to request for data)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation schedule
Allocation concealment (selection bias)	Unclear risk	Unclear risk - possibility of typographical error as the paper states that "All assessments were done by researcher who were [sic] not in the study" and it is unclear if this should read "allocations".
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (6%) and reasons reported. Only used completed cases in analysis
Selective reporting (reporting bias)	Low risk	Reported outcomes appear consistent with trial registry.

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Multanen 2014

Study characteristics	
Methods	Conducted in Finland
Participants	Knee OA recruitment 80 community volunteers Knee pain on most days and KL grade I or II
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Usual care; option to participate in social group meeting (4 synchronous contact occasions) 2. Exercise (mixed aerobic): 156 sessions (3 x 55 minutes per week over 52 weeks); 156 synchronous contact occasions
Outcomes	At 52 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated blocked randomisation list
Allocation concealment (selection bias)	Low risk	Randomly allocated by the statistician blinded for the study participants to each group
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (5%) and reasons reported. Only those with complete data were included in the analysis.
Selective reporting (reporting bias)	Low risk	Reported outcomes appear consistent with trial registry.

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Nahayatbin 2018

Study characteristics	
Methods	Conducted in Iran
Participants	Knee OA recruitment 48 community volunteers KL grade II or III
Interventions	Clinic, supervised setting, delivered entirely in-person 1. Routine physiotherapy; 1 synchronous contact occasion 2. Routine physiotherapy + exercise (strengthening): 12 sessions (3 x 20 minutes per week over 4 weeks); 12 synchronous contact occasions 3. Routine physiotherapy + exercise (Tai Chi): 12 sessions (3 x 20 minutes per week over 4 weeks); 12 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) and 8 weeks Pain (KOOS pain subscale) Physical function (KOOS ADL subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	Groups 2 and 3 were pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition not described
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registration or protocol

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Nelligan 2021

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 206 community volunteers NICE clinical criteria
Interventions	Home, unsupervised individual setting, delivered entirely digitally 1. Education; 0 synchronous contact occasions 2. Education + exercise (strengthening and general physical activity): 72 sessions (3 x 20 to 30 minutes per week over 24 weeks); 0 synchronous contact occasions
Outcomes	At 24 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (AQoL) Participant-reported treatment success (global improvement)
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Randomisation schedule was accessed via a password-protected computer programme by a researcher not involved in the study.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded to group allocation.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	Low risk	Limited disclosure was used to blind participants who were also assessors, as all outcomes were participant-reported. Participants were informed that the study was investigating a range of digital resources (e.g. computer, cell phone) to promote knee pain self-management that might include exercise and email or text messaging but were not given specific details of either the intervention or the control groups nor the hypothesis under investigation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (13%) and reasons are reported. ITT analysis performed
Selective reporting (reporting bias)	Low risk	Outcomes reported are consistent with published protocol.

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O'Reilly 1999

Study characteristics	
Methods	Conducted in the UK
Participants	Knee OA recruitment 191 community volunteers Knee pain past week
Interventions	Home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. Lifestyle advice; 0 synchronous contact occasions 2. Lifestyle advice + exercise (strengthening): 183 sessions (7x per week over 26 weeks); 4 synchronous contact occasions
Outcomes	At 26 weeks (immediately post-intervention) Pain (WOMAC) Function (WOMAC) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	Community sample, most with mild radiographic/symptomatic disease (only 41% > KL Grade I)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Unclear risk	Not reported whether sealed envelopes were opaque with sequential numbers for audit trail
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded partcipants
Incomplete outcome data (attrition bias) All outcomes	Low risk	6% loss to follow-up, ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

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Oh 2020

Study characteristics	
Methods	Conducted in South Korea
Participants	Knee OA recruitment 60 community volunteers KL Grade <III
Interventions	Mix clinic/home, mix supervised/unsupervised mix individual/group setting, delivered via mix of digital/in-person 1. Education; 5 synchronous contact occasions 2. Education + exercise (strengthening): 40 to 60 sessions (2 to 3 x 60 to 80 minutes per week over 20 weeks); 6 synchronous contact occasions
Outcomes	At 20 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcomes assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (46%) and reasons reported. Complete cases analysed
Selective reporting (reporting bias)	Unclear risk	Unable to verify against protocol or trial registration

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Ozlu 2023

Study characteristics	
Methods	Conducted in Turkey
Participants	Knee OA recruitment 82 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Conventional treatment; 15 synchronous contact occasions 2. Conventional treatment + exercise (gamification of strengthening and proprioception exercise): 15 sessions (5 x 15 minutes per week over 3 weeks); 15 synchronous contact occasions
Outcomes	At 3 weeks (immediately post-intervention) and 7 weeks Pain (global/overall pain VAS/NRS) Physical function (WOMAC global scale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (11%), reasons not reported. Complete cases analysed
Selective reporting (reporting bias)	Unclear risk	Unable to verify against published protocol or trial registry

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Pazit 2018

Study characteristics	
Methods	Conducted in Australia
Participants	Knee OA recruitment 30 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening): 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions 3. Exercise (strengthening + balance): 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 week (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) Quality of life (AQoL) No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	Opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcomes assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (7%) and reasons reported. Complete cases analysed

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Pazit 2018 *(Continued)*

Selective reporting (reporting bias)	Low risk	Outcomes appear consistent with published protocol

Peloquin 1999

Study characteristics	
Methods	Conducted in Canada
Participants	Knee OA recruitment 137 community volunteers KL < Grade IV
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; health education, 6 synchronous contact occasions 2. Exercise (strengthening, mixed aerobic, stretching): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (AIMS) Function (AIMS) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	High risk	Imbalance in missing data between allocation groups, efficacy analysis

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Peloquin 1999 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	Not registered

Qiu 2023

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 42 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. General treatment; 30 synchronous contact occasions 2. General treatment + exercise (hip muscle strengthening); 30 sessions (5x per week over 6 weeks); 30 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes were used.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were unblinded to group allocation.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (14%) with reasons reported

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Qiu 2023 *(Continued)*

Selective reporting (reporting bias)	High risk	Some inconsistencies between report and registry (e.g. 6 min walk test, single leg standing test) https://www.chictr.org.cn/showprojEN.html?proj=15444

Rafiq 2021a

Study characteristics	
Methods	Conducted in Pakistan
Participants	Knee OA recruitment 56 community volunteers KL grade II or III
Interventions	Home, unsupervised individual setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening): 36 sessions (3 x 60 to 85 minutes per week over 12 weeks); 0 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Physical function (WOMAC global scale) No pain measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Outcomes presented as median and IQR; converted by review authors to mean and standard deviation

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated simple randomisation
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias)	Unclear risk	Attrition (18%) and reasons reported. Complete cases analysed

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Rafiq 2021a *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Registered; reporting consistent with registry details

Rafiq 2021b

Study characteristics	
Methods	Conducted in Pakistan
Participants	Knee OA recruitment 114 community volunteers KL grade II or III diagnosed by an orthopaedic surgeon
Interventions	Home, unsupervised individual setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening with mobile health): 36 sessions (3 x 65 to 80 minutes per week over 12 weeks); 36 synchronous contact occasions 3. Exercise (strengthening without mobile health): 36 sessions (3 x 65 to 80 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (patient-specific functional scale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated numbers
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported	High risk	Participants were the assessors.

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Rafiq 2021b *(Continued)*

outcomes (pain, function, quality of life)		
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (16%) and reason was reported. Analysis on complete cases
Selective reporting (reporting bias)	Unclear risk	Unable to verify against registry

Rafiq 2021c

Study characteristics	
Methods	Conducted in Pakistan
Participants	Knee OA recruitment 50 community volunteers KL grade II - III
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. Education; 0 synchronous contact occasions 2. Exercise (strengthening): 12 sessions (3 x 65 to 80 minutes per week over 4 weeks); 12 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) Participant-reported treatment success (satisfaction 0 to 10 NRS) No quality of life measure
Notes	Authors contacted to request or clarify data Participant-reported treatment success measured on continuous scale (rather than binary); not included in meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.

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Rafiq 2021c *(Continued)*

Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (16%) and reasons were reported. Only complete cases analysed
Selective reporting (reporting bias)	Low risk	Outcomes are consistent with trial registration.

Rakhshani 2022

Study characteristics	
Methods	Conducted in Iran
Participants	Knee OA recruitment 36 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Routine physiotherapy; 18 synchronous contact occasions 2. Routine physiotherapy + hip strengthening exercise (hip strengthening exercises); 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants not blinded to group allocation

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Rakhshani 2022 *(Continued)*

Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition was not described - only described the 36 participants that were randomised.
Selective reporting (reporting bias)	Unclear risk	No registry or protocol available to assess

Rego 2023

Study characteristics	
Methods	Conducted in Brazil
Participants	Knee OA recruitment 18 community volunteers with knee OA KL grade II or III
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; phone calls to check health status and activities of daily living, 14 synchronous contact occasions 2. Exercise (pilates): 14 sessions (2 x 60 minutes per week over 7 weeks); 14 synchronous contact occasions
Outcomes	At 7 weeks (immediately post-intervention) and 52 weeks Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) Quality of life (SF-36 total) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Generated using website www.randomization.com
Allocation concealment (selection bias)	Low risk	Black opaque envelopes
Blinding of participants and personnel (performance bias)	High risk	Participants were not blinded to which intervention they received.

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Rego 2023 *(Continued)*

All outcomes		Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (10%) and reason reported. ITT analysis not performed.
Selective reporting (reporting bias)	Low risk	Outcomes consistent with trial registry

Robbins 2022

Study characteristics	
Methods	Conducted in Brazil
Participants	Knee OA recruitment 86 community volunteers KL grade II or III
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Education; 0 synchronous contact occasions 2. Exercise (stretching): 24 sessions (3 x 45 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (pain on walking VAS/NRS) Physical function (Lequesne OA index global score) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random number table
Allocation concealment (selection bias)	Low risk	Accessed the allocation schedule from a centrally located locked cabinet

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Robbins 2022 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were outcomes assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition reported
Selective reporting (reporting bias)	Unclear risk	Cannot access the trial registry

Rogind 1998

Study characteristics	
Methods	Conducted in Denmark
Participants	Knee OA recruitment 25 community volunteers ACR criteria x-ray (> KL Grade II)
Interventions	Mix clinic/home, mix supervised/unsupervised mix clinic/individual, delivered entirely in-person: 1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, balance): 36 sessions (3 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 52 weeks Pain (VAS with weight bearing) Function (AFI) No quality of life measure No participant-reported treatment success measure
Notes	Moderate to severe disease. Only median (IQR) provided. Baseline differences in pain scores. Very complex exercise programme (including venous, truncal muscles, balance)

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers table

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Rogind 1998 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Missing data minimal and balanced between allocation groups, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Rosadi 2023

Study characteristics	
Methods	Conducted in Indonesia
Participants	Knee OA recruitment 39 community volunteers KL grade I-IV and physician-diagnosed
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Usual physiotherapy; 18 synchronous contact occasions 2. Exercise (strengthening + walking programme + cycling): 24 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions 3. Exercise (strengthening + walking programme + cycling): 18 sessions (3 x 60 minutes per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) and 12 weeks Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Groups 2 and 3 pooled for meta-analysis
Risk of bias	

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Rosadi 2023 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (23%) reasons reported. Analysis on complete cases
Selective reporting (reporting bias)	Low risk	Reported outcomes are consistent with trial registry.

Rosedale 2014

Study characteristics	
Methods	Conducted in Canada
Participants	Knee OA recruitment 180 community volunteers Knee pain > 4 months and radiographically confirmed
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (strengthening + walking programme + cycling): variable number of sessions (approx 3 to 4 x 20 to 60 minutes per week over 2 weeks, but varied widely); 4 tp 6 synchronous contact occasions
Outcomes	At 2 weeks (immediately post-intervention) and 12 weeks Pain (VAS/NRS) Physical function (KOOS ADL/WOMAC function subscale) Quality of life (KOOS QoL) No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

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Rosedale 2014 *(Continued)*

	Participants were classified by knee "derangement" as either a responder or a nonresponder. Responders and non-responders were prescribed different programmes within the exercise group.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number generator
Allocation concealment (selection bias)	Low risk	Opaque envelopes were used.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (31%) and reasons were reported. ITT analysis performed.
Selective reporting (reporting bias)	Low risk	Reported outcomes consistent with trial registry

Sadeghi 2023

Study characteristics

Methods	Conducted in Iran
Participants	Knee OA recruitment 96 community volunteers KL Grade I-III
Interventions	Home, mix of supervised and unsupervised individual setting, delivered via mix of in-person and remotely 1. No intervention; 0 synchronous contact occasions 2. Exercise (quadriceps strengthening): 40 sessions (5x per week over 8 weeks); 8 synchronous contact occasions 3. Exercise (hamstring strengthening): 40 sessions (5x per week over 8 weeks); 8 synchronous contact occasions 4. Exercise (quadriceps and hamstring strengthening): 40 sessions (5x per week over 8 weeks); 8 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention)

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Sadeghi 2023 *(Continued)*

	Pain (VAS/NRS) Function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Groups 2 to 4 pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Microsoft Excel programme to generate random allocation sequence
Allocation concealment (selection bias)	Low risk	Concealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants not blinded to group allocation
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (0%)
Selective reporting (reporting bias)	Unclear risk	Could not find the trial on RCT20220206053950N2, 2022-09-07 - notably, it was retrospectively registered.

Salacinski 2012

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 37 community volunteers KL Grades I-III
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (cycling): 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention)

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Salacinski 2012 *(Continued)*

	Pain (WOMAC): reverse score
	Physical function (WOMAC): reverse score
	Quality of life (KOOS QoL)
	No participant-reported treatment success measure
Notes	Baseline incomparability between groups for BMI and age

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Preset randomisation scheme with computer assignment
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	High risk	17% (control) and 32% (intervention) lost to follow-up and not included in analysis
Selective reporting (reporting bias)	Low risk	Registered

Salli 2010

Study characteristics

Methods	Conducted in Turkey
Participants	Knee OA recruitment
	75 community volunteers
	ACR criteria, KL Grade I or II
Interventions	Clinic, supervised individual setting, delivered entirely in-person
	1. Paracetamol; 0 synchronous contact occasions
	2. Paracetamol + exercise (strengthening, concentric-eccentric): 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions

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Salli 2010 *(Continued)*

	3. Paracetamol + exercise (strengthening, isometric): 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) and 20 weeks Pain (VAS motion) Physical function (WOMAC) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information provided, major differences between allocation groups in physical function/SF-36 MCS at baseline
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only 4/75 lost fo follow-up (5%), balanced between allocation groups
Selective reporting (reporting bias)	Unclear risk	Not registered

Samuel 2014

Study characteristics

Methods	Conducted in India
Participants	Knee OA recruitment 73 community volunteers Chronic knee pain > 6 months and KL grade I-III
Interventions	Home, supervised individual setting, delivered entirely in-person 1. Interferential; 10 synchronous contact occasions

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Samuel 2014 *(Continued)*

	2. Interferential + exercise (strengthening): 20 sessions (5x per week over 4 weeks); 20 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was no attrition.
Selective reporting (reporting bias)	Unclear risk	No trial registry or protocol to verify against

Samut 2015

Study characteristics

Methods	Conducted in Turkey
Participants	Knee OA recruitment 42 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. No treatment; 0 synchronous contact occasions

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Samut 2015 *(Continued)*

	2. Exercise (strengthening): 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions 3. Exercise (walking): 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (KOOS ADL subscale) No quality of life measure No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (5%) reasons reported. Analysis on complete cases
Selective reporting (reporting bias)	Unclear risk	No trial registration or protocol to verify against

Santana 2022

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 34 community volunteers Clinical diagnosis by orthopaedist
Interventions	Clinic, supervised group setting, delivered entirely in-person

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Santana 2022 *(Continued)*

	1. Attention control; telephone calls to monitor symptoms and check not exercising, 12 synchronous contact occasions 2. Exercise (strengthening, proprioceptive/balance training): 12 sessions (2 x 50 minutes per week over 6 weeks); 12 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Pain data excluded from meta-analysis as very large difference between groups at baseline

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation took place digitally through www.random.org.
Allocation concealment (selection bias)	Unclear risk	Unclear risk if the schedule was stored online
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (20%) and reasons reported. Analysis on complete cases
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registration or protocol

Schilke 1996

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 20 community volunteers Self-reported OA
Interventions	Clinic, supervised individual setting, delivered entirely in-person

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Schilke 1996 *(Continued)*

	1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening): 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (OASI) Function (OASI) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Unclear risk	Not registered

Schlenk 2020

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 182 community volunteers ACR criteria
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered mix of digital/in-person

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Schlenk 2020 *(Continued)*

	1. Attention control; health education, 15 synchronous contact occasions 2. Exercise (strengthening, proprioceptive/balance training, walking, stretching) + nurse counselling: 78 sessions (2 to 3 x 60 to 65 minutes per week over 26 weeks); 15 synchronous contact occasions
Outcomes	At 26 weeks (immediately post-intervention) and 52 weeks Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Computer programme used to generate the random sequence
Allocation concealment (selection bias)	Low risk	Unclear how the sequence was concealed
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (21%) and reasons reported. ITT analysis performed
Selective reporting (reporting bias)	Low risk	Outcomes appear consistent with the trial registry.

Sedaghatnezhad 2021

Study characteristics

Methods	Conducted in Iran
Participants	Knee OA recruitment 30 community volunteers ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person

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Sedaghatnezhad 2021 *(Continued)*

	1. Physiotherapy; 15 synchronous contact occasions 2. Physiotherapy + exercise (walking): 10 sessions (5 x 30 minutes per week over 2 weeks); 10 synchronous contact occasions
Outcomes	At 2 weeks (immediately post-intervention) and 6 weeks Pain (global/overall pain VAS/NRS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Simple random assignment via an even-odd technique
Allocation concealment (selection bias)	High risk	Sequence allocation not concealed as it was performed concurrently with assigning participants
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Unclear risk	Unable to access registry

Sekir 2005

Study characteristics	
Methods	Conducted in Turkey
Participants	Knee OA recruitment 22 community volunteers with knee OA ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions

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Sekir 2005 *(Continued)*

	2. Exercise (proprioceptive/balance training): 12 sessions (1 to 3x per week over 6 weeks); 12 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global pain VAS/NRS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessor.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registry or protocol

Shabbir 2022

Study characteristics

Methods	Conducted in Pakistan
Participants	Knee OA recruitment 36 community volunteers KL grade I-III
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Conventional therapy; 18 synchronous contact occasions

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Shabbir 2022 *(Continued)*

	2. Conventional therapy + exercise (retrowalking): 18 sessions (3 x 30 minutes per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC global scale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registry or protocol

Shahine 2020

Study characteristics

Methods	Conducted in Egypt
Participants	Knee OA recruitment 66 community volunteers Diagnosis of OA (unspecified)
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered via mix of in-person/digital 1. Education; 0 synchronous contact occasions

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Shahine 2020 *(Continued)*

	2. Exercise (walking): variable number of sessions (unspecified frequency over 12 weeks); 16 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomly assigned by study nurse into two equal groups using Simple RAND function in excel
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were also the assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition not described
Selective reporting (reporting bias)	Unclear risk	Cannot verify against trial registration or protocol

Shen 2022

Study characteristics

Methods	Conducted in China
Participants	Knee OA recruitment 36 community volunteers with knee OA ACR Criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Attention control; health education; 18 synchronous contact occasions

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Shen 2022 *(Continued)*

	2. Exercise (stretching): 18 sessions (3 x 60 minutes per week over 6 weeks); 18 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) Pain (global pain VAS/NRS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (25%) reasons not described. ITT analysis not preformed
Selective reporting (reporting bias)	Unclear risk	Unable to access registry using number reported in publication

Shin 2017

Study characteristics	
Methods	Conducted in South Korea
Participants	Knee OA recruitment 25 community volunteers with knee OA KL Grade II-III
Interventions	Unknown whether clinic/home, supervised individual setting, delivered entirely in-person 1. Theracurmin; 0 synchronous contact occasions

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Shin 2017 *(Continued)*

	2. Theracurmin + exercise (strengthening, walking, stretching): 12 sessions (3 x 15 to 30 minutes per week over 4 weeks); 12 synchronous contact occasions
Outcomes	At 4 weeks (immediately post-intervention) Pain (global pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition not described
Selective reporting (reporting bias)	Unclear risk	Unable to verify against trial registry or published protocol

Simao 2012

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 23 community volunteers (in the 2 groups relevant to this study) ACR criteria, KL Grade II+
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; telephone calls to confirm adherence to daily activities, 12 synchronous contact occasions

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Simao 2012 *(Continued)*

	2. Exercise (cycling): 36 sessions (3 x 30 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (median, IQR provided) (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	The 2 allocation groups were incomparable at baseline for BMI (27 vs 30), WOMAC pain, and function. Only medians and IQR provided for outcome data; converted by review authors to mean and SD

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No procedure described
Allocation concealment (selection bias)	Low risk	Serially numbered opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only 1 participant in each allocation lost to follow-up at 12 weeks
Selective reporting (reporting bias)	Low risk	Registered

Song 2003

Study characteristics

Methods	Conducted in South Korea
Participants	Knee OA recruitment 72 community volunteers OA (confirmed by email), clinical and radiographic criteria
Interventions	Clinic, mix supervised/unsupervised mix group/individual setting, delivered entirely in-person

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Song 2003 *(Continued)*

	1. Attention control; telephone calls to ensure not exercising and provide attention, 12 synchronous contact occasions 2. Exercise (Tai Chi): 36 sessions (3 x 60 minutes per week over 12 weeks), 16 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (Korean WOMAC pain subscale) Physical function (Korean WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	Central allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	High risk	43% missing data, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Song 2020

Study characteristics

Methods	Conducted in China
Participants	Knee OA recruitment 36 community volunteers Diagnosis of OA (unspecified)
Interventions	Clinic, supervised group setting, delivered entirely in-person

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Song 2020 *(Continued)*

	1. Attention control; watched TV or read magazines together instead of exercising, 36 synchronous contact occasions 2. Exercise (PNF stretching): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12-weeks (immediately post-intervention) Pain (KOOS/WOMAC pain subscale) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (19%) and reasons are reported. Analysis on complete cases.
Selective reporting (reporting bias)	Unclear risk	Unable to verify against registry

Takacs 2017

Study characteristics

Methods	Conducted in Canada
Participants	Knee OA recruitment 40 community volunteers KL grade > 2
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person

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Takacs 2017 *(Continued)*

	1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening, proprioceptive/balance training): 40 sessions (4x per week over 10 weeks); 6 synchronous contact occasions
Outcomes	At 10 weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer random number generator
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition (10%) and reasons reported. Analysis on completed cases
Selective reporting (reporting bias)	Low risk	Outcomes reported are consistent with trial registry.

Talbot 2003

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 34 community volunteers ACR criteria
Interventions	Home, mix supervised/unsupervised individual setting, delivered entirely in-person

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Talbot 2003 *(Continued)*

	1. Education; 12 synchronous contact occasions 2. Education + exercise (walking): variable frequency across 12 weeks; 12 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) and 24 weeks Pain (McGill Pain Questionnaire) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Minimal missing data, balanced between allocation groups, efficacy analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Thomas 2002

Study characteristics

Methods	Conducted in the UK
Participants	Knee OA recruitment 786 community volunteers Self-reported persistent knee pain
Interventions	Home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. Education; 24 synchronous contact occasions via telephone

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Thomas 2002 *(Continued)*

	2. Education + exercise (strengthening): 728 sessions (7x per week over 104 weeks); 7 synchronous contact occasions
Outcomes	At 24 months (immediately post-intervention) Pain (WOMAC pain subscale) Function (WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	The trial authors presented results as if it were a 2-arm trial in the original report: 3 exercise groups pooled for meta-analysis and 3 control groups pooled as comparison (as presented in publication).

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central administration, sequential list audit trail
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Thorstensson 2005

Study characteristics	
Methods	Conducted in Sweden
Participants	Knee OA recruitment 61 community volunteers Radiographic knee OA (KL Grade III or higher) and longstanding knee pain
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions

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Thorstensson 2005 *(Continued)*

	2. Exercise (strengthening): 12 sessions (2 x 60 minutes per week over 6 weeks); 12 synchronous contact occasions
Outcomes	At 6 weeks (immediately post-intervention) and 26 weeks Pain (KOOS pain) Physical function (KOOS ADL) Quality of life (SF-36 MCS) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No sequence generation, sealed envelopes produced before randomisation
Allocation concealment (selection bias)	Low risk	Participants selected sealed envelope
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only 7% to 10% loss to follow-up, no ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Topp 2002

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 102 community volunteers ACR criteria
Interventions	Mix clinic/home, mix supervised/unsupervised mix group/individual setting, delivered entirely in-person

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Topp 2002 *(Continued)*

	1. No intervention; 0 synchronous contact occasions 2. Exercise (strengthening - dynamic): 48 sessions (3 x 60 minutes per week over 16 weeks); 16 synchronous contact occasions 3. Exercise (strengthening - isometric): 48 sessions (3 x 60 minutes per week over 16 weeks); 16 synchronous contact occasions
Outcomes	At 16 weeks (immediately post-intervention) Pain (WOMAC) Function (WOMAC) No quality of life measure No participant-reported treatment success measure
Notes	2 exercise groups (dynamic and isometric) pooled for meta-analysis

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information
Allocation concealment (selection bias)	Unclear risk	No information
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No information
Selective reporting (reporting bias)	Unclear risk	Not registered

Tsai 2013

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 55 community volunteers Diagnosed by healthcare provider

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Tsai 2013 *(Continued)*

Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Attention control; health education, 60 synchronous contact occasions 2. Exercise (Tai Chi): 60 sessions (3 x 20 to 40 minutes per week over 20 weeks); 60 synchronous contact occasions
Outcomes	At 20 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A random number table was used to randomly assign to each site to the Tai Chi or attention control according to the sequence of even/odd numbers from a random starting point in the table.
Allocation concealment (selection bias)	High risk	Allocation sequence was not concealed but was selected from a random starting point in a random numbers table.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants and personnel were not blinded to which intervention participants received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participant were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (18%) and reasons for attrition are reported. ITT analysis performed
Selective reporting (reporting bias)	Unclear risk	No trial protocol or registration to verify reported outcomes against

Van Baar 1998

Study characteristics

Methods	Conducted in the Netherlands
Participants	Knee and hip OA recruitment 113 community volunteers

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Van Baar 1998 *(Continued)*

	ACR criteria
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. Education; 2 synchronous contact occasions 2. Education + exercise (strengthening): 36 sessions (12 over 3 weeks); 17 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (VAS) Function (IRGL) No quality of life measure No participant-reported treatment success measure
Notes	Recruited participants with hip and knee OA. Separate results provided for knee OA

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random numbers table
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes, sequential numbering for audit trail
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

Vassao 2021

Study characteristics

Methods	Conducted in Brazil
Participants	Knee OA recruitment 28 community volunteers

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Vassao 2021 *(Continued)*

	ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (strengthening, stretching) + sham photobiomodulation: 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions
Outcomes	At 8 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer programme
Allocation concealment (selection bias)	Low risk	Opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (14%) and reasons were reported. Analysis on complete cases
Selective reporting (reporting bias)	Unclear risk	Unable to access registry

Vincent 2020

Study characteristics

Methods	Conducted in the USA
Participants	Knee OA recruitment 87 community volunteers

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Vincent 2020 *(Continued)*

	ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Attention control; telephone calls to encourage healthy behaviours, 16 synchronous contact occasions 2. Exercise (concentric strengthening): 32 sessions (2x per week over 16 weeks); 32 synchronous contact occasions 3. Exercise (eccentric strengthening): 32 sessions (2x per week over 16 weeks); 32 synchronous contact occasions
Outcomes	At 16 weeks (immediately post-intervention) Pain (pain on activities other than walking VAS/NRS) No physical function measure No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Group 2 and 3 pooled for meta-analysis Data taken from supplementary files in the publication

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated list
Allocation concealment (selection bias)	Low risk	Group assignments were placed in numbered sealed opaque envelopes.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (40%) and reasons reported. Analysis performed on complete cases
Selective reporting (reporting bias)	Unclear risk	Unable to verify against registry

Wallis 2017

Study characteristics

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Wallis 2017 *(Continued)*

Methods	Conducted in Australia
Participants	Knee OA recruitment 46 community volunteers KL grade II to IV
Interventions	Mix clinic/home, mix supervised/unsupervised individual setting, delivered via mix of in-person/digital 1. Usual care; 0 synchronous contact occasions 2. Exercise (walking): variable number of sessions (accumulation of 70 minutes walking per week over 12 weeks); 13 synchronous contact occasions
Outcomes	At 12-weeks (immediately post-intervention) Pain (global/overall pain VAS/NRS) Physical function (WOMAC function subscale) Quality of life (EuroQoL) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer random number generator
Allocation concealment (selection bias)	Low risk	Opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the blinded outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (2%) and reason reported. Analysis on complete cases
Selective reporting (reporting bias)	Low risk	Outcomes consistent with trial registry

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Wang 2011

Study characteristics	
Methods	Conducted in Taiwan
Participants	Knee OA recruitment 52 community volunteers Physician diagnosis of OA
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No intervention; 0 synchronous contact occasions 2. Exercise (flexibility and aerobic): 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (KOOS pain) Function (KOOS ADL) Quality of life (KOOS quality of life) No participant-reported treatment success measure
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	External, researcher not recruiting participants
Blinding of participants and personnel (performance bias) All outcomes	High risk	Unblinded
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Unblinded participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only 7% loss to follow-up, 2/28 in each allocation, no ITT analysis
Selective reporting (reporting bias)	Unclear risk	Not registered

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Wortley 2013

Study characteristics	
Methods	Conducted in the USA
Participants	Knee OA recruitment 39 community volunteers ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. No treatment; 1 synchronous contact occasions 2. Exercise (strengthening): 20 sessions (2 x 60 minutes per week over 10 weeks); 20 synchronous contact occasions 3. Exercise (Tai Ji): 20 sessions (2x 60 minutes per week over 10 weeks); 20 synchronous contact occasions
Outcomes	At 10 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	High risk	Attrition (21%) and reasons reported. Complete cases analysed
Selective reporting (reporting bias)	Unclear risk	Unable to verify against registration or published protocol

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Xiao 2021

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 284 community volunteers ACR criteria
Interventions	Unclear setting, delivered entirely in-person 1. No treatment; 0 synchronous contact occasions 2. Exercise (Wuqinxi-Qigong): 144 sessions (6 x 60 minutes per week over 24 weeks); unspecified number of synchronous contact occasions
Outcomes	At 24-weeks (immediately post-intervention) Pain (WOMAC pain subscale) Quality of life measure (SF-36) No physical function measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants are the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition (6%) reasons not reported. Analysis on complete cases
Selective reporting (reporting bias)	Unclear risk	Unable to verify against published protocol and trial registry

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Ye 2020

Study characteristics	
Methods	Conducted in China
Participants	Knee OA recruitment 50 community volunteers ACR criteria
Interventions	Mix clinic/home, mix supervised/unsupervised mix of group/individual setting, delivered mix of in-person/digital delivery 1. Attention control; telephone calls to remind not to engage in exercise, 6 synchronous contact occasions 2. Exercise (Baduanjin Qigong): 36 sessions (3 x 40 minutes per week over 12 weeks); 12 synchronous contact occasions
Outcomes	At 12 weeks (immediately post-intervention) Pain (WOMAC pain subscale) Physical function (WOMAC function subscale) No quality of life measure No participant-reported treatment success measure
Notes	Authors contacted to request or clarify data

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated. The program called "PLAN" was used to generate the randomisation sequence in the Statistical Analysis System (SAS).
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants were not blinded to which intervention they received. Personnel (care providers) were not blinded to which intervention a participant received.
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Outcomes consistent with registry

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Øiestad 2023

Study characteristics	
Methods	Conducted in Norway
Participants	Knee OA recruitment 168 community volunteers KL grade II and III
Interventions	Mix of clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person 1. Usual care; 0 synchronous contact occasions 2. Exercise (strengthening): 36 sessions (3x per week over 12 weeks); 24 synchronous contact occasions 3. Exercise (cycling): 36 sessions (3 x 45 minutes per week over 12 weeks); 24 synchronous contact occasions
Outcomes	At 20 weeks (immediately post-intervention) and 52 weeks: Pain (VAS/NRS) Function (KOOS ADL subscale) Quality of life (EuroQoL) No participant-reported treatment success measure
Notes	Groups 2 and 3 pooled for meta-analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Concealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Participants aware of group allocation
Blinding of outcome assessment (detection bias) - subjective self-reported outcomes (pain, function, quality of life)	High risk	Participants were the outcome assessors.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition was > 20%; ITT analyses were used.
Selective reporting (reporting bias)	Low risk	Trial authors describe why some outcomes described in the registry record were not reported in the manuscript (lack of funding).

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ACR: American College of Rheumatology; ADL: activities of daily living; AFI: Algofunctional Index; AIMS: Arthritis Impact Measurement Scales; AQoL-6D: Assessment of Quality of Life - six dimensions; ASMP: Arthritis Self-Management Programme; BMI: body mass index; EQ5D: standardised measure of health outcome; FAST: Fitness Arthritis and Seniors Trial; GP: general practitioner; HA: hyaluronic acid; HADS: Hospital Anxiety Depression Scale; IFC: International Functional Classification; IRGL: Influence of Rheumatic Disease on Health and Lifestyle scale; ITT: intention-to-treat; KL: Kellgren and Lawrence; KOOS: Knee Osteoarthritis Outcome Scale; LOCF: last observation carried forward; MCS: Mental Component Summary; MMT: manual and manipulative therapy; MRI: magnetic resonance imaging; MSK-HQ: Arthritis Research UK Musculoskeletal Health Questionnaire; NRS: numerical rating scale; NSAIDs: non-steroidal anti-inflammatory drugs; OA: osteoarthritis; OASI: Osteoarthritis Screening Index; OMERACT-OARSI: Outcome Measures in Rheumatology-Osteoarthritis Research Society International; PACE: Patient-centered Assessment and Counselling for Exercise; PCS: Physical Component Summary; PNF: proprioceptive neuromuscular facilitation; QoL: quality of life; RCT: randomised controlled trial; ROM: range of motion; SD: standard deviation; SF: Short Form; TENS: transcutaneous electrical nerve stimulation; US: ultrasound; VAS: visual analogue scale; WOMAC: Western Ontario and McMasters Universities Osteoarthritis Index; x: times.

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Actrn12615000227594	Does not include an eligible comparator intervention
Actrn12616000273482	Does not include an eligible comparator intervention
Actrn12618001280291	Does not report self-reported pain, function, or quality of life
Actrn12620001281987	Does not include an eligible exercise intervention
Actrn12621000535875	Does not include an eligible comparator intervention
Actrn12622000461796	Does not include an eligible exercise intervention
Actrn12622001231730	Does not include an eligible comparator intervention
Ageberg 2010	Does not include an eligible comparator intervention
Aglamis 2008	Large baseline differences between 2 small comparator groups in pain and physical function scores
Ahmad 2023	Does not include an eligible comparator intervention
Akyol 2010	Does not include an eligible comparator intervention
Alfredo 2012	Does not include an eligible comparator intervention
Alkhawajah 2019	Does not include an eligible exercise intervention
Allen 2021	Does not include an eligible comparator intervention
Anjum 2023	Does not include an eligible comparator intervention
Anwer 2011	Does not include an eligible comparator intervention
Aoki 2009	Is perioperative
Aree-Ue 2017	Does not include an eligible comparator intervention
Armagan 2015	Does not include an eligible comparator intervention
Atamaz 2006	Does not include an eligible exercise intervention

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Study	Reason for exclusion
Atamaz 2012	Does not include an eligible comparator intervention
Ay 2013	Does not include an eligible comparator intervention
Bell 2023c	Does not include an eligible exercise intervention
Bennell 2005	Does not include an eligible comparator intervention
Bezalel 2010	Does not include an eligible comparator intervention
Bhalekar 2017	Not an RCT
Bhore 2023	Does not include an eligible comparator intervention
Boocock 2009	Does not include an eligible comparator intervention
Borjesson 1996	Is perioperative
Brosseau 2012a	Does not report self-reported pain, function, or quality of life
Brosseau 2012b	Does not include an eligible comparator intervention
Bulthuis 2007	Does not include people with knee OA
Bulthuis 2008	Does not include people with knee OA
Cagnin 2020	Does not include an eligible comparator intervention
Callaghan 1995	Unable to ascertain effect size, as only provided with median % improvements without baseline scores and with extremely wide confidence intervals because of small sample size
Centor 2023	Not an RCT
Cetin 2008	Does not include an eligible comparator intervention
Chaipinyo 2009	Does not include an eligible comparator intervention
Chamberlain 1982	Does not include an eligible comparator intervention
Chang 2019	Does not include an eligible comparator intervention
Chao 2021	Does not include an eligible comparator intervention
Chen 2021b	Does not include an eligible comparator intervention
ChiCTR1800015203	Does not include an eligible comparator intervention
ChiCTR2000031887	Does not include an eligible comparator intervention
ChiCTR2000033370	Does not include an eligible comparator intervention
ChiCTR2000033397	Does not include an eligible comparator intervention
ChiCTR2000037083	Does not include an eligible comparator intervention

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Study	Reason for exclusion
ChiCTR2000037256	Does not include an eligible comparator intervention
ChiCTR2000037303	Does not include an eligible comparator intervention
ChiCTR2000038158	Does not include an eligible comparator intervention
ChiCTR2000039005	Does not include an eligible comparator intervention
ChiCTR2100041978	Does not include an eligible comparator intervention
ChiCTR2100044586	Does not include an eligible comparator intervention
ChiCTR2100045654	Does not report self-reported pain, function, or quality of life
ChiCTR2100052416	Does not include an eligible comparator intervention
ChiCTR2100053574	Does not include an eligible comparator intervention
ChiCTR2300067530	Does not include an eligible comparator intervention
ChiCTR2300067546	Does not include an eligible comparator intervention
ChiCTR2300070376	Does not include an eligible comparator intervention
ChiCTR2300070872	Does not include an eligible comparator intervention
ChiCTR2300071181	Does not include an eligible comparator intervention
ChiCTR2300072820	Does not include an eligible comparator intervention
ChiCTR2300073480	Does not include an eligible comparator intervention
ChiCTR2300073688	Does not include an eligible exercise intervention
ChiCTR-IIR-16009240	Does not include an eligible comparator intervention
ChiCTR-INR-17014012	Does not include an eligible comparator intervention
ChiCTR-IOR-16009124	Does not include an eligible comparator intervention
ChiCTR-IOR-17013324	Does not include an eligible comparator intervention
Cho 2015	Does not report self-reported pain, function, or quality of life
Ciolac 2011	Not an RCT
Crotty 2009	Is perioperative
Ctri/2014/01/004270	Does not include an eligible comparator intervention
Ctri/2015/01/005485	Does not include an eligible comparator intervention
Ctri/2018/03/012620	Does not include an eligible comparator intervention
Ctri/2019/01/017049	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Ctri/2019/09/021331	Does not include an eligible comparator intervention
Ctri/2020/03/023737	Does not include an eligible comparator intervention
Ctri/2020/03/024311	Does not include an eligible comparator intervention
Ctri/2020/06/025912	Does not include an eligible comparator intervention
Ctri/2020/06/025973	Does not include an eligible comparator intervention
Ctri/2020/07/026397	Does not include an eligible comparator intervention
Ctri/2020/11/028915	Does not include an eligible exercise intervention
Ctri/2020/11/029138	Does not include an eligible comparator intervention
Ctri/2021/01/030519	Does not include an eligible comparator intervention
Ctri/2021/04/033203	Does not include an eligible comparator intervention
Ctri/2021/09/036724	Does not include an eligible comparator intervention
Ctri/2022/03/041074	Does not report self-reported pain, function, or quality of life
Ctri/2022/05/042506	Does not include an eligible exercise intervention
Ctri/2022/10/046835	Does not include an eligible comparator intervention
Ctri/2022/12/048269	Does not include an eligible comparator intervention
Ctri/2023/01/049164	Does not include an eligible exercise intervention
Ctri/2023/04/051550	Does not include an eligible comparator intervention
Ctri/2023/04/051625	Does not include an eligible comparator intervention
Ctri/2023/05/053101	Does not include an eligible comparator intervention
Ctri/2023/06/053585	Does not report self-reported pain, function, or quality of life
Ctri/2023/07/054805	Does not include an eligible comparator intervention
Ctri/2023/07/054846	Does not include an eligible comparator intervention
Ctri/2023/07/055413	Does not include an eligible comparator intervention
Ctri/2023/07/055542	Does not report self-reported pain, function, or quality of life
Ctri/2023/08/056055	Does not include an eligible exercise intervention
Ctri/2023/08/056749	Does not include an eligible exercise intervention
Ctri/2023/08/056896	Does not include an eligible comparator intervention
Ctri/2023/08/057102	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Ctri/2023/08/057124	Does not report self-reported pain, function, or quality of life
Ctri/2023/10/058883	Does not include an eligible comparator intervention
Cui 2023	Not an RCT
Dantas 2023	Does not include an eligible comparator intervention
Darlow 2023	Does not include an eligible exercise intervention
De Oliveira 2020	Does not include an eligible comparator intervention
De Olivera 2017	Does not include an eligible comparator intervention
Deyle 2000	Does not include an eligible comparator intervention
Deyle 2005	Does not include an eligible comparator intervention
Dias 2003	Unable to extract change (SD) or post-treatment (SD) scores from published manuscript
Diracoglu 2005	Does not include an eligible comparator intervention
Doi 2008	Does not include an eligible comparator intervention
Drks00009257	Not an RCT
Drks00024160	Does not include an eligible comparator intervention
Drks00025033	Does not include an eligible comparator intervention
Duman 2012	Does not include an eligible comparator intervention
Durmus 2007	Does not include an eligible comparator intervention
Durmus 2012	Does not include an eligible comparator intervention
Ebnezar 2012a	Does not include an eligible comparator intervention
Ebnezar 2012b	Does not include an eligible comparator intervention
Evcik 2002	Not an RCT
Evgeniadis 2008	Is perioperative
Eyigor 2004	Does not include an eligible comparator intervention
Falck 2018	Does not include an eligible exercise intervention
Fan 2023	Not an RCT
Farr 2010	Does not report self-reported pain, function, or quality of life
Feinglass 2012	Does not include an eligible comparator intervention
Feng 2023	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Fitzgerald 2011	Does not include an eligible comparator intervention
Focht 2022	Does not include an eligible comparator intervention
Forestier 2010	Does not include an eligible exercise intervention
Foroughi 2011	Does not include an eligible comparator intervention
Foster 2007	Does not include an eligible comparator intervention
Foster 2023	Does not include an eligible comparator intervention
Gaal 2008	Does not include an eligible exercise intervention
Gaudreault 2011	Not an RCT
Ghandali 2017	Not an RCT
Gill 2009	Does not include an eligible comparator intervention
Gohir 2020	Does not include an eligible comparator intervention
Green 1993	Does not include an eligible comparator intervention
Gremion 2009	Does not include an eligible comparator intervention
Griessmann 2014	Does not include people with knee OA
Gustavsson 2023	Does not include an eligible exercise intervention
Hammami 2022	Does not include an eligible comparator intervention
Haslam 2001	Does not include an eligible comparator intervention
Helmark 2010	Does not report self-reported pain, function, or quality of life
Helmark 2012	Not an RCT
Hinman 2007	Does not include an eligible exercise intervention
Hiyama 2012	Does not include an eligible comparator intervention
Hoeksma 2004	Does not include an eligible comparator intervention
Holsgaard-Larsen 2017	Does not include an eligible comparator intervention
Huang 2018	Does not include an eligible comparator intervention
Huang 2021	Does not report self-reported pain, function, or quality of life
Hunt 2013	Does not report self-reported pain, function, or quality of life
Hunter 2023	Does not include an eligible exercise intervention
Hurley 1998	Not an RCT

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Study	Reason for exclusion
Hurley 2020	Does not include an eligible comparator intervention
Irct20100811004549n10	Does not report self-reported pain, function, or quality of life
Irct20101017004952n6	Does not include an eligible comparator intervention
Irct20101228005486n6	Does not include an eligible comparator intervention
Irct20131225015932n18	Does not include an eligible comparator intervention
Irct2014093019346n1	Does not report self-reported pain, function, or quality of life
Irct2015010120526n1	Does not include an eligible exercise intervention
Irct20150519022320n33	Does not report self-reported pain, function, or quality of life
Irct20150602022539n11	Does not include an eligible comparator intervention
Irct2015081123593n1	Does not include an eligible comparator intervention
Irct201604041552n6	Does not include an eligible comparator intervention
Irct2016061428406n2	Does not report self-reported pain, function, or quality of life
Irct2016070727359n3	Does not report self-reported pain, function, or quality of life
Irct201608044641n12	Does not include an eligible comparator intervention
Irct2016120415932n12	Does not include people with knee OA
Irct20161208031300n2	Does not include an eligible comparator intervention
Irct2017013032251n2	Does not include an eligible comparator intervention
Irct20180226038870n2	Does not include an eligible comparator intervention
Irct20180314039092n1	Does not report self-reported pain, function, or quality of life
Irct20180627040251n5	Does not include people with knee OA
Irct20181017041371n1	Does not include an eligible comparator intervention
Irct20190116042386n1	Does not include an eligible comparator intervention
Irct20190222042801n1	Does not include an eligible comparator intervention
Irct20200203046360n1	Does not include an eligible comparator intervention
Irct20200825048523n2	Does not include an eligible comparator intervention
Irct20201031049207n2	Does not include an eligible comparator intervention
Irct20210121050099n1	Does not include an eligible comparator intervention
Irct20210321050752n4	Does not include people with knee OA

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Study	Reason for exclusion
Irct20210417050997n1	Does not include an eligible comparator intervention
Irct20210815052194n1	Does not include an eligible comparator intervention
Irct20220216054038n1	Does not include an eligible comparator intervention
Irct20220830055831n1	Does not report self-reported pain, function, or quality of life
Irct20221115056509n1	Does not include an eligible comparator intervention
Irct20230101057015n1	Does not report self-reported pain, function, or quality of life
Irct20230202057310n2	Does not include an eligible comparator intervention
Irct20230306057633n2	Does not include an eligible comparator intervention
Irct20230427058007n1	Does not include an eligible comparator intervention
Irct20230606058395n1	Does not include an eligible exercise intervention
Irct20231015059728n1	Does not include an eligible comparator intervention
Isaramalai 2018	Does not include an eligible comparator intervention
Isrctn13307390	Does not include an eligible exercise intervention
Jan 1991	Not an RCT
Jessep 2009	Does not include an eligible comparator intervention
Jimenez 2014	Not an RCT
Jing 2019	Does not include an eligible comparator intervention
Jinnouchi 2023	Does not include an eligible comparator intervention
Jma-Iia00110	Does not include an eligible comparator intervention
Ju 2015	Not an RCT
Kao 2012	Does not include an eligible comparator intervention
Karagulle 2007	Does not include an eligible exercise intervention
Karimi 2021	Does not report self-reported pain, function, or quality of life
Karp 2016	Does not include an eligible comparator intervention
Karp 2019	Does not include an eligible comparator intervention
Kawasaki 2008	Does not include an eligible comparator intervention
Kawasaki 2009	Does not include an eligible comparator intervention
Kct0006037	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Khachian 2020	Does not include an eligible comparator intervention
Khan 2023	Does not include an eligible comparator intervention
King 2008	Not an RCT
Kletskova 2021	Not an RCT
Kloek 2016a	Does not include an eligible comparator intervention
Kloek 2018	Does not include an eligible comparator intervention
Knoop 2023	Not an RCT
Konishi 2009	Not an RCT
Kotteeswaran 2021	Does not include an eligible comparator intervention
Kreindler 1989	Does not report self-reported pain, function, or quality of life
Kumar 2017	Does not report self-reported pain, function, or quality of life
Kuptniratsaikul 2002	Not an RCT
Lai 2018	Does not report self-reported pain, function, or quality of life
Lall 2023	Does not include an eligible comparator intervention
Landsmeer 2016	Does not include an eligible comparator intervention
Lankhorst 1982	Does not report self-reported pain, function, or quality of life
Lee 2017a	Does not include an eligible comparator intervention
Lee 2018	Does not include an eligible comparator intervention
Li 2017a	Does not include an eligible exercise intervention
Liang 2019	Not an RCT
Lim 2002	Does not include an eligible comparator intervention
Lim 2010	Does not include an eligible comparator intervention
Lin 2004	Does not include an eligible exercise intervention
Lin 2020	Does not include an eligible comparator intervention
Liu 2008	Does not include an eligible comparator intervention
Liu 2023	Not an RCT
Lorbeer 2023	Does not include an eligible comparator intervention
Losciale 2023	Does not include people with knee OA

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Study	Reason for exclusion
Mahalle 2022	Does not report self-reported pain, function, or quality of life
Mahmoud 2023	Does not include an eligible exercise intervention
Mangione 1999	Does not include an eligible comparator intervention
Marra 2012	Does not include an eligible comparator intervention
Mascarin 2012	Does not include an eligible comparator intervention
Mat 2016	Does not include an eligible exercise intervention
Mat 2018	Does not include an eligible exercise intervention
McCarthy 2004	Does not include an eligible comparator intervention
McKnight 2010	Does not include an eligible comparator intervention
McQuade 2011	Not an RCT
Meiling 2023	Does not include an eligible exercise intervention
Mesa-Castrillon 2023	Does not include an eligible comparator intervention
Messier 2000a	Does not include an eligible comparator intervention
Messier 2007	Does not include an eligible comparator intervention
Messier 2008	Does not report self-reported pain, function, or quality of life
Messier 2017	Does not include an eligible comparator intervention
Mikesky 2006	Does not include an eligible comparator intervention
Minor 1989	Does not include an eligible comparator intervention
Miyagawa 2016	Not an RCT
Modarresi 2023	Does not include an eligible comparator intervention
Mohammedsadiq 2023	Does not include an eligible comparator intervention
Moss 2007	Does not include an eligible comparator intervention
Mulla 2018	Does not include people with knee OA
Murphy 2008	Does not include an eligible comparator intervention
Murphy 2016	Does not include an eligible exercise intervention
Nathani 2020	Not an RCT
Nct01945749	Does not include an eligible comparator intervention
Nct02024126	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Nct02043613	Does not include an eligible comparator intervention
Nct02098096	Does not include an eligible comparator intervention
Nct02350387	Does not include an eligible comparator intervention
Nct02454374	Does not include an eligible comparator intervention
Nct02598804	Does not include an eligible exercise intervention
Nct02600702	Does not include an eligible comparator intervention
Nct02698072	Does not include an eligible comparator intervention
Nct02901964	Does not include an eligible comparator intervention
Nct03050320	Does not include people with knee OA
Nct03167502	Does not include an eligible comparator intervention
Nct03171168	Does not include an eligible comparator intervention
Nct03379623	Does not include an eligible exercise intervention
Nct03470090	Does not include an eligible comparator intervention
Nct03571698	Does not include an eligible comparator intervention
Nct03621631	Does not include an eligible comparator intervention
Nct03666585	Does not include an eligible comparator intervention
Nct03676790	Does not include an eligible comparator intervention
Nct03750279	Does not include an eligible comparator intervention
Nct03798002	Does not include an eligible comparator intervention
Nct03893981	Does not include an eligible comparator intervention
Nct03921801	Does not include an eligible exercise intervention
Nct03951506	Does not include an eligible comparator intervention
Nct04058002	Does not include an eligible exercise intervention
Nct04099394	Does not include an eligible exercise intervention
Nct04138784	Does not include an eligible comparator intervention
Nct04183933	Does not include an eligible comparator intervention
Nct04198467	Does not include an eligible exercise intervention
Nct04225182	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Nct04323267	Does not include an eligible comparator intervention
Nct04390932	Does not include an eligible comparator intervention
Nct04423887	Does not include an eligible comparator intervention
Nct04445350	Does not include an eligible comparator intervention
Nct04528953	Does not include an eligible comparator intervention
Nct04536519	Does not include an eligible comparator intervention
Nct04540341	Does not include an eligible comparator intervention
Nct04542655	Does not include an eligible exercise intervention
Nct04574700	Does not include an eligible comparator intervention
Nct04600661	Does not include an eligible comparator intervention
Nct04654871	Does not include an eligible comparator intervention
Nct04678999	Does not include an eligible comparator intervention
Nct04747444	Does not include an eligible comparator intervention
Nct04819672	Does not include an eligible comparator intervention
Nct04857879	Does not include an eligible comparator intervention
Nct04925895	Does not include an eligible comparator intervention
Nct04980300	Does not report self-reported pain, function, or quality of life
Nct05026385	Does not include an eligible comparator intervention
Nct05131048	Does not include an eligible comparator intervention
Nct05169606	Does not include an eligible comparator intervention
Nct05209893	Does not include an eligible comparator intervention
Nct05274932	Does not include an eligible comparator intervention
Nct05403645	Does not include an eligible comparator intervention
Nct05601947	Does not include an eligible comparator intervention
Nct05777421	Does not include an eligible comparator intervention
Nct05810376	Does not include an eligible exercise intervention
Nct05839535	Does not include an eligible comparator intervention
Nct05850793	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Nct05877703	Does not include an eligible comparator intervention
Nct05878587	Does not include an eligible comparator intervention
Nct05889598	Does not include people with knee OA
Nct05893017	Does not include an eligible comparator intervention
Nct05941117	Does not include an eligible comparator intervention
Nct05955300	Does not include an eligible comparator intervention
Nct05955430	Does not include an eligible comparator intervention
Nct05958212	Does not include an eligible comparator intervention
Nct05995210	Does not include people with knee OA
Nct05997966	Does not include an eligible comparator intervention
Nct06090097	Does not include an eligible comparator intervention
Nct06105788	Does not include an eligible comparator intervention
Neves 2011	Does not include an eligible comparator intervention
Ng 2010	Does not include an eligible comparator intervention
Ntr6996	Does not include an eligible comparator intervention
O'Brien 2016	Does not include an eligible comparator intervention
Onwunzo 2021	Does not include an eligible comparator intervention
Osteras 2019	Does not report self-reported pain, function, or quality of life
Osteras 2023	Does not include an eligible comparator intervention
Osugi 2014	Does not include an eligible comparator intervention
Ozdincler 2005	Does not include an eligible comparator intervention
Ozudogru 2023	Does not include an eligible comparator intervention
Pactr201711002482359	Does not include an eligible comparator intervention
Pactr201804003266300	Does not include an eligible comparator intervention
Pactr202002763876912	Does not include an eligible comparator intervention
Pactr202003828737019	Does not include an eligible comparator intervention
Pactr202010869767272	Does not include an eligible comparator intervention
Pactr202202919624090	Not an RCT

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Study	Reason for exclusion
Paolillo 2018	Does not report self-reported pain, function, or quality of life
Park 2021	Does not include an eligible comparator intervention
Paterson 2022	Does not include an eligible comparator intervention
Peng 2021	Does not include an eligible comparator intervention
Penninx 2001	Does not include eligible comparator
Penninx 2002	Does not include eligible comparator
Pereira 2011	Does not include an eligible comparator intervention
Petersen 2010	Does not include an eligible comparator intervention
Petersen 2011	Does not include an eligible comparator intervention
Peterson 1993	Does not include eligible comparator intervention
Petrella 2000	Does not include an eligible comparator intervention
Pietrosimone 2010	Does not include an eligible comparator intervention
Pietrosimone 2012	Does not include an eligible comparator intervention
Pietrosimone 2020	Does not include an eligible comparator intervention
Pisters 2010a	Does not include an eligible comparator intervention
Pisters 2010b	Does not include eligible comparator intervention
Piva 2011	Does not include eligible comparator intervention
Piyakhachornrot 2011	Does not include an eligible comparator intervention
Quicke 2016	Does not include an eligible comparator intervention
Quilty 2003	Does not include an eligible comparator intervention
Quirk 1985	Does not include an eligible comparator intervention
Rafiq 2021d	Does not report self-reported pain, function, or quality of life
Raj 2018	Does not include an eligible comparator intervention
Rathwa 2019	Does not include an eligible comparator intervention
Rattanachaiyanont 2008	Does not include an eligible comparator intervention
Ravaud 2004	Not an RCT
RBR-10j4bw25	Does not include an eligible comparator intervention
RBR-2twzy65	Does not include an eligible comparator intervention

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Study	Reason for exclusion
RBR-4pzsfxw	Does not include an eligible comparator intervention
RBR-7p8wzyr	Does not include an eligible comparator intervention
RBR-7t6nzr	Does not include an eligible comparator intervention
RBR-8qby2wt	Does not include an eligible comparator intervention
RBR-9d7c7b	Does not include an eligible comparator intervention
Reid 2010	Does not include an eligible comparator intervention
Reid 2011	Does not report self-reported pain, function, or quality of life
Ren 2020	Does not include an eligible comparator intervention
Rizvi 2023	Does not include an eligible comparator intervention
Robbins 2021	Does not include an eligible comparator intervention
Robson 2022	Does not include an eligible comparator intervention
Runhaar 2019	Does not include an eligible comparator intervention
Sabharwal 2022	Does not include an eligible comparator intervention
Saji 2023	Does not include an eligible comparator intervention
Sandal 2016	Does not include an eligible comparator intervention
Sayers 2012	Does not include an eligible comparator intervention
Schlenk 2011	Not an RCT
Schmid 2013	Does not report self-reported pain, function, or quality of life
Scopaz 2009	Not an RCT
Selfe 2008	Does not include an eligible comparator intervention
Sen 2004	Does not include an eligible comparator intervention
Shakoor 2007	Does not include an eligible comparator intervention
Shakoor 2010	Not an RCT
Shaughnessy 2023	Not an RCT
Shen 2008	Not an RCT
Silva 2008	Does not include an eligible comparator intervention
Silva 2018	Not an RCT
Simao 2019	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Sled 2010	Not an RCT
Smith 2023	Not an RCT
Song 2010	Does not report self-reported pain, function, or quality of life
Song 2022	Does not include an eligible comparator intervention
Soni 2012	Does not include an eligible comparator intervention
Stitik 2007a	Not an RCT
Stitik 2007b	Not an RCT
Sudhesh 2013	Does not include an eligible exercise intervention
Sullivan 1998	Does not include eligible comparator intervention
Swank 2011	Perioperative
Sylvester 1989	Does not include an eligible comparator intervention
Tan 2020	Does not include an eligible comparator intervention
Tan 2023	Does not include an eligible comparator intervention
Tctr20170705003	Is perioperative
Tctr20210903007	Not an RCT
Tctr20221118001	Does not include an eligible comparator intervention
Tctr20230502005	Does not include an eligible comparator intervention
Teixeira 2011	Does not include eligible comparator intervention
Thiengwittayaporn 2009	Does not include an eligible comparator intervention
Thiengwittayaporn 2023	Does not include an eligible comparator intervention
Toda 2001	Not an RCT
Tok 2011	Does not include an eligible comparator intervention
Topp 2009	Is perioperative
Torstensen 2023a	Does not include an eligible comparator intervention
Torstensen 2023b	Not an RCT
Tsai 2009	Not an RCT
Tsai 2015	Does not include an eligible comparator intervention
Tsauo 2008	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Tunay 2010	Does not include an eligible comparator intervention
Tung 2022	Does not include an eligible comparator intervention
Tuzun 2004	Does not include an eligible comparator intervention
Umin000016326	Not an RCT
Umin000022060	Does not include an eligible comparator intervention
Umin000028944	Does not include an eligible comparator intervention
Umin000029045	Does not include an eligible comparator intervention
Umin000034385	Does not include an eligible comparator intervention
Umin000035225	Does not include an eligible comparator intervention
Umin000039579	Does not include an eligible comparator intervention
Umin000043356	Does not include an eligible comparator intervention
Umin000044440	Does not include an eligible comparator intervention
Umin000044884	Does not include an eligible comparator intervention
Umin000051857	Does not include an eligible comparator intervention
Van Gool 2005	Does not include eligible comparator intervention
Veenhof 2007	Does not include an eligible comparator intervention
Villadsen 2014	Is perioperative
Walls 2010	Is perioperative
Wan 2023	Does not include an eligible exercise intervention
Wang 2006	Does not include an eligible comparator intervention
Wang 2007a	Does not include an eligible exercise intervention
Wang 2007b	Does not report self-reported pain, function, or quality of life
Wang 2009	Does not include an eligible comparator intervention
Weng 2009	Does not include an eligible comparator intervention
Whitehurst 2011	Does not include an eligible comparator intervention
Williamson 2007	Is perioperative
Williamson 2007b	Is perioperative
Wong 2017	Does not include an eligible comparator intervention

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Study	Reason for exclusion
Wong 2018	Does not include an eligible comparator intervention
Wyatt 2001	Does not include an eligible comparator intervention
Xiao 20201	Does not include an eligible comparator intervention
Xiao 2020b	Does not include an eligible comparator intervention
Yilmaz 2010	Does not include an eligible comparator intervention
Yip 2007a	Does not include an eligible comparator intervention
Yip 2007b	Secondary analyses from an RCT (primary already excluded: Yip 2007a)
Yip 2008	Secondary analyses from an RCT (primary already excluded: Yip 2007a)
Yue 2023	Does not include an eligible comparator intervention
Zhang 2020	Does not report self-reported pain, function, or quality of life
Zhang 2022	Does not include an eligible comparator intervention
Zhao 2019	Does not include an eligible comparator intervention
Zhu 2023	Not an RCT

OA: osteoarthritis; **RCT:** randomised controlled trial; **SD:** standard deviation

Characteristics of studies awaiting classification *[ordered by study ID]*

Abou-Raya 2013

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Actrn12617000183381

Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 7/02/2020

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Actrn12617000183381 *(Continued)*

	Status: "Completed". Publication not found. Comparison: wait list

Bossen 2013

Methods	
Participants	
Interventions	
Outcomes	
Notes	Authors unable to provide data for knee OA participants only

Carlos 2012

Methods	
Participants	
Interventions	
Outcomes	
Notes	Not in English; translated version insufficient to enable extraction

Chang 2017

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Cheing 2002

Methods	
Participants	Knee OA recruitment 31 community volunteers with knee OA Other (KL grade II or above)
Interventions	Clinic, supervised individual setting, delivered entirely in person

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Cheing 2002 *(Continued)*

	1. Exercise (strengthening): 20 sessions (5 x 30 minutes per week over 4 weeks); 20 synchronous contact occasions 2. Control (placebo TENS): 20 synchronous contact occasions
Outcomes	At 4 weeks (immediately postintervention) Pain (global pain VAS/NRS) No physical function measure No quality-of-life measure No participant-reported treatment success measure
Notes	Excluded from meta-analysis for pain on suspicion of error in outcomes reported - e.g. measured on 0-100 VAS, yet some means reported as > 100. All 62 participants reported to have mean 100/100 pain at baseline. Authors non-responsive to attempts to confirm data

Coleman 2011

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

De Rooij 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Eungpinichpong 1997

Methods	
Participants	
Interventions	
Outcomes	
Notes	Record not found

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Gallego-Ariza 2013

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Gao 2023

Methods	
Participants	Knee OA recruitment 32 community volunteers Mean age 68 years, 60% female ACR criteria
Interventions	Clinic, supervised group setting, delivered entirely in person 1. Education; 24 synchronous contact occasions 2. Exercise (stretching): 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions
Outcomes	At 8 weeks (immediately postintervention): Pain (global/overall pain VAS/NRS) No physical function measure No quality-of-life measure No participant-reported treatment success meaure
Notes	Data unable to be extracted from publication; authors unresponsive to requests

Ghroubi 2008

Methods	
Participants	
Interventions	
Outcomes	
Notes	Not available in English. Google translated version difficult to understand

Gianoudis 2015

Methods	

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Gianoudis 2015 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Guerrero 2015

Methods	
Participants	
Interventions	
Outcomes	
Notes	Protocol; outcome publication not found

Hamad 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Hasan 2018

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Heggannavar 2015

Methods	

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Heggannavar 2015 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Hughes 2004

Methods	
Participants	
Interventions	
Outcomes	
Notes	Mixed sample; data for those with knee OA only unknown

Hurley 2018

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Islam 2018

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Keogan 2007

Methods	

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Keogan 2007 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Record not found

Kloek 2014

Methods	
Participants	
Interventions	
Outcomes	
Notes	Protocol; outcome publication not found

Kloek 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Lange 2009

Methods	
Participants	
Interventions	
Outcomes	
Notes	Protocol; outcome publication not found

Lee 2016

Methods	

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Lee 2016 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Lee 2017b

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Li 2017

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Loew 2015

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Manlapaz 2020

Methods	

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Manlapaz 2020 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Marcu 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Matei 2014

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Moonaz 2015

Methods	
Participants	
Interventions	
Outcomes	
Notes	Mixed sample; authors unable to provide data for those with knee OA

Morgan 2013

Methods	

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Morgan 2013 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Nct02560831

Methods	
Participants	
Interventions	
Outcomes	
Notes	Status: completed

Nct02620462

Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 17 May 2016 Unknown status Comparison: no intervention

Nct03421249

Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 5 February 2018 Status: "Completed" Comparison: placebo

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Nct04019366	
Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 15 July 2019 Status: "Complete". Publication not found Comparison: usual care

Nct04108650	
Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 11 January 2021 Status: "Completed" Comparison: no intervention

Nct04186143	
Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 29 June 2020 Status: "Completed". Publication not found Comparison: usual care

Nct04299568	
Methods	

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Nct04299568 *(Continued)*

Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 29 July 2020 Status: "Completed". Publication not found Comparison: intervention equally applied in exercise group

Nct04588558

Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 12/0/2021 Status: "Completed" Comparator: intervention equally applied in exercise group

Nejati 2015

Methods	
Participants	Knee OA recruitment 56 community volunteers with knee OA Mean age 62 years, 61% female ACR criteria
Interventions	Mix home/clinic, mix supervised/unsupervised individual setting, delivered entirely in person 1. Usual care, including electrotherapy + meds + supplements; 22 synchronous contact occasions 2. Usual care + exercise (strengthening, stretching): 84 sessions (7 x per week over 12 weeks); 6 synchronous contact occasions
Outcomes	At 12 weeks (immediately postintervention) and 52-weeks: Pain (global pain VAS) Physical function (KOOS function subscale) Quality of life (KOOS QoL subscale) No participant-reported treatment success measure
Notes	SDs not reported in publication; authors not responsive to requests

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O'Hanlon 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Obuchi 2011

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Oida 2008

Methods	
Participants	
Interventions	
Outcomes	
Notes	Not available in English. Google translated version difficult to understand/interpret

Paterson 2018

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

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Peeler 2019

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Pisters 2011

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Rahela 2012

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Raisanen 2021

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

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RBR-4zx26w

Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 4 September 2017 Status: "Analysis complete" Comparator: intervention equally applied in exercise group

Reid 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Rezende 2018

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Rosa 2012

Methods	
Participants	
Interventions	
Outcomes	
Notes	Not available in English. Google translated version difficult to understand/interpret

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Shellington 2019

Methods	
Participants	Knee OA recruitment 22 community volunteers Mean age 70 years, 68% female Diagnosis of OA (unspecified)
Interventions	Clinic, supervised group setting, delivered entirely in-person 1. Wait list; 0 synchronous contact occasions 2. Exercise (neuromuscular, proprioceptive/balance): 48 sessions (2 x 45 minutes per week over 24 weeks); 48 synchronous contact occasions
Outcomes	At 24 weeks (immediately postintervention) Pain (KOOS/WOMAC pain subscale) No physical function measure No quality-of-life measure No participant-reported treatment success meaure
Notes	Outcome data unable to be extracted from authors; unresponsive to contact

Umin000011402

Methods	
Participants	
Interventions	
Outcomes	
Notes	Trial registry; last update 7 August 2013 Registry status: "pre-initiation" Comparator: intervention equally applied in exercise group

Umin000043117

Methods	
Participants	
Interventions	
Outcomes	
Notes	Registry; last updated 25 January 2021 Status: "Completed". Publication not found Comparator: usual care

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Unal 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Walsh 2020

Methods	
Participants	
Interventions	
Outcomes	
Notes	Mixed sample; authors unable to provide data for those with knee OA only

Wang 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Wang 2020

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

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Yamada 2011

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

Yavuz 2013

Methods	
Participants	Knee OA recruitment 160 community volunteers Mean age 59 years, 77% female ACR criteria
Interventions	Clinic, supervised individual setting, delivered entirely in-person 1. Attention control (pseudoiontophoresis); 18 synchronous contact occasions 2. Exercise (strengthening): 18 sessions (3 x per week over 6 weeks); 18 synchronous contact occasions 3. Iontophoresis 4. Laser
Outcomes	At 6 weeks (immediately postintervention) Pain (global/overall pain VAS/NRS) Physical function (KOOS ADL/WOMAC function subscale) No quality-of-life measure No participant-reported treatment success meaure
Notes	Google translated but translation very difficult to understand/follow; data difficult to extract from outcome tables - therefore excluded from meta-analysis

Zaimi 2012

Methods	
Participants	
Interventions	
Outcomes	
Notes	Conference abstract; publication not found

ACR: American College of Rheumatology; **ADL:** activities of daily living; **KL:** Kellgren and Lawrence system for classification of osteoarthritis; **KOOS:** Knee Injury and Osteoarthritis Outcome Score; **NRS:** numerical rating scale; **OA:** osteoarthritis; **QoL:** quality of life; **SD:** standard deviation; **TENS:** transcutaneous electrical nerve stimulation; **VAS:** visual analogue scale; **WOMAC:** Western Ontario and McMaster Universities Osteoarthritis Index

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Characteristics of ongoing studies *[ordered by study ID]*

Actrn12615001101572

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 November 2018 Status: "Data analysis underway"

Actrn12618001494224

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 2 March 2022 Status: "Completed". Publication not found

Actrn12619001297112

Study name	
Methods	
Participants	
Interventions	
Outcomes	

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Actrn12619001297112 *(Continued)*

Starting date	
Contact information	
Notes	Registry; last updated 1 December 2020 Status: "Recruiting"

Actrn12623000780651

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Akgul 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Bilzon 2023

Study name	
Methods	

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Bilzon 2023 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Bozgeylk 1830

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Cagnin 2022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Conference abstract; publication not found

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ChiCTR1800019208	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 31 October 2018 Status: "Not yet recruiting"

ChiCTR2000039878	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 20 February 2021 Status: "Recruiting"

ChiCTR2000040769	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	

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ChiCTR2000040769 *(Continued)*

Contact information	
Notes	Registry; last updated 12 April 2021 Status: "Not yet recruiting"

ChiCTR2100046313

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 20 December 2021 Status: "Recruiting"

ChiCTR2100049907

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 17 April 2022 Status: "Not yet recruiting"

ChiCTR2100054191

Study name	
Methods	

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ChiCTR2100054191 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 13 November 2022 Status: "Not yet recruiting"

ChiCTR2300067559

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

ChiCTR2300069339

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

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ChiCTR2300071797	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

ChiCTR-IOR-14005414	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 4 November 2015 Status: "Recruiting"

ChiCTR-IPR-17013465	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 21 November 2017

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ChiCTR-IPR-17013465 *(Continued)*

	Status: "Not yet recruiting"

Ctri/2019/02/017422

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 January 2019 Status: "Not yet recruiting"

Ctri/2019/07/019989

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 11 July 2019 Status: "Not yet recruiting"

Ctri/2020/04/024415

Study name	
Methods	
Participants	
Interventions	

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Ctri/2020/04/024415 *(Continued)*

Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 20 March 2020 Status: "Not yet recruiting"

Ctri/2020/05/025097

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 14 December 2020 Status: "Recruiting"

Ctri/2021/06/034213

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 11 June 2021 Status: "Not yet recruiting"

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Ctri/2021/12/038685	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 1 November 2022 Status: "Completed"

Ctri/2022/02/040249	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 10 February 2022 Status: "Not yet recruiting"

Ctri/2022/05/042853	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	

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Ctri/2022/05/042853 *(Continued)*

Contact information	
Notes	Registry; last updated 4 September 2023 Status: "Completed"

Ctri/2022/12/047849

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 30 November 2022 Status: "Not yet recruiting"

Ctri/2023/02/049448

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Ctri/2023/06/054225

Study name	
Methods	

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Ctri/2023/06/054225 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Ctri/2023/08/056593

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Ctri/2023/10/059073

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

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Ctri/2023/11/059651	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Dieter 2022	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Conference abstract; publication not found

Drks00009677	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 5 July 2023

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Drks00009677 *(Continued)*

	Status: "Recruitment complete"

Drks00018847

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 25 April 2023 Status: "Recruitment complete". Publication not found

Drks00023269

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 8 November 2021 Status: "Study complete"

Drks00027164

Study name	
Methods	
Participants	
Interventions	

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Drks00027164 *(Continued)*

Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 22 May 2023 Status: "Complete"

Drks00028477

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 25 October 2023 Status: "Recruitment complete"

Drks00030932

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

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Fugaru 2022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Conference abstract; publication not found

Gholami 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Study protocol

Irct20101017004952n10

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 3 January 2020

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Irct20101017004952n10 *(Continued)*

	Status: "Recruitment complete"

Irct20101228005486n7

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 6 February 2020 Status: "Recruitment complete". Publication not found

Irct20150131020888n10

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 23 November 2018 Status: "Recruitment complete". Publication not found

Irct20160531028186n4

Study name	
Methods	
Participants	
Interventions	

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Irct20160531028186n4 *(Continued)*

Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 9 October 2019 Status: "Recruitment complete". Publication not found

Irct20161208031300n1

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 11 May 2018 Status: "Recruitment complete". Publication not found

Irct2017052027741n4

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 25 June 2017 Status: "Recruitment complete"

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Irct20170827035926n2	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 25 March 2018 Status: "Recruitment complete"

Irct2017100218117n2	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 19 November 2017 Status: "Recruitment complete". Publication not found

Irct20190322043099n1	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	

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Irct20190322043099n1 *(Continued)*

Contact information	
Notes	Ongoing trial

Irct20190816044538n1

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 6 December 2019 Status: "Recruitment complete". Publication not found

Irct20191221045846n2

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 June 2020 Status: "Recruitment complete"

Irct20200107046035n10

Study name	
Methods	

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Irct20200107046035n10 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 June 2022 Status: "Recruitment complete"

Irct20200204046368n8

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 18 November 2021 Status: "Recruitment complete". Publication not found

Irct20200807048331n1

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 6 February 2021 Status: "Recruitment complete". Publication not found

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Irct20210129050172n1	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 21 February 2021 Status: "Recruitment complete"

Irct20210529051427n1	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 25 July 2021 Status: "Recruitment complete". Publication not found

Irct20220206053950n2	
Study name	
Methods	
Participants	
Interventions	
Outcomes	

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Irct20220206053950n2 *(Continued)*

Starting date	
Contact information	
Notes	Registry; last updated 7 September 2022 Status: "Recruitment complete"

Irct20230124057196n1

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Irct20230216057434n3

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Isrctn12950684

Study name	
Methods	

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Isrctn12950684 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 12 September 2023 Status: "Completed"

Isrctn15564385

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 26 August 2022 Status: "Completed". Publication not found

Isrctn75983430

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 10 May 2023 Status: "Completed"

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Itmctr2200006721

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 31 October 2022 Status: "Not yet recruiting"

Jakiela 2022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Protocol; outcome publication not found

Jprn-Umin000048356

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	

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Jprn-Umin000048356 *(Continued)*

Contact information	
Notes	Registry; last updated 1 May 2023 Status: "Recruiting"

jRCTs052200069

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 18 January 2023 Status: "Recruitment complete"

Karaborklu 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Kct0006170

Study name	
Methods	

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Kct0006170 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 July 2022 Status: "Completed"

Kong 2022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Conference abstract; publication not found

Kong 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

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Kumar 2022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Conference abstract; publication not found

Moutzouri 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct02387034

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	

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Nct02387034 *(Continued)*

Notes	Registry; last updated 20 June 2017 Status: "Recruiting"

Nct02748291

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 11 July 2022 Status: "Completed"

Nct03064139

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 27 July 2021 Status: "Recruitment suspended"

Nct03363490

Study name	
Methods	
Participants	

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Nct03363490 *(Continued)*

Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 6 April 2018 Status: "Recruiting"

Nct03431298

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 August 2018 Status: "Recruiting"

Nct03920020

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 April 2023 Status: "Completed". Publication not found

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Nct04099017	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 10 May 2022 Status: "Completed"

Nct04154059	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 17 May 2022 Status: "Suspended due to COVID"

Nct04308967	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	

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Nct04308967 *(Continued)*

Contact information	
Notes	Trial registry; last update 12 October 2022 Marked as complete; publication not found

Nct04325334

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Being conducted in Vancouver, BC, Canada Registered in 2020

Nct04362618

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 30 May 2023 Status: "Recruiting"

Nct04615377

Study name	
Methods	

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Nct04615377 *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 16 August 2022 Status: "Completed"

Nct04645732

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 17 October 2023 Status: "Active, not recruiting"

Nct04669834

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 21 September 2022 Status: "Completed"

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Nct04678609	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 20 April 2021 Status: "Recruiting"

Nct04697667	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 15 February 2022 Status: "Completed". Publication not found

Nct04767854	
Study name	
Methods	
Participants	
Interventions	
Outcomes	

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Nct04767854 *(Continued)*

Starting date	
Contact information	
Notes	Registry; last updated 3 February 2023 Status: "Recruiting"

Nct04909086

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 30 September 2022 Status: "Recruiting"

Nct05197010

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 19 January 2022 Status: "Recruiting"

Nct05272124

Study name	

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Nct05272124 *(Continued)*

Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 9 March 2022 Status: "Completed"

Nct05272748

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 19 April 2022 Status: "Recruiting"

Nct05287802

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 18 March 2022

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Nct05287802 *(Continued)*	
	Status: "Completed". Publication not found

Nct05319964	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 30 January 2023 Status: "Recruiting"

Nct05334524	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 20 December 2022 Status: "Completed"

Nct05518890	
Study name	
Methods	
Participants	
Interventions	

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Nct05518890 *(Continued)*

Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 29 August 2022 Status: "Recruiting"

Nct05595317

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 7 September 2023 Status: "Completed"

Nct05670314

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 4 January 2023 Status: "Recruiting"

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Nct05827003	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct05836792	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct05839262	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

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Nct05855681	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct05857605	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct05997862	
Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	

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Nct05997862 *(Continued)*

Notes	Ongoing trial

Nct05999110

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct06010394

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Nct06084949

Study name	
Methods	
Participants	
Interventions	
Outcomes	

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Nct06084949 *(Continued)*

Starting date	
Contact information	
Notes	Ongoing trial

Nct06130774

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Ng 2022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	The protocol was registered on 22 December 2020.

Nicolson 2023

Study name	
Methods	
Participants	

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Nicolson 2023 *(Continued)*

Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Pactr202108619222230

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 6 November 2022 Status: "Recruitment complete"

RBR-29jhy2r

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 17 January 2022 Status: "Recruitment complete"

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RBR-2fkzb2s

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 12 July 2022 Status: "Data analysis complete"

RBR-2hxj58

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 18 August 2015 Status: "Data analysis complete"

RBR-4kryg4s

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	

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RBR-4kryg4s *(Continued)*

Contact information	
Notes	Registry; last updated 12 January 2023 Status: "Recruitment complete"

RBR-53trrwk

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

RBR-5g9n5g

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 2 February 2018 Status: "Recruitment complete". Publication not found

RBR-5zpphs

Study name	
Methods	

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RBR-5zpphs *(Continued)*

Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 16 February 2016 Status: "Recruiting"

RBR-747wrkx

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 20 December 2021 Status: "Recruitment complete"

RBR-7s6ymc

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 28 July 2015 Status: "Not yet recruiting"

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Slctr/2022/022

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 3 April 2023 Status: "Recruiting"

Tctr20230130003

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 30 January 2023 Status: "Recruitment complete"

Tctr20230130005

Study name	
Methods	
Participants	
Interventions	
Outcomes	

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Tctr20230130005 *(Continued)*

Starting date	
Contact information	
Notes	Registry; last updated 30 January 2023 Status: "Study complete". Publication not found

Umin000039723

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Registry; last updated 8 March 2021 Status: "Recruiting"

Wang 2022a

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Protocol; outcome publication not found

Wang 2022b

Study name	

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Wang 2022b *(Continued)*

Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Protocol; outcome publication not found

Weber 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Xie 2018

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Protocol; outcome publication not found

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Yang 2023

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Ongoing trial

Zheng 2021

Study name	
Methods	
Participants	
Interventions	
Outcomes	
Starting date	
Contact information	
Notes	Protocol; outcome publication not found

DATA AND ANALYSES

Comparison 1. Primary time point

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Knee pain severity immediately post-intervention	131		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
1.1.1 A) Exercise vs attention control/placebo	28	2873	Std. Mean Difference (IV, Random, 95% CI)	-0.58 [-0.78, -0.38]

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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1.2 B) Exercise vs no treatment/usual care/limited education	56	4184	Std. Mean Difference (IV, Random, 95% CI)	-0.71 [-0.86, -0.56]
1.1.3 C) Exercise when added to another co-intervention	47	4441	Std. Mean Difference (IV, Random, 95% CI)	-0.75 [-0.92, -0.58]
1.2 Self-reported physical function immediately post-intervention	122		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
1.2.1 A) Exercise vs attention control/placebo	24	2536	Std. Mean Difference (IV, Random, 95% CI)	-0.59 [-0.79, -0.40]
1.2.2 B) Exercise vs no treatment/usual care/limited education	54	4245	Std. Mean Difference (IV, Random, 95% CI)	-0.72 [-0.88, -0.56]
1.2.3 C) Exercise when added to another co-intervention	44	4381	Std. Mean Difference (IV, Random, 95% CI)	-0.71 [-0.88, -0.55]
1.3 Quality of life immediately post-intervention	46		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
1.3.1 A) Exercise vs attention control/placebo	6	454	Std. Mean Difference (IV, Random, 95% CI)	0.47 [-0.01, 0.95]
1.3.2 B) Exercise vs no treatment/usual care/limited education	28	2328	Std. Mean Difference (IV, Random, 95% CI)	0.37 [0.22, 0.52]
1.3.3 C) Exercise when added to another co-intervention	12	1660	Std. Mean Difference (IV, Random, 95% CI)	0.31 [0.10, 0.52]
1.4 Participant-reported treatment success immediately post-intervention	11		Risk Ratio (IV, Random, 95% CI)	Subtotals only
1.4.1 A) Exercise vs attention control/placebo	2	364	Risk Ratio (IV, Random, 95% CI)	1.46 [1.11, 1.92]
1.4.2 B) Exercise vs no treatment/usual care/limited education	3	405	Risk Ratio (IV, Random, 95% CI)	1.33 [0.71, 2.49]
1.4.3 C) Exercise when added to another co-intervention	6	1139	Risk Ratio (IV, Random, 95% CI)	1.63 [1.18, 2.24]
1.5 Study withdrawals at final follow-up	123		Risk Ratio (IV, Random, 95% CI)	Subtotals only
1.5.1 A) Exercise vs attention control/placebo	29	2907	Risk Ratio (IV, Random, 95% CI)	1.08 [0.92, 1.26]
1.5.2 B) Exercise vs no treatment/usual care/limited education	53	4301	Risk Ratio (IV, Random, 95% CI)	1.03 [0.88, 1.20]
1.5.3 C) Exercise when added to another co-intervention	41	3502	Risk Ratio (IV, Random, 95% CI)	0.82 [0.70, 0.97]

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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.6 Adverse events at final follow-up	48		Risk Ratio (IV, Random, 95% CI)	Subtotals only
1.6.1 A) Exercise vs attention control/placebo	11	1684	Risk Ratio (IV, Random, 95% CI)	2.02 [0.62, 6.58]
1.6.2 B) Exercise vs no treatment/usual care/limited education	18	1557	Risk Ratio (IV, Random, 95% CI)	3.17 [1.17, 8.57]
1.6.3 C) Exercise when added to another co-intervention	19	2187	Risk Ratio (IV, Random, 95% CI)	1.72 [1.07, 2.76]

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Analysis 1.1. Comparison 1: Primary time point, Outcome 1: Knee pain severity immediately post-intervention

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
1.1.1 A) Exercise vs attention control/placebo								
Baker 2001	128	95.968751	23	189	112.156251	23	3.4%	-0.57 [-1.17 , 0.02]
Bandak 2022	-10	15.149257	102	-7.3	15.297059	104	4.4%	-0.18 [-0.45 , 0.10]
Brismée 2007	2.41	2.05	22	3.37	1.78	19	3.3%	-0.49 [-1.11 , 0.14]
Cheung 2017	4.56	2.01423	60	5.2	1.85	23	3.7%	-0.32 [-0.80 , 0.16]
De Almeida 2020	1.78	1.474794	44	6.5	1.67	22	2.9%	-3.03 [-3.76 , -2.29]
Ettinger 1997	2.175241	0.665635	290	2.46	0.610328	149	4.6%	-0.44 [-0.64 , -0.24]
Foley 2003	8	3.86	35	10	3.09	35	3.8%	-0.57 [-1.04 , -0.09]
Imoto 2012	4.27	2.45	50	5.74	3.14	50	4.0%	-0.52 [-0.92 , -0.12]
Jenkinson 2009	5.7	3.96	82	7.04	4.21	76	4.3%	-0.33 [-0.64 , -0.01]
Kang 2022	8.58	3.75	15	7.87	3.02	15	3.0%	0.20 [-0.51 , 0.92]
Kim 2013	33.77	21.91	37	37.86	22.58	37	3.8%	-0.18 [-0.64 , 0.27]
Kovar 1992	3.77	1.73	52	4.77	2.12	51	4.0%	-0.51 [-0.91 , -0.12]
Kuntz 2018	-67.838095	13.166567	21	-49.9	24.7	10	2.7%	-0.99 [-1.79 , -0.20]
Lai 2021	2.41	2.1	27	2.74	2.16	27	3.6%	-0.15 [-0.69 , 0.38]
Liu 2019	-9.2	9.734129	105	-0.9	3.2	35	4.0%	-0.96 [-1.36 , -0.56]
Lu 2017	5.15	3.24	23	9.37	5.23	23	3.3%	-0.95 [-1.57 , -0.34]
Maurer 1999	-43.54	80.3	56	-28.49	80.3	57	4.1%	-0.19 [-0.56 , 0.18]
Messier 2021	4.700791	3.300649	253	4.8	2.9	124	4.5%	-0.03 [-0.25 , 0.18]
Peloquin 1999	3.09	1.54	69	3.94	2.22	68	4.2%	-0.44 [-0.78 , -0.10]
Rego 2023	2.75	3.01	9	10.11	3.55	9	1.7%	-2.13 [-3.34 , -0.92]
Schlenk 2020	4.25	4.801688	91	4.54	2.400844	91	4.4%	-0.08 [-0.37 , 0.21]
Shen 2022	2.6	2.1	18	6.9	1.7	18	2.6%	-2.20 [-3.05 , -1.35]
Simao 2012	159.19	148.442	11	175	125.79	12	2.7%	-0.11 [-0.93 , 0.71]
Song 2003	-2.45	3.9	38	0.61	5.1	34	3.8%	-0.67 [-1.15 , -0.20]
Song 2020	0.55	0.48	18	3.21	1.18	18	2.3%	-2.89 [-3.85 , -1.92]
Tsai 2013	-2.6	2.540234	28	-1.02	1.681047	27	3.5%	-0.72 [-1.27 , -0.17]
Vincent 2020	1.250345	1.630087	58	0.47	0.8	29	3.8%	0.55 [0.09 , 1.00]
Ye 2020	3.68	6.07	25	4.92	1.41	25	3.5%	-0.28 [-0.83 , 0.28]
Subtotal			**1662**			**1211**	**100.0%**	**-0.58 [-0.78 , -0.38]**
Test for overall effect: Z = 5.64 (P < 0.00001)								
Heterogeneity: Tau² = 0.22; Chi² = 158.52, df = 27 (P < 0.00001); I² = 83%								
1.1.2 B) Exercise vs no treatment/usual care/limited education								
Allen 2018	-1.59	3.405658	142	-0.66	3.098513	68	2.3%	-0.28 [-0.57 , 0.01]
An 2008	71.1	110.1	14	138.2	112.6	14	1.5%	-0.59 [-1.34 , 0.17]
Bennell 2010	2.6	2.1	45	3.9	2.6	44	2.1%	-0.55 [-0.97 , -0.12]
Braghin 2018	8	8.52	15	20.94	23.47	16	1.6%	-0.70 [-1.43 , 0.02]
Bruce-Brand 2012	10.78	4.31	14	8.33	4.36	13	1.5%	0.55 [-0.22 , 1.32]
Chen 2014	4.2	0.9	30	5.2	1.1	30	1.9%	-0.98 [-1.52 , -0.44]
Cheung 2014	5.8	2.842569	18	8.3	2.842569	18	1.6%	-0.86 [-1.55 , -0.17]
Chopp-Hurley 2017	-81	24.5	7	-70.8	15.5	11	1.2%	-0.50 [-1.47 , 0.46]
Christensen 2015	-5.6	19.816442	64	-5.5	20.016608	64	2.2%	-0.00 [-0.35 , 0.34]
Da Silva 2015	2.6	1.55	19	4	1.56	22	1.7%	-0.88 [-1.53 , -0.24]
De Matos 2019	8	10.99	28	22.33	23.59	28	1.9%	-0.77 [-1.31 , -0.22]
De Rooij 2017	4.3	2	63	5.8	2.2	63	2.2%	-0.71 [-1.07 , -0.35]
DeVita 2018	3.64	2.55	16	7.33	4.72	15	1.5%	-0.96 [-1.71 , -0.21]
Drummen 2023	16.6	20.085816	24	56.4	17.2	16	1.5%	-2.05 [-2.84 , -1.26]
Fransen 2001	-10.6	19.5	83	1.5	19.4	43	2.2%	-0.62 [-0.99 , -0.24]
Fransen 2007	-1.67	3.28	56	-0.5	2.37	41	2.1%	-0.40 [-0.80 , 0.01]
Gohir 2021	-1.8	2.254845	67	-0.3	2.232264	79	2.2%	-0.67 [-1.00 , -0.33]
Guo 2022	3.52	1.71	56	3.75	1.89	55	2.2%	-0.13 [-0.50 , 0.25]
Gur 2002	-20.9	8.3	17	0.7	4.6	6	0.9%	-2.74 [-4.02 , -1.47]
Hay 2006	7.36	4.3	109	8.99	3.7	108	2.3%	-0.40 [-0.67 , -0.14]
Henriksen 2014	-6.1	10.08716	31	0.7	10.252914	29	1.9%	-0.66 [-1.18 , -0.14]
Hopman-Rock 2000	-0.7	24.1	45	4	21.2	37	2.1%	-0.20 [-0.64 , 0.23]
Hu 2020	2.53	1.61	52	3.6	1.6	40	2.1%	-0.66 [-1.08 , -0.24]
Janyacharoen 2018	-69.2	5.1	20	-43.1	1.2	20	0.6%	-6.91 [-8.62 , -5.20]
Jorge 2015	4.3	3.1	29	6.6	1.5	31	1.9%	-0.94 [-1.48 , -0.41]
Karadag 2019	4.33	0.8	17	7.64	0.61	18	0.9%	-4.56 [-5.88 , -3.25]
Keefe 2004	3.19	1.85	16	4.03	2.08	18	1.6%	-0.42 [-1.10 , 0.27]
Krishnan 2021	-53.93	3.17	15	-50.93	6.04	15	1.6%	-0.61 [-1.34 , 0.13]
Lee 2009	4.6	4	29	5.9	3.7	15	1.7%	-0.33 [-0.95 , 0.30]
Lee 2023	34.5	21.69	18	43.9	17.85	18	1.7%	-0.46 [-1.13 , 0.20]
Li 2020	-73.1	15.3	26	-65.9	15.6	25	1.9%	-0.46 [-1.02 , 0.10]
Lim 2008	-9	12	53	-1.7	12.8	54	2.1%	-0.58 [-0.97 , -0.20]
Lin 2009	4.2	3	36	7.3	3.4	36	2.0%	-0.96 [-1.45 , -0.47]

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Analysis 1.1. (Continued)

Lim 2008	-9	12	53	-1.7	12.8	54	2.1%	-0.58 [-0.97 , -0.20]
Lin 2009	4.2	3	36	7.3	3.4	36	2.0%	-0.96 [-1.45 , -0.47]
Lund 2008	-51.5	20.5	25	-58.1	20.78461	27	1.9%	0.31 [-0.23 , 0.86]
Mazloum 2018	8.739394	1.950952	33	10.5	1.8	16	1.7%	-0.91 [-1.54 , -0.28]
Multanen 2014	-2	14.070612	40	0	7.817007	40	2.1%	-0.17 [-0.61 , 0.27]
O'Reilly 1999	-1.45	2.616812	78	0.42	3.594579	113	2.3%	-0.58 [-0.87 , -0.28]
Øiestad 2023	3.801786	2.51396	112	4.5	2.4	56	2.2%	-0.28 [-0.60 , 0.04]
Pazit 2018	107.15	126.818349	20	249.7	309.3	10	1.5%	-0.68 [-1.46 , 0.10]
Robbins 2022	-2.59	2.35	43	-0.27	2.1	43	2.0%	-1.03 [-1.48 , -0.58]
Rogind 1998	4.2	2.1	12	6	2.1	13	1.4%	-0.83 [-1.65 , -0.01]
Rosedale 2014	16	10	120	23	9	60	2.2%	-0.72 [-1.04 , -0.40]
Sadeghi 2023	35.133333	22.117119	72	67	21.22	24	1.9%	-1.44 [-1.95 , -0.94]
Salacinski 2012	-81.4	13.4	19	-65.7	15.9	18	1.6%	-1.05 [-1.74 , -0.35]
Samut 2015	3.160655	2.285789	29	6.192	2.393	13	1.6%	-1.28 [-2.00 , -0.57]
Schilke 1996	9.7	4.72	10	10.1	6.44	10	1.4%	-0.07 [-0.94 , 0.81]
Sekir 2005	1.9	2.47	12	4.1	2.63	10	1.3%	-0.83 [-1.71 , 0.05]
Shahine 2020	17.5	18.1	33	77	17.3	33	1.5%	-3.32 [-4.08 , -2.56]
Takacs 2017	2.8	1.7	20	4.6	2.3	20	1.7%	-0.87 [-1.52 , -0.22]
Thorstensson 2005	-1.8	13.390246	30	0.3	16.221243	31	1.9%	-0.14 [-0.64 , 0.36]
Topp 2002	10.552388	3.131421	67	10.77	3.194683	35	2.1%	-0.07 [-0.48 , 0.34]
Vassao 2021	3.36	2.87	14	7	4.27	14	1.5%	-0.97 [-1.76 , -0.18]
Wallis 2017	6.5	1.2	23	6.1	1.7	23	1.8%	0.27 [-0.31 , 0.85]
Wang 2011	-76	15	28	-68	18	28	1.9%	-0.48 [-1.01 , 0.06]
Wortley 2013	106	107.80506	30	157	96	9	1.5%	-0.47 [-1.23 , 0.28]
Xiao 2021	75.7	20.19	142	94.25	27.73	142	2.3%	-0.76 [-1.00 , -0.52]
Subtotal			**2286**			**1898**	**100.0%**	**-0.71 [-0.86 , -0.56]**
Test for overall effect: Z = 9.25 (P < 0.00001)								
Heterogeneity: Tau² = 0.24; Chi² = 270.28, df = 55 (P < 0.00001); I² = 80%								
1.1.3 C) Exercise when added to another co-intervention								
Alghadir 2019	4.25	1.242176	46	4.8	0.8	22	2.2%	-0.49 [-1.00 , 0.03]
Anadkat 2015	2.75	0.96	20	4.3	1.12	20	1.9%	-1.46 [-2.16 , -0.75]
Anwer 2014	1.24	0.76	21	4.24	0.88	21	1.4%	-3.58 [-4.59 , -2.57]
Assar 2020	3.4	2.2	12	6.7	2.5	12	1.6%	-1.35 [-2.26 , -0.45]
Bautch 1997	2.71	2.680019	17	3.51	3.009867	17	1.9%	-0.27 [-0.95 , 0.40]
Bennell 2016	26.4	18.4	73	33.2	22.3	74	2.5%	-0.33 [-0.66 , -0.01]
Bennell 2022a	3.3	1.9	172	4.2	2.1	68	2.6%	-0.46 [-0.74 , -0.17]
Bennell 2022b	4.2	2.4	107	4.9	2.4	105	2.6%	-0.29 [-0.56 , -0.02]
Bhagat 2017	1.933	0.8837	15	3.8	1.612	15	1.7%	-1.40 [-2.21 , -0.59]
Chang 2012	2	0.9	30	3.7	1.5	30	2.1%	-1.36 [-1.92 , -0.79]
Chen 2021	1.56	0.63	16	2.25	0.86	16	1.8%	-0.89 [-1.62 , -0.16]
Dincer 2016	1.7	0.9	19	3.6	2.8	16	1.9%	-0.93 [-1.63 , -0.22]
Dwyer 2015	97.7	86.780643	28	102.3	88.854206	27	2.2%	-0.05 [-0.58 , 0.48]
Gondhalekar 2013	3.53	1.33	15	4.07	1.18	15	1.9%	-0.42 [-1.14 , 0.31]
Hernandez 2019	2.42	2.35	53	4	2.83	60	2.4%	-0.60 [-0.98 , -0.22]
Hinman 2020	3.5	2.1	87	4.2	2.2	88	2.6%	-0.32 [-0.62 , -0.03]
Holm 2020	58.5	14.312663	45	61.2	13.314105	45	2.4%	-0.19 [-0.61 , 0.22]
Hsu 2021	4.48	1.44	22	4.33	1.65	22	2.1%	0.10 [-0.50 , 0.69]
Huang 2003	3.1	0.957143	99	4.4	0.4	33	2.4%	-1.51 [-1.95 , -1.08]
Huang 2005	4.1	0.6	35	4.9	1.2	35	2.3%	-0.83 [-1.32 , -0.34]
Jan 2008	4.8	3.102286	68	7.1	3.4	34	2.4%	-0.71 [-1.14 , -0.29]
Joshi 2019	2.85	0.88	21	5.2	1.17	21	1.8%	-2.23 [-3.01 , -1.44]
Kilic 2020	3.76	1.27	25	5.88	1.51	25	2.0%	-1.50 [-2.13 , -0.86]
Knoop 2013	2.8	2.1	80	3.3	2.1	79	2.5%	-0.24 [-0.55 , 0.08]
Kumar 2013	2.18	0.66	22	2.91	0.81	22	2.0%	-0.97 [-1.60 , -0.34]
Lee 2019	2.8	0.8	15	3.1	0.9	15	1.9%	-0.34 [-1.06 , 0.38]
Marconcin 2018	68.2	17.4	40	67.4	18.2	40	2.3%	0.04 [-0.39 , 0.48]
Mehwish 2023	5.77	1.2	22	5.24	1.09	22	2.1%	0.45 [-0.15 , 1.05]
Messier 2013	3.7	3.431956	152	4.8	3.431956	152	2.6%	-0.32 [-0.55 , -0.09]
Mete 2022	9.14	3.02	32	18.85	2.42	32	1.7%	-3.51 [-4.30 , -2.71]
Nahayatbin 2018	-69.94	12.471173	32	-53.06	9.36	16	1.9%	-1.44 [-2.11 , -0.77]
Nelligan 2021	3.5	2.2	103	5	2.4	103	2.6%	-0.65 [-0.93 , -0.37]
Oh 2020	5.06	4.39	40	10.33	5.22	20	2.1%	-1.11 [-1.69 , -0.54]
Ozlu 2023	4.11	1.34	41	5.05	1.43	41	2.3%	-0.67 [-1.12 , -0.23]
Qiu 2023	3.17	1.58	21	3.5	1.86	21	2.1%	-0.19 [-0.79 , 0.42]
Rafiq 2021b	7.785	2.377021	76	8.87	2.8	38	2.4%	-0.43 [-0.82 , -0.03]
Rafiq 2021c	8.85	1.49	25	9.52	2.08	25	2.1%	-0.36 [-0.92 , 0.19]
Rakhshani 2022	1.89	1.02	18	2.06	1	18	2.0%	-0.16 [-0.82 , 0.49]
Rosadi 2023	4.246154	0.818159	26	5.2	0.6	13	1.9%	-1.24 [-1.96 , -0.51]
Salli 2010	3.35	1.868782	50	6.5	1.8	25	2.1%	-1.69 [-2.24 , -1.13]
Samuel 2014	7.62	2.24	27	12.42	2.91	26	2.1%	-3.21 [-3.89 , -1.62]

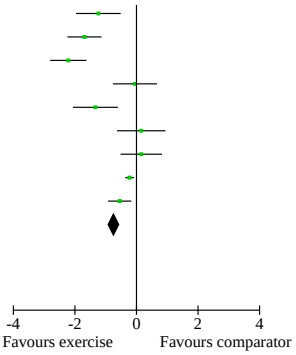
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Analysis 1.1. (Continued)

Rosadi 2023	4.246154	0.818159	26	5.2	0.6	13	1.9%	-1.24 [-1.96 , -0.51]
Salli 2010	3.35	1.868782	50	6.5	1.8	25	2.1%	-1.69 [-2.24 , -1.13]
Samuel 2014	7.62	2.24	37	13.42	2.91	36	2.1%	-2.21 [-2.80 , -1.62]
Sedaghatnezhad 2021	17.56	11.49	15	18.22	13.96	15	1.9%	-0.05 [-0.77 , 0.67]
Shabbir 2022	2.78	0.94	18	4.22	1.16	18	1.8%	-1.33 [-2.06 , -0.60]
Shin 2017	3.08	2.36	12	2.75	1.76	13	1.8%	0.15 [-0.63 , 0.94]
Talbot 2003	1.35	0.93	17	1.2	0.95	17	1.9%	0.16 [-0.52 , 0.83]
Thomas 2002	-1.27	3.6	470	-0.46	3.6	316	2.7%	-0.22 [-0.37 , -0.08]
Van Baar 1998	-27.4	28.7	54	-11.7	28.5	59	2.4%	-0.55 [-0.92 , -0.17]
Subtotal			**2464**			**1977**	**100.0%**	**-0.75 [-0.92 , -0.58]**

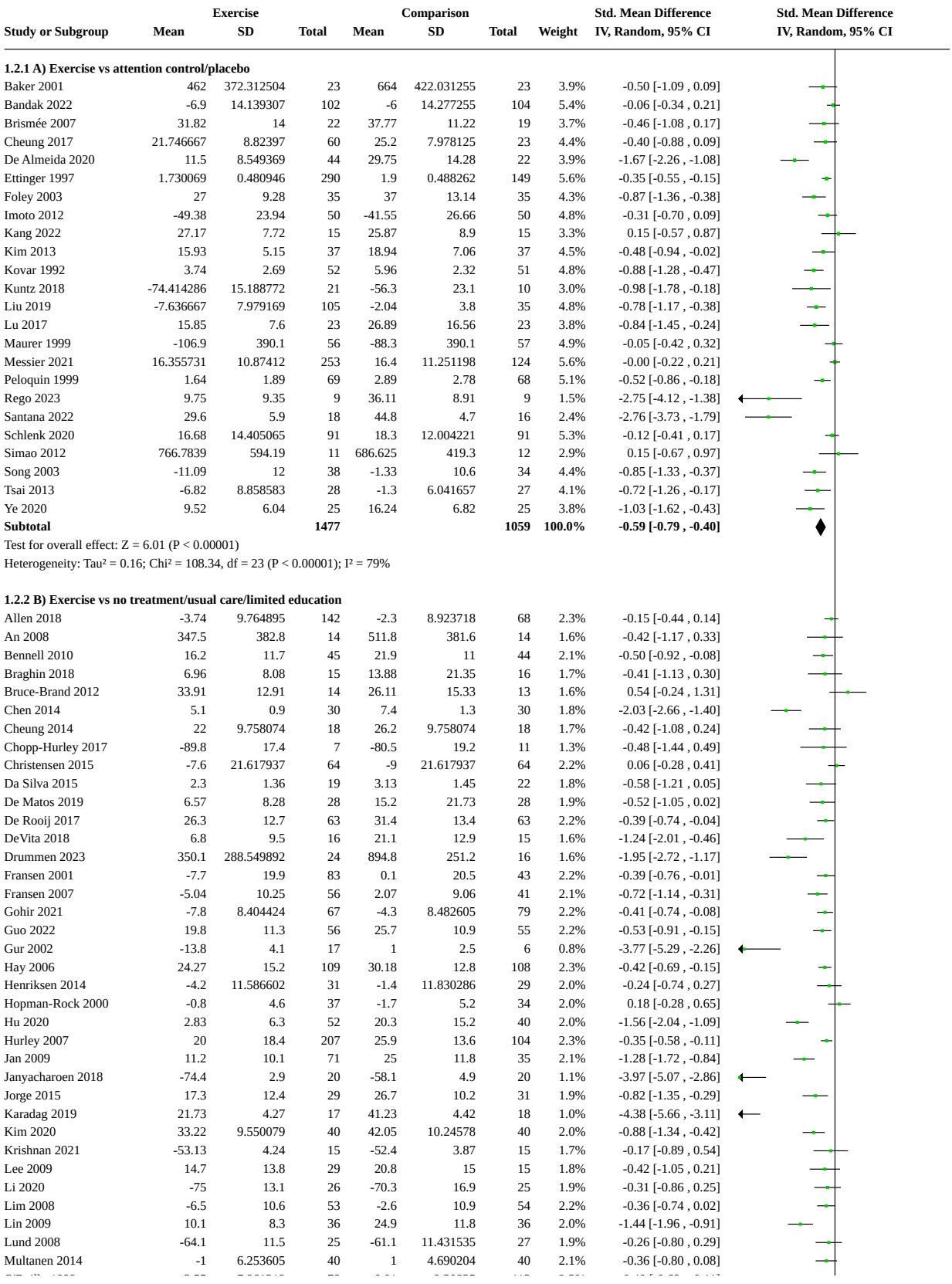
Test for overall effect: Z = 8.58 (P < 0.00001)
Heterogeneity: Tau² = 0.28; Chi² = 303.03, df = 46 (P < 0.00001); I² = 85%

Test for subgroup differences: Chi² = 1.76, df = 2 (P = 0.41), I² = 0%



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Analysis 1.2. Comparison 1: Primary time point, Outcome 2: Self-reported physical function immediately post-intervention

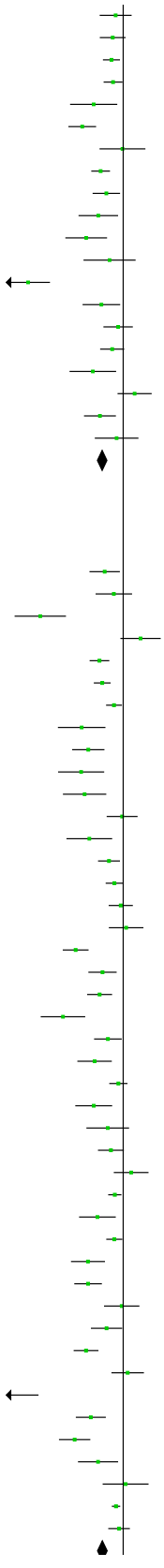


Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
1.2.1 A) Exercise vs attention control/placebo								
Baker 2001	462	372.312504	23	664	422.031255	23	3.9%	-0.50 [-1.09 , 0.09]
Bandak 2022	-6.9	14.139307	102	-6	14.277255	104	5.4%	-0.06 [-0.34 , 0.21]
Brismée 2007	31.82	14	22	37.77	11.22	19	3.7%	-0.46 [-1.08 , 0.17]
Cheung 2017	21.746667	8.82397	60	25.2	7.978125	23	4.4%	-0.40 [-0.88 , 0.09]
De Almeida 2020	11.5	8.549369	44	29.75	14.28	22	3.9%	-1.67 [-2.26 , -1.08]
Ettinger 1997	1.730069	0.480946	290	1.9	0.488262	149	5.6%	-0.35 [-0.55 , -0.15]
Foley 2003	27	9.28	35	37	13.14	35	4.3%	-0.87 [-1.36 , -0.38]
Imoto 2012	-49.38	23.94	50	-41.55	26.66	50	4.8%	-0.31 [-0.70 , 0.09]
Kang 2022	27.17	7.72	15	25.87	8.9	15	3.3%	0.15 [-0.57 , 0.87]
Kim 2013	15.93	5.15	37	18.94	7.06	37	4.5%	-0.48 [-0.94 , -0.02]
Kovar 1992	3.74	2.69	52	5.96	2.32	51	4.8%	-0.88 [-1.28 , -0.47]
Kuntz 2018	-74.414286	15.188772	21	-56.3	23.1	10	3.0%	-0.98 [-1.78 , -0.18]
Liu 2019	-7.636667	7.979169	105	-2.04	3.8	35	4.8%	-0.78 [-1.17 , -0.38]
Lu 2017	15.85	7.6	23	26.89	16.56	23	3.8%	-0.84 [-1.45 , -0.24]
Maurer 1999	-106.9	390.1	56	-88.3	390.1	57	4.9%	-0.05 [-0.42 , 0.32]
Messier 2021	16.355731	10.87412	253	16.4	11.251198	124	5.6%	-0.00 [-0.22 , 0.21]
Peloquin 1999	1.64	1.89	69	2.89	2.78	68	5.1%	-0.52 [-0.86 , -0.18]
Rego 2023	9.75	9.35	9	36.11	8.91	9	1.5%	-2.75 [-4.12 , -1.38]
Santana 2022	29.6	5.9	18	44.8	4.7	16	2.4%	-2.76 [-3.73 , -1.79]
Schlenk 2020	16.68	14.405065	91	18.3	12.004221	91	5.3%	-0.12 [-0.41 , 0.17]
Simao 2012	766.7839	594.19	11	686.625	419.3	12	2.9%	0.15 [-0.67 , 0.97]
Song 2003	-11.09	12	38	-1.33	10.6	34	4.4%	-0.85 [-1.33 , -0.37]
Tsai 2013	-6.82	8.858583	28	-1.3	6.041657	27	4.1%	-0.72 [-1.26 , -0.17]
Ye 2020	9.52	6.04	25	16.24	6.82	25	3.8%	-1.03 [-1.62 , -0.43]
Subtotal			**1477**			**1059**	**100.0%**	**-0.59 [-0.79 , -0.40]**
Test for overall effect: Z = 6.01 (P < 0.00001)								
Heterogeneity: Tau² = 0.16; Chi² = 108.34, df = 23 (P < 0.00001); I² = 79%								
1.2.2 B) Exercise vs no treatment/usual care/limited education								
Allen 2018	-3.74	9.764895	142	-2.3	8.923718	68	2.3%	-0.15 [-0.44 , 0.14]
An 2008	347.5	382.8	14	511.8	381.6	14	1.6%	-0.42 [-1.17 , 0.33]
Bennell 2010	16.2	11.7	45	21.9	11	44	2.1%	-0.50 [-0.92 , -0.08]
Braghin 2018	6.96	8.08	15	13.88	21.35	16	1.7%	-0.41 [-1.13 , 0.30]
Bruce-Brand 2012	33.91	12.91	14	26.11	15.33	13	1.6%	0.54 [-0.24 , 1.31]
Chen 2014	5.1	0.9	30	7.4	1.3	30	1.8%	-2.03 [-2.66 , -1.40]
Cheung 2014	22	9.758074	18	26.2	9.758074	18	1.7%	-0.42 [-1.08 , 0.24]
Chopp-Hurley 2017	-89.8	17.4	7	-80.5	19.2	11	1.3%	-0.48 [-1.44 , 0.49]
Christensen 2015	-7.6	21.617937	64	-9	21.617937	64	2.2%	0.06 [-0.28 , 0.41]
Da Silva 2015	2.3	1.36	19	3.13	1.45	22	1.8%	-0.58 [-1.21 , 0.05]
De Matos 2019	6.57	8.28	28	15.2	21.73	28	1.9%	-0.52 [-1.05 , 0.02]
De Rooij 2017	26.3	12.7	63	31.4	13.4	63	2.2%	-0.39 [-0.74 , -0.04]
DeVita 2018	6.8	9.5	16	21.1	12.9	15	1.6%	-1.24 [-2.01 , -0.46]
Drummen 2023	350.1	288.549892	24	894.8	251.2	16	1.6%	-1.95 [-2.72 , -1.17]
Fransen 2001	-7.7	19.9	83	0.1	20.5	43	2.2%	-0.39 [-0.76 , -0.01]
Fransen 2007	-5.04	10.25	56	2.07	9.06	41	2.1%	-0.72 [-1.14 , -0.31]
Gohir 2021	-7.8	8.404424	67	-4.3	8.482605	79	2.2%	-0.41 [-0.74 , -0.08]
Guo 2022	19.8	11.3	56	25.7	10.9	55	2.2%	-0.53 [-0.91 , -0.15]
Gur 2002	-13.8	4.1	17	1	2.5	6	0.8%	-3.77 [-5.29 , -2.26]
Hay 2006	24.27	15.2	109	30.18	12.8	108	2.3%	-0.42 [-0.69 , -0.15]
Henriksen 2014	-4.2	11.586602	31	-1.4	11.830286	29	2.0%	-0.24 [-0.74 , 0.27]
Hopman-Rock 2000	-0.8	4.6	37	-1.7	5.2	34	2.0%	0.18 [-0.28 , 0.65]
Hu 2020	2.83	6.3	52	20.3	15.2	40	2.0%	-1.56 [-2.04 , -1.09]
Hurley 2007	20	18.4	207	25.9	13.6	104	2.3%	-0.35 [-0.58 , -0.11]
Jan 2009	11.2	10.1	71	25	11.8	35	2.1%	-1.28 [-1.72 , -0.84]
Janyacharoen 2018	-74.4	2.9	20	-58.1	4.9	20	1.1%	-3.97 [-5.07 , -2.86]
Jorge 2015	17.3	12.4	29	26.7	10.2	31	1.9%	-0.82 [-1.35 , -0.29]
Karadag 2019	21.73	4.27	17	41.23	4.42	18	1.0%	-4.38 [-5.66 , -3.11]
Kim 2020	33.22	9.550079	40	42.05	10.24578	40	2.0%	-0.88 [-1.34 , -0.42]
Krishnan 2021	-53.13	4.24	15	-52.4	3.87	15	1.7%	-0.17 [-0.89 , 0.54]
Lee 2009	14.7	13.8	29	20.8	15	15	1.8%	-0.42 [-1.05 , 0.21]
Li 2020	-75	13.1	26	-70.3	16.9	25	1.9%	-0.31 [-0.86 , 0.25]
Lim 2008	-6.5	10.6	53	-2.6	10.9	54	2.2%	-0.36 [-0.74 , 0.02]
Lin 2009	10.1	8.3	36	24.9	11.8	36	2.0%	-1.44 [-1.96 , -0.91]
Lund 2008	-64.1	11.5	25	-61.1	11.431535	27	1.9%	-0.26 [-0.80 , 0.29]
Multanen 2014	-1	6.253605	40	1	4.690204	40	2.1%	-0.36 [-0.80 , 0.08]

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Analysis 1.2. (Continued)

Lund 2008	-64.1	11.5	25	-61.1	11.431535	27	1.9%	-0.26 [-0.80 , 0.29]
Multanen 2014	-1	6.253605	40	1	4.690204	40	2.1%	-0.36 [-0.80 , 0.08]
O'Reilly 1999	-3.55	7.961319	78	-0.01	9.30835	113	2.3%	-0.40 [-0.69 , -0.11]
Øiestad 2023	-71	23.891647	112	-63	21	56	2.2%	-0.35 [-0.67 , -0.02]
Pazit 2018	39.85	43.828049	20	88	51.4	10	1.5%	-1.01 [-1.82 , -0.20]
Robbins 2022	-4.63	3.86	43	0.22	2.94	43	2.0%	-1.40 [-1.87 , -0.93]
Rogind 1998	11.3	3.6	12	11.4	4.3	13	1.6%	-0.02 [-0.81 , 0.76]
Rosedale 2014	-65	20	120	-50	18	60	2.2%	-0.77 [-1.09 , -0.45]
Sadeghi 2023	28.79	4.057155	72	31.3	5.05	24	2.0%	-0.58 [-1.05 , -0.11]
Salacinski 2012	-84.2	13.9	19	-71.1	16.2	18	1.7%	-0.85 [-1.53 , -0.17]
Samut 2015	15.351034	11.317576	29	29.92	11.25	13	1.7%	-1.27 [-1.98 , -0.55]
Schilke 1996	7.64	3.5	10	9.48	4.04	10	1.4%	-0.47 [-1.36 , 0.42]
Shahine 2020	30	14.8	33	79.9	15.5	33	1.6%	-3.25 [-4.00 , -2.51]
Takacs 2017	20	11	20	28	10	20	1.8%	-0.75 [-1.39 , -0.10]
Thorstensson 2005	-2	11.515611	30	0.6	17.44806	31	2.0%	-0.17 [-0.68 , 0.33]
Topp 2002	35.62	10.739061	67	39.7	10.826426	35	2.1%	-0.38 [-0.79 , 0.04]
Vassao 2021	9.5	7.38	14	19.8	11.45	14	1.5%	-1.04 [-1.83 , -0.24]
Wallis 2017	38	9	23	34	11	23	1.9%	0.39 [-0.19 , 0.98]
Wang 2011	-82	14	28	-69	18	28	1.9%	-0.79 [-1.34 , -0.25]
Wortley 2013	396	359.568228	30	475	282	9	1.6%	-0.22 [-0.97 , 0.52]
Subtotal			**2375**			**1870**	**100.0%**	**-0.72 [-0.88 , -0.56]**
Test for overall effect: Z = 8.67 (P < 0.00001)								
Heterogeneity: Tau² = 0.28; Chi² = 308.03, df = 53 (P < 0.00001); I² = 83%								
1.2.3 C) Exercise when added to another co-intervention								
Alghadir 2019	49.15	3.924609	46	51.6	3.6	22	2.3%	-0.63 [-1.15 , -0.11]
Anadkat 2015	22.41	8.09	20	25.98	13.19	20	2.1%	-0.32 [-0.94 , 0.30]
Anwer 2014	8.05	2.33	21	18.05	4.3	21	1.6%	-2.84 [-3.72 , -1.96]
Bautch 1997	23.37	10.225302	17	17.88	7.627745	17	2.0%	0.59 [-0.09 , 1.28]
Bennell 2016	15.4	9.2	73	23.5	10.6	74	2.7%	-0.81 [-1.15 , -0.47]
Bennell 2022a	12.8	9.5	172	19.8	10.3	68	2.8%	-0.72 [-1.01 , -0.43]
Bennell 2022b	15.8	10.7	107	19.3	11.5	105	2.8%	-0.31 [-0.59 , -0.04]
Bhagat 2017	23.266	9.169	15	41.933	15.586	15	1.7%	-1.42 [-2.23 , -0.61]
Chang 2012	9.3	4.8	30	17.5	8.3	30	2.3%	-1.19 [-1.75 , -0.64]
Chen 2021	8.19	1.87	16	12.13	3.28	16	1.8%	-1.44 [-2.23 , -0.65]
Dincer 2016	6.7	3.7	19	19.8	13.8	16	1.9%	-1.32 [-2.06 , -0.58]
Dwyer 2015	378.9	328.073163	28	389.7	256.68993	27	2.3%	-0.04 [-0.56 , 0.49]
Gondhalekar 2013	37.13	12.68	15	52.6	13.29	15	1.8%	-1.16 [-1.94 , -0.38]
Hernandez 2019	7.15	6.21	53	9.95	5.27	60	2.6%	-0.49 [-0.86 , -0.11]
Hinman 2020	18.4	11.3	87	22	12.5	88	2.7%	-0.30 [-0.60 , -0.00]
Holm 2020	67	12.6484	45	68.1	13.646958	45	2.5%	-0.08 [-0.50 , 0.33]
Hsu 2021	19.95	4.79	22	19.43	5.29	22	2.2%	0.10 [-0.49 , 0.69]
Huang 2003	5.233333	0.990877	99	6.9	1.1	33	2.5%	-1.63 [-2.07 , -1.18]
Huang 2005	6.1	0.9	35	6.9	1.3	35	2.4%	-0.71 [-1.19 , -0.22]
Jan 2008	14.75	8.790718	68	22.5	10.9	34	2.5%	-0.81 [-1.23 , -0.38]
Joshi 2019	20.79	4.2	21	31.79	6.1	21	1.8%	-2.06 [-2.82 , -1.30]
Kholvadia 2019	65.6	9.9	32	70.8	9.8	40	2.4%	-0.52 [-1.00 , -0.05]
Kilic 2020	1.95	0.67	25	3.15	1.57	25	2.2%	-0.98 [-1.57 , -0.39]
Knoop 2013	17.4	11.6	80	19.3	11.4	79	2.7%	-0.16 [-0.48 , 0.15]
Kumar 2013	6.95	2.34	22	10	3.49	22	2.1%	-1.01 [-1.64 , -0.38]
Lee 2019	18.9	6	15	22.5	7.2	15	1.9%	-0.53 [-1.26 , 0.20]
Marconcin 2018	65.7	18.8	40	73.6	18.5	40	2.5%	-0.42 [-0.86 , 0.02]
Mehwish 2023	-39.04	11.01	22	-42.36	12.84	22	2.2%	0.27 [-0.32 , 0.87]
Messier 2013	14.2	11.543853	152	17.7	12.791837	152	2.9%	-0.29 [-0.51 , -0.06]
Nahayatbin 2018	-71.875	11.836708	32	-61.69	10.32	16	2.1%	-0.88 [-1.51 , -0.25]
Nelligan 2021	16.6	13	103	20.7	13.9	103	2.8%	-0.30 [-0.58 , -0.03]
Oh 2020	16.22	10.87	40	30.89	14.09	20	2.2%	-1.20 [-1.79 , -0.62]
Ozlu 2023	23.08	8.72	41	33.14	7.85	41	2.4%	-1.20 [-1.67 , -0.73]
Qiu 2023	15.44	11.65	21	16	10.07	21	2.1%	-0.05 [-0.66 , 0.55]
Rafiq 2021a	26	21.09	28	37	16.79	28	2.3%	-0.57 [-1.10 , -0.03]
Rafiq 2021b	-6.415	1.374707	76	-4.65	1.39	38	2.5%	-1.27 [-1.69 , -0.85]
Rafiq 2021c	23.9	12.37	25	22.07	11	25	2.2%	0.15 [-0.40 , 0.71]
Rosadi 2023	29.592308	4.729053	26	45.7	0.1	13	1.2%	-4.06 [-5.22 , -2.90]
Salli 2010	20.65	10.202166	50	32.6	11.6	25	2.3%	-1.11 [-1.62 , -0.59]
Samuel 2014	28.81	6.21	37	40.83	8.02	36	2.3%	-1.66 [-2.20 , -1.13]
Shabbir 2022	21.94	5.59	18	27.22	6.39	18	2.0%	-0.86 [-1.55 , -0.17]
Shin 2017	14.54	10.87	12	13.67	9.49	13	1.8%	0.08 [-0.70 , 0.87]
Thomas 2002	-2.59	10.5	470	-0.02	10.5	316	3.0%	-0.24 [-0.39 , -0.10]
Van Baar 1998	-1.3	5.7	54	-0.5	5.6	59	2.6%	-0.14 [-0.51 , 0.23]
Subtotal			**2430**			**1951**	**100.0%**	**-0.71 [-0.88 , -0.55]**



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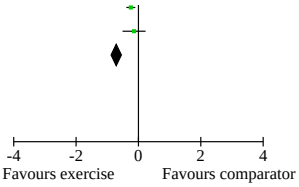
Analysis 1.2. (Continued)

Thomas 2002	-2.59	10.5	470	-0.02	10.5	316	3.0%	-0.24 [-0.39 , -0.10]
Van Baar 1998	-1.3	5.7	54	-0.5	5.6	59	2.6%	-0.14 [-0.51 , 0.23]
Subtotal			**2430**			**1951**	**100.0%**	**-0.71 [-0.88 , -0.55]**

Test for overall effect: Z = 8.47 (P < 0.00001)

Heterogeneity: Tau² = 0.23; Chi² = 260.41, df = 43 (P < 0.00001); I² = 83%

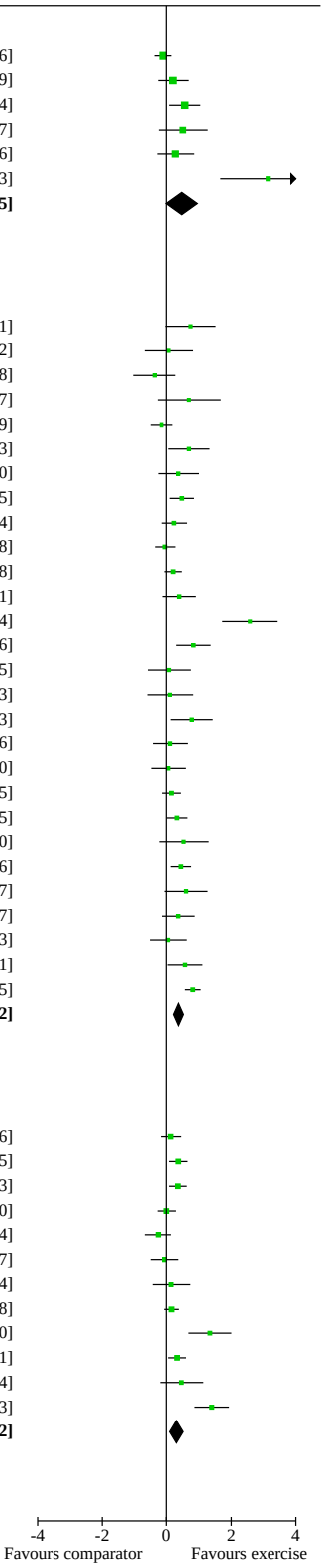
Test for subgroup differences: Chi² = 1.12, df = 2 (P = 0.57), I² = 0%



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Analysis 1.3. Comparison 1: Primary time point, Outcome 3: Quality of life immediately post-intervention

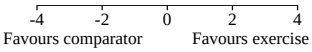
Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
1.3.1 A) Exercise vs attention control/placebo								
Bandak 2022	-8	15.149257	102	-6.2	15.297059	104	22.0%	-0.12 [-0.39 , 0.16]
Cheung 2017	54.546667	8.488751	60	52.8	8.325	23	19.1%	0.20 [-0.28 , 0.69]
Foley 2003	57.9	15.0731	35	50.4	10.8217	35	19.2%	0.57 [0.09 , 1.04]
Kuntz 2018	43.87619	17.390426	21	33.8	23.2	10	14.8%	0.51 [-0.26 , 1.27]
Lu 2017	4.01	9.654688	23	0.91	12.25625	23	17.6%	0.28 [-0.30 , 0.86]
Rego 2023	122.17	10.07	9	81.41	14.26	9	7.2%	3.14 [1.66 , 4.63]
Subtotal			**250**			**204**	**100.0%**	**0.47 [-0.01 , 0.95]**
Test for overall effect: Z = 1.93 (P = 0.05)								
Heterogeneity: Tau² = 0.25; Chi² = 23.18, df = 5 (P = 0.0003); I² = 78%								
1.3.2 B) Exercise vs no treatment/usual care/limited education								
An 2008	76.4	15.3	14	67	8.2	14	2.4%	0.74 [-0.03 , 1.51]
Bruce-Brand 2012	66.64	20.36	14	65	27.77	13	2.5%	0.07 [-0.69 , 0.82]
Cheung 2014	49.7	5.091169	18	51.7	5.091169	18	2.9%	-0.38 [-1.04 , 0.28]
Chopp-Hurley 2017	73.8	26.1	7	59.5	14.5	11	1.7%	0.69 [-0.29 , 1.67]
Christensen 2015	0.1	7.406145	64	1.3	7.406145	64	4.7%	-0.16 [-0.51 , 0.19]
Da Silva 2015	75.2	18.77	19	61.07	20.92	22	3.0%	0.69 [0.06 , 1.33]
Drummen 2023	0.74	0.09798	24	0.7	0.12	16	3.0%	0.37 [-0.27 , 1.00]
Fransen 2001	2	6.4	83	-0.7	3.7	43	4.6%	0.48 [0.10 , 0.85]
Fransen 2007	49.61	8.83	56	47.6	8.2	41	4.3%	0.23 [-0.17 , 0.64]
Gohir 2021	1.2	8.814396	67	1.6	8.929058	79	4.9%	-0.04 [-0.37 , 0.28]
Hay 2006	0.14	2	109	-0.28	2	108	5.3%	0.21 [-0.06 , 0.48]
Henriksen 2014	5.8	15.130739	31	-0.3	15.379371	29	3.7%	0.39 [-0.12 , 0.91]
Janyacharoen 2018	68.3	8.9	20	51.6	1.2	20	2.1%	2.58 [1.72 , 3.44]
Jorge 2015	76.4	18.7	29	59.5	21.2	31	3.6%	0.83 [0.30 , 1.36]
Keefe 2004	1.88	0.87	16	1.8	1.04	18	2.8%	0.08 [-0.59 , 0.75]
Krishnan 2021	3.8	5.34	15	3.27	4.01	15	2.6%	0.11 [-0.61 , 0.83]
Lee 2009	67.1	19.2	29	52.4	17.1	15	2.9%	0.78 [0.13 , 1.43]
Li 2020	48.7	17.5	26	46.9	13.6	25	3.4%	0.11 [-0.44 , 0.66]
Lund 2008	43.8	12.5	25	43.1	11.951151	27	3.5%	0.06 [-0.49 , 0.60]
O'Reilly 1999	-0.21	11.332128	78	-2.91	19.877486	113	5.1%	0.16 [-0.13 , 0.45]
Øiestad 2023	68.526786	20.490367	112	62	19	56	4.9%	0.32 [0.00 , 0.65]
Pazit 2018	0.715	0.171034	20	0.63	0.12	10	2.4%	0.53 [-0.24 , 1.30]
Rosedale 2014	37	21	120	28	18	60	5.0%	0.45 [0.13 , 0.76]
Salacinski 2012	59.2	17.6	19	46.7	22.6	18	2.9%	0.61 [-0.06 , 1.27]
Thorstensson 2005	4	11.917319	30	-0.7	13.494984	31	3.7%	0.36 [-0.14 , 0.87]
Wallis 2017	0.63	0.2	23	0.62	0.2	23	3.3%	0.05 [-0.53 , 0.63]
Wang 2011	74	11	28	67	13	28	3.5%	0.57 [0.04 , 1.11]
Xiao 2021	17.83	5.28	142	13.94	4.22	142	5.4%	0.81 [0.57 , 1.05]
Subtotal			**1238**			**1090**	**100.0%**	**0.37 [0.22 , 0.52]**
Test for overall effect: Z = 4.81 (P < 0.00001)								
Heterogeneity: Tau² = 0.10; Chi² = 75.88, df = 27 (P < 0.00001); I² = 64%								
1.3.3 C) Exercise when added to another co-intervention								
Bennell 2016	0.8	0.15	73	0.78	0.15	74	9.4%	0.13 [-0.19 , 0.46]
Bennell 2022a	0.76	0.16	172	0.7	0.17	68	10.0%	0.37 [0.08 , 0.65]
Bennell 2022b	0.78	0.13	107	0.73	0.15	105	10.1%	0.36 [0.08 , 0.63]
Hinman 2020	0.7	0.2	87	0.7	0.2	88	9.8%	0.00 [-0.30 , 0.30]
Holm 2020	0.72	0.116498	45	0.75	0.099856	45	8.2%	-0.27 [-0.69 , 0.14]
Marconcin 2018	79	14.9	40	80	13.2	40	7.9%	-0.07 [-0.51 , 0.37]
Mehwish 2023	39.4	12.69	22	37.68	10.46	22	6.1%	0.15 [-0.45 , 0.74]
Messier 2013	56.1	7.175908	152	54.9	7.7999	152	10.7%	0.16 [-0.07 , 0.38]
Nahayatbin 2018	64.695	18.415135	32	40.44	16.44	16	5.4%	1.34 [0.68 , 2.00]
Nelligan 2021	0.75	0.16	103	0.69	0.2	103	10.1%	0.33 [0.06 , 0.61]
Rosadi 2023	68.615385	12.981639	26	62.7	11.6	13	5.3%	0.46 [-0.21 , 1.14]
Salli 2010	68.3	12.243449	50	50.9	12.5	25	6.8%	1.40 [0.86 , 1.93]
Subtotal			**909**			**751**	**100.0%**	**0.31 [0.10 , 0.52]**
Test for overall effect: Z = 2.92 (P = 0.003)								
Heterogeneity: Tau² = 0.09; Chi² = 42.35, df = 11 (P < 0.0001); I² = 74%								
Test for subgroup differences: Chi² = 0.47, df = 2 (P = 0.79), I² = 0%								



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Analysis 1.3. (Continued)

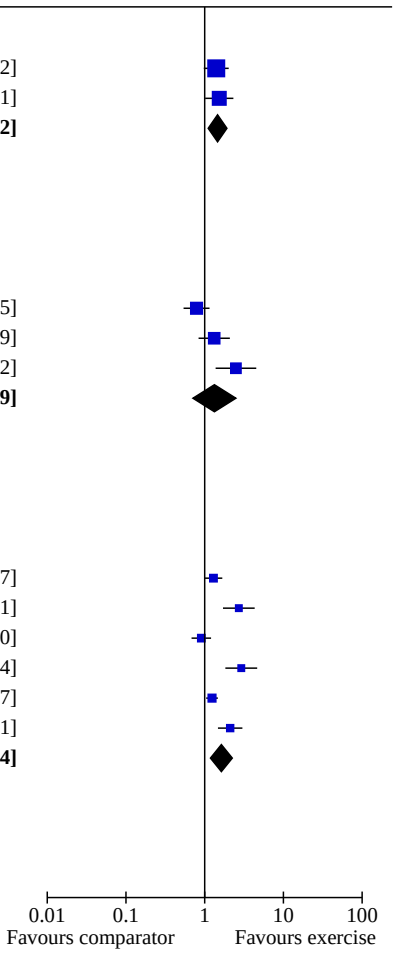
Test for subgroup differences: Chi² = 0.47, df = 2 (P = 0.79), I² = 0%



Analysis 1.4. Comparison 1: Primary time point, Outcome 4: Participant-reported treatment success immediately post-intervention

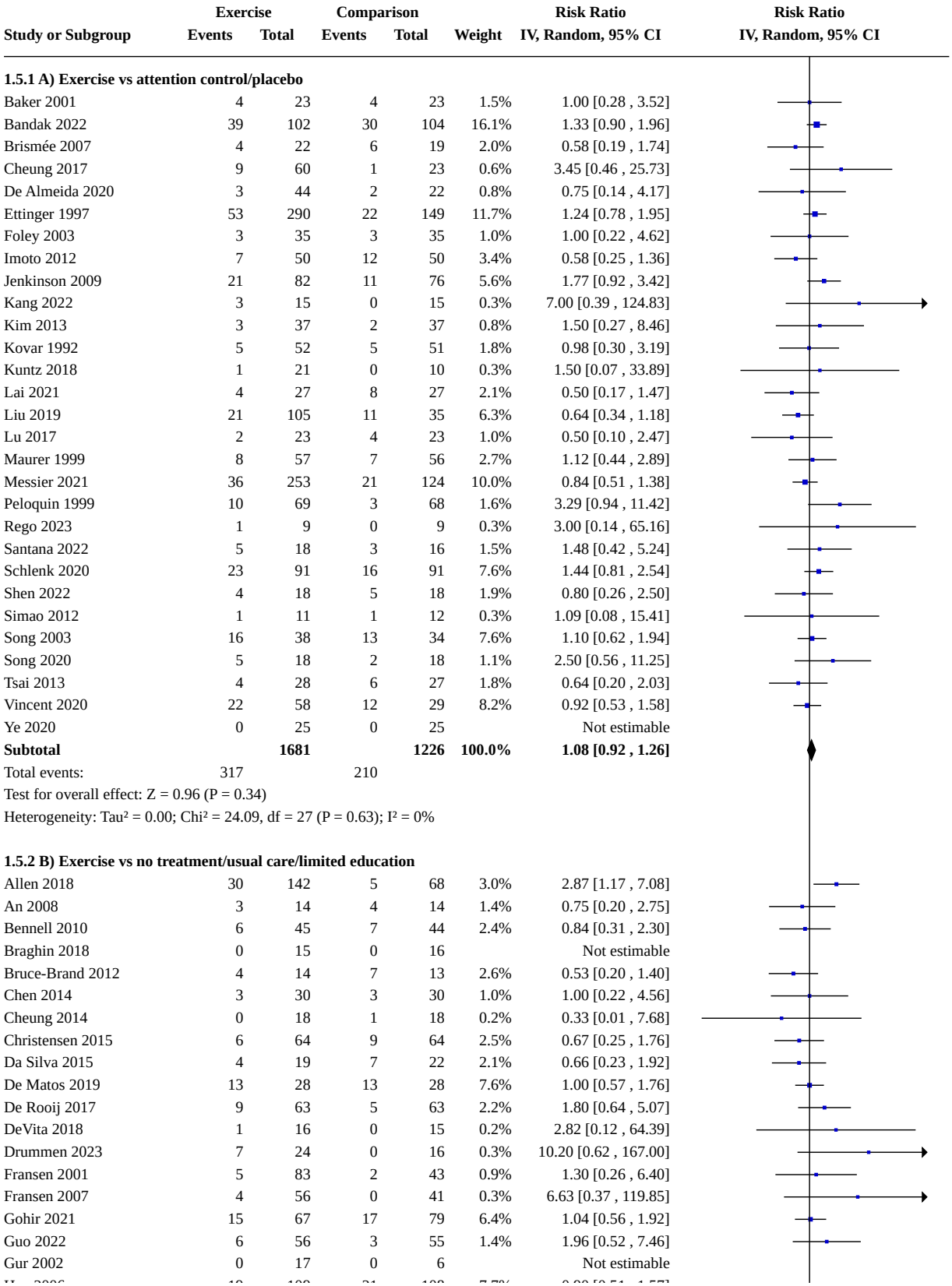
Study or Subgroup	Exercise Events	Exercise Total	Comparison Events	Comparison Total	Weight	Risk Ratio IV, Random, 95% CI
1.4.1 A) Exercise vs attention control/placebo						
Bandak 2022	44	102	32	104	56.3%	1.40 [0.97 , 2.02]
Jenkinson 2009	38	82	23	76	43.7%	1.53 [1.01 , 2.31]
Subtotal		**184**		**180**	**100.0%**	**1.46 [1.11 , 1.92]**
Total events:	82		55			
Test for overall effect: Z = 2.70 (P = 0.007)						
Heterogeneity: Tau² = 0.00; Chi² = 0.10, df = 1 (P = 0.75); I² = 0%						
1.4.2 B) Exercise vs no treatment/usual care/limited education						
Christensen 2015	26	64	33	64	35.9%	0.79 [0.54 , 1.15]
Hay 2006	32	109	24	108	33.9%	1.32 [0.84 , 2.09]
Jorge 2015	21	29	9	31	30.2%	2.49 [1.38 , 4.52]
Subtotal		**202**		**203**	**100.0%**	**1.33 [0.71 , 2.49]**
Total events:	79		66			
Test for overall effect: Z = 0.89 (P = 0.37)						
Heterogeneity: Tau² = 0.25; Chi² = 10.67, df = 2 (P = 0.005); I² = 81%						
1.4.3 C) Exercise when added to another co-intervention						
Bennell 2016	51	73	40	74	18.1%	1.29 [1.00 , 1.67]
Bennell 2022a	103	172	15	68	14.3%	2.71 [1.71 , 4.31]
Bennell 2022b	48	107	52	105	17.6%	0.91 [0.68 , 1.20]
Hinman 2020	49	87	17	88	14.3%	2.92 [1.83 , 4.64]
Knoop 2013	69	80	55	79	19.4%	1.24 [1.05 , 1.47]
Nelligan 2021	59	103	28	103	16.3%	2.11 [1.47 , 3.01]
Subtotal		**622**		**517**	**100.0%**	**1.63 [1.18 , 2.24]**
Total events:	379		207			
Test for overall effect: Z = 3.00 (P = 0.003)						
Heterogeneity: Tau² = 0.13; Chi² = 33.67, df = 5 (P < 0.00001); I² = 85%						

Test for subgroup differences: Chi² = 0.44, df = 2 (P = 0.80), I² = 0%



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Analysis 1.5. Comparison 1: Primary time point, Outcome 5: Study withdrawals at final follow-up

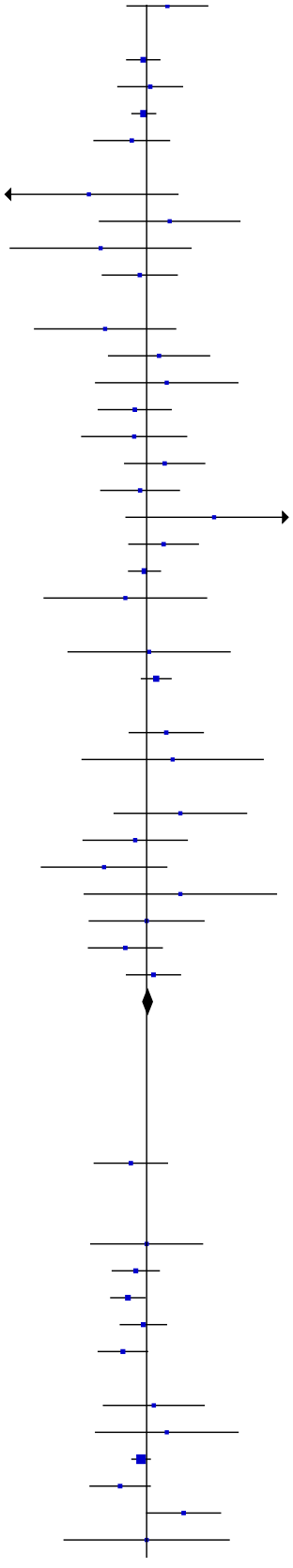


Study or Subgroup	Exercise Events	Exercise Total	Comparison Events	Comparison Total	Weight	Risk Ratio IV, Random, 95% CI
1.5.1 A) Exercise vs attention control/placebo						
Baker 2001	4	23	4	23	1.5%	1.00 [0.28 , 3.52]
Bandak 2022	39	102	30	104	16.1%	1.33 [0.90 , 1.96]
Brismée 2007	4	22	6	19	2.0%	0.58 [0.19 , 1.74]
Cheung 2017	9	60	1	23	0.6%	3.45 [0.46 , 25.73]
De Almeida 2020	3	44	2	22	0.8%	0.75 [0.14 , 4.17]
Ettinger 1997	53	290	22	149	11.7%	1.24 [0.78 , 1.95]
Foley 2003	3	35	3	35	1.0%	1.00 [0.22 , 4.62]
Imoto 2012	7	50	12	50	3.4%	0.58 [0.25 , 1.36]
Jenkinson 2009	21	82	11	76	5.6%	1.77 [0.92 , 3.42]
Kang 2022	3	15	0	15	0.3%	7.00 [0.39 , 124.83]
Kim 2013	3	37	2	37	0.8%	1.50 [0.27 , 8.46]
Kovar 1992	5	52	5	51	1.8%	0.98 [0.30 , 3.19]
Kuntz 2018	1	21	0	10	0.3%	1.50 [0.07 , 33.89]
Lai 2021	4	27	8	27	2.1%	0.50 [0.17 , 1.47]
Liu 2019	21	105	11	35	6.3%	0.64 [0.34 , 1.18]
Lu 2017	2	23	4	23	1.0%	0.50 [0.10 , 2.47]
Maurer 1999	8	57	7	56	2.7%	1.12 [0.44 , 2.89]
Messier 2021	36	253	21	124	10.0%	0.84 [0.51 , 1.38]
Peloquin 1999	10	69	3	68	1.6%	3.29 [0.94 , 11.42]
Rego 2023	1	9	0	9	0.3%	3.00 [0.14 , 65.16]
Santana 2022	5	18	3	16	1.5%	1.48 [0.42 , 5.24]
Schlenk 2020	23	91	16	91	7.6%	1.44 [0.81 , 2.54]
Shen 2022	4	18	5	18	1.9%	0.80 [0.26 , 2.50]
Simao 2012	1	11	1	12	0.3%	1.09 [0.08 , 15.41]
Song 2003	16	38	13	34	7.6%	1.10 [0.62 , 1.94]
Song 2020	5	18	2	18	1.1%	2.50 [0.56 , 11.25]
Tsai 2013	4	28	6	27	1.8%	0.64 [0.20 , 2.03]
Vincent 2020	22	58	12	29	8.2%	0.92 [0.53 , 1.58]
Ye 2020	0	25	0	25		Not estimable
Subtotal		**1681**		**1226**	**100.0%**	**1.08 [0.92 , 1.26]**
Total events:	317		210			
Test for overall effect: Z = 0.96 (P = 0.34)						
Heterogeneity: Tau² = 0.00; Chi² = 24.09, df = 27 (P = 0.63); I² = 0%						
1.5.2 B) Exercise vs no treatment/usual care/limited education						
Allen 2018	30	142	5	68	3.0%	2.87 [1.17 , 7.08]
An 2008	3	14	4	14	1.4%	0.75 [0.20 , 2.75]
Bennell 2010	6	45	7	44	2.4%	0.84 [0.31 , 2.30]
Braghin 2018	0	15	0	16		Not estimable
Bruce-Brand 2012	4	14	7	13	2.6%	0.53 [0.20 , 1.40]
Chen 2014	3	30	3	30	1.0%	1.00 [0.22 , 4.56]
Cheung 2014	0	18	1	18	0.2%	0.33 [0.01 , 7.68]
Christensen 2015	6	64	9	64	2.5%	0.67 [0.25 , 1.76]
Da Silva 2015	4	19	7	22	2.1%	0.66 [0.23 , 1.92]
De Matos 2019	13	28	13	28	7.6%	1.00 [0.57 , 1.76]
De Rooij 2017	9	63	5	63	2.2%	1.80 [0.64 , 5.07]
DeVita 2018	1	16	0	15	0.2%	2.82 [0.12 , 64.39]
Drummen 2023	7	24	0	16	0.3%	10.20 [0.62 , 167.00]
Fransen 2001	5	83	2	43	0.9%	1.30 [0.26 , 6.40]
Fransen 2007	4	56	0	41	0.3%	6.63 [0.37 , 119.85]
Gohir 2021	15	67	17	79	6.4%	1.04 [0.56 , 1.92]
Guo 2022	6	56	3	55	1.4%	1.96 [0.52 , 7.46]
Gur 2002	0	17	0	6		Not estimable

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Analysis 1.5. (Continued)

Guo 2022	6	56	3	55	1.4%	1.96 [0.52 , 7.46]
Gur 2002	0	17	0	6		Not estimable
Hay 2006	19	109	21	108	7.7%	0.90 [0.51 , 1.57]
Henriksen 2014	6	31	5	29	2.1%	1.12 [0.38 , 3.28]
Hurley 2007	49	207	27	104	14.6%	0.91 [0.61 , 1.37]
Jan 2009	5	71	4	35	1.5%	0.62 [0.18 , 2.15]
Janyacharoen 2018	0	20	0	20		Not estimable
Jorge 2015	0	29	3	31	0.3%	0.15 [0.01 , 2.83]
Karadag 2019	2	17	1	18	0.5%	2.12 [0.21 , 21.27]
Keefe 2004	0	16	2	18	0.3%	0.22 [0.01 , 4.34]
Kim 2020	4	40	5	40	1.6%	0.80 [0.23 , 2.76]
Krishnan 2021	0	15	0	15		Not estimable
Lee 2009	1	29	2	15	0.4%	0.26 [0.03 , 2.63]
Lee 2023	3	18	2	18	0.9%	1.50 [0.28 , 7.93]
Li 2020	2	26	1	25	0.4%	1.92 [0.19 , 19.90]
Lim 2008	4	53	6	54	1.7%	0.68 [0.20 , 2.27]
Lin 2009	2	36	3	36	0.8%	0.67 [0.12 , 3.75]
Lund 2008	5	25	3	27	1.4%	1.80 [0.48 , 6.76]
Mazloum 2018	5	33	3	16	1.4%	0.81 [0.22 , 2.97]
Multanen 2014	4	40	0	40	0.3%	9.00 [0.50 , 161.86]
O'Reilly 1999	6	78	5	113	1.8%	1.74 [0.55 , 5.50]
Øiestad 2023	28	112	15	56	8.3%	0.93 [0.54 , 1.60]
Pazit 2018	1	20	1	10	0.3%	0.50 [0.03 , 7.19]
Robbins 2022	0	43	0	43		Not estimable
Rogind 1998	1	12	1	13	0.3%	1.08 [0.08 , 15.46]
Rosedale 2014	41	120	15	60	9.5%	1.37 [0.83 , 2.26]
Sadeghi 2023	0	72	0	24		Not estimable
Salacinski 2012	6	19	3	18	1.6%	1.89 [0.56 , 6.46]
Samut 2015	2	29	0	13	0.3%	2.33 [0.12 , 45.45]
Sekir 2005	0	12	0	10		Not estimable
Takacs 2017	3	20	1	20	0.5%	3.00 [0.34 , 26.45]
Thorstensson 2005	2	30	3	31	0.8%	0.69 [0.12 , 3.84]
Vassao 2021	1	14	4	14	0.6%	0.25 [0.03 , 1.97]
Wallis 2017	1	23	0	23	0.2%	3.00 [0.13 , 70.02]
Wang 2011	2	28	2	28	0.7%	1.00 [0.15 , 6.61]
Wortley 2013	5	30	3	9	1.6%	0.50 [0.15 , 1.70]
Xiao 2021	10	142	8	142	3.0%	1.25 [0.51 , 3.07]
Subtotal		**2390**		**1911**	**100.0%**	**1.03 [0.88 , 1.20]**
Total events:	336		232			
Test for overall effect: Z = 0.38 (P = 0.70)						
Heterogeneity: Tau² = 0.00; Chi² = 34.31, df = 45 (P = 0.88); I² = 0%						
1.5.3 C) Exercise when added to another co-intervention						
Alghadir 2019	5	46	4	22	1.8%	0.60 [0.18 , 2.01]
Anwer 2014	0	21	0	21		Not estimable
Assar 2020	0	12	0	12		Not estimable
Bautch 1997	2	17	2	17	0.8%	1.00 [0.16 , 6.30]
Bennell 2016	9	73	13	74	4.3%	0.70 [0.32 , 1.54]
Bennell 2022a	22	172	16	68	7.8%	0.54 [0.30 , 0.97]
Bennell 2022b	11	107	12	105	4.4%	0.90 [0.42 , 1.95]
Chang 2012	6	30	13	30	3.9%	0.46 [0.20 , 1.05]
Chen 2021	0	16	0	16		Not estimable
Dincer 2016	3	19	2	16	1.0%	1.26 [0.24 , 6.65]
Dwyer 2015	2	28	1	27	0.5%	1.93 [0.19 , 20.05]
Hernandez 2019	28	53	38	60	25.9%	0.83 [0.61 , 1.15]
Hinman 2020	5	87	12	88	2.6%	0.42 [0.15 , 1.15]
Holm 2020	10	45	3	45	1.8%	3.33 [0.98 , 11.32]
Hsu 2021	1	22	1	22	0.4%	1.00 [0.07 , 15.00]



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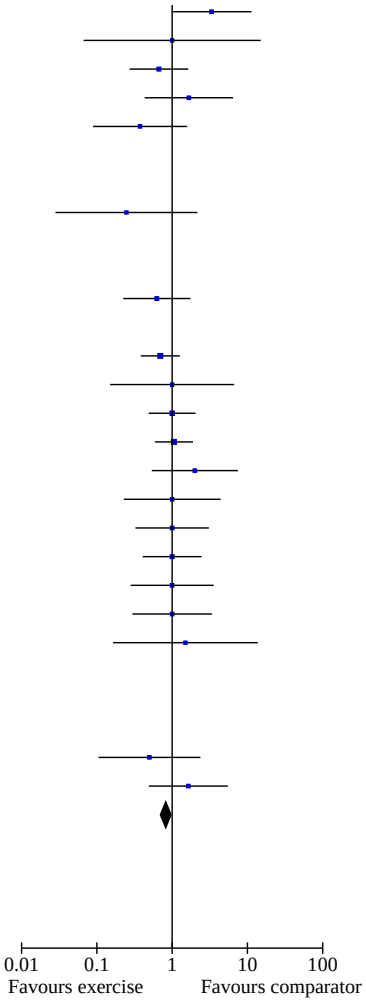
Analysis 1.5. (Continued)

Holm 2020	10	45	3	45	1.8%	3.33 [0.98 , 11.32]
Hsu 2021	1	22	1	22	0.4%	1.00 [0.07 , 15.00]
Huang 2003	12	99	6	33	3.3%	0.67 [0.27 , 1.64]
Huang 2005	5	35	3	35	1.4%	1.67 [0.43 , 6.45]
Jan 2008	3	68	4	34	1.3%	0.38 [0.09 , 1.58]
Joshi 2019	0	21	0	21		Not estimable
Kilic 2020	0	25	0	25		Not estimable
Knoop 2013	1	80	4	79	0.6%	0.25 [0.03 , 2.16]
Kumar 2013	0	22	0	22		Not estimable
Lee 2019	0	15	0	15		Not estimable
Marconcin 2018	5	40	8	40	2.5%	0.63 [0.22 , 1.75]
Mehwish 2023	0	22	0	22		Not estimable
Messier 2013	16	152	23	152	7.4%	0.70 [0.38 , 1.26]
Mete 2022	2	32	2	32	0.7%	1.00 [0.15 , 6.67]
Nelligan 2021	13	103	13	103	5.1%	1.00 [0.49 , 2.05]
Oh 2020	19	40	9	20	7.7%	1.06 [0.59 , 1.89]
Ozlu 2023	6	41	3	41	1.5%	2.00 [0.54 , 7.46]
Qiu 2023	3	21	3	21	1.2%	1.00 [0.23 , 4.40]
Rafiq 2021a	5	28	5	28	2.1%	1.00 [0.33 , 3.08]
Rafiq 2021b	12	76	6	38	3.3%	1.00 [0.41 , 2.46]
Rafiq 2021c	4	25	4	25	1.6%	1.00 [0.28 , 3.56]
Rosadi 2023	6	26	3	13	1.8%	1.00 [0.30 , 3.37]
Salli 2010	3	50	1	25	0.5%	1.50 [0.16 , 13.70]
Samuel 2014	0	37	0	36		Not estimable
Sedaghatnezhad 2021	0	15	0	15		Not estimable
Shabbir 2022	0	18	0	18		Not estimable
Talbot 2003	2	17	4	17	1.1%	0.50 [0.11 , 2.38]
Van Baar 1998	6	54	4	59	1.8%	1.64 [0.49 , 5.50]
Subtotal		**1910**		**1592**	**100.0%**	**0.82 [0.70 , 0.97]**
Total events:	227		222			

Test for overall effect: Z = 2.37 (P = 0.02)

Heterogeneity: Tau² = 0.00; Chi² = 21.32, df = 29 (P = 0.85); I² = 0%

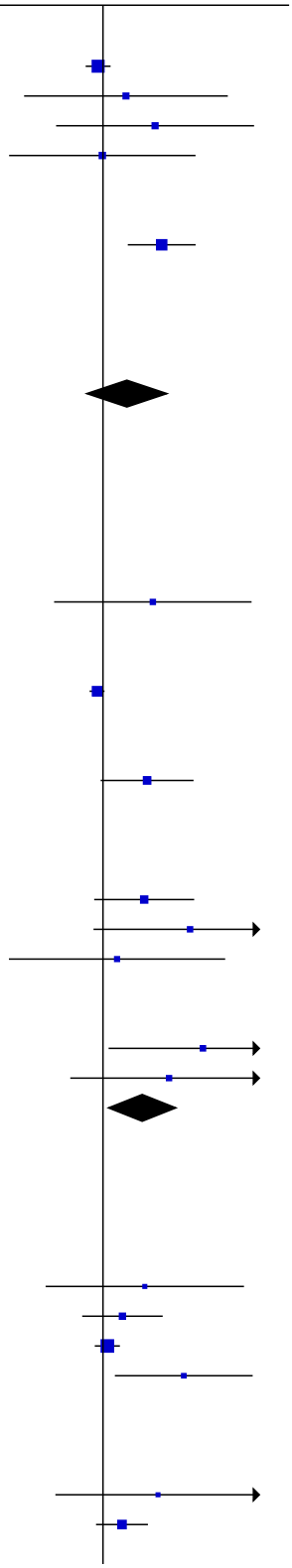
Test for subgroup differences: Chi² = 6.39, df = 2 (P = 0.04), I² = 68.7%



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Analysis 1.6. Comparison 1: Primary time point, Outcome 6: Adverse events at final follow-up

Study or Subgroup	Exercise Events	Exercise Total	Comparison Events	Comparison Total	Weight	Risk Ratio IV, Random, 95% CI
1.6.1 A) Exercise vs attention control/placebo						
Bandak 2022	34	102	40	104	35.9%	0.87 [0.60 , 1.25]
Cheung 2017	2	60	0	23	10.9%	1.97 [0.10 , 39.49]
Ettinger 1997	4	290	0	149	11.4%	4.64 [0.25 , 85.60]
Kovar 1992	1	52	1	51	12.4%	0.98 [0.06 , 15.26]
Kuntz 2018	0	21	0	10		Not estimable
Liu 2019	0	105	0	35		Not estimable
Messier 2021	46	253	4	124	29.4%	5.64 [2.08 , 15.30]
Rego 2023	0	9	0	9		Not estimable
Schlenk 2020	0	91	0	91		Not estimable
Tsai 2013	0	28	0	27		Not estimable
Ye 2020	0	25	0	25		Not estimable
Subtotal		**1036**		**648**	**100.0%**	**2.02 [0.62 , 6.58]**
Total events:	87		45			
Test for overall effect: Z = 1.17 (P = 0.24)						
Heterogeneity: Tau² = 0.98; Chi² = 12.99, df = 4 (P = 0.01); I² = 69%						
1.6.2 B) Exercise vs no treatment/usual care/limited education						
Abbott 2013	0	29	0	28		Not estimable
Allen 2018	4	142	0	68	8.1%	4.34 [0.24 , 79.52]
An 2008	0	14	0	14		Not estimable
Cheung 2014	0	18	0	18		Not estimable
Christensen 2015	42	64	50	64	25.8%	0.84 [0.67 , 1.05]
De Rooij 2017	0	63	0	63		Not estimable
DeVita 2018	0	16	0	15		Not estimable
Drummen 2023	11	24	2	16	17.5%	3.67 [0.93 , 14.39]
Fransen 2007	0	56	0	41		Not estimable
Gohir 2021	0	67	0	79		Not estimable
Hay 2006	0	109	0	109		Not estimable
Li 2020	7	26	2	25	16.6%	3.37 [0.77 , 14.67]
Multanen 2014	6	40	0	40	8.3%	13.00 [0.76 , 223.33]
Øiestad 2023	1	112	0	56	7.1%	1.51 [0.06 , 36.56]
Pazit 2018	0	20	0	10		Not estimable
Rogind 1998	0	12	0	13		Not estimable
Takacs 2017	9	20	0	20	8.6%	19.00 [1.18 , 305.88]
Wallis 2017	3	23	0	23	8.1%	7.00 [0.38 , 128.33]
Subtotal		**855**		**702**	**100.0%**	**3.17 [1.17 , 8.57]**
Total events:	83		54			
Test for overall effect: Z = 2.28 (P = 0.02)						
Heterogeneity: Tau² = 0.98; Chi² = 18.45, df = 7 (P = 0.01); I² = 62%						
1.6.3 C) Exercise when added to another co-intervention						
Alghadir 2019	3	46	0	22	2.5%	3.43 [0.18 , 63.57]
Bennell 2016	7	73	4	74	12.2%	1.77 [0.54 , 5.80]
Bennell 2022a	69	172	24	68	38.3%	1.14 [0.79 , 1.65]
Bennell 2022b	11	107	1	105	5.0%	10.79 [1.42 , 82.13]
Chen 2021	0	16	0	16		Not estimable
Dwyer 2015	0	28	0	27		Not estimable
Hernandez 2019	0	53	0	60		Not estimable
Hinman 2020	2	87	0	88	2.4%	5.06 [0.25 , 103.83]
Holm 2020	14	45	8	45	21.8%	1.75 [0.82 , 3.76]
Hsu 2021	0	22	0	22		Not estimable



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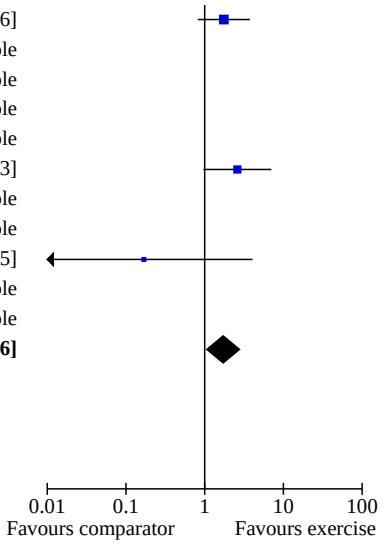
Analysis 1.6. (Continued)

Holm 2020	14	45	8	45	21.8%	1.75 [0.82 , 3.76]
Hsu 2021	0	22	0	22		Not estimable
Knoop 2013	0	80	0	79		Not estimable
Mehwish 2023	0	22	0	22		Not estimable
Messier 2013	0	152	0	152		Not estimable
Nelligan 2021	13	103	5	103	15.7%	2.60 [0.96 , 7.03]
Qiu 2023	0	21	0	21		Not estimable
Rafiq 2021a	0	28	0	28		Not estimable
Rafiq 2021b	0	76	1	38	2.2%	0.17 [0.01 , 4.05]
Rafiq 2021c	0	25	0	25		Not estimable
Shabbir 2022	0	18	0	18		Not estimable
Subtotal		**1174**		**1013**	**100.0%**	**1.72 [1.07 , 2.76]**
Total events:	119		43			

Test for overall effect: Z = 2.22 (P = 0.03)

Heterogeneity: Tau² = 0.12; Chi² = 9.82, df = 7 (P = 0.20); I² = 29%

Test for subgroup differences: Chi² = 1.20, df = 2 (P = 0.55), I² = 0%



Comparison 2. Sustainability up to and including 12 months post-intervention

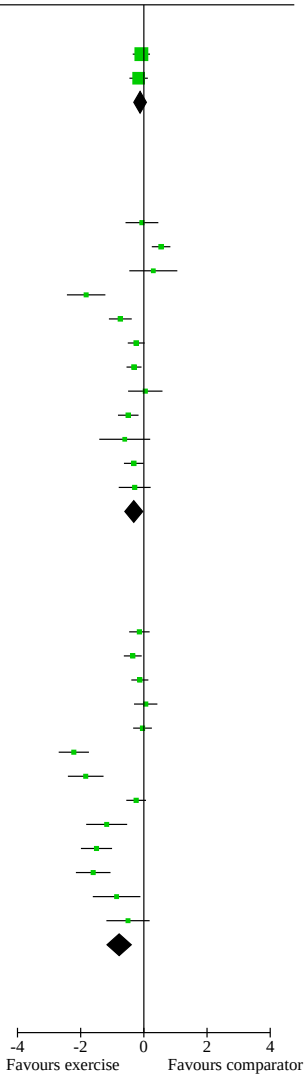
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Knee pain severity	27		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
2.1.1 A) Exercise vs attention control/placebo	2	388	Std. Mean Difference (IV, Random, 95% CI)	-0.12 [-0.32, 0.08]
2.1.2 B) Exercise vs no treatment/usual care/limited education	12	1494	Std. Mean Difference (IV, Random, 95% CI)	-0.32 [-0.60, -0.04]
2.1.3 C) Exercise when added to another co-intervention	13	1517	Std. Mean Difference (IV, Random, 95% CI)	-0.78 [-1.17, -0.40]
2.2 Self-reported physical function	26		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
2.2.1 A) Exercise vs attention control/placebo	2	388	Std. Mean Difference (IV, Random, 95% CI)	0.01 [-0.19, 0.21]
2.2.2 B) Exercise vs no treatment/usual care/limited education	12	1494	Std. Mean Difference (IV, Random, 95% CI)	-0.32 [-0.51, -0.14]
2.2.3 C) Exercise when added to another co-intervention	12	1525	Std. Mean Difference (IV, Random, 95% CI)	-0.61 [-0.93, -0.29]
2.3 Quality of life	13		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
2.3.1 A) Exercise vs attention control/placebo	1	206	Std. Mean Difference (IV, Random, 95% CI)	-0.06 [-0.33, 0.21]

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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.3.2 B) Exercise vs no treatment/usual care/limited education	6	799	Std. Mean Difference (IV, Random, 95% CI)	0.05 [-0.09, 0.20]
2.3.3 C) Exercise when added to another co-intervention	6	897	Std. Mean Difference (IV, Random, 95% CI)	0.39 [0.10, 0.68]

Analysis 2.1. Comparison 2: Sustainability up to and including 12 months post-intervention, Outcome 1: Knee pain severity

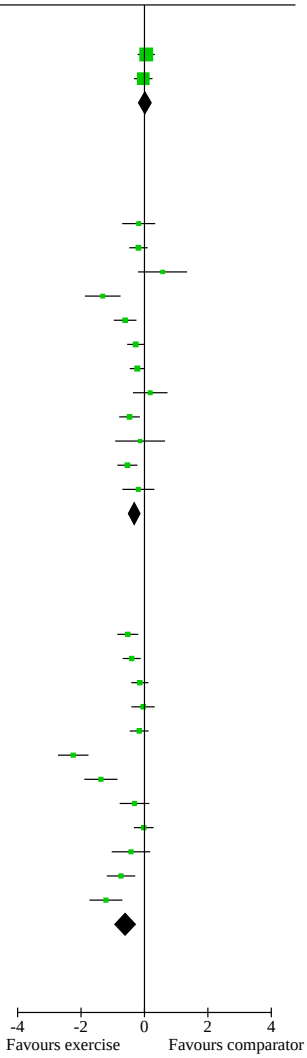
Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
2.1.1 A) Exercise vs attention control/placebo								
Bandak 2022	-8.4	18.179109	102	-7	17.336666	104	53.2%	-0.08 [-0.35 , 0.19]
Schlenk 2020	4.09	4.801688	91	4.72	2.400844	91	46.8%	-0.17 [-0.46 , 0.13]
Subtotal			**193**			**195**	**100.0%**	**-0.12 [-0.32 , 0.08]**
Test for overall effect: Z = 1.17 (P = 0.24)								
Heterogeneity: Tau² = 0.00; Chi² = 0.18, df = 1 (P = 0.67); I² = 0%								
2.1.2 B) Exercise vs no treatment/usual care/limited education								
Abbott 2013	14.9	11.2	29	15.6	11.4	28	7.9%	-0.06 [-0.58 , 0.46]
Allen 2018	1.15	3.375519	142	-0.64	3.036543	68	9.6%	0.55 [0.25 , 0.84]
Bruce-Brand 2012	9.6	4.14	14	8.33	4.08	13	6.0%	0.30 [-0.46 , 1.06]
Chen 2014	4	1.4	30	6.5	1.3	30	7.2%	-1.83 [-2.44 , -1.22]
De Rooij 2017	4.7	1.9	63	6.2	2.1	63	9.1%	-0.74 [-1.11 , -0.38]
Hay 2006	7.41	4.4	109	8.49	4.5	108	9.8%	-0.24 [-0.51 , 0.03]
Hurley 2007	5.7	3.16	207	6.7	3.28	104	10.0%	-0.31 [-0.55 , -0.07]
Lund 2008	-62	13	25	-62.6	12.990381	27	7.7%	0.05 [-0.50 , 0.59]
Øiestad 2023	3.852679	2.298529	112	5.1	2.9	56	9.4%	-0.49 [-0.82 , -0.17]
Rogind 1998	4.4	2.5	12	6.1	2.9	13	5.7%	-0.61 [-1.41 , 0.20]
Rosedale 2014	21	10	120	24	8	60	9.5%	-0.32 [-0.63 , -0.01]
Thorstensson 2005	-3.1	13.524148	30	1.1	14.994426	31	8.0%	-0.29 [-0.79 , 0.21]
Subtotal			**893**			**601**	**100.0%**	**-0.32 [-0.60 , -0.04]**
Test for overall effect: Z = 2.21 (P = 0.03)								
Heterogeneity: Tau² = 0.19; Chi² = 68.12, df = 11 (P < 0.00001); I² = 84%								
2.1.3 C) Exercise when added to another co-intervention								
Bennell 2016	31.7	22.6	73	34.8	21.2	74	8.2%	-0.14 [-0.46 , 0.18]
Bennell 2022a	3.4	2.3	172	4.2	2.2	68	8.3%	-0.35 [-0.63 , -0.07]
Bennell 2022b	4.3	2.4	107	4.6	2.3	105	8.4%	-0.13 [-0.40 , 0.14]
Hernandez 2019	3.8	2.97	53	3.63	2.8	60	8.1%	0.06 [-0.31 , 0.43]
Hinman 2020	3.9	2.4	87	4	2.3	88	8.3%	-0.04 [-0.34 , 0.25]
Huang 2003	2.566667	1.666905	99	6.1	1.3	33	7.7%	-2.22 [-2.70 , -1.74]
Huang 2005	3.9	1.4	35	6.6	1.5	35	7.3%	-1.84 [-2.40 , -1.28]
Knoop 2013	3.1	2.5	80	3.7	2.4	79	8.3%	-0.24 [-0.56 , 0.07]
Nahayatbin 2018	-64.285	12.099654	32	-50.31	10.77	16	7.0%	-1.18 [-1.83 , -0.53]
Ozlu 2023	4.05	0.72	41	5.36	0.99	41	7.6%	-1.50 [-1.99 , -1.01]
Salli 2010	3.5	1.829464	50	6.3	1.5	25	7.4%	-1.60 [-2.15 , -1.06]
Sedaghatnezhad 2021	13.67	10.06	15	25.89	16.69	15	6.5%	-0.86 [-1.62 , -0.11]
Talbot 2003	1.07	0.8	17	1.57	1.12	17	6.8%	-0.50 [-1.19 , 0.18]
Subtotal			**861**			**656**	**100.0%**	**-0.78 [-1.17 , -0.40]**
Test for overall effect: Z = 4.00 (P < 0.0001)								
Heterogeneity: Tau² = 0.44; Chi² = 141.31, df = 12 (P < 0.00001); I² = 92%								
Test for subgroup differences: Chi² = 9.16, df = 2 (P = 0.01), I² = 78.2%								



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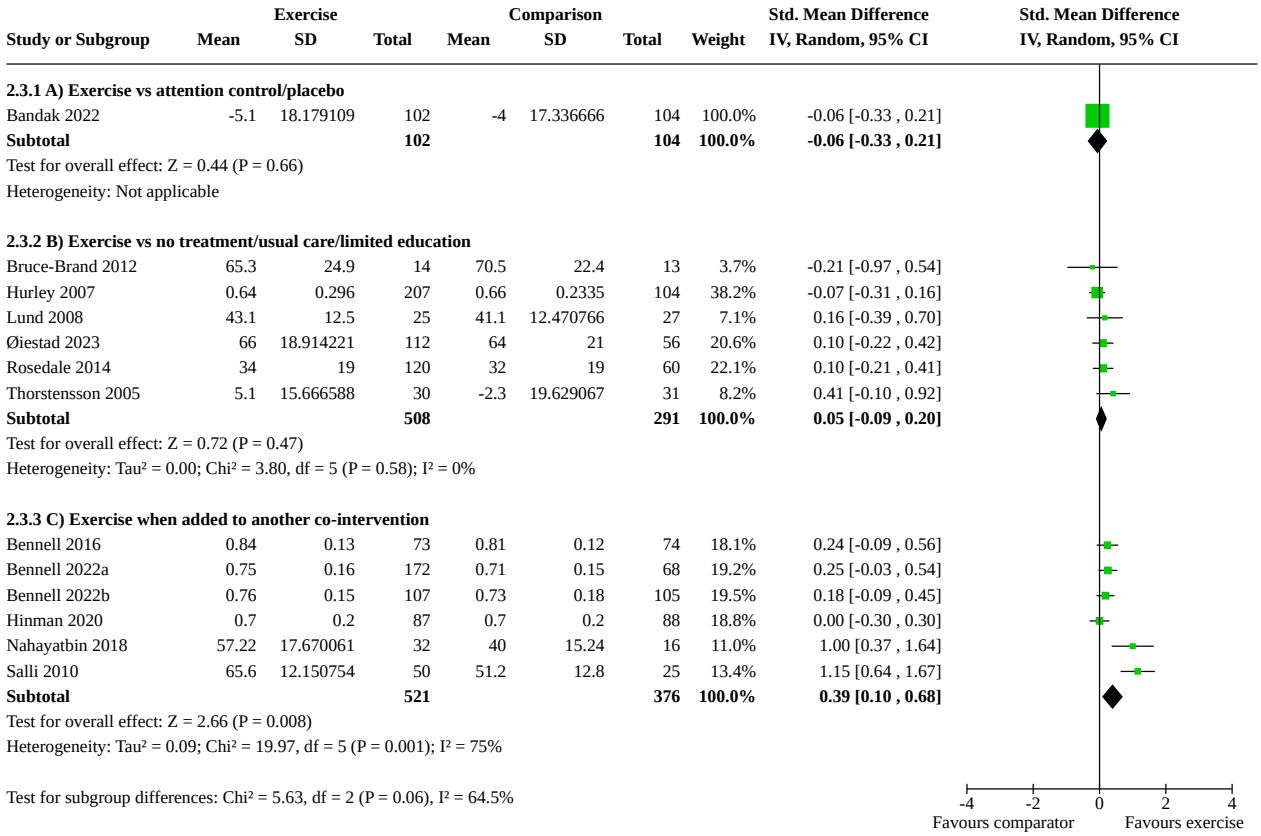
Analysis 2.2. Comparison 2: Sustainability up to and including 12 months post-intervention, Outcome 2: Self-reported physical function

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
2.2.1 A) Exercise vs attention control/placebo								
Bandak 2022	-2.1	18.179109	102	-3.1	17.336666	104	53.1%	0.06 [-0.22 , 0.33]
Schlenk 2020	17.02	14.405065	91	17.51	12.004221	91	46.9%	-0.04 [-0.33 , 0.25]
Subtotal			**193**			**195**	**100.0%**	**0.01 [-0.19 , 0.21]**
Test for overall effect: Z = 0.12 (P = 0.90)								
Heterogeneity: Tau² = 0.00; Chi² = 0.21, df = 1 (P = 0.65); I² = 0%								
2.2.2 B) Exercise vs no treatment/usual care/limited education								
Abbott 2013	50	34.2	29	56.8	40	28	6.8%	-0.18 [-0.70 , 0.34]
Allen 2018	-3.4	10.27725	142	-1.51	9.29554	68	11.1%	-0.19 [-0.48 , 0.10]
Bruce-Brand 2012	31.5	14.4	14	21.67	18.9	13	4.1%	0.57 [-0.20 , 1.34]
Chen 2014	5.4	1.7	30	7.6	1.6	30	6.2%	-1.32 [-1.88 , -0.75]
De Rooij 2017	23.5	13.1	63	31.4	12.6	63	9.7%	-0.61 [-0.97 , -0.25]
Hay 2006	24.83	15.3	109	28.95	14.4	108	11.5%	-0.28 [-0.54 , -0.01]
Hurley 2007	21.6	16.56	207	25	11.32	104	12.2%	-0.23 [-0.46 , 0.01]
Lund 2008	-61.4	13.509996	27	-63.9	13.5	25	6.5%	0.18 [-0.36 , 0.73]
Øiestad 2023	-70.526786	24.607169	112	-58	30	56	10.3%	-0.47 [-0.80 , -0.15]
Rogind 1998	9.7	2.9	12	10.44	6.7	13	4.0%	-0.14 [-0.92 , 0.65]
Rosedale 2014	-61	17	120	-52	16	60	10.5%	-0.54 [-0.85 , -0.22]
Thorstensson 2005	-0.9	12.586831	30	1.9	15.812304	31	7.1%	-0.19 [-0.70 , 0.31]
Subtotal			**895**			**599**	**100.0%**	**-0.32 [-0.51 , -0.14]**
Test for overall effect: Z = 3.47 (P = 0.0005)								
Heterogeneity: Tau² = 0.06; Chi² = 27.84, df = 11 (P = 0.003); I² = 60%								
2.2.3 C) Exercise when added to another co-intervention								
Bennell 2016	16	10.3	73	21.3	9.8	74	8.8%	-0.52 [-0.85 , -0.20]
Bennell 2022a	13.5	11	172	17.9	10.7	68	9.0%	-0.40 [-0.68 , -0.12]
Bennell 2022b	16.5	11.5	107	18.2	11.9	105	9.1%	-0.14 [-0.41 , 0.12]
Hernandez 2019	8.75	6.95	53	9.05	6.31	60	8.6%	-0.05 [-0.41 , 0.32]
Hinman 2020	18.1	11.4	87	20.1	12.5	88	8.9%	-0.17 [-0.46 , 0.13]
Huang 2003	3.966667	1.642484	99	7.6	1.5	33	8.0%	-2.25 [-2.73 , -1.77]
Huang 2005	5.8	1.8	35	8.1	1.5	35	7.7%	-1.37 [-1.90 , -0.85]
Kholvadia 2019	78	8.5	32	80.7	8.5	40	8.0%	-0.31 [-0.78 , 0.15]
Knoop 2013	18.9	13.3	80	19.2	13.2	79	8.9%	-0.02 [-0.33 , 0.29]
Nahayatbin 2018	-66.845	13.800561	32	-61.31	10.39	16	7.2%	-0.43 [-1.03 , 0.18]
Ozlu 2023	26.42	9.35	41	32.88	7.88	41	8.1%	-0.74 [-1.19 , -0.29]
Salli 2010	19.5	10.489956	50	32.7	11.3	25	7.7%	-1.21 [-1.73 , -0.69]
Subtotal			**861**			**664**	**100.0%**	**-0.61 [-0.93 , -0.29]**
Test for overall effect: Z = 3.71 (P = 0.0002)								
Heterogeneity: Tau² = 0.28; Chi² = 96.43, df = 11 (P < 0.00001); I² = 89%								
Test for subgroup differences: Chi² = 12.03, df = 2 (P = 0.002), I² = 83.4%								



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Analysis 2.3. Comparison 2: Sustainability up to and including 12 months post-intervention, Outcome 3: Quality of life



Comparison 3. Sensitivity analyses

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Selection bias: knee pain severity	53		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.1.1 A) Exercise vs attention control/placebo	13	1346	Std. Mean Difference (IV, Random, 95% CI)	-0.54 [-0.81, -0.28]
3.1.2 B) Exercise vs no treatment/usual care/limited education	23	1814	Std. Mean Difference (IV, Random, 95% CI)	-0.54 [-0.69, -0.39]
3.1.3 C) Exercise when added to another co-intervention	17	2454	Std. Mean Difference (IV, Random, 95% CI)	-0.39 [-0.54, -0.23]
3.2 Selection bias: self-reported physical function	50		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.2.1 A) Exercise vs attention control/placebo	9	1011	Std. Mean Difference (IV, Random, 95% CI)	-0.63 [-0.92, -0.34]
3.2.2 B) Exercise vs no treatment/usual care/limited education	25	2205	Std. Mean Difference (IV, Random, 95% CI)	-0.61 [-0.78, -0.43]

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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.2.3 C) Exercise when added to another co-intervention	16	2430	Std. Mean Difference (IV, Random, 95% CI)	-0.36 [-0.53, -0.20]
3.3 Attrition bias: knee pain severity	65		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.3.1 A) Exercise vs attention control/placebo	14	1679	Std. Mean Difference (IV, Random, 95% CI)	-0.36 [-0.51, -0.22]
3.3.2 B) Exercise vs no treatment/usual care/limited education	28	2486	Std. Mean Difference (IV, Random, 95% CI)	-0.59 [-0.77, -0.40]
3.3.3 C) Exercise when added to another co-intervention	23	3023	Std. Mean Difference (IV, Random, 95% CI)	-0.67 [-0.89, -0.46]
3.4 Attrition bias: self-reported physical function	61		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.4.1 A) Exercise vs attention control/placebo	12	1467	Std. Mean Difference (IV, Random, 95% CI)	-0.45 [-0.66, -0.24]
3.4.2 B) Exercise vs no treatment/usual care/limited education	27	2330	Std. Mean Difference (IV, Random, 95% CI)	-0.64 [-0.84, -0.43]
3.4.3 C) Exercise when added to another co-intervention	22	2999	Std. Mean Difference (IV, Random, 95% CI)	-0.55 [-0.73, -0.36]
3.5 Detection bias: knee pain severity	10		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.5.1 A) Exercise vs attention control/placebo	2	77	Std. Mean Difference (IV, Random, 95% CI)	-0.72 [-1.20, -0.25]
3.5.2 B) Exercise vs no treatment/usual care/limited education	1	111	Std. Mean Difference (IV, Random, 95% CI)	-0.13 [-0.50, 0.25]
3.5.3 C) Exercise when added to another co-intervention	7	1022	Std. Mean Difference (IV, Random, 95% CI)	-0.63 [-0.96, -0.30]
3.6 Detection bias: self-reported physical function	10		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
3.6.1 A) Exercise vs attention control/placebo	2	77	Std. Mean Difference (IV, Random, 95% CI)	-0.67 [-1.14, -0.20]
3.6.2 B) Exercise vs no treatment/usual care/limited education	1	111	Std. Mean Difference (IV, Random, 95% CI)	-0.53 [-0.91, -0.15]
3.6.3 C) Exercise when added to another co-intervention	7	1022	Std. Mean Difference (IV, Random, 95% CI)	-0.63 [-0.99, -0.28]

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Analysis 3.1. Comparison 3: Sensitivity analyses, Outcome 1: Selection bias: knee pain severity

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.1.1 A) Exercise vs attention control/placebo								
Baker 2001	128	95.968751	23	189	112.156251	23	7.2%	-0.57 [-1.17 , 0.02]
Bandak 2022	-10	15.149257	102	-7.3	15.297059	104	10.1%	-0.18 [-0.45 , 0.10]
Cheung 2017	4.56	2.01423	60	5.2	1.85	23	8.2%	-0.32 [-0.80 , 0.16]
Ettinger 1997	2.175241	0.665635	290	2.46	0.610328	149	10.6%	-0.44 [-0.64 , -0.24]
Foley 2003	8	3.86	35	10	3.09	35	8.3%	-0.57 [-1.04 , -0.09]
Jenkinson 2009	5.7	3.96	82	7.04	4.21	76	9.8%	-0.33 [-0.64 , -0.01]
Kuntz 2018	-67.838095	13.166567	21	-49.9	24.7	10	5.6%	-0.99 [-1.79 , -0.20]
Lai 2021	2.41	2.1	27	2.74	2.16	27	7.8%	-0.15 [-0.69 , 0.38]
Lu 2017	5.15	3.24	23	9.37	5.23	23	7.0%	-0.95 [-1.57 , -0.34]
Rego 2023	2.75	3.01	9	10.11	3.55	9	3.4%	-2.13 [-3.34 , -0.92]
Shen 2022	2.6	2.1	18	6.9	1.7	18	5.3%	-2.20 [-3.05 , -1.35]
Song 2003	-2.45	3.9	38	0.61	5.1	34	8.3%	-0.67 [-1.15 , -0.20]
Vincent 2020	1.250345	1.630087	58	0.47	0.8	29	8.5%	0.55 [0.09 , 1.00]
Subtotal			**786**			**560**	**100.0%**	**-0.54 [-0.81 , -0.28]**
Test for overall effect: Z = 4.01 (P < 0.0001)								
Heterogeneity: Tau² = 0.16; Chi² = 53.10, df = 12 (P < 0.00001); I² = 77%								
3.1.2 B) Exercise vs no treatment/usual care/limited education								
Bennell 2010	2.6	2.1	45	3.9	2.6	44	5.0%	-0.55 [-0.97 , -0.12]
Chen 2014	4.2	0.9	30	5.2	1.1	30	4.0%	-0.98 [-1.52 , -0.44]
Chopp-Hurley 2017	-81	24.5	7	-70.8	15.5	11	1.9%	-0.50 [-1.47 , 0.46]
Christensen 2015	-5.6	19.816442	64	-5.5	20.016608	64	5.8%	-0.00 [-0.35 , 0.34]
Da Silva 2015	2.6	1.55	19	4	1.56	22	3.3%	-0.88 [-1.53 , -0.24]
Fransen 2001	-10.6	19.5	83	1.5	19.4	43	5.5%	-0.62 [-0.99 , -0.24]
Fransen 2007	-1.67	3.28	56	-0.5	2.37	41	5.2%	-0.40 [-0.80 , 0.01]
Gohir 2021	-1.8	2.254845	67	-0.3	2.232264	79	5.9%	-0.67 [-1.00 , -0.33]
Henriksen 2014	-6.1	10.08716	31	0.7	10.252914	29	4.2%	-0.66 [-1.18 , -0.14]
Jorge 2015	4.3	3.1	29	6.6	1.5	31	4.1%	-0.94 [-1.48 , -0.41]
Lee 2009	4.6	4	29	5.9	3.7	15	3.4%	-0.33 [-0.95 , 0.30]
Lim 2008	-9	12	53	-1.7	12.8	54	5.4%	-0.58 [-0.97 , -0.20]
Lin 2009	4.2	3	36	7.3	3.4	36	4.4%	-0.96 [-1.45 , -0.47]
Lund 2008	-51.5	20.5	25	-58.1	20.78461	27	4.0%	0.31 [-0.23 , 0.86]
Multanen 2014	-2	14.070612	40	0	7.817007	40	4.9%	-0.17 [-0.61 , 0.27]
Øiestad 2023	3.801786	2.51396	112	4.5	2.4	56	6.0%	-0.28 [-0.60 , 0.04]
Pazit 2018	107.15	126.818349	20	249.7	309.3	10	2.6%	-0.68 [-1.46 , 0.10]
Robbins 2022	-2.59	2.35	43	-0.27	2.1	43	4.8%	-1.03 [-1.48 , -0.58]
Rosedale 2014	16	10	120	23	9	60	6.1%	-0.72 [-1.04 , -0.40]
Takacs 2017	2.8	1.7	20	4.6	2.3	20	3.3%	-0.87 [-1.52 , -0.22]
Vassao 2021	3.36	2.87	14	7	4.27	14	2.5%	-0.97 [-1.76 , -0.18]
Wallis 2017	6.5	1.2	23	6.1	1.7	23	3.7%	0.27 [-0.31 , 0.85]
Wang 2011	-76	15	28	-68	18	28	4.1%	-0.48 [-1.01 , 0.06]
Subtotal			**994**			**820**	**100.0%**	**-0.54 [-0.69 , -0.39]**
Test for overall effect: Z = 7.00 (P < 0.00001)								
Heterogeneity: Tau² = 0.07; Chi² = 49.59, df = 22 (P = 0.0007); I² = 56%								
3.1.3 C) Exercise when added to another co-intervention								
Assar 2020	3.4	2.2	12	6.7	2.5	12	2.3%	-1.35 [-2.26 , -0.45]
Bennell 2016	26.4	18.4	73	33.2	22.3	74	7.4%	-0.33 [-0.66 , -0.01]
Bennell 2022a	3.3	1.9	172	4.2	2.1	68	8.1%	-0.46 [-0.74 , -0.17]
Bennell 2022b	4.2	2.4	107	4.9	2.4	105	8.3%	-0.29 [-0.56 , -0.02]
Chen 2021	1.56	0.63	16	2.25	0.86	16	3.1%	-0.89 [-1.62 , -0.16]
Dincer 2016	1.7	0.9	19	3.6	2.8	16	3.3%	-0.93 [-1.63 , -0.22]
Dwyer 2015	97.7	86.780643	28	102.3	88.854206	27	4.8%	-0.05 [-0.58 , 0.48]
Hinman 2020	3.5	2.1	87	4.2	2.2	88	7.8%	-0.32 [-0.62 , -0.03]
Holm 2020	58.5	14.312663	45	61.2	13.314105	45	6.1%	-0.19 [-0.61 , 0.22]
Hsu 2021	4.48	1.44	22	4.33	1.65	22	4.2%	0.10 [-0.50 , 0.69]
Kilic 2020	3.76	1.27	25	5.88	1.51	25	3.8%	-1.50 [-2.13 , -0.86]
Knoop 2013	2.8	2.1	80	3.3	2.1	79	7.6%	-0.24 [-0.55 , 0.08]
Mehwish 2023	5.77	1.2	22	5.24	1.09	22	4.1%	0.45 [-0.15 , 1.05]
Nelligan 2021	3.5	2.2	103	5	2.4	103	8.1%	-0.65 [-0.93 , -0.37]
Qiu 2023	3.17	1.58	21	3.5	1.86	21	4.0%	-0.19 [-0.79 , 0.42]
Thomas 2002	-1.27	3.6	470	-0.46	3.6	316	10.2%	-0.22 [-0.37 , -0.08]
Van Baar 1998	-27.4	28.7	54	-11.7	28.5	59	6.7%	-0.55 [-0.92 , -0.17]
Subtotal			**1356**			**1098**	**100.0%**	**-0.39 [-0.54 , -0.23]**
Test for overall effect: Z = 4.97 (P < 0.00001)								

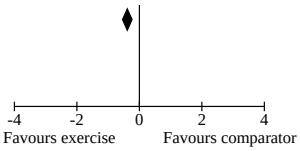
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Analysis 3.1. (Continued)

Subtotal	**1356**	**1098**	**100.0%**	**-0.39 [-0.54 , -0.23]**	

Test for overall effect: Z = 4.97 (P < 0.00001)
Heterogeneity: Tau² = 0.05; Chi² = 42.73, df = 16 (P = 0.0003); I² = 63%

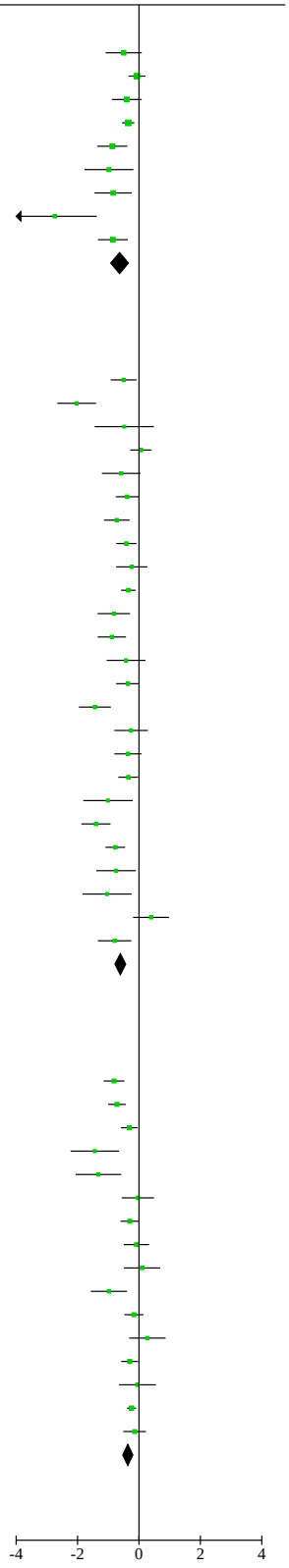
Test for subgroup differences: Chi² = 2.17, df = 2 (P = 0.34), I² = 7.7%



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Analysis 3.2. Comparison 3: Sensitivity analyses, Outcome 2: Selection bias: self-reported physical function

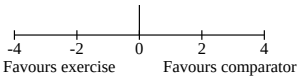
Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.2.1 A) Exercise vs attention control/placebo								
Baker 2001	462	372.312504	23	664	422.031255	23	10.4%	-0.50 [-1.09 , 0.09]
Bandak 2022	-6.9	14.139307	102	-6	14.277255	104	15.6%	-0.06 [-0.34 , 0.21]
Cheung 2017	21.746667	8.82397	60	25.2	7.978125	23	12.0%	-0.40 [-0.88 , 0.09]
Ettinger 1997	1.730069	0.480946	290	1.9	0.488262	149	16.7%	-0.35 [-0.55 , -0.15]
Foley 2003	27	9.28	35	37	13.14	35	11.9%	-0.87 [-1.36 , -0.38]
Kuntz 2018	-74.414286	15.188772	21	-56.3	23.1	10	7.6%	-0.98 [-1.78 , -0.18]
Lu 2017	15.85	7.6	23	26.89	16.56	23	10.1%	-0.84 [-1.45 , -0.24]
Rego 2023	9.75	9.35	9	36.11	8.91	9	3.6%	-2.75 [-4.12 , -1.38]
Song 2003	-11.09	12	38	-1.33	10.6	34	12.0%	-0.85 [-1.33 , -0.37]
Subtotal			**601**			**410**	**100.0%**	**-0.63 [-0.92 , -0.34]**
Test for overall effect: Z = 4.29 (P < 0.0001)								
Heterogeneity: Tau² = 0.12; Chi² = 28.12, df = 8 (P = 0.0005); I² = 72%								
3.2.2 B) Exercise vs no treatment/usual care/limited education								
Bennell 2010	16.2	11.7	45	21.9	11	44	4.4%	-0.50 [-0.92 , -0.08]
Chen 2014	5.1	0.9	30	7.4	1.3	30	3.4%	-2.03 [-2.66 , -1.40]
Chopp-Hurley 2017	-89.8	17.4	7	-80.5	19.2	11	2.1%	-0.48 [-1.44 , 0.49]
Christensen 2015	-7.6	21.617937	64	-9	21.617937	64	4.9%	0.06 [-0.28 , 0.41]
Da Silva 2015	2.3	1.36	19	3.13	1.45	22	3.4%	-0.58 [-1.21 , 0.05]
Fransen 2001	-7.7	19.9	83	0.1	20.5	43	4.7%	-0.39 [-0.76 , -0.01]
Fransen 2007	-5.04	10.25	56	2.07	9.06	41	4.5%	-0.72 [-1.14 , -0.31]
Gohir 2021	-7.8	8.404424	67	-4.3	8.482605	79	4.9%	-0.41 [-0.74 , -0.08]
Henriksen 2014	-4.2	11.586602	31	-1.4	11.830286	29	4.0%	-0.24 [-0.74 , 0.27]
Hurley 2007	20	18.4	207	25.9	13.6	104	5.4%	-0.35 [-0.58 , -0.11]
Jorge 2015	17.3	12.4	29	26.7	10.2	31	3.9%	-0.82 [-1.35 , -0.29]
Kim 2020	33.22	9.550079	40	42.05	10.24578	40	4.2%	-0.88 [-1.34 , -0.42]
Lee 2009	14.7	13.8	29	20.8	15	15	3.4%	-0.42 [-1.05 , 0.21]
Lim 2008	-6.5	10.6	53	-2.6	10.9	54	4.7%	-0.36 [-0.74 , 0.02]
Lin 2009	10.1	8.3	36	24.9	11.8	36	3.9%	-1.44 [-1.96 , -0.91]
Lund 2008	-64.1	11.5	25	-61.1	11.431535	27	3.8%	-0.26 [-0.80 , 0.29]
Multanen 2014	-1	6.253605	40	1	4.690204	40	4.3%	-0.36 [-0.80 , 0.08]
Øiestad 2023	-71	23.891647	112	-63	21	56	5.0%	-0.35 [-0.67 , -0.02]
Pazit 2018	39.85	43.828049	20	88	51.4	10	2.6%	-1.01 [-1.82 , -0.20]
Robbins 2022	-4.63	3.86	43	0.22	2.94	43	4.2%	-1.40 [-1.87 , -0.93]
Rosedale 2014	-65	20	120	-50	18	60	5.0%	-0.77 [-1.09 , -0.45]
Takacs 2017	20	11	20	28	10	20	3.3%	-0.75 [-1.39 , -0.10]
Vassao 2021	9.5	7.38	14	19.8	11.45	14	2.7%	-1.04 [-1.83 , -0.24]
Wallis 2017	38	9	23	34	11	23	3.6%	0.39 [-0.19 , 0.98]
Wang 2011	-82	14	28	-69	18	28	3.8%	-0.79 [-1.34 , -0.25]
Subtotal			**1241**			**964**	**100.0%**	**-0.61 [-0.78 , -0.43]**
Test for overall effect: Z = 6.80 (P < 0.00001)								
Heterogeneity: Tau² = 0.13; Chi² = 86.19, df = 24 (P < 0.00001); I² = 72%								
3.2.3 C) Exercise when added to another co-intervention								
Bennell 2016	15.4	9.2	73	23.5	10.6	74	7.4%	-0.81 [-1.15 , -0.47]
Bennell 2022a	12.8	9.5	172	19.8	10.3	68	8.0%	-0.72 [-1.01 , -0.43]
Bennell 2022b	15.8	10.7	107	19.3	11.5	105	8.3%	-0.31 [-0.59 , -0.04]
Chen 2021	8.19	1.87	16	12.13	3.28	16	3.1%	-1.44 [-2.23 , -0.65]
Dincer 2016	6.7	3.7	19	19.8	13.8	16	3.3%	-1.32 [-2.06 , -0.58]
Dwyer 2015	378.9	328.073163	28	389.7	256.68993	27	5.1%	-0.04 [-0.56 , 0.49]
Hinman 2020	18.4	11.3	87	22	12.5	88	7.9%	-0.30 [-0.60 , -0.00]
Holm 2020	67	12.6484	45	68.1	13.646958	45	6.4%	-0.08 [-0.50 , 0.33]
Hsu 2021	19.95	4.79	22	19.43	5.29	22	4.5%	0.10 [-0.49 , 0.69]
Kilic 2020	1.95	0.67	25	3.15	1.57	25	4.5%	-0.98 [-1.57 , -0.39]
Knoop 2013	17.4	11.6	80	19.3	11.4	79	7.7%	-0.16 [-0.48 , 0.15]
Mehwish 2023	-39.04	11.01	22	-42.36	12.84	22	4.4%	0.27 [-0.32 , 0.87]
Nelligan 2021	16.6	13	103	20.7	13.9	103	8.2%	-0.30 [-0.58 , -0.03]
Qiu 2023	15.44	11.65	21	16	10.07	21	4.3%	-0.05 [-0.66 , 0.55]
Thomas 2002	-2.59	10.5	470	-0.02	10.5	316	9.9%	-0.24 [-0.39 , -0.10]
Van Baar 1998	-1.3	5.7	54	-0.5	5.6	59	6.9%	-0.14 [-0.51 , 0.23]
Subtotal			**1344**			**1086**	**100.0%**	**-0.36 [-0.53 , -0.20]**
Test for overall effect: Z = 4.36 (P < 0.0001)								
Heterogeneity: Tau² = 0.07; Chi² = 46.81, df = 15 (P < 0.0001); I² = 68%								
Test for subgroup differences: Chi² = 4.89, df = 2 (P = 0.09), I² = 59.1%								



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Analysis 3.2. (Continued)

Test for subgroup differences: Chi² = 4.89, df = 2 (P = 0.09), I² = 59.1%



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Analysis 3.3. Comparison 3: Sensitivity analyses, Outcome 3: Attrition bias: knee pain severity

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.3.1 A) Exercise vs attention control/placebo								
Baker 2001	128	95.968751	23	189	112.156251	23	4.6%	-0.57 [-1.17 , 0.02]
Bandak 2022	-10	15.149257	102	-7.3	15.297059	104	12.2%	-0.18 [-0.45 , 0.10]
Brismée 2007	2.41	2.05	22	3.37	1.78	19	4.2%	-0.49 [-1.11 , 0.14]
Cheung 2017	4.56	2.01423	60	5.2	1.85	23	6.3%	-0.32 [-0.80 , 0.16]
Ettinger 1997	2.175241	0.665635	290	2.46	0.610328	149	15.4%	-0.44 [-0.64 , -0.24]
Foley 2003	8	3.86	35	10	3.09	35	6.4%	-0.57 [-1.04 , -0.09]
Jenkinson 2009	5.7	3.96	82	7.04	4.21	76	10.7%	-0.33 [-0.64 , -0.01]
Kuntz 2018	-67.838095	13.166567	21	-49.9	24.7	10	2.8%	-0.99 [-1.79 , -0.20]
Lai 2021	2.41	2.1	27	2.74	2.16	27	5.4%	-0.15 [-0.69 , 0.38]
Lu 2017	5.15	3.24	23	9.37	5.23	23	4.4%	-0.95 [-1.57 , -0.34]
Messier 2021	4.700791	3.300649	253	4.8	2.9	124	14.7%	-0.03 [-0.25 , 0.18]
Simao 2012	159.19	148.442	11	175	125.79	12	2.7%	-0.11 [-0.93 , 0.71]
Tsai 2013	-2.6	2.540234	28	-1.02	1.681047	27	5.2%	-0.72 [-1.27 , -0.17]
Ye 2020	3.68	6.07	25	4.92	1.41	25	5.1%	-0.28 [-0.83 , 0.28]
Subtotal			**1002**			**677**	**100.0%**	**-0.36 [-0.51 , -0.22]**
Test for overall effect: Z = 4.99 (P < 0.00001)								
Heterogeneity: Tau² = 0.02; Chi² = 20.92, df = 13 (P = 0.07); I² = 38%								
3.3.2 B) Exercise vs no treatment/usual care/limited education								
Allen 2018	-1.59	3.405658	142	-0.66	3.098513	68	4.6%	-0.28 [-0.57 , 0.01]
Bennell 2010	2.6	2.1	45	3.9	2.6	44	4.1%	-0.55 [-0.97 , -0.12]
Braghin 2018	8	8.52	15	20.94	23.47	16	2.9%	-0.70 [-1.43 , 0.02]
Cheung 2014	5.8	2.842569	18	8.3	2.842569	18	3.0%	-0.86 [-1.55 , -0.17]
Christensen 2015	-5.6	19.816442	64	-5.5	20.016608	64	4.4%	-0.00 [-0.35 , 0.34]
De Rooij 2017	4.3	2	63	5.8	2.2	63	4.4%	-0.71 [-1.07 , -0.35]
Fransen 2001	-10.6	19.5	83	1.5	19.4	43	4.3%	-0.62 [-0.99 , -0.24]
Fransen 2007	-1.67	3.28	56	-0.5	2.37	41	4.2%	-0.40 [-0.80 , 0.01]
Gur 2002	-20.9	8.3	17	0.7	4.6	6	1.5%	-2.74 [-4.02 , -1.47]
Hay 2006	7.36	4.3	109	8.99	3.7	108	4.7%	-0.40 [-0.67 , -0.14]
Janyacharoen 2018	-69.2	5.1	20	-43.1	1.2	20	1.0%	-6.91 [-8.62 , -5.20]
Jorge 2015	4.3	3.1	29	6.6	1.5	31	3.6%	-0.94 [-1.48 , -0.41]
Krishnan 2021	-53.93	3.17	15	-50.93	6.04	15	2.9%	-0.61 [-1.34 , 0.13]
Lee 2009	4.6	4	29	5.9	3.7	15	3.3%	-0.33 [-0.95 , 0.30]
Lee 2023	34.5	21.69	18	43.9	17.85	18	3.1%	-0.46 [-1.13 , 0.20]
Lim 2008	-9	12	53	-1.7	12.8	54	4.3%	-0.58 [-0.97 , -0.20]
Lin 2009	4.2	3	36	7.3	3.4	36	3.8%	-0.96 [-1.45 , -0.47]
Lund 2008	-51.5	20.5	25	-58.1	20.78461	27	3.6%	0.31 [-0.23 , 0.86]
O'Reilly 1999	-1.45	2.616812	78	0.42	3.594579	113	4.6%	-0.58 [-0.87 , -0.28]
Øiestad 2023	3.801786	2.51396	112	4.5	2.4	56	4.5%	-0.28 [-0.60 , 0.04]
Robbins 2022	-2.59	2.35	43	-0.27	2.1	43	4.0%	-1.03 [-1.48 , -0.58]
Schilke 1996	9.7	4.72	10	10.1	6.44	10	2.4%	-0.07 [-0.94 , 0.81]
Sekir 2005	1.9	2.47	12	4.1	2.63	10	2.4%	-0.83 [-1.71 , 0.05]
Thorstensson 2005	-1.8	13.390246	30	0.3	16.221243	31	3.8%	-0.14 [-0.64 , 0.36]
Vassao 2021	3.36	2.87	14	7	4.27	14	2.7%	-0.97 [-1.76 , -0.18]
Wallis 2017	6.5	1.2	23	6.1	1.7	23	3.5%	0.27 [-0.31 , 0.85]
Wang 2011	-76	15	28	-68	18	28	3.7%	-0.48 [-1.01 , 0.06]
Xiao 2021	75.7	20.19	142	94.25	27.73	142	4.8%	-0.76 [-1.00 , -0.52]
Subtotal			**1329**			**1157**	**100.0%**	**-0.59 [-0.77 , -0.40]**
Test for overall effect: Z = 6.22 (P < 0.00001)								
Heterogeneity: Tau² = 0.17; Chi² = 118.02, df = 27 (P < 0.00001); I² = 77%								
3.3.3 C) Exercise when added to another co-intervention								
Alghadir 2019	4.25	1.242176	46	4.8	0.8	22	4.4%	-0.49 [-1.00 , 0.03]
Anadkat 2015	2.75	0.96	20	4.3	1.12	20	3.6%	-1.46 [-2.16 , -0.75]
Anwer 2014	1.24	0.76	21	4.24	0.88	21	2.5%	-3.58 [-4.59 , -2.57]
Assar 2020	3.4	2.2	12	6.7	2.5	12	2.8%	-1.35 [-2.26 , -0.45]
Bennell 2016	26.4	18.4	73	33.2	22.3	74	5.2%	-0.33 [-0.66 , -0.01]
Bennell 2022a	3.3	1.9	172	4.2	2.1	68	5.3%	-0.46 [-0.74 , -0.17]
Bennell 2022b	4.2	2.4	107	4.9	2.4	105	5.4%	-0.29 [-0.56 , -0.02]
Bhagat 2017	1.933	0.8837	15	3.8	1.612	15	3.2%	-1.40 [-2.21 , -0.59]
Chen 2021	1.56	0.63	16	2.25	0.86	16	3.5%	-0.89 [-1.62 , -0.16]
Dwyer 2015	97.7	86.780643	28	102.3	88.854206	27	4.3%	-0.05 [-0.58 , 0.48]
Hinman 2020	3.5	2.1	87	4.2	2.2	88	5.3%	-0.32 [-0.62 , -0.03]
Holm 2020	58.5	14.312663	45	61.2	13.314105	45	4.8%	-0.19 [-0.61 , 0.22]
Knoop 2013	2.8	2.1	80	3.3	2.1	79	5.2%	-0.24 [-0.55 , 0.08]

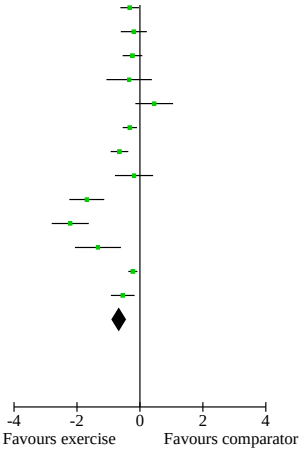
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Analysis 3.3. (Continued)

Hinman 2020	3.5	2.1	87	4.2	2.2	88	5.3%	-0.32 [-0.62 , -0.03]
Holm 2020	58.5	14.312663	45	61.2	13.314105	45	4.8%	-0.19 [-0.61 , 0.22]
Knoop 2013	2.8	2.1	80	3.3	2.1	79	5.2%	-0.24 [-0.55 , 0.08]
Lee 2019	2.8	0.8	15	3.1	0.9	15	3.5%	-0.34 [-1.06 , 0.38]
Mehwish 2023	5.77	1.2	22	5.24	1.09	22	4.0%	0.45 [-0.15 , 1.05]
Messier 2013	3.7	3.431956	152	4.8	3.431956	152	5.5%	-0.32 [-0.55 , -0.09]
Nelligan 2021	3.5	2.2	103	5	2.4	103	5.3%	-0.65 [-0.93 , -0.37]
Qiu 2023	3.17	1.58	21	3.5	1.86	21	4.0%	-0.19 [-0.79 , 0.42]
Salli 2010	3.35	1.868782	50	6.5	1.8	25	4.2%	-1.69 [-2.24 , -1.13]
Samuel 2014	7.62	2.24	37	13.42	2.91	36	4.0%	-2.21 [-2.80 , -1.62]
Shabbir 2022	2.78	0.94	18	4.22	1.16	18	3.5%	-1.33 [-2.06 , -0.60]
Thomas 2002	-1.27	3.6	470	-0.46	3.6	316	5.7%	-0.22 [-0.37 , -0.08]
Van Baar 1998	-27.4	28.7	54	-11.7	28.5	59	5.0%	-0.55 [-0.92 , -0.17]
Subtotal			**1664**			**1359**	**100.0%**	**-0.67 [-0.89 , -0.46]**

Test for overall effect: Z = 6.24 (P < 0.00001)
Heterogeneity: Tau² = 0.20; Chi² = 144.90, df = 22 (P < 0.00001); I² = 85%

Test for subgroup differences: Chi² = 6.94, df = 2 (P = 0.03), I² = 71.2%



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Analysis 3.4. Comparison 3: Sensitivity analyses, Outcome 4: Attrition bias: self-reported physical function

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.4.1 A) Exercise vs attention control/placebo								
Baker 2001	462	372.312504	23	664	422.031255	23	7.0%	-0.50 [-1.09 , 0.09]
Bandak 2022	-6.9	14.139307	102	-6	14.277255	104	12.2%	-0.06 [-0.34 , 0.21]
Brismée 2007	31.82	14	22	37.77	11.22	19	6.6%	-0.46 [-1.08 , 0.17]
Cheung 2017	21.746667	8.82397	60	25.2	7.978125	23	8.5%	-0.40 [-0.88 , 0.09]
Ettinger 1997	1.730069	0.480946	290	1.9	0.488262	149	13.5%	-0.35 [-0.55 , -0.15]
Foley 2003	27	9.28	35	37	13.14	35	8.4%	-0.87 [-1.36 , -0.38]
Kuntz 2018	-74.414286	15.188772	21	-56.3	23.1	10	4.8%	-0.98 [-1.78 , -0.18]
Lu 2017	15.85	7.6	23	26.89	16.56	23	6.8%	-0.84 [-1.45 , -0.24]
Messier 2021	16.355731	10.87412	253	16.4	11.251198	124	13.2%	-0.00 [-0.22 , 0.21]
Simao 2012	766.7839	594.19	11	686.625	419.3	12	4.6%	0.15 [-0.67 , 0.97]
Tsai 2013	-6.82	8.858583	28	-1.3	6.041657	27	7.6%	-0.72 [-1.26 , -0.17]
Ye 2020	9.52	6.04	25	16.24	6.82	25	6.9%	-1.03 [-1.62 , -0.43]
Subtotal			**893**			**574**	**100.0%**	**-0.45 [-0.66 , -0.24]**
Test for overall effect: Z = 4.17 (P < 0.0001)								
Heterogeneity: Tau² = 0.08; Chi² = 31.43, df = 11 (P = 0.0009); I² = 65%								
3.4.2 B) Exercise vs no treatment/usual care/limited education								
Allen 2018	-3.74	9.764895	142	-2.3	8.923718	68	4.6%	-0.15 [-0.44 , 0.14]
Bennell 2010	16.2	11.7	45	21.9	11	44	4.1%	-0.50 [-0.92 , -0.08]
Braghin 2018	6.96	8.08	15	13.88	21.35	16	3.1%	-0.41 [-1.13 , 0.30]
Cheung 2014	22	9.758074	18	26.2	9.758074	18	3.3%	-0.42 [-1.08 , 0.24]
Christensen 2015	-7.6	21.617937	64	-9	21.617937	64	4.4%	0.06 [-0.28 , 0.41]
De Rooij 2017	26.3	12.7	63	31.4	13.4	63	4.4%	-0.39 [-0.74 , -0.04]
Fransen 2001	-7.7	19.9	83	0.1	20.5	43	4.3%	-0.39 [-0.76 , -0.01]
Fransen 2007	-5.04	10.25	56	2.07	9.06	41	4.2%	-0.72 [-1.14 , -0.31]
Gur 2002	-13.8	4.1	17	1	2.5	6	1.3%	-3.77 [-5.29 , -2.26]
Hay 2006	24.27	15.2	109	30.18	12.8	108	4.6%	-0.42 [-0.69 , -0.15]
Jan 2009	11.2	10.1	71	25	11.8	35	4.1%	-1.28 [-1.72 , -0.84]
Janyacharoen 2018	-74.4	2.9	20	-58.1	4.9	20	2.0%	-3.97 [-5.07 , -2.86]
Jorge 2015	17.3	12.4	29	26.7	10.2	31	3.8%	-0.82 [-1.35 , -0.29]
Kim 2020	33.22	9.550079	40	42.05	10.24578	40	4.0%	-0.88 [-1.34 , -0.42]
Krishnan 2021	-53.13	4.24	15	-52.4	3.87	15	3.1%	-0.17 [-0.89 , 0.54]
Lee 2009	14.7	13.8	29	20.8	15	15	3.4%	-0.42 [-1.05 , 0.21]
Lim 2008	-6.5	10.6	53	-2.6	10.9	54	4.3%	-0.36 [-0.74 , 0.02]
Lin 2009	10.1	8.3	36	24.9	11.8	36	3.8%	-1.44 [-1.96 , -0.91]
Lund 2008	-64.1	11.5	25	-61.1	11.431535	27	3.7%	-0.26 [-0.80 , 0.29]
O'Reilly 1999	-3.55	7.961319	78	-0.01	9.30835	113	4.6%	-0.40 [-0.69 , -0.11]
Øiestad 2023	-71	23.891647	112	-63	21	56	4.5%	-0.35 [-0.67 , -0.02]
Robbins 2022	-4.63	3.86	43	0.22	2.94	43	4.0%	-1.40 [-1.87 , -0.93]
Schilke 1996	7.64	3.5	10	9.48	4.04	10	2.6%	-0.47 [-1.36 , 0.42]
Thorstensson 2005	-2	11.515611	30	0.6	17.44806	31	3.9%	-0.17 [-0.68 , 0.33]
Vassao 2021	9.5	7.38	14	19.8	11.45	14	2.8%	-1.04 [-1.83 , -0.24]
Wallis 2017	38	9	23	34	11	23	3.6%	0.39 [-0.19 , 0.98]
Wang 2011	-82	14	28	-69	18	28	3.7%	-0.79 [-1.34 , -0.25]
Subtotal			**1268**			**1062**	**100.0%**	**-0.64 [-0.84 , -0.43]**
Test for overall effect: Z = 6.11 (P < 0.00001)								
Heterogeneity: Tau² = 0.22; Chi² = 133.00, df = 26 (P < 0.00001); I² = 80%								
3.4.3 C) Exercise when added to another co-intervention								
Alghadir 2019	49.15	3.924609	46	51.6	3.6	22	4.3%	-0.63 [-1.15 , -0.11]
Anadkat 2015	22.41	8.09	20	25.98	13.19	20	3.8%	-0.32 [-0.94 , 0.30]
Anwer 2014	8.05	2.33	21	18.05	4.3	21	2.6%	-2.84 [-3.72 , -1.96]
Bennell 2016	15.4	9.2	73	23.5	10.6	74	5.4%	-0.81 [-1.15 , -0.47]
Bennell 2022a	12.8	9.5	172	19.8	10.3	68	5.7%	-0.72 [-1.01 , -0.43]
Bennell 2022b	15.8	10.7	107	19.3	11.5	105	5.8%	-0.31 [-0.59 , -0.04]
Bhagat 2017	23.266	9.169	15	41.933	15.586	15	2.9%	-1.42 [-2.23 , -0.61]
Chen 2021	8.19	1.87	16	12.13	3.28	16	3.0%	-1.44 [-2.23 , -0.65]
Dwyer 2015	378.9	328.073163	28	389.7	256.68993	27	4.3%	-0.04 [-0.56 , 0.49]
Hinman 2020	18.4	11.3	87	22	12.5	88	5.6%	-0.30 [-0.60 , -0.00]
Holm 2020	67	12.6484	45	68.1	13.646958	45	4.9%	-0.08 [-0.50 , 0.33]
Knoop 2013	17.4	11.6	80	19.3	11.4	79	5.5%	-0.16 [-0.48 , 0.15]
Lee 2019	18.9	6	15	22.5	7.2	15	3.2%	-0.53 [-1.26 , 0.20]
Mehwish 2023	-39.04	11.01	22	-42.36	12.84	22	3.9%	0.27 [-0.32 , 0.87]
Messier 2013	14.2	11.543853	152	17.7	12.791837	152	6.0%	-0.29 [-0.51 , -0.06]
Nelligan 2021	16.6	13	103	20.7	13.9	103	5.7%	-0.30 [-0.58 , -0.03]

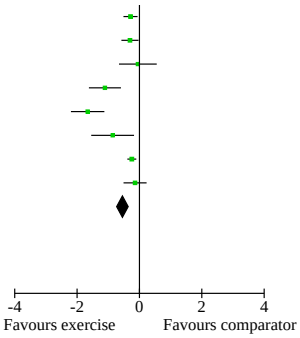
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Analysis 3.4. (Continued)

Messier 2013	14.2	11.543853	152	17.7	12.791837	152	6.0%	-0.29 [-0.51 , -0.06]
Nelligan 2021	16.6	13	103	20.7	13.9	103	5.7%	-0.30 [-0.58 , -0.03]
Qiu 2023	15.44	11.65	21	16	10.07	21	3.9%	-0.05 [-0.66 , 0.55]
Salli 2010	20.65	10.202166	50	32.6	11.6	25	4.4%	-1.11 [-1.62 , -0.59]
Samuel 2014	28.81	6.21	37	40.83	8.02	36	4.2%	-1.66 [-2.20 , -1.13]
Shabbir 2022	21.94	5.59	18	27.22	6.39	18	3.4%	-0.86 [-1.55 , -0.17]
Thomas 2002	-2.59	10.5	470	-0.02	10.5	316	6.3%	-0.24 [-0.39 , -0.10]
Van Baar 1998	-1.3	5.7	54	-0.5	5.6	59	5.2%	-0.14 [-0.51 , 0.23]
Subtotal			**1652**			**1347**	**100.0%**	**-0.55 [-0.73 , -0.36]**

Test for overall effect: Z = 5.79 (P < 0.00001)
Heterogeneity: Tau² = 0.14; Chi² = 105.04, df = 21 (P < 0.00001); I² = 80%

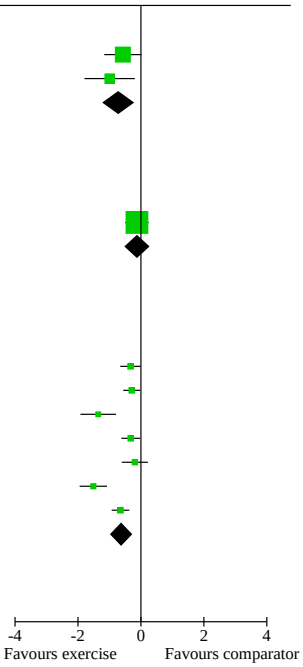
Test for subgroup differences: Chi² = 1.55, df = 2 (P = 0.46), I² = 0%



Analysis 3.5. Comparison 3: Sensitivity analyses, Outcome 5: Detection bias: knee pain severity

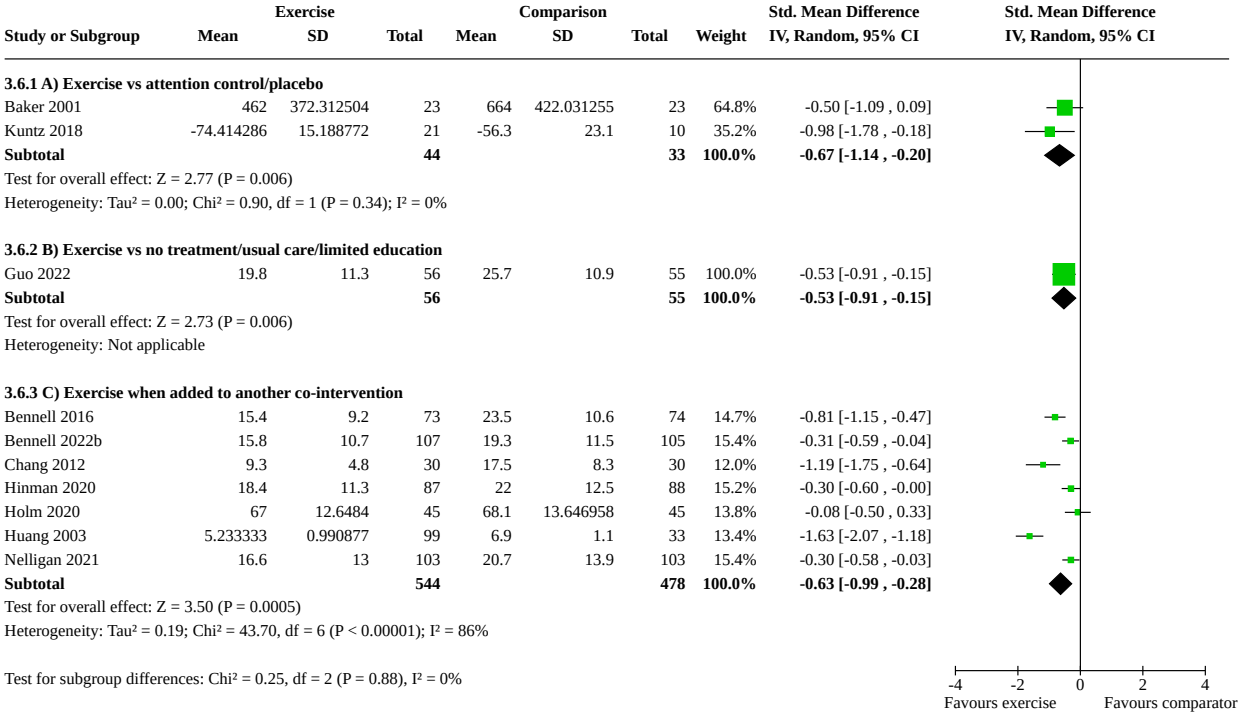
Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
3.5.1 A) Exercise vs attention control/placebo								
Baker 2001	128	95.968751	23	189	112.156251	23	64.6%	-0.57 [-1.17 , 0.02]
Kuntz 2018	-67.838095	13.166567	21	-49.9	24.7	10	35.4%	-0.99 [-1.79 , -0.20]
Subtotal			**44**			**33**	**100.0%**	**-0.72 [-1.20 , -0.25]**
Test for overall effect: Z = 2.98 (P = 0.003)								
Heterogeneity: Tau² = 0.00; Chi² = 0.69, df = 1 (P = 0.41); I² = 0%								
3.5.2 B) Exercise vs no treatment/usual care/limited education								
Guo 2022	3.52	1.71	56	3.75	1.89	55	100.0%	-0.13 [-0.50 , 0.25]
Subtotal			**56**			**55**	**100.0%**	**-0.13 [-0.50 , 0.25]**
Test for overall effect: Z = 0.67 (P = 0.50)								
Heterogeneity: Not applicable								
3.5.3 C) Exercise when added to another co-intervention								
Bennell 2016	26.4	18.4	73	33.2	22.3	74	14.9%	-0.33 [-0.66 , -0.01]
Bennell 2022b	4.2	2.4	107	4.9	2.4	105	15.6%	-0.29 [-0.56 , -0.02]
Chang 2012	2	0.9	30	3.7	1.5	30	11.5%	-1.36 [-1.92 , -0.79]
Hinman 2020	3.5	2.1	87	4.2	2.2	88	15.3%	-0.32 [-0.62 , -0.03]
Holm 2020	58.5	14.312663	45	61.2	13.314105	45	13.7%	-0.19 [-0.61 , 0.22]
Huang 2003	3.1	0.957143	99	4.4	0.4	33	13.4%	-1.51 [-1.95 , -1.08]
Nelligan 2021	3.5	2.2	103	5	2.4	103	15.5%	-0.65 [-0.93 , -0.37]
Subtotal			**544**			**478**	**100.0%**	**-0.63 [-0.96 , -0.30]**
Test for overall effect: Z = 3.77 (P = 0.0002)								
Heterogeneity: Tau² = 0.16; Chi² = 37.30, df = 6 (P < 0.00001); I² = 84%								

Test for subgroup differences: Chi² = 5.27, df = 2 (P = 0.07), I² = 62.0%



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Analysis 3.6. Comparison 3: Sensitivity analyses, Outcome 6: Detection bias: self-reported physical function



Comparison 4. Subgroup analysis: type of exercise

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1 Exercise vs attention control/placebo (pain severity)	23		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
4.1.1 Mixed	5	566	Std. Mean Difference (IV, Random, 95% CI)	-0.33 [-0.53, -0.13]
4.1.2 Stretching	2	72	Std. Mean Difference (IV, Random, 95% CI)	-2.50 [-3.17, -1.84]
4.1.3 Balance/proprioceptive	0	0	Std. Mean Difference (IV, Random, 95% CI)	Not estimable
4.1.4 Mind-body	6	294	Std. Mean Difference (IV, Random, 95% CI)	-0.52 [-0.81, -0.23]
4.1.5 Aerobic	1	23	Std. Mean Difference (IV, Random, 95% CI)	-0.11 [-0.93, 0.71]
4.1.6 Strengthening	9	1159	Std. Mean Difference (IV, Random, 95% CI)	-0.23 [-0.47, 0.01]
4.2 Exercise vs attention control/placebo (physical function)	19		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only

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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.2.1 Mixed	6	600	Std. Mean Difference (IV, Random, 95% CI)	-0.80 [-1.25, -0.35]
4.2.2 Stretching	0	0	Std. Mean Difference (IV, Random, 95% CI)	Not estimable
4.2.3 Balance/proprioceptive	0	0	Std. Mean Difference (IV, Random, 95% CI)	Not estimable
4.2.4 Mind-body	6	294	Std. Mean Difference (IV, Random, 95% CI)	-0.67 [-0.96, -0.37]
4.2.5 Aerobic	1	23	Std. Mean Difference (IV, Random, 95% CI)	0.15 [-0.67, 0.97]
4.2.6 Strengthening	6	860	Std. Mean Difference (IV, Random, 95% CI)	-0.26 [-0.56, 0.04]
4.3 Exercise vs no treatment/usual care/limited education (pain severity)	52		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
4.3.1 Mixed	18	1621	Std. Mean Difference (IV, Random, 95% CI)	-0.51 [-0.65, -0.38]
4.3.2 Stretching	1	18	Std. Mean Difference (IV, Random, 95% CI)	-0.50 [-1.47, 0.46]
4.3.3 Balance/proprioceptive	1	22	Std. Mean Difference (IV, Random, 95% CI)	-0.83 [-1.71, 0.05]
4.3.4 Mind-body	7	621	Std. Mean Difference (IV, Random, 95% CI)	-1.05 [-1.64, -0.46]
4.3.5 Aerobic	6	320	Std. Mean Difference (IV, Random, 95% CI)	-1.10 [-2.07, -0.14]
4.3.6 Strengthening	19	1303	Std. Mean Difference (IV, Random, 95% CI)	-0.76 [-1.04, -0.47]
4.4 Exercise vs no treatment/usual care/limited education (physical function)	50		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
4.4.1 Mixed	19	1967	Std. Mean Difference (IV, Random, 95% CI)	-0.43 [-0.55, -0.32]
4.4.2 Stretching	1	18	Std. Mean Difference (IV, Random, 95% CI)	-0.48 [-1.44, 0.49]
4.4.3 Balance/proprioceptive	0	0	Std. Mean Difference (IV, Random, 95% CI)	Not estimable

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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.4.4 Mind-body	6	337	Std. Mean Difference (IV, Random, 95% CI)	-1.16 [-1.90, -0.43]
4.4.5 Aerobic	6	320	Std. Mean Difference (IV, Random, 95% CI)	-1.03 [-1.97, -0.09]
4.4.6 Strengthening	18	1324	Std. Mean Difference (IV, Random, 95% CI)	-0.89 [-1.23, -0.54]
4.5 Exercise when added to another co-intervention (compared to that co-intervention alone) (pain severity)	46		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
4.5.1 Mixed	12	1300	Std. Mean Difference (IV, Random, 95% CI)	-0.42 [-0.61, -0.24]
4.5.2 Stretching	0	0	Std. Mean Difference (IV, Random, 95% CI)	Not estimable
4.5.3 Balance/proprioceptive	1	44	Std. Mean Difference (IV, Random, 95% CI)	-0.97 [-1.60, -0.34]
4.5.4 Mind-body	1	212	Std. Mean Difference (IV, Random, 95% CI)	-0.29 [-0.56, -0.02]
4.5.5 Aerobic	9	362	Std. Mean Difference (IV, Random, 95% CI)	-0.90 [-1.39, -0.41]
4.5.6 Strengthening	23	2475	Std. Mean Difference (IV, Random, 95% CI)	-0.87 [-1.16, -0.58]
4.6 Exercise when added to another co-intervention (compared to that co-intervention alone) (physical function)	43		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
4.6.1 Mixed	13	1372	Std. Mean Difference (IV, Random, 95% CI)	-0.57 [-0.87, -0.27]
4.6.2 Stretching	0	0	Std. Mean Difference (IV, Random, 95% CI)	Not estimable
4.6.3 Balance/proprioceptive	1	44	Std. Mean Difference (IV, Random, 95% CI)	-1.01 [-1.64, -0.38]
4.6.4 Mind-body	1	212	Std. Mean Difference (IV, Random, 95% CI)	-0.31 [-0.59, -0.04]
4.6.5 Aerobic	7	298	Std. Mean Difference (IV, Random, 95% CI)	-1.02 [-1.43, -0.62]

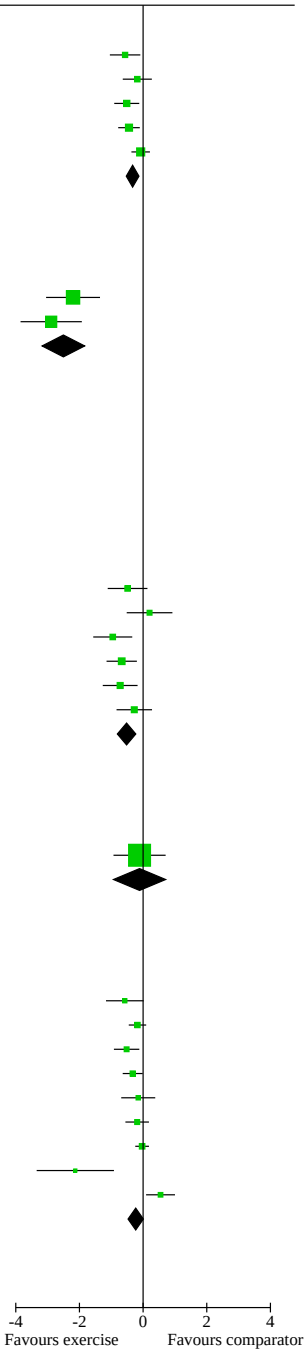
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Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.6.6 Strengthening	21	2407	Std. Mean Difference (IV, Random, 95% CI)	-0.71 [-0.97, -0.46]

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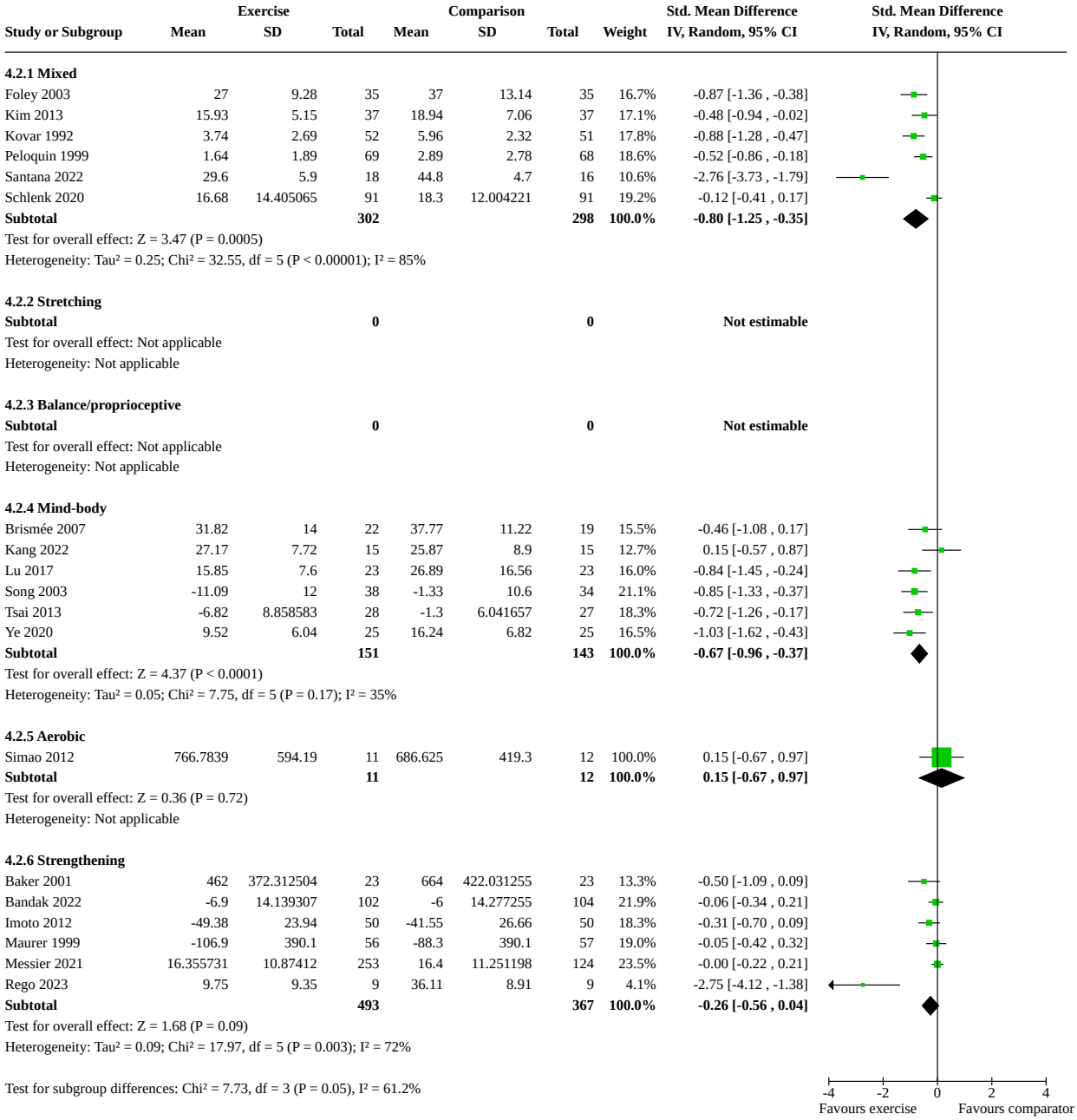
Analysis 4.1. Comparison 4: Subgroup analysis: type of exercise, Outcome 1: Exercise vs attention control/placebo (pain severity)

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
4.1.1 Mixed								
Foley 2003	8	3.86	35	10	3.09	35	14.0%	-0.57 [-1.04 , -0.09]
Kim 2013	33.77	21.91	37	37.86	22.58	37	15.1%	-0.18 [-0.64 , 0.27]
Kovar 1992	3.77	1.73	52	4.77	2.12	51	19.0%	-0.51 [-0.91 , -0.12]
Peloquin 1999	3.09	1.54	69	3.94	2.22	68	23.4%	-0.44 [-0.78 , -0.10]
Schlenk 2020	4.25	4.801688	91	4.54	2.400844	91	28.6%	-0.08 [-0.37 , 0.21]
Subtotal			**284**			**282**	**100.0%**	**-0.33 [-0.53 , -0.13]**
Test for overall effect: Z = 3.25 (P = 0.001)								
Heterogeneity: Tau² = 0.01; Chi² = 5.51, df = 4 (P = 0.24); I² = 27%								
4.1.2 Stretching								
Shen 2022	2.6	2.1	18	6.9	1.7	18	55.8%	-2.20 [-3.05 , -1.35]
Song 2020	0.55	0.48	18	3.21	1.18	18	44.2%	-2.89 [-3.85 , -1.92]
Subtotal			**36**			**36**	**100.0%**	**-2.50 [-3.17 , -1.84]**
Test for overall effect: Z = 7.34 (P < 0.00001)								
Heterogeneity: Tau² = 0.02; Chi² = 1.10, df = 1 (P = 0.29); I² = 9%								
4.1.3 Balance/proprioceptive								
Subtotal			**0**			**0**		**Not estimable**
Test for overall effect: Not applicable								
Heterogeneity: Not applicable								
4.1.4 Mind-body								
Brismée 2007	2.41	2.05	22	3.37	1.78	19	15.1%	-0.49 [-1.11 , 0.14]
Kang 2022	8.58	3.75	15	7.87	3.02	15	12.3%	0.20 [-0.51 , 0.92]
Lu 2017	5.15	3.24	23	9.37	5.23	23	15.5%	-0.95 [-1.57 , -0.34]
Song 2003	-2.45	3.9	38	0.61	5.1	34	21.4%	-0.67 [-1.15 , -0.20]
Tsai 2013	-2.6	2.540234	28	-1.02	1.681047	27	18.1%	-0.72 [-1.27 , -0.17]
Ye 2020	3.68	6.07	25	4.92	1.41	25	17.6%	-0.28 [-0.83 , 0.28]
Subtotal			**151**			**143**	**100.0%**	**-0.52 [-0.81 , -0.23]**
Test for overall effect: Z = 3.52 (P = 0.0004)								
Heterogeneity: Tau² = 0.04; Chi² = 7.45, df = 5 (P = 0.19); I² = 33%								
4.1.5 Aerobic								
Simao 2012	159.19	148.442	11	175	125.79	12	100.0%	-0.11 [-0.93 , 0.71]
Subtotal			**11**			**12**	**100.0%**	**-0.11 [-0.93 , 0.71]**
Test for overall effect: Z = 0.27 (P = 0.79)								
Heterogeneity: Not applicable								
4.1.6 Strengthening								
Baker 2001	128	95.968751	23	189	112.156251	23	8.6%	-0.57 [-1.17 , 0.02]
Bandak 2022	-10	15.149257	102	-7.3	15.297059	104	14.4%	-0.18 [-0.45 , 0.10]
Imoto 2012	4.27	2.45	50	5.74	3.14	50	11.9%	-0.52 [-0.92 , -0.12]
Jenkinson 2009	5.7	3.96	82	7.04	4.21	76	13.6%	-0.33 [-0.64 , -0.01]
Lai 2021	2.41	2.1	27	2.74	2.16	27	9.4%	-0.15 [-0.69 , 0.38]
Maurer 1999	-43.54	80.3	56	-28.49	80.3	57	12.5%	-0.19 [-0.56 , 0.18]
Messier 2021	4.700791	3.300649	253	4.8	2.9	124	15.5%	-0.03 [-0.25 , 0.18]
Rego 2023	2.75	3.01	9	10.11	3.55	9	3.2%	-2.13 [-3.34 , -0.92]
Vincent 2020	1.250345	1.630087	58	0.47	0.8	29	10.9%	0.55 [0.09 , 1.00]
Subtotal			**660**			**499**	**100.0%**	**-0.23 [-0.47 , 0.01]**
Test for overall effect: Z = 1.89 (P = 0.06)								
Heterogeneity: Tau² = 0.08; Chi² = 26.94, df = 8 (P = 0.0007); I² = 70%								
Test for subgroup differences: Chi² = 41.50, df = 4 (P < 0.00001), I² = 90.4%								



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Analysis 4.2. Comparison 4: Subgroup analysis: type of exercise, Outcome 2: Exercise vs attention control/placebo (physical function)

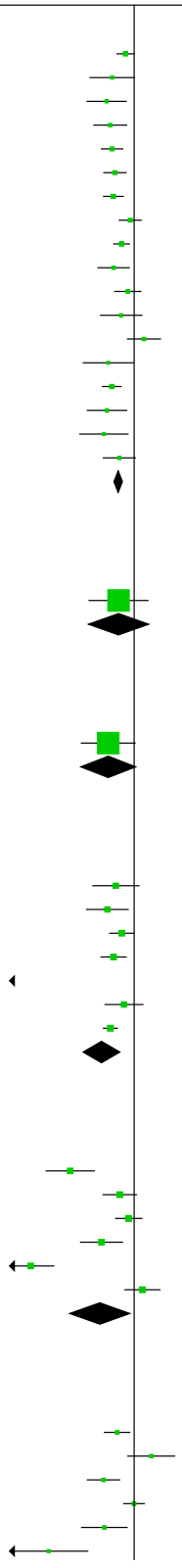


Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
4.2.1 Mixed								
Foley 2003	27	9.28	35	37	13.14	35	16.7%	-0.87 [-1.36 , -0.38]
Kim 2013	15.93	5.15	37	18.94	7.06	37	17.1%	-0.48 [-0.94 , -0.02]
Kovar 1992	3.74	2.69	52	5.96	2.32	51	17.8%	-0.88 [-1.28 , -0.47]
Peloquin 1999	1.64	1.89	69	2.89	2.78	68	18.6%	-0.52 [-0.86 , -0.18]
Santana 2022	29.6	5.9	18	44.8	4.7	16	10.6%	-2.76 [-3.73 , -1.79]
Schlenk 2020	16.68	14.405065	91	18.3	12.004221	91	19.2%	-0.12 [-0.41 , 0.17]
Subtotal			**302**			**298**	**100.0%**	**-0.80 [-1.25 , -0.35]**
Test for overall effect: Z = 3.47 (P = 0.0005)								
Heterogeneity: Tau² = 0.25; Chi² = 32.55, df = 5 (P < 0.00001); I² = 85%								
4.2.2 Stretching								
Subtotal			**0**			**0**		**Not estimable**
Test for overall effect: Not applicable								
Heterogeneity: Not applicable								
4.2.3 Balance/proprioceptive								
Subtotal			**0**			**0**		**Not estimable**
Test for overall effect: Not applicable								
Heterogeneity: Not applicable								
4.2.4 Mind-body								
Brismée 2007	31.82	14	22	37.77	11.22	19	15.5%	-0.46 [-1.08 , 0.17]
Kang 2022	27.17	7.72	15	25.87	8.9	15	12.7%	0.15 [-0.57 , 0.87]
Lu 2017	15.85	7.6	23	26.89	16.56	23	16.0%	-0.84 [-1.45 , -0.24]
Song 2003	-11.09	12	38	-1.33	10.6	34	21.1%	-0.85 [-1.33 , -0.37]
Tsai 2013	-6.82	8.858583	28	-1.3	6.041657	27	18.3%	-0.72 [-1.26 , -0.17]
Ye 2020	9.52	6.04	25	16.24	6.82	25	16.5%	-1.03 [-1.62 , -0.43]
Subtotal			**151**			**143**	**100.0%**	**-0.67 [-0.96 , -0.37]**
Test for overall effect: Z = 4.37 (P < 0.0001)								
Heterogeneity: Tau² = 0.05; Chi² = 7.75, df = 5 (P = 0.17); I² = 35%								
4.2.5 Aerobic								
Simao 2012	766.7839	594.19	11	686.625	419.3	12	100.0%	0.15 [-0.67 , 0.97]
Subtotal			**11**			**12**	**100.0%**	**0.15 [-0.67 , 0.97]**
Test for overall effect: Z = 0.36 (P = 0.72)								
Heterogeneity: Not applicable								
4.2.6 Strengthening								
Baker 2001	462	372.312504	23	664	422.031255	23	13.3%	-0.50 [-1.09 , 0.09]
Bandak 2022	-6.9	14.139307	102	-6	14.277255	104	21.9%	-0.06 [-0.34 , 0.21]
Imoto 2012	-49.38	23.94	50	-41.55	26.66	50	18.3%	-0.31 [-0.70 , 0.09]
Maurer 1999	-106.9	390.1	56	-88.3	390.1	57	19.0%	-0.05 [-0.42 , 0.32]
Messier 2021	16.355731	10.87412	253	16.4	11.251198	124	23.5%	-0.00 [-0.22 , 0.21]
Rego 2023	9.75	9.35	9	36.11	8.91	9	4.1%	-2.75 [-4.12 , -1.38]
Subtotal			**493**			**367**	**100.0%**	**-0.26 [-0.56 , 0.04]**
Test for overall effect: Z = 1.68 (P = 0.09)								
Heterogeneity: Tau² = 0.09; Chi² = 17.97, df = 5 (P = 0.003); I² = 72%								
Test for subgroup differences: Chi² = 7.73, df = 3 (P = 0.05), I² = 61.2%								

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Analysis 4.3. Comparison 4: Subgroup analysis: type of exercise, Outcome 3: Exercise vs no treatment/usual care/ limited education (pain severity)

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
4.3.1 Mixed								
Allen 2018	-1.59	3.405658	142	-0.66	3.098513	68	9.2%	-0.28 [-0.57 , 0.01]
Braghin 2018	8	8.52	15	20.94	23.47	16	2.9%	-0.70 [-1.43 , 0.02]
Da Silva 2015	2.6	1.55	19	4	1.56	22	3.5%	-0.88 [-1.53 , -0.24]
De Matos 2019	8	10.99	28	22.33	23.59	28	4.5%	-0.77 [-1.31 , -0.22]
De Rooij 2017	4.3	2	63	5.8	2.2	63	7.5%	-0.71 [-1.07 , -0.35]
Fransen 2001	-10.6	19.5	83	1.5	19.4	43	7.2%	-0.62 [-0.99 , -0.24]
Gohir 2021	-1.8	2.254845	67	-0.3	2.232264	79	8.1%	-0.67 [-1.00 , -0.33]
Guo 2022	3.52	1.71	56	3.75	1.89	55	7.3%	-0.13 [-0.50 , 0.25]
Hay 2006	7.36	4.3	109	8.99	3.7	108	9.8%	-0.40 [-0.67 , -0.14]
Henriksen 2014	-6.1	10.08716	31	0.7	10.252914	29	4.8%	-0.66 [-1.18 , -0.14]
Hopman-Rock 2000	-0.7	24.1	45	4	21.2	37	6.1%	-0.20 [-0.64 , 0.23]
Keefe 2004	3.19	1.85	16	4.03	2.08	18	3.2%	-0.42 [-1.10 , 0.27]
Lund 2008	-51.5	20.5	25	-58.1	20.78461	27	4.5%	0.31 [-0.23 , 0.86]
Rogind 1998	4.2	2.1	12	6	2.1	13	2.3%	-0.83 [-1.65 , -0.01]
Rosedale 2014	16	10	120	23	9	60	8.5%	-0.72 [-1.04 , -0.40]
Takacs 2017	2.8	1.7	20	4.6	2.3	20	3.4%	-0.87 [-1.52 , -0.22]
Vassao 2021	3.36	2.87	14	7	4.27	14	2.5%	-0.97 [-1.76 , -0.18]
Wang 2011	-76	15	28	-68	18	28	4.7%	-0.48 [-1.01 , 0.06]
Subtotal			**893**			**728**	**100.0%**	**-0.51 [-0.65 , -0.38]**
Test for overall effect: Z = 7.36 (P < 0.00001)								
Heterogeneity: Tau² = 0.03; Chi² = 27.51, df = 17 (P = 0.05); I² = 38%								
4.3.2 Stretching								
Chopp-Hurley 2017	-81	24.5	7	-70.8	15.5	11	100.0%	-0.50 [-1.47 , 0.46]
Subtotal			**7**			**11**	**100.0%**	**-0.50 [-1.47 , 0.46]**
Test for overall effect: Z = 1.02 (P = 0.31)								
Heterogeneity: Not applicable								
4.3.3 Balance/proprioceptive								
Sekir 2005	1.9	2.47	12	4.1	2.63	10	100.0%	-0.83 [-1.71 , 0.05]
Subtotal			**12**			**10**	**100.0%**	**-0.83 [-1.71 , 0.05]**
Test for overall effect: Z = 1.85 (P = 0.06)								
Heterogeneity: Not applicable								
4.3.4 Mind-body								
An 2008	71.1	110.1	14	138.2	112.6	14	13.7%	-0.59 [-1.34 , 0.17]
Cheung 2014	5.8	2.842569	18	8.3	2.842569	18	14.3%	-0.86 [-1.55 , -0.17]
Fransen 2007	-1.67	3.28	56	-0.5	2.37	41	16.4%	-0.40 [-0.80 , 0.01]
Hu 2020	2.53	1.61	52	3.6	1.6	40	16.3%	-0.66 [-1.08 , -0.24]
Janyacharoen 2018	-69.2	5.1	20	-43.1	1.2	20	7.1%	-6.91 [-8.62 , -5.20]
Lee 2009	4.6	4	29	5.9	3.7	15	14.8%	-0.33 [-0.95 , 0.30]
Xiao 2021	75.7	20.19	142	94.25	27.73	142	17.3%	-0.76 [-1.00 , -0.52]
Subtotal			**331**			**290**	**100.0%**	**-1.05 [-1.64 , -0.46]**
Test for overall effect: Z = 3.49 (P = 0.0005)								
Heterogeneity: Tau² = 0.51; Chi² = 54.67, df = 6 (P < 0.00001); I² = 89%								
4.3.5 Aerobic								
Drummen 2023	16.6	20.085816	24	56.4	17.2	16	16.1%	-2.05 [-2.84 , -1.26]
Li 2020	-73.1	15.3	26	-65.9	15.6	25	17.0%	-0.46 [-1.02 , 0.10]
Multanen 2014	-2	14.070612	40	0	7.817007	40	17.4%	-0.17 [-0.61 , 0.27]
Salacinski 2012	-81.4	13.4	19	-65.7	15.9	18	16.5%	-1.05 [-1.74 , -0.35]
Shahine 2020	17.5	18.1	33	77	17.3	33	16.2%	-3.32 [-4.08 , -2.56]
Wallis 2017	6.5	1.2	23	6.1	1.7	23	16.9%	0.27 [-0.31 , 0.85]
Subtotal			**165**			**155**	**100.0%**	**-1.10 [-2.07 , -0.14]**
Test for overall effect: Z = 2.24 (P = 0.03)								
Heterogeneity: Tau² = 1.35; Chi² = 74.81, df = 5 (P < 0.00001); I² = 93%								
4.3.6 Strengthening								
Bennell 2010	2.6	2.1	45	3.9	2.6	44	6.1%	-0.55 [-0.97 , -0.12]
Bruce-Brand 2012	10.78	4.31	14	8.33	4.36	13	4.6%	0.55 [-0.22 , 1.32]
Chen 2014	4.2	0.9	30	5.2	1.1	30	5.6%	-0.98 [-1.52 , -0.44]
Christensen 2015	-5.6	19.816442	64	-5.5	20.016608	64	6.3%	-0.00 [-0.35 , 0.34]
DeVita 2018	3.64	2.55	16	7.33	4.72	15	4.7%	-0.96 [-1.71 , -0.21]
Gur 2002	-20.9	8.3	17	0.7	4.6	6	2.9%	-2.74 [-4.02 , -1.47]



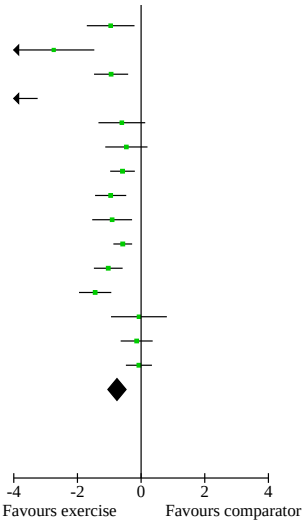
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Analysis 4.3. (Continued)

DeVita 2018	3.64	2.55	16	7.33	4.72	15	4.7%	-0.96 [-1.71 , -0.21]
Gur 2002	-20.9	8.3	17	0.7	4.6	6	2.9%	-2.74 [-4.02 , -1.47]
Jorge 2015	4.3	3.1	29	6.6	1.5	31	5.6%	-0.94 [-1.48 , -0.41]
Karadag 2019	4.33	0.8	17	7.64	0.61	18	2.8%	-4.56 [-5.88 , -3.25]
Krishnan 2021	-53.93	3.17	15	-50.93	6.04	15	4.8%	-0.61 [-1.34 , 0.13]
Lee 2023	34.5	21.69	18	43.9	17.85	18	5.1%	-0.46 [-1.13 , 0.20]
Lim 2008	-9	12	53	-1.7	12.8	54	6.2%	-0.58 [-0.97 , -0.20]
Lin 2009	4.2	3	36	7.3	3.4	36	5.8%	-0.96 [-1.45 , -0.47]
Mazloum 2018	8.739394	1.950952	33	10.5	1.8	16	5.2%	-0.91 [-1.54 , -0.28]
O'Reilly 1999	-1.45	2.616812	78	0.42	3.594579	113	6.5%	-0.58 [-0.87 , -0.28]
Robbins 2022	-2.59	2.35	43	-0.27	2.1	43	5.9%	-1.03 [-1.48 , -0.58]
Sadeghi 2023	35.133333	22.117119	72	67	21.22	24	5.7%	-1.44 [-1.95 , -0.94]
Schilke 1996	9.7	4.72	10	10.1	6.44	10	4.2%	-0.07 [-0.94 , 0.81]
Thorstensson 2005	-1.8	13.390246	30	0.3	16.221243	31	5.7%	-0.14 [-0.64 , 0.36]
Topp 2002	10.552388	3.131421	67	10.77	3.194683	35	6.1%	-0.07 [-0.48 , 0.34]
Subtotal			**687**			**616**	**100.0%**	**-0.76 [-1.04 , -0.47]**

Test for overall effect: Z = 5.20 (P < 0.00001)
Heterogeneity: Tau² = 0.30; Chi² = 98.53, df = 18 (P < 0.00001); I² = 82%

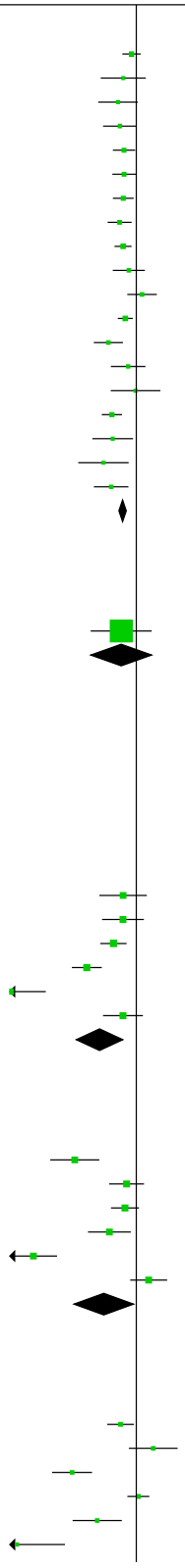
Test for subgroup differences: Chi² = 6.29, df = 5 (P = 0.28), I² = 20.6%



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Analysis 4.4. Comparison 4: Subgroup analysis: type of exercise, Outcome 4: Exercise vs no treatment/usual care/ limited education (physical function)

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
4.4.1 Mixed								
Allen 2018	-3.74	9.764895	142	-2.3	8.923718	68	8.6%	-0.15 [-0.44 , 0.14]
Braghin 2018	6.96	8.08	15	13.88	21.35	16	2.4%	-0.41 [-1.13 , 0.30]
Da Silva 2015	2.3	1.36	19	3.13	1.45	22	2.9%	-0.58 [-1.21 , 0.05]
De Matos 2019	6.57	8.28	28	15.2	21.73	28	3.8%	-0.52 [-1.05 , 0.02]
De Rooij 2017	26.3	12.7	63	31.4	13.4	63	6.9%	-0.39 [-0.74 , -0.04]
Fransen 2001	-7.7	19.9	83	0.1	20.5	43	6.4%	-0.39 [-0.76 , -0.01]
Gohir 2021	-7.8	8.404424	67	-4.3	8.482605	79	7.5%	-0.41 [-0.74 , -0.08]
Guo 2022	19.8	11.3	56	25.7	10.9	55	6.3%	-0.53 [-0.91 , -0.15]
Hay 2006	24.27	15.2	109	30.18	12.8	108	9.3%	-0.42 [-0.69 , -0.15]
Henriksen 2014	-4.2	11.586602	31	-1.4	11.830286	29	4.1%	-0.24 [-0.74 , 0.27]
Hopman-Rock 2000	-0.8	4.6	37	-1.7	5.2	34	4.7%	0.18 [-0.28 , 0.65]
Hurley 2007	20	18.4	207	25.9	13.6	104	10.4%	-0.35 [-0.58 , -0.11]
Kim 2020	33.22	9.550079	40	42.05	10.24578	40	4.8%	-0.88 [-1.34 , -0.42]
Lund 2008	-64.1	11.5	25	-61.1	11.431535	27	3.7%	-0.26 [-0.80 , 0.29]
Rogind 1998	11.3	3.6	12	11.4	4.3	13	2.0%	-0.02 [-0.81 , 0.76]
Rosedale 2014	-65	20	120	-50	18	60	7.7%	-0.77 [-1.09 , -0.45]
Takacs 2017	20	11	20	28	10	20	2.8%	-0.75 [-1.39 , -0.10]
Vassao 2021	9.5	7.38	14	19.8	11.45	14	1.9%	-1.04 [-1.83 , -0.24]
Wang 2011	-82	14	28	-69	18	28	3.7%	-0.79 [-1.34 , -0.25]
Subtotal			**1116**			**851**	**100.0%**	**-0.43 [-0.55 , -0.32]**
Test for overall effect: Z = 7.24 (P < 0.00001)								
Heterogeneity: Tau² = 0.02; Chi² = 26.26, df = 18 (P = 0.09); I² = 31%								
4.4.2 Stretching								
Chopp-Hurley 2017	-89.8	17.4	7	-80.5	19.2	11	100.0%	-0.48 [-1.44 , 0.49]
Subtotal			**7**			**11**	**100.0%**	**-0.48 [-1.44 , 0.49]**
Test for overall effect: Z = 0.97 (P = 0.33)								
Heterogeneity: Not applicable								
4.4.3 Balance/proprioceptive								
Subtotal			**0**			**0**		**Not estimable**
Test for overall effect: Not applicable								
Heterogeneity: Not applicable								
4.4.4 Mind-body								
An 2008	347.5	382.8	14	511.8	381.6	14	16.2%	-0.42 [-1.17 , 0.33]
Cheung 2014	22	9.758074	18	26.2	9.758074	18	16.8%	-0.42 [-1.08 , 0.24]
Fransen 2007	-5.04	10.25	56	2.07	9.06	41	18.3%	-0.72 [-1.14 , -0.31]
Hu 2020	2.83	6.3	52	20.3	15.2	40	18.0%	-1.56 [-2.04 , -1.09]
Janyacharoen 2018	-74.4	2.9	20	-58.1	4.9	20	13.5%	-3.97 [-5.07 , -2.86]
Lee 2009	14.7	13.8	29	20.8	15	15	17.0%	-0.42 [-1.05 , 0.21]
Subtotal			**189**			**148**	**100.0%**	**-1.16 [-1.90 , -0.43]**
Test for overall effect: Z = 3.09 (P = 0.002)								
Heterogeneity: Tau² = 0.73; Chi² = 43.32, df = 5 (P < 0.00001); I² = 88%								
4.4.5 Aerobic								
Drummen 2023	350.1	288.549892	24	894.8	251.2	16	16.1%	-1.95 [-2.72 , -1.17]
Li 2020	-75	13.1	26	-70.3	16.9	25	17.0%	-0.31 [-0.86 , 0.25]
Multanen 2014	-1	6.253605	40	1	4.690204	40	17.4%	-0.36 [-0.80 , 0.08]
Salacinski 2012	-84.2	13.9	19	-71.1	16.2	18	16.5%	-0.85 [-1.53 , -0.17]
Shahine 2020	30	14.8	33	79.9	15.5	33	16.2%	-3.25 [-4.00 , -2.51]
Wallis 2017	38	9	23	34	11	23	16.9%	0.39 [-0.19 , 0.98]
Subtotal			**165**			**155**	**100.0%**	**-1.03 [-1.97 , -0.09]**
Test for overall effect: Z = 2.14 (P = 0.03)								
Heterogeneity: Tau² = 1.28; Chi² = 72.34, df = 5 (P < 0.00001); I² = 93%								
4.4.6 Strengthening								
Bennell 2010	16.2	11.7	45	21.9	11	44	6.2%	-0.50 [-0.92 , -0.08]
Bruce-Brand 2012	33.91	12.91	14	26.11	15.33	13	5.1%	0.54 [-0.24 , 1.31]
Chen 2014	5.1	0.9	30	7.4	1.3	30	5.6%	-2.03 [-2.66 , -1.40]
Christensen 2015	-7.6	21.617937	64	-9	21.617937	64	6.4%	0.06 [-0.28 , 0.41]
DeVita 2018	6.8	9.5	16	21.1	12.9	15	5.1%	-1.24 [-2.01 , -0.46]
Gur 2002	-13.8	4.1	17	1	2.5	6	2.9%	-3.77 [-5.29 , -2.26]



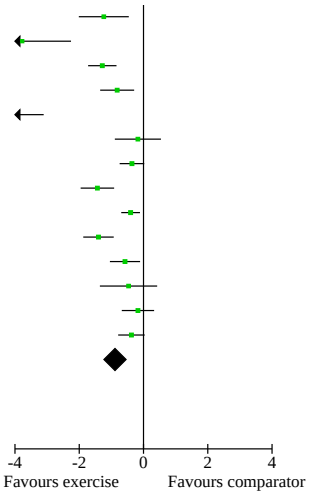
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Analysis 4.4. (Continued)

DeVita 2018	6.8	9.5	16	21.1	12.9	15	5.1%	-1.24 [-2.01 , -0.46]
Gur 2002	-13.8	4.1	17	1	2.5	6	2.9%	-3.77 [-5.29 , -2.26]
Jan 2009	11.2	10.1	71	25	11.8	35	6.2%	-1.28 [-1.72 , -0.84]
Jorge 2015	17.3	12.4	29	26.7	10.2	31	5.9%	-0.82 [-1.35 , -0.29]
Karadag 2019	21.73	4.27	17	41.23	4.42	18	3.5%	-4.38 [-5.66 , -3.11]
Krishnan 2021	-53.13	4.24	15	-52.4	3.87	15	5.3%	-0.17 [-0.89 , 0.54]
Lim 2008	-6.5	10.6	53	-2.6	10.9	54	6.3%	-0.36 [-0.74 , 0.02]
Lin 2009	10.1	8.3	36	24.9	11.8	36	5.9%	-1.44 [-1.96 , -0.91]
O'Reilly 1999	-3.55	7.961319	78	-0.01	9.30835	113	6.5%	-0.40 [-0.69 , -0.11]
Robbins 2022	-4.63	3.86	43	0.22	2.94	43	6.1%	-1.40 [-1.87 , -0.93]
Sadeghi 2023	28.79	4.057155	72	31.3	5.05	24	6.1%	-0.58 [-1.05 , -0.11]
Schilke 1996	7.64	3.5	10	9.48	4.04	10	4.7%	-0.47 [-1.36 , 0.42]
Thorstensson 2005	-2	11.515611	30	0.6	17.44806	31	6.0%	-0.17 [-0.68 , 0.33]
Topp 2002	35.62	10.739061	67	39.7	10.826426	35	6.2%	-0.38 [-0.79 , 0.04]
Subtotal			**707**			**617**	**100.0%**	**-0.89 [-1.23 , -0.54]**

Test for overall effect: Z = 5.05 (P < 0.00001)
Heterogeneity: Tau² = 0.45; Chi² = 134.42, df = 17 (P < 0.00001); I² = 87%

Test for subgroup differences: Chi² = 10.39, df = 4 (P = 0.03), I² = 61.5%



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Analysis 4.5. Comparison 4: Subgroup analysis: type of exercise, Outcome 5: Exercise when added to another co-intervention (compared to that co-intervention alone) (pain severity)

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
4.5.1 Mixed								
Bautch 1997	2.71	2.680019	17	3.51	3.009867	17	5.3%	-0.27 [-0.95 , 0.40]
Bennell 2022a	3.3	1.9	172	4.2	2.1	68	12.6%	-0.46 [-0.74 , -0.17]
Bhagat 2017	1.933	0.8837	15	3.8	1.612	15	4.1%	-1.40 [-2.21 , -0.59]
Dwyer 2015	97.7	86.780643	28	102.3	88.854206	27	7.3%	-0.05 [-0.58 , 0.48]
Hinman 2020	3.5	2.1	87	4.2	2.2	88	12.2%	-0.32 [-0.62 , -0.03]
Lee 2019	2.8	0.8	15	3.1	0.9	15	4.8%	-0.34 [-1.06 , 0.38]
Marconcin 2018	68.2	17.4	40	67.4	18.2	40	9.0%	0.04 [-0.39 , 0.48]
Messier 2013	3.7	3.431956	152	4.8	3.431956	152	14.1%	-0.32 [-0.55 , -0.09]
Nelligan 2021	3.5	2.2	103	5	2.4	103	12.7%	-0.65 [-0.93 , -0.37]
Ozlu 2023	4.11	1.34	41	5.05	1.43	41	8.8%	-0.67 [-1.12 , -0.23]
Rosadi 2023	4.246154	0.818159	26	5.2	0.6	13	4.8%	-1.24 [-1.96 , -0.51]
Shin 2017	3.08	2.36	12	2.75	1.76	13	4.2%	0.15 [-0.63 , 0.94]
Subtotal			**708**			**592**	**100.0%**	**-0.42 [-0.61 , -0.24]**
Test for overall effect: Z = 4.49 (P < 0.00001)								
Heterogeneity: Tau² = 0.05; Chi² = 23.93, df = 11 (P = 0.01); I² = 54%								
4.5.2 Stretching								
Subtotal			**0**			**0**		**Not estimable**
Test for overall effect: Not applicable								
Heterogeneity: Not applicable								
4.5.3 Balance/proprioceptive								
Kumar 2013	2.18	0.66	22	2.91	0.81	22	100.0%	-0.97 [-1.60 , -0.34]
Subtotal			**22**			**22**	**100.0%**	**-0.97 [-1.60 , -0.34]**
Test for overall effect: Z = 3.03 (P = 0.002)								
Heterogeneity: Not applicable								
4.5.4 Mind-body								
Bennell 2022b	4.2	2.4	107	4.9	2.4	105	100.0%	-0.29 [-0.56 , -0.02]
Subtotal			**107**			**105**	**100.0%**	**-0.29 [-0.56 , -0.02]**
Test for overall effect: Z = 2.10 (P = 0.04)								
Heterogeneity: Not applicable								
4.5.5 Aerobic								
Alghadir 2019	4.25	1.242176	46	4.8	0.8	22	12.3%	-0.49 [-1.00 , 0.03]
Anadkat 2015	2.75	0.96	20	4.3	1.12	20	11.0%	-1.46 [-2.16 , -0.75]
Chen 2021	1.56	0.63	16	2.25	0.86	16	10.8%	-0.89 [-1.62 , -0.16]
Gondhalekar 2013	3.53	1.33	15	4.07	1.18	15	10.9%	-0.42 [-1.14 , 0.31]
Joshi 2019	2.85	0.88	21	5.2	1.17	21	10.4%	-2.23 [-3.01 , -1.44]
Kilic 2020	3.76	1.27	25	5.88	1.51	25	11.5%	-1.50 [-2.13 , -0.86]
Sedaghatnezhad 2021	17.56	11.49	15	18.22	13.96	15	10.9%	-0.05 [-0.77 , 0.67]
Shabbir 2022	2.78	0.94	18	4.22	1.16	18	10.8%	-1.33 [-2.06 , -0.60]
Talbot 2003	1.35	0.93	17	1.2	0.95	17	11.2%	0.16 [-0.52 , 0.83]
Subtotal			**193**			**169**	**100.0%**	**-0.90 [-1.39 , -0.41]**
Test for overall effect: Z = 3.61 (P = 0.0003)								
Heterogeneity: Tau² = 0.44; Chi² = 37.02, df = 8 (P < 0.0001); I² = 78%								
4.5.6 Strengthening								
Anwer 2014	1.24	0.76	21	4.24	0.88	21	3.2%	-3.58 [-4.59 , -2.57]
Assar 2020	3.4	2.2	12	6.7	2.5	12	3.4%	-1.35 [-2.26 , -0.45]
Bennell 2016	26.4	18.4	73	33.2	22.3	74	4.8%	-0.33 [-0.66 , -0.01]
Chang 2012	2	0.9	30	3.7	1.5	30	4.3%	-1.36 [-1.92 , -0.79]
Dincer 2016	1.7	0.9	19	3.6	2.8	16	4.0%	-0.93 [-1.63 , -0.22]
Hernandez 2019	2.42	2.35	53	4	2.83	60	4.7%	-0.60 [-0.98 , -0.22]
Holm 2020	58.5	14.312663	45	61.2	13.314105	45	4.7%	-0.19 [-0.61 , 0.22]
Hsu 2021	4.48	1.44	22	4.33	1.65	22	4.2%	0.10 [-0.50 , 0.69]
Huang 2003	3.1	0.957143	99	4.4	0.4	33	4.6%	-1.51 [-1.95 , -1.08]
Huang 2005	4.1	0.6	35	4.9	1.2	35	4.5%	-0.83 [-1.32 , -0.34]
Jan 2008	4.8	3.102286	68	7.1	3.4	34	4.6%	-0.71 [-1.14 , -0.29]
Knoop 2013	2.8	2.1	80	3.3	2.1	79	4.9%	-0.24 [-0.55 , 0.08]
Mehwish 2023	5.77	1.2	22	5.24	1.09	22	4.2%	0.45 [-0.15 , 1.05]
Mete 2022	9.14	3.02	32	18.85	2.42	32	3.7%	-3.51 [-4.30 , -2.71]
Oh 2020	5.06	4.39	40	10.33	5.22	20	4.3%	-1.11 [-1.69 , -0.54]
Qiu 2023	3.17	1.58	21	3.5	1.86	21	4.2%	-0.19 [-0.79 , 0.42]
Rafia 2021b	7.795	2.277021	76	8.97	2.8	28	4.7%	-0.42 [-0.93 , 0.09]

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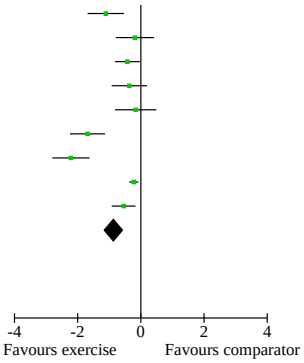
Analysis 4.5. (Continued)

Oh 2020	5.06	4.39	40	10.33	5.22	20	4.3%	-1.11 [-1.69 , -0.54]
Qiu 2023	3.17	1.58	21	3.5	1.86	21	4.2%	-0.19 [-0.79 , 0.42]
Rafiq 2021b	7.785	2.377021	76	8.87	2.8	38	4.7%	-0.43 [-0.82 , -0.03]
Rafiq 2021c	8.85	1.49	25	9.52	2.08	25	4.3%	-0.36 [-0.92 , 0.19]
Rakhshani 2022	1.89	1.02	18	2.06	1	18	4.1%	-0.16 [-0.82 , 0.49]
Salli 2010	3.35	1.868782	50	6.5	1.8	25	4.3%	-1.69 [-2.24 , -1.13]
Samuel 2014	7.62	2.24	37	13.42	2.91	36	4.3%	-2.21 [-2.80 , -1.62]
Thomas 2002	-1.27	3.6	470	-0.46	3.6	316	5.1%	-0.22 [-0.37 , -0.08]
Van Baar 1998	-27.4	28.7	54	-11.7	28.5	59	4.7%	-0.55 [-0.92 , -0.17]
Subtotal			**1402**			**1073**	**100.0%**	**-0.87 [-1.16 , -0.58]**

Test for overall effect: Z = 5.91 (P < 0.00001)

Heterogeneity: Tau² = 0.42; Chi² = 218.14, df = 22 (P < 0.00001); I² = 90%

Test for subgroup differences: Chi² = 13.66, df = 4 (P = 0.008), I² = 70.7%



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Analysis 4.6. Comparison 4: Subgroup analysis: type of exercise, Outcome 6: Exercise when added to another co-intervention (compared to that co-intervention alone) (physical function)

Study or Subgroup	Exercise Mean	Exercise SD	Exercise Total	Comparison Mean	Comparison SD	Comparison Total	Weight	Std. Mean Difference IV, Random, 95% CI
4.6.1 Mixed								
Bautch 1997	23.37	10.225302	17	17.88	7.627745	17	6.7%	0.59 [-0.09 , 1.28]
Bennell 2022a	12.8	9.5	172	19.8	10.3	68	9.5%	-0.72 [-1.01 , -0.43]
Bhagat 2017	23.266	9.169	15	41.933	15.586	15	5.9%	-1.42 [-2.23 , -0.61]
Dwyer 2015	378.9	328.073163	28	389.7	256.68993	27	7.8%	-0.04 [-0.56 , 0.49]
Hinman 2020	18.4	11.3	87	22	12.5	88	9.4%	-0.30 [-0.60 , -0.00]
Kholvadia 2019	65.6	9.9	32	70.8	9.8	40	8.2%	-0.52 [-1.00 , -0.05]
Lee 2019	18.9	6	15	22.5	7.2	15	6.4%	-0.53 [-1.26 , 0.20]
Marconcin 2018	65.7	18.8	40	73.6	18.5	40	8.4%	-0.42 [-0.86 , 0.02]
Messier 2013	14.2	11.543853	152	17.7	12.791837	152	9.8%	-0.29 [-0.51 , -0.06]
Nelligan 2021	16.6	13	103	20.7	13.9	103	9.5%	-0.30 [-0.58 , -0.03]
Ozlu 2023	23.08	8.72	41	33.14	7.85	41	8.2%	-1.20 [-1.67 , -0.73]
Rosadi 2023	29.592308	4.729053	26	45.7	0.1	13	4.1%	-4.06 [-5.22 , -2.90]
Shin 2017	14.54	10.87	12	13.67	9.49	13	6.1%	0.08 [-0.70 , 0.87]
Subtotal			**740**			**632**	**100.0%**	**-0.57 [-0.87 , -0.27]**
Test for overall effect: Z = 3.73 (P = 0.0002)								
Heterogeneity: Tau² = 0.23; Chi² = 72.84, df = 12 (P < 0.00001); I² = 84%								
4.6.2 Stretching								
Subtotal			**0**			**0**		**Not estimable**
Test for overall effect: Not applicable								
Heterogeneity: Not applicable								
4.6.3 Balance/proprioceptive								
Kumar 2013	6.95	2.34	22	10	3.49	22	100.0%	-1.01 [-1.64 , -0.38]
Subtotal			**22**			**22**	**100.0%**	**-1.01 [-1.64 , -0.38]**
Test for overall effect: Z = 3.13 (P = 0.002)								
Heterogeneity: Not applicable								
4.6.4 Mind-body								
Bennell 2022b	15.8	10.7	107	19.3	11.5	105	100.0%	-0.31 [-0.59 , -0.04]
Subtotal			**107**			**105**	**100.0%**	**-0.31 [-0.59 , -0.04]**
Test for overall effect: Z = 2.27 (P = 0.02)								
Heterogeneity: Not applicable								
4.6.5 Aerobic								
Alghadir 2019	49.15	3.924609	46	51.6	3.6	22	17.1%	-0.63 [-1.15 , -0.11]
Anadkat 2015	22.41	8.09	20	25.98	13.19	20	15.2%	-0.32 [-0.94 , 0.30]
Chen 2021	8.19	1.87	16	12.13	3.28	16	12.5%	-1.44 [-2.23 , -0.65]
Gondhalekar 2013	37.13	12.68	15	52.6	13.29	15	12.6%	-1.16 [-1.94 , -0.38]
Joshi 2019	20.79	4.2	21	31.79	6.1	21	12.9%	-2.06 [-2.82 , -1.30]
Kilic 2020	1.95	0.67	25	3.15	1.57	25	15.8%	-0.98 [-1.57 , -0.39]
Shabbir 2022	21.94	5.59	18	27.22	6.39	18	14.1%	-0.86 [-1.55 , -0.17]
Subtotal			**161**			**137**	**100.0%**	**-1.02 [-1.43 , -0.62]**
Test for overall effect: Z = 4.97 (P < 0.00001)								
Heterogeneity: Tau² = 0.18; Chi² = 15.33, df = 6 (P = 0.02); I² = 61%								
4.6.6 Strengthening								
Anwer 2014	8.05	2.33	21	18.05	4.3	21	3.4%	-2.84 [-3.72 , -1.96]
Bennell 2016	15.4	9.2	73	23.5	10.6	74	5.3%	-0.81 [-1.15 , -0.47]
Chang 2012	9.3	4.8	30	17.5	8.3	30	4.6%	-1.19 [-1.75 , -0.64]
Dincer 2016	6.7	3.7	19	19.8	13.8	16	3.9%	-1.32 [-2.06 , -0.58]
Hernandez 2019	7.15	6.21	53	9.95	5.27	60	5.2%	-0.49 [-0.86 , -0.11]
Holm 2020	67	12.6484	45	68.1	13.646958	45	5.1%	-0.08 [-0.50 , 0.33]
Hsu 2021	19.95	4.79	22	19.43	5.29	22	4.4%	0.10 [-0.49 , 0.69]
Huang 2003	5.233333	0.990877	99	6.9	1.1	33	5.0%	-1.63 [-2.07 , -1.18]
Huang 2005	6.1	0.9	35	6.9	1.3	35	4.8%	-0.71 [-1.19 , -0.22]
Jan 2008	14.75	8.790718	68	22.5	10.9	34	5.0%	-0.81 [-1.23 , -0.38]
Knoop 2013	17.4	11.6	80	19.3	11.4	79	5.4%	-0.16 [-0.48 , 0.15]
Mehwish 2023	-39.04	11.01	22	-42.36	12.84	22	4.4%	0.27 [-0.32 , 0.87]
Oh 2020	16.22	10.87	40	30.89	14.09	20	4.5%	-1.20 [-1.79 , -0.62]
Qiu 2023	15.44	11.65	21	16	10.07	21	4.4%	-0.05 [-0.66 , 0.55]
Rafiq 2021a	26	21.09	28	37	16.79	28	4.6%	-0.57 [-1.10 , -0.03]
Rafiq 2021b	-6.415	1.374707	76	-4.65	1.39	38	5.0%	-1.27 [-1.69 , -0.85]
Rafiq 2021c	23.9	12.37	25	22.07	11	25	4.6%	0.15 [-0.40 , 0.71]

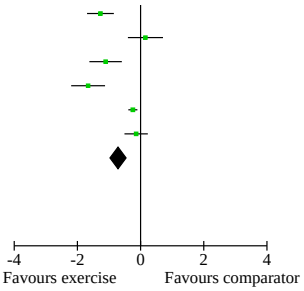
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Analysis 4.6. (Continued)

Rafiq 2021b	-6.415	1.374707	76	-4.65	1.39	38	5.0%	-1.27 [-1.69 , -0.85]
Rafiq 2021c	23.9	12.37	25	22.07	11	25	4.6%	0.15 [-0.40 , 0.71]
Salli 2010	20.65	10.202166	50	32.6	11.6	25	4.7%	-1.11 [-1.62 , -0.59]
Samuel 2014	28.81	6.21	37	40.83	8.02	36	4.6%	-1.66 [-2.20 , -1.13]
Thomas 2002	-2.59	10.5	470	-0.02	10.5	316	5.7%	-0.24 [-0.39 , -0.10]
Van Baar 1998	-1.3	5.7	54	-0.5	5.6	59	5.2%	-0.14 [-0.51 , 0.23]
Subtotal			**1368**			**1039**	**100.0%**	**-0.71 [-0.97 , -0.46]**

Test for overall effect: Z = 5.50 (P < 0.00001)
Heterogeneity: Tau² = 0.29; Chi² = 152.72, df = 20 (P < 0.00001); I² = 87%

Test for subgroup differences: Chi² = 10.81, df = 4 (P = 0.03), I² = 63.0%



ADDITIONAL TABLES

Table 1. GRADE certainty assessment rubric

Certainty assessment domain	Standard downgrade	A priori domain cut-offs
Risk of bias	**Serious** = -1 **Very serious** = -2	**Serious:** ≥ 50% of trials received "high" risk of bias ratings (≥ 1 out of 6 dimensions in the Cochrane risk of bias tool) (-1); 33% to 100% of trials received "unclear" risk of bias ratings (≥ 3 out of 6 dimensions in the Cochrane risk of bias tool) (-1) **Very serious:** > 25% of trials received multiple (≥ 2 out of 6 dimensions in the Cochrane risk of bias tool) "high" risk of bias ratings (-2); > 50% of trials received overall (≥ 4 out of 6 dimensions in the Cochrane risk of bias tool) "unclear" risk of bias ratings (-2)
Inconsistency	**Serious** = -1 **Very serious** = -2	**Serious:** I^2 > 50% and ≤ 75%, and judged to be unexplained heterogeneity (e.g. not due to differences in intensity/length/type of exercise intervention) (-1) **Very serious:** I^2 > 75%, and judged to be unexplained heterogeneity (-2)
Indirectness	**Serious** = -1 **Very serious** = -2	**Serious:** indirectness present in one of the four key extraction categories - population, intervention, comparator, outcome (-1) **Very serious:** indirectness present in more than one of the four key extraction categories - population, intervention, comparator, outcome (-2)
Imprecision	**Serious** = -1 **Very serious** = -2	**Serious:** for continuous outcomes: optimal information size (N = 400) was not met, and 95% confidence interval of an SMD extends between > 0.2 to ≤ 0.5 points in either direction (Cohen 1988*) (-1). For dichotomous outcomes: ratio of the upper and lower 95% confidence Intervals of the risk ratio is > 3.0$ (-1) **Very serious:** for continuous outcomes: optimal information size (N = 400) was not met, and 95% confidence Interval of an SMD extends > 0.5 points in either direction (Cohen 1988*) (-2)
Publication bias	**Strongly suspected** = -1	**Strongly suspected:** Egger's test P < 0.05 and asymmetric plot. We also considered the volume of trials that are unpublished (and 'awaiting classification').

I^2 = measure of heterogeneity, with 100% being the maximum possible heterogeneity

*Cohen 1998

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Table 1. GRADE certainty assessment rubric *(Continued)*

[$]Based on GRADE Guidance (Zeng 2022)

Created by the authors, adapted from Bannuru 2019
N = number; P: probability; SMD = standardised mean difference

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Table 2. Baseline demographic and clinical characteristics of the trial participants

Study ID (country)	Treatment groups	Sample size	Age (mean) years	Female (%)	Mean BMI (SD)	Osteoarthritis criteria	Mean pain at baseline (SD)	Mean function at baseline (SD)	Follow-up
Exercise versus attention control/placebo (30 trials)									
Baker 2001 **(USA)**	Exercise	23	69 (6.0)	74	31 (4.0)	ACR criteria	0 to 500 scale; higher worse 205 (87)	0 to 1700 scale; higher worse 725 (291)	16 weeks
	Attention control	23	68 (6.0)	82	32 (5.0)		204 (93)	774 (318)	
Bandak 2022 **(Denmark)**	Exercise	102	70.1 (8.3)	44	27.2 (3.7)	ACR criteria	0 to 100 scale; higher better 56.3 (14.9)	0 to 100 scale; higher better 65.7 (15.0)	12 and 52 weeks
	Saline injection	104	66.7 (8.2)	47	27.4 (3.6)		57.6 (13.1)	67.8 (14.7)	
Brismée 2007 **(USA)**	Exercise	22	70.8 (9.8)	86	27.96 (5.92)	ACR criteria	0 to 10 scale; higher worse 4.7 (2.6)	17 to 85 scale; higher worse 42.7 (12.1)	12 weeks
	Attention control	19	68.8 (8.9)	79	27.7 (6.57)		4.2 (1.8)	37.6 (10.6)	
Cheung 2017 **(USA)**	Exercise (yoga)	32	68.9 (7.7)	Not reported	29.8 (6.3)	Self-reported medical diagnosis of OA	0 to 10 scale; higher worse 6.2 (2.5)	0 to 68 scale; higher worse 27.1 (13.2)	8 weeks
	Exercise (aerobic, strengthening)	28	74.4 (7.5)	Not reported	29.2 (7.1)		4.9 (2.0)	29.9 (15.9)	
	Attention control	23	71.8 (8.0)	Not reported	27.8 (7.9)		5.0 (2.1)	24.3 (11.3)	
De Almeida 2020 **(Brazil)**	Exercise (circuit training)	22	55.6 (5.3)	75	26 (3.1)	ACR criteria	0 to 10 scale; higher worse 6.77 (1.79)	Not reported	14 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Exercise (conventional strength training)	22	55.2 (7.4)	76	26 (3.1)		6.74 (1.79)	Not reported	
	Attention control	22	53.8 (7.7)	80	27 (2.7)		6.72 (2.02)	Not reported	
Ettinger 1997 **(USA)**	Exercise (aerobic walking)	144	69 (6.0)	69	Not reported	ACR criteria	1 to 6 scale; higher worse 2.1 (SE 0.1)	Not reported	12, 36, 72 weeks
	Exercise (strengthening)	146	68 (6.0)	73	Not reported		2.2 (0.1)	Not reported	
	Attention control	149	69 (6.0)	69	Not reported		2.4 (0.1)	Not reported	
Foley 2003 **(Australia)**	Exercise	35	69.8 (9.2)	49	Not reported	Radiographic criteria	0 to 20 scale; higher worse 8.0 (4.0)	Not reported	6 weeks
	Attention control	35	69.8 (9.0)	57	Not reported		10.0 (4.0)	Not reported	
Imoto 2012 **(Brazil)**	Exercise	50	61.5 (6.9)	90	29.72 (4.1)	ACR criteria	0 to 10 scale; higher worse 7.4 (2.0)	0 to 100 scale; higher worse 31.0 (19.6)	8 weeks
	Attention control	50	58.8 (9.6)	90	30.00 (5.1)		6.9 (2.6)	34.5 (24.8)	
Jenkinson 2009 **(UK)**	Exercise	82	61.1 (9.8)	68	34.8 (6.6)	45 years of age and older, BMI > 28, knee pain on most days past month	0 to 20 scale; higher worse 7.6 (3.1)	0 to 68 scale; higher worse 27.1 (11.2)	24 months
	Attention control	76	61.5 (9.2)	65	33.0 (6.5)		7.3 (3.4)	26.3 (12.3)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Kang 2022 **(China)**	Exercise	15	63.4 (4.6)	100	25.6 (4.4)	ACR criteria	0 to 20 scale; higher worse 10.7 (4.0)	0 to 68 scale; higher worse 36.6 (8.8)	36 weeks
	Attention control	15	64.7 (6.1)	100	24.3 (4.8)		8.5 (3.7)	26.3 (9.0)	
Kim 2013 **(Japan)**	Exercise	37	80.9 (2.3)	100	23.7 (3.2)	Aged 75 years and older with knee pain	0 to 100 scale; higher worse 39.6 (24.1)	0 to 40 scale; higher worse 17.0 (4.9)	12 weeks
	Attention control	37	80.5 (2.7)	100	23.6 (3.1)		32.8 (19.8)	18.5 (6.8)	
Kovar 1992 **(USA)**	Exercise	52	70.38 (9.1)	77	Not reported	Diagnosis via pain and x-ray	0 to 10 scale; higher worse 5.25 (2.0)	0 to 10 scale; higher worse 6.2 (2.3)	8 weeks
	Attention control	51	68.5 (11.3)	90	Not reported		4.9 (2.3)	5.7 (2.5)	
Kuntz 2018 **(Canada)**	Exercise (yoga)	10	65.5 (5.6)	100	30.1 (3.8)	ACR criteria	0 to 100 scale; higher better 48.8 (12.4)	0 to 100 scale; higher better 56.2 (16.7)	12 weeks
	Exercise (strengthening)	11	63.7 (8.9)	100	28.9 (6.4)		57.3 (14.3)	66.0 (17.6)	
	Attention control	10	71.1 (9.3)	100	32.3 (5.7)		52.0 (17.9)	56.3 (15.2)	
Lai 2021 **(China)**	Exercise	27	64.8 (4.0)	93	23.1 (2.9)	Diagnosed by an orthopaedic surgeon based on patient's medical history, x-ray findings and physical	0 to 10 scale; higher worse 3.3 (2.4)	Not reported	8 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

						assessment			
	Attention control	27	63.7 (4.8)	85	23.5 (3.0)		3.5 (2.3)	Not reported	
Liu 2019 **(China)**	Exercise 1	35	58.6 (7.6)	79	22.8 (2.2)	ACR criteria	0 to 100 scale; higher better 70.5 (9.3)	0 to 100 scale; higher better 52.0 (13.4)	12 weeks
	Exercise 2	35	59.7 (7.4)	83	23.1 (2.7)		68.0 (10.3)	53.3 (11.4)	
	Exercise 3	35	61.3	85	23.4 (2.6)		72.0 (8.9)	53.8 (12.8)	
	Attention control	35	56.9 (6.5)	58	23.4 (3.3)		72.3 (7.6)	56.8 (13.1)	
Lu 2017 **(China)**	Exercise	23	64.6 (3.4)	100	25.2 (3.5)	ACR criteria	0 to 20 scale; higher worse 8.5 (4.9)	0 to 68 scale; higher worse 24.7 (13.0)	24 weeks
	Education	23	64.5 (3.4)	100	25.1 (3.4)		9.7 (5.6)	25.4 (14.6)	
Maurer 1999 **(USA)**	Exercise	57	66.3 (8.8)	47	Not reported	ACR criteria	0 to 200 scale; higher worse 191.3 (SD not reported)	0 to 680 scale; higher worse 643.4 (SD not reported)	8 weeks
	Attention control	56	64.5 (8.4)	36	Not reported		191.9	679.9	
Messier 2004 **(USA)**	Exercise	80	69.0 (0.8)	74	34.2 (0.6)	Knee pain on most days of month, pain with activity, KL Grade I to III	0 to 20 scale; higher worse 6.6 (0.4)	0 to 68 scale; higher worse 24.0 (1.3)	6 and 18 months
	Attention control	78	69.0 (0.1)	68	34.2 (0.6)		7.3 (0.4)	26.0 (1.3)	
Messier 2021 **(USA)**	Exercise (high intensity)	127	67.0 (9.0)	41	31.0 (6.0)	KL Grade II or III	0 to 20 scale; higher worse	0 to 68 scale; higher worse	78 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

							7.0 (2.7)	25.5 (10.3)	
	Exercise (low intensity)	126	64.0 (8.0)	41	31.0 (6.0)		7.4 (2.6)	26.6 (10.7)	
	Attention control	124	64.0 (7.0)	39	32.0 (5.0)		7.2 (2.7)	25.0 (10.3)	
Peloquin 1999 **(Canada)**	Exercise	69	65.6 (7.4)	71	29.8 (4.5)	KL < Grade IV	0 to 10 scale; higher worse 4.5 (2.0)	0 to 10 scale; higher worse 3.1 (2.4)	12 weeks
	Attention control	68	66.4 (8.3)	69	29.8 (4.8)		4.5 (2.2)	3.4 (2.6)	
Rego 2023 **(Brazil)**	Exercise	9	65.8 (2.8)	100	28.9 (1.4)	KL Grade II or III	0 to 20 scale; higher worse 8.0 (2.6)	0 to 68 scale; higher worse 29.8 (6.9)	7 and 52 weeks
	Attention control	9	64.8 (2.2)	100	32.0 (2.4)		8.8 (2.7)	33.3 (11.7)	
Santana 2022 **(Brazil)**	Exercise	18	60 (10.8)	100	35.3 (6.6)	Clinical diagnosis by orthopaedist	0 to 100 scale; higher worse 43.4 (8.3)	0 to 10 scale; higher worse 47.0 (5.6)	6 weeks
	Attention control	16	61 (6.6)	100	33.1 (7.3)		65.6 (5.6)	53.7 (4.8)	
Schlenk 2020 **(USA)**	Exercise	91	64.5 (8.5)	74	34.5 (7.5)	ACR criteria	0 to 20 scale; higher worse 5.8 (3.9)	0 to 68 scale; higher worse 22.5 (13.4)	26 and 52 weeks
	Attention control	91	65.0 (7.8)	73	33.6 (6.4)		4.8 (3.0)	19.3 (11.9)	
Shen 2022 **(China)**	Exercise	18	65.3 (4.6)	72	27.3 (2.7)	ACR criteria	0 to 10 scale; higher worse 5.7 (2.1)	Not reported	6 weeks
	Attention control	18	66.6 (7.0)	69	27.3 (2.7)		7.0 (1.8)	Not reported	
Simao 2012	Exercise	11	69 (3.7)	90	29.8 (2.53)	ACR criteria, KL Grade II+	0 to 500 scale; higher worse	0 to 1700 scale; higher worse	12 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

(Brazil)							275.0 (150 to 350) median (IQR)	900 (737.5 to 1213) median (IQR)	
	Attention control	12	71 (5.3)	91	26.7 (2.7)		175.0 (125 to 350) median (IQR)	625 (525 to 675) median (IQR)	
Song 2003 **(South Korea)**	Exercise	38	64.8 (6.0)	100	24.9 (2.6)	OA (confirmed by email), clinical and radiographic criteria	0 to 20 scale; higher worse 6.9 (4.1)	0 to 68 scale; higher worse 37.6 (10.6)	12 weeks
	Attention control	34	62.5 (5.6)	100	26.4 (3.5)		8.9 (5.1)	38.0 (12.6)	
Song 2020 **(China)**	Exercise	18	68.5 (4.3)	44	Not reported	Diagnosis of OA (unspecified)	0 to 10 scale; higher worse 3.2 (1.5)	Not reported	12 weeks
	Attention control	18	67.4 (3.4)	56	Not reported		3.2 (1.1)	Not reported	
Tsai 2013 **(USA)**	Exercise	28	78.9 (6.9)	79	Not reported	Diagnosed by healthcare provider	0 to 20 scale; higher worse 7.0 (3.3)	0 to 68 scale; higher worse 26.4 (8.6)	20 weeks
	Attention control	27	78.9 (8.3)	67	Not reported		7.7 (3.2)	26.7 (11.5)	
Vincent 2020 **(USA)**	Exercise (concentric)	28	69.5 (6.5)	67	32.8 (7.3)	ACR criteria	0 to 10 scale; higher worse 1.9 (2.3)	Not reported	16 weeks
	Exercise (eccentric)	30	66.8 (5.4)	70	28.7 (6.6)		1.4 (1.6)	Not reported	
	Attention control	29	68.6 (7.1)	66	32.8 (18.2)		0.5 (1.0)	Not reported	
Ye 2020 **(China)**	Exercise	25	64.5 (7.8)	52	24.2 (2.5)	ACR criteria	0 to 20 scale; higher worse	0 to 68 scale; higher worse	12 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

							6.6 (2.7)	20.8 (11.0)	
	Attention control	25	63.1 (3.7)	68	24.6 (2.3)		7.7 (9.3)	19.1 (8.4)	
Exercise versus no treatment/usual care/limited education (60 trials)									
Abbott 2013 **(New Zealand)**	Exercise	29	66.9 (8.2)	63	29.5 (6.8)	ACR criteria	0 to 20 scale; higher worse 3.5 (2.0)	0 to 240 scale; higher worse 95.5 (57.3)	52 and 104 weeks
	Usual care	28	66.1 (10.7)	49	29.3 (6.0)		3.1 (2.0)	93.8 (52.8)	
Allen 2018 **(USA)**	Exercise	142	65.3 (11.5)	69	31.5 (7.8)	ACR criteria	0 to 20 scale; higher worse 6.0 (3.9)	0 to 68 scale; higher worse 21.8 (12.7)	17 and 52 weeks
	Usual care	68	64.3 (12.2)	78	30.1 (7.3)		6.2 (4.0)	23.9 (13.8)	
An 2008 **(China)**	Exercise	14	65.4 (8.2)	100	25.7 (2.9)	ACR criteria	0 to 200 scale; higher worse 153.0 (88.7)	0 to 680 scale; higher worse 392.7 (296.5)	8 weeks
	No treatment	14	64.6 (6.7)	100	25.4 (2.9)		118.0 (72.6)	296.5 (179.4)	
Bennell 2010 **(Australia)**	Exercise	45	64.5 (9.1)	51	27.5 (4.7)	ACR criteria, KL Grade II +, medial tibiofemoral compartment disease	0 to 10 scale; higher worse 4.3 (2.0)	0 to 68 scale; higher worse 24.8 (10.9)	12 weeks
	Wait list	44	64.6 (7.6)	46	28.4 (4.1)		4.1 (2.5)	23.7 (11.8)	
Braghin 2018 **(Brazil)**	Exercise	15	59.4 (8.1)	67	30.2 (4.6)	Radiographic OA	0 to 100 scale; higher worse 31.3 (21.4)	0 to 100 scale; higher worse 25.8 (16.8)	8 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	No treatment	16	60.2 (9.3)	88	31.1 (7.0)		15.6 (28.4)	14.4 (26.1)	
Bruce-Brand 2012 **(Ireland)**	Exercise (resistance training)	14	63.4 (5.9)	60	33.9 (8.3)	KL Grade III+	0 to 20 scale; higher worse 11.1 (3.0)	0 to 68 scale; higher worse 31.7 (12.9)	8 and 14 weeks
	Standard care	13	65.2 (3.1)	50	31.7 (4.1)		9.0 (3.4)	31.7 (18.0)	
Chen 2014 **(Taiwan)**	Exercise	30	Not reported	Not reported	Not reported	Altman III criteria	0 to 10 scale; higher worse 5.5 (1.4)	0 to 10 scale; higher worse 7.8 (1.2)	8 and 26 weeks
	No treatment	30	Not reported	Not reported	Not reported		5.6 (1.4)	8.0 (1.1)	
Cheung 2014 **(USA)**	Exercise	18	71.9 (5.7)	100	29.1 (95% CI 26.7, 31.7)	ACR criteria	0 to 20 scale; higher worse 9.3 (7.4, 11.3) mean (95% CI)	0 to 68 scale; higher worse 35.0 (29.2, 40.9) mean (95% CI)	8 weeks
	Wait list	18	71.9 (6.5)	100	28.8 (95%CI 26.0 to 31.7)		7.7 (5.6, 9.8) mean (95% CI)	27.1 (19.6, 34.6) mean (95% CI)	
Chopp-Hurley 2017 **(Canada)**	Exercise	7	52.8 (6.4)	83	30.7 (6.8)	ACR criteria	Not reported for those with knee OA	Not reported for those with knee OA	12 weeks
	No treatment	11	54.9 (6.7)	75	27.6 (3.4)		Not reported for those with knee OA	Not reported for those with knee OA	
Christensen 2015 **(Denmark)**	Exercise	64	62.9 (5.8)	81	36.5 (4.4)	ACR criteria	0 to 100 scale; higher worse 43.6 (20.0)	0 to 100 scale; higher worse 41.5 (21.8)	52 weeks
	No treatment	64	61.7 (6.8)	80	37.9 (5.3)		46.7 (19.7)	47.7 (21.9)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Da Silva 2015 **(Brazil)**	Exercise	19	57.0 (6.0)	87	29.4 (4.1)	ACR criteria	0 to 10 scale; higher worse 4.9 (1.3)	0 to 10 scale; higher worse 9.6 (1.9)	8 weeks
	Education	22	60.0 (7.8)	87	29.3 (5.0)		4.5 (1.5)	8.8 (3.2)	
De Matos 2019 **(Brazil)**	Exercise	28	58.6 (7.4)	67	29.3 (4.7)	KL Grade I to III	0 to 100 scale; higher worse 24.6 (26.2)	0 to 100 scale; higher worse 19.2 (19.1)	8 weeks
	Education	28	60.8 (9.2)	80	31.5 (7.0)		16.7 (29.1)	15.49 (26.7)	
De Rooij 2017 **(Netherlands)**	Exercise	63	63.2 (8.4)	78	36.0 (6.8)	ACR criteria	0 to 10 scale; higher worse 6.4 (1.8)	0 to 68 scale; higher worse 35.1 (11.9)	20 weeks
	No treatment	63	63.9 (12.4)	73	35.0 (7.6)		5.9 (2.1)	31.0 (12.3)	
DeVita 2018 **(United States of America)**	Exercise	16	58.1 (6.5)	67	26.4 (4.0)	ACR criteria	0 to 20 scale; higher worse 6.4 (3.2)	0 to 68 scale; higher worse 17.7 (9.7)	12 weeks
	No treatment	15	56.2 (8.9)	53	27.9 (3.9)		6.7 (3.3)	22.9 (10.7)	
Drummen 2023 **(Australia)**	Exercise	24	65.1 (8.9)	635	33.6 (4.9)	ACR criteria	0 to 100 scale; higher worse 55.3 (17.3)	0 to 1700 scale; higher worse 915.8 (232.9)	24 weeks
	Usual care	16	67.4 (8.3)	56	31.8 (5.8)		52.2 (13.0)	893.4 (185.9)	
Fransen 2001 **(Australia)**	Exercise	83	68.5 (8.7)	74	30.0 (4.6)	ACR criteria	0 to 100 scale; higher worse 59.5 (20.0)	0 to 10 scale; higher worse 58.5 (18.8)	8 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Wait list	43	66.1 (10.3)	67	28.3 (4.5)		65.8 (19.4)	60.0 (20.5)	
Fransen 2007 **(Australia)**	Exercise	56	70.8 (6.3)	68	29.6 (5.9)	ACR criteria	Not reported for those with knee OA	Not reported for those with knee OA	12 weeks
	Wait list	41	69.6 (6.1)	83	30.7 (5.0)		Not reported for those with knee OA	Not reported for those with knee OA	
Gohir 2021 **(UK)**	Exercise	67	65.2 (9.7)	70.8	30.4 (5.5)	ACR criteria	0 to 10 scale; higher worse 4.4 (2.0)	0 to 68 scale; higher worse 26.8 (12.9)	6 weeks
	Education	79	68.0 (8.6)	65	31.9 (5.9)		4.7 (2.1)	28.3 (12.8)	
Guo 2022 **(China)**	Exercise	56	63.2 (8.0)	54	27.4 (6.0)	ACR criteria	0 to 10 scale; higher worse 5.8 (1.2)	0 to 68 scale; higher worse 30.7 (7.3)	16 weeks
	Usual care	55	61.5 (7.2)	54	27.6 (5.4)		5.8 (1.1)	31.2 (7.7)	
Gur 2002 **(Turkey)**	Exercise (concentric)	9	56.0 (12.0)	Not reported	Not reported	Radiographic, bilateral KL Grade II-III	0 to 70 scale; higher worse 33.9 (7.7)	0 to 50 scale; higher worse 33.0 (5.3)	8 weeks
	Exercise (eccentric and concentric)	8	55.0 (12.0)	Not reported	Not reported		35.8 (6.1)	29.9 (6.1)	
	No treatment	6	57.0 (9.0)	Not reported	Not reported		27.3 (3.8)	33.8 (3.4)	
Hay 2006 **(UK)**	Exercise	109	67.9 (8.5)	65	Not reported	Presenting with persistent knee pain	0 to 12 scale; higher worse 9.1 (4.1)	0 to 68 scale; higher worse 30.0 (14.1)	12 and 26 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Education	108	68.2 (8.0)	65	Not reported		9.2 (3.3)	30.6 (12.0)	
Henriksen 2014 **(Denmark)**	Exercise	31	65.9 (8.5)	87	28.7 (4.2)	Clinical diagnosis confirmed by radiography	0 to 100 scale; higher better 56.5 (14.8)	0 to 100 scale; higher better 65.0 (14.0)	12 weeks
	No treatment	29	61.3 (7.1)	72	28.1 (4.5)		63.3 (12.4)	74.2 (13.9)	
Hopman-Rock 2000 **(Netherlands)**	Exercise	37	65.4 (5.3)	78	28.4 (4.8)	ACR criteria	Not reported for those with knee OA	Not reported for those with knee OA	6 weeks
	Wait list	34	65.2 (5.7)	88	26.8 (3.5)		Not reported for those with knee OA	Not reported for those with knee OA	
Hu 2020 **(China)**	Exercise	52	66.3 (4.2)	100	36.5 (9.0)	Clinical assessment - KL Grade 1 or greater plus physical examination	0 to 10 scale; higher worse 4.2 (3.3)	0 to 68 scale; higher worse 38.4 (15.1)	24 weeks
	Education	40	65.5 (3.6)	100	26.4 (3.1)		3.8 (2.1)	34.2 (23.0)	
Hurley 2007 **(United Kingdom)**	Exercise (individual)	146	66.0 (range 50 to 91)	71	30.0 (range 18 to 45)	> 50 years of age who consulted a primary care physician for knee pain of > 6 months' duration	0 to 20 scale; higher worse 7.4 (4.0)	0 to 68 scale; higher worse 26.4 (14.7)	6 and 26 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Exercise (group)	132	68.0 (range 51 to 84)	71	30.2 (range 20 to 50)		7.69 (4.1)	27.9 (14.7)	
	Usual care	140	67.0 (range 51 to 89)	69	30.3 (range 20 to 51)		7.7 (4.0)	27.2 (14.6)	
Jan 2009 **(Taiwan)**	Exercise (strengthening – weight bearing)	36	62.0 (6.7)	67	Not reported	ACR criteria, KL Grade < IV (most KL II)	Not reported	0 to 68 scale; higher worse 22.6 (10.1)	8 weeks
	Exercise (strengthening – non weight bearing)	35	63.2 (6.8)	72	Not reported		Not reported	27.3 (9.5)	
	No treatment	35	62.2 (6.7)	69	Not reported		Not reported	24.8 (10.7)	
Jany-acharoen 2018 **(Thailand)**	Education	20	59.0 (7.8)	90	Not reported	ACR criteria	0 to 100 scale; higher better 43.0 (1.2)	0 to 100 scale; higher better 58.0 (5.0)	12 weeks
	Exercise	20	60.3 (6.8)	90	Not reported		45.3 (1.6)	63.4 (3.5)	
Jorge 2015 **(Brazil)**	Exercise	29	61.7 (6.4)	100	30.6 (5.8)	ACR criteria	0 to 10 scale; higher worse 7.0 (1.3)	0 to 68 scale; higher worse 27.7 (9.3)	12 weeks
	Wait list	31	59.9 (7.5)	100	31.4 (4.42)		7.0 (1.2)	28.4 (10.6)	
Karadag 2019 **(Turkey)**	Exercise	17	58.7 (10.3)	73	29.8 (5.7)	ACR criteria	0 to 10 scale; higher worse 7.5 (0.6)	0 to 68 scale; higher worse 42.1 (3.4)	4 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Usual care	18	58.5 (11.0)	88	33.7 (7.1)		8.0 (0.4)	40.7 (3.7)	
Keefe 2004 **(United States of America)**	Exercise	16	60.3 (8.7)	38	Not reported	Persistent knee pain	0 to 10 scale; higher worse 3.9 (1.6)	Not reported	12 weeks
	Standard care	18	57.56 (14.27)	64.7	Not reported		3.9 (1.7)	Not reported	
Kim 2020 **(South Korea)**	Exercise	40	76.15 (5.70)	77.5	26.77 (2.38)	KL Grade II to IV	0 to 100 scale; higher worse 41.7 (17.0)	Not reported	12 weeks
	Usual care	40	73.4 (5.0)	91	27.0 (2.5)		39.2 (16.7)	Not reported	
Krishnan 2021 **(Malaysia)**	Exercise	15	49.4 (5.1)	50	34.4 (2.2)	KL Grade II	0 to 100 scale; higher better 46.6 (4.2)	0 to 100 scale; higher better 47.2 (5.1)	8 weeks
	No treatment	15	50.3 (6.3)	50	34.1 (53.0)		46.5 (7.2)	47.7 (7.6)	
Lee 2009 **(Korea)**	Exercise	29	70.2 (4.8)	93	26.0 (3.8)	KL Grade II+	0 to 20 scale; higher worse 6.8 (4.2)	0 to 68 scale; higher worse 24.2 (14.7)	8 weeks
	No treatment	15	66.9 (6.0)	93	26.0 (2.8)		6.1 (3.4)	23.5 (13.4)	
Lee 2023 **(Korea)**	Exercise	18	65.6 (3.7)	100	24.3 (SD not reported)	KL Grade II+	0 to 100 scale; higher worse 34.5 (21.7)	Not reported	8 weeks
	Usual care	18	68.3 (4.8)	100	25.5 (SD not reported)		43.9 (17.9)	Not reported	
Li 2020 **(Canada)**	Exercise	26	65.0 (8.0)	89	29.8 (9.0)	Physician-informed diagnosis	0 to 100 scale; higher better 72.6 (13.5)	0 to 100 scale; higher better 75.5 (14.7)	12 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Wait list	25	64.8 (9.0)	76	28.9 (6.2)		65.1 (13.7)	72.2 (15.8)	
Lim 2008 **(Australia)**	Exercise (more malaligned)	26	67.2 (6.7)	50	28.2 (3.7)	ACR criteria	0 to 100 scale; higher worse 33.1 (15.4)	0 to 100 scale; higher worse 31.4 (17.4)	13 weeks
	Exercise (more neutrally aligned)	27	64.1 (9.3)	63	29.0 (5.2)		35.7 (14.6)	33.2 (15.4)	
	No intervention (more malaligned)	26	66.6 (8.9)	46	30.3 (5.3)		39.2 (14.0)	38.5 (14.5)	
	No intervention (more neutrally aligned)	28	60.8 (7.8)	61	28.4 (5.0)		34.6 (16.2)	36.1 (15.7)	
Lin 2009 **(Taiwan)**	Exercise (strength training)	36	61.6 (7.2)	67	Not reported	50 years of age and older, KL Grade < IV, history of knee pain > 6 months	0 to 20 scale; higher worse 8.8 (3.6)	0 to 68 scale; higher worse 27.3 (9.5)	8 weeks
	No intervention	36	62.2 (6.7)	72	Not reported		8.5 (4.6)	24.8 (10.7)	
Lund 2008 **(Denmark)**	Exercise	25	68.0 (9.4)	88	Not reported	ACR criteria	0 to 100 scale; higher worse 53.0 (32.6)	0 to 100 scale; higher better 39.6 (13.2)	8 and 20 weeks
	No intervention	27	70.0 (9.9)	66	Not reported		48.5 (31.9)	39.6 (13.2)	
Mazloum 2018 **(Iran)**	Exercise (conventional exercise)	16	50.3 (8.3)	Not reported	26.6 (SD not reported)	ACR criteria	0 to 24 scale; higher worse 10.4 (1.9)	Not reported	8 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Exercise (pilates)	17	55.0 (8.2)	Not reported	26.81 (SD not reported)		10.4 (1.7)	Not reported	
	Usual care	16	50.8 (9.9)	Not reported	27.25 (SD not reported)		10.3 (1.9)	Not reported	
Multanen 2014 **(Finland)**	Exercise	40	58.0 (4.0)	100	27.1 (3.1)	Knee pain on most days and KL Grade I or II	0 to 100 scale; higher worse 8 (8)	0 to 100 scale; higher worse 5 (5)	52 weeks
	Social group meeting	40	59.0 (4.0)	100	26.7 (4.2)		6 (6)	4 (4)	
Øiestad 2023 **(Norway)**	Exercise (strengthening)	57	57.6 (6.6)	56	28.9 (4.3)	KL Grade II or III	0 to 10 scale; higher worse 4.9 (2.1)	0 to 100 scale; higher better 66 (20)	20 and 52 weeks
	Exercise (aerobic)	55	57.3 (7.1)	53	29.4 (4.4)		4.6 (1.9)	64 (21)	
	Usual care	56	57.8 (7.4)	44	28.4 (4.1)		5.3 (2.3)	61 (21)	
O'Reilly 1999 **(UK)**	Lifestyle advice + exercise	78	61.9 (10.0)	65	Not reported	Knee pain past week	0 to 20 scale; higher worse 6.5 (3.5)	0 to 68 scale; higher worse 20.4 (12.5)	26 weeks
	Lifestyle advice	113	62.2 (9.7)	68	Not reported		6.8 (2.8)	19.5 (11.5)	
Pazit 2018 **(Australia)**	Exercise (strengthening)	10	67.8 (6.2)	44	28.2 (5.6)	ACR criteria	0 to 500 scale; higher worse 181.6 (111.4)	0 to 1700 scale; higher worse 105.7 (59.2)	8 weeks
	Exercise (strengthening + balance)	10	65.1 (4.7)	60	34.0 (8.7)		127.3 (107.8)	78.7 (60.3)	
	Usual care	10	70.4 (7.8)	56	28.4 (3.9)		153.7 (91.1)	81.0 (35.3)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Robbins 2022 **(Brazil)**	Exercise	43	61.4 (7.8)	63	29.6 (4.9)	KL Grade II or III	0 to 100 scale; higher worse 61 (50 to 70) median (IQR)	0 to 100 scale; higher worse 50 (39.7 to 58.8) median (IQR)	8 weeks
	Education	43	62.4 (3.3)	88	31.7 (3.8)		61.5 (50 to 77.5)	50 (39.0 to 60.3)	
Rogind 1998 **(Denmark)**	Exercise	12	69.3 (8.2)	91	27.4 (4.0)	ACR criteria x-ray (> KL Grade II)	0 to 10 scale; higher worse 7.0 (4.5 to 7.5) median (IQR)	0 to 24 scale; higher worse 13.5 (11.5 to 15.3) median (IQR)	12 and 52 weeks
	No treatment	13	73.0 (6.5)	92	26.8 (3.2)		5.0 (5.0 to 6.5)	13.5 (10.5 to 15.0)	
Rosedale 2014 **(Canada)**	Exercise	120	66.0 (10.0)	56	31.4 (7.7)	Knee pain > 4 months and radiographically confirmed	0 to 100 scale; higher better 51 (17)	0 to 100 scale; higher better 56 (17)	2 and 12 weeks
	Wait list	60	64.0 (11.0)	60	30.7 (5.3)		46 (17)	51 (18)	
Sadeghi 2023 **(Iran)**	Exercise (quadriceps strengthening)	24	52.8 (9.6)	83	29.2 (3.7)	KL Grade I to III	0 to 100 scale; higher worse 54.6 (20.2)	0 to 68 scale; higher worse 51.2 (22.6)	8 weeks
	Exercise (hamstring strengthening)	24	49.6 (7.4)	75	28.5 (4.1)		50.2 (21.7)	48.1 (25.4)	
	Exercise (quadriceps and hamstring strengthening)	24	52.8 (9.7)	88	28.6 (4.3)		56.0 (19.8)	52.8 (18.7)	
	No treatment	24	55.7 (9.2)	71	31.3 (5.0)		55.0 (21.7)	57.1 (23.6)	
Salacinski 2012	Exercise	19	55.1 (10.5)	79	22.4 (3.3)	KL Grades I to III	0 to 100 scale; higher better	0 to 100 scale; higher worse	12 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

(USA)							70.8 (13.0)	72.3 (17.9)	
	Wait list	18	60.6 (8.4)	67	25.7 (6.3)		70.0 (16.7)	70.3 (15.8)	
Samut 2015 **(Turkey)**	Exercise (strengthening)	15	62.5 (7.7)	90	30.5 (4.5)	ACR criteria	0 to 10 scale; higher worse 6.8 (6.8)	0 to 68 scale; higher worse 33.9 (7.1)	6 weeks
	Exercise (walking)	14	57.6 (5.8)	90	33.9 (7.3)		6.8 (2.2)	26.3 (12.1)	
	No treatment	13	60.9 (8.9)	90	30.4 (5.7)		6.6 (2.1)	30.0 (10.7)	
Schilke 1996 **(USA)**	Exercise	10	64.5 (SD not reported)	Not reported	Not reported	Self-reported OA	OASI; scale not reported 15.8 (4.9)	OASI; scale not reported 11.3 (3.6)	8 weeks
	No treatment	10	68.4 (SD not reported)	Not reported	Not reported		9.7 (6.7)	9.5 (4.0)	
Sekir 2005 **(Turkey)**	Exercise	12	59.0 (8.9)	75	Not reported	ACR criteria	0 to 10 scale; higher worse 4.4 (2.4, 6.7) mean (IQR)	Not reported	6 weeks
	No treatment	10	62.0 (8.1)	70	Not reported		4.8 (2.8, 6.7) mean (IQR)	Not reported	
Shahine 2020 **(Egypt)**	Exercise	33	65.6 (5.0)	64	Not reported	Diagnosis of OA (unspecified)	0 to 100 scale; higher worse 74.8 (13.7)	0 to 100 scale; higher worse 80 (12.9)	12 weeks
	Education	33	66.9 (5.8)	49	Not reported		75.9 (16.7)	79 (15)	
Takacs 2017 **(Canada)**	Exercise	20	66.1 (8.7)	95	28.5 (5.4)	KL Grade > 2	0 to 10 scale; higher worse 5.0 (1.8)	0 to 68 scale; higher worse 30 (7)	10 weeks
	No treatment	20	67.1 (5.4)	65	28.9 (4.5)		4.8 (2.2)	30 (10)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Thorstensson 2005 **(Sweden)**	Exercise	30	54.8 (7.1)	50	29.6 (4.5)	Radiographic knee OA (KL Grade III or higher) and long-standing knee pain	0 to 100 scale; higher better 60 (18)	0 to 100 scale; higher better 69 (18)	6 and 26 weeks
	Wait list	31	57.3 (4.7)	52	29.5 (5.1)		64 (19)	71 (21)	
Topp 2002 **(USA)**	Exercise (dynamic)	35	65.57 (1.82)	71	Not reported	ACR criteria	0 to 20 scale; higher worse mean (SEM) 12.4 (0.5)	0 to 68 scale; higher worse mean (SEM) 41.1 (1.9)	16 weeks
	Exercise (isometric)	32	63.5 (1.9)	66	Not reported		11.8 (0.6)	38.1 (1.9)	
	No treatment	35	60.9 (1.8)	80	Not reported		10.8 (0.5)	38.9 (1.8)	
Vassao 2021 **(Brazil)**	Exercise	14	62.3 (4.4)	100	30.1 (3.6)	ACR criteria	0 to 20 scale; higher worse 7.4 (3.5)	0 to 68 scale; higher worse 23 (11.4)	8 weeks
	Wait list	14	66.5 (4.1)	100	27.2 (3.0)		9.1 (4.3)	21.3 (9.3)	
Wallis 2017 **(Australia)**	Exercise	23	68.0 (8.0)	39	34.0 (5.0)	KL Grade II to IV	0 to 10 scale; higher worse 6.1 (1.9)	0 to 68 scale; higher worse 37 (10)	12 weeks
	Usual care	23	61.0 (7.0)	48	34.0 (7.0)		5.3 (1.6)	34 (9)	
Wang 2011 **(Taiwan)**	Exercise	28	68.3 (6.4)	89	25.4 (2.4)	Physician diagnosis of OA	0 to 100 scale; higher better 65 (14)	0 to 100 scale; higher better 75 (16)	12 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	No treatment	28	67.9 (5.9)	85	26.6 (2.1)		66 (18)	70 (19)	
Wortley 2013 **(USA)**	Exercise (strengthening)	15	69.5 (6.7)	69	30.5 (6.0)	ACR criteria	WOMAC; scale range not reported; higher worse 155 (110)	WOMAC; scale range not reported; higher worse 494 (265)	10 weeks
	Exercise (Tai Ji)	15	68.1 (5.3)	75	35.1 (5.9)		169 (135)	694 (361)	
	No treatment	9	70.5 (5.0)	67	30.0 (6.2)		170 (86)	547 (396)	
Xiao 2021 **(China)**	Exercise	142	71.0 (2.9)	100	29.8 (7.1)	ACR criteria	WOMAC; scale range not reported 98.5 (36.0)	Not measured	24 weeks
	No treatment	142	69 (3.7)	100	28.4 (3.7)		98.0 (25.6)		
Exercise when added to another co-intervention (compared to that co-intervention alone) (49 trials)									
Alghadir 2019 **(Saudi Arabia)**	Exercise (retrowalking)	23	54.6 (SD not reported)	48	26.2 (SD not reported	ACR criteria	0 to 10 scale; higher worse 5.7 (0.8)	0 to 10 scale; higher worse 52.9 (3.3)	6 weeks
	Exercise (walking)	23	55.3 (SD not reported	61	26.1 (SD not reported		6.1 (0.7)	53.3 (3.1)	
	Routine physiotherapy	22	56. 8 (SD not reported	59	26.1 (SD not reported		5.8 (0.8)	53.8 (3.3)	
Anadkat 2015 **(India)**	Conventional exercise + retrowalking	20	51.3 (6.5)	60	Not reported	Diagnosed by clinician; KL Grade III	0 to 10 scale; higher worse 8.4 (1.3)	0 to 100 scale; higher worse 57.9 (7.5)	3 weeks
	Conventional exercise	20	50.0 (6.9)	45	Not reported		8.6 (1.1)	54.6 (11.0)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Anwer 2014 **(Country not reported)**	Ultrasound + exercise	21	54.9 (7.7)	Not reported	26.5 (1.8)	ACR criteria	0 to 10 scale; higher worse 6.1 (0.9)	0 to 68 scale; higher worse 24.7 (3.4)	5 weeks
	Ultrasound	21	56.0 (6.8)	Not reported	27.1 (1.3)		6.0 (1.1)	24.5 (4.4)	
Assar 2020 **(Iran)**	Drug regimen + exercise	12	55.9 (8.6)	100	29.8 (7.2)	ACR criteria	0 to 10 scale; higher worse 7.6 (2.4)	Not reported	8 weeks
	Drug regimen	12	63.8 (7.5)	100	23.1 (11.6)		7.5 (2.0)	Not reported	
Bautch 1997 **(Country not reported)**	Education + exercise	17	66.0 (2.4)	67	32.6 (1.1)	ACR criteria	0 to 10 scale; higher worse 4.2 (5.6)	0 to 100 scale; higher worse 26.2 (2.0)	12 weeks
	Education	17	70.0 (2.1)	80	24.9 (1.4)		2.1 (0.3)	21.4 (2.1)	
Bennell 2016 **(Australia)**	Pain coping skills training + exercise	73	64.6 (8.3)	60	31.0 (6.0)	ACR criteria	0 to 100 scale; higher worse 58.4 (12.8)	0 to 68 scale; higher worse 35.6 (7.3)	12 weeks
	Pain coping skills training	74	63.0 (7.9)	61	30.8 (6.4)		58.7 (12.6)	35.0 (7.4)	
Bennell 2022a **(Australia)**	Education + exercise	172	65.4 (8.2)	54	33.1 (3.6)	NICE clinical criteria	0 to 10 scale; higher worse 5.9 (1.3)	0 to 68 scale; higher worse 22.8 (10.6)	26 and 52 weeks
	Education	68	65.3 (8.7)	67	33.8 (3.6)		5.9 (1.3)	22.7 (10.2)	
Bennell 2022b	Education + exercise	107	62.8 (8.2)	65	29.9 (4.8)	NICE clinical criteria	0 to 10 scale; higher worse	0 to 68 scale; higher worse	12 and 24 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

(Australia)							6.3 (1.3)	25.5 (9.1)	
	Education	105	61.8 (7.2)	74	30.7 (6.1)		6.3 (1.3)	24.0 (9.9)	
Bhagat 2017 **(India)**	Interferential therapy + exercise	15	57.4 (2.4)	47	Not reported	ACR criteria	0 to 10 scale; higher worse 6.2 (2.1)	0 to 100 scale; higher worse 59.9 (19.0)	4 weeks
	Interferential therapy	15	55.4 (3.5)	33	Not reported		5.5 (1.7)	55.7 (18.5)	
Chang 2012 **(Taiwan)**	General physiotherapy + exercise	30	65.0 (8.4)	100	24.9 (3.3)	KL Grade II or III, knee flexion > 90 degrees	0 to 20 scale; higher worse 4.3 (1.7)	0 to 68 scale; higher worse 20.0 (8.9)	8 weeks
	General physiotherapy	30	70.8 (8.4)	100	25.7 (3.6)		4.5 (1.7)	22.0 (8.6)	
Chen 2021 **(China)**	Conventional treatment + exercise	16	60.3 (7.9)	81	24.1 (2.2)	ACR criteria	0 to 10 scale; higher worse 3.7 (0.8)	0 to 68 scale; higher worse 14.6 (3.6)	4 weeks
	Conventional treatment	16	60.9 (6.9)	81	24.4 (2.3)		3.6 (1.0)	15.0 (3.3)	
Dincer 2016 **(Turkey)**	TENS + exercise	19	49.5 (7.0)	81	28.0 (5.0)	ACR criteria	0 to 10 scale; higher worse 6.6 (1.7)	0 to 68 scale; higher worse 24.5 (14.0)	12 weeks
	TENS	16	52.9 (3.7)	79	29.2 (3.8)		5.4 (1.5)	27.7 (11.0)	
Dwyer 2015 **(South Africa and USA)**	MMT + exercise	28	62.2 (11.8)	58	30.6 (7.6)	ACR criteria	WOMAC; scale range not reported 216.8 (17.0) mean (SE)	WOMAC; scale range not reported 411.7 (52.0) mean (SE)	4 weeks
	MMT	27	63.5 (10.9)	73	28.6 (5.2)		227.3 (17.2)	759.0 (47.6)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Gondhalekar 2013 **(India)**	Conventional treatment + exercise	15	Not reported	47	Not reported	ACR criteria	0 to 10 scale; higher worse 7.7 (1.0)	0 to 100 scale; higher worse 62.8 (15.0)	4 weeks
	Conventional treatment	15	Not reported	53	Not reported		7.5 (1.1)	67.3 (14.7)	
Hernandez 2019 **(Argentina)**	Conventional exercises + core stability exercises	53	64.2 (11.0)	68	28.6 (6.0)	Confirmed by clinician	0 to 10 scale; higher worse 6.9 (2.6)	Not reported	12 and 24 weeks
	Conventional exercises	60	60.1 (9.8)	73	27.6 (6.1)		6.1 (2.1)	Not reported	
Hinman 2020 **(Australia)**	Phone advice + exercise	87	62.4 (9.1)	63	31.1 (6.8)	NICE clinical criteria	0 to 10 scale; higher worse 6.0 (1.5)	0 to 68 scale; higher worse 29.3 (10.1)	26 and 52 weeks
	Phone advice	88	62.5 (8.1)	63	31.2 (7.6)		6.0 (1.5)	27.8 (12.0)	
Holm 2020 **(Denmark)**	Education and neuromuscular exercise + strength training	45	63.2 (10.7)	56	32.2 (6.5)	Symptomatic and radiographic (KL II or more) knee OA	0 to 100 scale; higher better 43.4 (16.3)	0 to 100 scale; higher better 48.2 (14.5)	12 weeks
	Education and neuromuscular exercise	45	66.4 (9.3)	60	29.6 (5.4)		49.1 (12.8)	54.3 (11.8)	
Hsu 2021 **(Taiwan)**	Diet + exercise	22	65.6 (3.9)	71	31.1 (2.6)	KL Grade III+	0 to 20 scale; higher worse 7.4 (2.0)	0 to 68 scale; higher worse 28.6 (5.8)	12 weeks
	Diet	22	66.0 (3.9)	57	30.8 (2.6)		6.5 (2.2)	25.2 (5.6)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Huang 2003 **(Taiwan)**	Hotpack/ROM + exercise	99	Not reported	Not reported	Not reported	ACR criteria	0 to 10 scale; higher worse 4.7 (1.5)	1 to 26 scale; higher worse 6.9 (1.6)	8 weeks
	Hotpack/ROM	33	Not reported	Not reported	Not reported		4.6 (1.3)	7.2 (1.5)	
Huang 2005 **(Taiwan)**	Hotpack/ROM + exercise	35	Not reported	Not reported	Not reported	ACR criteria	0 to 10 scale; higher worse 5.3 (1.5)	1 to 26 scale; higher worse 7.6 (1.2)	8 and 52 weeks
	Hotpack/ROM	35	Not reported	Not reported	Not reported		5.4 (1.7)	7.4 (1.1)	
Jan 2008 **(Taiwan)**	Exercise (high resistance training)	34	63.3 (6.6)	79	Not reported	ACR criteria, KL Grade < IV	0 to 20 scale; higher worse 8.5 (3.8)	0 to 68 scale; higher worse 26.4 (9.0)	8 weeks
	Exercise (low resistance training)	34	61.8 (7.1)	79	Not reported		7.8 (3.3)	26.1 (8.1)	
	Education	34	62.8 (6.3)	83	Not reported		8.3 (4.6)	25.4 (11.3)	
Joshi 2019 **(India)**	Conventional exercise + retrowalking	21	50.7 (7.3)	43	28.6 (10.2)	Diagnosed by clinician	0 to 10 scale; higher worse 7.8 (1.1)	0 to 100 scale; higher worse 46.8 (6.4)	6 weeks
	Conventional exercise	21	54.3 (10.2)	62	28.1 (8.9)		7.9 (1.0)	51.7 (7.3)	
Kholvadia 2019 **(South Africa)**	Low level laser therapy + exercise	32	Not reported	Not reported	29.6 (4.7)	Diagnostic criteria unclear	Not reported	0 to 100 scale; higher worse 56 (10.8)	12 weeks and 24 months
	Low level laser therapy	40	Not reported	Not reported	31.6 (4.5)		Not reported	59.1 (10.2)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Kilic 2020 **(Turkey)**	Physiotherapy + exercise	25	59.5 (8.6)	100	30.6 (2.9)	KL Grade II or III	0 to 10 scale; higher worse 6.8 (1.1)	0 to 68 scale; higher worse 3.0 (0.8)	6 weeks
	Physiotherapy	25	60.5 (7.4)	100	31.2 (3.3)		6.9 (1.2)	8.0 (12.7)	
Knoop 2013 **(Netherlands)**	Exercise with feedback (neuromuscular/neuromotor exercise with personalised feedback from PT) + exercise for knee position	80	62.1 (7.6)	66	28.8 (4.8)	ACR criteria	0 to 10 scale; higher worse 4.8 (2.2)	0 to 68 scale; higher worse 25.2 (11.8)	12 and 38 weeks
	Exercise without feedback	79	61.8 (6.6)	56	28.3 (4.5)		5.2 (2.0)	27.1 (12.7)	
Kumar 2013 **(India)**	Supervised exercise + ultrasound + proprioceptive/balance training	22	53.2 (6.9)	Not reported	25.7 (2.7)	Diagnostic criteria unclear	0 to 10 scale; higher worse 4.9 (1.0)	0 to 68 scale; higher worse 23.3 (1.7)	4 weeks
	Supervised exercise + ultrasound	22	53.3 (5.4)	Not reported	25.2 (3.1)		5.1 (0.8)	23.6 (2.6)	
Lee 2019 **(South Korea)**	HA injection + exercise	15	70.1 (3.9)	20	24.6 (1.7)	ACR criteria	0 to 10 scale; higher worse 5.5 (1.2)	0 to 68 scale; higher worse 28.9 (5.9)	4 weeks
	HA injection	15	71.0 (3.8)	13	25 (3.6)		5.4 (1.1)	28.5 (7.4)	
Marconcin 2018 **(Portugal)**	Education + exercise	40	70.3 (6.1)	80	32.3 (5.0)	ACR criteria	0 to 100 scale; higher better 52.1 (18.9)	0 to 100 scale; higher better 64.9 (19.7)	12 weeks
	Education	40	67.8 (5.3)	59	30.1 (5.3)		61.4 (20.4)	49.7 (18.5)	
Messier 2013	Diet + exercise	152	65.0 (6.0)	72	33.6 (3.7)	KL Grade II or III	0 to 20 scale; higher worse	0 to 68 scale; higher worse	78 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

(USA)							6.7 (3.4)	24.6 (11.7)	
	Diet	152	66.0 (6.0)	71	33.7 (3.8)		6.6 (3.0)	24.8 (10.4)	
Mete 2022 **(Turkey)**	Physiotherapy + exercise	32	59.5 (55 to 64) median (IQR)	80	30.3 (24.96 to 33.16) median (IQR)	KL Grade II or III	0 to 100 scale; higher worse 32.2 (IQR 20.8 to 40.0)	0 to 10 scale; higher worse 6.74 (6.4 to 7.2)	6 weeks
	Physiotherapy	32	57 (51/65) median (IQR)	77	32.14 (31.8/32.42) median (IQR)		36.6 (30.0 to 40.0)	5.2 (5.7 to 6.4)	
Mehwish 2023 **(Pakistan)**	Knee exercises + hip strengthening	22	53.6 (4.7)	77	29.1 (SD not reported)	KL Grade I, II or III	0 to 10 scale; higher worse 6.8 (1.1)	0 to 100 scale; higher better 34.7 (8.4)	4 weeks
	Knee exercises	22	54.9 (4.3)	71	28.1 (SD not reported)		6.5 (1.0)	36.5 (10.7)	
Nahayatbin 2018 **(Iran)**	Routine physiotherapy + exercise (strengthening)	16	56.1 (6.0)	Not reported	29.0 (4.2)	KL Grade II or III	0 to 100 scale; higher better 45.3 (11.5)	0 to 100 scale; higher better 49.3 (10.7)	4 and 8 weeks
	Routine physiotherapy + exercise (Tai Chi)	16	55.3 (5.7)	Not reported	29.0 (4.7)		46.1 (11.5)	51.9 (12.2)	
	Routine physiotherapy	16	56.1 (6.1)	Not reported	29.0 (3.5)		43.7 (9.1)	52.4 (10.4)	
Nelligan 2021 **(Australia)**	Education + exercise	103	60.3 (8.2)	58	31.1 (26.6 to 34.9) median (IQR)	NICE clinical criteria	0 to 10 scale; higher worse 6.3 (1.5)	0 to 68 scale; higher worse 26.7 (11.8)	24 weeks
	Education	103	59.0 (8.5)	64	31.6 (26.9 to 36.4) median (IQR)		6.2 (1.5)	25.0 (12.2)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Oh 2020 **(South Korea)**	Education + exercise	40	72.4 (6.3)	Not reported	24.8 (2.5)	KL Grade < III	0 to 20 scale; higher worse 5.0 (4.5)	0 to 68 scale; higher worse 15.1 (11.2)	20 weeks
	Education	20	71.1 (5.4)	Not reported	25.7 (3.8)		6.7 (5.2)	22.1 (19.3)	
Ozlu 2023 **(Turkey)**	Conventional treatment + exercise	41	53.3 (10.4)	49	26.3 (4.1)	ACR criteria	0 to 10 scale; higher worse 5.6 (0.9)	0 to 100 scale; higher worse 31.7 (6.8)	3 and 7 weeks
	Conventional treatment	41	53.7 (9.7)	68	27.1 (5.6)		5.8 (0.7)	33.0 (7.9)	
Qiu 2023 **(China)**	General treatment + exercise	21	71.3 (7.8)	72	24.5 (3.4)	ACR criteria	0 to 10 scale; higher worse 3.5 (1.9)	0 to 68 scale; higher worse 19.7 (12.4)	6 weeks
	General treatment	21	74.4 (4.8)	78	24.7 (3.1)		4.8 (1.9)	17.2 (9.9)	
Rafiq 2021a **(Pakistan)**	Usual care + exercise	28	54.2 (5.2)	61	33.9 (5.7)	KL Grade II or III	Not reported	0 to 100 scale; higher worse 41.0 (17.0) Median (IQR)	12 weeks
	Usual care	28	55.0 (4.9)	68	31.0 (2.8)		Not reported	39.6 (13.6)	
Rafiq 2021b **(Pakistan)**	Usual care + exercise (strengthening with mobile health)	38	54.0 (4.4)	55	33.1 (4.3)	KL Grade II or III diagnosed by an orthopaedic surgeon	0 to 20 scale; higher worse 10.6 (2.5)	0 to 10 scale; higher worse 4.7 (1.2)	12 weeks
	Usual care + exercise (strengthening without mobile health)	38	51.7 (4.9)	55	32.0 (3.6)		9.1 (2.3)	4.6 (0.9)	
	Usual care	38	52.9 (4.6)	58	32.0 (4.7)		9.3 (2.6)	4.4 (1.4)	

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

Rafiq 2021c **(Pakistan)**	Education + exercise	25	53.4 (5.2)	56	32.2 (4.5)	KL Grade II or III	0 to 20 scale; higher worse 9.8 (1.5)	0 to 68 scale; higher worse 24.5 (12.8)	4 weeks
	Education	25	52.8 (5.7)	52	32.0 (3.9)		10.0 (1.9)	22.7 (11.7)	
Rakhshani 2022 **(Iran)**	Routine physiotherapy + exercise	18	53.6 (8.8)	67 (across whole sample)	26.3 (SD not reported)	ACR criteria	0 to 10 scale; higher worse 6.3 (1.5)	Not reported	6 weeks
	Routine physiotherapy	18	53.1 (7.1)	67 (across whole sample)	25.8 (SD not reported)		6.3 (1.5)	Not reported	
Rosadi 2023 **(Indonesia)**	Exercise (12 weeks)	12	63.3 (8.3)	40	25.2 (3.6)	KL Grade I to IV and physician-diagnosed	0 to 10 scale; higher worse 6.2 (1.1)	0 to 68 scale; higher worse 39.8 (5.7)	6 and 12 weeks
	Exercise (6 weeks)	14	63.0 (6.9)	60	28.5 (4.6)		5.9 (0.9)	37.2 (8.8)	
	Usual physiotherapy	13	64.0 (7.4)	90	25.4 (3.1)		5.9 (0.9)	38.2 (6.9)	
Salli 2010 **(Turkey)**	Paracetamol + exercise (concentric-eccentric)	25	55.7 (8.2)	83	31.5 (4.4)	ACR criteria, KL Grade I or II	0 to 10 scale; higher worse 7.1 (1.2)	0 to 68 scale; higher worse 29.4 (10.8)	8 and 20 weeks
	Paracetamol + exercise (isometric)	25	57.1 (6.8)	83	32.65 (4.3)		7.5 (1.4)	35.2 (8.5)	
	Paracetamol	25	58.3 (6.7)	79	32.8 (5.1)		7.1 (1.3)	33.5 (12.3)	
Samuel 2014 **(India)**	Interferential + exercise	37	Reported separately for M and F	65	Reported separately for M and F	Chronic knee pain > 6 months and KL	0 to 20 scale; higher worse 13.7 (2.5)	0 to 68 scale; higher worse 40.0 (6.3)	4 weeks

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

						Grade I to III			
	Interferential	36	Reported separately for M and F	69	Reported separately for M and F		13.7 (2.8)	41.8 (8.6)	
Sedaghat-nezhad 2021 **(Iran)**	Physiotherapy + exercise	15	53.8 (7.4)	93	26.6 (2.1)	ACR criteria	0 to 100 scale; higher worse 42.7 (10.5)	Not reported	2 and 6 weeks
	Physiotherapy	15	59.6 (7.4)	73	27.7 (1.9)		39 (14.8)	Not reported	
Shabbir 2022 **(Pakistan)**	Conventional therapy + exercise	18	52.2 (7.4)	78	Not reported	KL Grade I to III	0 to 10 scale; higher worse 7.4 (1.0)	0 to 100 scale; higher worse 37.6 (8.0)	6 weeks
	Conventional therapy	18	53.8 (7.4)	83	Not reported		7.5 (0.9)	37.3 (8.0)	
Shin 2017 **(South Korea)**	Theracurmin + exercise	12	40.6 (7.7)	Not reported	25.3 (2.7)	KL Grade II or III	0 to 10 scale; higher worse 5.0 (1.9)	0 to 68 scale; higher worse 23.2 (13.1)	4 weeks
	Theracurmin	13	44.8 (5.4)	Not reported	25.6 (2.1)		4.4 (1.2)	17.4 (7.8)	
Talbot 2003 **(USA)**	Education + exercise	17	69.6 (6.7)	77	31.0 (5.9)	ACR criteria	McGill pain questionnaire; score range not reported 1.47 (0.72)	Not reported	12 and 24 weeks
	Education	17	70.8 (4.7)	77	32.6 (6.9)		1.20 (1.13)	Not reported	
Thomas 2002 **(UK)**	Exercise	235	61.5 (9.6)	64	Not reported	Self-reported persistent knee pain	0 to 20 scale; higher worse 6.9 (3.6)	Not reported by group. Combined into exercise (n = 466) 23.15 and non-exercise (n = 316) 22.97	24 months

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Table 2. Baseline demographic and clinical characteristics of the trial participants *(Continued)*

	Exercise plus telephone	121	62.6 (9.4)	61	Not reported		7.0 (3.5)		
	Exercise, telephone plus placebo	114	62.5 (8.9)	62	Not reported		7.8 (3.6)		
	Telephone	160	61.5 (9.0)	66	Not reported		7.4 (3.5)		
	Placebo	78	61.9 (9.8)	65	Not reported		7.5 (3.6)		
	No intervention	78	61.9 (9.4)	65	Not reported		7.0 (3.7)		
Van Baar 1998 **(Netherlands)**	Education + exercise	54	68.3 (8.4)	78	Not reported	ACR criteria	0 to 100 scale; higher worse 46.9 (27.7)	-28 to -7 scale -20.0 (5.8)	12 weeks
	Education	59	67.7 (9.2)	79	Not reported		43.1 (26.8)	-20.6 (5.4)	

ACR: American College of Rheumatology; BMI: body mass index; F: female; IQR: interquartile range; KL: Kellgren and Lawrence system; M; male; MMT: manual and manipulative therapy; NICE: National Institute for Health and Care Excellence (UK); n: number; OA: osteoarthritis; OASI: Osteoarthritis Screening Index; ROM: range of motion; SD: standard deviation; SE: standard error; SEM: standard error of the mean; TENS: transcutaneous electrical nerve stimulation; WOMAC: Western Ontario and McMaster Universities Arthritis Index

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Table 3. Characteristics of interventions in included trials

Study ID	Treatment groups	Description of exercise	Description of comparison
Exercise versus attention control/placebo (30 trials)			
Baker 2001	Attention control Exercise	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person Strengthening 48 sessions (3 x per week over 16 weeks); 12 synchronous contact occasions	Attention control; 7 synchronous contact occasions (home visits where received general nutrition education)
Bandak 2022	Saline injection Exercise	Clinic, supervised group setting, entirely in-person Neuromuscular exercise 14 sessions (2 x 60 minutes per week over 12 weeks); 14 synchronous contact occasions	Intra-articular saline injections; 4 synchronous contact occasions
Brismée 2007	Attention control Exercise	Mix of home/clinic, mix of supervised/unsupervised mix of individual/group, entirely in-person Tai Chi 36 sessions (3 x 40 minutes per week over 12 weeks); 18 synchronous contact occasions	Attention control; education programme delivered via 18 synchronous contact occasions
Cheung 2017	Attention control Exercise 1 (yoga) Exercise 2 (mixed)	Mix of home/clinic, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person Group 1: yoga 40 sessions (5 x 30 to 45 minutes per week over 8 weeks); 8 synchronous contact occasions Group 2: aerobic, strengthening 64 sessions (8 x 45 minutes per week over 8 weeks); 8 synchronous contact occasions	Attention control; education brochures + 8 synchronous contact occasions (telephone calls, asked about symptoms and general health)
De Almeida 2020	Attention control Exercise 1 (circuit training) Exercise 2 (conventional strength training)	Clinic, supervised group setting, delivered entirely in-person Exercise 1: circuit training 42 sessions (3x per week over 14 weeks); 42 synchronous contact occasions Exercise 2: conventional strength training 42 sessions (3 x 60 minutes per week over 14 weeks); 42 synchronous contact occasions	Attention control; education sessions, 6 synchronous contact occasions
Ettinger 1997	Attention control Exercise 1 (aerobic walking)	Clinic, supervised group setting, delivered entirely in-person Exercise 1: aerobic walking	Attention control; education, 3 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

	Exercise 2 (strengthening)	36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions Exercise 2: strengthening 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	
Foley 2003	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening + range of motion 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions	Attention control; fortnightly telephone calls to check condition, drug use, or injuries, 3 synchronous contact occasions
Imoto 2012	Attention control Exercise	Clinic, supervised group setting, delivered entirely in person Strengthening 16 sessions (2 x 30 to 40 minutes per week over 8 weeks); 16 synchronous contact occasions	Attention control; biweekly phone calls to encourage health education; 3 synchronous contact occasions
Jenkinson 2009	Attention control Exercise	Home, unsupervised individual setting, delivered entirely in-person Strengthening 728 sessions (7 x per week over 104 weeks); 6 synchronous contact occasions	Attention control; home visits and support telephone calls, 6 synchronous contact occasions
Kang 2022	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Tai Chi 108 sessions (3 x 60 minutes per week over 36 weeks); 108 synchronous contact occasions	Attention control; wellness education: 9 synchronous contact occasions
Kim 2013	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, proprioceptive/balance training 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions	Attention control; education, 3 synchronous contact occasions
Kovar 1992	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Walking, stretching 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions	Attention control; weekly telephone call regarding activities of daily living, 8 synchronous contact occasions
Kuntz 2018	Attention control Exercise 1 (yoga) Exercise 2 (strengthening)	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: yoga	Attention control; meditation 12 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

		36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions Exercise 2: strengthening 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	
Lai 2021	Attention control Exercise	Mix of home/clinic, entirely supervised group setting, delivered entirely in-person Strengthening 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions	Attention control; health education, 8 synchronous contact occasions
Liu 2019	Attention control Exercise 1 (Tai Chi) Exercise 2 (Baduanjin) Exercise 3 (cycling)	Clinic, supervised group setting, delivered entirely in-person Exercise 1: Tai Chi 60 sessions (5 x 60 minutes per week over 12 weeks); 60 synchronous contact occasions Exercise 2: Baduanjin 60 sessions (5 x 60 minutes per week over 12 weeks); 60 synchronous contact occasions Exercise 3: cycling 60 sessions (5 x 60 minutes per week over 12 weeks); 60 synchronous contact occasions	Attention control; health education, 12 synchronous contact occasions
Lu 2017	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Tai Ji Quan 72 sessions (3 x 60 minutes per week over 24 weeks); 72 synchronous contact occasions	Attention control; health education, 12 synchronous contact occasions
Maurer 1999	Attention control Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions	Attention control; health education, 4 synchronous contact occasions
Messier 2004	Attention control Exercise	Clinic, mix supervised/unsupervised mix group/individual setting, delivered entirely in-person Strengthening, walking 78 sessions (3x per week for 26 weeks); 51 synchronous contact occasions	Attention control; health education, 6 synchronous contact occasions
Messier 2021	Attention control Exercise 1 (high intensity)	Clinic, supervised group setting, delivered entirely in-person Exercise 1: high-intensity strengthening	Attention control; health education, 24 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

	Exercise 2 (low intensity)	234 sessions (3 x 60 minutes per week over 78 weeks); 234 synchronous contact occasions Exercise 2: low-intensity strengthening 234 sessions (3 x 60 minutes per week over 78 weeks); 234 synchronous contact occasions	
Peloquin 1999	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, mixed aerobic, stretching 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	Attention control; health education, 6 synchronous contact occasions
Rego 2023	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Pilates 14 sessions (2 x 60 minutes per week over 7 weeks); 14 synchronous contact occasions	Attention control; phone calls to check health status and activities of daily living, 14 synchronous contact occasions
Santana 2022	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, proprioceptive/balance training 12 sessions (2 x 50 minutes per week over 6 weeks); 12 synchronous contact occasions	Attention control; telephone calls to monitor symptoms and check not exercising, 12 synchronous contact occasions
Schlenk 2020	Attention control Exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered mix of digital/in-person Strengthening, proprioceptive/balance training, walking, stretching 78 sessions (2 to 3 x 60 to 65 minutes per week over 26 weeks); 15 synchronous contact occasions	Attention control; health education, 15 synchronous contact occasions
Shen 2022	Attention control Exercise	Clinic, supervised individual setting, delivered entirely in-person Stretching 18 sessions (3 x 60 minutes per week over 6 weeks); 18 synchronous contact occasions	Attention control; health education; 18 synchronous contact occasions
Simao 2012	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Cycling 36 sessions (3 x 30 minutes per week over 12 weeks); 36 synchronous contact occasions	Attention control; telephone calls to confirm adherence to daily activities, 12 synchronous contact occasions
Song 2003	Attention control Exercise	Clinic, mix supervised/unsupervised mix group/individual setting, delivered entirely in-person Tai Chi	Attention control; telephone calls to ensure not exercising and provide at-

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Table 3. Characteristics of interventions in included trials *(Continued)*

		36 sessions (3 x 60 minutes per week over 12 weeks), 16 synchronous contact occasions	nous contact occasions
Song 2020	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person PNF stretching 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	Attention control; watched TV or read magazines together instead of exercising, 36 synchronous contact occasions
Tsai 2013	Attention control Exercise	Clinic, supervised group setting, delivered entirely in-person Tai Chi 60 sessions (3 x 20 to 40 minutes per week over 20 weeks); 60 synchronous contact occasions	Attention control; health education, 60 synchronous contact occasions
Vincent 2020	Attention control Exercise 1 (concentric strengthening) Exercise 2 (eccentric strengthening)	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: concentric strengthening 32 sessions (2x per week over 16 weeks); 32 synchronous contact occasions Exercise 2: eccentric strengthening 32 sessions (2x per week over 16 weeks); 32 synchronous contact occasions	Attention control; telephone calls to encourage healthy behaviours, 16 synchronous contact occasions
Ye 2020	Attention control Exercise	Mix clinic/home, mix supervised/unsupervised mix of group/individual setting, delivered mix of in-person/digital delivery Baduanjin Qigong 36 sessions (3 x 40 minutes per week over 12 weeks); 12 synchronous contact occasions	Attention control; telephone calls to remind not to engage in exercise, 6 synchronous contact occasions
Exercise versus no treatment/usual care/limited education (60 trials)			
Abbott 2013	Exercise Usual care	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person Aerobic + strengthening + neuromuscular 156 sessions (3 x 50 minutes over 16 weeks); 9 synchronous contact occasions	All participants continued to receive routine care offered by their own GP and other healthcare providers. No trial interventions were provided.
Allen 2018	Wait list Exercise	Home-based, unsupervised individual setting, entirely digitally-delivered Strengthening, mixed aerobic, stretching 48 sessions (3 x per week over 16 weeks); 0 synchronous contact occasions	Wait list; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

An 2008	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Baduanjin (type of Qigong, less physically demanding than Tai Chi) 40 sessions (5 x 30 minutes per week over 8 weeks); 40 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Bennell 2010	Wait list Exercise	Clinic, mix of supervised/unsupervised individual, entirely in-person Strengthening 60 sessions (5 x 15 to 30 minutes per week over 12 weeks); 7 synchronous contact occasions	Waiting list; 0 synchronous contact occasions
Braghin 2018	No intervention Exercise	Clinic, supervised group setting, entirely in-person Strengthening, mixed aerobic, stretching 16 sessions (2 x 50 to 60 minutes per week over 8 weeks); 16 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Bruce-Brand 2012	Standard care Exercise	Home, mixed supervised/unsupervised individual setting, delivered entirely in-person Strengthening 18 sessions (2 x 30 minutes per week over 6 weeks); 12 synchronous contact occasions	Standard care; unclear number synchronous contact occasions
Chen 2021	No treatment Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3 x per week over 8 weeks); 24 synchronous contact occasions	No treatment; 0 synchronous contact occasions
Cheung 2014	Wait list Exercise	Mix of home/clinic, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person Yoga 40 sessions (5 x 30 to 60 minutes per week over 8 weeks); 8 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Chopp-Hurley 2017	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Yoga 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Christensen 2015	No intervention Exercise	Mix of clinic/home, mix of supervised/unsupervised mix of individual/group setting, delivered entirely in-person Neuromuscular 196 sessions (3 to 4x per week over 52 weeks); 52 synchronous contact occasions	No intervention; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Da Silva 2015	Education Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, proprioceptive, body mobility 16 sessions (2 x 60 minutes per week over 8 weeks); 17 synchronous contact occasions	Education; 1 synchronous contact occasions
De Matos 2019	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, proprioceptive/balance, circuit training, cycling, stretching 15 sessions (2 x 50 minutes per week over 8 weeks); 15 synchronous contact occasions	No intervention; 0 synchronous contact occasions
De Rooij 2017	No intervention Exercise	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person Strengthening, proprioceptive/balance, mixed aerobic, stretching 140 sessions (7 x 30 to 60 minutes per week over 20 weeks); 40 synchronous contact occasions	No intervention; 0 synchronous contact occasions
DeVita 2018	No intervention Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Drummen 2023	Usual care Exercise	Mix of home/clinic, mix of supervised/unsupervised mix of individual/group setting, delivered entirely in-person Walking 72 sessions (3 x 60 minutes per week over 24 weeks); 48 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Fransen 2001	Wait list Exercise	Mix of clinic/home, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person Strengthening + mixed aerobic 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Fransen 2007	Wait list Exercise	Clinic, supervised group setting, delivered entirely in-person Tai Chi 24 sessions (2 x 60 minutes per week for 12 weeks); 24 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Gohir 2021	Education Exercise	Home, unsupervised individual setting, delivered entirely digitally Strengthening, neuromuscular, proprioceptive/balance 42 sessions (7x per week over 6 weeks); 0 synchronous contact occasions	Education; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Guo 2022	Usual care Exercise	Home, unsupervised individual setting, delivered entirely digitally-delivered Strengthening, mixed aerobic 24 sessions (3x per week over 16 weeks); 24 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Gur 2002	No treatment Exercise 1 (concentric strengthening) Exercise 2 (eccentric and concentric strengthening)	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions	No treatment; 0 synchronous contact occasions
Hay 2006	Education Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening + mixed aerobic unclear number of sessions over 10 weeks; 3 to 6 synchronous contact occasions	Education; 1 synchronous contact occasion
Henriksen 2014	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, neuromuscular, circuit training 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Hopman-Rock 2000	Wait list Exercise	Mix of clinic/home, mix of supervised/unsupervised mix of group/individual setting, delivered entirely in-person Strengthening + mixed aerobic 24 sessions (4 x 60 minutes per week for 6 weeks); 6 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Hu 2020	Education Exercise	Clinic, supervised group setting, delivered entirely in-person Tai Chi 72 sessions (3x per week over 24 weeks); 73 synchronous contact occasions	Education; 1 synchronous contact occasion
Hurley 2007	Usual care Exercise - same intervention delivered two ways - either on individual or class basis	Clinic, supervised mix of individual/group setting, delivered entirely in-person Strengthening, balance, aerobic, and motor control 12 sessions (2 x 45 minutes per week over 6 weeks); 12 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Jan 2009	No intervention Exercise 1 (strengthening – weight bearing)	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: strengthening - weight bearing	No intervention; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

	Exercise 2 (strengthening – non-weight bearing)	24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions Exercise 2: strengthening - non-weight bearing 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions	
Janyacharoen 2018	Education Exercise	Clinic, supervised group setting, delivered entirely in-person Sakon Nakhom ancient boxing 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	Education; 1 synchronous contact occasions
Jorge 2015	Wait list Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (2x per week over 12 weeks); 24 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Karadag 2019	Usual care Exercise	Home, mix of supervised/unsupervised individual setting, delivered entirely in-person Strengthening 40 sessions (10 x 10 minutes per week over 4 weeks); 13 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Keefe 2004	Standard care Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, mixed aerobic 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	Standard care; 0 synchronous contact occasions
Kim 2020	Usual care Exercise	Mix of home/clinic, supervised/unsupervised individual/group setting, delivered via mix of in-person and digital delivery Strengthening, stretching, dance 48 sessions (4 x 60 minutes per week over 12 weeks); 12 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Krishnan 2021	No treatment Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening 16 sessions (2 x 60 minutes per week over 8 weeks); 16 synchronous contact occasions	No treatment; 0 synchronous contact occasions
Lee 2009	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Tai Chi Qigong 16 sessions (2 x 45 minutes per week over 8 weeks); 16 synchronous contact occasions	No intervention control; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Lee 2023	Usual care Exercise	Home, mix of supervised/unsupervised individual setting, delivered via mix of in-person and remotely Strengthening 24 sessions (3x per week over 8 weeks); 5 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Li 2020	Wait list Exercise	Home, mix of supervised/unsupervised individual setting, delivered entirely digitally-delivered Physical activity Variable number of sessions (variable frequency/duration over 12 weeks); 5 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Lim 2008	No intervention (1 group malaligned and 1 group neutrally aligned) Exercise (1 group malaligned and 1 group neutrally aligned)	Home, mix of supervised/unsupervised individual setting, delivered entirely in-person Strengthening 60 sessions (5x per week over 12 weeks); 7 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Lin 2009	No intervention Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3 x 50 minutes per week over 8 weeks); 24 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Lund 2008	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, balance, stretching 16 sessions (2 x 50 minutes per week over 8 weeks); 16 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Mazloum 2018	Usual care Exercise 1 (conventional exercise) Exercise 2 (Pilates)	Clinic, supervised group setting, delivered entirely in-person Exercise 1: conventional therapeutic exercise 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions Exercise 2: Pilates 24 sessions (3x per week over 8 weeks); 24 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Multanen 2014	Usual care Exercise	Clinic, supervised group setting, delivered entirely in-person Mixed aerobic	Usual care; option to participate in social group meeting (4 synchronous contact occasions)

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Table 3. Characteristics of interventions in included trials *(Continued)*

		156 sessions (3 x 55 minutes per week over 52 weeks); 156 synchronous contact occasions	
Øiestad 2023	Usual care Exercise 1 (strengthening) Exercise 2 (cycling)	Mix of clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person Strengthening 36 sessions (3x per week over 12 weeks); 24 synchronous contact occasions Cycling 36 sessions (3x 45 minutes per week over 12 weeks); 24 synchronous contact occasions	Usual care; 0 synchronous contact occasions
O'Reilly 1999	Lifestyle advice Lifestyle advice + exercise	Home, mix supervised/unsupervised individual setting, delivered entirely-in-person Strengthening 183 sessions (7x per week over 26 weeks); 4 synchronous contact occasions	Lifestyle advice; 0 synchronous contact occasions
Pazit 2018	Usual care Exercise 1 (strengthening) Exercise 2 (strengthening + balance)	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: strengthening 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions Exercise 2: strengthening + balance 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Robbins 2022	Education Exercise	Clinic, supervised group setting, delivered entirely in-person Stretching 24 sessions (3 x 45 minutes per week over 8 weeks); 24 synchronous contact occasions	Education; 0 synchronous contact occasions
Rogind 1998	No intervention Exercise	Mix clinic/home, mix supervised/unsupervised mix clinic/individual, delivered entirely in-person Strengthening, balance 36 sessions (3 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Rosedale 2014	Wait list Exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person Strengthening + walking program + cycling	Wait list; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

		Variable number of sessions (approx 3 to 4 x 20 to 60 minutes per week over 2 weeks, but varied widely); 4 to 6 synchronous contact occasions	
Sadeghi 2023	No intervention Exercise 1 (quad strengthening) Exercise 2 (hamstring strengthening) Exercise 3 (quad and hamstring strengthening)	Home, mix of supervised and unsupervised individual setting, delivered via mix of in-person and remotely Quad strengthening exercise Hamstring strengthening exercise Quad and hamstring strengthening exercise 40 sessions (5x per week over 8 weeks); 8 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Salacinski 2012	Wait list Exercise	Clinic, supervised group setting, delivered entirely in-person Cycling 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Samut 2015	No treatment Exercise 1 (strengthening) Exercise 2 (walking)	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: strengthening 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions Exercise 2: walking 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions	No treatment; 0 synchronous contact occasions
Schilke 1996	No intervention Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Sekir 2005	No intervention Exercise	Clinic, supervised individual setting, delivered entirely in-person Proprioceptive/balance training 12 sessions (1 to 3x per week over 6 weeks); 12 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Shahine 2020	Education Exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered via mix of in-person/digital Walking	Education; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

		variable number of sessions (unspecified frequency over 12 weeks); 16 synchronous contact occasions	
Takacs 2017	No intervention Exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person Strengthening, proprioceptive/balance training 40 sessions (4 x per week over 10 weeks); 6 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Thorstensson 2005	Wait list Exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening 12 sessions (2 x 60 minutes per week over 6 weeks); 12 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Topp 2002	No intervention Exercise 1 strengthening - dynamic Exercise 2 strengthening - isometric	Mix clinic/home, mix superviseds/unsupervised mix group/individual setting, delivered entirely in-person Strengthening 48 sessions (3 x 60 minutes per week over 16 weeks); 16 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Vassao 2021	Wait list Exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening, stretching + sham photobiomodulation 16 sessions (2x per week over 8 weeks); 16 synchronous contact occasions	Wait list; 0 synchronous contact occasions
Wallis 2017	Usual care Exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered via mix of in-person/digital Walking Variable number of sessions (accumulation of 70 minutes walking per week over 12 weeks); 13 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Wang 2011	No intervention Exercise	Clinic, supervised group setting, delivered entirely in-person Flexibility and aerobic 36 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions	No intervention; 0 synchronous contact occasions
Wortley 2013	No treatment Exercise 1 (strengthening) Exercise 2 (Tai Ji)	Clinic, supervised group setting, delivered entirely in-person Both exercise groups: 20 sessions (2 x 60 minutes per week over 10 weeks); 20 synchronous contact occasions	No treatment; 1 synchronous contact occasions
Xiao 2021	No treatment Exercise	Unclear setting, delivered entirely in-person Wuqinxi-Qigong	No treatment; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

		144 sessions (6 x 60 minutes per week over 24 weeks); unspecified number of synchronous contact occasions	
Exercise when added to another co-intervention (compared to that co-intervention alone) (49 trials)			
Alghadir 2019	Routine physiotherapy Routine physiotherapy + retrowalking Routine physiotherapy + forward walking	Clinic, supervised group setting, delivered entirely in-person Retrowalking and forward walking 18 sessions (3 x 10 to 30 minutes per week over 6 weeks); 18 synchronous contact occasions in both groups	Routine physiotherapy; 18 synchronous contact occasions
Anadkat 2015	Conventional exercises Conventional exercises + retrowalking	Clinic, supervised individual setting, delivered entirely in-person Retrowalking 51 sessions (5 x 10 minutes per week over 3 weeks); 15 synchronous contact occasions	Conventional exercises: 15 synchronous contact occasions
Anwer 2014	Ultrasound Ultrasound + exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 50 sessions (10x per week over 5 weeks); 50 synchronous contact occasions	Ultrasound: 6 synchronous contact occasions
Assar 2020	Drug regimen Drug regimen + exercise	Clinic, supervised group setting, entirely in-person Total Resistance eXercises (TRX) 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions	Drug regimen: 0 synchronous contact occasions
Bautch 1997	Education Education + exercise	Clinic, supervised individual, entirely-in-person Range of motion + walking 36 sessions (3x per week over 12 weeks); 36 synchronous contact occasions	Education; 12 synchronous contact occasions
Bennell 2016	Pain coping skills training Pain coping skills training + exercise	Mix of home/clinic, mix of supervised/unsupervised individual setting, entirely in-person Neuromuscular exercise 48 sessions (4 x 25 minutes per week over 12 weeks); 10 synchronous contact occasions	Pain coping skills training; 10 synchronous contact occasions
Bennell 2022a	Education Education + exercise	Home, unsupervised individual setting, entirely digitally-delivered Strengthening, physical activity plan 78 sessions (3 x per week over 26 weeks); 6 synchronous contact occasions	Education website; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Bennell 2022b	Education Education + exercise	Home, unsupervised individual setting, entirely digitally-delivered Yoga 36 sessions (3 x 30 minutes per week over 12 weeks); 0 synchronous contact occasions	Education website; 0 synchronous contact occasions
Bhagat 2017	Interferential therapy Interferential therapy + exercise	Clinic, supervised individual setting, delivered entirely in-person Circuit training 12 sessions (3 x 20 minutes per week over 4 weeks); 12 synchronous contact occasions	Interferential therapy; 12 synchronous contact occasions
Chang 2012	General physiotherapy General physiotherapy + exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 16 sessions (2 x 60 minutes per week over 8 weeks); 16 synchronous contact occasions	General physiotherapy; 16 synchronous contact occasions
Chen 2021	Conventional treatment Conventional treatment + exercise	Mix of home/clinic, mix of supervised/unsupervised individual setting, delivered entirely in-person Retrowalking 12 sessions (3 x 10 minutes per week over 4 weeks); 5 synchronous contact occasions	Conventional treatment; 4 synchronous contact occasions
Dincer 2016	TENS TENS + exercise	Mix of clinic/home, mix of supervised/unsupervised mix of individual/group setting, delivered entirely in-person Strengthening 60 sessions (5 x 30 minutes per week over 12 weeks); 15 synchronous contact occasions	TENS; 15 synchronous contact occasions
Dwyer 2015	MMT MMT + exercise	Mix of home/clinic, mix of supervised/unsupervised mix of individual/group setting, entirely in-person Strengthening + cycling + stretching 12 sessions (3 x 20 minutes per week over 4 weeks); 9 synchronous contact occasions	MMT; 6 synchronous contact occasions
Gondhalekar 2013	Conventional treatment Conventional treatment + exercise	Mix of clinic/home, Mix of supervised/unsupervised setting, delivered entirely in-person Retrowalking 63 sessions (10 minutes per session, 3x per day over 3 weeks); unknown number of synchronous contact occasions	Conventional treatment; unknown number of synchronous contact occasions
Hernandez 2019	Conventional exercises Conventional exercises + core stability exercises	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered entirely in-person Core stability exercises	Conventional exercises; 24 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

		36 sessions (3x per week over 12 weeks); 24 synchronous contact occasions	
Hinman 2020	Phone advice Phone advice + exercise	Home, unsupervised individual setting, delivered entirely digitally-delivered Strengthening, physical activity 78 sessions (3x per week over 26 weeks); 6 to 11 synchronous contact occasions	Phone advice; 1 synchronous contact occasion
Holm 2020	Education + neuromuscular exercise Education + neuromuscular exercise + strengthening exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions	Education + neuromuscular exercise; 24 synchronous contact occasions
Hsu 2021	Diet Diet + exercise	Mix of clinic/home, mix of supervised/unsupervised individual setting, delivered via mix of in-person and digital Strengthening 36 sessions (3 x per week over 12 weeks); 24 synchronous contact occasions	Diet; 12 synchronous contact occasions
Huang 2003	Hotpack/ROM Hotpack/ROM + exercise 1 (isokinetic strengthening) Hotpack/ROM + exercise 2 (isometric strengthening) Hotpack/ROM + exercise 3 (isotonic strengthening)	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3 x 60 minutes per week for 8 weeks); 24 synchronous contact occasions	Hotpack/ROM; unclear number of contact occasions
Huang 2005	Hotpack/ROM Hotpack/ROM + exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 24 sessions (3 x 60 minutes per week for 8 weeks); 24 synchronous contact occasions	Hotpack/ROM; unclear number of contact occasions
Jan 2008	Education Exercise 1 (high resistance training) Exercise 2 (low resistance training)	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: high resistance training 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions Exercise 2: low resistance training 24 sessions (3 x 30 minutes per week over 8 weeks); 24 synchronous contact occasions	Education; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Joshi 2019	Conventional exercise Conventional exercise + retrowalking	Clinic, supervised individual setting, delivered entirely in-person Retrowalking 18 sessions (3 x 10 minutes per week over 6 weeks); 18 synchronous contact occasions	Conventional exercise: 0 synchronous contact occasions
Kholvadia 2019	Low level laser therapy Low level laser therapy + exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, neuromuscular, stretching 36 sessions (3 x per week over 12 weeks); 12 synchronous contact occasions	Low level laser therapy; 12 synchronous contact occasions
Kilic 2020	Physiotherapy Physiotherapy + exercise	Clinic, supervised individual setting, delivered entirely in-person Walking 40 sessions (5 x 40 minutes per week over 6 weeks); 30 synchronous contact occasions	Physiotherapy: 30 synchronous contact occasions
Knoop 2013	Exercise without feedback Exercise with feedback (neuromuscular/neuromotor exercise with personalised feedback from PT) + exercise for knee position	Mix of home/clinic, mix of supervised/unsupervised group setting, delivered entirely in-person Neuromuscular/neuromotor exercise with personalised feedback from PT) + exercise for knee position 84 sessions (7 x 420 minutes per week over 12 weeks); 24 synchronous contact occasions	Exercise without feedback; 24 synchronous contact occasions
Kumar 2013	Supervised exercise + ultrasound Supervised exercise + ultrasound + proprioceptive/balance training	Clinic, supervised individual setting, delivered entirely in-person Proprioceptive/balance training 12 sessions (3 x per week over 4 weeks); 24 synchronous contact occasions	Supervised exercise + ultrasound; 12 synchronous contact occasions
Lee 2019	HA injection HA injection + exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening, proprioceptive, core stability 20 sessions (5 x 30 minutes per week over 4 weeks); 20 synchronous contact occasions	HA injection; 1 synchronous contact occasion
Marconcin 2018	Education Education + exercise	Clinic, supervised group setting, delivered entirely in-person Strengthening, proprioceptive/balance 24 sessions (2 x 60 minutes per week over 12 weeks); 24 synchronous contact occasions	Education; 6 synchronous contact occasions
Messier 2013	Diet Diet + exercise	Mix clinic/home, mix supervised/unsupervised mix individual/group setting, delivered entirely in-person Strengthening + mixed aerobic	Diet; 54 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

		234 sessions (3 x 60 minutes per week over 78 weeks); 90 synchronous contact occasions	
Mete 2022	Physiotherapy Physiotherapy + exercise	Clinic, supervised individual setting, delivered entirely in-person Strengthening 30 sessions (5 x 20 minutes per week over 6 weeks); 30 synchronous contact occasions	Physiotherapy; 30 synchronous contact occasions
Mehwish 2023	Knee exercises Knee exercises + hip strengthening	Clinic, supervised individual setting, delivered entirely in-person Strengthening 18 sessions (5 x 30 to 40 minutes per week over 4 weeks); 18 synchronous contact occasions	Knee exercises; 18 synchronous contact occasions
Nahayatbin 2018	Routine physiotherapy Routine physiotherapy + exercise 1 Routine physiotherapy + exercise 2	Clinic, supervised setting, delivered entirely in-person Exercise 1: strengthening 12 sessions (3 x 20 minutes per week over 4 weeks); 12 synchronous contact occasions Exercise 2: Tai Chi 12 sessions (3 x 20 minutes per week over 4 weeks); 12 synchronous contact occasions	Routine physiotherapy; 1 synchronous contact occasion
Nelligan 2021	Education Education + exercise	Home, unsupervised individual setting, delivered entirely digitally Strengthening and general physical activity 72 sessions (3 x 20 to 30 minutes per week over 24 weeks); 0 synchronous contact occasions	Education; 0 synchronous contact occasions
Oh 2020	Education Education + exercise	Mix clinic/home, mix supervised/unsupervised mix individual/group setting, delivered via mix of digital/in-person Strengthening 40 to 60 sessions (2 to 3 x 60 to 80 minutes per week over 20 weeks); 6 synchronous contact occasions	Education; 5 synchronous contact occasions
Ozlu 2023	Conventional treatment Conventional treatment + exercise	Clinic, supervised individual setting, delivered entirely in-person Gamification of strengthening and proprioception exercise 15 sessions (5 x 15 minutes per week over 3 weeks); 15 synchronous contact occasions	Conventional treatment; 15 synchronous contact occasions
Qiu 2023	General treatment General treatment + exercise	Clinic, supervised group setting, delivered entirely in-person Hip muscle strengthening 90 sessions (3 x per day, 5x per week over 6 weeks); 90 synchronous contact occasions	General treatment; 30 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Rafiq 2021a	Usual care Usual care + exercise	Home, unsupervised individual setting, delivered entirely in-person Strengthening 36 sessions (3 x 60 to 85 minutes per week over 12 weeks); 0 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Rafiq 2021b	Usual care Usual care + exercise 1 Usual care + exercise 2	Home, unsupervised individual setting, delivered entirely in-person Exercise 1: strengthening with mobile health 36 sessions (3 x 65 to 80 minutes per week over 12 weeks); 36 synchronous contact occasions Exercise 2: strengthening without mobile health 36 sessions (3 x 65 to 80 minutes per week over 12 weeks); 36 synchronous contact occasions	Usual care; 0 synchronous contact occasions
Rafiq 2021c	Education Education + exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person Strengthening 12 sessions (3 x 65 to 80 minutes per week over 4 weeks); 12 synchronous contact occasions	Education; 0 synchronous contact occasions
Rakhshani 2022	Routine physiotherapy Routine physiotherapy + hip strengthening exercise	Clinic, supervised group setting, delivered entirely in-person Hip strengthening exercises 18 sessions (3x per week over 6 weeks); 18 synchronous contact occasions	Routine physiotherapy; 18 synchronous contact occasions
Rosadi 2023	Usual physiotherapy Usual physiotherapy + exercise 1 Usual physiotherapy + exercise 2	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: strengthening + walking programme + cycling 24 sessions (3 x 60 minutes per week over 12 weeks); 36 synchronous contact occasions Exercise 2: strengthening + walking programme + cycling 18 sessions (3 x 60 minutes per week over 6 weeks); 18 synchronous contact occasions	Usual physiotherapy; 18 synchronous contact occasions
Salli 2010	Paracetamol Paracetamol + exercise 1 Paracetamol + exercise 2	Clinic, supervised individual setting, delivered entirely in-person Exercise 1: strengthening, concentric-eccentric 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions Exercise 2: strengthening, isometric 24 sessions (3 x 60 minutes per week over 8 weeks); 24 synchronous contact occasions	Paracetamol; 0 synchronous contact occasions

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Table 3. Characteristics of interventions in included trials *(Continued)*

Samuel 2014	Interferential Interferential + exercise	Home, supervised individual setting, delivered entirely in-person Strengthening 20 sessions (5 x per week over 4 weeks); 20 synchronous contact occasions	Interferential; 10 synchronous contact occasions
Sedaghatnezhad 2021	Physiotherapy Physiotherapy + exercise	Clinic, supervised individual setting, delivered entirely in-person Walking 10 sessions (5 x 30 minutes per week over 2 weeks); 10 synchronous contact occasions	Physiotherapy; 15 synchronous contact occasions
Shabbir 2022	Conventional therapy Conventional therapy + exercise	Clinic, supervised individual setting, delivered entirely in-person Retrowalking 18 sessions (3 x 30 minutes per week over 6 weeks); 18 synchronous contact occasions	Conventional therapy; 18 synchronous contact occasions
Shin 2017	Theracurmin Theracurmin + exercise	Unknown whether clinic/home, supervised individual setting, delivered entirely in-person Strengthening, walking, stretching 12 sessions (3 x 15 to 30 minutes per week over 4 weeks); 12 synchronous contact occasions	Theracurmin; 0 synchronous contact occasions
Talbot 2003	Education Education + exercise	Home, mix supervised/unsupervised individual setting, delivered entirely in-person Walking Variable frequency across 12 weeks; 12 synchronous contact occasions	Education; 12 synchronous contact occasions
Thomas 2002	Education Education + exercise	Home, mix supervised/unsupervised individual setting, delivered entirely in-person Strengthening 728 sessions (7x per week over 104 weeks); 7 synchronous contact occasions	Education; 24 synchronous contact occasions via telephone
Van Baar 1998	Education Education + exercise	Mix clinic/home, mix supervised/unsupervised individual setting, delivered entirely in-person Strengthening 36 sessions (12 over 3 weeks); 17 synchronous contact occasions	Education; 2 synchronous contact occasions

MMT: manual and manipulative therapy; PNF: proprioceptive neuromuscular facilitation; PT: personal trainer; TENS: transcutaneous electrical nerve stimulation; x: times
Synchronous contact occasions include live consultations with healthcare providers who delivered the intervention.

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix

Study ID	Pain	Function	Quality of life	Participant-reported treatment success	Adverse events
Exercise versus attention control/placebo (30 trials)					
Baker 2001	WOMAC	WOMAC	No quality of life measure	No participant-reported treatment success measure	Only reported for exercise group
Bandak 2022	KOOS pain subscale	KOOS ADL subscale	KOOS QoL	OMERACT-OARSI	Reported
Brismée 2007	VAS/NRS	WOMAC	No quality of life measure	No participant-reported treatment success measure	Only reported for exercise group
Cheung 2017	VAS/NRS	WOMAC function subscale	SF-12 MCS	No participant-reported treatment success measure	Reported
De Almeida 2020	VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Ettinger 1997	Score 1 to 6	Score 1 to 6	No quality of life measure	No participant-reported treatment success measure	Reported
Foley 2003	WOMAC pain	WOMAC function	SF-12 MCS	No participant-reported treatment success measure	Not reported
Imoto 2012	VAS/NRS	SF-36 functional capacity	No quality of life measure	No participant-reported treatment success measure	Not reported
Jenkinson 2009	WOMAC pain subscale	WOMAC function sub-scale	No quality of life measure	30% or greater pain relief	Not reported
Kang 2022	VAS/NRS	WOMAC function subscale	SF-36 MCS	No participant-reported treatment success measure	Not reported
Kim 2013	VAS/NRS	Global disability score	No quality of life measure	No participant-reported treatment success measure	Not reported
Kovar 1992	AIMS	AIMS	No quality of life measure	No participant-reported treatment success measure	Reported
Kuntz 2018	KOOS pain subscale	KOOS ADL	KOOS QoL	No participant-reported treatment success measure	Not reported
Lai 2021	VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Liu 2019	KOOS pain subscale	KOOS ADL function subscale	KOOS QoL	No participant-reported treatment success measure	Reported
Lu 2017	WOMAC pain subscale	WOMAC function subscale	SF-36 MCS	No participant-reported treatment success measure	Not reported
Maurer 1999	WOMAC pain subscale	WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

Messier 2004	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Only reported for exercise group
Messier 2021	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Peloquin 1999	AIMS	AIMS	No quality of life measure	No participant-reported treatment success measure	Not reported
Rego 2023	WOMAC pain subscale	WOMAC function subscale	SF-36 total	No participant-reported treatment success measure	Reported
Santana 2022	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Schlenk 2020	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Shen 2022	VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Simao 2012	WOMAC pain subscale	WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported
Song 2003	Korean WOMAC pain subscale	Korean WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported
Song 2020	KOOS/WOMAC pain subscale	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Tsai 2013	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Vincent 2020	Pain on activities other than walking VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Ye 2020	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Exercise versus no treatment/usual care/limited education (60 trials)					
Abbott 2013	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Allen 2018	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	Global assessment of change	Reported
An 2008	WOMAC	WOMAC	SF-36 MCS	No participant-reported treatment success measure	Reported
Bennell 2010	NRS pain on walking	WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

Braghin 2018	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Bruce-Brand 2012	WOMAC pain subscale	WOMAC function subscale	SF-36 MCS	No participant-reported treatment success measure	Not reported
Chen 2014	On activities other than walking VAS/NRS	Lequesne OA Index global score	No quality of life measure	No participant-reported treatment success measure	Not reported
Cheung 2014	WOMAC pain subscale	WOMAC function subscale	SF-12 MCS	No participant-reported treatment success measure	Reported
Chopp-Hurley 2017	KOOS pain subscale	KOOS ADL subscale	KOOS QoL	No participant-reported treatment success measure	Not reported
Christensen 2015	Global/overall pain VAS/NRS	Global disability score	SF-36 MCS	OARSI OMERACT	Reported
Da Silva 2015	Lequesne Index Pain subscale	Lequesne OA index global scale	SF-36 MCS	No participant-reported treatment success measure	Not reported
De Matos 2019	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
De Rooij 2017	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	Global change	Not reported
DeVita 2018	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Drummen 2023	Global pain VAS/NRS	WOMAC function subscale	EuroQoL	No participant-reported treatment success measure	Reported
Fransen 2001	WOMAC pain	WOMAC function	SF-36 MCS	No participant-reported treatment success measure	Not reported
Fransen 2007	WOMAC pain	WOMAC function	SF-12 MCS	No participant-reported treatment success measure	Reported
Gohir 2021	Global pain VAS/NRS	WOMAC function subscale	MSK-HQ	No participant-reported treatment success measure	Reported
Guo 2022	Global pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Gur 2002	VAS	VAS	No quality of life measure	No participant-reported treatment success measure	Not reported
Hay 2006	WOMAC pain	WOMAC physical function	Hospital Anxiety and Depression Scale	OMERACT-OARSI	Not reported
Henriksen 2014	KOOS pain subscale	KOOS ADL subscale	KOOS QoL	No participant-reported treatment success measure	Not reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

Hopman-Rock 2000	VAS	IRGL mobility	No quality of life measure	No participant-reported treatment success measure	Not reported
Hu 2020	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Hurley 2007	WOMAC pain subscale	WOMAC function subscale	EQ5D	No participant-reported treatment success measure	Not reported
Jan 2009	No pain measure	WOMAC function sub-scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Janyacharoen 2018	KOOS pain subscale	KOOS ADL sub-scale	KOOS QoL	No participant-reported treatment success measure	Not reported
Jorge 2015	Global/overall pain VAS/NRS	WOMAC function subscale	SF-36 mental health domain	Global change	Not reported
Karadag 2019	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Keefe 2004	AIMS pain subscales	No physical function measure	AIMS psychological	No participant-reported treatment success measure	Not reported
Kim 2020	No pain measure	WOMAC global scale	EQ-5D Korean version	No participant-reported treatment success measure	Not reported
Krishnan 2021	KOOS pain subscale	KOOS ADL sub-scale	KOOS QoL	No participant-reported treatment success measure	Not reported
Lee 2009	WOMAC pain subscale	WOMAC function subscale	SF-36 MCS	No participant-reported treatment success measure	Not reported
Lee 2023	Global/overall pain VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Li 2020	KOOS pain subscale	KOOS ADL sub-scale	KOOS QoL	No participant-reported treatment success measure	Reported
Lim 2008	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Lin 2009	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Lund 2008	VAS/NRS	KOOS ADL	KOOS QoL	No participant-reported treatment success measure	Not reported
Mazloum 2018	Lequesne OA Index	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Multanen 2014	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Øiestad 2023	Pain last week VAS/NRS	KOOS ADL sub-scale	EuroQoL	No participant-reported treatment success measure	Reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

O'Reilly 1999	WOMAC	WOMAC	SF-36 MCS	No participant-reported treatment success measure	Not reported
Pazit 2018	WOMAC pain subscale	WOMAC function subscale	AQoL	No participant-reported treatment success measure	Not reported
Robbins 2022	Pain on walking VAS/NRS	Lequesne OA index global score	No quality of life measure	No participant-reported treatment success measure	Not reported
Rogind 1998	VAS	AFI	No quality of life measure	No participant-reported treatment success measure	Not reported
Sadeghi 2023	Global/overall pain VAS/NRS	WOMAC function subsccale	No quality of life measure	No participant-reported treatment success measure	Not reported
Rosedale 2014	Global/overall pain VAS/NRS	KOOS ADL subscale	KOOS QoL	No participant-reported treatment success measure	Not reported
Salacinski 2012	WOMAC	WOMAC	KOOS QoL	No participant-reported treatment success measure	Not reported
Samut 2015	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Schilke 1996	OASI	OASI	No quality of life measure	No participant-reported treatment success measure	Not reported
Sekir 2005	Global pain VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Shahine 2020	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Takacs 2017	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Thorstensson 2005	KOOS pain	KOOS ADL	SF-36 MCS	No participant-reported treatment success measure	Not reported
Topp 2002	WOMAC	WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported
Vassao 2021	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Wallis 2017	Global/overall pain VAS/NRS	WOMAC function subscale	EuroQoL	No participant-reported treatment success measure	Reported
Wang 2011	KOOS pain	KOOS ADL	KOOS quality of life	No participant-reported treatment success measure	Not reported
Wortley 2013	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Xiao 2021	WOMAC pain subscale	No physical function measure	SF-36	No participant-reported treatment success measure	Not reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

Exercise when added to another co-intervention (compared to that co-intervention alone) (49 trials)

Alghadir 2019	Global pain VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Reported
Anadkat 2015	Global pain VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Anwer 2014	Global/overall pain VAS/NRS	Reduced WOMAC function scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Assar 2020	Global pain VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Bautch 1997	VAS	AIMS	No quality of life measure	No participant-reported treatment success measure	Not reported
Bennell 2016	Global pain VAS/NRS	WOMAC function subscale	AQoL-6D	Global rating of change	Not reported
Bennell 2022a	Global pain VAS/NRS	WOMAC function subscale	AQoL-8D	Global rating of change	Reported
Bennell 2022b	Global pain VAS/NRS	WOMAC function subscale	AQoL-6D	Global rating of change	Reported
Bhagat 2017	Global pain VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Chang 2012	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Chen 2021	Global pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Dincer 2016	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Dwyer 2015	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Gondhalekar 2013	Global VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Hernandez 2019	Global pain VAS/NRS	Short form WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Hinman 2020	Global pain VAS/NRS	WOMAC function subscale	AQoL-8D	Global improvement	Reported
Holm 2020	KOOS pain subscale	KOOS ADL subscale	EuroQoL	No participant-reported treatment success measure	Reported
Hsu 2021	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

Huang 2003	VAS/NRS	Lequesne function	No quality of life measure	No participant-reported treatment success measure	Not reported
Huang 2005	VAS/NRS	Lequesne function	No quality of life measure	No participant-reported treatment success measure	Not reported
Jan 2008	WOMAC pain subscale	WOMAC function subscale	No quality of life assessment	No participant-reported treatment success measure	Not reported
Joshi 2019	Global/overall pain VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Kholvadia 2019	No pain measure	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Kilic 2020	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Knoop 2013	Global pain NRS	WOMAC function subscale	No quality of life measure	Global perceived effect	Reported
Kumar 2013	Global/overall pain VAS/NRS	Reduced WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported
Lee 2019	Global VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Marconcin 2018	KOOS pain subscale	KOOS ADL subscale	EuroQoL	No participant-reported treatment success measure	Not reported
Messier 2013	WOMAC pain subscale	WOMAC function subscale	SF-36 MCS	No participant-reported treatment success measure	Reported
Mete 2022	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Mehwish 2023	Global/overall pain VAS/NRS	KOOS ADL subscale	KOOS QoL	No participant-reported treatment success measure	Reported
Nahayatbin 2018	KOOS pain subscale	KOOS ADL subscale	KOOS QoL	No participant-reported treatment success measure	Not reported
Nelligan 2021	Global/overall pain VAS/NRS	WOMAC function subscale	AQoL	Global improvement	Reported
Oh 2020	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Ozlu 2023	Global/overall pain VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Not reported
Qiu 2023	Global/overall pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Reported
Rafiq 2021a	No pain measure	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Reported

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Table 4. Outcome Reporting Bias In Trials (ORBIT) matrix *(Continued)*

Rafiq 2021b	WOMAC pain subscale	Patient-specific functional scale	No quality of life measure	No participant-reported treatment success measure	Reported
Rafiq 2021c	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	Satisfaction 0-10 NRS	Reported
Rakhshani 2022	Global/overall pain VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Rosadi 2023	Global/overall pain VAS/NRS	WOMAC function subscale	SF-36 MCS	No participant-reported treatment success measure	Not reported
Salli 2010	VAS motion	WOMAC	SF-36 MCS	No participant-reported treatment success measure	Not reported
Samuel 2014	WOMAC pain subscale	WOMAC function subscale	No quality of life measure	No participant-reported treatment success meaure	Not reported
Sedaghat-nezhad 2021	Global/overall pain VAS/NRS	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Shabbir 2022	Global/overall pain VAS/NRS	WOMAC global scale	No quality of life measure	No participant-reported treatment success measure	Reported
Shin 2017	Global pain VAS/NRS	WOMAC function subscale	No quality of life measure	No participant-reported treatment success measure	Not reported
Talbot 2003	McGill Pain Questionnaire	No physical function measure	No quality of life measure	No participant-reported treatment success measure	Not reported
Thomas 2002	WOMAC pain subscale	WOMAC	No quality of life measure	No participant-reported treatment success measure	Not reported
Van Baar 1998	VAS	IRGL	No quality of life measure	No participant-reported treatment success measure	Not reported

AFI: Algofunctional Index; AIMS: Arthritis Impact Measurement Scales; AQoL: Assessment of Quality of Life; AQoL-6D: Assessment of Quality of Life - 6 Dimensions; AQoL-8D: Assessment of Quality of Life - 8 Dimensions; IRGL: Influence of Rheumatic Disease on General Health and Lifestyle questionnaire; KOOS ADL: Knee Injury and Osteoarthritis Outcome Score Funcion in Daily Living subscale; KOOS QoL: Knee Injury and Osteoarthritis Outcome Score knee-related Quality of Life subscale; MSK-HQ: Arthritis Research UK Musculoskeletal Health Questionnaire; NRS: numeric rating scale; OA: osteoarthritis; OASI: Osteoarthritis Screening Index; OMERACT-OARSI: Outcome Measures in Rheumatology-Osteoarthritis Research Society International; SF-12 MCS: Short Form 12 Health Survey Mental Component Summary; SF-36: Short Form 36 Health Survey; SF-36 MCS: Short Form 36 Health Survey Mental Component Summary; VAS: visual analogue scale; WOMAC: Western Ontario and McMaster Universities Arthritis Index

Table 5. Summary of adverse events

Study ID	Study arms	Withdrawals due to serious adverse events	Serious adverse events	Non-serious adverse events
Exercise versus attention control/placebo (30 trials)				
Baker 2001	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	None	None

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Table 5. Summary of adverse events *(Continued)*

Bandak 2022	Saline injection	Reported	5 participants; 5 events: 3 not related to trial treatment, 2 probably not related to trial treatment	40 participants (39%); 48 events: 6 general disorders and administrative site conditions, 38 musculoskeletal and connective tissue disorders, 3 skin and subcutaneous tissue disorders, 1 injury poisoning and procedural complications
	Exercise	Reported	5 participants; 9 events: 1 not related to trial treatment, 8 probably not related to trial treatment	34 participants (34%); 49 events: 3 infections and infestations, 5 general disorders and administrative site conditions, 34 musculoskeletal and connective tissue disorders, 7 injury, poisoning, and procedural complications
Brismée 2007	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	None	Sporadic complaints of minor muscle soreness and foot and knee pain were made mainly during the first few days of the intervention. No other adverse effect associated with the practice of Tai Chi was reported by the participants.
Cheung 2017	Attention control	Reported	None	None
	Exercise (Yoga)	Reported	None	None
	Exercise (aerobic, strengthening)	Reported	None	3 participants. 3 events: 1 fell on ice and experienced back pain (non-study), 1 participant with a history of herniated disk had recurring low back pain (non-serious), 1 participant developed leg pain halfway through the programme, which may have been related to a pre-existing condition of left patella femoral pain syndrome (non-serious)
De Almeida 2020	Attention control	Reported	Not reported	Not reported
	Exercise (circuit training)	Reported	Not reported	Not reported
	Exercise (conventional strength training)	Reported	Not reported	Not reported
Ettinger 1997	Attention control	Not reported	1 person in the health education group had sudden death while walking from her car to an intervention session	Not reported
	Exercise (Aerobic walking)	Not reported	5 occurred in persons participating in the exercise intervention: 2 people in the aerobic	Not reported

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Table 5. Summary of adverse events *(Continued)*

			group fell while walking (1 fall resulted in a fracture of the distal radius);	
	Exercise (Strengthening)	Not reported	2 people fell during participation in the weight-training program; and 1 dropped a dumbbell on her foot resulting in a fracture.	Not reported
Foley 2003	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Jenkinson 2009	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Imoto 2012	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Kang 2022	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Kim 2013	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Kovar 1992	Attention control	Reported	One patient also suffered a fall-related hip fracture during the study period	Not reported
	Exercise	Reported	One patient fell and fractured a hip during one of the supervised sessions (requiring subsequent surgical repair)	Not reported
Kuntz 2018	Attention control	Reported	None	None
	Exercise (yoga)	Reported	None	None
	Exercise (strengthening)	Reported	None	None
Lai 2021	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Liu 2019	Attention control	Reported	None	None
	Exercise 1 (Tai Chi)	Reported	None	None

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Table 5. Summary of adverse events *(Continued)*

	Exercise 2 (Baduanjin)	Reported	None	None
	Exercise 3 (cycling)	Reported	None	None
Lu 2017	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Maurer 1999	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Messier 2004	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	One serious adverse event related to participation in the trial occurred when a participant tripped while exercising and sustained a laceration to his head.	Not reported
Messier 2021	Attention control	Reported	5 participants; 5 events: 1 cancer, 1 knee replacement, 2 hip replacement, 1 myocardial infarction (All unrelated to study)	4 participants: 1 body pain, 2 fall, 1 fracture
	Exercise 1 (high intensity)	Reported	5 particpants; 5 events: 2 cancer, 1 knee replacement, 1 myocardial infarction, 1 TIA (all unrelated to study)	31 participants: 12 body pain, 11 fall, 8 muscle strain
	Exercise 2 (low intensity)	Reported	3 people; 3 events: 1 knee replacement,1 high/uncontrollable blood pressure, 1 stroke (All unrelated to study)	15 people: 7 body pain, 6 fall, 2 muscle strain
Peloquin 1999	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Rego 2023	Attention control	Reported	None	None
	Exercise	Reported	None	None
Santana 2022	Attention control	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Schlenk 2020	Attention control	Reported	None	None
	Exercise	Reported	None	None

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Table 5. Summary of adverse events *(Continued)*

Shen 2022	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Simao 2012	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Song 2003	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Song 2020	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Tsai 2013	Attention control	Reported	None	None
	Exercise	Reported	None	None
Vincent 2020	Attention control	Reported	Not reported	Not reported
	Exercise 1 (Concentric strengthening)	Reported	Not reported	Not reported
	Exercise 2 (Eccentric strengthening)	Reported	Not reported	Not reported
Ye 2020	Attention control	Reported	None	None
	Exercise	Reported	None	None
Exercise versus no treatment/usual care/limited education (60 trials)				
Abbott 2013	Exercise	Reported	None	None
	Usual care	Reported	1 death (Non-trial related)	None
Allen 2018	Wait list	Reported	None	None
	Exercise	Reported	None	4 events: 2 increased knee pain, 1 shoulder pain, 1 ankle pain
An 2008	No intervention	Reported	None	None
	Exercise	Reported	None	None
Bennell 2010	Wait list	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	5 participants (3 = back pain, 1 = back and hip pain, 1 = aggravated varicose veins and knee pain)
Braghin 2018	No intervention	Not reported	Not reported	Not reported

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Table 5. Summary of adverse events *(Continued)*

	Exercise	Not reported	Not reported	Not reported
Bruce-Brand 2012	Standard care	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Chen 2014	No treatment	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Cheung 2014	Wait list	Reported	None	None
	Exercise	Reported	None	None
Chopp-Hurley 2017	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Christensen 2015	No intervention	Not reported	Not reported	50 = musculoskeletal symptoms
	Exercise	Not reported	Not reported	42 = musculoskeletal symptoms
Da Silva 2015	Education	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
De Matos 2019	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
De Rooij 2017	No intervention	Reported	None	Not reported
	Exercise	Reported	None	Not reported
DeVita 2018	No intervention	Reported	None	None
	Exercise	Reported	None	None
Drummen 2023	Usual care	Reported	1 event (Non-study related): angioplasty (n = 1)	Total number of adverse events: 8 Musculoskeletal = 2 (1 increased knee pain, 1 calf strain) Non-musculoskeletal = 4 (1 knee VTE, 1 angioplasty, 1 PMR, 1 flu)
	Exercise	Reported	2 events (Non-study related): pneumonia (n = 1); complications due to anaemia (n = 1)	Total number of adverse events: 26 Musculoskeletal = 11 (3 increased knee pain, 2 foot pain, 3 upper body pain, 1 plantar fasciitis, 2 acute knee injury, unrelated) Non-musculoskeletal = 6 (1 anaemia, 1 pneumonia, 1 bronchi, 3 chest infection)

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Table 5. Summary of adverse events *(Continued)*

Fransen 2001	Wait list	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Fransen 2007	Wait list	Not reported	None	None
	Exercise	Reported	None	None
Gohir 2021	Education	Reported	None	None
	Exercise	Reported	None	None
Guo 2022	Usual care	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Gur 2002	No treatment	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Hay 2006	Education	Reported	None	Not reported
	Exercise	Reported	None	Not reported
Henriksen 2014	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Hopman-Rock 2000	Wait list	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Hu 2020	Education	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Hurley 2007	Usual care	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Jan 2009	No intervention	Not reported	Not reported	Not reported
	Exercise (strengthening – weight bearing)	Reported	Not reported	Not reported
	Exercise (strengthening – non weight bearing)	Reported	Not reported	Not reported
Janyacharoen 2018	Education	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Jorge 2015	Wait list	Not reported	Not reported	Not reported

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Table 5. Summary of adverse events *(Continued)*

	Exercise	Not reported	Not reported	3 people - increased pain in the knees following the intervention
Karadag 2019	Usual care	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Keefe 2004	Standard care	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Kim 2020	Usual care	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Krishnan 2021	No treatment	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Lee 2009	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Lee 2023	Usual care	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Li 2020	Wait list	Not reported	Three participants from reported a fall (unrelated to intervention); 1 occurred while being physically active. Of the remaining 2 participants who had a fall, 1 sustained a vertebral compression fracture	Muscle pain (n=2)
	Exercise	Not reported	Falls were reported by 3 in the IG; of those, 2 fell while being physically active (1 had an ankle sprain)	Muscle pain (n=5)
Lim 2008	No intervention	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Amongst strengthening participants with more malaligned knees, 4 reported increased knee pain and 2 reported increased hip/groin pain that were attributed to the intervention. Amongst strengthening participants with more neutral alignment, 3 had increased knee pain and 1 reported neck pain due to awkward exercise positioning.
Lin 2009	No intervention	Reported	Not reported	Not reported

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Table 5. Summary of adverse events *(Continued)*

	Exercise	Reported	Not reported	Not reported
Lund 2008	No intervention	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	11 patients reported adverse effects of the exercise (44%), 8 patients (32%) reported increased pain during and after exercise, 3 patients (12%) reported swollen knees
Mazloum 2018	Usual care	Not reported	Not reported	Not reported
	Exercise 1 (conventional exercise)	Not reported	Not reported	Not reported
	Exercise 2 (pilates)	Not reported	Not reported	Not reported
Multanen 2014	Social group meeting	Not reported	Not reported	2 visits to the attending physician due to previous meniscal tear injury and cardiac dysrhythmia (unrelated to intervention)
	Exercise	Not reported	Not reported	6 medical consultations due to musculoskeletal injuries or other symptoms: (1) knee swelling; (2) distension of the hamstring muscles; (3) ankle sprain; (4) low back pain; (5) Achilles tendon pain; and (6) asthma-like symptoms during exercise classes
Øiestad 2023	Usual care	Reported	None	None
	Exercise (Strengthening)	Reported	None	None
	Exercise (Aerobic)	Reported	None	1 participant experienced swelling during cycling
O'Reilly 1999	Lifestyle advice	Reported	Not reported	Not reported
	Lifestyle advice + exercise	Reported	Not reported	Not reported
Pazit 2018	Usual care	Reported	None	Not reported
	Exercise 1	Reported	None	Not reported
	Exercise 2	Reported	None	Not reported
Robbins 2022	Education	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Rogind 1998	No intervention	Reported	None	None
	Exercise	Reported	None	None

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Table 5. Summary of adverse events *(Continued)*

Rosedale 2014	Wait list	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Sadeghi 2023	No intervention	Not reported	Not reported	Not reported
	Exercise 1	Not reported	Not reported	Not reported
	Exercise 2	Not reported	Not reported	Not reported
	Exercise 3	Not reported	Not reported	Not reported
Salacinski 2012	Wait list	Reported	Not reported	Not reported
	Exercise	Reported	None	None
Samut 2015	No treatment	Not reported	Not reported	Not reported
	Exercise 1	Not reported	Not reported	Not reported
	Exercise 2	Not reported	Not reported	Not reported
Schilke 1996	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Sekir 2005	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Shahine 2020	Education	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Takacs 2017	No intervention	Not reported	Not reported	None
	Exercise	Not reported	Not reported	9 events: 5 postexercise soreness, 2 back pain, 1 hip soreness, 1 foot pain
Thorstensson 2005	Wait list	Reported	Not reported	Not reported
	Exercise	Reported	Not reported	Not reported
Topp 2002	No intervention	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Vassao 2021	Wait list	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Wallis 2017	Usual care	Reported	None	None
	Exercise	Reported	None	3 events: 2 increased knee pain and were not able to continue the program after week 1, 1 tripped during a

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Table 5. Summary of adverse events *(Continued)*

				supervised walking session and had to seek medical management due to knee pain
Wang 2011	No intervention	Reported	Not reported	Not reported
	Exercise	Reported	None	Two participants in the land group reported increased pain after exercise
Wortley 2013	No treatment	Not reported	Not reported	Not reported
	Exercise 1	Not reported	Not reported	Not reported
	Exercise 2	Not reported	Not reported	Not reported
Xiao 2021	No treatment	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	Not reported
Exercise when added to another co-intervention (compared to that co-intervention alone) (49 trials)				
Alghadir 2019	Routine physiotherapy	Reported	None	None
	Routine physiotherapy + retrowalking	Reported	None	None
	Routine physiotherapy + forward walking	Reported	None	3 events: 3 mild increase of pain
Anadkat 2015	Conventional exercises	Not reported	Not reported	Not reported
	Conventional exercises + Retrowalking	Not reported	Not reported	Not reported
Anwer 2014	Ultrasound	Not reported	Not reported	Not reported
	Ultrasound + exercise	Not reported	Not reported	Not reported
Assar 2020	Drug regimen	Not reported	Not reported	Not reported
	Drug regimen + exercise	Not reported	Not reported	Not reported
Bautch 1997	Education	Not reported	Not reported	Not reported
	Education + exercise	Not reported	Not reported	Not reported
Bennell 2016	Pain coping skills training	Reported	None	Total number of participants reporting adverse events during follow up: 4 Total number of adverse events during follow up: 4 (4 increased knee pain)
	Pain coping skills training + exercise	Reported	None	Total number of participants reporting adverse events during follow up: 7 Total number of adverse events during follow up: 8 (3 increased

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Table 5. Summary of adverse events *(Continued)*

				knee pain, 2 pain in other region, 2 swelling/inflammation, 1 increased stiffness)
Bennell 2022a	Education	Reported	Serious adverse event: 2 (Unrelated to study)	Non-serious adverse event: n=24. (8 knee joint pain, 2 knee stiffness, 9 other lower-limb musculoskeletal symptoms (non-knee), 7 upper-body musculoskeletal symptoms, 2 medical occurrence)
	Education + exercise	Reported	Serious adverse event: 3 (Unrelated to study)	Non-serious adverse event: n=69. (25 knee joint pain, 2 knee stiffness, 23 other lower-limb musculoskeletal symptoms (non-knee), 19 upper-body musculoskeletal symptoms, 19 medical occurrence)
Bennell 2022b	Education	Reported	None	Non-serious related adverse event: n=1. (1 knee pain, 1 back pain, 1 foot pain)
	Education + exercise	Reported	None	Non-serious related adverse event: n=11. (3 knee pain, 4 upper-limb pain, 2 back pain, 2 foot pain, 2 hip pain)
Bhagat 2017	Interferential therapy	Not reported	Not reported	Not reported
	Interferential therapy + exercise	Not reported	Not reported	Not reported
Chang 2012	General physiotherapy	Not reported	Not reported	Not reported
	General physiotherapy + exercise	Not reported	Not reported	Not reported
Chen 2021	Conventional treatment	Reported	None	None
	Conventional treatment + exercise	Reported	None	None
Dincer 2016	TENS	Reported	Not reported	Not reported
	TENS + exercise	Reported	Not reported	Not reported
Dwyer 2015	MMT	Reported	None	None
	MMT + Exercise	Reported	None	None
Gondhalekar 2013	Conventional treatment	Not reported	Not reported	Not reported
	Conventional treatment + exercise	Not reported	Not reported	Not reported
Hernandez 2019	Conventional exercises	Reported	None	None

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Table 5. Summary of adverse events *(Continued)*

	Conventional exercises + Core stability exercises	Reported	None	None
Hinman 2020	Phone advice	Reported	None	None
	Phone advice + exercise	Reported	None	N = 2 (1 ankle/foot pain, 1 knee pain)
Holm 2020	Education	Reported	5 unrelated events (1 renal system, 1 deep venous thrombosis, 3 other "involving other sites")	8 events ("5 involving index knee"; "3 involving other sites")
	Education + exercise	Reported	3 unrelated events (2 renal system, 1 other "involving other sites")	14 events ("3 involving index knee"; "11 involving other sites")
Hsu 2021	Diet	Reported	None	None
	Diet + exercise	Reported	None	None
Huang 2003	Hotpack/ROM	Not reported	Not reported	Not reported
	Hotpack/ROM + exercise	Not reported	Not reported	Not reported
Huang 2005	Hotpack/ROM	Not reported	Not reported	Not reported
	Hotpack/ROM + exercise	Not reported	Not reported	Not reported
Jan 2008	Education	Not reported	Not reported	Not reported
	Exercise (high resistance training)	Not reported	Not reported	Not reported
	Exercise (low resistance training)	Not reported	Not reported	Not reported
Joshi 2019	Conventional exercise + retrowalking	Not reported	Not reported	Not reported
	Conventional exercise	Not reported	Not reported	Not reported
Kholvadia 2019	Low level laser therapy	Not reported	Not reported	Not reported
	Low level laser therapy + Exercise	Not reported	Not reported	Not reported
Kilic 2020	Physiotherapy	Not reported	Not reported	Not reported
	Physiotherapy + exercise	Not reported	Not reported	Not reported
Knoop 2013	Exercise without feedback	Reported	None	None

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Table 5. Summary of adverse events *(Continued)*

	Exercise with feedback (neuromuscular/neuromotor exercise with personalised feedback from PT) + Exercise for knee position	Reported	None	None
Kumar 2013	Supervised exercise + ultrasound	Not reported	Not reported	Not reported
	Supervised exercise + ultrasound + Proprioceptive/balance training	Not reported	Not reported	Not reported
Lee 2019	HA injection	Not reported	Not reported	Not reported
	HA injection + exercise	Not reported	Not reported	Not reported
Marconcin 2018	Education	Not reported	Not reported	Not reported
	Education + exercise	Not reported	Not reported	Not reported
Messier 2013	Diet	Reported	1 event: 1 cancer (Unrelated to study)	Not reported
	Diet + exercise	Reported	6 events: 1 ALS, 1 stroke, 1 lung infection, 2 cancer, 1 staph infection (Unrelated to study)	Not reported
Mete 2022	Physiotherapy	Not reported	Not reported	Not reported
	Physiotherapy + exercise	Not reported	Not reported	Not reported
Mehwish 2023	Knee exercises	Reported	None	None
	Knee exercises + hip strengthening	Reported	None	None
Nahayatbin 2018	Routine physiotherapy	Not reported	Not reported	Not reported
	Routine physiotherapy + exercise 1	Not reported	Not reported	Not reported
	Routine physiotherapy + exercise 2	Not reported	Not reported	Not reported
Nelligan 2021	Education	Reported	None	N = 5 (1 knee pain, 4 pain in other areas)
	Education + exercise	Reported	None	N = 13 (8 knee pain, 6 pain in other areas)
Oh 2020	Education	Not reported	Not reported	Not reported

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Table 5. Summary of adverse events *(Continued)*

	Education + exercise	Not reported	Not reported	Not reported
Ozlu 2023	Conventional treatment	Not reported	Not reported	Not reported
	Conventional treatment + exercise	Not reported	Not reported	4 events: (3 mild nausea, 1 headache)
Qiu 2023	General treatment	Not reported	None	None
	General treatment + hip strengthening exercise	Not reported	None	None
Rafiq 2021a	Usual care	Reported	None	None
	Usual care + exercise	Reported	None	None
Rafiq 2021b	Usual care	Reported	None	1 non-serious adverse event of muscle spasm related to the study.
	Usual care + exercise 1	Reported	2 serious adverse events (1 participant had appendix surgery, 1 underwent gallbladder surgery) (unrelated to study)	None
	Usual care + exercise 2	Reported	None	None
Rafiq 2021c	Education	Reported	None	None
	Education + exercise	Reported	None	None
Rakhshani 2022	Routine physiotherapy	Not reported	Not reported	Not reported
	Routine physiotherapy + exercise	Not reported	Not reported	Not reported
Rosadi 2023	Usual physiotherapy	Not reported	Not reported	Not reported
	Exercise 1	Not reported	Not reported	Not reported
	Exercise 2	Not reported	Not reported	Not reported
Salli 2010	Paracetamol	Not reported	Not reported	Not reported
	Paracetamol + exercise	Not reported	Not reported	Not reported
Samuel 2014	Interferential	Not reported	Not reported	Not reported
	Interferential + exercise	Not reported	Not reported	Not reported
Sedaghatnezhad 2021	Physiotherapy	Not reported	Not reported	Not reported
	Physiotherapy + exercise	Not reported	Not reported	Not reported

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Table 5. Summary of adverse events *(Continued)*

Shabbir 2022	Conventional therapy	Reported	None	Not reported
	Conventional therapy + exercise	Reported	None	Not reported
Shin 2017	Theracurmin	Not reported	Not reported	Not reported
	Theracurmin + exercise	Not reported	Not reported	Not reported
Talbot 2003	Education	Not reported	Not reported	Not reported
	Education + exercise	Not reported	Not reported	Not reported
Thomas 2002	Attention control	Not reported	Not reported	Not reported
	Exercise	Not reported	Not reported	52 of those exercising reported side effects, but these were generally minor (for example, the exercise band was painful around their ankle)
Van Baar 1998	Education	Reported	Not reported	Not reported
	Education + exercise	Reported	Not reported	Not reported

APPENDICES

Appendix 1. Search strategy

Cochrane Central Register of Controlled Trials (CENTRAL)

Search run by D Horrigan

EBM Reviews - Cochrane Central Register of Controlled Trials <February 2023>

1 exp Osteoarthritis/ 9563

2 osteoarthr$.tw. 18520

3 (degenerative adj2 arthritis).tw. 124

4 arthrosis.tw. 669

5 or/1-4 21029

6 exp knee/ 976

7 exp Knee Joint/ 4240

8 knee$.tw. 35383

9 or/6-8 35793

10 exp exercise/ 32730

11 exercis*.tw,kf. 111639

12 exp Exercise Therapy/ 18361

13 exp Exercise Movement Techniques/ 2855

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14 exp Physical Therapy Modalities/ 33335

15 exp Recreation/ 21027

16 exp Physical Fitness/ 4270

17 pilates.tw,kf. 963

18 (Tai Chi or Tai Ji or Taiji or Taijiquan).tw,kf. 1653

19 run$.tw,kf. 25082

20 jog$.tw,kf. 684

21 treadmill$.tw,kf. 9365

22 bicycl$.tw,kf. 3407

23 (cycle$ or cycling).tw,kf. 64152

24 walk$.tw,kf. 36548

25 or/10-24 230786

26 5 and 9 and 25 4803

Embase

Search run by D Horrigan

Embase Classic+Embase <1947 to 2023 March 13>

1 exp osteoarthritis/ 167813

2 osteoarthr$.tw. 131996

3 (degenerative adj2 arthritis).tw. 2204

4 arthrosis.tw. 9597

5 or/1-4 199917

6 Knee/ 89046

7 knee$.tw. 246105

8 or/6-7 258083

9 exp EXERCISE/ 450047

10 exercis*.tw,kf. 508495

11 exp fitness/ 45208

12 exp physiotherapy/ 115766

13 physical therapy.tw,kf. 39985

14 exp pilates/ 1016

15 exp Tai Chi/ 3911

16 (Tai Chi or Tai Ji or Taiji or Taijiquan).tw,kf. 3544

17 jog$.tw,kf. 3545

18 treadmill$.tw,kf. 49538

19 bicycl$.tw,kf. 38314

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20 (cycle$ or cycling).tw,kf. 1018983

21 walk$.tw,kf. 213063

22 or/9-21 1948086

23 exp randomized controlled trial/ 780132

24 controlled clinical trial/ 469209

25 random$.ti,ab. 1956079

26 exp randomization/ 99082

27 intermethod comparison/ 295798

28 placebo.ti,ab. 367792

29 (compare or compared or comparison).ti. 630096

30 ((evaluated or evaluate or evaluating or assessed or assess) and (compare or compared or comparing or comparison)).ab. 2738746

31 (open adj label).ti,ab. 107471

32 ((double or single or doubly or singly) adj (blind or blinded or blindly)).ti,ab. 277744

33 double blind procedure/ 211094

34 parallel group$1.ti,ab. 31888

35 (crossover or cross over).ti,ab. 124850

36 ((assign$ or match or matched or allocation) adj5 (alternate or group$1 or intervention$1 or patient$1 or subject$1 or participant $1)).ti,ab. 412492

37 (assigned or allocated).ti,ab. 486213

38 (controlled adj7 (study or design or trial)).ti,ab. 449108

39 (volunteer or volunteers).ti,ab. 286817

40 human experiment/ 642731

41 trial.ti. 405511

42 or/23-41 6318516

43 5 and 8 and 22 and 42 6494

MEDLINE

Search run by D Horrigan

Ovid MEDLINE(R) ALL <1946 to March 13, 2023>

1 exp Osteoarthritis/ 76014

2 osteoarthr$.tw. 76462

3 (degenerative adj2 arthritis).tw. 1350

4 arthrosis.tw. 5303

5 or/1-4 103151

6 exp knee/ 15923

7 exp Knee Joint/ 71000

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8 knee$.tw. 150866

9 or/6-8 168724

10 exp exercise/ 242910

11 exercis*.tw,kf. 309840

12 exp Exercise Therapy/ 62382

13 exp Exercise Movement Techniques/ 10140

14 exp Physical Therapy Modalities/ 176607

15 exp Recreation/ 243165

16 exp Physical Fitness/ 36000

17 pilates.tw,kf. 598

18 (Tai Chi or Tai Ji or Taiji or Taijiquan).tw,kf. 1927

19 run$.tw,kf. 190135

20 jog$.tw,kf. 2331

21 treadmill$.tw,kf. 33006

22 bicycl$.tw,kf. 22247

23 (cycle$ or cycling).tw,kf. 624576

24 walk$.tw,kf. 121243

25 or/10-24 1487005

26 randomized controlled trial.pt. 588993

27 controlled clinical trial.pt. 95202

28 randomized.ab. 522436

29 placebo.ab. 216698

30 drug therapy.fs. 2575003

31 randomly.ab. 342937

32 trial.ab. 560238

33 groups.ab. 2129412

34 or/19-26 1991207

35 exp animals/ not humans.sh. 5105329

36 34 not 35 1733008

37 5 and 9 and 25 and 36 7019

ClinicalTrials.gov and WHO ICTRP portal

“knee osteoarthritis” and “exercise”

WHAT'S NEW

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Date	Event	Description
3 December 2024	New citation required and conclusions have changed	Authors changed; methods updated; 96 new studies included; certainty of evidence downgraded for some comparisons and outcomes (due to risk of publication bias, number of trials at unclear or high risk of bias, and imprecision); conclusion updated (to "We found low to moderate certainty evidence that exercise probably results in an improvement in pain, physical function, and quality of life in the short-term. However, based on the thresholds for minimal important differences that we used, benefits were of uncertain clinical importance.")
3 December 2024	New search has been performed	New search run 4 January 2024; 10,314 new records identified; 96 new studies added to this update

HISTORY

Review first published: Issue 3, 2003

Date	Event	Description
30 September 2014	New search has been performed	Methods were updated in accordance with current recommendations of The Cochrane Collaboration: 'Risk of bias' assessment and 'Summary of findings' tables were added. Quality of life assessment and study withdrawal rates were added to the update. Pain and physical function outcomes were further disaggregated into immediate post-treatment effects and sustainability (2 to 6 months and > 6 months post treatment).
29 October 2013	New search has been performed	Twenty-three new studies were added to this update: Brismée 2007; Hurley 2007; Yip 2007; An 2008; Doi 2008; Jan 2008; Lim 2008; Lund 2008; Jan 2009; Jenkinson 2009; Lee 2009; Lin 2009; Bennell 2010; Bezalel 2010; Salli 2010; Foroughi 2011; Wang 2011; Bruce-Brand 2012; Chang 2012; Kao 2012; Salacinski 2012; Simao 2012; Abbott 2013. One study that was included in the original review was excluded from this update: Petrella 2000.
12 May 2009	Amended	Minor amendment; see Published notes
13 August 2008	Amended	CMSG ID A007-R
11 August 2008	Amended	Substantive amendment
3 June 2008	Amended	Converted to new review format
3 June 2008	New search has been performed	This updated review is 1 of 2 Cochrane reviews replacing an earlier review, 'Exercise for osteoarthritis of the hip or knee.' Since the time of the original review, the editors decided to subdivide the review into separate conditions. The Background section has been revised to provide information on the specific disorder only, and the search strategy has been revised accordingly. The Methods section has been updated to

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Date	Event	Description
		reflect current methods of the Cochrane Musculoskeletal Review Group. A total of 15 new studies were added to this updated review: Gur 2002; Foley 2003; Huang 2003; Quilty 2003; Song 2003; Talbot 2003; Hughes 2004; Keefe 2004; Messier 2004; Bennell 2005; Huang 2005a; Thorstensson 2005; Hay 2006; Mikesky 2006; Fransen 2007.

CONTRIBUTIONS OF AUTHORS

- Conceptualisation: all review authors
- Screening and data extraction: all review authors independently screened studies and extracted data from those studies that were eligible for inclusion. BJL, RSH, KLB, AK, and LS were not involved in decisions regarding the eligibility of studies that they authored, nor did they conduct data extraction, risk of bias assessments, or grading of certainty of evidence for those studies.
- Data import and analysis: BJL
- Assessment of risk of bias and certainty of evidence: BJL and MH. AD assessed risk of bias and certainty of evidence for the trials authored by BJL.
- Writing original draft: BJL
- Writing review and editing: all
- Some authors involved in previous published versions of this review in 2002, 2008, and 2015 are no longer included in the author byline: Marlene Fransen, Milena Simic, and Sara McConnell. Some of the content retained in this review reflects their contributions.

DECLARATIONS OF INTEREST

- Kim Bennell
  - Royalties from UpToDate knee osteoarthritis guidelines
  - Chair of Osteoarthritis Research Society International Rehabilitation Group
- Andrea Dell'Isola
  - Papers published on osteoarthritis management
  - Affiliation to Lund University and University of Genoa, organisations that have declared an opinion or position on the topic
- Michelle Hall
  - Grant funds from Medical Research Future Fund
  - Grant funds from National Health and Medical Research Council (NHMRC)
  - Publication of opinion relevant to this work
- Rana Hinman
  - Publication of papers and social media posts expressing opinions related to the topic
  - Editorial Board Member role for the Journal of Physiotherapy
  - Grant funds from NHMRC
- Belinda Lawford
  - Steering Committee member of Osteoarthritis Research Society International Rehabilitation Group

Kim Bennell, Rana Hinman, Alexander Kimp, Belinda Lawford and Libby Spiers are authors of studies included in this review. None of these authors were involved in assessment of study eligibility, data extraction, assessment of risk bias and the GRADE assessments for their own studies.

SOURCES OF SUPPORT

Internal sources

- National Health and Medical Research Council, Australia

  Grant awarded to KL Bennell from National Health and Medical Research Council

External sources

- National Health and Medical Research Council, Australia

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KL Bennell, RS Hinman, and M Hall are supported by National Health and Medical Research Council Investigator Grants.

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

- We did not include quasi-randomised trials as there are a lot of relevant RCTs.
- Given the availability of language translation programmes (e.g. Google Translate), we decided to include studies in any language.
- We only included comparators that were:
  - attention control or placebo;
  - no treatment, usual care, or limited education;
  - an intervention that was also equally applied in the exercise group.
- We analysed results depending on comparator, whereas the previous version of the review put all studies together irrespective of the comparator. We did this because of the volume of new studies that we anticipated finding and the fact that the type of comparator can affect the magnitude of the observed effect.
- We did not assess outcomes at 2 to 6 months after treatment because of the anticipated volume of studies, our interest in longer-term outcomes, and to minimise the number of additional analyses.
- We examined the effects of exercise in subgroups that were not in the previous review, including the number of exercise sessions prescribed and the ratio (between exercise and control) of synchronous (i.e., live) contact occasions with a healthcare provider.

INDEX TERMS

Medical Subject Headings (MeSH)

Bias; Exercise; *Exercise Therapy [methods]; *Osteoarthritis, Knee [therapy]; Pain Measurement; *Quality of Life; *Randomized Controlled Trials as Topic

MeSH check words

Aged; Female; Humans; Male; Middle Aged

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